ETSI TS 134 108 V8.6.0 (2009-04)

Technical Specification

Universal Mobile Telecommunications System (UMTS);
LTE;
Common test environments for User Equipment (UE);
Conformance testing
(3GPP TS 34.108 version 8.6.0 Release 8)



Reference
RTS/TSGR-0534108v860

Keywords
LTE, UMTS

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Important notice

Individual copies of the present document can be downloaded from:
http://www.etsi.org

The present document may be made available in more than one electronic version or in print. In any case of existing or perceived difference in contents between such versions, the reference version is the Portable Document Format (PDF). In case of dispute, the reference shall be the printing on ETSI printers of the PDF version kept on a specific network drive within ETSI Secretariat.

Users of the present document should be aware that the document may be subject to revision or change of status.
Information on the current status of this and other ETSI documents is available at
http://portal.etsi.org/tb/status/status.asp

If you find errors in the present document, please send your comment to one of the following services:
http://portal.etsi.org/chaircor/ETSI_support.asp

Copyright Notification

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 2009.
All rights reserved.

DECT™, **PLUGTESTS**™, **UMTS**™, **TIPHON**™, the TIPHON logo and the ETSI logo are Trade Marks of ETSI registered for the benefit of its Members.
3GPP™ is a Trade Mark of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners.
LTE™ is a Trade Mark of ETSI currently being registered
for the benefit of its Members and of the 3GPP Organizational Partners.
GSM® and the GSM logo are Trade Marks registered and owned by the GSM Association.

Intellectual Property Rights

IPRs essential or potentially essential to the present document may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (http://www.etsi.org/legal/home.htm).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Foreword

This Technical Specification (TS) has been produced by the ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities, UMTS identities or GSM identities. These should be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between GSM, UMTS, 3GPP and ETSI identities can be found under http://webapp.etsi.org/key/queryform.asp .

Contents

Intellectual Property Rights 2
Foreword 2
Foreword 42
Introduction 42
1 Scope 43
2 References 43
3 Definitions, abbreviations and symbols 44
3.1 Definitions 44
3.2 Abbreviations 45
3.3 Symbols 45
4 Common requirements of test equipment 46
4.1 General Functional Requirements 46
4.2 Minimum performance levels 46
4.2.1 Supported Cell Configuration 46
4.2.1.1 Supported Channels for FDD Mode 47
4.2.1.1.1 Logical channels 47
4.2.1.1.2 Transport channels 47
4.2.1.1.3 Physical channels 47
4.2.1.2 Supported Channels for TDD Mode 48
4.2.1.2.1 Logical channels 48
4.2.1.2.2 Transport channels 48
4.2.1.2.3 Physical channels (3.84 Mcps option) 49
4.2.1.2.4 Physical channels (1.28 Mcps option) 49
4.2.1.2A Supported Channels for MBSFN (FDD and TDD Mode) 49
4.2.1.2A.1 Logical channels 49
4.2.1.2A.2 Transport channels 50
4.2.1.2A.3 Physical channels (3.84/7.68 Mcps options) 50
4.2.1.3 Support of T_{cell} timing offset 50
4.2.2 RF Performance 50
4.2.2.1 Frequency of Operation 50
4.2.2.2 Power Level Setting Accuracy 50
4.2.2.3 Uplink Power Control 51
4.2.2.4 Uplink Signal Handling 51
4.2.2.5 Uplink Sensitivity 51
4.2.3 Timers Tolerances 51
5 Reference test conditions 51
5.1 Test frequencies 51
5.1.1 FDD Mode Test frequencies 51
5.1.1.1 FDD reference test frequencies for Operating Band I 51
5.1.1.2 FDD reference test frequencies for Operating Band II 52
5.1.1.3 FDD reference test frequencies for Operating Band III 52
5.1.1.4 FDD reference test frequencies for Operating Band IV 52
5.1.1.5 FDD reference test frequencies for Operating Band V 52
5.1.1.6 FDD reference test frequencies for Operating Band VI 52
5.1.1.7 FDD reference test frequencies for Operating Band VII 52
5.1.1.8 FDD reference test frequencies for Operating Band VIII 52
5.1.1.9 FDD reference test frequencies for Operating Band IX 53
5.1.1.10 FDD reference test frequencies for Operating Band X 53
5.1.1.11 FDD reference test frequencies for Operating Band XI 53
5.1.1.12 FDD reference test frequencies for Operating Band XII 53
5.1.1.13 FDD reference test frequencies for Operating Band XIII 53
5.1.1.14 FDD reference test frequencies for Operating Band XIV 53

5.1.2 TDD Mode Test frequencies....53
5.1.2.1 Standard TDD reference test frequencies (3.84 Mcps option)....53
5.1.2.2 Standard TDD reference test frequencies (1.28 Mcps option)....54
5.1.2.3 Standard TDD reference test frequencies (7.68 Mcps option)....54
5.2 Radio conditions....54
5.2.1 Normal propagation condition....54
5.2.2 Static propagation condition....54
5.2.3 Multi-path fading propagation conditions....54
5.2.4 Moving propagation conditions....54
5.2.5 Birth-Death propagation conditions....54
5.2.6 High speed train conditions....54
5.3 Standard test signals....55
5.4 Signal levels....55
5.4.1 Downlink signal levels....55
5.4.2 Uplink signal levels....55
5.5 Downlink Physical Channels Code Allocation....55
5.5.1 Downlink physical channels code allocation for Signalling (FDD)....55
5.5.1.1 Downlink physical channels code allocation for non-HSDPA test cases....55
5.5.1.2 Downlink physical channels code allocation for HSDPA test cases....55
5.5.1.3 Downlink physical channels code allocation for E-DCH test cases....56
5.5.1.4 Downlink physical channels code allocation for MBMS test cases....57
5.5.2 Downlink physical channels code allocation for Signalling (TDD)....59
5.5.3 Downlink physical channels code allocation for RF....59

6 Reference system configurations....60
6.1 Simulated network environments....60
6.1.0a Default Master Information Block and Scheduling Block messages....60
6.1.0a.1 Grouping SIBs for testing....60
6.1.0a.2 SIB configurations....61
6.1.0a.3 SIB default schedule....61
Contents of Master Information Block PLMN type is the case of GSM-MAP....63
Contents of Scheduling Block 1 (FDD and 1.28 Mcps TDD)....64
Contents of Scheduling Block 1 (3.84 Mcps TDD and 7.68 Mcps TDD)....64
6.1.0a.4 SIB special schedules....65
6.1.0a.4.1 SIB schedule for two S-CCPCH or two PRACH (For FDD)....65
6.1.0a.4.2 SIB schedule for Idle Mode, Measurement and Inter RAT UTRAN to GERAN test cases....66
6.1.0a.4.3 SIB schedule for Inter RAT handover GERAN to UTRAN test cases....67
6.1.0a.4.4 SIB schedule for MBMS test cases....68
6.1.0a.4.5 SIB schedule for MBMS MBSFN test cases (FDD, 3.84 Mcps TDD and 7.68 Mcps TDD)....69
Contents of Master Information Block in the case where PLMN type is GSM-MAP....69
6.1.0b Default System Information Block Messages....70
Contents of System Information Block type 1 (supported PLMN type is GSM-MAP)....70
Contents of System Information Block type 2....70
Contents of System Information Block type 3 (FDD)....71
Contents of System Information Block type 3 (3.84 Mcps TDD, 1.28 Mcps TDD and 7.68 Mcps TDD)....72
Contents of System Information Block type 4 in connected mode (FDD)....72
Contents of System Information Block type 4 in connected mode (similar to SIB type3) (3.84 Mcps TDD, 1.28 Mcps TDD and 7.68 Mcps TDD)....73
Contents of System Information Block type 5 (FDD)....73
Contents of System Information Block type 5 (3.84 Mcps TDD)....79
Contents of System Information Block type 5 (1.28 Mcps TDD)....82
Contents of System Information Block type 5 (7.68 Mcps TDD)....85
Contents of System Information Block type 5bis (FDD)....89
Contents of System Information Block type 6 in connected mode (FDD)....89
Contents of System Information Block type 6 in connected mode (similar to SIB type 5) (3.84 Mcps TDD)....89
Contents of System Information Block type6 In connected mode (1.28 Mcps TDD)....92
Contents of System Information Block type 6 in connected mode (7.68 Mcps TDD)....92
Contents of System Information Block type 7 (FDD)....92
Contents of System Information Block type 7 (TDD)....93
Contents of System Information Block type 8, 9 (only for FDD R99 and Rel-4)....93
Contents of System Information Block type 10 (only for FDD R99 and Rel-4)....93
Contents of System Information Block type 11 (FDD)....93

Contents of System Information Block type 11 (3.84 Mcps, 1.28 Mcps and 7.68 Mcps TDD)....97
Contents of System Information Block type 12 in connected mode (FDD)....100
Contents of System Information Block type 12 in connected mode (3.84 Mcps, 1.28 Mcps and 7.68 Mcps TDD)....100
Contents of System Information Block type 13 (used when supported PLMN type is ANSI-41)....100
Contents of System Information Block type 14 (3.84 Mcps and 7.68 Mcps TDD)....101
Contents of System Information Block type 16 (FDD)....101
Contents of System Information Block type17 (3.84 Mcsps TDD and 1.28 Mcps TDD)....107
Contents of System Information Block type 18....107
6.1.1 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second SCCPCH....107
Contents of System Information Block type 5 (FDD)....107
Contents of System Information Block type 5bis (FDD)....111
Contents of System Information Block type 5 (3.84 Mcps TDD)....111
Contents of System Information Block type 5 (1.28 Mcps TDD)....115
Contents of System Information Block type 5 (3.84 Mcps TDD)....115
Contents of System Information Block type 6 in connected mode (FDD)....119
Contents of System Information Block type 6 in connected mode (3.84 Mcps TDD)....119
Contents of System Information Block type 6 in connected mode (1.28 Mcps TDD)....119
Contents of System Information Block type 6 in connected mode (7.68 Mcps TDD)....119
6.1.2 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH, RB for CTCH + SRBs for CCCH/BCCH in the second SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the third SCCPCH (FDD only)....119
Contents of System Information Block type 5 (FDD)....119
Contents of System Information Block type 5bis (FDD)....123
Contents of System Information Block type 6 in connected mode (FDD)....123
6.1.3 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second and third SCCPCHs....124
Contents of Scheduling Block 1 (FDD)....125
Contents of System Information Block type 5 (FDD)....125
Contents of System Information Block type 5bis (FDD)....130
Contents of System Information Block type 5 (3.84 Mcps TDD)....130
Contents of System Information Block type 5 (1.28 Mcps TDD)....130
6.1.4 Default parameters for 1 to 8 cell environments....130
Default settings for cell No.1 (FDD)....130
Contents of System Information Block type 11 for cell No.1 (FDD)....130
Contents of System Information Block type 12 in connected mode for cell No.1 (FDD)....130
Default settings for cell No.1 (TDD)....131
Contents of System Information Block type 11 for cell No.1 (TDD)....131
Contents of System Information Block type 12 in connected mode for cell No.1 (TDD)....131
Cell No.2....131
Default settings for cell No.2 (FDD)....131
Contents of System Information Block type 11 for cell No.2 (FDD)....131
Default settings for cell No.2 (TDD)....132
Contents of System Information Block type 11 for cell No.2 (TDD)....132
Cell No.3....133
Default settings for cell No.3 (FDD)....133
Contents of System Information Block type 11 for cell No.3 (FDD)....133
Default settings for cell No.3 (TDD)....135
Contents of System Information Block type 11 for cell No.3 (TDD)....135
Cell No.4....135
Default settings for cell No.4 (FDD)....136
Contents of System Information Block type 11 for cell No.4 (FDD)....136
Default settings for cell No.4 (TDD)....137
Contents of System Information Block type 11 for cell No.4 (TDD)....137
Cell No.5....138
Default settings for cell No.5 (FDD)....138
Contents of System Information Block type 11 for cell No.5 (FDD)....139
Default settings for cell No.5 (TDD)....140
Contents of System Information Block type 11 for cell No.5 (TDD)....140
Cell No.6....141
Default settings for cell No.6 (FDD)....141

Contents of System Information Block type 11 for cell No.6 (FDD) 141
Default settings for cell No.6 (TDD) 143
Contents of System Information Block type 11 for cell No.6 (TDD) 143
Cell No.7 144
Default settings for cell No.7 (FDD) 144
Contents of System Information Block type 11 for cell No.7 (FDD) 144
Default settings for cell No.7 (TDD) 145
Contents of System Information Block type 11 for cell No.7 (TDD) 145
Cell No.8 146
Default settings for cell No.8 (FDD) 146
Contents of System Information Block type 11 for cell No.8 (FDD) 146
Default settings for cell No.8 (TDD) 147
Contents of System Information Block type 11 for cell No.8 (TDD) 147
Cell No.9 148
Contents of System Information for cell No.9 (GSM) 148
Default settings for cell No.9 (GSM) 148
Cell No.10 148
Contents of System Information for cell No.10 (GSM) 148
Default settings for cell No.10 (GSM) 148
Cell No.11 148
Default settings for cell No.11 (FDD) 148
Contents of System Information Block type 11 for cell No.11 (FDD) 149
6.1.4.1 Default Cell parameters Two PLMN in UTRAN test scenario 149
Contents of System Information Block type 18 for cell No.1, 2, 3, 7, 8 149
Contents of System Information Block type 18 for cell No.4, 5, 6 150
6.1.4.2 Default Cell parameters Three PLMN in UTRAN test scenario 150
Contents of System Information Block type 18 for cell No.1, 2, 3 150
Contents of System Information Block type 18 for cell No.4, 5, 6 150
Contents of System Information Block type 18 for cell No.7, 8 150
6.1.4.3 Default Cell parameters for MBMS 21 to 28 cell environnements 151
Cell No.21 151
Default settings for cell No.21 (FDD) 151
Contents of System Information Block type 11 for cell No.21 (FDD) 151
Default settings for cell No.21 (TDD) 153
Contents of System Information Block type 11 for cell No.21 (TDD) 153
Cell No.22 154
Default settings for cell No.22 (FDD) 154
Contents of System Information Block type 11 for cell No.22 (FDD) 154
Default settings for cell No.22 (TDD) 156
Contents of System Information Block type 11 for cell No.22 (TDD) 156
Cell No.23 158
Default settings for cell No.23 (FDD) 158
Contents of System Information Block type 11 for cell No.23 (FDD) 158
Default settings for cell No.23 (TDD) 160
Contents of System Information Block type 11 for cell No.23 (TDD) 160
Cell No.24 161
Default settings for cell No.24 (FDD) 161
Contents of System Information Block type 11 for cell No.24 (FDD) 161
Default settings for cell No.24 (TDD) 164
Contents of System Information Block type 11 for cell No.24 (TDD) 164
Cell No.25 165
Default settings for cell No.25 (FDD) 165
Contents of System Information Block type 11 for cell No.25 (FDD) 166
Default settings for cell No.25 (TDD) 168
Contents of System Information Block type 11 for cell No.25 (TDD) 168
Cell No.26 170
Default settings for cell No.26 (FDD) 170
Contents of System Information Block type 11 for cell No.26 (FDD) 170
Default settings for cell No.26 (TDD) 172
Contents of System Information Block type 11 for cell No.26 (TDD) 172
Cell No.27 174
Default settings for cell No.27 (FDD) 174

Contents of System Information Block type 11 for cell No.27 (FDD) 174
Default settings for cell No.27 (TDD) 176
Contents of System Information Block type 11 for cell No.27 (TDD) 176
Cell No.28 178
Default settings for cell No.28 (FDD) 178
Contents of System Information Block type 11 for cell No.28 (FDD) 178
Default settings for cell No.28 (TDD) 180
Contents of System Information Block type 11 for cell No.28 (TDD) 180
6.1.4.4 Default Cell parameters for MBSFN 31 to 38 cell environments 181
Cell No.31 181
Default settings for cell No.31 (3.84 Mcps TDD and 7.68 Mcps TDD) 181
Default settings for cell No.31 (FDD) 182
Contents of System Information Block type 3 for cell No.31 (FDD, 3.84 Mcps TDD and 7.68 Mcps TDD) 182
Contents of System Information Block type 5 for cell No.31 (FDD) 182
Contents of System Information Block type 5 for cell No.31 (3.84 Mcps TDD) 183
Contents of System Information Block type 5 for cell No.31 (7.68 Mcps TDD) 186
Contents of System Information Block type 11 for cell No.31 (FDD) 189
Contents of System Information Block type 11 for cell No.31 (3.84 Mcps and 7.68 Mcps TDD) 189
Cell No.32 190
Default settings for cell No.32 (FDD) 191
Default settings for cell No.32 (3.84/7.68 Mcps TDD) 191
Contents of System Information Block type 5 for cell No.32 (FDD) 191
Contents of System Information Block type 5 for cell No.32 (3.84 Mcps TDD) 191
Contents of System Information Block type 5 for cell No.32 (7.68 Mcps TDD) 191
Contents of System Information Block type 11 for cell No.32 (FDD) 191
Contents of System Information Block type 11 for cell No.32 (3.84 Mcps and 7.68 Mcps TDD) 192
Cell No.33 192
Default settings for cell No.33 (FDD) 192
Default settings for cell No.33 (3.84/7.68 Mcps TDD) 192
Contents of System Information Block type 5 for cell No.33 (FDD) 193
Contents of System Information Block type 5 for cell No.33 (3.84 Mcps and 7.68 Mcps TDD) 193
Contents of System Information Block type 11 for cell No.33 (FDD) 194
Contents of System Information Block type 11 for cell No.33 (3.84 Mcps and 7.68 Mcps TDD) 194
Cell No.34 195
Default settings for cell No.34 (FDD) 195
Default settings for cell No.34 (3.84/7.68 Mcps TDD) 195
Contents of System Information Block type 5 for cell No.34 (FDD) 195
Contents of System Information Block type 5 for cell No.34 (3.84 Mcps and 7.68 Mcps TDD) 195
Contents of System Information Block type 11 for cell No.34 (FDD) 196
Contents of System Information Block type 11 for cell No.34 (3.84 Mcps and 7.68 Mcps TDD) 197
Cell No.35 197
Default settings for cell No.35 (FDD) 197
Default settings for cell No.35 (3.84/7.68 Mcps TDD) 197
Contents of System Information Block type 5 for cell No.35 (FDD) 197
Contents of System Information Block type 5 for cell No.35 (3.84 Mcps and 7.68 Mcps TDD) 198
Contents of System Information Block type 11 for cell No.35 (FDD) 199
Contents of System Information Block type 11 for cell No.35 (3.84 Mcps and 7.68 Mcps TDD) 199
Cell No.36 199
Default settings for cell No.36 (FDD) 200
Default settings for cell No.36 (3.84/7.68 Mcps TDD) 200
Contents of System Information Block type 5 for cell No.36 (FDD) 200
Contents of System Information Block type 5 for cell No.36 (3.84 Mcps and 7.68 Mcps TDD) 200
Contents of System Information Block type 11 for cell No.36 (FDD) 201
Contents of System Information Block type 11 for cell No.36 (3.84 Mcps and 7.68 Mcps TDD) 201
Cell No.37 202
Default settings for cell No.37 (FDD) 202
Default settings for cell No.37 (3.84/7.68 Mcps TDD) 202
Contents of System Information Block type 5 for cell No.37 (FDD) 202
Contents of System Information Block type 5 for cell No.37 (3.84 Mcps TDD) 202
Contents of System Information Block type 5 for cell No.37 (7.68 Mcps TDD) 202
Contents of System Information Block type 11 for cell No.37 (FDD) 203
Contents of System Information Block type 11 for cell No.37 (3.84 Mcps and 7.68 Mcps TDD) 203

Cell No.38 204
Default settings for cell No.38 (FDD) 204
Default settings for cell No.38 (3.84/7.68 Mcps TDD) 204
Contents of System Information Block type 5 for cell No.38 (FDD) 204
Contents of System Information Block type 5 for cell No.38 (3.84 Mcps TDD) 204
Contents of System Information Block type 5 for cell No.38 (7.68 Mcps TDD) 204
Contents of System Information Block type 11 for cell No.38 (FDD) 204
Contents of System Information Block type 11 for cell No.38 (3.84 Mcps and 7.68 Mcps TDD) 205
6.1.5 Reference Radio Conditions (FDD) 205
6.1.5.1 HARQ Transmission Parameters (FDD) 207
6.1.5.2 Inter-band testing (FDD) 207
6.1.6 Reference Radio Conditions (TDD) 207
6.1.7 Reference Radio Conditions (GSM) 208
6.2 Number of neighbour cells 208
6.2.1 Basic Network 208
6.2.2 Soft Handover Network (FDD) 208
6.2.3 Hard Handover Network 209
6.2.4 'Roaming' Network 209
6.3 Cell/BS codes etc 209
6.4 Routing/location area 209
6.5 Network options settings 209
6.6 Power control mode 209
6.6.1 Downlink Power Control 209
6.6.1.1 Outer Loop Power Control 209
6.6.1.2 Inner Loop Power Control 209
6.6.2 Uplink Power Control 209
6.6.2.1 Outer Loop Power Control 209
6.6.2.2 Inner Loop Power Control (FDD) 209
6.7 Tx Diversity modes 209
6.7.1 Non-Diverse Operation 210
6.7.2 Diverse Operation 210
6.7.2.1 Diverse Operation (FDD mode) 210
6.7.2.2 Diverse Operation (TDD mode) 210
6.7.2.2.1 3.84 Mcps option 210
6.7.2.2.2 1.28 Mcps option 210
6.7.2.2.1 7.68 Mcps option 210
6.8 Compressed mode parameters 211
6.8.1 Single compressed mode pattern 211
6.8.1.1 Inter Frequency FDD measurement 211
6.8.1.2 Inter Frequency TDD measurement 211
6.8.1.3 Inter RAT measurement (GSM - Carrier RSSI) 212
6.8.1.4 Inter RAT measurement (GSM - Initial BSIC Identification) 212
6.8.1.5 Inter RAT measurement (GSM - BSIC re-confirmation) 213
6.8.2 Multiple compressed mode patterns 213
6.8.2.1 Inter RAT measurement GSM 213
6.8.2.2 Inter Frequency FDD measurement & Inter RAT measurement GSM 214
6.8.2.3 Inter Frequency FDD measurement & Inter Frequency TDD measurement 215
6.8.2.4 Inter Frequency TDD measurement & Inter RAT measurement GSM 215
6.8.2.5 Inter Frequency FDD measurement & Inter Frequency TDD measurement & Inter RAT measurement GSM 215
6.9 BCCH parameters 215
6.10 Reference Radio Bearer configurations used in Radio Bearer interoperability testing 216
6.10.1 QoS Architecture and RAB attributes 216
6.10.2 RAB and signalling RB for FDD 217
6.10.2.1 RABs and signalling RBs 217
6.10.2.2 Combinations of RABs and Signalling RBs 219
6.10.2.3 Example of linkage between RABs and services 225
6.10.2.4 Typical radio parameter sets 225
6.10.2.4.1 Combinations on DPCH 226
6.10.2.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH 226
6.10.2.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH 227
6.10.2.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH 229

6.10.2.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....230
6.10.2.4.1.4a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....232
6.10.2.4.1.4b Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH....233
6.10.2.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....235
6.10.2.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....237
6.10.2.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....239
6.10.2.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....241
6.10.2.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....242
6.10.2.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....244
6.10.2.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....246
6.10.2.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....248
6.10.2.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....250
6.10.2.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....252
6.10.2.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....253
6.10.2.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....255
6.10.2.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....256
6.10.2.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....257
6.10.2.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....259
6.10.2.4.1.18 Void....260
6.10.2.4.1.19 Void....260
6.10.2.4.1.20 Void....260
6.10.2.4.1.21 Void....260
6.10.2.4.1.22 Void....260
6.10.2.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....260
6.10.2.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....262
6.10.2.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....263
6.10.2.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....265
6.10.2.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH....266
6.10.2.4.1.24 Void....268
6.10.2.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....268
6.10.2.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....269
6.10.2.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....269
6.10.2.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....270
6.10.2.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....271

6.10.2.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....272
6.10.2.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....273
6.10.2.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....274
6.10.2.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....275
6.10.2.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....275
6.10.2.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....276
6.10.2.4.1.36 Void....277
6.10.2.4.1.37 Void....277
6.10.2.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....277
6.10.2.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....278
6.10.2.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....280
6.10.2.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....282
6.10.2.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....283
6.10.2.4.1.38e Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....285
6.10.2.4.1.38f Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....286
6.10.2.4.1.38g Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....287
6.10.2.4.1.38h Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....288
6.10.2.4.1.38i Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....290
6.10.2.4.1.38j Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....292
6.10.2.4.1.38k Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing)....293
6.10.2.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....295
6.10.2.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....296
6.10.2.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....296
6.10.2.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....297
6.10.2.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....298
6.10.2.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....299
6.10.2.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....302
6.10.2.4.1.46 Void....303
6.10.2.4.1.47 Void....303
6.10.2.4.1.48 Void....303
6.10.2.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....303
6.10.2.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....304

6.10.2.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 305
6.10.2.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 306
6.10.2.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 307
6.10.2.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 308
6.10.2.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 309
6.10.2.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 310
6.10.2.4.1.54 Void 310
6.10.2.4.1.55 Void 310
6.10.2.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 310
6.10.2.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 312
6.10.2.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 314
6.10.2.4.1.58a Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 315
6.10.2.4.1.59 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 316
6.10.2.4.1.60 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 319
6.10.2.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 320
6.10.2.4.1.62 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH 323
6.10.2.4.1.63 Interactive or background / UL:64 DL:768 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH 325
6.10.2.4.2 Combinations on PDSCH and DPCH 326
6.10.2.4.2.1 Void 326
6.10.2.4.2.2 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH 326
6.10.2.4.2.3 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH 327
6.10.2.4.2.4 Void 329
6.10.2.4.2.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 329
6.10.2.4.2.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 329
6.10.2.4.3 Combinations on SCCPCH 330
6.10.2.4.3.1 Stand-alone signalling RB for PCCH 330
6.10.2.4.3.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 331
6.10.2.4.3.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 332
6.10.2.4.3.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 333
6.10.2.4.3.4 RB for CTCH + SRB for CCCH + SRB for BCCH 334
6.10.2.4.3.5 64.8kbps RB for MTCH with 80 ms TTI 335
6.10.2.4.3.6 129.6 kbps RB for MTCH with 80 ms TTI 336
6.10.2.4.3.7 259.2 kbps RB for MTCH with 40 ms TTI 337
6.10.2.4.3.8 7.6 kbps signalling RB for MCCH 338
6.10.2.4.3.9 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH + SRB for MCCH 338
6.10.2.4.4 Combinations on PRACH 339
6.10.2.4.4.1 Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH 339

6.10.2.4.4.2 Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH 340
6.10.2.4.5 Combinations on DPCH and HS-PDSCH 342
6.10.2.4.5.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 343
6.10.2.4.5.1a Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 346
6.10.2.4.5.2 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 347
6.10.2.4.5.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 347
6.10.2.4.5.3a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 349
6.10.2.4.5.4 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 349
6.10.2.4.5.4a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 351
6.10.2.4.5.5 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 351
6.10.2.4.5.5.1 Uplink 351
6.10.2.4.5.5a Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 352
6.10.2.4.5.6 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 354
6.10.2.4.5.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 356
6.10.2.4.5.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Interactive or Background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH 357
6.10.2.4.5.9 Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 360
6.10.2.4.5.10 Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 363
6.10.2.4.6 Combinations on HS-PDSCH and E-DPDCH 364
6.10.2.4.6.1a Stand-alone UL: [max bit rate depending on UE category and TTI] DL:[max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 364
6.10.2.4.6.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 364
6.10.2.4.6.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH 366
6.10.2.4.6.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 367
6.10.2.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 369

6.10.2.4.6.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH 369
6.10.2.4.6.6 Conversational / unknown or speech / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)....370
6.10.2.4.6.7 Conversational / unknown or speech / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6) 372
6.10.2.4.6.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH 374
6.10.2.4.6.9 Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) kbps DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB on E-DCH and HS-DSCH + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH....375
6.10.3 RAB and signalling RB for TDD....378
6.10.3.1 RABs and signalling RBs....378
6.10.3.2 Combinations of RABs and Signalling RBs 380
6.10.3.3 Example of linkage between RABs and services....385
6.10.3.4 Typical radio parameter sets 386
6.10.3.4.1 Combinations on DPCH....386
6.10.3.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH 386
6.10.3.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe) 387
6.10.3.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH 389
6.10.3.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH 391
6.10.3.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....392
6.10.3.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....394
6.10.3.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....396
6.10.3.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....397
6.10.3.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....399
6.10.3.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH 401
6.10.3.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH 403
6.10.3.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 404
6.10.3.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 406
6.10.3.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH....408
6.10.3.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH....409
6.10.3.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....411

6.10.3.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....412
6.10.3.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....414
6.10.3.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....415
6.10.3.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....417
6.10.3.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....418
6.10.3.4.1.18 Void....420
6.10.3.4.1.19 Void....420
6.10.3.4.1.20 Void....420
6.10.3.4.1.21 Void....420
6.10.3.4.1.22 Void....420
6.10.3.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....420
6.10.3.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....422
6.10.3.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....422
6.10.3.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....424
6.10.3.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH....426
6.10.3.4.1.24 Void....427
6.10.3.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....427
6.10.3.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....428
6.10.3.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....429
6.10.3.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....430
6.10.3.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....431
6.10.3.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....432
6.10.3.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....432
6.10.3.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....433
6.10.3.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....434
6.10.3.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....435
6.10.3.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....436
6.10.3.4.1.36 Void....437
6.10.3.4.1.37 Void....437
6.10.3.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....437
6.10.3.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....438
6.10.3.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....440
6.10.3.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....441
6.10.3.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....443

6.10.3.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....445
6.10.3.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....446
6.10.3.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....447
6.10.3.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....449
6.10.3.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....450
6.10.3.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....452
6.10.3.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....453
6.10.3.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....454
6.10.3.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....456
6.10.3.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....457
6.10.3.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....459
6.10.3.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....460
6.10.3.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....463
6.10.3.4.1.46 Void....464
6.10.3.4.1.47 Void....464
6.10.3.4.1.48 Void....464
6.10.3.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....464
6.10.3.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....465
6.10.3.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....466
6.10.3.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....467
6.10.3.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....469
6.10.3.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....470
6.10.3.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....470
6.10.3.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....471
6.10.3.4.1.54 Void....472
6.10.3.4.1.55 Void....472
6.10.3.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....472
6.10.3.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....473
6.10.3.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....475
6.10.3.4.1.59 Reserved for future use....476
6.10.3.4.1.60 Reserved for future use....476
6.10.3.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....476
6.10.3.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH....478
6.10.3.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH....478

6.10.3.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH 484
6.10.3.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 486
6.10.3.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 487
6.10.3.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH 489
6.10.3.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH.....489
6.10.3.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.....491
6.10.3.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH ..492
6.10.3.4.4 Combinations on SCCPCH 493
6.10.3.4.4.1 Stand-alone signalling RB for PCCH 493
6.10.3.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 494
6.10.3.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 495
6.10.3.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH 496
6.10.3.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 497
6.10.3.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 497
6.10.3.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH 498
6.10.3.4.4.5 64.8kbps RB for MTCH with 80 ms TTI.....499
6.10.3.4.4.6 129.6kbps RB for MTCH with 80 ms TTI.....500
6.10.3.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI.....501
6.10.3.4.4.8 7.6 kbps signalling RB for MCCH.....502
6.10.3.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI 503
6.10.3.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI 504
6.10.3.4.4.11 497.6kbps RB for MBSFN MTCH with 80 ms TTI 505
6.10.3.4.4.12 7.2 kbps signalling RB for MBSFN MCCH 506
6.10.3.4.5 Combinations on PRACH 506
6.10.3.4.5.1 SRB for CCCH + SRB for DCCH 506
6.10.3.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH 507
6.10.3.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH.....507
6.10.3.4.6 Combinations on DPCH and HS-PDSCH 508
6.10.3.4.6.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 508
6.10.3.4.6.1.1 Uplink.....508
6.10.3.4.6.1.2 Downlink.....508
6.10.3.4.6.2 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 510
6.10.3.4.6.3 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 511
6.10.3.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.....511
6.10.3.4.6.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.....513
6.10.3.4.6.6 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.....513

6.10.3.4.6.7 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....515
6.10.3.4.6.8 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....515
6.10.3.4.6.9 Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....517
6.10.3.4.6.10 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....518
6.10.3.4.6.11 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....519
6.10.3.4.7 Combinations on HS-PDSCH and E-PUCH....521
6.10.3.4.7.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH....521
6.10.3.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH....523
6.10.3.4.7.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....523
6.10.3.4.7.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH....524
6.11 Common Radio Bearer configurations for other test purposes....526
6.11.1 Unacknowledged Mode Radio Bearer configuration (7 bit Length Indicator)....526
6.11.1.1 Transport channel parameters for the Uplink RAB....526
6.11.1.2 TFCS....526
6.11.1.3 Transport channel parameters for the Downlink RAB....526
6.11.1.4 TFCS....527
6.11.1a Streaming / unknown / DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....527
6.11.1b Streaming / unknown / DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....528
6.11.1c 8kbps RB for MBSFN MTCH (3.84 Mcps TDD)....529
6.11.1d 8kbps RB for MBSFN MTCH (7.68 Mcps TDD)....529
6.11.2 Unacknowledged Mode Radio Bearer configuration (15 bit Length Indicator)....530
6.11.2.1 Transport channel parameters for the Uplink RAB....530
6.11.2.2 Transport channel parameters for the Downlink RAB....530
6.11.2a 64kbps RB for MBSFN MTCH (3.84 Mcps TDD)....531
6.11.2b 64kbps RB for MBSFN MTCH (7.68 Mcps TDD)....531
6.11.3 Acknowledged Mode Radio Bearer configuration (7 bit Length Indicator)....532
6.11.3.1 Transport channel parameters for the Uplink RAB....532
6.11.3.2 TFCS....532
6.11.3.3 Transport channel parameters for the Downlink RAB....532
6.11.3.4 TFCS....532
6.11.4 Acknowledged Mode Radio Bearer configuration (15 bit Length Indicator)....533
6.11.4.1 Transport channel parameters for the Uplink RAB....533
6.11.4.2 Transport channel parameters for the Downlink RAB....533
6.11.4a Reference Radio Bearer configurations used in MAC-hs testing....533
6.11.4a.1 5 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB....533
6.11.4a.1.1 Uplink....533
6.11.4a.1.1.1 Uplink Transport channel parameters for DCH....533
6.11.4a.1.1.2 Uplink physical channel parameters....534

6.11.4a.1.2 Downlink 534
6.11.4a.1.2.1 Transport channel parameters for HS-DSCH 534
6.11.4a.1.2.2 Downlink Transport channel parameters for DCH 535
6.11.4a.1.2.3 Downlink physical channel parameters 535
6.11.4b Interactive or background / UL: 0 kbps DL: 0 kbps PS RAB 536
6.11.4b.1 Uplink 536
6.11.4b.1.1 Uplink Transport channel parameters for DCH 536
6.11.4b.2 Downlink 536
6.11.4b.2.1 Downlink Transport channel parameters for DCH 536
6.11.4c Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 537
6.11.4c.1 Uplink 537
6.11.4c.1.1 Transport channel parameters 537
6.11.4c.1.1.1 Transport channel parameters for E-DCH 537
6.11.4c.1.2 Physical channel parameters 538
6.11.4c.1.2.1 Physical channel parameters on E-DPDCH 538
6.11.4c.1.2.2 Physical channel parameters for DPCH 538
6.11.4c.2 Downlink 538
6.11.4c.2.1 Transport channel parameters 538
6.11.4c.2.1.1 Transport channel parameters for HS-DSCH 538
6.11.4c.2.2 Physical channel parameters 538
6.11.4c.2.2.1 Physical channel parameters on DPCH 538
6.11.4c.2.2.2 Physical channel parameters on HS-PDSCH 538
6.11.4d Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + DL: 3.4 kbps SRBs for DCCH on E-DCH and DCH 539
6.11.4d.1 Uplink 539
6.11.4d.1.1 Transport channel parameters 539
6.11.4d.1.1.1 Transport channel parameters for E-DCH 539
6.11.4d.1.2 Physical channel parameters 539
6.11.4d.1.2.1 Physical channel parameters on E-DPDCH 539
6.11.4d.2 Downlink 539
6.11.4d.2.1 Transport channel parameters 539
6.11.4d.2.1.1 Transport channel parameters for HS-DSCH 539
6.11.4d.2.1.2 Transport channel parameters for DCH 539
6.11.4d.2.2 Physical channel parameters 540
6.11.4d.2.2.1 Physical channel parameters on DPCH 540
6.11.4d.2.2.2 Physical channel parameters on HS-PDSCH 540
6.11.4e Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 540
6.11.4e.1 Uplink 540
6.11.4e.1.1 Transport channel parameters 540
6.11.4e.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB 540
6.11.4e.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH 540
6.11.4e.1.1.3 TFCS 540
6.11.4e.2 Downlink 541
6.11.4e.2.1 Transport channel parameters 541
6.11.4e.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB 541
6.11.4e.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH 541
6.11.4e.2.1.3 TFCS 541
6.11.4e.2.2 Physical channel parameters 541
6.11.4f Reference Radio Bearer configurations used in MAC-ehs testing 541
6.11.4f.1 3 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB 541
6.11.4f.1.1 Uplink 541
6.11.4f.1.1.1 Uplink Transport channel parameters for DCH 541
6.11.4f.1.1.2 Uplink physical channel parameters 542
6.11.4f.1.2 Downlink 542
6.11.4f.1.2.1 Transport channel parameters for HS-DSCH 542
6.11.4f.1.2.2 Downlink Transport channel parameters for DCH 543
6.11.4f.1.2.3 Downlink physical channel parameters 543

6.11.4f.2 1 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB 543
6.11.4f.2.1 Uplink 543
6.11.4f.2.1.1 Uplink Transport channel parameters for DCH 543
6.11.4f.2.1.2 Uplink physical channel parameters 544
6.11.4f.2.2 Downlink 544
6.11.4f.2.2.1 Transport channel parameters for HS-DSCH 544
6.11.4f.2.2.2 Downlink Transport channel parameters for DCH 544
6.11.4f.2.2.3 Downlink physical channel parameters 544
6.11.4g Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 545
6.11.4g.1 Uplink 545
6.11.4g.1.1 Transport channel parameters 545
6.11.4g.1.1.1 Transport channel parameters for E-DCH 545
6.11.4g.1.2 Physical channel parameters 545
6.11.4g.1.2.1 Physical channel parameters on E-PUCH 545
6.11.4g.1.2.2 Physical channel parameters for DPCH 545
6.11.4g.2 Downlink 545
6.11.4g.2.1 Transport channel parameters 545
6.11.4g.2.1.1 Transport channel parameters for HS-DSCH 545
6.11.4g.2.2 Physical channel parameters 546
6.11.4g.2.2.1 Physical channel parameters on DPCH 546
6.11.4g.2.2.2 Physical channel parameters on HS-PDSCH 546
6.11.4h Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + DL: 3.4 kbps SRBs for DCCH on E-DCH and DCH for 3.84Mcps TDD 546
6.11.4h.1 Uplink 546
6.11.4h.1.1 Transport channel parameters 546
6.11.4h.1.1.1 Transport channel parameters for E-DCH 546
6.11.4h.1.2 Physical channel parameters 546
6.11.4h.1.2.1 Physical channel parameters on E-PUCH 546
6.11.4h.2 Downlink 547
6.11.4h.2.1 Transport channel parameters 547
6.11.4h.2.1.1 Transport channel parameters for HS-DSCH 547
6.11.4h.2.1.2 Transport channel parameters for DCH 547
6.11.4h.2.2 Physical channel parameters 547
6.11.4h.2.2.1 Physical channel parameters on DPCH 547
6.11.4h.2.2.2 Physical channel parameters on HS-PDSCH 547
6.11.4i Reference Radio Bearer configurations used in CPC testing 547
6.11.4i.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / UM PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 547
6.11. 4i.1.1 Uplink 547
6.11. 4i.1.1.1 Transport channel parameters 547
6.11. 4i.1.1.1.1 Transport channel parameters for E-DCH 547
6.11. 4i.1.1.2 Physical channel parameters 548
6.11. 4i.1.1.2.1 Physical channel parameters on E-DPDCH 548
6.11. 4i.1.2 Downlink 548
6.11. 4i.1.2.1 Transport channel parameters 548
6.11. 4i.1.2.1.1 Transport channel parameters for HS-DSCH 548
6.11. 4i.1.2.2 Physical channel parameters 548
6.11. 4i.1.2.2.2 Physical channel parameters on HS-PDSCH 548
6.11.4j Reference Radio Bearer configurations used in Improved L2 testing 548
6.11.4j.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / UM PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 548
6.11. 4j.1.2.1 Transport channel parameters 549
6.11. 4j.1.2.1.1 Transport channel parameters for HS-DSCH 549
6.11.4j.1.2.2 Physical channel parameters 549
6.11.4j.1.2.2.1 Physical channel parameters on DPCH 549
6.11.4j.1.2.2.2 Physical channel parameters on HS-PDSCH 549

6.11.4j.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] with Flexible RLC and MAC-ehs SRBs for DCCH on E-DCH and HS-DSCH 549
6.11.4k Reference Radio Bearer configurations used in Improved L2 UL testing 551
6.11.4k.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 551
6.11.4k.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 552
6.11.4k.2.1.1.1 Transport channel parameters for E-DCH 552
6.11.4k.2.2.2.2 Physical channel parameters on HS-PDSCH 552
6.11.4k.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] with Flexible RLC and MAC-ehs SRBs for DCCH on E-DCH and HS-DSCH 553
6.11.4k.4 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + 3.4 kbps SRBs for DCCH on E-DCH and DL DCH 554
6.11.4k.4.1.2.1 Physical channel parameters on E-DPDCH 554
6.11.5 Reference Radio Bearer configurations used in Radio Bearer testing for 1.28 Mcps TDD 555
6.11.5.1 RABs and signalling RBs 555
6.11.5.2 Combinations of RABs and Signalling RBs 555
6.11.5.3 Example of linkage between RABs and services 561
6.11.5.4 Typical radio parameter sets 561
6.11.5.4.1 Combinations on DPCH 561
6.11.5.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH 561
6.11.5.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe) 562
6.11.5.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH 562
6.11.5.4.1.2a Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH (multiframe) 563
6.11.5.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH 565
6.11.5.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 565
6.11.5.4.1.4a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 566
6.11.5.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 567
6.11.5.4.1.5a Conversational / speech / UL:10.2 6.7 5.9 4.75 DL:10.2 6.7 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 568
6.11.5.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 569
6.11.5.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH 570
6.11.5.4.1.7a Conversational / speech / UL:7.4 6.7 5.9 4.75 DL:7.4 6.7 5.9 4.75 / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH 570
6.11.5.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 571
6.11.5.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 572
6.11.5.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH 573
6.11.5.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH 574
6.11.5.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 575
6.11.5.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 575

6.11.5.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....576
6.11.5.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....577
6.11.5.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....578
6.11.5.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....579
6.11.5.4.1.18 Void....580
6.11.5.4.1.19 Void....580
6.11.5.4.1.20 Void....580
6.11.5.4.1.21 Void....580
6.11.5.4.1.22 Void....580
6.11.5.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....580
6.11.5.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....581
6.11.5.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....581
6.11.5.4.1.23c Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....582
6.11.5.4.1.23d Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....583
6.11.5.4.1.24 Void....584
6.11.5.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....584
6.11.5.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....584
6.11.5.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....585
6.11.5.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....585
6.11.5.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....586
6.11.5.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....586
6.11.5.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....587
6.11.5.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....587
6.11.5.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....588
6.11.5.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....588
6.11.5.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....588
6.11.5.4.1.36 Void....589
6.11.5.4.1.37 Void....589
6.11.5.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....589
6.11.5.4.1.38a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....590
6.11.2.5.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....591
6.11.5.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....592
6.11.5.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....593
6.11.5.4.1.38e Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....594

6.11.5.4.1.38f Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....595
6.11.5.4.1.38g Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....596
6.11.5.4.1.38h Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....597
6.11.5.4.1.38i Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....598
6.11.5.4.1.38j Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....599
6.11.5.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....600
6.11.5.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....600
6.11.5.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....601
6.11.5.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....602
6.11.5.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....603
6.11.5.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....603
6.11.5.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....605
6.11.5.4.1.46 Void....606
6.11.5.4.1.47 Void....606
6.11.5.4.1.48 Void....606
6.11.5.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....606
6.11.5.4.1.49a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL: 12.2 7.95 5.9 4.75 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....607
6.11.5.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....608
6.11.5.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....609
6.11.5.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....610
6.11.5.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....610
6.11.5.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....611
6.11.5.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....612
6.11.5.4.1.54 Void....612
6.11.5.4.1.55 Void....612
6.11.5.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....612
6.11.5.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....613
6.11.5.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....614
6.11.5.4.1.59 Reserved for future use....615
6.11.5.4.1.60 Reserved for future use....615
6.11.5.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....615
6.11.5.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH....616

6.11.5.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 616
6.11.5.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH 619
6.11.5.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 620
6.11.5.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH 621
6.11.5.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....621
6.11.5.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....623
6.11.5.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH 623
6.11.5.4.4 Combinations on SCCPCH 624
6.11.5.4.4.1 Stand-alone signalling RB for PCCH 624
6.11.5.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 624
6.11.5.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 625
6.11.5.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH 626
6.11.5.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 626
6.11.5.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 627
6.11.5.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH 627
6.11.5.4.4.5 64.8kbps RB for MTCH with 40 ms TTI....628
6.11.5.4.4.6 129.6 kbps RB for MTCH with 40 ms TTI....629
6.11.5.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI....630
6.11.5.4.4.8 7.6 kbps signalling RB for MCCH....630
6.11.5.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI 631
6.11.5.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI 632
6.11.5.4.4.11 497.6 kbps RB for MBSFN MTCH with 80 ms TTI 633
6.11.5.4.4.12 7.2 kbps signalling RB for MBSFN MCCH 634
6.11.5.4.4.13 8kbps RB for MBSFN MTCH with 40 ms TTI 635
6.11.5.4.4.10 64kbps RB for MBSFN MTCH with 40 ms TTI 636
6.11.5.4.5 Combinations on PRACH 636
6.11.5.4.5.1 SRB for CCCH + SRBs for DCCH 636
6.11.5.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH....637
6.11.5.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH....637
6.11.5.4.6 Combinations on DPCH and HS-PDSCH 638
6.11.5.4.6.1 Interactive or background / UL:8 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 638
6.11.5.4.6.1a Interactive or background / UL:8 (multiframe) DL: [max bit rate depending on UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH (multiframe) (REL-5)....641
6.11.5.4.6.2 Interactive or background / UL:16 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)....642
6.11.5.4.6.2a Interactive or background / UL:16(multiframe) DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)....643
6.11.5.4.6.3 Interactive or background / UL:32 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)....643
6.11.5.4.6.3a Interactive or background / UL:32(multiframe) DL: [max bit rate depending on UE category] / PS RAB +UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)....644
6.11.5.4.6.4 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)....644
6.11.5.4.6.5 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)....645

6.11.5.4.6.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5) 645
6.11.5.4.6.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5) 645
6.11.5.4.6.8 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5) 646
6.11.5.4.6.9 Interactive or background / UL:64 DL: [bit rate depending on UE category] / PS RAB + Interactive or background / UL:64 DL: [bit rate depending on UE category] / PS RAB+UL:3.4 DL:3.4 kbps SRBs for DCCH 646
6.11.5.4.6.10 Conversational/Speech/UL:12.2 DL:12.2kbps/CS RAB + interactive or Background / UL:64 kbps DL: [bit rate depending on UE category]/PS RAB + interactive or Background / UL:64 kbps DL: [bit rate depending on UE category]/PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH 647
6.11.5.4.6.11 Streaming/ UL:32 DL:[Rate depending on UE category]/ PS RAB + interactive or Background/ UL:8 kbps DL:[Rate depending on UE category]/ PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH 648
6.11.5.4.6.12 Streaming/ UL:16 DL:[Rate depending on UE category]/ PS RAB + interactive or Background/ UL:8 kbps DL:[Rate depending on UE category]/ PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH 649
6.11.5.4.7 Combinations on HS-PDSCH and E-PUCH 649
6.11.5.4.7.1 Stand-alone UL: [max bit rate depending on UE category and TTI] DL:[max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 649
6.11.5.4.7.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 650
6.11.5.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH 651
6.11.5.4.7.4 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 652
6.11.5.4.7.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 652
6.11.5.4.7.6 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 653
6.11.5.4.7.7 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH 654
6.11.6 Reference Radio Bearer configurations used in Radio Bearer testing for 7.68 Mcps TDD 655
6.11.6.1 RABs and signalling RBs 655
6.11.6.2 Combinations of RABs and Signalling RBs 656
6.11.6.3 Example of linkage between RABs and services 662
6.11.6.4 Typical radio parameter sets 663
6.11.6.4.1 Combinations on DPCH 663
6.11.6.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH 663
6.11.6.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe) 664
6.11.6.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH 666
6.11.6.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH 667
6.11.6.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 669
6.11.6.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 671

6.11.6.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....672
6.11.6.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....674
6.11.6.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....676
6.11.6.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....678
6.11.6.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....679
6.11.6.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....681
6.11.6.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....683
6.11.6.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH....684
6.11.6.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH....686
6.11.6.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....688
6.11.6.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....689
6.11.6.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....691
6.11.6.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....692
6.11.6.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....694
6.11.6.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....695
6.11.6.4.1.18 Void....697
6.11.6.4.1.19 Void....697
6.11.6.4.1.20 Void....697
6.11.6.4.1.21 Void....697
6.11.6.4.1.22 Void....697
6.11.6.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....697
6.11.6.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....698
6.11.6.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....699
6.11.6.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....701
6.11.6.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH....702
6.11.6.4.1.24 Void....704
6.11.6.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....704
6.11.6.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....705
6.11.6.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....706
6.11.6.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....706
6.11.6.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....707
6.11.6.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....708
6.11.6.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....709
6.11.6.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....710

6.11.6.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....711
6.11.6.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....711
6.11.6.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....712
6.11.6.4.1.36 Void....714
6.11.6.4.1.37 Void....714
6.11.6.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....714
6.11.6.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....715
6.11.6.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....717
6.11.6.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....718
6.11.6.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....719
6.11.6.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....721
6.11.6.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....722
6.11.6.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....724
6.11.6.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....725
6.11.6.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....727
6.11.6.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....728
6.11.6.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....729
6.11.6.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....730
6.11.6.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....732
6.11.6.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....734
6.11.6.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....735
6.11.6.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....737
6.11.6.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....740
6.11.6.4.1.46 Void....741
6.11.6.4.1.47 Void....741
6.11.6.4.1.48 Void....741
6.11.6.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....741
6.11.6.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....742
6.11.6.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....743
6.11.6.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....744
6.11.6.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....746
6.11.6.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....747

6.11.6.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 747
6.11.6.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 748
6.11.6.4.1.54 Void 749
6.11.6.4.1.55 Void 749
6.11.6.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 749
6.11.6.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 750
6.11.6.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....752
6.11.6.4.1.59 Reserved for future use 753
6.11.6.4.1.60 Reserved for future use 753
6.11.6.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 753
6.11.6.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH 755
6.11.6.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH....755
6.11.6.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH 761
6.11.6.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 763
6.11.6.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH 764
6.11.6.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH 766
6.11.6.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH....766
6.11.6.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....768
6.11.6.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH ..769
6.11.6.4.4 Combinations on SCCPCH 770
6.11.6.4.4.1 Stand-alone signalling RB for PCCH 770
6.11.6.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 771
6.11.6.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH 772
6.11.6.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH 773
6.11.6.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 774
6.11.6.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH 774
6.11.6.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH 775
6.11.6.4.4.5 64.8kbps RB for MTCH with 80 ms TTI....776
6.11.6.4.4.6 129.6kbps RB for MTCH with 80 ms TTI....777
6.11.6.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI....778
6.11.6.4.4.8 7.6 kbps signalling RB for MCCH....779
6.11.6.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI 780
6.11.6.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI 781
6.11.6.4.4.11 497.6 kbps RB for MBSFN MTCH with 80 ms TTI 782
6.11.6.4.4.12 7.2 kbps signalling RB for MBSFN MCCH 783
6.11.6.4.5 Combinations on PRACH 783
6.11.6.4.5.1 SRB for CCCH + SRB for DCCH 783
6.11.6.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH 784
6.11.6.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH....784
6.11.6.4.6 Combinations on DPCH and HS-PDSCH 785

6.11.6.4.6.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 785
6.11.6.4.6.1.1 Uplink 785
6.11.6.4.6.1.2 Downlink 785
6.11.6.4.6.2 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 787
6.11.6.4.6.3 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 788
6.11.6.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 788
6.11.6.4.6.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 790
6.11.6.4.6.6 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 790
6.11.6.4.6.7 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 792
6.11.6.4.6.8 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 792
6.11.6.4.6.9 Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 794
6.11.6.4.6.10 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 795
6.11.6.4.6.11 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 796
6.11.6.4.7 Combinations on HS-PDSCH and E-PUCH 798
6.11.6.4.7.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH 798
6.11.6.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH 800
6.11.6.4.7.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH 801
6.11.6.4.7.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH 801

7 Generic setup procedures 803
7.1 Basic Generic Procedures 803
7.1.1 UE Test States for Basic Generic Procedures 803
7.1.2 Mobile terminated establishment of Radio Resource Connection 804
7.1.2.1 Initial conditions 804
7.1.2.2 Definition of system information messages 804
7.1.2.3 Procedure 804
7.1.2.4 Specific message contents 804
7.1.2.4.1 PAGING TYPE 1 804
7.1.2.4.2 RRC CONNECTION REQUEST 805
7.1.2.4.3 RRC CONNECTION SETUP 805
7.1.2.4.4 RRC CONNECTION SETUP COMPLETE 805

7.1.3 Radio Bearer Setup Procedure 807
7.1.3.1 Initial conditions 807
7.1.3.2 Definition of system information messages 807
7.1.3.3 Procedure 807
7.1.3.4 Specific message contents 807
7.1.3.4.1 RADIO BEARER SETUP 807
7.1.3.4.2 RADIO BEARER SETUP COMPLETE 808
7.2 Generic setup procedures 808
7.2.1 UE Test States for Generic setup procedures 808
7.2.2 Registration of UE 809
7.2.2.1 Registration on CS 809
7.2.2.1.1 Initial condition 809
7.2.2.1.2 Definition of system information messages 809
7.2.2.1.3 Procedure 809
7.2.2.1.4 Specific message contents 810
7.2.2.2 Registration on PS 810
7.2.2.2.1 Initial condition 810
7.2.2.2.2 Definition of system information messages 810
7.2.2.2.3 Procedure 810
7.2.2.2.4 Specific message contents 810
7.2.2.3 Registration on CS / PS combined environment 811
7.2.2.3.1 Initial condition 811
7.2.2.3.2 Definition of system information messages 811
7.2.2.3.3 Procedure UE establish PS registration immediately after the UE has been switched on 811
7.2.2.3.3a Procedure UE establish PS registration later the user decides to use the PS services 811
7.2.2.3.4 Specific message contents 812
7.2.2.4 Registration on CS / PS non-combined environment 812
7.2.2.4.1 Initial condition 812
7.2.2.4.2 Definition of system information messages 812
7.2.2.4.3 Procedure 812
7.2.2.4.4 Specific message contents 812
7.2.3 Call setup 812
7.2.3.1 Generic call set up procedure for mobile terminating circuit switched calls 812
7.2.3.1.1 Initial conditions 812
7.2.3.1.2 Definition of system information messages 812
7.2.3.1.3 Procedure 812
7.2.3.1.4 Specific message contents 813
7.2.3.2 Generic call set-up procedure for mobile originating circuit switched calls 813
7.2.3.2.1 Initial conditions 813
7.2.3.2.2 Definition of system information messages 813
7.2.3.2.3 Procedure 813
7.2.3.2.4 Specific message contents 814
7.2.4 Session setup 814
7.2.4.1 Generic session set up procedure for mobile terminating packet switched sessions 814
7.2.4.1.1 Initial conditions 814
7.2.4.1.2 Definition of system information messages 814
7.2.4.1.3 Procedure 814
7.2.4.1.4 Specific message contents 814
7.2.4.2 Generic session set up procedure for mobile originating packet switched sessions 814
7.2.4.2.1 Initial conditions 814
7.2.4.2.2 Definition of system information messages 815
7.2.4.2.3 Procedure 815
7.2.4.2.4 Specific message contents 815
7.3 Test procedures for RF test 815
7.3.1 UE Test States for RF testing 815
7.3.2 Test procedure for TX, RX and Performance Requirement (without handover) 815
7.3.2.1 Initial conditions 815
7.3.2.2 Definition of system information messages 816
Contents of System information block type 1: RRC 816
Contents of System information block type 3 and 4: RRC 816
Contents of System Information Block type 5 (FDD) 816
7.3.2.3 Procedure 816

7.3.2.3.1 For UE supporting CS 816
7.3.2.3.2 For UE supporting PS only 817
7.3.2.4 Specific message contents 817
7.3.2.4.1 ATTACH ACCEPT 817
Contents of Attach Accept message: GMM 817
7.3.2.4.2 Reference measurement channels 817
7.3.2.4.3 Void 818
7.3.2.4.4 Compressed mode 818
7.3.2.4.5 Transmit diversity mode 818
7.3.3 Test procedure for test cases using Cell_PCH or URA_PCH state 818
7.3.3.1 Initial conditions 818
7.3.3.2 Definition of system information messages 818
Contents of System information block type 1: RRC 818
Contents of System information block type 3 and 4: RRC 818
Contents of System Information Block type 5 (FDD) 818
7.3.3.3 Procedure 819
7.3.3.3.1 For UE supporting PS 819
7.3.3.4 Specific message contents 819
Contents of Attach Accept message: GMM 819
7.3.4 Test procedure for Handover 819
7.3.4.1 Initial conditions 820
7.3.4.2 Definition of system information messages 820
Contents of System information block type 1: RRC 820
Contents of System information block type 3 and 4: RRC 820
Contents of System Information Block type 5 (FDD) 820
7.3.4.3 Procedure 821
7.3.4.3.1 For UE supporting CS 821
7.3.4.3.2 For UE supporting PS only 821
7.3.4.4 Specific message contents 822
Contents of Attach Accept message: GMM 822
7.3.5 Test procedure for test cases using CELL_FACH state 822
7.3.5.1 Initial conditions 822
7.3.5.2 Definition of system information messages 822
Contents of System information block type 1: RRC 822
Contents of System information block type 3 and 4: RRC 822
Contents of System Information Block type 5 (FDD) 823
7.3.5.3 Procedure 823
7.3.5.3.1 For UE supporting CS 823
7.3.5.3.2 For UE supporting PS only 823
7.3.5.4 Specific message contents 823
Contents of Attach Accept message: GMM 824
7.3.6 Test procedure for HSDPA RF Performance Requirement 824
7.3.6.1 Initial conditions 824
7.3.6.2 Definition of system information messages 824
Contents of System information block type 1: RRC 824
Contents of System information block type 3 and 4: RRC 824
Contents of System Information Block type 5 (FDD) 824
7.3.6.3 Procedure 825
7.3.6.4 Specific message contents 825
7.3.6.4.1 ATTACH ACCEPT 826
Contents of Attach Accept message: GMM 826
7.3.6.4.2 RADIO BEARER SETUP 826
7.3.6.4.3 RRC CONNECTION SETUP 826
Contents of RRC CONNECTION SETUP message: UM 826
7.3.7 Test procedure for inter-RAT handover used in RRM testing 826
7.3.7.1 Initial conditions 826
7.3.7.2 Definition of system information messages 826
Contents of System information block type 1: RRC 826
Contents of System information block type 3 and 4: RRC 827
Contents of System Information Block type 5 (FDD) 827
7.3.7.3 Procedure 827
7.3.7.4 Specific message contents 827

Contents of Attach Accept message: GMM 827
7.3.8 Test procedure for inter-RAT cell FACH reselection used in RRM testing 828
7.3.8.1 Initial conditions 828
7.3.8.2 Definition of system information messages 828
Contents of System information block type 1: RRC 828
Contents of System information block type 3 and 4: RRC 828
Contents of System Information Block type 5 (FDD) 828
7.3.8.3 Procedure 829
7.3.8.4 Specific message contents 829
Contents of Attach Accept message: GMM 829
7.3.9 Test procedure for E-DCH RF test cases 829
7.3.9.1 Initial conditions 829
7.3.9.2 Definition of system information messages 829
Contents of System information block type 3 and 4: RRC 830
7.3.9.3 Procedure 830
7.3.9.3.1 For UE not transmitting E-DPDCH with SF4x2 and SF2x2 configuration 830
7.3.9.3.2 For UE transmitting E-DPDCH with SF4x2 and SF2x2 configuration 831
7.3.9.4 Specific message contents 831
7.3.9.4.1 ATTACH ACCEPT 831
Contents of Attach Accept message: GMM 831
7.3.9.4.2 RADIO BEARER SETUP 832
7.3.9.4.3 RRC CONNECTION SETUP 832
Contents of RRC CONNECTION SETUP message: UM 832
7.3.10 Test procedure for MBMS RF/RRM test cases 832
7.3.10.1 Initial conditions 832
7.3.10.2 Definition of system information messages 832
Contents of System information block type 1: RRC 832
Contents of System information block type 3 and 4: RRC 832
Contents of System Information Block type 5 (FDD) 833
7.3.10.3 Procedure 833
7.3.10.4 Specific message contents 834
Contents of MBMS GENERAL INFORMATION message: UM (Step 4) 834
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (Step 5) 834
Contents of MBMS CURRENT P-T-M RB INFORMATION message: UM (Step 6) 834
Contents of ATTACH ACCEPT message: GMM 834
7.3.11 Test procedure for HSDPA with F-DPCH RF Performance Requirement 835
7.3.11.1 Initial conditions 835
7.3.11.2 Definition of system information messages 835
Contents of System information block type 1: RRC 835
Contents of System information block type 3 and 4: RRC 835
Contents of System Information Block type 5 (FDD) 835
7.3.11.3 Procedure 836
7.3.11.4 Specific message contents 836
7.3.11.4.1 ATTACH ACCEPT 836
Contents of Attach Accept message: GMM 836
7.3.11.4.2 RADIO BEARER SETUP 836
7.3.11.4.3 RRC CONNECTION SETUP 836
Contents of RRC CONNECTION SETUP message: UM 836
7.4 Common generic procedures for AS testing 837
7.4.1 UE RRC Test States for common procedures 837
7.4.2 Generic Setup Procedure for RRC test cases 839
7.4.2.1 RRC connection establishment procedure for circuit-switched calls (procedure P3 and P4) 839
7.4.2.1.1 Mobile terminating call 839
7.4.2.1.1.1 Initial conditions 839
7.4.2.1.1.2 Definition of system information messages 839
7.4.2.1.1.3 Procedure 839
7.4.2.1.1.4 Specific message contents 839
7.4.2.1.2 Mobile originating calls 839
7.4.2.1.2.1 Initial conditions 839
7.4.2.1.2.2 Definition of system information messages 839
7.4.2.1.2.3 Procedure 839
7.4.2.1.2.4 Specific message contents 840

7.4.2.2 RRC connection establishment procedure for packet switched sessions (procedure P5 and P6) 840
7.4.2.2.1 Mobile terminating session 840
7.4.2.2.1.1 Initial conditions 840
7.4.2.2.1.2 Definition of system information messages 840
7.4.2.2.1.3 Procedure 840
7.4.2.2.1.4 Specific message contents 840
7.4.2.2.2 Mobile originating sessions 840
7.4.2.2.2.1 Initial conditions 840
7.4.2.2.2.2 Definition of system information messages 841
7.4.2.2.2.3 Procedure 841
7.4.2.2.2.4 Specific message contents 841
7.4.2.3 NAS call set up procedure for circuit switched calls (procedure P7 and P8) 841
7.4.2.3.1 Mobile terminating call 841
7.4.2.3.1.1 Initial conditions 841
7.4.2.3.1.2 Definition of system information messages 841
7.4.2.3.1.3 Procedure 841
7.4.2.3.1.4 Specific message contents 841
7.4.2.3.2 Mobile originating calls 842
7.4.2.3.2.1 Initial conditions 842
7.4.2.3.2.2 Definition of system information messages 842
7.4.2.3.2.3 Procedure 842
7.4.2.3.2.4 Specific message contents 842
7.4.2.4 NAS session activation procedure for packet switched sessions (procedure P9 and P10) 842
7.4.2.4.1 Mobile terminating session 842
7.4.2.4.1.1 Initial conditions 842
7.4.2.4.1.2 Definition of system information messages 842
7.4.2.4.1.3 Procedure 842
7.4.2.4.1.4 Specific message contents 843
7.4.2.4.2 Mobile originating sessions 843
7.4.2.4.2.1 Initial conditions 843
7.4.2.4.2.2 Definition of system information messages 843
7.4.2.4.2.3 Procedure 843
7.4.2.4.2.4 Specific message contents 843
7.4.2.5 Radio access bearer establishment procedure for circuit switched calls (procedure P11 and P12) 843
7.4.2.5.1 Mobile terminating call 843
7.4.2.5.1.1 Initial conditions 843
7.4.2.5.1.2 Definition of system information messages 843
7.4.2.5.1.3 Procedure 843
7.4.2.5.1.4 Specific message contents 844
7.4.2.5.2 Mobile originating calls 844
7.4.2.5.2.1 Initial conditions 844
7.4.2.5.2.2 Definition of system information messages 844
7.4.2.5.2.3 Procedure 844
7.4.2.5.2.4 Specific message contents 844
7.4.2.5a Test loop activation and radio access bearer establishment procedure for circuit switched calls (procedure P7a) 844
7.4.2.5a.1 Initial conditions 844
7.4.2.5a.2 Definition of system information messages 844
7.4.2.5a.3 Procedure 845
7.4.2.5a.4 Specific message contents 845
7.4.2.6 Radio access bearer establishment procedure for packet switched sessions (procedure P13, P14, P25 and P26) 845
7.4.2.6.1 Mobile terminating session 845
7.4.2.6.1.1 Initial conditions 845
7.4.2.6.1.2 Definition of system information messages 845
7.4.2.6.1.3 Procedure 845
7.4.2.6.1.4 Specific message contents 845
7.4.2.6.2 Mobile originating sessions 846
7.4.2.6.2.1 Initial conditions 846
7.4.2.6.2.2 Definition of system information messages 846
7.4.2.6.2.3 Procedure 846
7.4.2.6.2.4 Specific message contents 846

7.4.2.6a Test loop activation and radio access bearer establishment procedure for packet switched sessions (procedure P4a and P9a) 846
7.4.2.6a.1 Initial conditions 846
7.4.2.6a.2 Definition of system information messages 846
7.4.2.6a.3 Procedure 846
7.4.2.6a.4 Specific message contents 847
7.4.2.7 Procedure for transitions to CELL_PCH or URA_PCH state (procedure P15, P16, P17 and P18) 847
7.4.2.7.1 Transition to CELL_PCH (procedure P15 and P16) 847
7.4.2.7.1.1 Initial conditions 847
7.4.2.7.1.2 Definition of system information messages 847
7.4.2.7.1.3 Procedure 847
7.4.2.7.1.4 Specific message contents 847
Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1) 847
7.4.2.7.2 Transition to URA_PCH (procedure P17 and P18) 847
7.4.2.7.2.1 Initial conditions 847
7.4.2.7.2.2 Definition of system information messages 848
7.4.2.7.2.3 Procedure 848
7.4.2.7.2.4 Specific message contents 848
Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1) 848
7.4.2.8 Radio access bearer establishment procedure with packet switched sessions for transitions to Multi Call state (procedure P19, 20 and 21) 848
7.4.2.8.1 Transition to PS+CS-DCCH+DTCH DCH (procedure P19) 848
7.4.2.8.1.1 Mobile terminating session 848
7.4.2.8.1.2 Mobile originating sessions 849
7.4.2.8.2 Transition to PS+PS-DCCH+DTCH DCH (procedure P20 and P21) 849
7.4.2.8.2.1 Mobile terminating session 849
7.4.2.8.2.2 Mobile originating sessions 850
7.4.2.9 Radio access bearer establishment procedure with circuit switched calls for transitions to Multi Call state (procedure P22, P23 and P24) 850
7.4.2.9.1 Transition to CS+CS-DCCH+DTCH DCH (procedure P22) 850
7.4.2.9.1.1 Mobile terminating call 850
7.4.2.9.1.2 Mobile originating calls 851
7.4.2.9.2 Transition to PS+CS-DCCH+DTCH DCH (procedure P23 and 24) 852
7.4.2.9.2.1 Mobile terminating call 852
7.4.2.9.2.2 Mobile originating calls 852
7.5 Test procedures for A-GPS Performance requirements testing 853
7.5.1 Normal UE based A-GPS procedure 853
7.5.1.1 Initial conditions 853
7.5.1.2 Procedure 853
7.5.1.3 Specific message contents 854
Contents of RESET UE POSITIONING STORED INFORMATION message: TC 854
Contents of MEASUREMENT CONTROL messages: RRC 854
7.5.2 UE based A-GPS procedure for moving scenario and periodic update test case 856
7.5.2.1 Initial conditions 856
7.5.2.2 Procedure 856
7.5.2.3 Specific message contents 857
Contents of RESET UE POSITIONING STORED INFORMATION message: TC 857
Contents of MEASUREMENT CONTROL message: RRC 857
7.5.3 Void 857
7.5.4 Normal UE assisted GPS procedure 857
7.5.4.1 Initial conditions 857
7.5.4.2 Procedure 857
7.5.4.3 Specific message contents 858
Contents of RESET UE POSITIONING STORED INFORMATION message: TC 858
Contents of MEASUREMENT CONTROL messages: RRC 858
7.5.5 UE assisted A-GPS procedure for moving scenario and periodic update test case 860
7.5.5.1 Initial conditions 860
7.5.5.2 Procedure 861
7.5.5.3 Specific message contents 861
Contents of RESET UE POSITIONING STORED INFORMATION message: TC 861
Contents of MEASUREMENT CONTROL message: RRC 861
7.6 Test procedures for MBMS testing 861

7.6.1 GMM-REGISTERED with 1 MBMS Service Activated 861
7.6.1.1 Initial conditions 861
7.6.1.2 Procedure 862
7.6.1.3 Specific message contents 862
7.6.2 IDLE with 1 MBMS Service Activated 864
7.6.2.1 Initial conditions 864
7.6.2.2 Procedure 865
7.6.2.3 Specific message contents 865
7.6.3 MBSFN IDLE (3.84/FDD, 7.68 Mcps TDD) 867
7.6.3.1 Initial conditions 867
7.6.3.2 Procedure 867
7.6.3.3 Specific message contents 868
7.6.4 MBSFN IDLE with 1 MBMS Service Activated (3.84/FDD, 7.68 Mcps TDD) 868
7.6.4.1 Initial conditions 868
7.6.4.2 Procedure 868
7.6.4.3 Specific message contents 869

8 Test USIM Parameters 869
8.1 Introduction 869
8.1.1 Definitions 869
8.1.2 Definition of the test algorithm for authentication 869
8.1.2.1 Authentication and key derivation in the test USIM and SS 870
8.1.2.2 Generation of re-synchronization parameters in the USIM 871
8.1.2.3 Using the authentication test algorithm for UE conformance testing 872
8.1.2.3.1 Authentication accept case 872
8.1.2.3.2 MAC failure case 872
8.1.2.3.3 SQN failure case 873
8.1.3 Definition of the test algorithm for VGCS/VBS VSTK generation 873
8.1.3.1 VSTK generation in the test USIM and SS 874
8.2 Default Parameters for the test USIM 874
8.3 Default settings for the Elementary Files (EFs) 875
8.3.1 Contents of the EFs at the MF level 875
8.3.1.1 EF_{DIR} 875
8.3.1.2 EF_{ICCID} (ICC Identity) 875
8.3.1.3 EF_{PL} (Preferred Languages) 875
8.3.1.4 EF_{ARR} (Access rule reference) 875
8.3.2 Contents of files at the USIM ADF (Application DF) level 875
8.3.2.1 EF_{LI} (Language Indication) 875
8.3.2.2 EF_{IMSI} (IMSI) 875
8.3.2.3 EF_{Keys} (Ciphering and Integrity Keys) 875
8.3.2.4 EF_{KeysPS} (Ciphering and Integrity Keys for Packet Switched domain) 875
8.3.2.5 $EF_{PLMNwAcT}$ (User controlled PLMN selector with Access Technology) 876
8.3.2.6 EF_{HPPLMN} (Higher Priority PLMN search period) 876
8.3.2.7 EF_{ACMmax} (ACM maximum value) 876
8.3.2.8 EF_{UST} (USIM Service Table) 876
8.3.2.9 EF_{ACM} (Accumulated Call Meter) 878
8.3.2.10 EF_{GID1} (Group Identifier Level 1) 878
8.3.2.11 EF_{GID2} (Group Identifier Level 2) 878
8.3.2.12 EF_{SPN} (Service Provider Name) 878
8.3.2.13 EF_{PUCT} (Price per Unit and Currency Table) 878
8.3.2.14 EF_{CBMI} (Cell Broadcast Message identifier selection) 878
8.3.2.15 EF_{ACC} (Access Control Class) 878
8.3.2.16 EF_{FPLMN} (Forbidden PLMNs) 879
8.3.2.17 EF_{LOCI} (Location Information) 879
8.3.2.18 EF_{AD} (Administrative Data) 879
8.3.2.19 Void 879
8.3.2.20 EF_{CBMID} (Cell Broadcast Message Identifier for Data Download) 879
8.3.2.21 EF_{ECC} (Emergency Call Codes) 879

8.3.2.22 EF_{CBMIR} (Cell Broadcast Message Identifier Range selection) 879
8.3.2.23 EF_{PSLOCI} (Packet Switched location information) 879
8.3.2.24 EF_{FDN} (Fixed Dialling Numbers) 880
8.3.2.25 EF_{SMS} (Short messages) 880
8.3.2.26 EF_{MSISDN} (MSISDN) 880
8.3.2.27 EF_{SMSP} (Short message service parameters) 880
8.3.2.28 EF_{SMSS} (SMS status) 880
8.3.2.29 EF_{SDN} (Service Dialling Numbers) 880
8.3.2.30 EF_{EXT2} (Extension2) 880
8.3.2.31 EF_{EXT3} (Extension3) 880
8.3.2.32 EF_{SMSR} (Short message status reports) 880
8.3.2.33 EF_{ICI} (Incoming Call Information) 880
8.3.2.34 EF_{OCI} (Outgoing Call Information) 880
8.3.2.35 EF_{ICT} (Incoming Call Timer) 880
8.3.2.36 EF_{OCT} (Outgoing Call Timer) 880
8.3.2.37 EF_{EXT5} (Extension5) 881
8.3.2.38 EF_{CCP2} (Capability Configuration Parameters 2) 881
8.3.2.39 EF_{eMLPP} (enhanced Multi Level Precedence and Pre-emption) 881
8.3.2.40 EF_{AAeM} (Automatic Answer for eMLPP Service) 881
8.3.2.41 Void 881
8.3.2.42 $EF_{Hiddenkey}$ (Key for hidden phone book entries) 881
8.3.2.43 Void 881
8.3.2.44 EF_{BDN} (Barred dialling numbers) 881
8.3.2.45 EF_{EXT4} (Extension 4) 881
8.3.2.46 EF_{CMI} (Comparison method information) 881
8.3.2.47 EF_{EST} (Enabled service table) 881
8.3.2.48 EF_{ACL} (Access point name control list) 881
8.3.2.49 EF_{DCK} (Depersonalization control keys) 881
8.3.2.50 EF_{CNL} (Co-operative network list) 881
8.3.2.51 $EF_{START-HFN}$ (Initialisation values for Hyperframe number) 881
8.3.2.52 $EF_{THRESHOLD}$ (Maximum value of START) 881
8.3.2.53 $EF_{OPLMNwACT}$ (Operator controlled PLMN selector with Access Technology) 881
8.3.2.54 $EF_{HPLMNwAcT}$ (HPLMN selector with Access Technology) 882
8.3.2.55 EF_{ARR} (Access rule reference) 882
8.3.2.56 Void 882
8.3.2.57 EF_{NETPAR} (Network Parameters) 882
8.3.2.58 EF_{PNN} (PLMN Network Name) 882
8.3.2.59 EF_{OPL} (Operator PLMN List) 882
8.3.2.60 EF_{MBDN} (Mailbox Dialling Numbers) 882
8.3.2.61 EF_{EXT6} (Extension6) 882
8.3.2.62 EF_{MBI} (Mailbox Identifier) 882
8.3.2.63 EF_{MWIS} (Message Waiting Indication Status) 882
8.3.2.64 EF_{CFIS} (Call Forwarding Indication Status) 882
8.3.2.65 EF_{EXT7} (Extension7) 882
8.3.2.66 EF_{SPDI} (Service Provider Display Information) 882
8.3.2.67 EF_{MMSN} (MMS Notification) 882
8.3.2.68 EF_{EXT8} (Extension 8) 882
8.3.2.69 EF_{MMSICP} (MMS Issuer Connectivity Parameters) 882
8.3.2.70 EF_{MMSUP} (MMS User Preferences) 882
8.3.2.71 EF_{MMSUCP} (MMS User Connectivity Parameters) 883
8.3.2.72 EF_{NIA} (Network's Indication of Alerting) 883
8.3.2.74 EF_{VGCSS} (Voice Group Call Service Status) 885
8.3.2.75 EF_{VBS} (Voice Broadcast Service) 885
8.3.2.76 EF_{VBSS} (Voice Broadcast Service Status) 886
8.3.2.77 EF_{VGCSCA} (Voice Group Call Service Ciphering Algorithm) 887
8.3.2.78 EF_{VBSCA} (Voice Broadcast Service Ciphering Algorithm) 887
8.3.2.79 EF_{GBABP} (GBA Bootstrapping parameters) 887

8.3.2.80 EF_{MSK} (MBMS Service Keys List) 887
8.3.2.81 EF_{MUK} (MBMS User Key) 887
8.3.2.82 Void 887
8.3.2.83 EF_{GBANL} (GBA NAF List) 887
8.3.2.84 EF_{EHPLMN} (Equivalent HPLMN) 887
8.3.2.85 $EF_{EHPLMNPI}$ (Equivalent HPLMN Presentation Indication) 887
8.3.2.86 $EF_{LRPLMNSI}$ (Last RPLMN Selection Indication) 888
8.3.3 Contents of DFs at the USIM ADF (Application DF) level 888
8.3.3.1 Contents of files at the USIM SoLSA level 888
8.3.3.1.1 EF_{SAI} (SoLSA Access Indicator) 888
8.3.3.1.2 EF_{SLL} (SoLSA LSA List) 888
8.3.3.1.3 LSA Descriptor files 888
8.3.3.1.4 Contents of files at the MExE level 888
8.3.3.1.4.1 $EF_{MExE-ST}$ (MExE Service table) 888
8.3.3.1.4.2 EF_{ORPK} (Operator Root Public Key) 888
8.3.3.1.4.3 EF_{ARPK} (Administrator Root Public Key) 888
8.3.3.1.4.4 EF_{TPRPK} (Third Party Root Public Key) 888
8.3.3.1.4.5 EF_{TKCDF} (Trusted Key/Certificates Data Files) 888
8.3.3.2 Contents of files at the DF PHONEBOOK level 888
8.3.3.2.1 EF_{PBR} (Phone Book Reference file) 888
8.3.3.2.2 EF_{IAP} (Index Administration Phone book) 888
8.3.3.2.3 EF_{ADN} (Abbreviated dialling numbers) 888
8.3.3.2.4 EF_{EXT1} (Extension1) 888
8.3.3.2.5 EF_{PBC} (Phone Book Control) 888
8.3.3.2.6 EF_{GRP} (Grouping file) 889
8.3.3.2.7 EF_{AAS} (Additional number Alpha String) 889
8.3.3.2.8 EF_{GAS} (Grouping information Alpha String) 889
8.3.3.2.9 EF_{ANR} (Additional Number) 889
8.3.3.2.10 EF_{SNE} (Second Name Entry) 889
8.3.3.2.11 EF_{CCP1} (Capability Configuration Parameters 1) 889
8.3.3.2.12 Phone Book Synchronization 889
8.3.3.2.12.1 EF_{UID} (Unique Identifier) 889
8.3.3.2.12.2 EF_{PSC} (Phone book Synchronization Counter) 889
8.3.3.2.12.3 EF_{CC} (Change Counter) 889
8.3.3.2.12.4 EF_{PUID} (Previous Unique Identifier) 889
8.3.3.2.13 EF_{EMAIL} (e-mail address) 889
8.3.3.3 Contents of files at the DF GSM-ACCESS level (Files required for GSM Access) 889
8.3.3.3.1 EF_{Kc} (GSM Ciphering key Kc) 889
8.3.3.3.2 EF_{KcGPRS} (GPRS Ciphering key KcGPRS) 889
8.3.3.3.3 Void 890
8.3.3.3.4 EF_{CPBCCH} (CPBCCH Information) 890
8.3.3.3.5 $EF_{InvScan}$ (Investigation Scan) 890
8.3.4 Contents of EFs at the TELECOM level 890
8.3.4.1 EF_{ADN} (Abbreviated dialling numbers) 890
8.3.4.2 EF_{EXT1} (Extension1) 890
8.3.4.3 EF_{ECCP} (Extended Capability Configuration Parameter) 890
8.3.4.4 EF_{SUME} (SetUpMenu Elements) 890
8.3.4.5 EF_{ARR} (Access rule reference) 890
8.3.5 Contents of DFs at the TELECOM level 890
8.3.5.1 Contents of files at the $DF_{GRAPHICS}$ level 890
8.3.5.1.1 EF_{IMG} (Image) 890
8.3.5.1.2 Image Instance Data Files 890
8.3.5.2 Contents of files at the $DF_{PHONEBOOK}$ under the $DF_{TELECOM}$ 890

9 Default Message Contents 891
9.1 Default Message Contents for Signalling 891

9.1.1 Default RRC Message Contents (FDD) 891
Default SYSTEM INFORMATION: 891
Contents of ACTIVE SET UPDATE message: AM 891
Contents of ACTIVE SET UPDATE COMPLETE message: AM 892
Contents of ACTIVE SET UPDATE FAILURE message: AM 892
Contents of CELL UPDATE message: TM 892
Contents of CELL UPDATE CONFIRM message: UM 893
Contents of DOWNLINK DIRECT TRANSFER message: AM 894
Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM 894
Contents of HANDOVER FROM UTRAN FAILURE message: AM 894
Contents of INITIAL DIRECT TRANSFER message: AM 895
Contents of MBMS ACCESS INFORMATION message: UM 895
Contents of MBMS GENERAL INFORMATION message: UM 895
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM 896
Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM 899
Contents of MBMS MODIFIED SERVICES INFORMATION message: UM 900
Contents of MBMS NEIGHBOURING CELL P-T-M RB INFORMATION message: UM 900
Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM 901
Contents of MEASUREMENT CONTROL message: AM 902
Contents of MEASUREMENT CONTROL FAILURE message: AM 903
Contents of MEASUREMENT REPORT message: AM 903
Contents of PAGING TYPE 1 message: TM (Speech in CS) 904
Contents of PAGING TYPE 1 message: TM (The others of speech in CS) 904
Contents of PAGING TYPE 1 message: TM (Packet in PS) 904
Contents of PAGING TYPE 1 message: TM (SMS in CS) 904
Contents of PAGING TYPE 1 message: TM (SMS in PS) 905
Contents of PAGING TYPE 2 message: AM (Speech in CS) 905
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM 905
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM 909
Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM 909
Contents of RADIO BEARER SETUP message: AM or UM 909
Contents of RADIO BEARER SETUP message: AM or UM, for MBMS PtP Radio Bearer Setup 965
Contents of RADIO BEARER SETUP COMPLETE message: AM 967
Contents of RADIO BEARER SETUP FAILURE message: AM 967
Contents of RADIO BEARER RECONFIGURATION message: AM or UM 968
Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM 976
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM 976
Contents of RADIO BEARER RELEASE message: AM or UM 977
Contents of RADIO BEARER RELEASE COMPLETE message: AM 982
Contents of RADIO BEARER RELEASE FAILURE message: AM 983
Contents of RRC CONNECTION REQUEST message: TM 983
Contents of RRC CONNECTION REJECT message: UM 984
Contents of RRC CONNECTION RELEASE message: UM 984
Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM 984
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH or HS-PDSCH in CELL_FACH) 984
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) 1006
Contents of RRC CONNECTION SETUP COMPLETE message: AM 1012
Contents of RRC STATUS message: AM 1012
Contents of SECURITY MODE COMMAND message: AM 1012
Contents of SECURITY MODE COMPLETE message: AM 1013
Contents of SECURITY MODE FAILURE message: AM 1013
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM 1014
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM 1020
Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM 1020
Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH) 1021
Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM 1021
Contents of UE CAPABILITY ENQUIRY message: AM or UM 1021
Contents of UE CAPABILITY INFORMATION message: AM 1021
Contents of UE CAPABILITY INFORMATION CONFIRM message: AM or UM 1022
Contents of URA UPDATE message: TM 1022
Contents of URA UPDATE CONFIRM message: UM 1022

Contents of UPLINK DIRECT TRANSFER message: AM 1023
Contents of UTRAN MOBILITY INFORMATION message: AM or UM 1023
Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM 1024
Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM 1024
9.1.2 Default Message Contents for Signalling (TDD) 1025
Contents of RRC STATUS message: AM 1025
Contents of SECURITY MODE FAILURE message: AM 1025
Contents of URA UPDATE message: TM 1025
Contents of URA UPDATE CONFIRM message: UM 1025
Contents of UPLINK DIRECT TRANSFER message: AM 1026
Contents of UTRAN MOBILITY INFORMATION message: AM or UM 1026
Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM 1027
Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM 1027
Contents of UE CAPABILITY ENQUIRY message 1027
Contents of UE CAPABILITY INFORMATION message (1.28 Mpcs TDD) 1027
Contents of UE CAPABILITY INFORMATION CONFIRM message 1029
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) 1029
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD) 1039
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) 1047
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) 1056
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) 1056
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD) 1057
Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM 1057
Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH) 1057
Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM 1057
Contents of RRC CONNECTION REJECT message: UM 1058
Contents of CELL UPDATE message: TM 1058
Contents of CELL UPDATE CONFIRM message: UM 1058
Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM 1059
Contents of HANDOVER FROM UTRAN FAILURE message: AM 1059
Contents of MEASUREMENT CONTROL Message: AM (Intra-frequence measurement) (1.28 Mcps TDD) 1060
Contents of MEASUREMENT CONTROL Message: AM (Inter-frequence measurement) (1.28 Mcps TDD) 1061
Contents of MEASUREMENT CONTROL FAILURE Message: AM 1063
Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement (3.84 Mcps TDD) 1064
Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement）(1.28 Mcps TDD) 1064
Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement (7.68 Mcps TDD) 1065
Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (3.84 Mcps TDD) 1066
Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (1.28 Mcps TDD) 1066
Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (7.68 Mcps TDD) 1067
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) 1067
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD) 1071
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (7.68 Mcps TDD) 1075
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) 1079
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) 1079
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD) 1080
Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM 1080
Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD) 1080
Contents of RADIO BEARER RECONFIGURATION message: AM or UM (1.28 Mcps TDD) 1089
Contents of RADIO BEARER RECONFIGURATION message: AM or UM (7.68 Mcps TDD) 1098
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) 1107
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) 1108
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD) 1108
Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM 1108
Contents of RADIO BEARER RELEASE message: AM or UM (3.84 Mcps TDD) 1109
Contents of RADIO BEARER RELEASE message: AM or UM (1.28 Mcps TDD) 1114
Contents of RADIO BEARER RELEASE message: AM or UM (7.68 Mcps TDD) 1120
Contents of DOWNLINK DIRECT TRANSFER message: AM 1125
Contents of INITIAL DIRECT TRANSFER message: AM 1126
Contents of MBMS GENERAL INFORMATION message: UM (3.84 Mcps TDD) 1126
Contents of MBMS GENERAL INFORMATION message: UM (1.28 Mcps TDD) 1126

Contents of MBMS GENERAL INFORMATION message: UM (7.68 Mcps TDD)....1127
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (3.84 Mcps)....1127
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (1.28 Mcps TDD)....1130
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (7.68 Mcps)....1134
Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM....1137
Contents of MBMS MODIFIED SERVICES INFORMATION message: UM....1138
Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM....1139
Contents of PAGING TYPE 1 message: TM (Speech in CS)....1140
Contents of PAGING TYPE 1 message: TM (The others of speech in CS)....1140
Contents of PAGING TYPE 1 message: TM (Packet in PS)....1141
Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD)....1141
Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)....1176
Contents of RADIO BEARER SETUP message: AM or UM (7.68 Mcps TDD)....1212
Contents of RADIO BEARER SETUP COMPLETE message: AM....1231
Contents of RADIO BEARER SETUP FAILURE message: AM....1231
Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD)....1231
Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD)....1232
Contents of RADIO BEARER RELEASE FAILURE message: AM....1232
Contents of RRC CONNECTION REQUEST message: TM....1233
Contents of RRC CONNECTION REJECT message: UM....1233
Contents of RRC CONNECTION RELEASE message: UM....1233
Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM....1233
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (3.84 Mcps TDD option)....1234
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD option)....1240
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (7.68 Mcps TDD option)....1255
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (3.84 Mcps TDD)....1262
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (1.28 Mcps TDD)....1267
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (7.68 Mcps TDD)....1271
Contents of RRC CONNECTION SETUP COMPLETE message: AM....1275
Contents of SECURITY MODE COMMAND message: AM....1276
Contents of SECURITY MODE COMPLETE message: AM....1277
Contents of UPLINK DIRECT TRANSFER message: AM....1277
9.1.3 Default RRC Message Contents for MBSFN (TDD)....1277
Contents of MBMS GENERAL INFORMATION message: UM (3.84 Mcps TDD)....1277
Contents of MBMS GENERAL INFORMATION message: UM (1.28 Mcps TDD)....1278
Contents of MBMS GENERAL INFORMATION message: UM (7.68 Mcps TDD)....1279
Contents of MBMS MODIFIED SERVICES INFORMATION message: UM....1279
Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM (Mixed Local/National carrier)....1280
Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM (Dedicated National carrier)....1282
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (3.84 Mcps)....1283
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (1.28 Mcps)....1286
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (7.68 Mcps)....1289
Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM....1292
9.2 Default Message Contents for RF....1293
9.2.1 Default Message Contents for RF (FDD)....1293
Contents of Activate RB Test Mode message....1293
Contents of Close UE Test Loop message (UE test loop mode 1 without Dummy DCCH transmission)....1293
Contents of Close UE Test Loop message (UE test loop mode 2 without Dummy DCCH transmission)....1293
Contents of Open UE Test Loop message....1293
Contents of MBMS COMMON P-T-M RB INFORMATION message: UM....1294
Contents of PAGING TYPE 1 message: TM (CS)....1295
Contents of PAGING TYPE 1 message: TM (PS)....1295
Contents of PAGING TYPE 2 message: TM (PS)....1295
Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)....1296
Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only)....1305
Contents of RADIO BEARER SETUP message: AM or UM (UE supports CS RAB for Test Loop Mode 2)....1310
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)....1319
Contents of RADIO BEARER SETUP message: BTFD RMC for Test Loop Mode 2....1322
Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)....1328
Contents of RRC CONNECTION RELEASE message: UM....1340
Contents of RRC CONNECTION SETUP message: UM....1340
Contents of SECURITY MODE COMMAND message: AM....1346

9.2.2 Default Message Contents for RF (TDD) 1348
Contents of Activate RB Test Mode message 1348
Contents of Close UE Test Loop message 1348
Contents of Open UE Test Loop message 1348
Contents of PAGING TYPE 1 message: TM (CS) 1348
Contents of PAGING TYPE 1 message: TM (PS) 1348
Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD) 1348
Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD) 1353
Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD) 1358
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) 1364
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) (1.28 Mcps TDD) 1368
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) (7.68 Mcps TDD) 1373
Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA) (3.84Mcps TDD) 1377
Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA) (7.68Mcps TDD) 1381
Contents of RRC CONNECTION RELEASE message: UM 1384
Contents of RRC CONNECTION SETUP message: UM (3.84 Mcps TDD) 1385
Contents of RRC CONNECTION SETUP message: UM (1.28 Mcps TDD) 1390
Contents of RRC CONNECTION SETUP message: UM (7.68 Mcps TDD) 1396
Contents of SECURITY MODE COMMAND message: AM 1401

10 A-GPS GPS Scenarios and Assistance Data 1404
10.1 General 1404
10.1.1 Satellite constellations and assistance data for performance testing 1404
10.1.2 GPS Scenarios for performance testing 1404
10.1.2.1 GPS Scenario #1 1404
10.1.2.2 GPS Scenario #2 1405
10.1.2.3 GPS Scenario #3 1405
10.1.2.4 UE Location for TTFF test cases 1405
10.1.2.4.1 UE Location Offset 1405
10.1.2.4.2 UE Altitude 1406
10.1.2.5 Satellites to be simulated in each test case 1406
10.2 Information elements required for normal UE based testing 1406
10.3 Information elements required for UE based Sensitivity Fine Time Assistance test case 1407
10.4 Information elements available for normal UE assisted testing 1407
10.5 Information elements available for UE assisted Sensitivity Fine Time Assistance test case 1408
10.6 Contents of Information elements for performance testing 1409
10.6.1 General 1409
10.6.2 IE Random Offset Values 1409
10.6.2.1 GPS TOW msec 1410
10.6.2.1 UTRAN GPS timing of cell frames 1410
10.6.3 Assistance Data Reference Time 1410
10.6.4 Assistance Data Reference Position 1412
10.6.5 Assistance Data Navigation Model 1412
10.6.6 Assistance Data Ionospheric Model 1413
10.6.7 Assistance Data Almanac 1413
10.6.8 Assistance Data Acquisition Assistance 1414
10.7 GPS Scenario and values of Information Elements for signalling testing 1416
10.7.1 General 1416
10.7.2 GPS Scenario 1416
10.7.3 Assistance Data Reference Time 1417
10.7.4 Assistance Data Reference Position 1417
10.7.5 Assistance Data Navigation Model 1417
10.7.6 Assistance Data Ionospheric Model 1418
10.7.7 Assistance Data Almanac 1418
10.7.8 Assistance Data Acquisition Assistance 1419

11 MBMS configurations for signalling test 1419
11.1 MCCH configurations 1420
11.1.1 MCCH configuration parameters 1420
11.1.1.1 Default1 MCCH information scheduling (mp 5.12s) 1420
11.1.1.2 Default2 MCCH information scheduling (mp 2.56s) 1420
11.1.1.3 Longest MCCH information scheduling (mp 10.24s) 1420

11.1.1.4 Shortest MCCH information scheduling (mp 1.28s) 1420
11.1.2 MCCH messages transmission in test 1421
11.1.3 Combinations and transmission order of critical MCCH messages 1422
11.2 MCCH configurations for MBSFN (3.84 Mcps and 7.68 Mcps TDD) 1423
11.2.1 MCCH configuration parameters 1423
11.2.1.1 Default1 MCCH information scheduling (mp 5.12s) 1423
11.1.1.2 Default2 MCCH information scheduling (mp 2.56s) 1423
11.1.1.3 Longest MCCH information scheduling (mp 10.24s) 1423
11.1.1.4 Shortest MCCH information scheduling (mp 1.28s) 1423
11.2.2 MCCH messages transmission in test 1423
11.2.3 Combinations and transmission order of critical MCCH messages 1424
11.2.4 MBSFN service availability 1424

Annex A (informative): Void 1426

Annex B (informative): Void 1427

Annex C (normative): GPS Data Files 1428

C.1 GPS data files for signalling tests 1428

C.2 GPS data files for performance tests 1428

Annex D (informative): Change history 1429

History 1445

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

Introduction

The definition of the Conformance Tests for UE in 3G will be a complex task as the complete test suite covers RF, EMC and Protocol aspects of the UE.

Each test requires a Test Environment to be defined in which the UE has to operate to defined standards, constraints and performance. The overall task can be simplified if there are a number of well defined and agreed Common Test Environments where every one can be used for a number of tests. Hence the present documents defines testing conditions that are common to several tests avoiding the need to duplicate the same information for every single test.

The present document defines default values for a variety of common areas. Where values are not specified in test cases, the defaults in the present document will apply. If specified, the test case values will take precedence.

The present document addresses the FDD mode as well as the TDD mode.

1 Scope

The present document contains definitions of reference conditions and test signals, default parameters, reference radio bearer configurations used in radio bearer interoperability testing, common radio bearer configurations for other test purposes, common requirements for test equipment and generic set-up procedures for use in UE conformance tests.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TS 34.123-1: "User Equipment (UE) conformance specification; Part 1: Protocol conformance specification".

[2] 3GPP TS 34.121-1: " User Equipment (UE) conformance specification; Radio transmission and reception (FDD);Part 1: Conformance specification ".

[2a] 3GPP TS 34.121-2: "User Equipment (UE) conformance specification; Radio transmission and reception (FDD); Part 2: Implementation Conformance Statement (ICS)".

[3] 3GPP TS 34.123-2: "User Equipment (UE) conformance specification; Part 2: Implementation Conformance Statement (ICS) proforma specification".

[4] 3GPP TS 34.124: "ElectroMagnetic Compatibility (EMC) requirements for Mobile terminals and ancillary equipment".

[5] 3GPP TS 34.122: "Terminal Conformance Specification; Radio Transmission and Reception (TDD)".

[6] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[7] 3GPP TS 25.301 "Radio interface protocol architecture".

[8] 3GPP TS 25.214: "Physical layer procedures (FDD)".

[9] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[10] 3GPP TR 25.990: "Vocabulary".

[11] 3GPP TS 25.101: "User Equipment (UE) radio transmission and reception (FDD)".

[12] 3GPP TS 25.102: "User Equipment (UE) radio transmission and reception (TDD)".

[13] 3GPP TS 25.211: "Physical channels and mapping of transport channels onto physical channels (FDD)".

[14] 3GPP TS 25.212: "Multiplexing and channel coding (FDD)".

[15] 3GPP TS 23.107: "Quality of Service (QoS) concept and architecture".

[16] 3GPP TS 26.110: "Codec for circuit switched multimedia telephony service; General description".

[17] 3GPP TS 29.007: "General requirements on interworking between the Public Land Mobile Network (PLMN) and the Integrated Services Digital Network (ISDN) or Public Switched Telephone Network (PSTN)".

[18] 3GPP TR 23.910: "Circuit switched data bearer service".

[19] Void.

[20] 3GPP TS 25.104: "Base Station (BS) radio Transmission and Reception (FDD)".

[21] 3GPP TS 25.105: "Base Station (BS) radio Transmission and Reception (TDD)".

[22] 3GPP TS 31.101: "UICC-terminal interface; Physical and logical characteristics".

[23] 3GPP TS 31.102: "Characteristics of the USIM application".

[24] 3GPP TS 33.102: "3G security; Security architecture".

[25] 3GPP TS 33.103: "3G security; Integration guidelines".

[26] 3GPP TS 33.105: "Cryptographic algorithm requirements".

[27] 3GPP TS 25.224: "Physical layer procedures (TDD)".

[28] 3GPP TS 25.221: "Physical channels and mapping of transport channels onto physical channels (TDD)".

[29] 3GPP TS 25.222: "Multiplexing and channel coding (TDD)".

[30] 3GPP TS 25.133: "Requirements for support of radio resource management (FDD)".

[31] 3GPP TS 51.010-1: "Mobile Station (MS) conformance specification; Part 1: Conformance specification".

[32] 3GPP TS 24.008: "Mobile radio interface layer 3 specification; Core network protocols; Stage 3".

[33] 3GPP TS 25.171: "Requirements for support of Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)".

[34] 3GPP TS 25.331: "Radio Resource Control (RRC) protocol specification".

[35] 3GPP TS 25.223: "Spreading and modulation (TDD)".

[36] 3GPP TS 25.304: "User Equipment (UE) procedures in idle mode and procedures for cell reselection in connected mode".

[37] 3GPP TS 25.123: "Requirements for support of radio resource management (TDD)".

[38] 3GPP TS 25.321: "Medium Access Control (MAC) protocol specification".

[39] 3GPP TS 31.120: "UICC-terminal interface; Physical, electrical and logical test specification".

[40] 3GPP TS 31.121: "Base Station System (BSS) equipment specification; Radio aspects".

[41] 3GPP TS 34.171: "Terminal conformance specification; Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)".

[42] 3GPP TS 23.032: "Universal Geographical Area Description (GAD)".

[43] NATO Standard Agreement STANAG 4294 Issue 1

[44] 3GPP TS 43.020: "Security related network functions"

3 Definitions, abbreviations and symbols

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [9], 3GPP TR 25.990 [10] and the following apply:

maximum average power: average transmitter output power obtained over any specified time interval, including periods with no transmission, when the transmit time slots are at the maximum power setting

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [9], 3GPP TR 25.990 [10] and the following apply:

AFC	Automatic Frequency Control
A-GPS	Assisted - Global Positioning System
AM	Acknowledgement Mode
ATT	ATTenuator
BCCH	Broadcast Control CHannel
CBS	Cell Broadcast Service
CC	Convolutional Coding
CCCH	Common Control CHannel
CCTrCH	Coded Composite Transport CHannel
CS	Circuit Switching
DCCH	Dedicated Control CHannel
DL	DownLink
DPCH	Dedicated Physical CHannel
DT	Direct transfer
DTCH	Dedicated Traffic CHannel
FTM	File Tunnelling Mode
GPS	Global Positioning System
GSS	GPS System Simulator
HYB	HYBrid
NAS	Non-Access Stratum
OBW	Occupied BandWidth
OCNS	Orthogonal Channel Noise Simulator

NOTE: A mechanism used to simulate the users or control signals on the other orthogonal channels of a downlink.

PRACH	Physical Randome Access CHannel
PS	Packet Switching
RAB	Radio Access Bearer
RB	Radio Bearer
RRC	Radio Resource Control

NOTE: (for sub-Layer of layer 3) but also Root-Raised Cosine (for Filter shape).

SCCPCH	Secondary Common Control Physical CHannel
SMS	Short Message Service
SRB	Signalling Radio Bearer
SS	System Simulator
SSD	Source Statistics Descriptor
TC	Turbo Coding
TLM	TeLeMetry word

NOTE: It contains an 8-bits preamble (10001011).

TM	Transparent Mode
TOW	Time Of Week
TTFF	Time To First Fix
UL	UpLink
UM	Unacknowledgement Mode

3.3 Symbols

For the purposes of the present document, the following symbols apply:

I_{oc} The power spectral density of a band limited white noise source (simulating interference from other cells) as measured at the UE antenna connector.

4 Common requirements of test equipment

Mobile conformance testing can be categorized into 3 distinct areas:

- RF Conformance Testing.
- EMC Conformance Testing.
- Signalling Conformance Testing.

The test equipment required for each category of testing may or not be different, depending on the supplier of the test equipment. However, there will be some generic requirements of the test equipment that are essential for all three categories of test, and these are specified in this clause.

In addition, there will be requirements to test operation in multi-system configurations (eg UTRA plus GSM/DCS1800). However, these would not form a common test equipment requirement for the three test areas and are not considered in the present document.

4.1 General Functional Requirements

NOTE: This clause has been written such that it does not constrain the implementation of different architectures and designs of test equipment.

All test equipment used to perform conformance testing on a UE shall provide a platform suitable for testing UE's that are either:

a) FDD Mode; or

b) TDD Mode; or

c) both FDD/TDD Modes.

All test equipment shall provide (for the mode(s) supported) the following minimum functionality.

- The capability of emulating a single UTRA cell with the appropriate channels to allow the UE to register on the cell.
- The capability to allow the UE to set up an RRC connection with the System Simulator, and to maintain the connection for the duration of the test.
- The capability (for the specific test):
  - to select and support an appropriate Radio Bearer for the downlink;
  - to set the appropriate downlink power levels;
  - to set up and support the appropriate Radio Bearer for the uplink;
  - to set and control the uplink power levels.

4.2 Minimum performance levels

4.2.1 Supported Cell Configuration

The System Simulator shall provide the capability to simulate a minimum number of cells (of the appropriate UTRA Mode) whose number and capabilities are governed by the test cases that need to be performed (test cases are defined in 3GPP TS 34.123-1 [1] (Signalling), 3GPP TS 34.121 [2] (RF-FDD) and 3GPP TS 34.122 [5] (RF-TDD)). For this purpose test cases can be split into two different categories: Tests that require only one cell and Tests that require several cells.

To perform test cases requiring one cell, the system simulator must provide a Cell offering the capabilities to perform all the test cases in this category.

To perform test cases requiring several cells, additional cells must be provided by the system simulator. The additional cells, however, need only provide a minimum set of capabilities so as to support the first cell in carrying out the multi-cell test cases.

To perform test cases for MBMS in MBSFN mode the SS must provide simultaneous support for both the MBSFN cell(s) (TDD or FDD) directly applicable to the test cases and must also provide support for cells(s) on a unicast carrier providing other necessary services to the UE such as PLMN registration. The choice of supporting unicast carrier cell(s) may be TDD or FDD decided by the capabilities of the UE under test, and the choice may be independent of the mode (TDD or FDD) of the MBSFN cell(s).

The type and number of channels (especially physical channels) constitute an important set of capabilities for a cell. The following clauses list possible channels that may be supported by the SS. Each channel type, however, and the minimum number of channels needed are only mandatory if specific test cases require them.

The mapping between Logical and Transport channels is as described in 3GPP TS 25.301 [7]. Similarly the mapping between Transport channels and Physical channels is as described in 3GPP TS 25.211 [13] for the FDD mode, and 3GPP TS 25.221 [28] for the TDD mode. The reference measurement channels (mapping between Transport channels and Physical channels for DTCH/DCCH to be tested) are defined in 3GPP TS 34.121 [2] annex C for FDD and 3GPP TS 34.122 [5] annex C for TDD.

4.2.1.1 Supported Channels for FDD Mode

4.2.1.1.1 Logical channels

Logical channel	Minimum number	Comments
BCCH	1	
CCCH	1	
DCCH	4	2 for RRC testing, 2 for NAS testing
PCCH	1	
DTCH	n <FFS>	Depending on SS's support for RB service testing (See clause 14 of 3GPP TS 34.123-1 [1])

4.2.1.1.2 Transport channels

Transport channel	Minimum number	Comments
BCH	1	
FACH	1	
PCH	1	
DCH	n <FFS>	
DSCH	1	Release 99 and Release 4 only.
RACH	2	
CPCH	1	Release 99 and Release 4 only.
FAUSCH	N/A	Not in Release 1999

4.2.1.1.3 Physical channels

Physical channel	Minimum number	Comments
P-CCPCH	1	Primary Common Control Physical channel. This is used by the Cell to Broadcast System Information messages, it is transmitted using the Primary Scrambling Code for the Cell.
P-CPICH	1	Primary Common Pilot Channel using the Primary Scrambling Code for the Cell.
S-CPICH	1 (For RF Tests)	Secondary Common Pilot Channel. This signal is used as the phase reference for some RF tests.
SCH	1	Synchronization Channel (includes P-SCH and S-SCH)
S-CCPCH	2	Secondary Common Control Physical channel.
PICH	1	To identify when the UE should access the PCCH for Paging Messages.

Physical channel	Minimum number	Comments
AICH	1	General Acquisition Indicator Channel that can be used for: - Aquisition Indicator Channel, for PRACH - Access Preamble Acquisition Indicator Channel (AP-ICH), for PCPCH. (For release 99 and release 4 only) - Collision-Detection/Channel-Assignment Indicator Channel (CD/CA-ICH), for PCPCH. (For release 99 and release 4 only)
DPDCH	3	Downlink Physical Data Channel. There will be a single DPCCH associated with all the DPDCHs used for Layer 1 signalling. This number is for the First Cell. Additional Cells may define a lower number which should be at least 1.
PDSCH	1	Physical Downlink Shared Channel. Release 99 and Release 4 only.
DPCH	1	Uplink Dedicated Physical channel
PRACH	2	Physical Random Access Channel.
PCPCH	1	Physical Common Packet Channel. (For release 99 and release 4 only)
CSICH	1	CPCH Status Indicator Channel. (For release 99 and release 4 only)

4.2.1.2 Supported Channels for TDD Mode

4.2.1.2.1 Logical channels

Logical channel	Minimum number	Comments
Control channels		
BCCH	1	Broadcast Control Channel: DL channel for broadcasting system control information.
CCCH	1	Common Control Channel: Bi-directional channel for transmitting control information between network and UEs. This channel is commonly used by the UEs having no RRC connection with the network and by the UEs using common transport channels when accessing a new cell after cell reselection.
DCCH	4	Dedicated Control Channel: A point-to-point bi-directional channel that transmits dedicated control information between a UE and the network. This channel is established through RRC connection setup procedure. 2 channels for RRC testing and 2 channels for NAS testing estimated.
PCCH	1	Paging Control Channel: DL channel that transfers paging information. This channel is used when the network does not know the location cell of the UE, or, the UE is in the cell connected state
SHCCH	1	Shared Channel Control Channel: Bi-directional channel that transmits control information for uplink and downlink shared channels between network and UEs. This channel is for TDD only.
Traffic channels		
DTCH	1	Dedicated Traffic Channel is a point-to-point channel, dedicated to one UE, for the transfer of user information. A DTCH can exist in both UL and DL.
CTCH	1	Common Traffic Channel is a point-to-multipoint unidirectional channel for transfer of dedicated user information for all or a group of specified UEs.

4.2.1.2.2 Transport channels

Transport channel	Minimum number	Comments
BCH	1	Broadcast Channel: DL channel used to broadcast system and cell-specific information.
FACH	1	Forward Access Channel: DL channel used to carry control information to a mobile station when the system knows the location cell of the mobile station (may also carry short user packets).
PCH	1	Paging Channel: DL channel used to carry control information to a mobile station when the system does not know the location cell of the mobile station.
DCH	2	Dedicated Channel:UL or DL channel used to carry user or control information between the UTRAN and a UE
DSCH	1	DL shared channel: DL channel shared by several UEs carrying dedicated control or traffic data.
USCH	1	UL shared channel: UL channel shared by several UEs carrying dedicated control or traffic data.

RACH	1	Random Access Channel: UL channel used to carry control information from mobile station. The RACH may also carry short user packets.

4.2.1.2.3 Physical channels (3.84 Mcps option)

Physical channel	Minimum number	Comments
P-CCPCH	1	Primary Common Control Physical channel. . The BCH as described in clause 4.2.1 is mapped onto the P-CCPCH. The position (time slot / code) of the P-CCPCH is known from PSCH.
SCH	1	Synchronization Channel. Code group of a cell can be derived from the synchronization channel. In order not to limit the uplink/downlink asymmetry the SCH is mapped on one or two downlink slots per frame only.
S-CCPCH	2	Secondary Common Control Physical channel. PCH and FACH as described in clause 4.2.1 are mapped onto one or more S-CCPCH.
PICH		Paging Indicator Channel is a physical channel used to carry the paging indicators.
DPCH (DL)	3	Downlink Dedicated Physical channel. DCH channels are mapped onto DPCH
PDSCH	1	Physical Downlink Shared Channel. DSCH as desribed in clause 4.2.1 is mapped onto one or more PDSCH.
DPCH (UL)	1	Uplink Dedicated Physical channel. DCH channels are mapped onto DPCH.
PUSCH	1	Physical Uplink Shared Channel. The USCH as desribed in clause 4.2.1 is mapped onto one or more PUSCH. Timing advance, as described in 3GPP TS 25.224 [27], clause 4.3, is applied to the PUSCH.
PRACH	2	Physical Random Access Channel. The RACH as described in clause 4.2.1 is mapped onto PRACH
PNBSCH	1	Physical node B synchronization channel: In case cell sync bursts are used for Node B synchronization the PNBSCH shall be used for the transmission of the cell sync burst 3GPP TS 25.223 [35]. The PNBSCH shall be mapped on the same timeslot as the PRACH.

4.2.1.2.4 Physical channels (1.28 Mcps option)

Physical channel	Minimum number	Comments
P-CCPCH	2	Primary Common Control Physical channel.The BCH as described in clause 4.2.1 is mapped onto the P-CCPCH1 and P-CCPCH2. The position (time slot / code) of the P-CCPCHs is fixed in the 1.28 Mcps TDD. The P-CCPCHs are mapped onto the first two code channels of timeslot#0 with spreading factor of 16.
DwPCH	1	Synchronization Channel for DL. Present in each 5 ms subframe.
UpPCH	1	Synchronization Channel for UL. Present in each 5 ms subframe.
S-CCPCH	2	Secondary Common Control Physical channel. PCH and FACH as described in clause 4.2.1 are mapped onto one or more S-CCPCH.
PICH		Paging Indicator Channel is a physical channel used to carry the paging indicators.
DPCH (DL)	3	Downlink Dedicated Physical channel.DCH channels are mapped onto DPCH
PDSCH	1	Physical Downlink Shared Channel. PDSCH provides the possibility for transmission of TFCI, SS, and TPC in downlink.
DPCH (UL)	1	Uplink Dedicated Physical channel. DCH channels are mapped onto DPCH.
PUSCH	1	Physical Uplink Shared Channel. PUSCH provides the possibility for transmission of TFCI, SS, and TPC in uplink.
FPACH	1	Fast Physical Access Channel. FPACH is used by the Node B to carry, in a single burst, the acknowledgement of a detected signature with timing and power level adjustment indication to a user equipment.
PRACH	2	Physical Random Access Channel. The RACH as described in clause 4.2.1 is mapped onto one or more uplink Physical Random Access Channels (PRACH).

4.2.1.2A Supported Channels for MBSFN (FDD and TDD Mode)

4.2.1.2A.1 Logical channels

Logical channel	Minimum number	Comments
Control channels		

BCCH	1	Broadcast Control Channel: DL channel for broadcasting system control information.
MCCH	1	MBMS point-to-multipoint Control Channel: A point-to-multipoint downlink channel used for transmitting control information from the network to the UE. This channel is only used by UEs that receive MBMS.
MSCH	1	MBMS point-to-multipoint Scheduling Channel: A point-to-multipoint downlink channel used for transmitting scheduling control information, from the network to the UE, for one or several MTCHs carried on a CCTrCH. This channel is only used by UEs that receive MBMS.
Traffic channels		
MTCH	1	MBMS point-to-multipoint Traffic Channel: A point-to-multipoint downlink channel used for transmitting traffic data from the network to the UE. This channel is only used for MBMS.

4.2.1.2A.2 Transport channels

Transport channel	Minimum number	Comments
BCH	1	Broadcast Channel: DL channel used to broadcast system and cell-specific information.
FACH	2	Forward Access Channel: Common downlink channel without closed-loop power control used for transmission of relatively small amounts of data. In addition FACH is used to carry broadcast and multicast data.

4.2.1.2A.3 Physical channels (3.84/7.68 Mcps options)

Physical channel	Minimum number	Comments
P-CCPCH	1	Primary Common Control Physical channel: The BCH as described in clause 4.2.1 is mapped onto the P-CCPCH. The position (time slot / code) of the P-CCPCH is known from PSCH.
SCH	1	Synchronization Channel: Code group of a cell can be derived from the synchronization channel. In order not to limit the uplink/downlink asymmetry the SCH is mapped on one or two downlink slots per frame only.
S-CCPCH	2	Secondary Common Control Physical channel: FACH as described in clause 4.2.1 is mapped onto one or more S-CCPCH.
MICH	1	MBMS Indicator Channel: Used to carry the MBMS notification indicators

4.2.1.3 Support of T_{cell} timing offset

In test case parameter declarations, the parameter T_{cell} may be specified between 0 to 38 399, to allow for extensibility. However, the system simulator is required only to support a maximum T_{cell} value of 2 304, with a step resolution of 256. The SS may limit a T_{cell} value of greater than 2 304, and may round T_{cell} to the nearest multiple of 256.

4.2.2 RF Performance

4.2.2.1 Frequency of Operation

The System Simulator shall be capable of adjusting the Carrier Frequency of the DL channels to any frequency allowed in the DL frequency band. The DL frequency shall be accurate to the level of accuracy set by the core specications 3GPP TS 25.104 [20] for FDD and 3GPP TS 25.105 [21] for TDD.

For RF tests, the requirement of Test Equipment is described in 3GPP TS 34.121 [2] annex F for FDD and 3GPP TS 34.122 [5] annex F for TDD respectively.

4.2.2.2 Power Level Setting Accuracy

The system simulator shall be able to adjust the average power output of the DL Channels to meet the absolute accuracy of the system simulator DL power levels covered in clause 5.4.1 Downlink Signal Levels.

For RF tests, the requirement of Test Equipment is described in 3GPP TS 34.121 [2] annex F for FDD and 3GPP TS 34.122 [5] annex F for TDD respectively.

The system simulator shall be capable of altering the power of the DL Dedicated channels under control of the UE Layer 1 Signalling information.

4.2.2.3 Uplink Power Control

The system simulator shall be able to command the UE to transmit at the maximum level for its power class or a lower level required for specific tests. The system simulator shall also provide the capability of generating the Layer 1 Signalling information to set the power levels of the Uplink Dedicated Channels from the UE to lower levels if required.

4.2.2.4 Uplink Signal Handling

For FDD mode, the System Simulator shall not be damaged by a Power Class 1 UE transmitting at the maximum power level permitted in 3GPP TS 25.101 [11] and for TDD mode by a Power Class 2 UE transmitting at the maximum power level permitted in 3GPP TS 25.102 [12].

4.2.2.5 Uplink Sensitivity

The simulator shall be able to receive uplink transmissions from the UE when it is transmitting at the minimum power level defined in 3GPP TS 25.101 [11] for FDD mode, and 3GPP TS 25.102 [12] for TDD mode.

Editor's note: this is obviously a useful feature for the system simulator; however it is <ffs> if it should be an essential common requirement for a protocol test system.

4.2.3 Timers Tolerances

All the timers used during testing are within a tolerance margin given by the equation below. If for a specific test a different tolerance value is required then this should be specified in the relevant test document (i.e. the document where the test is described).

Timer tolerance = 10%, or $2 \times TTI + t_{delta}$, whichever value is the greater.

Where t_{delta} is 55 ms.

5 Reference test conditions

5.1 Test frequencies

The test frequencies are based the UMTS frequency bands defined in the core specifications.

To avoid interference with adjacent frequency bands the lowest test frequency (downlink and uplink) needs to be offset upwardly by at least 2.6 MHz since the channel's width is 5 MHz for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option since the channel's width is 1.6 MHz. The raster spacing is 200KHz. Similarly the highest test frequency (downlink and uplink) needs to be offset downwardly by at least 2.6 MHz for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option.

NOTE1: Additional regulations concerning interferences to frequency bands used by different systems may also exist. Those regulations are specific to the country where the test equipment is used and need to be taken into account if they require a higher offset than 2.6 MHz from the edge frequencies for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option.

NOTE2: In Band VI, to avoid interference with adjacent frequency bands the lowest test frequency (downlink and uplink) needs to be offset upwardly by at least 2.5 MHz, highest test frequency (downlink and uplink) needs to be offset downwardly by at least 2.5 MHz from the edge frequencies since additional center frequencies are specified according to 3GPP TS 25.101 [11].

5.1.1 FDD Mode Test frequencies

UTRA/FDD is designed to operate in one of three paired bands 3GPP TS 25.101 [11]. The reference test frequencies for the common test environment for each of the 6 operating bands are defined in the following tables:

5.1.1.1 FDD reference test frequencies for Operating Band I

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	9 613	1 922.6 MHz	10 563	2 112.6 MHz
Mid Range	9 750	1 950.0 MHz	10 700	2 140.0 MHz
High Range	9 887	1 977.4 MHz	10 837	2 167.4 MHz

5.1.1.2 FDD reference test frequencies for Operating Band II

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	9 263	1 852.6 MHz	9 663	1 932.6 MHz
Mid Range	9 400	1 880 MHz	9 800	1 960 MHz
High Range	9 537	1 907.4 MHz	9 937	1 987.4 MHz

5.1.1.3 FDD reference test frequencies for Operating Band III

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	938	1 712.6 MHz	1163	1 807.6 MHz
Mid Range	1112	1 747.4 MHz	1337	1 842.4 MHz
High Range	1287	1 782.4 MHz	1512	1 877.4 MHz

5.1.1.4 FDD reference test frequencies for Operating Band IV

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	1313	1712.6 MHz	1538	2112.6 MHz
Mid Range	1450	1740.0 MHz	1675	2140.0 MHz
High Range	1512	1752.4 MHz	1737	2152.4 MHz

5.1.1.5 FDD reference test frequencies for Operating Band V

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	4 133	826.6 MHz	4 358	871.6 MHz
Mid Range	4 175	835 MHz	4 400	880 MHz
High Range	4 232	846.4 MHz	4 457	891.4 MHz

5.1.1.6 FDD reference test frequencies for Operating Band VI

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	812	832.5 MHz	1037	877.5 MHz
Mid Range	4 175	835.0MHz	4 400	880.0 MHz
High Range	837	837.5 MHz	1062	882.5 MHz

NOTE: For Band VI testing, the Mobile Country Code shall be set to (MCC = '442/443').

5.1.1.7 FDD reference test frequencies for Operating Band VII

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	2013	2502.6 MHz	2238	2622.6 MHz
Mid Range	2175	2535.0 MHz	2400	2655.0 MHz
High Range	2337	2567.4 MHz	2562	2687.4 MHz

5.1.1.8 FDD reference test frequencies for Operating Band VIII

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	2713	882.6 MHz	2938	927.6 MHz
Mid Range	2788	897.6 MHz	3013	942.6 MHz
High Range	2862	912.4 MHz	3087	957.4 MHz

5.1.1.9 FDD reference test frequencies for Operating Band IX

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	8 762	1752.4 MHz	9 237	1847.4 MHz
Mid Range	8 837	1767.4MHz	9 312	1862.4 MHz
High Range	8 912	1782.4 MHz	9 387	1877.4 MHz

5.1.1.10 FDD reference test frequencies for Operating Band X

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	2888	1712.6 MHz	3113	2112.6 MHz
Mid Range	3025	1740.0 MHz	3250	2140.0 MHz
High Range	3162	1767.4 MHz	3387	2167.4 MHz

5.1.1.11 FDD reference test frequencies for Operating Band XI

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	3487	1430.4 MHz	3712	1478.4 MHz
Mid Range	3537	1440.4 MHz	3762	1488.4 MHz
High Range	3587	1450.4 MHz	3812	1498.4 MHz

5.1.1.12 FDD reference test frequencies for Operating Band XII

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	3613	700.6 MHz	3838	730.6 MHz
Mid Range	3645	707.0 MHz	3870	737.0 MHz
High Range	3677	713.4 MHz	3902	743.4 MHz

5.1.1.13 FDD reference test frequencies for Operating Band XIII

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	3793	779.6 MHz	4018	748.6 MHz
Mid Range	3805	782.0 MHz	4030	751.0 MHz
High Range	3817	784.4 MHz	4042	753.4 MHz

5.1.1.14 FDD reference test frequencies for Operating Band XIV

Test Frequency ID	UARFCN	Frequency of Uplink	UARFCN	Frequency of Downlink
Low Range	3893	790.6 MHz	4118	760.6 MHz
Mid Range	3905	793.0 MHz	4130	763.0 MHz
High Range	3917	795.4 MHz	4142	765.4 MHz

5.1.2 TDD Mode Test frequencies

UTRA/TDD is designed to operate in one of three unpaired bands (3GPP TS 25.102 [12]). The reference test frequencies for the common test environment for each of the 3 operating bands are defined in the following tables:

5.1.2.1 Standard TDD reference test frequencies (3.84 Mcps option)

Test Frequency ID	Band a		Band b		Band c	
	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)
Low Range	9 513	1 902.6 MHz	9 263	1 852.6 MHz	9 563	1 912.6 MHz
Mid Range	9 550	1 910 MHz	9 400	1 880 MHz	9 600	1 920 MHz
High Range	9 587	1 917.4 MHz	9 537	1 907.4 MHz	9 637	1 927.4 MHz
Low Range	10 063	2 012.6 MHz	9 663	1 932.6 MHz		
Mid Range	10 087	2 017.4 MHz	9 800	1 960 MHz		
High Range	10 112	2 022.4 MHz	9 937	1 987.4 MHz		

5.1.2.2 Standard TDD reference test frequencies (1.28 Mcps option)

Test Frequency ID	Band a		Band b		Band c	
	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)
Low Range	9 504	1 900.8 MHz	9 254	1 850.8 MHz	9 554	1 910.8 MHz
Mid Range	9 550	1 910 MHz	9 400	1 880 MHz	9 600	1 920 MHz
High Range	9 596	1 919.2 MHz	9 546	1 909.2 MHz	9 646	1 929.2 MHz
Low Range	10 054	2 010.8 MHz	9 654	1 930.8 MHz		
Mid Range	10 087	2 017.4 MHz	9 800	1 960 MHz		
High Range	10 121	2 024.2 MHz	9 946	1 989.2 MHz		

5.1.2.3 Standard TDD reference test frequencies (7.68 Mcps option)

Test Frequency ID	Band a		Band b		Band c	
	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)	UARFCN	Frequency (UL and DL)
Low Range	9 513	1 905 MHz	9 275	1 855 MHz	9 575	1 915 MHz
Mid Range	9 550	1 910 MHz	9 400	1 880 MHz	9 600	1 920 MHz
High Range	9 575	1 915 MHz	9 525	1 905 MHz	9 625	1 925 MHz
Low Range	10 075	2 015 MHz	9 675	1 935 MHz		
Mid Range	10 087	2 017.4 MHz	9 800	1 960 MHz		
High Range	10 100	2 020 MHz	9 925	1 985 MHz		

5.2 Radio conditions

There are a number of radio propagation conditions defined in 3GPP TS 34.121 [2] for FDD mode and 3GPP TS 34.122 [5] for TDD mode, which may be required for a number of tests and hence can be considered as Common Conditions for FDD mode and TDD mode respectively.

NOTE: The System Simulator is required to support at least the normal Propagation Condition; support of the other propagation conditions is optional, depending on the specific test supported by the simulator.

5.2.1 Normal propagation condition

This condition provides a connection between the System Simulator that is effectively free from Additive White Gaussian Noise, and where there are no fading or multipath effects. This condition will be used for Signalling tests.

5.2.2 Static propagation condition

See 3GPP TS 34.121 [2], annex D for FDD.

For TDD mode, the propagation for the static performance measurement is an Additive White Gaussian Noise (AWGN) environment. No fading and multi-paths exist for this propagation model.

5.2.3 Multi-path fading propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD and 3GPP TS 34.122 [5], annex D for TDD.

5.2.4 Moving propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD. There are no currently defined Moving propagation conditions for TDD.

5.2.5 Birth-Death propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD. There are no currently defined Birth-Death propagation conditions for TDD.

5.2.6 High speed train conditions

See 3GPP TS 34.121 [2], annex D for FDD. There are no currently defined High speed train conditions for TDD.

5.3 Standard test signals

Reference 3GPP TS 25.101 [11] and 3GPP TS 25102 [12] for definitions of standard test signals.

5.4 Signal levels

The power levels given in clauses 5.4.1 and 5.4.2 apply for Signalling tests only. For RF tests power levels are given in 3GPP TS 34.121 [2], annex E for FDD and 3GPP TS 34.122 [5], annex E for TDD.

5.4.1 Downlink signal levels

The default signal levels are defined in clauses 6.1.5, 6.1.6, and 6.1.7 of this document. The SS shall be capable of setting these downlink signal levels, and any levels specifically defined in a test case. within a maximum tolerance of +/-3dB. If a test case fails due to inaccurate setting of the downlink signal levels by the SS, then the SS is adjusted in order that it provides the correct level, measured at the UE antenna, for the specific test case.

5.4.2 Uplink signal levels

The SS shall be capable of transmitting uplink TPC commands in order to meet the requirements specified in 3GPP TS 34.123-3. clause 7.3.2.2.14a.

5.5 Downlink Physical Channels Code Allocation

5.5.1 Downlink physical channels code allocation for Signalling (FDD)

5.5.1.1 Downlink physical channels code allocation for non-HSDPA test cases

Table 5.5.1.1.1 shows details of the downlink code tree for the Primary Scrambling Code, SF=16 & Code=0 used in the non-HSDPA test cases. The numbers in the Code columns indicate the code number with the respective spreading factor (SF). The Note column refers to specifications where the code allocation is defined.

Table 5.5.1.1.1: Non-HSDPA Downlink Physical Channels Code Allocation for SF=16 Code=0

<table>
<tr><th>Code with SF=256</th><th>Code with SF=128</th><th>Code with SF=64</th><th>Note</th></tr>
<tr><td>0: P-CPICH</td><td rowspan="2">0: -</td><td rowspan="4">0: -</td><td>TS 25.213</td></tr>
<tr><td>1: P-CCPCH</td><td>TS 25.213</td></tr>
<tr><td>2: PICH</td><td rowspan="2">1: -</td><td>Section 6.1.0b (SIB5)</td></tr>
<tr><td>3: AICH</td><td>Section 6.1.0b (SIB5)</td></tr>
<tr><td>4: -</td><td rowspan="2">2: -</td><td rowspan="4">1: S-CCPCH</td><td rowspan="4">Sections 6.1.0b, 6.1.1 & 6.1.3 (SIB5)</td></tr>
<tr><td>5: -</td></tr>
<tr><td>6: -</td><td rowspan="2">3: -</td></tr>
<tr><td>7: -</td></tr>
<tr><td>8: -</td><td rowspan="2">4: S-CCPCH</td><td rowspan="4">2: S-CCPCH</td><td rowspan="4">Code 2: Section 6.1.3 (SIB5)
Code 4: Sections 6.1.1 & 6.1.2 (SIB5)
Code 5: Section 6.1.2 (SIB5)
See Note.</td></tr>
<tr><td>9: -</td></tr>
<tr><td>10: -</td><td rowspan="2">5: S-CCPCH</td></tr>
<tr><td>11: -</td></tr>
<tr><td>12: -</td><td rowspan="2">6: S-CCPCH</td><td rowspan="4">3: -</td><td>Section 6.1.3 (SIB5)</td></tr>
<tr><td>13: -</td><td rowspan="3">-</td></tr>
<tr><td>14: -</td><td rowspan="2">7: -</td></tr>
<tr><td>15: -</td></tr>
<tr><td colspan="4">Note: The default code allocation is extracted from section 6.1.0b. The S-CCPCH channels on codes 2, 4 & 5 are defined in specific cell configurations, as per sections 6.1.1, 6.1.2 & 6.1.3. For each configuration described above, the orthogonality is respected.</td></tr>
</table>

5.5.1.2 Downlink physical channels code allocation for HSDPA test cases

Table 5.5.1.2.1 shows details of the downlink code tree for the Primary Scrambling Code, SF=16 & Code=0 used in the HSDPA test cases. Table 5.5.1.2.2 shows the downlink code tree used for 64QAM HSDPA test cases. The numbers in

the Code columns indicate the code number with the respective spreading factor (SF). The Note column refers to specifications where the code allocation is defined.

The HS-PDSCH channels are allocated dynamically by the SS during execution of the HSDPA test case, under the same Scrambling Code as the HS-SCCH channel, on SF=16, in the range Code=1 to Code=15.

Table 5.5.1.2.1: HSDPA Downlink Physical Channels Code Allocation for SF=16 Code=0

Code with SF=256	Code with SF=128	Code with SF=64	Note
0: P-CPICH	0: -	0: -	TS 25.213
1: P-CCPCH			TS 25.213
2: PICH	1: -		Section 6.1.0b (SIB5)
3: AICH			Section 6.1.0b (SIB5)
4: -	2: -	1: S-CCPCH	Section 6.1.0b (SIB5)
5: -			
6: -	3: -		
7: -			
8: -	4: -	2: -	-
9: -			
10: -	5: -		
11: -			
12: -	6: -	3: -	-
13: -			
14: -	7: HS-SCCH		Section 9.1.1 RB Setup message
15: -			

Table 5.5.1.2.2: HSDPA [64QAM] Downlink Physical Channels Code Allocation for SF=16 Code=0

Code with SF=256	Code with SF=128	Code with SF=64	Note
0: P-CPICH	0: -	0: -	TS 25.213
1: P-CCPCH			TS 25.213
2: PICH	1: -		Section 6.1.0b (SIB5)
3: AICH			Section 6.1.0b (SIB5)
4: -	2: -	1: S-CCPCH	Section 6.1.0b (SIB5)
5: -			
6: -	3: -		
7: -			
8: -	4: -	2: -	-
9: -			
10: -	5: -		
11: -			
12: -	6: -	3: -	-
13: DPCH-			Section 9.1.1 RB Setup msg, condition A17a
14: -	7: HS-SCCH		Section 9.1.1 RB Setup message
15: -			

5.5.1.3 Downlink physical channels code allocation for E-DCH test cases

Table 5.5.1.3.1 shows details of the downlink code tree for the Primary Scrambling Code, SF=16 & Code=0 used in the E-DCH test cases. The numbers in the Code columns indicate the code number with the respective spreading factor (SF). The Note column refers to specifications where the code allocation is defined.

The HS-PDSCH channels are allocated dynamically by the SS during execution of the HSDPA test case, under the same Scrambling Code as the HS-SCCH channel, on SF=16, in the range Code=1 to Code=15.

Table 5.5.1.3.1: E-DCH Downlink Physical Channels Code Allocation for SF=16 Code=0

Code with SF=256	Code with SF=128	Code with SF=64	Note
0: P-CPICH	0: -	0: -	TS 25.213
1: P-CCPCH			TS 25.213
2: PICH	1: -		Section 6.1.0b (SIB5)
3: AICH			Section 6.1.0b (SIB5)
4: -	2: -	1: S-CCPCH	Section 6.1.0b (SIB5)
5: -			
6: -	3: -		
7: -			
8: -	4: E-HICH/E-RGCH	2: -	Section 9.1.1 RB Setup message
9: -			
10: E-AGCH	5: -		Section 9.1.1 RB Setup message
11: -			
12: F-DPCH	6: -	3: -	Section 9.1.1 RB Setup message, condition A14
13: -			-
14: -	7: HS-SCCH		Section 9.1.1 RB Setup message
15: -			

5.5.1.4 Downlink physical channels code allocation for MBMS test cases

Table 5.5.1.4 shows details of the downlink code tree for the Primary Scrambling Code used in the MBMS test cases. The numbers in the Code columns indicate the code number with the respective spreading factor (SF). The Note column refers to specifications where the code allocation is defined.

Table 5.5.1.4: MBMS Downlink Physical Channels Code Allocation

Code with SF=256	Code with SF=128	Code with SF=64	Code with SF=32	Code with SF=16	Code with SF=8	Note
0: P-CPICH	0: -	0: -	0: -	0: -	0: -	TS 25.213
1: P-CCPCH						TS 25.213
2: PICH	1: -					Clause 6.1.0b (SIB5)
3: AICH						Clause 6.1.0b (SIB5)
4: -	2: -	1: S-CCPCH1				Clause 6.1.0b (SIB5)
5: -						
6: -	3: -					
7: -						
8: MICH	4: -	2	1: -			
9: S-CCPCH2						
10: -	5: -					
11: -						
12: -	6: -	3				
13: -						
14: -	7: -					
15: -						
16: -	8: -	4	2: S-CCPCH3	1: S-CCPCH3		Code 1: 129.6 kbps RB for MTCH Code 2: 64.8kbps RB for MTCH
17: -						
18: -	9: -					
19: -						
20: -	10:-	5: -				
21: -						
22: -	11: -					
23: -						
24: -	12: -	6: -	3: S-CCPCH4			64.8kbps RB for MTCH
25: -						
26: -	13: -					
27: -						
28: -	14: -	7: -				
29: -						
30: -	15: -					
31: -						
32: -	16: -	8: -	4: -	2: S-CCPCH4	1: S-CCPCH3	Code 1: 259.2 kbps RB for MTCH Code 2: 129.6 kbps RB for MTCH
33: -						
34: -	17: -					
35: -						
36: -	18: -	9: -				
37: -						
38: -	19: -					
39: -						
40: -	20: -	10: -	5: -			
41: -						
42: -	21: -					
43: -						
44: -	22: -	11: -				
45: -						
46: -	23: -					
47: -						
48: -	24: -	12: -	6: -	3: -		
49: -						
50: -	25: -					
51: -						
52: -	26: -	13: -				
53: -						
54: -	27: -					

55: -						
56: -	28: -	14: -	7: -			
57: -						
58: -	29: -					
59: -						
60: -	30: -	15: -				
61: -						
62: -	31: -					
63: -						
64: -	32: -	16: -	8: -	4: -	2: S-CCPCH4	259.2 kbps RB for MTCH
65: -						
66: -	33: -					
67: -						
68: -	34: -	17: -				
69: -						
70: -	35: -					
71: -						
72: -	36: -	18: -	9: -			
73: -						
74: -	37: -					
75: -						
76: -	38: -	19: -				
77: -						
78: -	39: -					
79: -						
80: -	40: -	20: -	10: -	5: -		
81: -						
82: -	41: -					
83: -						
84: -	42: -	21: -				
85: -						
86: -	43: -					
87: -						
88: -	44: -	22: -	11: -			
89: -						
90: -	45: -					
91: -						
92: -	46: -	23: -				
93: -						
94: -	47: -					
95: -						

5.5.2 Downlink physical channels code allocation for Signalling (TDD)

<FFS>

5.5.3 Downlink physical channels code allocation for RF

The downlink physical channels code allocation for RF tests is defined in 3GPP TS 34.121 [2] Annex E.6.

6 Reference system configurations

This clause defines a number of Reference System Configurations which can be used for different tests.

6.1 Simulated network environments

The UE will eventually have to operate in either single mode networks (FDD or TDD), dual mode networks (FDD+TDD), or inter-RAT networks (FDD or TDD + GSM).

The following tables list the default parameters for 1 to 8 cell environments for testing.

To simplify TTCN implementation the total number of simultaneous cells in intra-frequency, inter-frequency and inter-RAT cell information lists (SIB11) have been limited to 8 (or 16 in MBMS test cases) and a specific cell numbering scheme have been defined to associate cell identifiers with type of cell.

- Cell 1, Cell 2, Cell 3, Cell 7, Cell 8 and Cell 11 are associated with FDD/TDD cells using frequency f1; Note that Cell 7 and Cell 8 can be configured on frequency f3 in some cases.
- Cell 4, Cell 5 and Cell 6 are associated with FDD/TDD cells using frequency f2;
- Cell 9 and Cell 10 are associated with GSM cells;
- Cell 21, Cell 22, Cell 23, Cell 27 and Cell 28 are associated with MBMS cells using frequency f1; Note that Cell 27 and Cell 28 can be configured on frequency f3 in some cases.
- Cell 24, Cell 25 and Cell 26 are associated with MBMS cells using frequency f2.
- Cell 31, Cell 32, Cell 37 and Cell 38 are associated with MBMS in MBSFN mode cells (clusters) using frequency f1 (FDD and TDD).
- Cell 33, Cell 34, Cell 35 and Cell 36 are associated with MBMS in MBSFN mode clusters using frequency f2. Note that Cell 36 and/or Cell 37 can be configured on frequency f3 in some cases (FDD and TDD).

Note: For the purpose of protocol conformance testing the simulation of an MBSFN cluster may be achieved with a single MBSFN cell.

For protocol testing in FDD and TDD intra- and inter-frequency cell environment Cell 1 to Cell 8 are used.

For RF and RRM in FDD and TDD intra- and inter-frequency cell environment Cell 1 to Cell 8 and Cell 11 are used.

For FDD/GSM and TDD/GSM inter-RAT cell environment Cell 1 to Cell 6, Cell 9 and Cell 10 are used.

For FDD inter-band testing the cells using frequency f1 are on one supported FDD band and the cells using frequency f2 are on a different supported FDD band. FDD inter-band testing only applies for UEs supporting multiple FDD bands simultaneously.

For MBMS testing intra- and inter-frequency cell environment Cell 21 to Cell 28 are used.

For MBSFN testing intra- and inter-frequency cell environment Cell 31 to Cell 38 are used (FDD and TDD).

In this clause, decimal values are normally used. However, sometimes a hexadecimal value, indicated by an "H", or a binary value, indicated by a "B" is used.

6.1.0a Default Master Information Block and Scheduling Block messages

6.1.0a.1 Grouping SIBs for testing

Mandatory in 34.108	**Used in Idle Mode**	MIB, SB1, (SB2), SIB1, SIB2, SIB3, SIB5/SIB5bis, SIB7, SIB11
	Used in Connected Mode	SIB4, SIB6, SIB12

Mandatory for FDD CPCH (R99 and Rel-4 only)	SIB8, SIB9
Mandatory for FDD DRAC	SIB10
Mandatory for TDD	SIB14, SIB17
Mandatory for LCS	SIB15, SIB15.1, SIB15.2, SIB15.3
Mandatory for ANSI-41 system	SIB13, SIB13.1, SIB13.2, SIB13.3, SIB13.4
Mandatory for InterSys HO from GERAN To UTRAN	SIB16
Mandatory for Cell reselection	SIB18

6.1.0a.2 SIB configurations

Currently five SIB configurations are used.

Configuration 1 is the default. It is used for the following test case scenarios:

- UTRAN/FDD only SYSTEM.
- UTRAN/FDD + GERAN SYSTEM (not involving inter-RAT handover from GERAN to UTRAN).
- UTRAN/TDD only SYSTEM.
- UTRAN/TDD + GERAN SYSTEM (not involving inter-RAT handover from GERAN to UTRAN).
- inter-RAT handover from GERAN to UTRAN test cases.

Configuration 2 is for test cases which need two S_CCPCH or two PRACH.

Configuration 3 is for inter-RAT handover from GERAN to UTRAN test cases.

Configuration 4 is applied to MBMS test cases.

Configuration 5 is applied to MBMS MBSFN test cases.

Configuration 1	MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5/SIB5bis, SIB6, SIB7, SIB11, SIB12, SIB18
Configuration 2	MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5/SIB5bis, SIB7, SIB11, SIB12, SIB18
Configuration 3	MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5/SIB5bis, SIB7, SIB11, SIB16, SIB18
Configuration 4	MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5/SIB5bis, SIB7, SIB11, SIB11bis (empty segment), SIB16, SIB18
Configuration 5	MIB, SIB3, SIB5/SIB5bis, SIB11

6.1.0a.3 SIB default schedule

Block Type	MIB	SB1	SIB1	SIB2	SIB3	SIB4	SIB5/SIB5bis	SIB6	SIB7	SIB11	SIB12	SIB18
SIB_REP	8	16	64	64	64	64	64	64	16	64	64	64
SEG_ COUNT	1	1	1	1	1	1	4	4	1	3	3	1

Frame No / SIB_POS	0	2	4	6	8	10	12	14
Block Type	MIB	SB1	SIB7	SIB6	MIB	SIB6	SIB6	SIB6
Frame No / SIB_POS	16	18	20	22	24	26	28	30
Block Type	MIB	SB1	SIB7/SIB3	SIB1/SIB2	MIB	SIB12	SIB12	SIB12
Frame No / SIB_POS	32	34	36	38	40	42	44	46
Block Type	MIB	SB1	SIB7/SIB18	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis
Frame No / SIB_POS	48	50	52	54	56	58	60	62
Block Type	MIB	SB1	SIB7/SIB4	-	MIB	SIB11	SIB11	SIB11

The SEG_COUNT in the table specifies the maximum possible transport BCH blocks scheduled for broadcasting. The more contents a SIB has, the more transport BCH blocks are needed for broadcasting. In order to keep SIB repetition

period, SIB_REP, unchanged in different test cases, each specific SIB in the individual test cases after the PER encoding shall not exceed the SEG_COUNT scheduled.

If the transport BCH blocks actually required for a SIB is less than the scheduled SEG_COUNT, the no_segment blocks shall be placed at the rest scheduled transport BCH blocks. In addition, the corresponding SEG_COUNT IE value in MIB or in SB1 shall be set to the number of transport BCH blocks actually required.

Contents of Master Information Block PLMN type is the case of GSM-MAP

- MIB value tag	A valid MIB value tag value as defined in TS 25.331 [34]
- Supported PLMN types	
- PLMN type	GSM-MAP
- PLMN identity	
- MCC digit	Set to the same Mobile Country Codes stored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)).
- MNC digit	Set to the same Mobile Network Codesstored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)).
- ANSI-41 Core Network information	Not Present
- References to other system information blocks and scheduling blocks	
- References to other system information blocks	
- Scheduling information	
- CHOICE Value tag	Cell Value Tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- Scheduling	
- SEG_COUNT	1
- SIB_REP	16
- SIB_POS	2
- SIB_POS offset info	Not Present - use default
- SIB and SB type	Scheduling Block 1
- Scheduling information	
- CHOICE Value tag	PLMN Value tag
- PLMN Value tag	A valid PLMN value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	22
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 1
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	22
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 2
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	1
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	20
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 3
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	52
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 4
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	4
- SIB_REP	64
- SIB_POS	38
- SIB_POS offset info	
- SIB_OFF	4
- SIB_OFF	2
- SIB_OFF	2
- SIB and SB type	System Information Type 5 / System Information Type 5bis

NOTE: System Information Type 5 or System Information Type 5bis are used dependent on the frequency band variant used by the SS.

Contents of Scheduling Block 1 (FDD and 1.28 Mcps TDD)

- References to other system information blocks	
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	4
- SIB_REP	64
- SIB_POS	6
- SIB_POS offset info	
- SIB_OFF	4
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 6
- Scheduling information	
- CHOICE Value tag	Not Present
- SEG_COUNT	1
- SIB_REP	16
- SIB_POS	4
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 7
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	58
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 11
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	26
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 12
- Scheduling information	
- CHOICE Value tag	Cell Value tag
Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	36
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 18

Contents of Scheduling Block 1 (3.84 Mcps TDD and 7.68 Mcps TDD)

- References to other system information blocks	
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	4
- SIB_REP	128
- SIB_POS	3
- SIB_POS offset info	

- SIB_OFF	4
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 6
- Scheduling information	
- CHOICE Value tag	Not Present
- SEG_COUNT	1
- SIB_REP	16
- SIB_POS	2
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 7
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	29
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 11
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	13
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 12
- Scheduling information	
- CHOICE Value tag	Not Present
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	54
- SIB_POS offset info	Not Present - use default
- SIB type SIBs only	System Information Type 14
- Scheduling information	
- CHOICE Value tag	PLMN Value tag
- PLMN Value tag	A valid PLMN value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	6
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 18

6.1.0a.4 SIB special schedules

6.1.0a.4.1 SIB schedule for two S-CCPCH or two PRACH (For FDD)

Table 1

Frame No.	0	2	4	6	8	10	12	14
REP-POS	0	1	2	3	4	5	6	7
Block Type	MIB	SB1	SB1		MIB	SIB1	SIB18	SIB2

Frame No.	16	18	20	22	24	26	28	30
REP-POS	8	9	10	11	12	13	14	15
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB3		SIB4

Frame No.	32	34	36	38	40	42	44	46
REP-POS	16	17	18	19	20	21	22	23
Block Type	MIB	SB1	SB1	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis

Frame No.	48	50	52	54	56	58	60	62

REP-POS	24	25	26	27	28	29	30	31
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB11	SIB11	SIB11

Frame No.	64	66	68	70	72	74	76	78
REP-POS	32	33	34	35	36	37	38	39
Block Type	MIB	SB1	SB1	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis

Frame No.	80	82	84	86	88	90	92	94
REP-POS	40	41	42	43	44	45	46	47
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB3		SIB4

Frame No.	96	98	100	102	104	106	108	110
REP-POS	48	49	50	51	52	53	54	55
Block Type	MIB	SB1	SB1		MIB			

Frame No.	112	114	116	118	120	122	124	126
REP-POS	56	57	58	59	60	61	62	63
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB12	SIB12	SIB12

SIB-repeat period (in frame)

Table 2

Block Type	MIB	SB1	SIB1	SIB2	SIB3	SIB4	SIB5/ SIB5bis	SIB7	SIB11	SIB12	SIB18
SIB Rep	8	16	128	128	64	64	128	32	128	128	128
Max. No of seg.	1	2	1	1	1	1	8	1	3	3	1

6.1.0a.4.2 SIB schedule for Idle Mode, Measurement and Inter RAT UTRAN to GERAN test cases

Frame No.	0	2	4	6	8	10	12	14
REP-POS	0	1	2	3	4	5	6	7
Block Type	MIB	SB1	SIB6	SIB6	MIB	SIB6	SIB6	SIB7/ SIB3

Frame No.	16	18	20	22	24	26	28	30
REP-POS	8	9	10	11	12	13	14	15
Block Type	MIB	SB1	SIB1/SIB2	SIB12	MIB	SIB12	SIB12	SIB7/ SIB12

Frame No.	32	34	36	38	40	42	44	46
REP-POS	16	17	18	19	20	21	22	23
Block Type	MIB	SB1	SIB5/ SIB5bis	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB7/ SIB18

Frame No.	48	50	52	54	56	58	60	62
REP-POS	24	25	26	27	28	29	30	31
Block Type	MIB	SB1	SIB11	SIB11	MIB	SIB11	SIB11	SIB7/SIB 4

SIB-repeat period (in frame)

Block Type	MIB	SB1	SIB1	SIB2	SIB3	SIB4	SIB5/ SIB5bis	SIB6	SIB7	SIB11	SIB12	SIB18
SIB Rep	8	16	64	64	64	64	64	64	16	64	64	64
Max. No of seg.	1	1	1	1	1	1	4	4	1	4	4	1

6.1.0a.4.3 SIB schedule for Inter RAT handover GERAN to UTRAN test cases

Frame No.	0	2	4	6	8	10	12	14
REP-POS	0	1	2	3	4	5	6	7
Block Type	MIB	SB1	SB1		MIB	SIB1	SIB18	SIB2

Frame No.	16	18	20	22	24	26	28	30
REP-POS	8	9	10	11	12	13	14	15
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB3		SIB4

Frame No.	32	34	36	38	40	42	44	46
REP-POS	16	17	18	19	20	21	22	23
Block Type	MIB	SB1	SB1	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis

Frame No.	48	50	52	54	56	58	60	62
REP-POS	24	25	26	27	28	29	30	31
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB11	SIB11	SIB11

Frame No.	64	66	68	70	72	74	76	78
REP-POS	32	33	34	35	36	37	38	39
Block Type	MIB	SB1	SB1	SIB16	MIB	SIB16	SIB16	SIB16

Frame No.	80	82	84	86	88	90	92	94
REP-POS	40	41	42	43	44	45	46	47
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB3		SIB4

Frame No.	96	98	100	102	104	106	108	110
REP-POS	48	49	50	51	52	53	54	55
Block Type	MIB	SB1	SB1	SIB16	MIB	SIB16	SIB16	SIB16

Frame No.	112	114	116	118	120	122	124	126
REP-POS	56	57	58	59	60	61	62	63
Block Type	MIB	SB1	SB1	SIB7	MIB			

SIB-repeat period (in frame)

Block Type	MIB	SB1	SIB1	SIB2	SIB3	SIB4	SIB5/ SIB5bis	SIB7	SIB11	SIB16	SIB18
SIB Rep	8	16	128	128	64	64	128	32	128	128	128
Max. No of seg.	1	2	1	1	1	1	4	1	3	8	1

6.1.0a.4.4 SIB schedule for MBMS test cases

Table 3

Frame No.	0	2	4	6	8	10	12	14
REP-POS	0	1	2	3	4	5	6	7
Block Type	MIB	SB1	SB1	SIB6	MIB	SIB1	SIB18	SIB2

Frame No.	16	18	20	22	24	26	28	30
REP-POS	8	9	10	11	12	13	14	15
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB3	SIB6	SIB4

Frame No.	32	34	36	38	40	42	44	46
REP-POS	16	17	18	19	20	21	22	23
Block Type	MIB	SB1	SB1	SIB5/ SIB5bis	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis

Frame No.	48	50	52	54	56	58	60	62
REP-POS	24	25	26	27	28	29	30	31
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB5/ SIB5bis	SIB5/ SIB5bis	SIB5/ SIB5bis

Frame No.	64	66	68	70	72	74	76	78
REP-POS	32	33	34	35	36	37	38	39
Block Type	MIB	SB1	SB1	SIB11	MIB	SIB11	SIB11	SIB11

Frame No.	80	82	84	86	88	90	92	94
REP-POS	40	41	42	43	44	45	46	47
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB11	SIB11	SIB11

Frame No.	96	98	100	102	104	106	108	110
REP-POS	48	49	50	51	52	53	54	55
Block Type	MIB	SB1	SB1	SIB12	MIB	SIB12		

Frame No.	112	114	116	118	120	122	124	126
REP-POS	56	57	58	59	60	61	62	63
Block Type	MIB	SB1	SB1	SIB7	MIB	SIB11bis	SIB11bis	SIB11bis

SIB-repeat period (in frame)

Table 4

Block Type	MIB	SB1	SIB1	SIB2	SIB3	SIB4	SIB5/ SIB5bis	SIB6	SIB7	SIB11	SIB11 bis	SIB12	SIB18
SIB Rep	8	16	128	128	128	128	128	128	32	128	128	128	128
Max. No of seg.	1	2	1	1	1	1	7	2	1	7	3	2	1

6.1.0a.4.5 SIB schedule for MBMS MBSFN test cases (FDD, 3.84 Mcps TDD and 7.68 Mcps TDD)

Contents of Master Information Block in the case where PLMN type is GSM-MAP

- MIB value tag	A valid MIB value tag value as defined in TS 25.331 [34]
- Supported PLMN types	
- PLMN type	GSM-MAP
- PLMN identity	
- MCC digit	Set to the same Mobile Country Codes stored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)).
- MNC digit	Set to the same Mobile Network Codesstored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)).
- ANSI-41 Core Network information	Not Present
- References to other system information blocks and scheduling blocks	
- References to other system information blocks	
- Scheduling information	
- CHOICE Value tag	Cell Value Tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- Scheduling	
- SEG_COUNT	1
- SIB_REP	16
- SIB_POS	2
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 3
- Scheduling information	
- CHOICE Value tag	Cell Value Tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- Scheduling	
- SEG_COUNT	2
- SIB_REP	16
- SIB_POS	4
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 5
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- Scheduling	
- SEG_COUNT	2
- SIB_REP	16
- SIB_POS	10
- SIB_POS offset info	Not Present - use default
- SIB and SB type	System Information Type 11

SIB schedule

Frame No / SIB_POS	0	2	4	6	8	10	12	14
Block Type	MIB	SIB3	SIB5	SIB5	SIB11	SIB11	SIB11	-

SIB-repeat period (in frame)

Block Type	MIB	SIB3	SIB5	SIB11
SIB Rep	16	16	16	16
Max. No of seg.	1	1	2	3

6.1.0b Default System Information Block Messages

Contents of System Information Block type 1 (supported PLMN type is GSM-MAP)

- CN common GSM-MAP NAS system information	A1	
- GSM-MAP NAS system information		00 01H
- CN domain system information		
- CN domain identity		PS
- CHOICE CN Type		GSM-MAP
- CN domain specific NAS system information		
- GSM-MAP NAS system information		05 00H
- CN domain specific DRX cycle length coefficient		7
- CN domain identity		CS
- CHOICE CN Type		GSM-MAP
- CN domain specific NAS system information		
- GSM-MAP NAS system information		1E 01H
- CN domain specific DRX cycle length coefficient		7
- CN common GSM-MAP NAS system information	A2	
- GSM-MAP NAS system information		00 80H (see note)
- CN domain system information		
- CN domain identity		PS
- CHOICE CN Type		GSM-MAP
- CN domain specific NAS system information		
- GSM-MAP NAS system information		00 00H (see note)
- CN domain specific DRX cycle length coefficient		7
- CN domain identity		CS
- CHOICE CN Type		GSM-MAP
- CN domain specific NAS system information		
- GSM-MAP NAS system information		1E 01H
- CN domain specific DRX cycle length coefficient		7
- UE Timers and constants in idle mode	A1, A2	
-T300		4 000 milliseconds
-N300		3
-T312		10 seconds
- N312		1
- UE Timers and constants in connected mode		
- T301		Not Present (2 000 milliseconds: default value)
- N301		Not Present (2: default value)
- T302		Not Present (4 000 milliseconds: default value)
- N302		Not Present (3: default value)
- T304		Not Present (2 000 milliseconds: default value)
- N304		Not Present (2: default value)
- T305		Not Present (30 minutes: default value)
- T307		Not Present (30 seconds: default value)
- T308		Not Present (160 milliseconds: default value)
- T309		Not Present (5 seconds: default value)
- T310		Not Present (160 milliseconds: default value)
- N310		Not Present (4: default value)
- T311		Not Present (2 000 milliseconds: default value)
- T312		Not Present (1 seconds: default value)
- N312		Not Present (1: default value)
- T313		Not Present (3 seconds: default value)
- N313		Not Present (20: default value)
- T314		Not Present (12 seconds: default value)
- T315		Not Present (180 seconds: default value)
- N315		Not Present (1: default value)
- T316		Not Present (30 seconds: default value)
- T317		Not Present (infinity: default value)
NOTE: For Inter-RAT test cases GERAN and UTRAN cells use different LAC and RAC.		

Condition	Explanation
A1	UTRAN cell environment
A2	UTRAN/GSM inter-RAT cell environment

Contents of System Information Block type 2

- URA identity list - URA identity	*Only 1 URA identity broadcasted* 0000 0000 0000 0001B

Contents of System Information Block type 3 (FDD)

Information Element	Value/remark	Version
- SIB4 indicator	TRUE	
- Cell identity	0000 0000 0000 0000 0000 0000 0001B	
- Cell selection and re-selection info		
- Mapping info	Not Present	
- Cell selection and reselection quality measure	CPICH RSCP	
- CHOICE mode	FDD	
- Sintrasearch	8 (16 dB)	
- Sintersearch	8 (16 dB)	
- SsearchHCS	Not Present	
- RAT List	This parameter is configurable	
- RAT identifier	GSM	
- Ssearch,RAT	-16 (-32 dB)	
- SHCS,RAT	Not Present	
- Slimit,SearchRAT	0 (0dB)	
- Qqualmin	Reference to table 6.1.1	
- Qrxlevmin	Reference to table 6.1.1	
- Qhyst1s	1 (2 dB)	
- Qhyst2s	Not Present	
- Treselections	0 seconds	
- HCS Serving cell information	Not Present	
- Maximum allowed UL TX power	Reference to table 6.1.1	
- Cell Access Restriction		
- Cell barred	Not barred	
- Intra-frequency cell re-selection indicator	Not present	
- T_{barred}	Not present	
- Cell Reserved for operator use	Not reserved	
- Cell Reservation Extension	Not reserved	
- Access Class Barred List		
- Access Class Barred0	Not barred	
- Access Class Barred1	Not barred	
- Access Class Barred2	Not barred	
- Access Class Barred3	Not barred	
- Access Class Barred4	Not barred	
- Access Class Barred5	Not barred	
- Access Class Barred6	Not barred	
- Access Class Barred7	Not barred	
- Access Class Barred8	Not barred	
- Access Class Barred9	Not barred	
- Access Class Barred10	Not barred	
- Access Class Barred11	Not barred	
- Access Class Barred12	Not barred	
- Access Class Barred13	Not barred	
- Access Class Barred14	Not barred	
- Access Class Barred15	Not barred	
- Domain Specific Access Restriction Parameters For PLMN Of MIB	Not present	REL-6
- Domain Specific Access Restriction For Shared Network	Not present	REL-6

Contents of System Information Block type 3 (3.84 Mcps TDD, 1.28 Mcps TDD and 7.68 Mcps TDD)

Information Element	Value/remark	Version
- SIB4 Indicator	TRUE	
- Cell identity	0000 0000 0000 0000 0000 0000 0001B	
- Cell selection and re-selection info		
- Mapping info	Not present	
- Cell selection and reselection quality measure	(no data)	
- CHOICE mode	TDD	
- Sintrasearch	10 dB	
- Sintersearch	10 dB	
- SsearchHCS	Not present	
- RAT List	This parameter is configurable	
- RAT identifier	GSM	
- Ssearch,RAT	-32 dB	
- SHCS,RAT	Not present	
- $S_{limit,ShearchRAT}$	Not Present	
- Qrxlevmin	-103 dBm	
- $Qhyst1_S$	0 dB	
- Treselections	0 seconds	
- HCS Serving cell information	Not present	
- Maximum allowed UL TX power	30dBm	
- Cell Access Restriction		
- Cell barred	Not barred	
- Intra-frequency cell re-selection indicator	Not present	
- T_{barred}	Not present	
- Cell Reserved for operator use	Not reserved	
- Cell Reservation Extension	Not reserved	
- Access Class Barred List		
- Access Class Barred0	Not barred	
- Access Class Barred1	Not barred	
- Access Class Barred2	Not barred	
- Access Class Barred3	Not barred	
- Access Class Barred4	Not barred	
- Access Class Barred5	Not barred	
- Access Class Barred6	Not barred	
- Access Class Barred7	Not barred	
- Access Class Barred8	Not barred	
- Access Class Barred9	Not barred	
- Access Class Barred10	Not barred	
- Access Class Barred11	Not barred	
- Access Class Barred12	Not barred	
- Access Class Barred13	Not barred	
- Access Class Barred14	Not barred	
- Access Class Barred15	Not barred	
- Domain Specific Access Restriction Parameters For PLMN Of MIB	Not present	REL-6
- Domain Specific Access Restriction For Shared Network	Not present	REL-6

Contents of System Information Block type 4 in connected mode (FDD)

- Cell identity	0000 0000 0000 0000 0000 0000 0001B
- Cell selection and re-selection info	
- Mapping Info	Not present
- Cell selection and reselection quality measure	CPICH RSCP
- CHOICE mode	FDD
- Sintrasearch	8 (16 dB)
- Sintersearch	8 (16 dB)
- SsearchHCS	Not present
- RAT List	This parameter is configurable
- RAT identifier	GSM
- Ssearch,RAT	-16 (-32 dB)
- SHCS,RAT	Not Present
- $S_{limit,SearchRAT}$	0 (0dB)
- Qqualmin	Reference to table 6.1.1
- Qrxlevmin	Reference to table 6.1.1

- Qhyst1s	1 (2 dB)
- Qhyst2s	Not Present
- Treselections	0 seconds
- HCS Serving cell information	Not Present
- Maximum allowed UL TX power	Reference to table 6.1.1
- Cell Access Restriction	
- Cell barred	Not barred
- Intra-frequency cell re-selection indicator	Not present
- T_{barred}	Not present
- Cell Reserved for operator use	Not reserved
- Cell Reservation Extension	Not reserved
- Access Class Barred List	Not present

Contents of System Information Block type 4 in connected mode (similar to SIB type3) (3.84 Mcps TDD, 1.28 Mcps TDD and 7.68 Mcps TDD)

- Cell identity	0000 0000 0000 0000 0000 0000 0001B
- Cell selection and re-selection info	
- Mapping info	Not Present
- Cell selection and reselection quality measure	CPICH RSCP
- CHOICE mode	TDD
- Sintrasearch	11 dB
- Sintersearch	11dB
- SsearchHCS	Not present
- RAT List	This parameter is configurable
- RAT identifier	GSM
- Ssearch,RAT	-32 dB
- SHCS,RAT	Not present
- $S_{limit,ShearchRAT}$	Not Present
- Qrxlevmin	-103 dBm
- Qhyst1s	0 dB
- Treselections	0 seconds
- HCS Serving cell information	Not present
- Maximum allowed UL TX power	30dBm
- Cell Access Restriction	
- Cell barred	Not barred
- Intra-frequency cell re-selection indicator	Not present
- T_{barred}	Not present
- Cell Reserved for operator use	Not reserved
- Cell Reservation Extension	Not reserved
- Access Class Barred List	Not present

Contents of System Information Block type 5 (FDD)

Information Element	Conditrions	Value/remark	Version
- SIB6 indicator		TRUE	
- PICH Power offset		-5 dB	
- CHOICE Mode		FDD	
- AICH Power offset		-5 dB	
- Primary CCPCH info		Not present	
- PRACH system information list			
- PRACH system information			
- PRACH info			
- CHOICE mode		FDD	
- Available Signature		'0000 0000 1111 1111'B	
- Available SF		64	
- Preamble scrambling code number		0	
- Puncturing Limit		1.00	
- Available Sub Channel number		'1111 1111 1111'B	
- Transport channel Identity		15	
- RACH TFS			
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC size		168	
- Number of TB and TTI List			

- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- RLC size		360	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		150	
- CRC size		16	
- Additional RACH TFS for CCCH			Rel6
- RLC size		240	
- Number of Transport blocks		1	
- RACH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factor	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- CTFC information		1	
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- Additional RACH TFCS for CCCH			Rel-6
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- PRACH partitioning			
- Access Service Class			
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#1)	
- Available signature End Index		7 (ASC#1)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#3)	
- Available signature End Index		7 (ASC#3)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	

- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#5)	
- Available signature End Index		7 (ASC#5)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#7)	
- Available signature End Index		7 (ASC#7)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- Persistence scaling factor			
- Persistence scaling factor		0.9 (for ASC#2)	
- Persistence scaling factor		0.9 (for ASC#3)	
- Persistence scaling factor		0.9 (for ASC#4)	
- Persistence scaling factor		0.9 (for ASC#5)	
- Persistence scaling factor		0.9 (for ASC#6)	
- Persistence scaling factor		0.9 (for ASC#7)	
- AC-to-ASC mapping table			
- AC-to-ASC mapping		6 (AC0-9)	
- AC-to-ASC mapping		5 (AC10)	
- AC-to-ASC mapping		4 (AC11)	
- AC-to-ASC mapping		3 (AC12)	
- AC-to-ASC mapping		2 (AC13)	
- AC-to-ASC mapping		1 (AC14)	
- AC-to-ASC mapping		0 (AC15)	
- CHOICE mode		FDD	
- Primary CPICH TX power		31	
- Constant value		-10	
- PRACH power offset			
- Power Ramp Step		3dB	
- Preamble Retrans Max		4	
- RACH transmission parameters			
- Mmax		2	
- NB01min		3 slot	
- NB01max		10 slot	
- AICH info			
- Channelisation code		3	
- STTD indicator		FALSE	
- AICH transmission timing		0	
- Secondary CCPCH system information			
- Secondary CCPCH info			
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		64	
- Code number		1	
- Pilot symbol existence		FALSE	
- TFCI existence		TRUE (default value)	
- Fixed or Flexible position		Flexible (default value)	
- Timing offset		Not Present Absence of this IE is equivalent to default value 0	
- TFCS		(This IE is repeated for TFC number for PCH and FACH.)	
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size	M2	6 bit	
- CHOICE CTFC Size	A1,A2,A3,M1	4 bit	
- CTFC information		0	

- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- CTFC information		2	
- Power offset information		Not Present	
- CTFC information		3	
- Power offset information		Not Present	
- CTFC information		4	
- Power offset information		Not Present	
- CTFC information	A1,A2,A3,M1	5	
- Power offset information		Not Present	
- CTFC information		6	
- Power offset information		Not Present	
- CTFC information		8	
- Power offset information		Not Present	
- CTFC information	M2	12	
- Power offset information	M2	Not Present	
- CTFC information	M2	13	
- Power offset information	M2	Not Present	
- CTFC information	M2	14	
- Power offset information	M2	Not Present	
- CTFC information	M2	15	
- Power offset information	M2	Not Present	
- CTFC information	M2	16	
- Power offset information	M2	Not Present	
- CTFC information	M2	18	
- Power offset information	M2	Not Present	
- FACH/PCH information			
- TFS		(PCH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		240	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		230	
- CRC size		16 bit	
- Transport channel Identity		12 (for PCH)	
- CTCH indicator		FALSE	
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		168	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- Number of Transport blocks		2	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		220	
- CRC size		16 bit	
- Transport channel Identity		13 (for FACH)	
- CTCH indicator		FALSE	
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		360	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	

- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Turbo	
- Rate matching attribute		130	
- CRC size		16bit	
- Transport channel Identity		14 (for FACH)	
- TFS	M2	(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		160	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		225	
- CRC size		16bit	
- Transport channel Identity		16 (for FACH)	
- CTCH indicator		FALSE	
- PICH info			
- CHOICE mode		FDD	
- Channelisation code		2	
- Number of PI per frame		18	
- STTD indicator		FALSE	
- MCCH configuration information	M1	Not Present	Rel-6
- MCCH configuration information	M2		Rel-6
- Access Info Period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- Repetition Period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- Modification period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- RLC info			
- DL UM RLC LI size		7	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info			
- Timer_OSD		Not Present	
- Window size OSD		48	
- TCTF presence		Not Present	
- CBS DRX Level 1 information		Not Present	
- Frequency Band Indicator	A1	Not Present	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A2	FDD Band under test	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A3	Extension indicator	
- Frequency Band Indicator 2		FDD Band under test	
- Secondary CCPCH system information MBMS	M2	Not Present	Rel-6
- Secondary CCPCH system information MBMS	M1		Rel-6
- Secondary CCPCH info MBMS			
- CHOICE Mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		Reference to clause 5.5.1.4 “Downlink physical channels code allocation for MBMS test cases”	
- Code number		Reference to clause 5.5.1.4 “Downlink physical channels code allocation for MBMS test cases”	
- Timing Offset		Set to (Cell No, – 21) * 9 for MBMS Cell Nos 21-28. (actual value = IE value * 256 chips)	

- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- FACH carrying MCCH			
- TFS			
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		160	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		160	
- CRC size		16bit	
- MCCH configuration information			
- Access Info Period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- Repetition Period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- Modification period coefficient		Reference to clause 11.1.1 “MCCH configuration parameters”	
- RLC info MBMS			
- DL UM RLC LI size		7	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info			
- Timer_OSD		Not Present	
- Window size OSD		48	
- TCTF presence		FALSE	
- FACH carrying MTCH list		Not Present	
- FACH carrying MSCH		Not Present	
- CHOICE Mode	B1	FDD	Rel-7
- HS-DSCH common system information			Rel-7
- CCCH mapping info			
- Logical channel identity		2	
- MAC-ehs queue identity		1	
- SRB1 mapping info		Not Present	
- Common MAC-ehs reordering queue list			
- MAC-ehs queue to configure list		Configure 1 queue	
- MAC-ehs queue Id		1	
- T1		50ms	
- Treset		Not Present	
- MAC-ehs window size		16	
- HS-SCCH system info			
- DL Scrambling Code		Not Present	
- HS-SCCH Channelisation Code Information		Use 1 HS-SCCH	
- HS-SCCH Channelisation Code		7	
- HARQ system Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Common H-RNTI Information		Use 4	
- Common H-RNTI		'1111 1010 1010 1010'	

- BCCH specific H-RNTI		'1111 1010 1110 1010'	
- Common H-RNTI		'1111 1010 1010 1011'	
- BCCH specific H-RNTI		'1111 1010 1110 1011'	
- Common H-RNTI		'1111 1010 1010 1100'	
- BCCH specific H-RNTI		'1111 1010 1110 1100'	
- Common H-RNTI		'1111 1010 1010 1110'	
- BCCH specific H-RNTI		'1111 1010 1110 1110'	
- HS-DSCH paging system information			Rel-7
- DL Scrambling Code		Not Present	
- PICH for HSDPA supported paging list		Use value 1	
- HSDPA associated PICH info			
- CHOICE mode		FDD	
- Channelisation code		13	
- Number of PI per frame		18	
- STTD Indicator		False	
- HS-PDSCH Channelisation Code		1	
- Number of PCCH transmissions		3	
- Transport Block Size List		1	
- Transport Block Size Index		1	

Condition	Explanation
A1	Band I, Band II, Band III
A2	Band V, Band VI, Band VII
A3	Band VIII & bands beyond Band X
B1	Only for cells which configure HS-DSCH reception in CELL_FACH
M1	Only for MBMS cells with MCCH mapped on an S-CCPCH used for MBMS purposes only
M2	Only for MBMS cells with MCCH mapped on an S-CCPCH also used for non- MBMS purposes

Note: for non-MBMS cell MBMS specific IEs should be set to not present.

Contents of System Information Block type 5 (3.84 Mcps TDD)

- SIB6 indicator	TRUE
- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Alpha	(1/8)
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	-10
- UE positioning related parameters	Not Present /REL-4/
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- CHOICE SyncCase	Sync Case 2
- Timeslot	0
- Cell parameters ID	Not Present
- SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Timeslot number	14
- PRACH Channelisation Code List	
- CHOICE SF	SF8
- Channelisation Code List	
- Channelisation Code	8/1
- Channelisation Code	8/2
- Channelisation Code	8/3
- Channelisation Code	8/4

- PRACH Midamble	Direct
- PNBSCH allocation	Not Present /REL-4/
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- RACH TFCS	Not present
- PRACH partitioning	
- Access Service Class	
- ASC Settings	(ASC#0)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#1)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#2)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#3)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#4)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#5)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#6)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- Persistence scaling factors	
- Access Service Class	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)
- Persistence scaling factor	0.9 (for ASC#4)

- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- AC-to-ASC mapping	
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE *mode*	TDD
- Offset	0
- Common timeslot info	
- 2nd interleaving mode	Frame
- TFCI coding	Reference clause 6.10 "Parameter Set"
- Puncturing limit	Reference clause 6.10 "Parameter Set"
- Repetition period	Not Present (MD "1")
- Repetition length	Not present (empty)
- Individual timeslot info	
- CHOICE TDD option	3.84 Mcps TDD
- Timeslot number	1
- TFCI existence	Reference clause 6.10 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE *TDD option*	3.84 Mcps TDD
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	4
- Midamble Shift	Not Present
- CHOICE *TDD option*	3.84 Mcps TDD
- no data	
- Code List	
- Channelisation Code	(This IE is repeated for Code number for PCH and FACH)
- TFCS	(This IE is repeated for TFC number for PCH and FACH)
-CHOICE *TFCI signalling*	
- Normal	
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete information	
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.
- CTFC information	Reference clause 6.10 "Parameter Set"
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	12 (for PCH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	

- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	13 (for FACH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	14 (for FACH)
- CTCH indicator	FALSE
- PICH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	8
- Midamble Shift	Not Present
- Channelisation code	16/16
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- N_{GAP}	4
- N_{PCH}	2
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 5 (1.28 Mcps TDD)

- SIB6 indicator	TRUE
- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- no data	
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- TSTD indicator	FALSE
- Cell parameters ID	Not Present
- Block SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	

- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- SYNC_UL info	
- SYNC_UL codes bitmap	"11111111"
- UL Target SIR	10 dB
- Power Ramping Step	3 dB
- Max SYNC_UL Transmissions	8
- Mmax	32
- PRACH definition	
- Timeslot number	
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- Timeslot number	1
- PRACH Channelisation Code	
- Channelisation Code List	
- Channelisation Code	(8/1)
- Midamble Shift and burst type	
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- Midamble Allocation Mode	Default midamble
- Midamble configuration	8
- Midamble Shift	Not present
- FPACH info	
- Timeslot number	6
- Channelisation code	(16/16)
- Midamble Shift and burst type	
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- Midamble Allocation Mode	Common Midamble
- Midamble configuration	8
- Midamble Shift	Not present
- WT	4
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	Reference clause 6.11 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.11 "Parameter Set"
- Number of Transport blocks	Reference clause 6.11 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.11 "Parameter Set"
- Type of channel coding	Reference clause 6.11 "Parameter Set"
- Coding Rate	Reference clause 6.11 "Parameter Set"
- Rate matching attribute	Reference clause 6.11 "Parameter Set"
- CRC size	Reference clause 6.11 "Parameter Set"
- RACH TFCS	Not present
- PRACH partitioning	
- Access Service Class	
- ASC Settings	(ASC#0)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#1)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#2)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#3)
- CHOICE mode	TDD

- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#4)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#5)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- ASC Settings	(ASC#6)
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
- Available SYNC_UL codes indices	"11111111"
- CHOICE subchannel size	Size1
- Available Subchannels	Null
- Access Service Class	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)
- Persistence scaling factor	0.9 (for ASC#4)
- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- AC-to-ASC mapping	
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE *mode*	TDD
- Offset	0
- Common timeslot info	
- 2nd interleaving mode	Frame
- TFCI coding	Reference clause 6.11 "Parameter Set"
- Puncturing limit	Reference clause 6.11 "Parameter Set"
- Repetition period	1
- Repetition length	0
- Individual timeslot info	
- CHOICE *TDD option*	1.28 Mcps TDD
- Timeslot number	0
- TFCI existence	Reference clause 6.11 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE *TDD option*	1.28 Mcps TDD
- Midamble Allocation Mode	Default midamble
- Midamble configuration	4
- Midamble Shift	Not Present
- CHOICE *TDD option*	1.28 Mcps TDD
- Modulation	Reference clause 6.11 "Parameter Set"
- SS-TPC Symbols	Reference clause 6.11 "Parameter Set"
- Code List	
- Channelisation Code	Reference clause 6.11 "Parameter Set"
- TFCS	Reference clause 6.11 "Parameter Set"
- CHOICE TFCI *signalling*	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Addition
- TFCS addition information	
- CHOICE CTFC Size	Number of bits used must be enough to cover all

	combinations of CTFC from clause 6.11.
- CTFC information	Reference clause 6.11 "Parameter Set"
- Power offset information	Not Present
- FACH/PCH information	
- Transport channel Identity	12 (for PCH)
- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	(This IE is repeated for TFI number.)
- RLC Size	Reference clause 6.11 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.11 "Parameter Set"
- Number of Transport blocks	Reference clause 6.11 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.11 "Parameter Set"
- Type of channel coding	Reference clause 6.11 "Parameter Set"
- Coding Rate	Reference clause 6.11 "Parameter Set"
- Rate matching attribute	Reference clause 6.11 "Parameter Set"
- CRC size	Reference clause 6.11 "Parameter Set"
- Transport channel Identity	13 (for FACH)
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	(This IE is repeated for TFI number.)
- RLC Size	Reference clause 6.11 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.11 "Parameter Set"
- Number of Transport blocks	Reference clause 6.11 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.11 "Parameter Set"
- Type of channel coding	Reference clause 6.11 "Parameter Set"
- Coding Rate	Reference clause 6.11 "Parameter Set"
- Rate matching attribute	Reference clause 6.11 "Parameter Set"
- CRC size	Reference clause 6.11 "Parameter Set"
- CTCH indicator	FALSE
- PICH info	
- CHOICE *mode*	TDD
- CHOICE *TDD option*	1.28 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	
- Midamble Allocation Mode	Default midamble
- Midamble configuration	8
- Midamble Shift	Not Present
- Channelisation code list	
- Channelisation code	(16/1)
- Channelisation code	(16/2)
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- N_{GAP}	4
- N_{PCH}	2
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 5 (7.68 Mcps TDD)

- SIB6 indicator	TRUE
- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PUSCH system information VHCR	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	7.68 Mcps TDD
- Alpha	(1/8)

- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	-10
- UE positioning related parameters	Not Present
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	7.68 Mcps TDD
- CHOICE SyncCase	Sync Case 2
- Timeslot	0
- Cell parameters ID	Not Present
- SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Timeslot number	14
- PRACH Channelisation Code List VHCR	
- CHOICE SF	SF16
- Channelisation Code List	
- Channelisation Code	16/1
- Channelisation Code	16/2
- Channelisation Code	16/3
- Channelisation Code	16/4
- PRACH Midamble	Direct
- PNBSCH allocation	Not Present /REL-4/
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- RACH TFCS	Not present
- PRACH partitioning	
- Access Service Class	
- ASC Settings	(ASC#0)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#1)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#2)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#3)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1

- Available Subchannels	null
- ASC Settings	(ASC#4)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#5)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#6)
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- Persistence scaling factors	
- Access Service Class	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)
- Persistence scaling factor	0.9 (for ASC#4)
- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- AC-to-ASC mapping	
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE *mode*	7.68 Mcps TDD
- Offset	0
- Common timeslot info	
- 2nd interleaving mode	Frame
- TFCI coding	Reference clause 6.10 "Parameter Set"
- Puncturing limit	Reference clause 6.10 "Parameter Set"
- Repetition period	Not Present (MD "1")
- Repetition length	Not present (empty)
- Individual timeslot info	
- CHOICE TDD option	7.68 Mcps TDD
- Timeslot number	1
- TFCI existence	Reference clause 6.10 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE *TDD option*	7.68 Mcps TDD
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	4
- Midamble Shift	Not Present
- CHOICE *TDD option*	7.68 Mcps TDD
- no data	
- Code List	
- Channelisation Code	(This IE is repeated for Code number for PCH and FACH)
- TFCS	(This IE is repeated for TFC number for PCH and FACH)
-CHOICE *TFCI signalling*	
- Normal	
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete information	
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations

	of CTFC from clause 6.10.
- CTFC information	Reference clause 6.10 "Parameter Set"
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	12 (for PCH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	13 (for FACH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	14 (for FACH)
- CTCH indicator	FALSE
- PICH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	8
- Midamble Shift	Not Present
- Channelisation code	32/32
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- N_{GAP}	4

- N_{PCH}	2
- MCCH configuration information	Not Present
- CBS DRX Level 1 information	Not Present
- Frequency band indicator	Not Present
- Frequency band indicator 2	Not Present
- HSDPA cell Indicator	Not Present
- E-DCH cell Indicator	Not Present
- Secondary CCPCH system information MBMS	Not Present

Contents of System Information Block type 5bis (FDD)

The message structure of the System information block type 5bis should be the same as System information block type 5 with the following exceptions as given below.

- Frequency Band Indicator	A1	FDD Band under test
- Frequency Band Indicator 2		Not Present
- Frequency Band Indicator	A2	Extension indicator
- Frequency Band Indicator 2		FDD Band under test

Condition	Explanation
A1	Band IV
A2	Band IX, Band X

Contents of System Information Block type 6 in connected mode (FDD)

- PICH power offset	-5 dB
- CHOICE Mode	FDD
- AICH power offset	-5 dB
- Primary CCPCH info	Not Present
- PRACH system information list	Not present
- Secondary CCPCH system info	Not Present
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 6 in connected mode (similar to SIB type 5) (3.84 Mcps TDD)

- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Alpha	(1/8)
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	-10
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- CHOICE SyncCase	Sync Case 2
- Timeslot	0
- Cell parameters ID	Not Present
- SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Timeslot number	14
- PRACH Channelisation Code List	
- CHOICE SF	SF8
- Channelisation Code List	
- Channelisation Code	8/1

- Channelisation Code	8/2
- Channelisation Code	8/3
- Channelisation Code	8/4
- PRACH Midamble	Direct
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- RACH TFCS	Not present
- PRACH partitioning	
- Access Service Class	
- ASC Settings	(ASC#0)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#1)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#2)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#3)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#4)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#5)
- CHOICE mode	TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Settings	(ASC#6)
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- Persistence scaling factors	
- Access Service Class	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)

- Persistence scaling factor	0.9 (for ASC#4)
- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- AC-to-ASC mapping	Not Present
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE *mode*	TDD
- Offset	0
- Common timeslot info	
- 2nd interleaving mode	Not Present (MD "Frame")
- TFCI coding	Reference clause 6.10 "Parameter Set"
- Puncturing limit	Reference clause 6.10 "Parameter Set"
- Repetition period	Not Present (MD "1")
- Repetition length	Not present
- Individual timeslot info	
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Timeslot number	1
- TFCI existence	Reference clause 6.10 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	4
- Midamble Shift	Not Present
- Code List	
- Channelisation Code	Reference clause 6.10 "Parameter Set"
- TFCS	(This IE is repeated for TFC number for PCH and FACH.)
- Normal	
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.
- CTFC information	Reference clause 6.10 "Parameter Set"
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	12 (for PCH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- Transmission Time Interval	Reference clause 6.10 "Parameter Set"
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"

- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	13 (for FACH)
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	(This IE is repeated for TFI number.)
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TB and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Reference clause 6.10 "Parameter Set"
- Coding Rate	Reference clause 6.10 "Parameter Set"
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- Transport channel Identity	14 (for FACH)
- CTCH indicator	FALSE
- CTCH indicator	FALSE
- PICH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	8
- Midamble Shift	Not Present
- Channelisation code	16/16
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- N_{GAP}	4
- N_{PCH}	2
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type6 In connected mode (1.28 Mcps TDD)

- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dBm
- CHOICE TDD option	1.28 Mcps TDD /REL-4/
- Primary CCPCH info	Not Present
- PRACH system information list	Not Present
- Secondary CCPCH system information	Not Present
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 6 in connected mode (7.68 Mcps TDD)

- PICH Power offset	-5 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	Not present
- Primary CCPCH info	Not Present
- PRACH system information list	Not Present
- Secondary CCPCH system information	Not Present
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 7 (FDD)

CHOICE Mode	FDD

- UL interference	-100 dBm
- PRACHs listed in system information block type5	
- Dynamic persistence level	2
- PRACHs listed in system information block type6	Not Present
- Expiration Time Factor	Not Present - use default value of 1

Contents of System Information Block type 7 (TDD)

CHOICE Mode	TDD
PRACHs listed in system information block type5	
- Dynamic persistence level	2
PRACHs listed in system information block type6	
- Dynamic persistence level	2
Expiration Time Factor	Not Present - use default value of 1

Contents of System Information Block type 8, 9 (only for FDD R99 and Rel-4)

This information is used for static CPCH in the cell, so this is not present.

Contents of System Information Block type 10 (only for FDD R99 and Rel-4)

This information is used for DRAC, so this is not present.

Contents of System Information Block type 11 (FDD)

This is the default message content of SIB 11 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 11 (FDD) for cell 2 to 8.

See clause 6.1.4.3 for the difference in message contents of System informationBlock type 11(FDD) for cell 21 to 28

- SIB12 indicator	A1, A2, A3	TRUE
- FACH measurement occasion info		Not Present
- Measurement control system information		
- Use of HCS		Not used
- Cell selection and reselection quality measure		CPICH RSCP
- Intra-frequency measurement system information	A1, A2, A3	
- Intra-frequency measurement identity		Not Present Absence of this IE is equivalent to default value 1
- Intra-frequency cell info list		
- CHOICE intra-frequency cell removal		Not present (This IE shall be ignored by the UE for SIB11)
- New intra-frequency cells		
- Intra-frequency cell id		1
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0 dB
- Reference time difference to cell		Not Present
- Read SFN indicator		FALSE
- CHOICE mode		FDD
- Primary CPICH info		
- Primary scrambling code		Refer to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Primary CPICH TX power		Not Present
- TX Diversity indicator		FALSE
- Cell Selection and Re-selection info		Not Present (The IE shall be absent as this is the serving cell)
- Intra-frequency cell id		2
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0dB
- Reference time difference to cell		Not present

- Read SFN indicator		TRUE
- CHOICE mode		FDD
- Primary CPICH info		
- Primary scrambling code		Refer to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Primary CPICH TX power		Not Present
- TX Diversity indicator		FALSE
- Cell Selection and Re-selection info		Not present For neighbouring cell, if HCS is not used and all the parameters in cell selection and re-selection info are Default value, this IE is absent.
- Intra-frequency cell id		3
- Cell info		Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Intra-frequency cell id	A1, A3	7
- Cell info		Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 Note that this cell can also be configured as an inter-frequency cell on f3.
- Intra-frequency cell id	A1, A3	8
- Cell info		Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 Note that this cell can also be configured as an inter-frequency cell on f3.
- Intra-frequency cell id	A3	11
- Cell info		Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4
- Cells for measurement	A1, A2, A3	Not Present
- Intra-frequency measurement quantity	A1, A2, A3	
- Filter coefficient		Not present Absence of this IE is equivalent to the default value 0
- CHOICE mode		FDD
- Measurement quantity		CPICH RSCP
- Intra-frequency reporting quantity for RACH Reporting		Not Present
- Maximum number of reported cells on RACH		Not Present
- Reporting information for state CELL_DCH		
- Intra-frequency reporting quantity		
- Reporting quantities for active set cells		
- Cell synchronization information reporting indicator		FALSE
- Cell identity reporting indicator		TRUE
- CHOICE mode		FDD
- CPICH Ec/N0 reporting indicator		FALSE
- CPICH RSCP reporting indicator		TRUE
- Pathloss reporting indicator		FALSE
- Reporting quantities for monitored set cells		
- Cell synchronization information reporting indicator		TRUE
- Cell identity reporting indicator		TRUE
- CHOICE mode		FDD
- CPICH Ec/N0 reporting indicator		FALSE
- CPICH RSCP reporting indicator		TRUE
- Pathloss reporting indicator		FALSE
- Reporting quantities for detected set cells		Not Present

- Measurement reporting mode		
- Measurement Report Transfer Mode		Acknowledged mode RLC
- Periodic Reporting/Event Trigger Reporting Mode		Event trigger
- CHOICE report criteria		Intra-frequency measurement reporting criteria
- Intra-frequency measurement reporting criteria		
- Parameters required for each event		3 kinds
- Intra-frequency event identity		1a
- Triggering condition 1		Not Present
- Triggering condition 2		Monitored set cells
- Reporting Range Constant		10 (5dB)
- Cells forbidden to affect Reporting range		Not Present
- W		1(0.1): 34.123 test cases 10(1.0): 34.121 test cases
- Hysteresis		0 (0.0)
- Threshold Used Frequency		Not Present
- Reporting deactivation threshold		2
- Replacement activation threshold		Not Present
- Time to trigger		640
- Amount of reporting		4
- Reporting interval		4 000
- Reporting cell status		
- CHOICE reported cell		Report cell within active set and/or monitored set cells on used frequency
- Maximum number of reported cells		3
- Intra-frequency event identity		1b
- Triggering condition 1		Active set cells
- Triggering condition 2		Not Present
- Reporting Range Constant		10 (5dB)
- Cells forbidden to affect Reporting range		Not Present
- W		1 (0.1): 34.123 test cases 10(1.0): 34.121 test cases
- Hysteresis		0 (0.0)
- Threshold Used Frequency		Not Present
- Reporting deactivation threshold		Not Present
- Replacement activation threshold		Not Present
- Time to trigger		640
- Amount of reporting		Not Present
- Reporting interval		Not Present
- Reporting cell status		
- CHOICE reported cell		Report cell within active set and/or monitored set cells on used frequency
- Maximum number of reported cells		3
- Intra-frequency event identity		1c
- Triggering condition 1		Not Present
- Triggering condition 2		Not Present
- Reporting Range Constant		Not Present
- Cells forbidden to affect Reporting range		Not Present
- W		Not Present
- Hysteresis		0 (0.0)
- Threshold Used Frequency		Not Present
- Reporting deactivation threshold		Not Present
- Replacement activation threshold		3
- Time to trigger		640
- Amount of reporting		4
- Reporting interval		4 000
- Reporting cell status		
- CHOICE reported cell		Report cell within active set and/or monitored set cells on used frequency
- Maximum number of reported cells		3
- Inter-frequency measurement system information	A1, A2	
- Inter-frequency cell info list		
- CHOICE Inter-frequency cell removal		Not present (This IE shall be ignored by the UE for SIB11)
- New inter-frequency cells		
- Inter frequency cell id		4
- Frequency info		
- CHOICE mode		FDD

- UARFCN uplink(Nu)		Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]
- UARFCN downlink(Nd)		Reference to table 6.1.2 for Cell 4
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0 dB
- Reference time difference to cell		Not present
- Read SFN indicator		FALSE
- CHOICE mode		FDD
- Primary CPICH info		
- Primary scrambling code		Refer to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Primary CPICH Tx power		Not present
- TX Diversity Indicator		FALSE
- Cell Selection and Re-selection Info		Not present (same values as for serving cell applies)
- Inter frequency cell id		5
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
- Inter frequency cell id		6
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
- Cell for measurement		Not present
- Inter-RAT measurement system information	A1, A3	Not Present
- Inter-RAT measurement system information	A2	
- Inter-RAT cell info list		
- CHOICE *Inter-RAT cell removal*		Not Present (This IE shall be ignored by the UE for SIB11)
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		
- Cell individual offset		0
- Cell selection and re-selection info		Not Present
- BSIC		
- Base transceiver Station Identity Code (BSIC)		Reference to table 6.1.10 for Cell 9
- Band indicator		According to PICS/PIXIT
- BCCH ARFCN		Reference to table 6.1.10 for Cell 9
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		
- Cell individual offset		0
- Cell selection and re-selection info		Not Present
- BSIC		
- Base transceiver Station Identity Code (BSIC)		Reference to table 6.1.10 for Cell 10
- Band indicator		According to PICS/PIXITs
- BCCH ARFCN		Reference to table 6.1.10 for Cell 10
- Cell for measurement		Not present
- Traffic volume measurement system information	A1, A2, A3	Not Present

Condition	Explanation
A1	FDD cell environment

A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

Contents of System Information Block type 11 (3.84 Mcps, 1.28 Mcps and 7.68 Mcps TDD)

This is the default message content of SIB 11 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 11 (TDD) for cell 2 to 8.

- SIB 12 Indicator	A1, A2	TRUE
- FACH measurement occasion info		Not Present
- Measurement control system information		
- Use of HCS		Not used
- Cell selection and reselection quality measureCell		CPICH RSCP
- Intra-frequency measurement system information	A1, A2	
- Intra-frequency measurement identity		Not Present Absence of this IE is equivalent to default value 1
- Intra-frequency cell info list		
- CHOICE intra-frequency cell removal		Not present (This IE shall be ignored by the UE for SIB11)
- New intra-frequency cells		
- Intra-frequency cell id		1
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0dB
- Reference time difference to cell		Not Present
- Read SFN Indicator		FALSE
- CHOICE mode		TDD
- Primary CCPCH info		
- Cell parameters ID		Reference clause 6.1.4 Default settings for cell
- Primary CCPCH TX power		Not Present
- Timeslot list		Not Present
- CHOICE TDD option		
- 3.84 Mcps TDD		
- Timeslot number		Not Present
- Burst type		Not Present
- 1.28 Mcps TDD		
- Timeslot number		Not Present
- Cell Selection and Re-selection info		Not Present (The IE shall be absent as this is the serving cell)
- Intra-frequency cell id		2
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0dB
- Reference time difference to cell		Not Present
- Read SFN Indicator		FALSE
- CHOICE mode		TDD
- Primary CCPCH info		
- Cell parameters ID		Refer to clause titled “Default setting for cell No.2 (TDD)” in clause 6.1.4
- Primary CCPCH TX power		Not Present
- Timeslot list		Not Present
- CHOICE TDD option		
- 3.84 Mcps TDD		
- Timeslot number		Not Present
- Burst type		Not Present
- 1.28 Mcps TDD		
- Timeslot number		Not Present
- Cell Selection and Re-selection info		Not Present
- Intra-frequency cell id		3
- Cell info		Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.3(TDD)” in clause 6.1.4
- Intra-frequency cell id		7

- Cell info		Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.7(TDD)” in clause 6.1.4
- Intra-frequency cell id		8
- Cell info		Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.8(TDD)” in clause 6.1.4
- Cell for measurement	A1, A2	Not Present
- Intra-frequency measurement quantity	A1, A2	
- Filter coefficient		Not present Absence of this IE is equivalent to the default value 0
- CHOICE mode		TDD
- Measurement quantity list		
- Measurement quantity		P-CCPCH RSCP
- Intra-frequency reporting quantity for RACH Reporting		Not Present
- Maximum number of reported cells on RACH		Not Present
- Reporting information for state CELL_DCH		
- Intra-frequency reporting quantity		
- Reporting quantities for active set cells		
- Cell synchronization information reporting indicator		TRUE
- Cell identity reporting indicator		TRUE
- CHOICE mode		TDD
- Timeslot ISCP reporting indicator		FALSE
- Proposed TSGN reporting required		FALSE
- P-CCPCH RSCP reporting indicator		TRUE
- Pathloss reporting indicator		FALSE
- Reporting quantities for monitored set cells		
- Cell synchronization information reporting indicator		FALSE
- Cell identity reporting indicator		TRUE
- CHOICE mode		TDD
- Timeslot ISCP reporting indicator		FALSE
- Proposed TSGN reporting required		FALSE
- P-CCPCH RSCP reporting indicator		TRUE
- Pathloss reporting indicator		FALSE
- Reporting quantities for detected set cells		Not Present
- Measurement reporting mode		
- Measurement Report Transfer Mode		Acknowledged mode RLC
- Periodical Reporting / Event Trigger Reporting Mode		Event trigger
-CHOICE report criteria		
- Intra-frequency measurement reporting criteria		
- Parameters required for each event		
- Intra-frequency event identity		1g
- Triggering condition1		Not Present
- Triggering condition2		Not Present
- Reporting Range Constant		Not Present
- cells forbidden to affect reporting range		Not Present
- W(optional in case of 1a,1b)		Not Present
- Hysteresis		0.0
- Threshold used frequency		Not Present
- Reporting deactivation threshold		3
- Replacement activation threshold		Not Present
- Time to trigger		640
- Amount of reporting		4
- Reporting interval		4000
- Reporting cell status		
- CHOICE reported cells		Report cell within active set and/or monitored cells on used frequency
- Maximum number of reported cells		3
- Inter-frequency measurement system information	A1, A2	
- Inter-frequency cell info list		
- CHOICE Inter-frequency cell removal		Not present (This IE shall be ignored by the UE for SIB11)

- New inter-frequency cells		
- Inter frequency cell id		4
- Frequency info		
- CHOICE mode		TDD
- UARFCN (Nt)		Reference to table 6.1.7 for Cell 4
- Cell info		
- Cell individual offset		Not present Absence of this IE is equivalent to default value 0dB
- Reference time difference to cell		Not present
- Read SFN indicator		FALSE
- CHOICE mode		TDD
- Primary CCPCH info		Refer to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Primary CCPCH Tx power		Not present
- TX Diversity Indicator		FALSE
- Cell Selection and Re-selection Info		Not present (same values as for serving cell applies)
- Inter frequency cell id		5
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
- Inter frequency cell id		6
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
- Cell for measurement		Not present
- Inter-RAT measurement system information	A1	Not Present
- Inter-RAT measurement system information	A2	
- Inter-RAT cell info list		
- CHOICE *Inter-RAT cell removal*		Not Present (This IE shall be ignored by the UE for SIB11)
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		
- Cell individual offset		0
- Cell selection and re-selection info		Not Present
- BSIC		
- Base transceiver Station Identity Code (BSIC)		Reference to table 6.1.10 for Cell 9
- Band indicator		According to PICS/PIXIT
- BCCH ARFCN		Reference to table 6.1.10 for Cell 9
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		
- Cell individual offset		0
- Cell selection and re-selection info		Not Present
- BSIC		
- Base transceiver Station Identity Code (BSIC)		Reference to table 6.1.10 for Cell 10
- Band indicator		According to PICS/PIXITs
- BCCH ARFCN		Reference to table 6.1.10 for Cell 10
- Cell for measurement		Not present
- Traffic volume measurement system information	A1, A2	Not Present

Condition	Explanation
A1	TDD cell environment
A2	TDD/GSM inter-RAT cell environment

Contents of System Information Block type 12 in connected mode (FDD)

This is the default message content of SIB 12 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 12 (FDD) for cell 2 to 8.

- FACH measurement occasion info	Not Present
- Measurement control system information	
- Use of HCS	Not used
- Cell selection and reselection quality measure	CPICH RSCP
- Intra-frequency measurement system information	Not Present
- Inter-frequency measurement system information	Not Present
- Inter-RAT measurement system information	Not Present
- Traffic volume measurement system information	Not Present

Contents of System Information Block type 12 in connected mode (3.84 Mcps, 1.28 Mcps and 7.68 Mcps TDD)

This is the default message content of SIB 12 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 12 (TDD) for cell 2 to 8.

- FACH measurement occasion info	Not Present
- Measurement control system information	
- Use of HCS	Not used
- Cell selection and reselection quality measure	CPICH RSCP
- Intra-frequency measurement system information	Not Present
- Inter-RAT measurement system information	Not Present
- Traffic volume measurement system information	Not Present

Contents of System Information Block type 13 (used when supported PLMN type is ANSI-41)

- CN Domain system information list	
- CN Domain system information	*For Packet-Switched domain*
- CN domain identity	PS
- CHOICE CN Type	ANSI-41
- CN domain specific NAS system information	
- NAS (ANSI-41) system information	T.B.D
- CN domain specific DRX cycle length coefficient	7
- CN Domain system information	*For Circuit-Switched domain*
- CN domain identity	CS
- CHOICE CN Type	ANSI-41
- CN domain specific NAS system information	
- NAS (ANSI-41) system information	T.B.D
- CN domain specific DRX cycle length coefficient	7
- UE timers and constants in idle mode	
- T300	400 milliseconds
- N300	3
- T312	10 seconds
- N312	200
- Capability update requirement	
- UE radio access FDD capability update requirement	TRUE
- UE radio access TDD capability update requirement	FALSE
- System specific capability update requirement list	Not Present

Contents of System Information Block type 14 (3.84 Mcps and 7.68 Mcps TDD)

- Individual Timeslot interference list	
- Individual Timeslot interference	
- Timeslot number	2
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	3
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	4
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	5
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	6
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	7
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	9
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	10
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	11
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	12
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	13
- UL Timeslot Interference	-90 dbm
- Individual Timeslot interference	
- Timeslot number	14
- UL Timeslot Interference	-90 dbm
- Expiration Time Factor	Not Present (MD "1")

Contents of System Information Block type 16 (FDD)

- Pre-Defined Radio Configuration	(12.2 KBPS AMR)
- Pre-defined RB configuration	
- Re-establishment timer	useT315
- SRB InformationList	
- Signalling RB information to setup	(UM DCCH for RRC)
- RB identity	1
- CHOICE RLC info type	
- RLC info	
- CHOICE Uplink RLC mode	UM RLC
- Transmission RLC discard	timerBasedNoExplicit : dt100
- CHOICE Downlink RLC mode	UM RLC
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	5
- Logical channel identity	1
- CHOICE RLC size list	Configured
- MAC logical channel priority	1
- Downlink RLC logical channel info	
- Number of RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	10
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	1
- Signalling RB information to setup	(AM DCCH for RRC)
- RB identity	2
- CHOICE RLC info type	
- RLC info	
- CHOICE Uplink RLC mode	AM RLC
- Transmission RLC discard	
- SDU discard mode	Max DAT retransmissions
- MAX_DAT	4
- Timer_MRW	100
- MaxMRW	4
- Transmission window size	8
- Timer_RST	500
- Max_RST	4
- Polling info	
- Timer_poll_prohibit	200
- Timer_poll	200
- Poll_PDU	Not Present
- Poll_SDU	1
- Last transmission PDU poll	TRUE
- Last retransmission PDU poll	TRUE
- Poll_Window	99
- Timer_poll_periodic	Not Present
- CHOICE Downlink RLC mode	AM RLC
- In-sequence delivery	TRUE
- Receiving window size	8
- Downlink RLC status info	
- Timer_status_prohibit	200
- Timer_EPC	200
- Missing PDU indicator	TRUE
- Timer_STATUS_periodic	Not Present
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	5
- Logical channel identity	2
- CHOICE RLC size list	Configured
- MAC logical channel priority	2
- Downlink RLC logical channel info	
- Number of RLC logical channels	1
- Downlink transport channel type	DCH

- DL DCH Transport channel identity	10
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	2
- Signalling RB information to setup	(AM DCCH for RRC)
- RB identity	3
- CHOICE RLC info type	
- RLC info	
- CHOICE Uplink RLC mode	AM RLC
- Transmission RLC discard	
- SDU discard mode	Max DAT retransmissions
- MAX_DAT	4
- Timer_MRW	100
- MaxMRW	4
- Transmission window size	8
- Timer_RST	500
- Max_RST	4
- Polling info	
- Timer_poll_prohibit	200
- Timer_poll	200
- Poll_PDU	Not Present
- Poll_SDU	1
- Last transmission PDU poll	TRUE
- Last retransmission PDU poll	TRUE
- Poll_Window	99
- Timer_poll_periodic	Not Present
- CHOICE Downlink RLC mode	AM RLC
- In-sequence delivery	TRUE
- Receiving window size	8
- Downlink RLC status info	
- Timer_status_prohibit	200
- Timer_EPC	200
- Missing PDU indicator	TRUE
- Timer_STATUS_periodic	Not Present
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	5
- Logical channel identity	3
- CHOICE RLC size list	Configured
- MAC logical channel priority	3
- Downlink RLC logical channel info	
- Number of RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	10
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	3
- Signalling RB information to setup	(AM DCCH for RRC)
- RB identity	4
- CHOICE RLC info type	
- RLC info	
- CHOICE Uplink RLC mode	AM RLC
- Transmission RLC discard	
- SDU discard mode	Max DAT retransmissions
- MAX_DAT	4
- Timer_MRW	100
- MaxMRW	4
- Transmission window size	8
- Timer_RST	500
- Max_RST	4
- Polling info	
- Timer_poll_prohibit	200
- Timer_poll	200
- Poll_PDU	Not Present
- Poll_SDU	1
- Last transmission PDU poll	TRUE
- Last retransmission PDU poll	TRUE

- Poll_Window	99
- Timer_poll_periodic	Not Present
- CHOICE Downlink RLC mode	AM RLC
- In-sequence delivery	TRUE
- Receiving window size	8
- Downlink RLC status info	
- Timer_status_prohibit	200
- Timer_EPC	200
- Missing PDU indicator	TRUE
- Timer_STATUS_periodic	Not Present
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	5
- Logical channel identity	4
- CHOICE RLC size list	Configured
- MAC logical channel priority	4
- Downlink RLC logical channel info	
- Number of RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	10
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	4
- RAB information for setup	
- RB information to setup	
- RB identity	10
- PDCP info	Not Present
- CHOICE RLC info type	RLC info
- CHOICE Uplink RLC mode	TM RLC
- Transmission RLC discard	Not Present
- Segmentation indication	TRUE
- CHOICE Downlink RLC mode	TM RLC
- Segmentation indication	TRUE
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of uplink RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	1
- Logical channel identity	7
- CHOICE RLC size list	Configured
- MAC logical channel priority	6
- Downlink RLC logical channel info	
- Number of downlink RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	6
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	7
- RB identity	11
- PDCP info	Not Present
- CHOICE RLC info type	RLC info
- CHOICE Uplink RLC mode	TM RLC
- Transmission RLC discard	Not Present
- Segmentation indication	TRUE
- CHOICE Downlink RLC mode	TM RLC
- Segmentation indication	TRUE
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of uplink RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	2
- Logical channel identity	8
- CHOICE RLC size list	Configured
- MAC logical channel priority	6
- Downlink RLC logical channel info	

- Number of downlink RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	7
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	8
- RB identity	12
- PDCP info	Not Present
- CHOICE RLC info type	RLC info
- CHOICE Uplink RLC mode	TM RLC
- Transmission RLC discard	Not Present
- Segmentation indication	TRUE
- CHOICE Downlink RLC mode	TM RLC
- Segmentation indication	TRUE
- RB mapping info	
- Information for each multiplexing option	
- RLC logical channel mapping indicator	Not Present
- Number of uplink RLC logical channels	1
- Uplink transport channel type	DCH
- UL Transport channel identity	3
- Logical channel identity	9
- CHOICE RLC size list	Configured
- MAC logical channel priority	6
- Downlink RLC logical channel info	
- Number of downlink RLC logical channels	1
- Downlink transport channel type	DCH
- DL DCH Transport channel identity	8
- DL DSCH Transport channel identity	Not Present
- Logical channel identity	9
- Pre-Defined Transport Channel Configuration	
- UL CommonTransChInfo	
- UL TFCS	
- TFC subset	Default value is the complete existing set of transport format combinations
- Allowed Transport Format combination	0,1,2,3,4,5
- PRACH TFCS	Not Present
- CHOICE mode	FDD
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Addition
- TFCS addition configure information	
- CHOICE TFCS Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4.1.4.1 Parameter Set.
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.10.2.4.1.4.1 Parameter Set
- Power offset information	
- CHOICE Gain Factors	Signalled Gain Factor
- CHOICE mode	FDD
- Gain factor ßc	0
- Gain factor ßd	0
- Reference TFC ID	0
- Power offset Pp-m	0 dB
- Reference TFC ID	0
- Power offset Pp-m	0 dB
- Added or Reconfigured UL TrCH information	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)
- Uplink transport channel type	DCH
- UL Transport channel identity	1
- TFS	
- CHOICE Transport channel type	Dedicated transport channels
- Dynamic Transport format information	
- RLC Size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Number of TBs and TTI List	(This IE is repeated for TFI number.)
- Transmission Time Interval	Not Present
- Number of Transport blocks	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CHOICE Logical channel list	All
- Semi-static Transport Format information	
- Transmission time interval	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Type of channel coding	Reference to clause 6.10.2.4.1.4.1 Parameter Set

- Coding Rate	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Rate matching attribute	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CRC size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Uplink transport channel type	DCH
- UL Transport channel identity	2
- TFS	
- CHOICE Transport channel type	Dedicated transport channels
- Dynamic Transport format information	
- RLC Size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Number of TBs and TTI List	(This IE is repeated for TFI number.)
- Transmission Time Interval	Not Present
- Number of Transport blocks	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CHOICE Logical channel list	All
- Semi-static Transport Format information	
- Transmission time interval	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Type of channel coding	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Coding Rate	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Rate matching attribute	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CRC size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Uplink transport channel type	DCH
- UL Transport channel identity	3
- TFS	
- CHOICE Transport channel type	Dedicated transport channels
- Dynamic Transport format information	
- RLC Size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Number of TBs and TTI List	(This IE is repeated for TFI number.)
- Transmission Time Interval	Not Present
- Number of Transport blocks	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CHOICE Logical channel list	All
- Semi-static Transport Format information	
- Transmission time interval	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Type of channel coding	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Coding Rate	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Rate matching attribute	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- CRC size	Reference to clause 6.10.2.4.1.4.1 Parameter Set
- Uplink transport channel type	DCH
- UL Transport channel identity	5
- TFS	
- CHOICE Transport channel type	Dedicated transport channels
- Dynamic Transport format information	
- RLC Size	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- Number of TBs and TTI List	(This IE is repeated for TFI number.)
- Transmission Time Interval	Not Present
- Number of Transport blocks	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- CHOICE Logical channel list	All
- Semi-static Transport Format information	
- Transmission time interval	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- Type of channel coding	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- Coding Rate	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- Rate matching attribute	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- CRC size	Reference to clause 6.10.2.4.1.2.1 Parameter Set
- DL CommonTransChInfo	
- SCCPCH TFCS	Not Present
- CHOICE mode	FDD
- CHOICE DL parameters	SameasUL
- Added or Reconfigured DL TrCH information	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)
- Downlink transport channel type	DCH
- DL Transport channel identity	6
- CHOICE DL parameters	Same as UL
- Uplink transport channel type	DCH
- UL TrCH identity	1
- DCH quality target	
- BLER Quality value	0
- Downlink transport channel type	DCH
- DL Transport channel identity	7
- CHOICE DL parameters	Same as UL
- Uplink transport channel type	DCH
- UL TrCH identity	2

- DCH quality target	
- BLER Quality value	0
- Downlink transport channel type	DCH
- DL Transport channel identity	8
- CHOICE DL parameters	Same as UL
- Uplink transport channel type	DCH
- UL TrCH identity	3
- DCH quality target	
- BLER Quality value	0
- Downlink transport channel type	DCH
- DL Transport channel identity	10
- CHOICE DL parameters	Same as UL
- Uplink transport channel type	DCH
- UL TrCH identity	5
- DCH quality target	
- BLER Quality value	0
- Pre-Defined Physical Channel Configuration	
- Uplink DPCH power control info Predef	
- CHOICE mode	FDD
- Power Control Algorithm	Algorithm1
- CHOICE mode	FDD
- TFCI existence	FALSE
- Puncturing Limit	0.88
- Downlink DPCH power control info Predef	
- CHOICE mode	FDD
- Spreading factor	128
- Fixed or Flexible Position	Fixed
- TFCI existence	FALSE

Contents of System Information Block type17 (3.84 Mcsps TDD and 1.28 Mcps TDD)

This system information block contains fast changing parameters for the configuration of the shared physical channels to be used in connected mode, so this is not present.

Contents of System Information Block type 18

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not present
- PLMNs of inter-frequency cells list	Not present
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

6.1.1 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second SCCPCH

Two SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH and the second SCCPCH carries the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs.

Contents of System Information Block type 5 (FDD)

Information Element	Condition	Value/remark	Version
- SIB6 indicator		TRUE	
- PICH Power offset		-5 dB	
- CHOICE Mode		FDD	
- AICH Power offset		-5 dB	
- Primary CCPCH info		Not Present	
- PRACH system information list			
- PRACH system information			
- PRACH info			
- CHOICE mode		FDD	
- Available Signature		'0000 0000 1111 1111'B	

- Available SF		64	
- Preamble scrambling code number		0	
- Puncturing Limit		1.00	
- Available Sub Channel number		'1111 1111 1111'B	
- Transport channel Identity		15	
- RACH TFS			
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC size		168	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- RLC size		360	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		150	
- CRC size		16	
- Additional RACH TFS for CCCH			Rel6
- RLC size		240	
- Number of Transport blocks		1	
- RACH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factor	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- CTFC information		1	
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- Additional RACH TFCS for CCCH			Rel-6
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- PRACH partitioning			
- Access Service Class			
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#1)	
- Available signature End Index		7 (ASC#1)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	

- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#3)	
- Available signature End Index		7 (ASC#3)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#5)	
- Available signature End Index		7 (ASC#5)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#7)	
- Available signature End Index		7 (ASC#7)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- Persistence scaling factor			
- Persistence scaling factor		0.9 (for ASC#2)	
- Persistence scaling factor		0.9 (for ASC#3)	
- Persistence scaling factor		0.9 (for ASC#4)	
- Persistence scaling factor		0.9 (for ASC#5)	
- Persistence scaling factor		0.9 (for ASC#6)	
- Persistence scaling factor		0.9 (for ASC#7)	
- AC-to-ASC mapping table			
- AC-to-ASC mapping		6 (AC0-9)	
- AC-to-ASC mapping		5 (AC10)	
- AC-to-ASC mapping		4 (AC11)	
- AC-to-ASC mapping		3 (AC12)	
- AC-to-ASC mapping		2 (AC13)	
- AC-to-ASC mapping		1 (AC14)	
- AC-to-ASC mapping		0 (AC15)	
- CHOICE mode		FDD	
- Primary CPICH TX power		31	
- Constant value		-10	
- PRACH power offset			
- Power Ramp Step		3dB	
- Preamble Retrans Max		4	
- RACH transmission parameters			
- Mmax		2	
- NB01min		3 slot	
- NB01max		10 slot	
- AICH info			
- Channelisation code		3	
- STTD indicator		FALSE	
- AICH transmission timing		0	
- Secondary CCPCH system information		(For 2 SCCPCHs)	
- Secondary CCPCH info		(SCCPCH for standalone PCH)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		128	
- Code number		4	
- Pilot symbol existence		FALSE	
- TFCI existence		FALSE	
- Fixed or Flexible position		Fixed	
- Timing offset		30 (7680 Chip)	
- TFCS			

- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(PCH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		240	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		230	
- CRC size		16 bit	
- Transport channel Identity		12 (for PCH)	
- CTCH indicator		FALSE	
- PICH info			
- CHOICE mode		FDD	
- Channelisation code		2	
- Number of PI per frame		18	
- STTD indicator		FALSE	
- Secondary CCPCH info		(SCCPCH including two FACHs)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		64	
- Code number		1	
- Pilot symbol existence		FALSE	
- TFCI existence		TRUE (default value)	
- Fixed or Flexible position		Flexible (default value)	
- Timing offset		Not Present Absence of this IE is equivalent to default value 0	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		4 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- CTFC information		2	
- Power offset information		Not Present	
- CTFC information		3	
- Power offset information		Not Present	
- CTFC information		4	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		168	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	

- Number of Transport blocks		2	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		220	
- CRC size		16 bit	
- Transport channel Identity		13 (for FACH)	
- CTCH indicator		FALSE	
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		360	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Turbo	
- Rate matching attribute		130	
- CRC size		16bit	
- Transport channel Identity		14 (for FACH)	
- CTCH indicator		FALSE	
- CBS DRX Level 1 information		Not Present	
- Frequency Band Indicator	A1	Not Present	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A2	FDD Band under test	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A3	Extension indicator	
- Frequency Band Indicator 2		FDD Band under test	

Condition	Explanation
A1	Band I, Band II, Band III
A2	Band V, Band VI, Band VII
A3	Band VIII & bands beyond Band X

Contents of System Information Block type 5bis (FDD)

The message structure of the System information block type 5bis should be the same as System information block type 5 with the following exceptions as given below.

- Frequency Band Indicator	A1	FDD Band under test
- Frequency Band Indicator 2		Not Present
- Frequency Band Indicator	A2	Extension indicator
- Frequency Band Indicator 2		FDD Band under test

Condition	Explanation
A1	Band IV
A2	Band IX, Band X

Contents of System Information Block type 5 (3.84 Mcps TDD)

- SIB6 indicator	FALSE
- CHOICE Mode	TDD
- TDD open loop power control	
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	3.84 Mcps TDD /REL-4/

- Alpha	(1/8)
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	-10
- UE positioning related parameters	Not Present /REL-4/
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- CHOICE SyncCase	Sync Case 2
- Timeslot	0
- Cell parameters ID	Not Present
- SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Timeslot number	14
- PRACH Channelisation Code List	
- CHOICE SF	SF8
- Channelisation Code List	
- Channelisation Code	8/1
- Channelisation Code	8/2
- Channelisation Code	8/3
- Channelisation Code	8/4
- PRACH Midamble	Direct
- PNBSCH allocation	Not Present /REL-4/
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	168
- Number of TB and TTI List	
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	Configured
- RLC size	360
- Number of TB and TTI List	
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	20 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	150
- CRC size	16
- RACH TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	2 bit
- CTFC information	0
- Power offset information	
- CHOICE Gain Factors	Computed Gain Factor
- Reference TFC ID	0
- CHOICE Mode	TDD
- Power offset Pp-m	0 dB
- CTFC information	1
- Power offset information	
- CHOICE Gain Factors	Signalled Gain Factor
- CHOICE mode	TDD
- Gain factor ßc	11
- Gain factor ßd	15
- Reference TFC ID	0
- CHOICE Mode	TDD
- Power offset Pp-m	0 dB

- PRACH partitioning	
- Access Service Class	
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- Persistence scaling factor	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)
- Persistence scaling factor	0.9 (for ASC#4)
- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- Persistence scaling factor	0.9 (for ASC#7)
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE mode	TDD (no data)
- Secondary CCPCH system information	(For 2 SCCPCHs)
- Secondary CCPCH info	(SCCPCH for standalone PCH)
- CHOICE mode	TDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	128
- Code number	4
- Pilot symbol existence	FALSE
- TFCI existence	FALSE
- Fixed or Flexible position	Fixed
- Timing offset	30 (7680 Chip)
- TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	2 bit
- CTFC information	0
- Power offset information	Not Present
- CTFC information	1
- Power offset information	Not Present
- FACH/PCH information	

- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	240
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	230
- CRC size	16 bit
- Transport channel Identity	12 (for PCH)
- CTCH indicator	FALSE
- PICH info	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	4
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	8
- Midamble Shift	Not Present
- Channelisation code	16/16
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- NGAP	4
- NPCH	2
- Secondary CCPCH info	(SCCPCH including two FACHs)
- CHOICE mode	TDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	1
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0
- TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	4 bit
- CTFC information	0
- Power offset information	Not Present
- CTFC information	1
- Power offset information	Not Present
- CTFC information	2
- Power offset information	Not Present
- CTFC information	3
- Power offset information	Not Present
- CTFC information	4
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	168
- Number of TB and TTI List	

- Number of Transport blocks	0
- Number of Transport blocks	1
- Number of Transport blocks	2
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	220
- CRC size	16 bit
- Transport channel Identity	13 (for FACH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	360
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Turbo
- Rate matching attribute	130
- CRC size	16bit
- Transport channel Identity	14 (for FACH)
- CTCH indicator	FALSE
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 5 (1.28 Mcps TDD)

<FFS>

Contents of System Information Block type 5 (3.84 Mcps TDD)

- SIB6 indicator	FALSE
- CHOICE Mode	TDD
- TDD open loop power control	
- PUSCH system information	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Alpha	(1/8)
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	-10
- UE positioning related parameters	Not Present /REL-4/
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- CHOICE SyncCase	Sync Case 2
- Timeslot	0
- Cell parameters ID	Not Present
- SCTD indicator	FALSE
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD /REL-4/
- Timeslot number	14
- PRACH Channelisation Code List	
- CHOICE SF	SF8
- Channelisation Code List	
- Channelisation Code	8/1

- Channelisation Code	8/2
- Channelisation Code	8/3
- Channelisation Code	8/4
- PRACH Midamble	Direct
- PNBSCH allocation	Not Present /REL-4/
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	168
- Number of TB and TTI List	
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	Configured
- RLC size	360
- Number of TB and TTI List	
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	Configured
- Semi-static Transport Format information	
- Transmission time interval	20 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	150
- CRC size	16
- RACH TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	2 bit
- CTFC information	0
- Power offset information	
- CHOICE Gain Factors	Computed Gain Factor
- Reference TFC ID	0
- CHOICE Mode	TDD
- Power offset Pp-m	0 dB
- CTFC information	1
- Power offset information	
- CHOICE Gain Factors	Signalled Gain Factor
- CHOICE mode	TDD
- Gain factor ßc	11
- Gain factor ßd	15
- Reference TFC ID	0
- CHOICE Mode	TDD
- Power offset Pp-m	0 dB
- PRACH partitioning	
- Access Service Class	
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1

- Available Subchannels	null
- ASC Setting	Not Present
- ASC Setting	
- CHOICE mode	TDD
- CHOICE TDD option	3.84 Mcps TDD
- Available Channelisation codes indices	Not Present (Default all)
- CHOICE subchannel size	Size1
- Available Subchannels	null
- Persistence scaling factor	
- Persistence scaling factor	0.9 (for ASC#2)
- Persistence scaling factor	0.9 (for ASC#3)
- Persistence scaling factor	0.9 (for ASC#4)
- Persistence scaling factor	0.9 (for ASC#5)
- Persistence scaling factor	0.9 (for ASC#6)
- Persistence scaling factor	0.9 (for ASC#7)
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE mode	TDD (no data)
- Secondary CCPCH system information	(For 2 SCCPCHs)
- Secondary CCPCH info	(SCCPCH for standalone PCH)
- CHOICE mode	TDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	128
- Code number	4
- Pilot symbol existence	FALSE
- TFCI existence	FALSE
- Fixed or Flexible position	Fixed
- Timing offset	30 (7680 Chip)
- TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	2 bit
- CTFC information	0
- Power offset information	Not Present
- CTFC information	1
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(PCH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	240
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	230
- CRC size	16 bit
- Transport channel Identity	12 (for PCH)
- CTCH indicator	FALSE
- PICH info	
- CHOICE mode	TDD

- CHOICE TDD option	3.84 Mcps TDD
- Timeslot number	0
- Midamble shift and burst type	4
- CHOICE Burst Type	Type 1
- Midamble Allocation Mode	Default midamble
- Midamble configuration burst type 1 and 3	8
- Midamble Shift	Not Present
- Channelisation code	16/16
- Repetition period/length	64/2
- Offset	0
- Paging indicator length	4
- NGAP	4
- NPCH	2
- Secondary CCPCH info	(SCCPCH including two FACHs)
- CHOICE mode	TDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	1
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0
- TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	4 bit
- CTFC information	0
- Power offset information	Not Present
- CTFC information	1
- Power offset information	Not Present
- CTFC information	2
- Power offset information	Not Present
- CTFC information	3
- Power offset information	Not Present
- CTFC information	4
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	168
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- Number of Transport blocks	2
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	220
- CRC size	16 bit
- Transport channel Identity	13 (for FACH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	360
- Number of TB and TTI List	
- Number of Transport blocks	0

- Number of Transport blocks	1
- CHOICE Mode	TDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Turbo
- Rate matching attribute	130
- CRC size	16bit
- Transport channel Identity	14 (for FACH)
- CTCH indicator	FALSE
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 6 in connected mode (FDD)

- PICH Power offset	-5 dB
- CHOICE Mode	FDD
- AICH Power offset	-5 dB
- Primary CCPCH info	Not Present
- PRACH system information list	Not Present
- Secondary CCPCH system information	Not Present
- CBS DRX Level 1 information	Not Present

Contents of System Information Block type 6 in connected mode (3.84 Mcps TDD)

None

Contents of System Information Block type 6 in connected mode (1.28 Mcps TDD)

<FFS>

Contents of System Information Block type 6 in connected mode (7.68 Mcps TDD)

<FFS>

6.1.2 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH, RB for CTCH + SRBs for CCCH/BCCH in the second SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the third SCCPCH (FDD only)

Three SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH. The second SCCPCH carries the FACH for CTCH (Cell Broadcast Service) and the FACH for SRBs on CCCH/ BCCH for idle mode UEs. The third SCCPCH carries the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH for connected mode UEs.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs.

Contents of System Information Block type 5 (FDD)

Information Element	Condition	Value/remark	Version
- SIB6 indicator		TRUE	
- PICH Power offset		-5 dB	
- CHOICE Mode		FDD	
- AICH Power offset		-5 dB	
- Primary CCPCH info		Not Present	
- PRACH system information list			
- PRACH system information			
- PRACH info			
- CHOICE mode		FDD	
- Available Signature		'0000 0000 1111 1111'B	
- Available SF		64	
- Preamble scrambling code number		0	
- Puncturing Limit		1.00	
- Available Sub Channel number		'1111 1111 1111'B	
- Transport channel Identity		15	

- RACH TFS			
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC size		168	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- RLC size		360	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		150	
- CRC size		16	
- Additional RACH TFS for CCCH			Rel6
- RLC size		240	
- Number of Transport blocks		1	
- RACH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factor	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		0 dB	
- CTFC information		1	
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- Additional RACH TFCS for CCCH			Rel-6
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßd		15	
- Gain factor ßc		11	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- PRACH partitioning			
- Access Service Class			
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#1)	
- Available signature End Index		7 (ASC#1)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#3)	
- Available signature End Index		7 (ASC#3)	
- Assigned Sub-Channel Number		'1111'B	

		The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#5)	
- Available signature End Index		7 (ASC#5)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#7)	
- Available signature End Index		7 (ASC#7)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- Persistence scaling factor			
- Persistence scaling factor		0.9 (for ASC#2)	
- Persistence scaling factor		0.9 (for ASC#3)	
- Persistence scaling factor		0.9 (for ASC#4)	
- Persistence scaling factor		0.9 (for ASC#5)	
- Persistence scaling factor		0.9 (for ASC#6)	
- Persistence scaling factor		0.9 (for ASC#7)	
- AC-to-ASC mapping table			
- AC-to-ASC mapping		6 (AC0-9)	
- AC-to-ASC mapping		5 (AC10)	
- AC-to-ASC mapping		4 (AC11)	
- AC-to-ASC mapping		3 (AC12)	
- AC-to-ASC mapping		2 (AC13)	
- AC-to-ASC mapping		1 (AC14)	
- AC-to-ASC mapping		0 (AC15)	
- CHOICE mode		FDD	
- Primary CPICH TX power		31	
- Constant value		-10	
- PRACH power offset			
- Power Ramp Step		3dB	
- Preamble Retrans Max		4	
- RACH transmission parameters			
- Mmax		2	
- NB01min		3 slot	
- NB01max		10 slot	
- AICH info			
- Channelisation code		3	
- STTD indicator		FALSE	
- AICH transmission timing		0	
- Secondary CCPCH system information		(For 2 SCCPCHs)	
- Secondary CCPCH info		(SCCPCH for standalone PCH)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		128	
- Code number		4	
- Pilot symbol existence		FALSE	
- TFCI existence		FALSE	
- Fixed or Flexible position		Fixed	
- Timing offset		30 (7680 Chip)	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	

- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(PCH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		240	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		230	
- CRC size		16 bit	
- Transport channel Identity		12 (for PCH)	
- CTCH indicator		FALSE	
- PICH info			
- CHOICE mode		FDD	
- Channelisation code		2	
- Number of PI per frame		18	
- STTD indicator		FALSE	
- Secondary CCPCH info		(SCCPCH including two FACHs)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		128	
- Code number		5	
- Pilot symbol existence		FALSE	
- TFCI existence		TRUE (default value)	
- Fixed or Flexible position		Flexible (default value)	
- Timing offset		Not Present Absence of this IE is equivalent to default value 0	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- CTFC information		2	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		168	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		220	
- CRC size		16 bit	

- Transport channel Identity - CTCH indicator - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TB and TTI List - Number of Transport blocks - Number of Transport blocks		13 (for FACH) FALSE (FACH) Common transport channels 168 0 1	
- CHOICE Mode - CHOICE Logical channel List - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - Transport channel Identity - CTCH indicator - CBS DRX Level 1 information - Period of CTCH allocation (N) - CBS frame offset (K) - Frequency Band Indicator	A1	FDD ALL 10 ms Convolutional 1/3 220 16bit 14 (for FACH) TRUE 2 0 Not Present	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A2	FDD Band under test	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A3	Extension indicator	
- Frequency Band Indicator 2		FDD Band under test	

Condition	Explanation
A1	Band I, Band II, Band III
A2	Band V, Band VI, Band VII
A3	Band VIII & bands beyond Band X

Contents of System Information Block type 5bis (FDD)

The message structure of the System information block type 5bis should be the same as System information block type 5 with the following exceptions as given below.

- Frequency Band Indicator - Frequency Band Indicator 2 - Frequency Band Indicator - Frequency Band Indicator 2	A1 A2 	FDD Band under test Not Present Extension indicator FDD Band under test

Condition	Explanation
A1	Band IV
A2	Band IX, Band X

Contents of System Information Block type 6 in connected mode (FDD)

- PICH Power offset	-5 dB
- CHOICE Mode	FDD
- AICH Power offset	-5 dB
- Primary CCPCH info	Not present
- PRACH system information list	Not Present
- Secondary CCPCH system information	
- Secondary CCPCH info	(SCCPCH including two FACHs)
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	1
- Pilot symbol existence	FALSE

- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	90 (23040 Chip)
- TFCS	
- CHOICE TFCI signalling	Normal
- TFCI Field 1 information	
- CHOICE TFCS representation	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE CTFC Size	4 bit
- CTFC information	0
- Power offset information	Not Present
- CTFC information	1
- Power offset information	Not Present
- CTFC information	2
- Power offset information	Not Present
- CTFC information	3
- Power offset information	Not Present
- CTFC information	4
- Power offset information	Not Present
- FACH/PCH information	
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	168
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- Number of Transport blocks	2
- CHOICE Mode	FDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Convolutional
- Coding Rate	1/2
- Rate matching attribute	220
- CRC size	16 bit
- Transport channel Identity	16 (for FACH)
- CTCH indicator	FALSE
- TFS	(FACH)
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC Size	360
- Number of TB and TTI List	
- Number of Transport blocks	0
- Number of Transport blocks	1
- CHOICE Mode	FDD
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10 ms
- Type of channel coding	Turbo
- Rate matching attribute	130
- CRC size	16bit
- Transport channel Identity	17 (for FACH)
- CTCH indicator	FALSE
- CBS DRX Level 1 information	Not Present

6.1.3 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second and third SCCPCHs

Three SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH and both the second and third SCCPCHs carry the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs. (SIB6 is not used in this configuration.)

Contents of Scheduling Block 1 (FDD)

- References to other system information blocks	
- Scheduling information	
- CHOICE Value tag	Not Present
- SEG_COUNT	1
- SIB_REP	16
- SIB_POS	4
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 7
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	58
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 11
- Scheduling information	
- CHOICE Value tag	Cell Value tag
- Cell Value tag	A valid Cell value tag value as defined in TS 25.331 [34]
- SEG_COUNT	3
- SIB_REP	64
- SIB_POS	26
- SIB_POS offset info	
- SIB_OFF	2
- SIB_OFF	2
- SIB type SIBs only	System Information Type 12
- Scheduling information	
- CHOICE Value tag	PLMN Value tag
- PLMN Value tag	A valid PLMN value tag value as defined in TS 25.331 [34]
- SEG_COUNT	1
- SIB_REP	64
- SIB_POS	36
- SIB_POS offset info	Not Present
- SIB type SIBs only	System Information Type 18

Contents of System Information Block type 5 (FDD)

Information Element	Condition	Value/remark	Version
- SIB6 indicator		FALSE	
- PICH Power offset		-5 dB	
- CHOICE Mode		FDD	
- AICH Power offset		-5 dB	
- Primary CCPCH info		Not Present	
- PRACH system information list			
- PRACH system information			
- PRACH info			
- CHOICE mode		FDD	
- Available Signature		'0000 0000 1111 1111'B	
- Available SF		64	
- Preamble scrambling code number		0	
- Puncturing Limit		1.00	
- Available Sub Channel number		'1111 1111 1111'B	
- Transport channel Identity		15	
- RACH TFS			
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC size		168	

- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- RLC size		360	
- Number of TB and TTI List			
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		Configured	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		150	
- CRC size		16	
- Additional RACH TFS for CCCH			Rel6
- RLC size		240	
- Number of Transport blocks		1	
- RACH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factor	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		0 dB	
- CTFC information		1	
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- Additional RACH TFCS for CCCH			Rel-6
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factor	
- CHOICE mode		FDD	
- Gain factor ßc		11	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE Mode		FDD	
- Power offset Pp-m		0 dB	
- PRACH partitioning			
- Access Service Class			
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#1)	
- Available signature End Index		7 (ASC#1)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#3)	
- Available signature End Index		7 (ASC#3)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	

- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#5)	
- Available signature End Index		7 (ASC#5)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- ASC Setting		Not Present	
- ASC Setting			
- CHOICE mode		FDD	
- Available signature Start Index		0 (ASC#7)	
- Available signature End Index		7 (ASC#7)	
- Assigned Sub-Channel Number		'1111'B The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number.	
- Persistence scaling factor			
- Persistence scaling factor		0.9 (for ASC#2)	
- Persistence scaling factor		0.9 (for ASC#3)	
- Persistence scaling factor		0.9 (for ASC#4)	
- Persistence scaling factor		0.9 (for ASC#5)	
- Persistence scaling factor		0.9 (for ASC#6)	
- Persistence scaling factor		0.9 (for ASC#7)	
- AC-to-ASC mapping table			
- AC-to-ASC mapping		6 (AC0-9)	
- AC-to-ASC mapping		5 (AC10)	
- AC-to-ASC mapping		4 (AC11)	
- AC-to-ASC mapping		3 (AC12)	
- AC-to-ASC mapping		2 (AC13)	
- AC-to-ASC mapping		1 (AC14)	
- AC-to-ASC mapping		0 (AC15)	
- CHOICE mode		FDD	
- Primary CPICH TX power		31	
- Constant value		-10	
- PRACH power offset			
- Power Ramp Step		3dB	
- Preamble Retrans Max		4	
- RACH transmission parameters			
- Mmax		2	
- NB01min		3 slot	
- NB01max		10 slot	
- AICH info			
- Channelisation code		3	
- STTD indicator		FALSE	
- AICH transmission timing		0	
- Secondary CCPCH system information		(For 3 SCCPCHs)	
- Secondary CCPCH info		(SCCPCH for standalone PCH)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		128	
- Code number		6	
- Pilot symbol existence		FALSE	
- TFCI existence		FALSE	
- Fixed or Flexible position		Fixed	
- Timing offset		30 (7680 Chip)	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	

- FACH/PCH information			
- TFS		(PCH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		240	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		230	
- CRC size		16 bit	
- Transport channel Identity		12 (for PCH)	
- CTCH indicator		FALSE	
- PICH info			
- CHOICE mode		FDD	
- Channelisation code		2	
- Number of PI per frame		18	
- STTD indicator		FALSE	
- Secondary CCPCH info		(SCCPCH including two FACHs)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		64	
- Code number		1	
- Pilot symbol existence		FALSE	
- TFCI existence		TRUE (default value)	
- Fixed or Flexible position		Flexible (default value)	
- Timing offset		Not Present Absence of this IE is equivalent to default value 0	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		4 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- CTFC information		2	
- Power offset information		Not Present	
- CTFC information		3	
- Power offset information		Not Present	
- CTFC information		4	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		168	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- Number of Transport blocks		2	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		220	
- CRC size		16 bit	

- Transport channel Identity		13 (for FACH)	
- CTCH indicator		FALSE	
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		360	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Turbo	
- Rate matching attribute		130	
- CRC size		16bit	
- Transport channel Identity		14 (for FACH)	
- CTCH indicator		FALSE	
- Secondary CCPCH info		(SCCPCH including two FACHs)	
- CHOICE mode		FDD	
- Secondary scrambling code		Not Present	
- STTD indicator		FALSE	
- Spreading factor		64	
- Code number		2	
- Pilot symbol existence		FALSE	
- TFCI existence		TRUE (default value)	
- Fixed or Flexible position		Flexible (default value)	
- Timing offset		90 (23040 Chip)	
- TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		4 bit	
- CTFC information		0	
- Power offset information		Not Present	
- CTFC information		1	
- Power offset information		Not Present	
- CTFC information		2	
- Power offset information		Not Present	
- CTFC information		3	
- Power offset information		Not Present	
- CTFC information		4	
- Power offset information		Not Present	
- FACH/PCH information			
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		168	
- Number of TB and TTI List			
- Number of Transport blocks		0	
- Number of Transport blocks		1	
- Number of Transport blocks		2	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		220	
- CRC size		16 bit	
- Transport channel Identity		16 (for FACH)	
- CTCH indicator		FALSE	
- TFS		(FACH)	
- CHOICE Transport channel type		Common transport channels	
- Dynamic Transport format information			
- RLC Size		360	
- Number of TB and TTI List			

- Number of Transport blocks		0	
- Number of Transport blocks		1	
- CHOICE Mode		FDD	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10 ms	
- Type of channel coding		Turbo	
- Rate matching attribute		130	
- CRC size		16bit	
- Transport channel Identity		17 (for FACH)	
- CTCH indicator		FALSE	
- CBS DRX Level 1 information		Not Present	
- Frequency Band Indicator	A1	Not Present	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A2	FDD Band under test	
- Frequency Band Indicator 2		Not Present	
- Frequency Band Indicator	A3	Extension indicator	
- Frequency Band Indicator 2		FDD Band under test	

Condition	Explanation
A1	Band I, Band II, Band III
A2	Band V, Band VI, Band VII
A3	Band VIII & bands beyond Band X

Contents of System Information Block type 5bis (FDD)

The message structure of the System information block type 5bis should be the same as System information block type 5 with the following exceptions as given below.

- Frequency Band Indicator	A1	FDD Band under test
- Frequency Band Indicator 2		Not Present
- Frequency Band Indicator	A2	Extension indicator
- Frequency Band Indicator 2		FDD Band under test

Condition	Explanation
A1	Band IV
A2	Band IX, Band X

Contents of System Information Block type 5 (3.84 Mcps TDD)

<FFS>

Contents of System Information Block type 5 (1.28 Mcps TDD)

<FFS>

6.1.4 Default parameters for 1 to 8 cell environments

Default settings for cell No.1 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	100

Contents of System Information Block type 11 for cell No.1 (FDD)

See clause 6.1.0b for contents of System Information Block type 11 (FDD) for cell 1.

Contents of System Information Block type 12 in connected mode for cell No.1 (FDD)

See clause 6.1.0b for contents of System Information Block type 12 (FDD) for cell 1.

Default settings for cell No.1 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	0

Contents of System Information Block type 11 for cell No.1 (TDD)

See clause 6.1.0b for contents of System Information Block type 11 (TDD) for cell 1.

Contents of System Information Block type 12 in connected mode for cell No.1 (TDD)

See clause 6.1.0b for contents of System Information Block type 12 (TDD) for cell 1.

Cell No.2

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.2 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 0010B
URA identity	0000 0000 0000 0001B

Default settings for cell No.2 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	150

Contents of System Information Block type 11 for cell No.2 (FDD)

- Intra-frequency measurement system information	A1, A2, A3	
….		
- New intra-frequency cells		
- Intra-frequency cell id		2
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Intra-frequency cell id		1
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id		3
- Cell info		Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	A1, A3	7
- Cell info		Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		8

- Cell info		Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	A3	11
- Cell info		Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4
…..		
- Inter-frequency measurement system information	A1, A2	
…..		
- New inter-frequency cells		
- Inter frequency cell id		4
- Frequency info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		5
- Frequency info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		6
- Frequency info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..		
- Inter-RAT cell info list	A2	
….		
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b
….		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

Default settings for cell No.2 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	4

Contents of System Information Block type 11 for cell No.2 (TDD)

- Intra-frequency measurement system information	

….	
- New intra-frequency cells	
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	4
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.3

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.3 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 0011B
URA identity	0000 0000 0000 0010B

Default settings for cell No.3 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	200

Contents of System Information Block type 11 for cell No.3 (FDD)

- Intra-frequency measurement system information	A1, A2, A3	
….		
- New intra-frequency cells		

- Intra-frequency cell id - Cell info		3 Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Intra-frequency cell id - Cell info		1 Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id - Cell info		2 Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id - Cell info	A1, A3	7 Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id - Cell info		8 Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id - Cell info	A3	11 Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b
….. **- Inter-frequency measurement system information** ….. - New inter-frequency cells - Inter frequency cell id - Frequency info - Cell info - Inter frequency cell id - Frequency info - Cell info - Inter frequency cell id - Frequency info - Cell info …..	A1, A2	 4 Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b 5 Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list. Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b 6 Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list. Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Inter-RAT cell info list …. - New inter-RAT cells - Inter-RAT cell id - CHOICE *Radio Access Technology* - GSM - Inter-RAT cell id - CHOICE *Radio Access Technology* - GSM ….	A2	 9 GSM Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b 10 GSM Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

Default settings for cell No.3 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	8

Contents of System Information Block type 11 for cell No.3 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	4
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.4

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.4 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 0100B
URA identity	0000 0000 0000 0010B

Default settings for cell No.4 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	250

Contents of System Information Block type 11 for cell No.4 (FDD)

- Intra-frequency measurement system information	A1, A2	
….		
- New intra-frequency cells		
- Intra-frequency cell id		4
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id		5
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
- Intra-frequency cell id		6
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
…..		
- Inter-frequency measurement system information	A1, A2	
…..		
- New inter-frequency cells		
- Inter-frequency cell id		1
- Frequency info		
- UARFCN uplink(Nu)		Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- UARFCN downlink(Nd)		Reference to table 6.1.2 for Cell 1
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Inter-frequency cell id		2
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Inter-frequency cell id		3
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.

- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Inter-frequency cell id	A1	7
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Inter-frequency cell id		8
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
- Inter-RAT cell info list	A2	
….		
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b
….		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment

Default settings for cell No.4 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	12

Contents of System Information Block type 11 for cell No.4 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4

- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter-frequency cell id	1
- Frequency info	
- UARFCN downlink(Nt)	Reference to table 6.1.7 for Cell 1
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter-frequency cell id	2
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter-frequency cell id	3
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter-frequency cell id	7
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter-frequency cell id	8
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4

Cell No.5

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.5 are identical to those of cell No.4 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 0101B
URA identity	0000 0000 0000 0011B

Default settings for cell No.5 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	300

Contents of System Information Block type 11 for cell No.5 (FDD)

- Intra-frequency measurement system information	A1, A2	
….		
- New intra-frequency cells		
- Intra-frequency cell id		5
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
- Intra-frequency cell id		4
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id		6
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
…..		
- Inter-frequency measurement system information	A1, A2	
…..		
- New inter-frequency cells		
- Inter-frequency cell id		1
- Frequency info		
- UARFCN uplink(Nu)		Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- UARFCN downlink(Nd)		Reference to table 6.1.2 for Cell 1
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Inter-frequency cell id		2
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Inter-frequency cell id		3
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Inter-frequency cell id	A1	7
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Inter-frequency cell id		8

- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
…..		
- Inter-RAT cell info list	A2	
….		
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b
….		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment

Default settings for cell No.5 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	114

Contents of System Information Block type 11 for cell No.5 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter-frequency cell id	1
- Frequency info	
- UARFCN downlink(Nt)	Reference to table 6.1.7 for Cell 1

- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter-frequency cell id	2
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter-frequency cell id	3
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter-frequency cell id	7
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter-frequency cell id	8
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4

Cell No.6

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.6 are identical to those of cell No.4 with the following exceptions:

Cell identity	0000 0000 0000 0000 0000 0000 0110B
URA identity	0000 0000 0000 0011B

Default settings for cell No.6 (FDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	350

Contents of System Information Block type 11 for cell No.6 (FDD)

- Intra-frequency measurement system information	A1, A2	
….		
- New intra-frequency cells		
- Intra-frequency cell id		6

- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
- Intra-frequency cell id		4
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id		5
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
…..		
- Inter-frequency measurement system information	A1, A2	
…..		
- New inter-frequency cells		
- Inter-frequency cell id		1
- Frequency info		
- UARFCN uplink(Nu)		Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]
- UARFCN downlink(Nd)		Reference to table 6.1.2 for Cell 1
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Inter-frequency cell id		2
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Inter-frequency cell id		3
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Inter-frequency cell id	A1	7
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Inter-frequency cell id		8
- Frequency info		Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.

- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
…..		
- Inter-RAT cell info list	A2	
….		
- New inter-RAT cells		
- Inter-RAT cell id		9
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b
- Inter-RAT cell id		10
- CHOICE *Radio Access Technology*		GSM
- GSM		Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b
….		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment

Default settings for cell No.6 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	119

Contents of System Information Block type 11 for cell No.6 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter-frequency cell id	1
- Frequency info	
- UARFCN downlink(Nt)	Reference to table 6.1.7 for Cell 1
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter-frequency cell id	2

- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter-frequency cell id	3
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter-frequency cell id	7
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter-frequency cell id	8
- Frequency info	Not Present Absence of this IE is equivalent to value of the previous "frequency info" in the list.
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4
…..	

Cell No.7

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.7 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 0111B
URA identity	0000 0000 0000 0100B

Default settings for cell No.7 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	400

Contents of System Information Block type 11 for cell No.7 (FDD)

- Intra-frequency measurement system information	A1, A3	
….		
- New intra-frequency cells		
- Intra-frequency cell id		7
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Intra-frequency cell id		1

- Cell info		Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id		2
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		3
- Cell info		Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		8
- Cell info		Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	A3	11
- Cell info		Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b
…..		
- Inter-frequency measurement system information	A1	
…..		
- New inter-frequency cells		
- Inter frequency cell id		4
- Frequency info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		5
- Frequency info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		6
- Frequency info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

Default settings for cell No.7 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	123

Contents of System Information Block type 11 for cell No.7 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Intra-frequency cell id	1

- Cell info	Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	4
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.8

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.8 are identical to those of cell No.1 with the following exceptions:

Cell identity	0000 0000 0000 0000 0000 0000 1000B
URA identity	0000 0000 0000 0100B

Default settings for cell No.8 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	450

Contents of System Information Block type 11 for cell No.8 (FDD)

- Intra-frequency measurement system information	A1, A3	
….		
- New intra-frequency cells		
- Intra-frequency cell id		8
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
- Intra-frequency cell id		1

- Cell info		Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id		2
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		3
- Cell info		Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		7
- Cell info		Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	A3	11
- Cell info		Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b
…..		
- Inter-frequency measurement system information	A1	
…..		
- New inter-frequency cells		
- Inter frequency cell id		4
- Frequency info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		5
- Frequency info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id		6
- Frequency info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info		Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

Default settings for cell No.8 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	127

Contents of System Information Block type 11 for cell No.8 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4
- Intra-frequency cell id	1

- Cell info	Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	4
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.9

Contents of System Information for cell No.9 (GSM)

See 3GPP TS 51.010-1 [31], clause 10.1.2.

Default settings for cell No.9 (GSM)

See table 6.1.10.

Cell No.10

Contents of System Information for cell No.10 (GSM)

See 3GPP TS 51.010-1 [31], clause 10.1.2.

Default settings for cell No.10 (GSM)

See table 6.1.10

Cell No.11

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.11 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0000 1011B
URA identity	0000 0000 0000 0010B

Default settings for cell No.11 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"

Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	500

Contents of System Information Block type 11 for cell No.11 (FDD)

- Intra-frequency measurement system information	A3	
….		
- New intra-frequency cells		
- Intra-frequency cell id		11
- Cell info		Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4
- Intra-frequency cell id		1
- Cell info		Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id		2
- Cell info		Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		3
- Cell info		Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		7
- Cell info		Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id		8
- Cell info		Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..		

Condition	Explanation
A1	FDD cell environment
A2	FDD/GSM inter-RAT cell environment
A3	FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells)

6.1.4.1 Default Cell parameters Two PLMN in UTRAN test scenario

In this scenario two cell groups belong to two different PLMN, Cell 1, 2, 3, 7, 8 (for PLMN1) and Cell 4,5,6 (for PLMN2) shall be configured on two different frequencies.

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.1 to 8 are identical to those of cell No.1-8 in clause 6.1.4. Exceptions are found in SYSTEM INFORMATION BLOCK TYPE 11:

- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.1, 2, 3, 7, 8 contains cell No.1, 2, 3, 7, 8 in Intra-frequency measurement system information, and cell No.4, 5, 6 in Inter-frequency measurement system information.
- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.4,5,6 contains cell No.4,5,6 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 7, 8 in Inter-frequency measurement system information.
- All other parameters in SYSTEM INFORMATION BLOCK TYPE 11 are set to identical to clause 6.1.4.

Contents of System Information Block type 18 for cell No.1, 2, 3, 7, 8

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not Present
- PLMNs of inter-frequency cells list	

- PLMN identity	Set to PLMN2
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

Contents of System Information Block type 18 for cell No.4, 5, 6

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not Present
- PLMNs of inter-frequency cells list	
- PLMN identity	Set to PLMN1
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

6.1.4.2 Default Cell parameters Three PLMN in UTRAN test scenario

In this scenario three cell groups belong to three different PLMN, Cell 1, 2, 3 (for PLMN1), Cell 4, 5, 6 (for PLMN2) and Cell 7, 8 (for PLMN3) shall be configured on three different frequencies.

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.1 to 8 are identical to those of cell No.1-8 in clause 6.1.4. Exceptions are found in SYSTEM INFORMATION BLOCK TYPE 11:

- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.1, 2, 3 contains cell No.1, 2, 3 in Intra-frequency measurement system information, and cell No.4, 5, 6, 7, 8 in Inter-frequency measurement system information.
- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.4, 5, 6 contains cell No. 4, 5, 6 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 7, 8 in Inter-frequency measurement system information.
- SYSTEM INFORMATION BLOCK TYPE 11 for cell No. 7, 8 contains cell No. 7, 8 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 4, 5, 6 in Inter-frequency measurement system information.
- All other parameters in SYSTEM INFORMATION BLOCK TYPE 11 are set to identical to clause 6.1.4.

Contents of System Information Block type 18 for cell No.1, 2, 3

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not Present
- PLMNs of inter-frequency cells list	
- PLMN identity	Set to PLMN2
- PLMN identity	Set to PLMN2
- PLMN identity	Set to PLMN2
- PLMN identity	Set to PLMN3
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

Contents of System Information Block type 18 for cell No.4, 5, 6

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not Present
- PLMNs of inter-frequency cells list	
- PLMN identity	Set to PLMN1
- PLMN identity	Set to PLMN1
- PLMN identity	Set to PLMN1
- PLMN identity	Set to PLMN3
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

Contents of System Information Block type 18 for cell No.7, 8

- Idle mode PLMN identities	
- PLMNs of intra-frequency cells list	Not Present
- PLMNs of inter-frequency cells list	
- PLMN identity	Set to PLMN1
- PLMN identity	Set to PLMN1

- PLMN identity	Set to PLMN1
- PLMN identity	Set to PLMN2
- PLMNs of inter-RAT cells list	Not present
- Connected mode PLMN identities	Not present

6.1.4.3 Default Cell parameters for MBMS 21 to 28 cell environnements

Cell No.21

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.21 are identical to those of cell No.1 with the following exceptions.

Cell identity	00000000000000000000000010101B
URA identity	0000 0000 0000 0001B

Default settings for cell No.21 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	120

Contents of System Information Block type 11 for cell No.21 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4

- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4
- Inter frequency cell id	25
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=5 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4
- Inter frequency cell id	26
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=6 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4
- Inter frequency cell id	4
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101

- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Default settings for cell No.21 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	2

Contents of System Information Block type 11 for cell No.21 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b

- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	25
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	26
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	4
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.22

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.22 are identical to those of cell No.1 with the following exceptions.

Cell identity	00000000000000000000000010110B
URA identity	0000 0000 0000 0001B

Default settings for cell No.22 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	170

Contents of System Information Block type 11 for cell No.22 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4
- Inter frequency cell id	25
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101

- Cell info	Same content as specified for Intra-frequency cell id=5 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4
- Inter frequency cell id	26
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=6 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4
- Inter frequency cell id	4
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Default settings for cell No.22 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	6

Contents of System Information Block type 11 for cell No.22 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	23

- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	25
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	26
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	4
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b

- Inter frequency cell id	6
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.23

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.23 are identical to those of cell No.1 with the following exceptions.

Cell identity	00000000000000000000000010111B
URA identity	0000 0000 0000 0010B

Default settings for cell No.23 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	220

Contents of System Information Block type 11 for cell No.23 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	1

- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b
.....	
- Inter-frequency measurement system information	
.....	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4
- Inter frequency cell id	25
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=5 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4
- Inter frequency cell id	26
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=6 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4
- Inter frequency cell id	4
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
.....	

Default settings for cell No.23 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	10

Contents of System Information Block type 11 for cell No.23 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	

- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	25
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	26
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	4
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.24

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.24 are identical to those of cell No.1 with the following exceptions.

Cell identity	00000000000000000000000011000B
URA identity	0000 0000 0000 0010B

Default settings for cell No.24 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	270

Contents of System Information Block type 11 for cell No.24 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	25

- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	26
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4
- Inter frequency cell id	22
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4
- Inter frequency cell id	23
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	27
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101

- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	28
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	1
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	7
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
..	

Default settings for cell No.24 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	14

Contents of System Information Block type 11 for cell No.24 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	25
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	26
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	22
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	23

- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	27
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	28
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	1
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter frequency cell id	7
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4
…..	

Cell No.25

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.25 are identical to those of cell No.4 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0001 1001B
URA identity	0000 0000 0000 0011B

Default settings for cell No.25 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	320

Contents of System Information Block type 11 for cell No.25 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	25
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	26
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4
- Inter frequency cell id	22
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101

- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4
- Inter frequency cell id	23
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	27
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	28
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	1
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	7

- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
..	

Default settings for cell No.25 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	116

Contents of System Information Block type 11 for cell No.25 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	25
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	26
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
- Intra-frequency cell id	6

- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	22
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	23
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	27
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	28
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	1
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter frequency cell id	7
- Frequency info	Not present

- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4
…..	

Cell No.26

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.26 are identical to those of cell No.4 with the following exceptions:

Cell identity	0000 0000 0000 0000 0000 0001 1010B
URA identity	0000 0000 0000 0011B

Default settings for cell No.26 (FDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	370

Contents of System Information Block type 11 for cell No.26 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	26
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	25
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4
- Intra-frequency cell id	4

- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4
- Inter frequency cell id	22
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4
- Inter frequency cell id	23
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	27
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	28
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	1
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101

- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	7
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=4 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4
..	

Default settings for cell No.26 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	121

Contents of System Information Block type 11 for cell No.26 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	26

- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	24
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	25
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	4
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4
- Intra-frequency cell id	5
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4
- Intra-frequency cell id	6
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	21
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	22
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	23
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	27
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	28
- Frequency info	Not present

- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	1
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Inter frequency cell id	2
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4
- Inter frequency cell id	3
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4
- Inter frequency cell id	7
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4
- Inter frequency cell id	8
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4
…..	

Cell No.27

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.27 are identical to those of cell No.1 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0001 1011B
URA identity	0000 0000 0000 0100B

Default settings for cell No.27 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	420

Contents of System Information Block type 11 for cell No.27 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	27

- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b

- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4
- Inter frequency cell id	25
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=5 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4
- Inter frequency cell id	26
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=6 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4
- Inter frequency cell id	4
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Default settings for cell No.27 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	125

Contents of System Information Block type 11 for cell No.27 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	27

- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.27 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	25
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	26
- Frequency info	Not present

- Cell info	Same content as specified for Intra-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	4
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Cell No.28

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.28 are identical to those of cell No.1 with the following exceptions:

Cell identity	0000 0000 0000 0000 0000 0001 1000B
URA identity	0000 0000 0000 0100B

Default settings for cell No.28 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	470

Contents of System Information Block type 11 for cell No.28 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.28 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.21 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.22 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.23 (FDD)" in clause 6.1.4.3

- Intra-frequency cell id	27
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.27 (FDD)" in clause 6.1.4.3
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (FDD)" in clause 6.1.4
- Inter frequency cell id	25
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=5 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (FDD)" in clause 6.1.4
- Inter frequency cell id	26
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Intra-frequency cell id=6 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (FDD)" in clause 6.1.4

- Inter frequency cell id	4
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

Default settings for cell No.28 (TDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	129

Contents of System Information Block type 11 for cell No.28 (TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	28
- Cell info	Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.28 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	21
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.21 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	22
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.22 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	23
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.23 (TDD)" in clause 6.1.4.3
- Intra-frequency cell id	27

- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.27(TDD)" in clause 6.1.4.3
- Intra-frequency cell id	1
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4
- Intra-frequency cell id	2
- Cell info	Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	3
- Cell info	Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	7
- Cell info	Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b
- Intra-frequency cell id	8
- Cell info	Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b
…..	
- Inter-frequency measurement system information	
…..	
- New inter-frequency cells	
- Inter frequency cell id	24
- Frequency info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Cell info	Same content as specified for Intra-frequency cell id=1 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.24 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	25
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.25 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	26
- Frequency info	Not present
- Cell info	Same content as specified for Intra-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.26 (TDD)" in clause 6.1.4.3
- Inter frequency cell id	4
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	5
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b
- Inter frequency cell id	6
- Frequency info	Not present
- Cell info	Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b
…..	

6.1.4.4 Default Cell parameters for MBSFN 31 to 38 cell environments

Cell No.31

Default settings for cell No.31 (3.84 Mcps TDD and 7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	1

Default settings for cell No.31 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	120

Contents of System Information Block type 3 for cell No.31 (FDD, 3.84 Mcps TDD and 7.68 Mcps TDD)

Information Element	Value/remark	Version
- SIB4 Indicator	FALSE	
- Cell identity	0000 0000 0000 0000 0000 0001 1111B	
- Cell selection and re-selection info		
- Mapping info	Not present	
- Cell selection and reselection quality measure	CPICH RSCP	
- CHOICE *mode*	TDD	
- Sintrasearch	Not present	
- Sintersearch	Not present	
- SsearchHCS	Not present	
- RAT List	Not present	
- Qrxlevmin	-103 (dBm)	
- DeltaQrxlevmin	Not Present	
- $Qhyst1_s$	1 (dB = value*2 (step size))	
- $Qhyst1_{s,PCH}$	Not Present	
- $Qhyst1_{s,FACH}$	Not Present	
- $Qhyst2_s$	Not Present	
- $Qhyst2_{s,PCH}$	Not Present	
- $Qhyst2_{s,FACH}$	Not Present	
- Treselections	1 (second)	
- $Treselections_{,PCH}$	Not Present	
- $Treselections_{,FACH}$	Not Present	
- Speed dependent ScalingFactor for Treselection	Not Present	
- Inter-frequency ScalingFactor for Treselection	Not Present	
- Inter-RAT ScalingFactor for Treselection	Not Present	
- Non-HCS_T_{CRmax}	Not Present (MD, default = 'not used')	
- Non-HCS_N_{CR}	Not Present (MD)	
- Non-HCS_$T_{CRmaxHyst}$	Not Present	
- HCS Serving cell information	Not present	
- Maximum allowed UL TX power	1 (dBm)	
- Cell Access Restriction		
- Cell barred	barred	
- Intra-frequency cell re-selection indicator	not allowed	
- T_{barred}	1280	
- Cell Reserved for operator use	not reserved	
- Cell Reservation Extension	not reserved	
- Access Class Barred List	Not Present (MD - no access class barred)	
- Domain Specific Access Restriction Parameters For PLMN Of MIB	Not Present	REL-6
- Domain Specific Access Restriction For Shared Network	Not Present	REL-6
- Deferred measurement control UTRAN support	Not Present	REL-7
- MBSFN only service	true	REL-7

Contents of System Information Block type 5 for cell No.31 (FDD)

FFS

Contents of System Information Block type 5 for cell No.31 (3.84 Mcps TDD)

- SIB6 indicator	FALSE
- PICH Power offset	0 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PUSCH system information VHCR	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE *TDD option*	3.84 Mcps TDD
- Alpha	Not present
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	Not present
- UE positioning related parameters	Not Present
- Primary CCPCH info	Not Present
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE *mode*	TDD
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	14
- PRACH Channelisation Code List	
- CHOICE SF	SF8
- Channelisation Code List	
- Channelisation Code	8/1
- PRACH Midamble	Direct
- PNBSCH allocation	Not Present
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	16
- Number of TBs and TTI List	
- Number of Transport blocks	0
- CHOICE *mode*	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10
- Type of channel coding	No coding
- Coding Rate	Not Present
- Rate matching attribute	1
- CRC size	0
- Additional RACH TFS for CCCH	Not present
- RACH TFCS	Not present
- Additional RACH TFCS for CCCH	Not present
- PRACH partitioning	
- Access Service Class	
- ASC Settings	Not Present (Default all)
- Persistence scaling factors	Not Present
- AC-to-ASC mapping	
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	(MP - but treated as if not received by UE)
- Secondary CCPCH system information list	(MP - but treated as if not received by UE)
- Secondary CCPCH info	
- CHOICE *mode*	3.84 Mcps TDD
- Offset	0
- Common timeslot info	

- 2nd interleaving mode	Not Present (MD "Frame")
- TFCI coding	Not Present (MD)
- Puncturing limit	1.0
- Repetition period	Not Present (MD "1")
- Repetition length	Not present (empty)
- Individual timeslot info	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	1
- TFCI existence	FALSE
- Midamble Shift and burst type	
- CHOICE *TDD option*	3.84 Mcps TDD
- CHOICE *Burst Type*	MBSFN Burst Type
- no data	
- CHOICE *TDD option*	3.84 Mcps TDD
- no data	
- Code List	
- Channelisation Code	16/1
- TFCS	(MP - but treated as if not received by UE)
-CHOICE *TFCI signalling*	Normal TFCI signalling
- TFCI Field 1 information	
- CHOICE *TFCS representation*	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE *CTFC Size*	2 bit CTFC
- CTFC information	
- 2 bit CTFC	0
- Power offset information	Not Present
- FACH/PCH information list	(MP - but treated as if not received by UE)
- TFS	
- CHOICE *Transport channel type*	Common transport channels
- Dynamic Transport format information	
- RLC Size	16
- Number of TBs and TTI List	
- Number of Transport blocks	0
- CHOICE *mode*	TDD
- Transmission Time Interval	10
- CHOICE *Logical Channel List*	ALL
- Semi-static Transport Format information	
- Transmission time interval	10
- Type of channel coding	No coding
- Coding Rate	Not Present
- Rate matching attribute	1
- CRC size	0
- Transport channel Identity	1
- CTCH indicator	FALSE
- PICH info	Not Present
- MCCH configuration information	Not Present
- CBS DRX Level 1 information	Not Present
- Frequency band indicator	Not Present
- Frequency band indicator 2	Not Present
- HSDPA cell Indicator	Not Present (Default 'HSDPA capability not indicated')
- E-DCH cell Indicator	Not Present (Default 'E-DCH capability not indicated')
- Secondary CCPCH system information MBMS	
- Secondary CCPCH system information	
- Secondary CCPCH info MBMS	
- CHOICE *mode*	3.84 Mcps TDD
- Common timeslot info MBMS	
- 2nd interleaving mode	Frame
- TFCI coding	Reference clause 6.10 "Parameter Set"
- Puncturing limit	Reference clause 6.10 "Parameter Set"
- Downlink Timeslots and Codes	
- First Individual timeslot info	
- Timeslot number	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	0
- TFCI existence	Reference clause 6.10 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE *TDD option*	3.84 Mcps TDD
- CHOICE *Burst Type*	MBSFN Burst Type

- no data	
- CHOICE *TDD option*	3.84 Mcps TDD
- no data	
- First timeslot channelisation codes	
- CHOICE *codes representation*	Reference clause 5.5.2 "Downlink physical channels code allocation for Signalling"
- CHOICE *more timeslots*	No more timeslots
- no data	
- Modulation	QPSK
- TFCS	
- CHOICE *TFCI signalling*	Normal TFCI signalling
- TFCI Field 1 information	
- CHOICE *TFCS representation*	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE *CTFC Size*	2 bit
- CTFC information	
- 2bit CTFC	0
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	1
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	2
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	3
- Power offset information	Not Present
- FACH carrying MCCH	
- TFS	
- CHOICE *Transport channel type*	Common transport channels
- Dynamic Transport Format Information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TBs and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE *mode*	TDD
- Transmission Time Interval	Not Present
- CHOICE *Logical Channel List*	ALL
- no data	
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Turbo
- Coding Rate	Not Present
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- MCCH configuration information	
- Access Info Period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- Repetition Period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- Modification period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- RLC info	
- DL UM RLC LI size	7
- DL Duplication Avoidance and Reordering info	Not Present
- DL Out of sequence delivery info	
- Timer_OSD	Not Present
- Window size OSD	48
- TCTF presence	false
- FACH carrying MTCH list	Not Present
- Scheduling information	Not Present
- CHOICE *mode*	TDD
- no data	
- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	0
- Cell parameters ID	1
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	3.84 Mcps TDD

- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 5 for cell No.31 (7.68 Mcps TDD)

- SIB6 indicator	FALSE
- PICH Power offset	0 dB
- CHOICE Mode	TDD
- PUSCH system information	Not Present
- PUSCH system information VHCR	Not Present
- PDSCH system information	Not Present
- TDD open loop power control	
- Primary CCPCH Tx Power	30 dbm
- CHOICE TDD option	7.68 Mcps TDD
- Alpha	Not Present
- PRACH Constant Value	-10
- DPCH Constant Value	-10
- PUSCH Constant Value	Not Present
- UE positioning related parameters	Not Present
- Primary CCPCH info	Not Present
- PRACH system information list	
- PRACH system information	
- PRACH info	
- CHOICE mode	TDD
- CHOICE TDD option	7.68 Mcps TDD
- Timeslot number	14
- PRACH Channelisation Code List VHCR	
- CHOICE SF	SF16
- Channelisation Code List	
- Channelisation Code	16/1
- PRACH Midamble	Direct
- PNBSCH allocation	Not Present
- Transport channel Identity	15
- RACH TFS	
- CHOICE Transport channel type	Common transport channels
- Dynamic Transport format information	
- RLC size	16
- Number of TBs and TTI List	
- Number of Transport blocks	0
- CHOICE Mode	TDD
- Transmission Time Interval	Not Present
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10
- Type of channel coding	No coding
- Coding Rate	Not Present
- Rate matching attribute	1
- CRC size	0
- Additional RACH TFS for CCCH	Not present
- RACH TFCS	Not present
- Additional RACH TFCS for CCCH	Not present
- Persistence scaling factors	Not Present
- AC-to-ASC mapping	
- AC-to-ASC mapping table	
- AC-to-ASC mapping	6 (AC0-9)
- AC-to-ASC mapping	5 (AC10)
- AC-to-ASC mapping	4 (AC11)
- AC-to-ASC mapping	3 (AC12)
- AC-to-ASC mapping	2 (AC13)
- AC-to-ASC mapping	1 (AC14)
- AC-to-ASC mapping	0 (AC15)
- CHOICE *mode*	TDD (no data)
- Secondary CCPCH system information	(MP - but treated as if not received by UE)
- Secondary CCPCH system information list	(MP - but treated as if not received by UE)
- Secondary CCPCH info	
- CHOICE *mode*	7.68 Mcps TDD
- Offset	0

- Common timeslot info	
- 2nd interleaving mode	Not Present (MD "Frame")
- TFCI coding	Not Present (MD)
- Puncturing limit	1.0
- Repetition period	Not Present (MD "1")
- Repetition length	Not present (empty)
- Individual timeslot info	
- CHOICE TDD option	7.68 Mcps TDD
- Timeslot number	1
- TFCI existence	FALSE
- Midamble Shift and burst type	
- CHOICE *TDD option*	7.68 Mcps TDD
- CHOICE Burst Type	MBSFN Burst Type
- no data	
- CHOICE *TDD option*	7.68 Mcps TDD
- no data	
- Code List	
- Channelisation Code	32/1
- TFCS	(MP - but treated as if not received by UE)
-CHOICE *TFCI signalling*	Normal TFCI signalling
- TFCI Field 1 information	
- CHOICE *TFCS representation*	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE *CTFC Size*	2 bit CTFC
- CTFC information	
- 2 bit CTFC	0
- Power offset information	Not Present
- FACH/PCH information	(MP - but treated as if not received by UE)
- TFS	(PCH)
- CHOICE *Transport channel type*	Common transport channels
- Dynamic Transport format information	
- RLC Size	16
- Number of TBs and TTI List	
- Number of Transport blocks	0
- CHOICE *mode*	TDD
- Transmission Time Interval	10
- CHOICE Logical channel List	ALL
- Semi-static Transport Format information	
- Transmission time interval	10
- Type of channel coding	No coding
- Coding Rate	Not Present
- Rate matching attribute	1
- CRC size	0
- Transport channel Identity	1
- CTCH indicator	FALSE
- PICH info	Not Present
- MCCH configuration information	Not Present
- CBS DRX Level 1 information	Not Present
- Frequency band indicator	Not Present
- Frequency band indicator 2	Not Present
- HSDPA cell Indicator	Not Present (Default 'HSDPA capability not indicated')
- E-DCH cell Indicator	Not Present (Default 'E-DCH capability not indicated')
- Secondary CCPCH system information MBMS	
- Secondary CCPCH system information	
- Secondary CCPCH info MBMS	
- CHOICE *mode*	7.68 Mcps TDD
- Common timeslot info MBMS	
- 2nd interleaving mode	Frame
- TFCI coding	Reference clause 6.10 "Parameter Set"
- Puncturing limit	Reference clause 6.10 "Parameter Set"
- Downlink Timeslots and Codes VHCR-	
- First Individual timeslot info	
- Timeslot number	
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	0
- TFCI existence	Reference clause 6.10 "Parameter Set"
- Midamble Shift and burst type	
- CHOICE *TDD option*	7.68 Mcps TDD

- CHOICE *Burst Type*	MBSFN Burst Type
- no data	
- CHOICE *TDD option*	7.68 Mcps TDD
- no data	
- First timeslot channelisation codes VHCR	
- CHOICE *codes representation*	Reference clause 5.5.2 "Downlink physical channels code allocation for Signalling"
- CHOICE *more timeslots*	No more timeslots
- no data	
- Modulation	QPSK
- TFCS	
- CHOICE *TFCI signalling*	Normal TFCI signalling
- TFCI Field 1 information	
- CHOICE *TFCS representation*	Complete reconfiguration
- TFCS complete reconfiguration information	
- CHOICE *CTFC Size*	2 bit
- CTFC information	
- 2bit CTFC	0
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	1
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	2
- Power offset information	Not Present
- CTFC information	
- 2bit CTFC	3
- Power offset information	Not Present
- FACH carrying MCCH	
- TFS	
- CHOICE *Transport channel type*	Common transport channels
- Dynamic Transport Format Information	
- RLC Size	Reference clause 6.10 "Parameter Set"
- Number of TBs and TTI List	Reference clause 6.10 "Parameter Set"
- Number of Transport blocks	Reference clause 6.10 "Parameter Set"
- CHOICE *mode*	TDD
- Transmission Time Interval	Not Present
- CHOICE *Logical Channel List*	ALL
- no data	
- Semi-static Transport Format information	
- Transmission time interval	Reference clause 6.10 "Parameter Set"
- Type of channel coding	Turbo
- Coding Rate	Not Present
- Rate matching attribute	Reference clause 6.10 "Parameter Set"
- CRC size	Reference clause 6.10 "Parameter Set"
- MCCH configuration information	
- Access Info Period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- Repetition Period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- Modification period coefficient	Reference clause 11.1.1 "MCCH configuration parameters"
- RLC info	
- DL UM RLC LI size	7
- DL Duplication Avoidance and Reordering info	Not Present
- DL Out of sequence delivery info	
- Timer_OSD	Not Present
- Window size OSD	48
- TCTF presence	false
- FACH carrying MTCH list	Not Present
- Scheduling information	Not Present
- CHOICE *mode*	TDD
- no data	
- TDD MBSFN information	
- Time slot list	(This IE is repeated for all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	0
- Cell parameters ID	1
- Timeslot Number	(Repeated for each Timeslot (1…14)

- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 11 for cell No.31 (FDD)

- SIB 12 Indicator	FALSE
- FACH measurement occasion info	Not Present
- Measurement control system information	
- Use of HCS	Not used
- Cell selection and reselection quality measureCell	CPICH RSCP
- Intra-frequency measurement system information	
- Intra-frequency measurement identity	Not Present
- Intra-frequency cell info list	
- CHOICE *intra-frequency cell removal*	Not present
- New intra-frequency cells	
- Intra-frequency cell id	31
- Cell info	
- Cell individual offset	Not present (MD) Absence of this IE is equivalent to default value 0 dB
- Reference time difference to cell	Not Present
- Read SFN Indicator	FALSE
- CHOICE *mode*	FDD
- Primary CCPCH info	
- Primary scrambling code	Refer to clause titled "Default settings for cell No.31 (FDD)" in clause 6.1.4.4
- Primary CCPCH TX power	Not Present
- TX Diversity indicator	FALSE
- Cell Selection and Re-selection info	Not Present (The IE shall be absent as this is the serving cell)
- Intra-frequency cell id	32
- Cell info	
- Cell individual offset	Not present (MD) Absence of this IE is equivalent to default value 0 dB
- Reference time difference to cell	Not Present
- Read SFN Indicator	FALSE
- CHOICE *mode*	FDD
- Primary CCPCH info	
- Primary scrambling code	Refer to clause titled "Default settings for cell No.32 (FDD)" in clause 6.1.4.4
- Primary CCPCH TX power	Not Present
- TX Diversity indicator	FALSE
- Cell Selection and Re-selection info	Not Present (The IE shall be absent as this is the serving cell)
- Intra-frequency cell id	37
- Cell info	Same content as specified for intra-frequency cell id=32 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.37 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (FDD)” in clause 6.1.4.4
- Cells for measurement	Not Present
- Intra-frequency measurement quantity	Not Present
- Intra-frequency reporting quantity for RACH Reporting	Not Present
- Maximum number of reported cells on RACH	Not Present
- Reporting information for state CELL_DCH	Not Present
- Inter-frequency measurement system information	Not Present
- Inter-RAT measurement system information	Not Present
- Traffic volume measurement system information	Not Present
- MBSFN frequency list	Not Present-

Contents of System Information Block type 11 for cell No.31 (3.84 Mcps and 7.68 Mcps TDD)

- SIB 12 Indicator	FALSE
- FACH measurement occasion info	Not Present
- Measurement control system information	
- Use of HCS	Not used
- Cell selection and reselection quality measureCell	CPICH RSCP
- Intra-frequency measurement system information	
- Intra-frequency measurement identity	Not Present
- Intra-frequency cell info list	
- CHOICE *intra-frequency cell removal*	Not present
- New intra-frequency cells	
- Intra-frequency cell id	31
- Cell info	
- Cell individual offset	Not present (MD)
- Reference time difference to cell	Not Present
- Read SFN Indicator	FALSE
- CHOICE *mode*	TDD
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE *TDD option*	3.84 and 7.68 Mcps TDD
- CHOICE *SyncCase*	Not Present
- Cell parameters ID	Refer to clause titled “Default settings for cell No.31 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- SCTD indicator	FALSE
- Primary CCPCH TX power	Not Present
- Timeslot list	Not Present
- Cell Selection and Re-selection info	Not Present (The IE shall be absent as this is the serving cell)
- Intra-frequency cell id	32
- Cell info	
- Cell individual offset	Not present (MD)
- Reference time difference to cell	Not Present
- Read SFN Indicator	FALSE
- CHOICE mode	TDD
- Primary CCPCH info	
- CHOICE *mode*	TDD
- CHOICE *TDD option*	3.84 and 7.68 Mcps TDD
- CHOICE *Synccase*	Not Present
- Cell parameters ID	Refer to clause titled "Default settings for cell No.32 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Primary CCPCH TX power	Not Present
- Timeslot list	Not Present
- Cell Selection and Re-selection info	Not Present
- Intra-frequency cell id	37
- Cell info	Same content as specified for intra-frequency cell id=32 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.37 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Cells for measurement	Not Present
- Intra-frequency measurement quantity	Not Present
- Intra-frequency reporting quantity for RACH Reporting	Not Present
- Maximum number of reported cells on RACH	Not Present
- Reporting information for state CELL_DCH	Not Present
- Inter-frequency measurement system information	Not Present
- Inter-RAT measurement system information	Not Present
- Traffic volume measurement system information	Not Present
- MBSFN frequency list	Not Present-

Cell No.32

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.32 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0000B

Default settings for cell No.32 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	170

Default settings for cell No.32 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	9

Contents of System Information Block type 5 for cell No.32 (FDD)

FFS

Contents of System Information Block type 5 for cell No.32 (3.84 Mcps TDD)

- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	0
- Cell parameters ID	9
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 5 for cell No.32 (7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	0
- Cell parameters ID	9
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 11 for cell No.32 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	32
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.32 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	31

- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	37
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.37 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.32 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	32
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.32 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	31
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	37
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.37 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.33

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.33 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0001B

Default settings for cell No.33 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	220

Default settings for cell No.33 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	126

Contents of System Information Block type 5 for cell No.33 (FDD)

FFS

Contents of System Information Block type 5 for cell No.33 (3.84 Mcps and 7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	0
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	1
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	2
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	3
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	4
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	5
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	6
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	7
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	8
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	9
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	10
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	11
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	12

- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	13
- Cell parameters ID	126
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	14
- Cell parameters ID	126

Contents of System Information Block type 11 for cell No.33 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	33
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.33 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	36
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.33 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	33
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.33 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	36

- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.34

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.34 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0010B

Default settings for cell No.34 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	270

Default settings for cell No.34 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	118

Contents of System Information Block type 5 for cell No.34 (FDD)

FFS

Contents of System Information Block type 5 for cell No.34 (3.84 Mcps and 7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	0
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	1
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	2
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	3
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	4
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	5

- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	6
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	7
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	8
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	9
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	10
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	11
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	12
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	13
- Cell parameters ID	118
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	14
- Cell parameters ID	118

Contents of System Information Block type 11 for cell No.34 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	34
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.34 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	36

- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.34 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	34
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.34 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	36
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.35

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.35 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0011B

Default settings for cell No.35 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	320

Default settings for cell No.35 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	110

Contents of System Information Block type 5 for cell No.35 (FDD)

FFS

Contents of System Information Block type 5 for cell No.35 (3.84 Mcps and 7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	0
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	1
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	2
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	3
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	4
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	5
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	6
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	7
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	8
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	9
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	10
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	11
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	12
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	13
- Cell parameters ID	110
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	14
- Cell parameters ID	110

Contents of System Information Block type 11 for cell No.35 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	35
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.35 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	36
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.35 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	35
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.35 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	36
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.36 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.36

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.36 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0100B

Default settings for cell No.36 (FDD)

Downlink input level	Reference clause 6 Parameter Set
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6 Parameter Set
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	370

Default settings for cell No.36 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	102

Contents of System Information Block type 5 for cell No.36 (FDD)

FFS

Contents of System Information Block type 5 for cell No.36 (3.84 Mcps and 7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	0
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	1
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	2
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	3
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	4
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	5
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	6
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	7
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	8
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)

- Timeslot number	9
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	10
- Cell parameters ID	122
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	11
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	12
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	13
- Cell parameters ID	102
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD or 7.68 Mcps TDD (as appropriate)
- Timeslot number	14
- Cell parameters ID	102

Contents of System Information Block type 11 for cell No.36 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	36
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.36 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.36 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	36
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.36 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4

- Intra-frequency cell id	33
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.33 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	34
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.34 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	35
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.35 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.37

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.37 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0101B

Default settings for cell No.37 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	420

Default settings for cell No.37 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	17

Contents of System Information Block type 5 for cell No.37 (FDD)

FFS

Contents of System Information Block type 5 for cell No.37 (3.84 Mcps TDD)

- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	0
- Cell parameters ID	17
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 5 for cell No.37 (7.68 Mcps TDD)

- TDD MBSFN information	

- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	0
- Cell parameters ID	17
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 11 for cell No.37 (FDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	37
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.37 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	31
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (FDD)” in clause 6.1.4.4
- Intra-frequency cell id	32
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.37 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	37
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.37 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	31
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4
- Intra-frequency cell id	32
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4
- Intra-frequency cell id	38
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.38 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

Cell No.38

The contents of SYSTEM INFORMATION BLOCK TYPE 3, 5 and 11 messages for cell No.38 are identical to those of cell No.31 with the following exceptions.

Cell identity	0000 0000 0000 0000 0000 0010 0110B

Default settings for cell No.38 (FDD)

Downlink input level	Reference clause 6.10 "Parameter Set"
Uplink output power	Minimum supported by the UE's power class.
PCCPCH/PCPICH carrier number	Reference clause 6.10 "Parameter Set"
Cell Channel Description	
- Primary CPICH info	
- Primary scrambling code	470

Default settings for cell No.38 (3.84/7.68 Mcps TDD)

Downlink input level	Reference clause 6.1.6
PCCPCH/PCPICH carrier number	Reference clause 5.1.2
Cell Channel Description	
- Primary CCPCH info	
- Cell parameters ID	25

Contents of System Information Block type 5 for cell No.38 (FDD)

FFS

Contents of System Information Block type 5 for cell No.38 (3.84 Mcps TDD)

- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	0
- Cell parameters ID	25
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	3.84 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 5 for cell No.38 (7.68 Mcps TDD)

- TDD MBSFN information	
- Time slot list	(This list describes all Timeslots (0…14) in the frame)
- Timeslot Number	
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	0
- Cell parameters ID	25
- Timeslot Number	(Repeated for each Timeslot (1…14)
- CHOICE *TDD option*	7.68 Mcps TDD
- Timeslot number	(1…14)
- Cell parameters ID	5 (Repeated for each Timeslot (1…14)

Contents of System Information Block type 11 for cell No.38 (FDD)

- Intra-frequency measurement system information	

….	
- New intra-frequency cells	
- Intra-frequency cell id	38
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.38 (FDD)" in clause 6.1.4.4
- Intra-frequency cell id	31
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (FDD)” in clause 6.1.4.4

Contents of System Information Block type 11 for cell No.38 (3.84 Mcps and 7.68 Mcps TDD)

- Intra-frequency measurement system information	
….	
- New intra-frequency cells	
- Intra-frequency cell id	38
- Cell info	Same content as specified for Intra-frequency cell id=31 (serving cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.38 (3.84/7.68 Mcps TDD)" in clause 6.1.4.4
- Intra-frequency cell id	31
- Cell info	Same content as specified for intra-frequency cell id=32 (neighbour cell) in SIB11 for Cell 31 in clause 6.1.4.4 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.31 (3.84/7.68 Mcps TDD)” in clause 6.1.4.4

6.1.5 Reference Radio Conditions (FDD)

The following transmission parameters shall be used unless otherwise stated in the description of the individual test case.

Table 6.1.3 are the default settings for a non-suitable cell which is configured and always present whereas table 6.1.4 is for a cell that is switched off. Cells configured according to table 6.1.3 are for test cases in which it is necessary to make a cell unsuitable, and then subsequently make it suitable. This could be achieved by switching the cell off and then reconfiguration as in table 6.1.4, but this takes a lot of time to do.

Table 6.1.1: Default settings for a serving cell in a single cell environment

Parameter	Unit	Cell 1/Cell 21
Cell type		Serving cell
UTRA RF Channel Number (Note 3)		Mid Range Test Frequency
Qqualmin	dB	-24
Qrxlevmin	dBm	-79
UE_TXPWR_MAX_RACH	dBm	21
CPICH Ec (see notes 1 and 2)	dBm/3.84 MHz	-60
NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS. NOTE 2: The cell fulfils 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.133 [30], clause 8.1.2.2.1. NOTE 3: The Test Frequencies are selected from the Tables in section 5.1.1 for the band under test.		

Table 6.1.2: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment

Parameter	Unit	Cell 1/Cell 21	Cell 2/Cell 22	Cell 4/Cell 24

Cell type		Serving cell	Suitable neighbour intra-frequency cell	Suitable neighbour inter-frequency cell
UTRA RF Channel Number (Note 3)		Mid Range Test Frequency	Mid Range Test Frequency	High Range Test Frequency
Qqualmin	dB	-24	-24	
Qrxlevmin	dBm	-79	-79	
UE_TXPWR_MAX_RACH	dBm	21	21	
CPICH Ec (see notes 1 and 2)	dBm/3.84 MHz	-60	-70	

NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.
NOTE 2: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.133 [30], clause 8.1.2.2.1.
NOTE 3: The Test Frequencies are selected from the Tables in section 5.1.1 for the band(s) under test. For the test frequencies for low and high ranges for serving cell, the mid range is used for suitable neighbour of inter-frequency cell in SIB11. For Band VI the Low Range Test Frequencies are used for Cell 1 and Cell 2 because of the small bandwidth available. For FDD interband testing the Test Frequencies will be selected from different Bands.

Table 6.1.3: Default settings for a non-suitable cell

Parameter	Unit	Level
Qqualmin	dB	-24
Qrxlevmin	dBm	-79
UE_TXPWR_MAX_RACH	dBm	21
CPICH_Ec	dBm/3.84 MHz	-90

NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.
NOTE 2: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2.

Table 6.1.4: Default settings for a non-suitable "Off" cell

Parameter	Unit	Level
Qqualmin	dB	-24
Qrxlevmin	dBm	-79
UE_TXPWR_MAX_RACH	dBm	21
CPICH_Ec	dBm/3.84 MHz	≤ -122

NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.
NOTE 2: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2.

Table 6.1.5: Default power levels of physical channels relative to CPICH_Ec

Parameter	Unit	Level Idle mode	Level Connected mode	Version
HS-SCCH_Ec	dB	+3		Rel-5
HS-PDSCH_Ec	dB	+7		Rel-5
DPCH_Ec	dB	(see note)	See table 6.1.6	
PCCPCH_Ec	dB	-2		
SCCPCH_Ec	dB	-2		
AICH_Ec	dB	-5		
SCH_Ec	dB	-5		
PICH_Ec	dB	-5		

NOTE: This shall be less than -122 dBm to ensure the channel is considered as "off".

Table 6.1.6: Default power levels of DPCH_Ec relative to CPICH_Ec

Data transmission rate	Level
12.2 kbps	-5
64 kbps	-2
144 kbps	+1
384 kbps	+5

6.1.5.1 HARQ Transmission Parameters (FDD)

The following HARQ transmission parameters shall be used for test cases in 34.123-1 configuring HS-DSCH channels.

Table 6.1.5.1 : HARQ transmission parameters without MIMO

Parameter	QPSK modulation	16QAM modulation	64QAM modulation
Redundancy and constellation version coding sequence	{0,2,5,6,1,3,7,4}	{6,2,1,5,3,4,7,0}	{6,2,1,5,3,4,7,0}
Maximum number of HARQ transmission	8	8	8

Table 6.1.5.1a : HARQ transmission parameters with MIMO

Parameter	QPSK modulation	16QAM modulation
Redundancy and constellation version coding sequence	{0,3,2,1,3,2,1,3}	{0,3,2,1,3,2,1,3}
Maximum number of HARQ transmission	8	8

6.1.5.2 Inter-band testing (FDD)

FDD inter-band testing only applies for UEs supporting multiple FDD bands simultaneously. In this case the UE can perform cell (re-)selection or inter-frequency mobility between a primary band and a secondary band. The primary and secondary FDD bands are selected according to PIXIT parameters. If a UE supports more than 2 FDD frequency bands, then the test may be executed for various band combinations.

6.1.6 Reference Radio Conditions (TDD)

The following transmission parameters shall be used for unless otherwise stated in the description of the individual test case.

Table 6.1.6a: Default settings for a serving cell in a single cell environment

Parameter	Unit	Cell 1/Cell 21/Cell 31
Cell type		Serving cell
UTRA RF Channel Number		Mid Range Test Frequency
Qrxlevmin	dBm	-81
UE_TXPWR_MAX_RACH	dBm	21
PCCPCH RSCP	dBm	-60
NOTE: The cell fulfils 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.123 [37]. The Test Frequencies are selected from the Tables in section 5.1.2 for the band under test.		

Table 6.1.7: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment

Parameter	Unit	Cell 1/Cell 21/Cell 31	Cell 2/Cell 22	Cell 4/Cell 24/Cell 32
Cell type		Serving cell	Suitable neighbour intra-frequency cell	Suitable neighbour inter-frequency cell
UTRA RF Channel Number		Mid Range Test Frequency	Mid Range Test Frequency	High Range Test Frequency
Qrxlevmin	dBm	-81	-81	
UE_TXPWR_MAX_RACH	dBm	21	21	
PCCPCH RSCP	dBm	-60	-70	
NOTE: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.123 [37]. The Test Frequencies are selected from the Tables in section 5.1.2 for the band under test.				

Table 6.1.8: Default settings for a non-suitable cell

Parameter	Unit	Level

Qrxlevmin	dBm	-81
UE_TXPWR_MAX_RACH	dBm	21
PCCPCH RSCP	dBm	-91
NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2.		

Table 6.1.9: Default settings for a non-suitable "Off" cell

Parameter	Unit	Level
Qrxlevmin	dBm	-81
UE_TXPWR_MAX_RACH	dBm	21
PCCPCH RSCP	dBm	≤ -110
NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2.		

6.1.7 Reference Radio Conditions (GSM)

The following transmission parameters shall be used unless otherwise stated in the description of the individual test case.

Table 6.1.10: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment

Parameter	Unit	Cell 9	Cell 10
Cell type		Serving cell	Suitable neighbour cell
BCCH ARFCN		As defined in the initial conditions in clause 26.6.5.1 of TS 51.010-1 [31] for cell A and the GSM band under test.	As defined in the initial conditions in clause 26.6.5.1 of TS 51.010-1 [31] for cell B and the GSM band under test.
Base transceiver Station Identity Code (BSIC)		BSIC1	BSIC2
Qrxlevmin	dBm	-81	-81
MS_TXPWR_MAX_CCH	dBm	According to maximum output power for the power class of the MS under test	
RF level	dBm	-48	-54
NOTE: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.6.1.4 and 3GPP TS 25.133 [37], clause 8.1.2.5.			

Table 6.1.11: Default settings for a non-suitable cell

Parameter	Unit	Level
Qrxlevmin	dBm	-81
MS_TXPWR_MAX_CCH	dBm	According to maximum output power for the power class of the MS under test
RF level	dBm	-90
NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.6.1.4		

6.2 Number of neighbour cells

The options for the number of neighbour cells (ie the total number of active cells in the simulated network) are given below. See clause 6.1 for cell configurations.

6.2.1 Basic Network

Number of Cells	Use of Network Configuration
1	Basic UE registration; RRC Connection Establishment and Release; operation of dedicated channels in non-handover modes; general RF and EMC testing

6.2.2 Soft Handover Network (FDD)

Number of Cells	Use of Network Configuration/Constraints

2	Can be used in place of basic network, plus offering operation of dedicated channels in 2 way soft handover or in 2 way SSDT (R99 and Rel-4 only) handover for RF or signalling tests; simple cell reselection tests

6.2.3 Hard Handover Network

Number of Cells	Use of Network Configuration
2	Can be used in place of basic network, plus offering operation in 2 cell hard handover (inter-frequency)

6.2.4 'Roaming' Network

Number of Cells	Use of Network Configuration
6	This configuration is intended to provide the capability for extensive cell selection and reselection testing, as defined under Idle Mode Testing. The maximum number of separate RF test channels is 4 in order to limit the test equipment complexity.

6.3 Cell/BS codes etc

See clause 6.1.

6.4 Routing/location area

See clause 6.1.

6.5 Network options settings

See clause 6.1.

6.6 Power control mode

6.6.1 Downlink Power Control

6.6.1.1 Outer Loop Power Control

This is used to set the SIR requirements from the given BER/BLER requirements for the dedicated channel - the reference configuration is for the BER/BLER and SIR requirements to be fixed, ie Outer Loop Power Control is disabled.

6.6.1.2 Inner Loop Power Control

The inner loop power control adjusts the power of the dedicated channel to meet the SIR requirements. The reference condition is for the Inner Loop Power Control to be disabled.

6.6.2 Uplink Power Control

6.6.2.1 Outer Loop Power Control

This is used to set the SIR requirements from the given BER/BLER requirements for the dedicated channel - the reference configuration is for the BER/BLER and SIR requirements to be fixed, ie Outer Loop Power Control is disabled.

6.6.2.2 Inner Loop Power Control (FDD)

The inner loop power control adjusts the power of the dedicated channel to meet the SIR requirements.

6.7 Tx Diversity modes

The reference settings for Tx Diversity Mode shall be:

6.7.1 Non-Diverse Operation

DL Transmit Diversity shall be disabled on all cells in the simulated network.

6.7.2 Diverse Operation

6.7.2.1 Diverse Operation (FDD mode)

The diversity options applied to the DL channels shall be as below for all cells in the simulated network.

Channel	Open loop mode		Closed loop Mode
	TSTD	STTD	
P-CCPCH	-	X	-
SCH	X	-	-
S-CCPCH	-	X	-
DPCH	-	X	-
PICH	-	X	-
AICH	-	X	-

6.7.2.2 Diverse Operation (TDD mode)

The diversity options applied to the DL channels shall be as below for all cells in the simulated network.

6.7.2.2.1 3.84 Mcps option

Table 6.7.1: Application of Tx diversity schemes on downlink physical channel types in 3.84 Mcps TDD "X" - can be applied, "-" - must not be applied

Physical channel type	Open loop TxDiversity		Closed loop TxDiversity
	TSTD	SCTD (see note)	
P-CCPCH	-	X	-
S-CCPCH	--	X	--
SCH	X	-	-
DPCH	-	-	X
PDSCH	-	X	X
PICH	-	X	-
NOTE: SCTD may only be applied to physical channels when they are allocated to beacon locations.			

6.7.2.2.2 1.28 Mcps option

Table 6.7.2: Application of Tx diversity schemes on downlink physical channel types in 1.28 Mcps TDD "X" - can be applied, "-" - must not be applied

Physical channel type	Open loop TxDiversity		Closed loop TxDiversity
	TSTD	SCTD (see note)	
P-CCPCH	X	X	-
S-CCPCH	X	X	-
DwPCH	X	-	-
DPCH	X	-	X
PDSCH	X	X	X
PICH	X	X	-
NOTE: SCTD may only be applied to physical channels when they are allocated to beacon locations.			

6.7.2.2.1 7.68 Mcps option

Table 6.7.1: Application of Tx diversity schemes on downlink physical channel types in 7.68 Mcps TDD "X" - can be applied, "-" - must not be applied

Physical channel type	Open loop TxDiversity		Closed loop TxDiversity
	TSTD	SCTD (see note)	
P-CCPCH	-	X	-
S-CCPCH	--	X	--

SCH	X	-	-
DPCH	-	-	X
PDSCH	-	X	X
PICH	-	X	-
NOTE: SCTD may only be applied to physical channels when they are allocated to beacon locations.			

6.8 Compressed mode parameters

In this clause, Parameters for reference compressed mode patterns are defined which are used in signalling test cases such as inter frequency FDD measurement, inter frequency TDD measurement and inter RAT measurement in 3GPP TS 34.123-1 [1]. These parameters are defined in 3GPP TS 25.133 [30] for measurment performance tests.

Depending on UE capability, there are four methods constructed of three types using of compressed mode such as UL only, DL only and both UL and DL, and using without application of compressed for the above measurement purposes. As test requirement is the same even if the test methods are different, ICS/IXIT statement is applied to the test cases so that the test procedure and specific message contents specified in 3GPP TS 34.123-1 [1] can be distinguished.

6.8.1 Single compressed mode pattern

Configuration parameters in single compressed mode pattern for one type of measurement objects are described in the following clauses.

6.8.1.1 Inter Frequency FDD measurement

The configuration parameters for an inter frequency FDD measurement is shown in table 6.8.1.

Table 6.8.1: Compressed mode parameters (Inter Frequency FDD measurement)

Parameter	Value	Note
TGSN (Transmission Gap Starting Slot Number)	4	
TGL1 (Transmission Gap Length 1)	7	
TGL2 (Transmission Gap Length 2)	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	
TGPL1 (Transmission Gap Pattern Length)	3	
TGPL2 (Transmission Gap Pattern Length)	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number)	(Current CFN + (256 - TTI/10msec))mod 256	
UL/DL compressed mode selection	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	
DL compressed mode method	SF/2	
Scrambling code change	No	
RPP (Recovery period power control mode)	0	
ITP (Initial transmission power control mode)	0	

6.8.1.2 Inter Frequency TDD measurement

The configuration parameters for an inter frequency TDD measurement is shown in table 6.8.2.

Table 6.8.2: Compressed mode parameters (Inter Frequency TDD measurement)

Parameter	Value	Note
TGSN (Transmission Gap Starting Slot Number)	10	
TGL1 (Transmission Gap Length 1)	10	
TGL2 (Transmission Gap Length 2)	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	
TGPL1 (Transmission Gap Pattern Length)	11	
TGPL2 (Transmission Gap Pattern Length)	-	R99 and Rel-4: Only one pattern in use.

Parameter	Value	Note
		Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number)	(Current CFN + (256 - TTI/10msec))mod 256	
UL/DL compressed mode selection	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	
DL compressed mode method	Puncturing	
Scrambling code change	No	
RPP (Recovery period power control mode)	0	
ITP (Initial transmission power control mode)	0	

6.8.1.3 Inter RAT measurement (GSM - Carrier RSSI)

The configuration parameters for an Inter RAT measurement (GSM - Carrier RSSI) is shown in table 6.8.3.

Table 6.8.3: Compressed mode parameters (Inter RAT measurement - GSM Carrier RSSI)

Parameter	Value	Note
TGSN (Transmission Gap Starting Slot Number)	4	
TGL1 (Transmission Gap Length 1)	7	
TGL2 (Transmission Gap Length 2)	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	
TGPL1 (Transmission Gap Pattern Length)	12	
TGPL2 (Transmission Gap Pattern Length)	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number)	(Current CFN + (256 - TTI/10msec))mod 256	
UL/DL compressed mode selection	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	
DL compressed mode method	SF/2	
Scrambling code change	No	
RPP (Recovery period power control mode)	0	
ITP (Initial transmission power control mode)	0	

6.8.1.4 Inter RAT measurement (GSM - Initial BSIC Identification)

The configuration parameters for an inter frequency RAT measurement (GSM - Initial BSIC Identification) is shown in table 6.8.4.

Table 6.8.4: Compressed mode parameters (Inter RAT measurement - GSM Initial BSIC Identification)

Parameter	Value	Note
TGSN (Transmission Gap Starting Slot Number)	4	
TGL1 (Transmission Gap Length 1)	7	
TGL2 (Transmission Gap Length 2)	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	
TGPL1 (Transmission Gap Pattern Length)	8	
TGPL2 (Transmission Gap Pattern Length)	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number)	(Current CFN + (256 - TTI/10msec))mod 256	
UL/DL compressed mode selection	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	
DL compressed mode method	SF/2	

Scrambling code change	No	
RPP (Recovery period power control mode)	0	
ITP (Initial transmission power control mode)	0	

6.8.1.5 Inter RAT measurement (GSM - BSIC re-confirmation)

The configuration parameters for an inter RAT measurement (GSM - BSIC re-confirmation) is shown in table 6.8.5.

Table 6.8.5: Compressed mode parameters (Inter RAT measurement - GSM BSIC re-confirmation)

Parameter	Value	Note
TGSN (Transmission Gap Starting Slot Number)	4	
TGL1 (Transmission Gap Length 1)	7	
TGL2 (Transmission Gap Length 2)	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	
TGPL1 (Transmission Gap Pattern Length)	8	
TGPL2 (Transmission Gap Pattern Length)	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number)	(Current CFN + (256 - TTI/10msec))mod 256	
UL/DL compressed mode selection	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	
DL compressed mode method	SF/2	
Scrambling code change	No	
RPP (Recovery period power control mode)	0	
ITP (Initial transmission power control mode)	0	

6.8.2 Multiple compressed mode patterns

Configuration parameters in multiple compressed mode patterns for several types of measurement objects are described in the following clauses.

6.8.2.1 Inter RAT measurement GSM

The configuration parameters for an inter RAT measurement (GSM - Carrier RSSI, Initial BSIC Identification and BSIC Re-confirmation) is shown in table 6.8.6.

Table 6.8.6: Compressed mode parameters (Inter RAT measurement - GSM Carrier RSSI and Initial BSIC identification and BSIC re-confirmation)

Parameter	GSM Carrier RSSI	GSM Initial BSIC identification	GSM BSIC re-confirmation	Note
TGSN (Transmission Gap Starting Slot Number)	4	4	4	
TGL1 (Transmission Gap Length 1)	7	7	7	
TGL2 (Transmission Gap Length 2)	-	-	-	Only one gap in use.
TGD (Transmission Gap Distance)	undefined	undefined	undefined	
TGPL1 (Transmission Gap Pattern Length)	12	8	8	
TGPL2 (Transmission Gap Pattern Length)	-	-	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable.
TGCFN (Transmission Gap Connection Frame Number):	(Current CFN + (252 - TTI/10msec))mod 256	(Current CFN + (254 - TTI/10msec))mod 256	(Current CFN + (250 - TTI/10msec))mod 256	Defined by higher layers
UL/DL compressed mode selection	DL, UL or DL & UL	DL, UL or DL & UL	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	SF/2	SF/2	
DL compressed mode method	SF/2	SF/2	SF/2	
Scrambling code change	No	No	No	
RPP (Recovery period power control mode)	0	0	0	
ITP (Initial transmission power control mode)	0	0	0	

6.8.2.2 Inter Frequency FDD measurement & Inter RAT measurement GSM

The configuration parameters for Inter Frequency FDD measurement and Inter RAT measurement (GSM - Carrier RSSI, Initial BSIC Identification and BSIC Re-confirmation) is shown in table 6.8.7.

The pattern is illustrated by Figure 6.8.2.2.

Table 6.8.7: Compressed mode parameters (Inter Frequency and Inter RAT measurement - GSM Carrier RSSI and Initial BSIC identification and BSIC re-confirmation)

Parameter	Inter Frequency FDD	GSM Carrier RSSI	GSM Initial BSIC identification	GSM BSIC re-confirmation	Note
TGSN (Transmission Gap Starting Slot Number)	8	8	8	8	
TGL1 (Transmission Gap Length 1)	14	14	14	14	
TGL2 (Transmission Gap Length 2)	14	14	14	14	
TGD (Transmission Gap Distance)	0	60	45	0	
TGPL1 (Transmission Gap Pattern Length)	12	24	24	24	
TGPL2 (Transmission Gap Pattern Length)	-	-	-	-	R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable
TGCFN (Transmission Gap Connection Frame Number):	(Current CFN + (238 - TTI/10msec)) mod 256	(Current CFN + (242 - TTI/10msec)) mod 256	(Current CFN + (256 - TTI/10msec)) mod 256	(Current CFN + (253 - TTI/10msec)) mod 256	Defined by higher layers
UL/DL compressed mode selection	DL, UL or DL & UL	DL, UL or DL & UL	DL, UL or DL & UL	DL, UL or DL & UL	3 configurations possible. DL, UL or both DL and UL
UL compressed mode method	SF/2	SF/2	SF/2	SF/2	

Parameter	Inter Frequency FDD	GSM Carrier RSSI	GSM Initial BSIC identification	GSM BSIC re-confirmation	Note
DL compressed mode method	SF/2	SF/2	SF/2	SF/2	
Scrambling code change	No	No	No	No	
RPP (Recovery period power control mode)	0	0	0	0	
ITP (Initial transmission power control mode)	0	0	0	0	

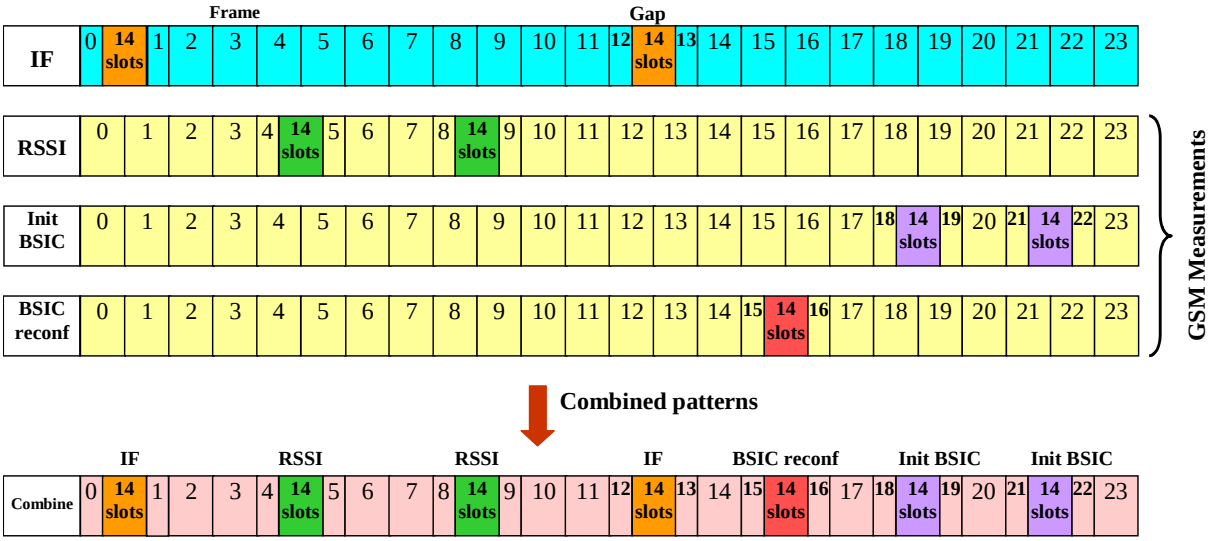


Figure 6.8.2.2: Inter-frequency (IF) and Inter-RAT (IRAT) measurement gaps during 24 frames cycle for the compressed mode pattern as specified in Table 6.8.7

6.8.2.3 Inter Frequency FDD measurement & Inter Frequency TDD measurement

FFS

6.8.2.4 Inter Frequency TDD measurement & Inter RAT measurement GSM

FFS

6.8.2.5 Inter Frequency FDD measurement & Inter Frequency TDD measurement & Inter RAT measurement GSM

FFS

6.9 BCCH parameters

See clause 6.1.

6.10 Reference Radio Bearer configurations used in Radio Bearer interoperability testing

The reference radio bearer configurations are typical configurations of the radio interface. This sub-set of the mandatory set of radio bearer configurations supported by the UE is intended to be used as test configurations for testing of the UE. The purpose of the reference radio bearer configurations is to ensure interoperability of UE's in different regions and networks.

The reference radio bearer configurations are used in the radio bearer interoperability test cases, clause 14 of 3GPP TS 34.123-1 [1]. The reference radio bearer configurations are also intended to be the first choice for other test cases where a radio bearer configuration is needed. For test cases requiring alternative configurations not provided by the reference radio bearer configurations then these specific radio bearer configurations are either specified in the actual test case itself; or in case the configurations are used by more than one test case then these common radio bearer configurations are specified in clause 6.11 of the present document.

NOTE If not specifically specified then the mid-value of the RM attribute value range as specified by the actual reference radio bearer configuration shall be applied for testing. However, in the case of UL and DL:3.4 kbps SRBs for DCCH and where the Choice "Same As UL" is used for the IE "DL Transport channel information common for all transport channel", the RM attribute for the "DL:3.4 kbps SRBs for DCCH" shall be set to the same value as that used in the Uplink.

6.10.1 QoS Architecture and RAB attributes

From a user point-of-view services are considered end-to-end, this means from a Terminal Equipment (TE) to another TE. An End-to-End Service may have a certain Quality of Service (QoS) which is provided for the user through the different networks. In UMTS, it is the UMTS Bearer Service that provides the requested QoS through the use of different QoS classes as defined in 3GPP TS 23.107 [15].

The UMTS Bearer Service consists of two parts, the Radio Access Bearer Service, RAB, and the Core Network Bearer Service. The Radio Access Bearer Service is realized by a Radio Bearer Service and an Iu-Bearer Service. The relationship between the services is illustrated in figure 6.10.1.1.

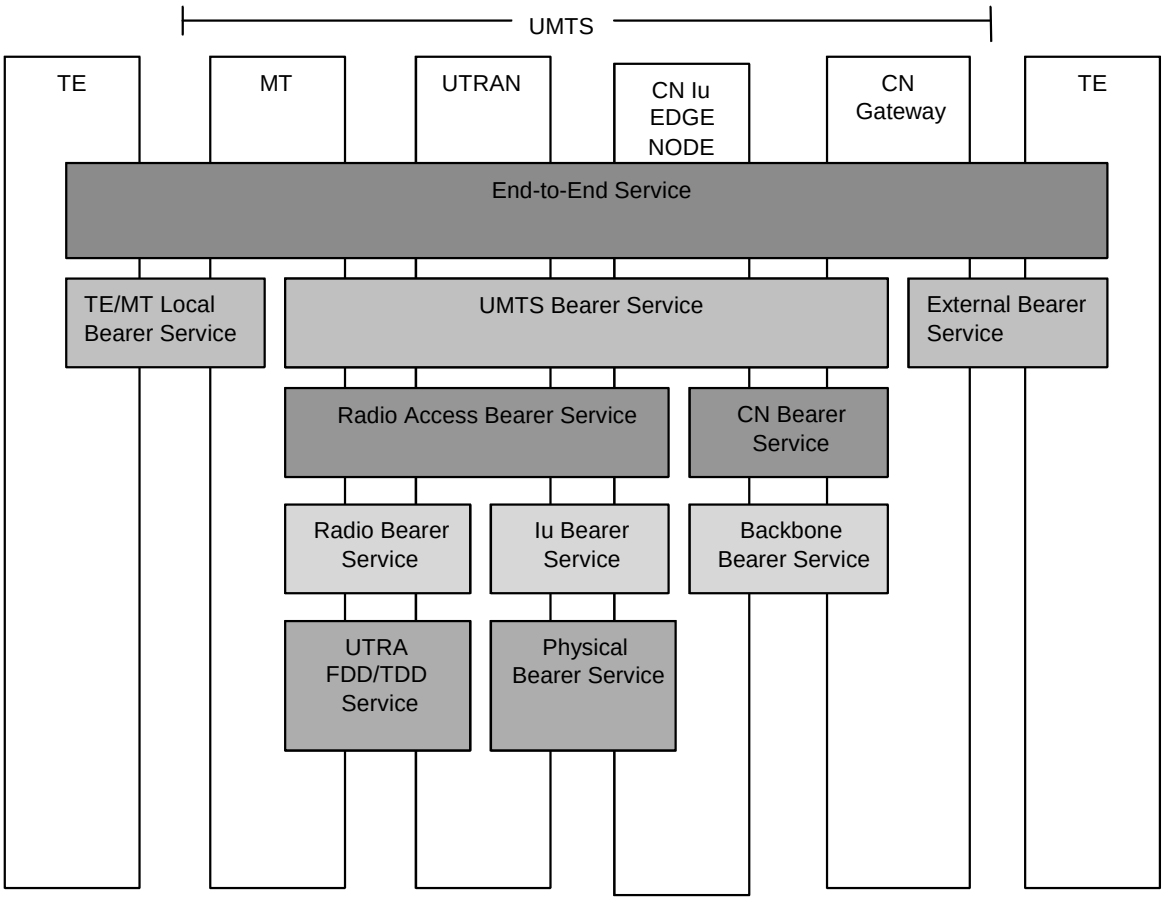


Figure 6.10.1.1: UMTS QoS Architecture

The Radio Access Bearer Service is characterized by a number of attributes such as Traffic class, Maximum bit rate, Guaranteed bit rate, SDU error ratio, Residual BER, Transfer Delay etc. As a first approach the four following attributes have been considered to come up with the parameter settings in clause 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode:

- Traffic class;
- SSD;
- Maximum bit rate;
- Residual BER.

The Traffic classes are explained in table 6.10.1.1. The Maximum bit rate has been considered at RLC layer and Physical Layer for the acknowledged and unacknowledged modes respectively. The Residual BER is understood as BER at RLC layer and Transport BLER for the acknowledged and unacknowledged modes respectively.

NOTE: The maximum bit rate in clause 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode is one of the RAB attribute as described above. For Interactive/Background PS RABs, however, the maximum bit rate of Radio Bearer can be lower than the maximum bit rate of RAB attributes due to radio resource management. Bit rates of Interactive/Background PS RABs described in 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode may represent the maximum bit rate of Radio Bearer taking account into this management.

Table 6.10.1.1: Traffic classes

Traffic class	**Conversational class** conversational RT	**Streaming class** streaming RT	**Interactive class** Interactive best effort	**Background** Background best effort
Fundamental characteristics	- Preserve time relation (variation) between information entities of the stream Conversational pattern (stringent and low delay)	- Preserve time relation (variation) between information entities of the stream (i.e. some but constant delay)	Request response pattern Preserve payload content	Destination is not expecting the data within a certain time Preserve payload content
Example of the application	- speech, video, etc.	- facsimile (NT) - streaming audio and video	- Web browsing	- background download of emails

6.10.2 RAB and signalling RB for FDD

6.10.2.1 RABs and signalling RBs

In the following clauses, the typical parameter sets are presented for reference RABs, signalling RBs and important combinations of them. The data rate given for each RAB is the maximum data rate that can be supported by that RAB.

NOTE: The granularity for each RAB needs to be clarified.

Table 6.10.2.1.1: Prioritized RABs

#	**Traffic class 3GPP TS 23.107 [15]**	**SSD 3GPP TS 23.107 [15]**	**Max. rate, kbps**	**CS/PS**	**Version**
1	Conversational	Speech	UL:12.2 DL:12.2	CS	R99
1a	Conversational	Speech	UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75)	CS	R99
1b	Conversational	Speech	UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75)	CS	R99
2	Conversational	Speech	UL:10.2 DL:10.2	CS	R99
2a	Conversational	Speech	UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75)	CS	R99
3	Conversational	Speech	UL:7.95 DL:7.95	CS	R99
4	Conversational	Speech	UL:7.4 DL:7.4	CS	R99
4a	Conversational	Speech	UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75)	CS	R99

#	Traffic class 3GPP TS 23.107 [15]	SSD 3GPP TS 23.107 [15]	Max. rate, kbps	CS/PS	Version
5	Conversational	Speech	UL:6.7 DL:6.7	CS	R99
6	Conversational	Speech	UL:5.9 DL:5.9	CS	R99
7	Conversational	Speech	UL:5.15 DL:5.15	CS	R99
8	Conversational	Speech	UL:4.75 DL:4.75	CS	R99
9	Conversational	Unknown	UL:28.8 DL:28.8	CS	R99
10	Conversational	Unknown	UL:64 DL:64	CS	R99
11	Conversational	Unknown	UL:32 DL:32	CS	R99
11a	Conversational	Unknown	UL:8 DL:8	PS	R99
12	Streaming	Unknown	UL:14.4 DL:14.4	CS	R99
13	Streaming	Unknown	UL:28.8 DL:28.8	CS	R99
14	Streaming	Unknown	UL:57.6 DL:57.6	CS	R99
15	Void				
15a	Streaming	Unknown	UL:16 DL:64	PS	R99
15b	Streaming	Unknown	UL:16 DL:128	PS	R99
16	Void				
17	Void				
18	Void				
19	Void				
20	Interactive or Background	N/A	UL:32 DL:8	PS	R99
20a	Interactive or Background	N/A	UL:8 DL:8	PS	R99
20b	Interactive or Background	N/A	UL:16 DL:16	PS	R99
20c	Interactive or Background	N/A	UL:32 DL:32	PS	R99
21	Void				
22	Interactive or Background	N/A	UL:32 DL:64	PS	R99
23	Interactive or Background	N/A	UL:64 DL:64	PS	R99
24	Interactive or Background	N/A	UL:64 DL:128	PS	R99
25	Interactive or Background	N/A	UL:128 DL:128	PS	R99
26	Interactive or Background	N/A	UL:64 DL:384	PS	R99
27	Interactive or Background	N/A	UL:128 DL:384	PS	R99
28	Interactive or Background	N/A	UL:384 DL:384	PS	R99
29	Interactive or Background	N/A	UL:64 DL:2048	PS	R99
30	Interactive or Background	N/A	UL:128 DL:2048	PS	R99
31	Void				
32	Interactive or Background	N/A	UL:64 DL:256	PS	R99
33	Interactive or Background	N/A	UL:0 DL:32	PS	R99
34	Interactive or Background	N/A	UL:32 DL: 0	PS	R99
35	Interactive or Background	N/A	UL:64 DL:144	PS	R99
36	Interactive or Background	N/A	UL:144 DL:144	PS	R99
37	Conversational	N/A	UL:42.8 DL:42.8	PS	REL-5
38	Conversational	Speech	UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6)	CS	REL-5
39	Interactive or Background	N/A	UL:64 DL:768	PS	REL-5

Table 6.10.2.1.2: Signalling RBs

#	Maximum rate, kbps	Logical channel	PhyCh onto which SRBs are mapped	Version
1	UL:1.7 DL:1.7	DCCH	DPCH	R99 and Rel-4 only
2	UL:3.4 DL:3.4	DCCH	DPCH	R99
3	UL:13.6 DL:13.6	DCCH	DPCH	R99
4	DL:27.2 (alt. 40.8)	DCCH	SCCPCH	R99
5	UL:16.6/23.8	CCCH	PRACH	R99/Rel-6
6	DL:30.4 (alt. 45.6)	CCCH	SCCPCH	R99
7	DL:33.2 (alt. 49.8)	BCCH	SCCPCH	R99
8	DL:24 (alt. 6.4)	PCCH	SCCPCH	R99
9	DL: 0.15	DCCH	DPCH	REL-5
10	UL: [max bit rate depending on UE category and TTI], DL: [max bit rate depending on UE category]	DCCH	E-DPCH/HS-DSCH	REL-6

6.10.2.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (R99 and Rel-4 only).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

4b) Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-4).

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void

19) Void.

20) Void.

21) Void.

22) Void.

23) Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI) + UL:3.4 DL:3.4 kbps SRBs for DCCH.

24) Void

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Void

37) Void

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38e) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38f) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38g) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38h) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38i) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38j) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38k) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing).

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void

47) Void.

48) Void.

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void

55) Void.

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

57) Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58) Streaming / unknown / UL:16 DL:64 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58a) Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

59) Conversational / Speech / UL:42.8 DL:42.8 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5).

60) Conversational / Speech / UL:42.8 DL:42.8 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5).

61) Conversational / unknown / UL:8 DL:8 kbps / PS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB +
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

62) Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-5).

63) Interactive or background / UL:64 DL:768 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (REL-5).

Combinations on DSCH and DPCH

1) Void

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

4) Void

5) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

6) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

Combinations on SCCPCH

1) Stand-alone 24 kbps SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

4) RB for CTCH
+ SRB for CCCH
+SRB for BCCH

5) 64.8kbps RB for MTCH with 80 ms TTI

6) 129.6 kbps RB for MTCH with 80 ms TTI

7) 259.2 kbps RB for MTCH with 40 ms TTI

8) 7.6 kbps signalling RB for MCCH

9) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH
+ SRB for MCCH.

Combinations on PRACH

1) Interactive or background / UL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

Combinations on DPCH and HS-PDSCH

1) Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

1a) Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

2) Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5) Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5a) Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6) Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

7) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

8) Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Interactive or Background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-5)

9) Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-6)

10) Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-6)

Combinations on HS-PDSCH and E-DPDCH

0) Stand-alone UL: [max bit rate depending on UE category and TTI] DL:[max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

1) Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH (REL-6)

2) Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH (REL-6)

3) Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6) Conversational / unknown or speech / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

7) Conversational / unknown or speech / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

8) Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-6)

9) Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) kbps DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB on E-DCH and HS-DSCH + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-8 + NOTE1)

10) Conversational / speech / UL:(12.65, 8.85, 6.6) kbps DL: (12.65, 8.85, 6.6) kbps / CS RAB on E-DCH and HS-DSCH + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-8 + NOTE1)

NOTE1: Support depends on the UE capability: Support for CS voice over HSPA. This is supported in rel-8 and may be supported in rel-7.

6.10.2.3 Example of linkage between RABs and services

RABs, which are included in the present document, can provide the services as shown in table 6.10.1.1. Furthermore, the required BER for each RAB, which is assumed in the present document, is shown in table 6.10.2.3.1.

Table 6.10.2.3.1: Example of linkage between RABs and services

RAB				Residual BER [15]	Services
Traffic class [15]	SSD [15]	Max. rate, kbps	CS/PS		
Conversational	Speech	UL:4.75-12.2 DL:4.75-12.2	CS	$5x10^{-4}$, $1x10^{-3}$, $5x10^{-3}$	AMR speech
Conversational	Unknown	UL:64 DL:64	CS	$1x10^{-4}$ or $1x10^{-6}$	UDI 1B, 64k 3G-324M [15]
Conversational	Unknown	UL:32 DL:32	CS	$1x10^{-4}$ or $1x10^{-6}$	32k 3G-324M [15]
Conversational	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	Transparent modem
Streaming	Unknown	UL:14.4 DL:14.4	CS	$1x10^{-3}$	FAX[6]
Streaming	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	FAX [18] PIAFS 32 kbps
Streaming	Unknown	UL:57.6 DL:57.6	CS	$1x10^{-3}$	Modem [18], FTM [17] PIAFS 64 kbps
Streaming	Unknown	UL:64-128 or DL:64-384	CS	$1x10^{-3}$ or $1x10^{-4}$	Streaming video, uni-directional
Interactive or Background	N/A	UL:32-384 DL:8-2048	PS	$1x10^{-3}$ or $1x10^{-4}$	Packet

NOTE 1: SMS can be provided via the signalling RB (DCCH) on DPCH or SCCPCH.

NOTE 2: CBS can be provided via the signalling RB (CTCH) on SCCPCH.

NOTE 3: UDI *n*B can be provided via *n* RABs of conversational 64 kbps.

6.10.2.4 Typical radio parameter sets

NOTE The order of tables and MAC-d flow numbering in this section may be different than the RB IDs and MAC-d flow IDs as defined in default messages in section 9.

6.10.2.4.1 Combinations on DPCH

6.10.2.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.2.4.1.1.1 Uplink

6.10.2.4.1.1.1.1 Transport channel parameters

6.10.2.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt 0, 148)			
	TFS	TF0, bits	0x148 (alt 1x0)			
		TF1, bits	1x148			
	TTI, ms		80			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Uplink: Max number of bits/radio frame before rate matching		65			
	RM attribute		155 to 185			

6.10.2.4.1.1.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.1.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	256
	Max number of DPDCH data bits/radio frame	150
	Puncturing Limit	1

6.10.2.4.1.1.2 Downlink

6.10.2.4.1.1.2.1 Transport channel parameters

6.10.2.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit	8	16	16	16
MAC	MAC header, bit	4	4	4	4
	MAC multiplexing	4 logical channel multiplexing			
Layer 1	TrCH type	DCH			

	TB sizes, bit		148 (alt 0, 148) (note)
	TFS	TF0, bits	0 x148 (alt 1x0) (note)
		TF1, bits	1x148
	TTI, ms		80
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		516
	RM attribute		155 to 185
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.			

6.10.2.4.1.1.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.1.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		512
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	4
		Number of data bits/frame	60

6.10.2.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.2.1 Uplink

6.10.2.4.1.2.1.1 Transport channel parameters

6.10.2.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt 0, 148)			
	TFS	TF0, bits	0x148 (alt 1x0)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Uplink: Max number of bits/radio frame before rate matching		129			
	RM attribute		155 to 185			

6.10.2.4.1.2.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.2.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	256
	Max number of DPDCH data bits/radio frame	150
	Puncturing Limit	1

6.10.2.4.1.2.2 Downlink

6.10.2.4.1.2.2.1 Transport channel parameters

6.10.2.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt 0, 148) (note)			
	TFS	TF0, bits	0x148 (alt 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	RM attribute		155 to 230			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.10.2.4.1.2.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.2.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	14
		Number of data bits/frame	210

6.10.2.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.10.2.4.1.3.1 Uplink

6.10.2.4.1.3.1.1 Transport channel parameters

6.10.2.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt 0, 148)			
	TFS	TF0, bits	0x148 (alt 1x0)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Uplink: Max number of bits/radio frame before rate matching		516			

6.10.2.4.1.3.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.3.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1

6.10.2.4.1.3.2 Downlink

6.10.2.4.1.3.2.1 Transport channel parameters

6.10.2.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt 0, 148) (note)			
	TFS	TF0, bits	0x148 (alt 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			

	Max number of bits/TTI before rate matching	516
NOTE: alternative parameters enable the measurement "transport channel BLER" in the UE.		

6.10.2.4.1.3.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.1.3.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.4.1 Uplink

6.10.2.4.1.4.1.1 Transport channel parameters

6.10.2.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Uplink: Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.

6.10.2.4.1.4.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=

	(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.1.4.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.84

6.10.2.4.1.4.2 Downlink

6.10.2.4.1.4.2.1 Transport channel parameters

6.10.2.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0 39 81	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 81	103	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.4.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.1.4.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.4a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.4a.1 Uplink

6.10.2.4.1.4a.1.1 Transport channel parameters

6.10.2.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 81)	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2 bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Uplink: Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.4a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)

6.10.2.4.1.4a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.84

6.10.2.4.1.4a.2 Downlink

6.10.2.4.1.4a.2.1 Transport channel parameters

6.10.2.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0, 39, 42, 55, 75, 81	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0, 39, 42, 55, 75, 81	53, 63, 84, 103	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.4a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)

6.10.2.4.1.4a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.4b Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.1.4b.1.1 Transport channel parameters

6.10.2.4.1.4b.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.4 5.9 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 61, 81 (alt. 0, 39, 42, 55, 61, 81)	53, 63, 87, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 61, 81 (alt. 0, 39, 42, 55, 61, 81)	53, 63, 87, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2 bits	1x42	1x63	N/A
		TF3, bits	1x55	1x87	N/A
		TF4, bits	1x61	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Uplink: Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.4b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See subclause 6.10.2.4.1.2.1.1.1 of [1].

6.10.2.4.1.4b.1.1.3 TFCS

See subclause 6.10.2.4.1.4a.1.1.3 of [1].

6.10.2.4.1.4b.1.1.4 TFC subset list

TFC subset list size	4
TFC subset list	0 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0, TF0,TF1), (TF2,TF1,TF0,TF1)}, 1 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1)}, 2 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1)} 3 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)}

6.10.2.4.1.4b.1.2 Physical channel parameters

See subclause 6.10.2.4.1.4a.1.2 of [1].

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.84

6.10.2.4.1.4b.2 Downlink

6.10.2.4.1.4b.2.1 Transport channel parameters

6.10.2.4.1.4b.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.4 5.9 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0, 39, 42, 55, 61, 81	53, 63, 87, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0, 39, 42, 55, 61, 81	53, 63, 87, 103	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x87	N/A
		TF4, bits	1x61	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.4b.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See subclause 6.10.2.4.1.2.2.1.1 of [1].

6.10.2.4.1.4b.2.1.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

See subclause 6.10.2.4.1.62.2.1.3 of [1].

6.10.2.4.1.4b.2.1.3 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH, DCCH 0.15)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)

6.10.2.4.1.4b.2.2 Physical channel parameters

See subclause 6.10.2.4.1.4a.2.2 of [1].

6.10.2.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.5.1 Uplink

6.10.2.4.1.5.1.1 Transport channel parameters

6.10.2.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Uplink: Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.5.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.1.5.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.5.2 Downlink

6.10.2.4.1.5.2.1 Transport channel parameters

6.10.2.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type	DTCH		
	RLC mode	TM	TM	TM
	Payload sizes, bit	0 39 65	99	40
	Max data rate, bps	10 200		
	TrD PDU header, bit	0		
MAC	MAC header, bit	0		
	MAC multiplexing	N/A		

Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 65	99	40
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.5.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.1.5.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.5a.1 Uplink

6.10.2.4.1.5a.1.1 Transport channel parameters

6.10.2.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A

		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Uplink: Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.5a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)

6.10.2.4.1.5a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.5a.2 Downlink

6.10.2.4.1.5a.2.1 Transport channel parameters

6.10.2.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0, 39, 42, 55, 58, 65	0, 53, 63, 76, 99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0, 39, 42, 55, 58, 65	0, 53, 63, 76, 99	40
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1:	The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2:	CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.5a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)

6.10.2.4.1.5a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.6.1 Uplink

6.10.2.4.1.6.1.1 Transport channel parameters

6.10.2.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39,75 (alt. 0, 39, 75)	84	60
	Max data rate, bps		7950		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39,75 (alt. 0, 39, 75)	84	60
	TFS (note 1)	TF0, bits	0x75 (alt. 1x0) (note)	0x84	0x60
		TF1, bits	1x39	1x84	N/A
		TF2, bits	1x75	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		285	276	0
	Uplink: Max number of bits/radio frame before rate matching		143	138	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE:	In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]).				

6.10.2.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.6.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow #3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.6.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.6.2 Downlink

6.10.2.4.1.6.2.1 Transport channel parameters

6.10.2.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0, 39, 75	84	60
	Max data rate, bps		7950		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 75	84	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x84	0x60
		TF1, bits	1x39	1x84	N/A
		TF2, bits	1x75	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		285	276	0
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.6.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.6.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2

		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.7.1 Uplink

6.10.2.4.1.7.1.1 Transport channel parameters

6.10.2.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 61 (alt. 0, 39, 61)	87	60
	Max data rate, bps		7400		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 61 (alt. 0, 39, 61)	87	60
	TFS (note 1)	TF0, bits	0x61 (alt. 1x0) (note)	0x87	0x60
		TF1, bits	1x39	1x87	N/A
		TF2, bits	1x61	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		243	285	0
	Uplink: Max number of bits/radio frame before rate matching		122	143	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.7.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.7.2 Downlink

6.10.2.4.1.7.2.1 Transport channel parameters

6.10.2.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0 39 61	87	60
	Max data rate, bps		7400		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 61	87	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x87	0x60
		TF1, bits	1x39	1x87	N/A
		TF2, bits	1x61	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		243	285	0
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.7.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.7.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.7a.1 Uplink

6.10.2.4.1.7a.1.1 Transport channel parameters

6.10.2.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Uplink: Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).				

6.10.2.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)

6.10.2.4.1.7a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.7a.2 Downlink

6.10.2.4.1.7a.2.1 Transport channel parameters

6.10.2.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2
RLC	Logical channel type	DTCH	
	RLC mode	TM	TM
	Payload sizes, bit	0, 39, 42, 55, 58, 61	53, 63, 76, 87
	Max data rate, bps	7 400	
	TrD PDU header, bit	0	
MAC	MAC header, bit	0	
	MAC multiplexing	N/A	
Layer 1	TrCH type	DCH	DCH
	TB sizes, bit	0, 39, 42, 55, 58, 61	53, 63, 76, 87

	TFS (note 1)	TF0, bits	1x0 (note 2)	0x87
		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	RM attribute		180 to 220	170 to 210
NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).				
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB #1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).				

6.10.2.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.7a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)

6.10.2.4.1.7a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.8.1 Uplink

6.10.2.4.1.8.1.1 Transport channel parameters

6.10.2.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 58 (alt. 0, 39, 58)	76	60
	Max data rate, bps		6700		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 58 (alt. 0, 39, 58)	76	60
	TFS (note 1)	TF0, bits	0x58 (alt. 1x0) (note)	0x76	0x60
		TF1, bits	1x39	1x76	N/A
		TF2, bits	1x58	N/A	N/A

	TTI, ms	20	20	20
	Coding type	CC 1/3	CC 1/3	CC 1/2
	CRC, bit	12	N/A	N/A
	Max number of bits/TTI after channel coding	234	252	0
	Uplink: Max number of bits/radio frame before rate matching	117	126	0
	RM attribute	180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]).				

6.10.2.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.8.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.8.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.8.2 Downlink

6.10.2.4.1.8.2.1 Transport channel parameters

6.10.2.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0 39 58	76	60
	Max data rate, bps		6700		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 58	76	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x76	0x60
		TF1, bits	1x39	1x76	N/A
		TF2, bits	1x58	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		234	252	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.8.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.8.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.9.1 Uplink

6.10.2.4.1.9.1.1 Transport channel parameters

6.10.2.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 55 (alt. 0, 39, 55)	63	60
	Max data rate, bps		5900		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 55 (alt. 0, 39, 55)	63	60
	TFS	TF0, bits	0x55 (alt. 1x0) (note)	0x63	0x60
		TF1, bits	1x39	1x63	N/A
		TF2, bits	1x55	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		225	213	0
	Uplink: Max number of bits/radio frame before rate matching		113	107	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.9.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.9.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.96

6.10.2.4.1.9.2 Downlink

6.10.2.4.1.9.2.1 Transport channel parameters

6.10.2.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0 39 55	63	60
	Max data rate, bps		5900		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 55	63	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x63	0x60
		TF1, bits	1x39	1x63	N/A
		TF2, bits	1x55	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC ½
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		225	213	0
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.9.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.9.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	0

		Number of TPC bits/slot	2
		Number of Pilot bits/slot	2
	DPDCH	Number of data bits/slot	16
		Number of data bits/frame	240

6.10.2.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.10.1 Uplink

6.10.2.4.1.10.1.1 Transport channel parameters

6.10.2.4.1.10.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 49 (alt. 0, 39, 49)	54	60
	Max data rate, bps		4750		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 49 (alt. 0, 39, 49)	54	60
	TFS	TF0, bits	0x49 (alt. 1x0) (note)	0x54	0x60
		TF1, bits	1x39	1x54	N/A
		TF2, bits	1x49	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	12	N/A
	Max number of bits/TTI after channel coding		207	186	0
	Uplink: Max number of bits/radio frame before rate matching		104	93	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of usign this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.10.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.10.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.10.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	128
	Max number of DPDCH data bits/radio frame	300
	Puncturing Limit	0.72

6.10.2.4.1.10.2 Downlink

6.10.2.4.1.10.2.1 Transport channel parameters

6.10.2.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0 39 49	54	60
	Max data rate, bps		5150		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 49	54	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x54	0x60
		TF1, bits	1x39	1x54	N/A
		TF2, bits	1x49	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		207	186	0
	RM attribute		180 to 220	170 to 210	215 to 256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.10.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.10.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.10.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	14
		Number of data bits/frame	210

6.10.2.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.11.1 Uplink

6.10.2.4.1.11.1.1 Transport channel parameters

6.10.2.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42 (alt. 0, 39, 42)	53	60
	Max data rate, bps		4750		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42 (alt. 0, 39, 42)	53	60
	TFS	TF0, bits	0x42 (alt. 1x0) (note)	0x53	0x60
		TF1, bits	1x39	1x53	N/A
		TF2, bits	1x42	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		186	183	0
	Uplink: Max number of bits/radio frame before rate matching		93	92	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of usign this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.11.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.11.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.11.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	128
	Max number of DPDCH data bits/radio frame	300
	Puncturing Limit	0.76

6.10.2.4.1.11.2 Downlink

6.10.2.4.1.11.2.1 Transport channel parameters

6.10.2.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type	DTCH		
	RLC mode	TM	TM	TM

	Payload sizes, bit		0 39 42	53	60
	Max data rate, bps		4750		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0 39 42	53	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x53	0x60
		TF1, bits	1x39	1x53	N/A
		TF2, bits	1x42	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		186	183	0
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).					

6.10.2.4.1.11.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.11.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1)

6.10.2.4.1.11.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	14
		Number of data bits/frame	210

6.10.2.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.12.1 Uplink

6.10.2.4.1.12.1.1 Transport channel parameters

6.10.2.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Uplink: Max number of bits/radio frame before rate matching		891
	RM attribute		160 to 200

6.10.2.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.12.1.1.3 TFCS

TFCS size	6
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.12.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.92

6.10.2.4.1.12.2 Downlink

6.10.2.4.1.12.2.1 Transport channel parameters

6.10.2.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576

		TF2, bits	2x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	RM attribute		160 to 200

6.10.2.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.12.2.1.3 TFCS

TFCS size	6
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.12.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.13.1 Uplink

6.10.2.4.1.13.1.1 Transport channel parameters

6.10.2.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640(alt. 4x640)
	TTI, ms		20(alt. 40)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948(alt. 7 884)
	Uplink: Max number of bits/radio frame before rate matching		1 974(alt. 1 971)
	RM attribute		150 to 195

6.10.2.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.13.1.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.13.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.88

6.10.2.4.1.13.2 Downlink

6.10.2.4.1.13.2.1 Transport channel parameters

6.10.2.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640(alt. 4x640)
	TTI, ms		20(alt. 40)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948(alt. 7 884)
	RM attribute		150 to 195

6.10.2.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.13.2.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.13.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.14.1 Uplink

6.10.2.4.1.14.1.1 Transport channel parameters

6.10.2.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		32 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	1x640(alt. 2x640)
	TTI, ms		20(alt. 40)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 980 (alt. 3 948)
	Uplink: Max number of bits/radio frame before rate matching		990 (alt. 987)
	RM attribute		165 to 210

6.10.2.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.13.1.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.14.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.80

6.10.2.4.1.14.2 Downlink

6.10.2.4.1.14.2.1 Transport channel parameters

6.10.2.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		32 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	1x640(alt. 2x640)
	TTI, ms		20(alt. 40)
	Coding type		TC
	CRC, bit		16

	Max number of bits/TTI after channel coding	1 980(alt. 3 948)
	RM attribute	165 to 210

6.10.2.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.14.2.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.14.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.15.1 Uplink

6.10.2.4.1.15.1.1 Transport channel parameters

6.10.2.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	Uplink: Max number of bits/radio frame before rate matching		447
	RM attribute		145 to 185

6.10.2.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.15.1.1.3 TFCS

TFCS size	4
TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.15.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64

	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.88

6.10.2.4.1.15.2 Downlink

6.10.2.4.1.15.2.1 Transport channel parameters

6.10.2.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	RM attribute		145 to 185

6.10.2.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.15.2.1.3 TFCS

TFCS size	4
TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.15.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	28
		Number of data bits/frame	420

6.10.2.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.16.1 Uplink

6.10.2.4.1.16.1.1 Transport channel parameters

6.10.2.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	576
	Max data rate, bps	28 800
	TrD PDU header, bit	0
MAC	MAC header, bit	0
	MAC multiplexing	N/A

Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Uplink: Max number of bits/radio frame before rate matching		891
	RM attribute		135 to 175

6.10.2.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.16.1.1.3 TFCS

TFCS size	6
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.16.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.96

6.10.2.4.1.16.2 Downlink

6.10.2.4.1.16.2.1 Transport channel parameters

6.10.2.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576 (alt. 1x0) (note)
		TF1, bits	1x576
		TF2, bits	2x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	RM attribute		135 to 175
NOTE: Alternative 1x0 is used to have CRC present in all transport formats.			

6.10.2.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.16.2.1.3 TFCS

TFCS size	6
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.16.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.17.1 Uplink

6.10.2.4.1.17.1.1 Transport channel parameters

6.10.2.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		57 600
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	Uplink: Max number of bits/radio frame before rate matching		1 779

6.10.2.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.17.1.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.17.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.96

6.10.2.4.1.17.2 Downlink

6.10.2.4.1.17.2.1 Transport channel parameters

6.10.2.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		57 600
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	RM attribute		125 to 165

6.10.2.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.17.2.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.17.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.18 Void

6.10.2.4.1.19 Void

6.10.2.4.1.20 Void

6.10.2.4.1.21 Void

6.10.2.4.1.22 Void

6.10.2.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23.1 Uplink

6.10.2.4.1.23.1.1 Transport channel parameters

6.10.2.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336 (alt. N/A)
	TTI, ms		20 (alt. 10)
	Coding type		TC (alt. CC 1/3)
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 1 080)
	Uplink: Max number of bits/radio frame before rate matching		1 062 (alt. 1 080)
	RM attribute		135 to 175

6.10.2.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23.1.1.3 TFCS

TFCS size	6 (alt. 4)
TFCS	(32 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1))

6.10.2.4.1.23.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.88

6.10.2.4.1.23.2 Downlink

6.10.2.4.1.23.2.1 Transport channel parameters

6.10.2.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC (alt. CC 1/3)
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068 (alt. 1 080)
	RM attribute		135 to 175

6.10.2.4.1.23.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.23.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23a.1 Uplink

6.10.2.4.1.23a.1.1 Transport channel parameters

6.10.2.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		CC 1/3 (alt. TC)
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 080 (alt. 1 068)
	Uplink: Max number of bits/radio frame before rate matching		270 (alt. 267)
	RM attribute		135 to 175

6.10.2.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23a.1.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.23a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.10.2.4.1.23a.2 Downlink

6.10.2.4.1.23a.2.1 Transport channel parameters

6.10.2.4.1.23a.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		CC 1/3 (alt. TC)
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 080 (alt. 1 068)
	RM attribute		135 to 175

6.10.2.4.1.23a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23a.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.1.23a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23b.1 Uplink

6.10.2.4.1.23b.1.1 Transport channel parameters

6.10.2.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336

		TF2, bits	2x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Uplink: Max number of bits/radio frame before rate matching		531
	RM attribute		135 to 175

6.10.2.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23b.1.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.23b.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	1.0

6.10.2.4.1.23b.2 Downlink

6.10.2.4.1.23b.2.1 Transport channel parameters

6.10.2.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	RM attribute		135 to 175

6.10.2.4.1.23b.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23b.2.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)

6.10.2.4.1.23b.2.2 Physical channel parameters

DPCH Downlink	DTX position	Flexible

	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23c.1 Uplink

6.10.2.4.1.23c.1.1 Transport channel parameters

6.10.2.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Uplink: Max number of bits/radio frame before rate matching		1 059
	RM attribute		135 to 175

6.10.2.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23c.1.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)

6.10.2.4.1.23c.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.88

6.10.2.4.1.23c.2 Downlink

6.10.2.4.1.23c.2.1 Transport channel parameters

6.10.2.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH

	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	RM attribute		135 to 175

6.10.2.4.1.23c.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23c.2.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)

6.10.2.4.1.23c.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23d.1 Uplink

6.10.2.4.1.23d.1.1 Transport channel parameters

6.10.2.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		20

	Coding type	TC
	CRC, bit	16
	Max number of bits/TTI after channel coding	2 124
	Uplink: Max number of bits/radio frame before rate matching	1 062
	RM attribute	135 to 175

6.10.2.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23d.1.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)

6.10.2.4.1.23d.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.88

6.10.2.4.1.23d.2 Downlink

6.10.2.4.1.23d.2.1 Transport channel parameters

6.10.2.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	RM attribute		135 to 175

6.10.2.4.1.23d.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23d.2.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)

6.10.2.4.1.23d.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8

		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.24 Void

6.10.2.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.25.1 Uplink

See clause 6.10.2.4.1.23.1.

6.10.2.4.1.25.2 Downlink

6.10.2.4.1.25.2.1 Transport channel parameters

6.10.2.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	RM attribute		130 to 170

6.10.2.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.25.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.25.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.26.1 Uplink

6.10.2.4.1.26.1.1 Transport channel parameters

6.10.2.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Uplink: Max number of bits/radio frame before rate matching		2 118
	RM attribute		130 to 170

6.10.2.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.26.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.26.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.96

6.10.2.4.1.26.2 Downlink

See clause 6.10.2.4.1.25.2.

6.10.2.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.27.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.27.2 Downlink

6.10.2.4.1.27.2.1 Transport channel parameters

6.10.2.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		128 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460
	RM attribute		120 to 160

6.10.2.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.27.2.1.3 TFCS

TFCS size	10
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.27.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.28.1 Uplink

6.10.2.4.1.28.1.1 Transport channel parameters

6.10.2.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	128 000
	AMD PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A

Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460
	Uplink: Max number of bits/radio frame before rate matching		4 230
	RM attribute		120 to 160

6.10.2.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.28.1.1.3 TFCS

TFCS size	10
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.2.4.1.28.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	8
	Max number of DPDCH data bits/radio frame	4 800
	Puncturing Limit	0.96

6.10.2.4.1.28.2 Downlink

See clause 6.10.2.4.1.27.2.

6.10.2.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.29.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.29.2 Downlink

6.10.2.4.1.29.2.1 Transport channel parameters

6.10.2.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		144 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336

		TF5, bits	9x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516
	RM attribute		140 to 180

6.10.2.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.29.2.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)

6.10.2.4.1.29.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.30.1 Uplink

6.10.2.4.1.30.1.1 Transport channel parameters

6.10.2.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		144 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	9x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516
	Uplink: Max number of bits/radio frame before rate matching		4 758
	RM attribute		140 to 180

6.10.2.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.30.1.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)

6.10.2.4.1.30.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	8
	Max number of DPDCH data bits/radio frame	4 800
	Puncturing Limit	0.84

6.10.2.4.1.30.2 Downlink

See clause 6.10.2.4.1.29.2.

6.10.2.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.31.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.31.2 Downlink

6.10.2.4.1.31.2.1 Transport channel parameters

6.10.2.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		256 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	N/A (alt. 12x336)
		TF6, bits	N/A (alt. 16x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 16 920)
	RM attribute		135 to 175

6.10.2.4.1.31.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.31.2.1.3 TFCS

TFCS size	10 (alt.14)
TFCS	(256 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0)

	(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1))

6.10.2.4.1.31.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	Number od DPDCH		1
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.32.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.32.2 Downlink

6.10.2.4.1.32.2.1 Transport channel parameters

6.10.2.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	N/A (alt. 16x336)
		TF7, bits	N/A (alt. 20x336)
		TF8, bits	N/A (alt. 24x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684 (alt. 25 368)
	RM attribute		110 to 150

6.10.2.4.1.32.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.32.2.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),

	(TF8, TF1))

6.10.2.4.1.32.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	Number of DPDCH		1
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.33.1 Uplink

See clause 6.10.2.4.1.28.1.

6.10.2.4.1.33.2 Downlink

See clause 6.10.2.4.1.32.2.

6.10.2.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.34.1 Uplink

6.10.2.4.1.34.1.1 Transport channel parameters

6.10.2.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	16x336(alt. N/A)
		TF7, bits	20x336(alt. N/A)
		TF8, bits	24x336 (alt. N/A)
	TTI, ms		20 (alt. 10)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		25 368
	Uplink: Max number of bits/radio frame before rate matching		12 684
	RM attribute		110-180

6.10.2.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.34.1.1.3 TFCS

TFCS size	18 (alt.12)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1))

6.10.2.4.1.34.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Number of DPDCH	1
	Puncturing Limit	0.64

6.10.2.4.1.34.2 Downlink

See clause 6.10.2.4.1.32.2.

6.10.2.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.35.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.35.2 Downlink

6.10.2.4.1.35.2.1 Transport channel parameters

6.10.2.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		2 048 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
		TF4, bits	8x656
		TF5, bits	12x656
		TF6, bits	16x656
		TF7, bits	20x656
		TF8, bits	24x656
		TF9, bits	28x656
		TF10, bits	32x656
		TF11, bits	N/A (alt. 36x656)
		TF12, bits	N/A (alt. 40x656)
		TF13, bits	N/A (alt. 44x656)
		TF14, bits	N/A (alt. 48x656)
		TF15, bits	N/A (alt. 52x656)
		TF16, bits	N/A (alt. 56x656)
		TF17, bits	N/A (alt. 60x656)
		TF18, bits	N/A (alt. 64x656)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16

	Max number of bits/TTI after channel coding	64 575 (alt. 129 141)
	RM attribute	130 to 170

6.10.2.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.35.2.1.3 TFCS

TFCS size	22 (alt.38)
TFCS	(2 048 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1) (alt. TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0))

6.10.2.4.1.35.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		4
	Number of DPCH		3
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	1 248
		Number of data bits/frame	18 720

6.10.2.4.1.36 Void

6.10.2.4.1.37 Void

6.10.2.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38.1 Uplink

6.10.2.4.1.38.1.1 Transport channel parameters

6.10.2.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38.1.1.4 TFCS

TFCS size	18 (alt. 12)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1))

6.10.2.4.1.38.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.96

6.10.2.4.1.38.2 Downlink

6.10.2.4.1.38.2.1 Transport channel parameters

6.10.2.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.23.2.1.1.

6.10.2.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.

6.10.2.4.1.38.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,8kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)

6.10.2.4.1.38.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38a.1 Uplink

6.10.2.4.1.38a.1.1 Transport channel parameters

6.10.2.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		CC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Uplink: Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.10.2.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38a.1.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)

6.10.2.4.1.38a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.64

6.10.2.4.1.38a.2 Downlink

6.10.2.4.1.38a.2.1 Transport channel parameters

6.10.2.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		CC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0

	RM attribute	130 to 170

6.10.2.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38a.2.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)

6.10.2.4.1.38a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38b.1 Uplink

6.10.2.4.1.38b.1.1 Transport channel parameters

6.10.2.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Uplink: Max number of bits/radio frame before rate matching		267
	RM attribute		135 to 175

6.10.2.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38b.1.1.4 TFCS

TFCS size	12

TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)

6.10.2.4.1.38b.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	1.0

6.10.2.4.1.38b.2 Downlink

6.10.2.4.1.38b.2.1 Transport channel parameters

6.10.2.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	RM attribute		135 to 175

6.10.2.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38b.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)

6.10.2.4.1.38b.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38c.1 Uplink

6.10.2.4.1.38c.1.1 Transport channel parameters

6.10.2.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.2.4.1.23c.1.1.1.

6.10.2.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38c.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF4,TF1)

6.10.2.4.1.38c.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	1.0

6.10.2.4.1.38c.2 Downlink

6.10.2.4.1.38c.2.1 Transport channel parameters

6.10.2.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.2.4.1.23c.2.1.1.

6.10.2.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38c.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF3,TF0),

	(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF4,TF1)

6.10.2.4.1.38c.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38d.1 Uplink

6.10.2.4.1.38d.1.1 Transport channel parameters

6.10.2.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	Uplink: Max number of bits/radio frame before rate matching		2 142	
	RM attribute		130 to 170	

6.10.2.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38d.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)

6.10.2.4.1.38d.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.76

6.10.2.4.1.38d.2 Downlink

6.10.2.4.1.38d.2.1 Transport channel parameters

6.10.2.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	RM attribute		130 to 170	

6.10.2.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38d.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0),

	(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)

6.10.2.4.1.38d.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.38e Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38e.1 Uplink

6.10.2.4.1.38e.1.1 Transport channel parameters

6.10.2.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.2.4.1.38a.1.1.2.

6.10.2.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38e.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1)

6.10.2.4.1.38e.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.84

6.10.2.4.1.38e.2 Downlink

6.10.2.4.1.38e.2.1 Transport channel parameters

6.10.2.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.2.4.1.38a.2.1.2

6.10.2.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38e.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),

6.10.2.4.1.38e.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.38f Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38f.1 Uplink

6.10.2.4.1.38f.1.1 Transport channel parameters

6.10.2.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38f.1.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)

6.10.2.4.1.38f.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	1.0

6.10.2.4.1.38f.2 Downlink

6.10.2.4.1.38f.2.1 Transport channel parameters

6.10.2.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2

6.10.2.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38f.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)

6.10.2.4.1.38f.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.38g Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38g.1 Uplink

6.10.2.4.1.38g.1.1 Transport channel parameters

6.10.2.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.2.4.1.23b.1.1.1.

6.10.2.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38g.1.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),

	(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1)

6.10.2.4.1.38g.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	0.88

6.10.2.4.1.38g.2 Downlink

6.10.2.4.1.38g.2.1 Transport channel parameters

6.10.2.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.2.4.1.23b.2.1.1.

6.10.2.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38g.2.1.4 TFCS

TFCS size	36
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1)

6.10.2.4.1.38g.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.38h Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38h.1 Uplink

6.10.2.4.1.38h.1.1 Transport channel parameters

6.10.2.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23c.1.1.1.

6.10.2.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38h.1.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF0,TF0,TF0,TF4,TF0), (TF5,TF4,TF1,TF0,TF0), (TF5,TF4,TF1,TF1,TF0), (TF5,TF4,TF1,TF2,TF0), (TF5,TF4,TF1,TF4,TF0), (TF4,TF3,TF0,TF0,TF0), (TF4,TF3,TF0,TF1,TF0), (TF3,TF2,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF5,TF4,TF1,TF0,TF1), (TF5,TF4,TF1,TF1,TF1), (TF5,TF4,TF1,TF2,TF1), (TF5,TF4,TF1,TF4,TF1), (TF4,TF3,TF0,TF0,TF1), (TF4,TF3,TF0,TF1,TF1), (TF3,TF2,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF4,TF1)

6.10.2.4.1.38h.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	1.0

6.10.2.4.1.38h.2 Downlink

6.10.2.4.1.38h.2.1 Transport channel parameters

6.10.2.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23c.2.1.1.

6.10.2.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38h.2.1.4 TFCS

TFCS size	48
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF0,TF0,TF0,TF4,TF0), (TF5,TF4,TF1,TF0,TF0), (TF5,TF4,TF1,TF1,TF0), (TF5,TF4,TF1,TF2,TF0), (TF5,TF4,TF1,TF4,TF0), (TF4,TF3,TF0,TF0,TF0), (TF4,TF3,TF0,TF1,TF0), (TF4,TF3,TF0,TF2,TF0), (TF4,TF3,TF0,TF4,TF0), (TF3,TF2,TF0,TF0,TF0), (TF3,TF2,TF0,TF1,TF0), (TF3,TF2,TF0,TF2,TF0), (TF3,TF2,TF0,TF4,TF0), (TF2,TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF1,TF0), (TF2,TF1,TF0,TF2,TF0), (TF2,TF1,TF0,TF4,TF0), (TF1,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF5,TF4,TF1,TF0,TF1), (TF5,TF4,TF1,TF1,TF1), (TF5,TF4,TF1,TF2,TF1), (TF5,TF4,TF1,TF4,TF1), (TF4,TF3,TF0,TF0,TF1), (TF4,TF3,TF0,TF1,TF1), (TF4,TF3,TF0,TF2,TF1), (TF4,TF3,TF0,TF4,TF1), (TF3,TF2,TF0,TF0,TF1), (TF3,TF2,TF0,TF1,TF1), (TF3,TF2,TF0,TF2,TF1), (TF3,TF2,TF0,TF4,TF1), (TF2,TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1,TF1), (TF2,TF1,TF0,TF2,TF1), (TF2,TF1,TF0,TF4,TF1), (TF1,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF4,TF1)

6.10.2.4.1.38h.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.38i Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38i.1 Uplink

6.10.2.4.1.38i.1.1 Transport channel parameters

6.10.2.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38i.1.1.4 TFCS

TFCS size	48
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)

6.10.2.4.1.38i.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.76

6.10.2.4.1.38i.2 Downlink

6.10.2.4.1.38i.2.1 Transport channel parameters

6.10.2.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38i.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)

6.10.2.4.1.38i.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.38j Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38j.1 Uplink

6.10.2.4.1.38j.1.1 Transport channel parameters

See clause 6.10.2.4.1.38i.1.1

6.10.2.4.1.38j.2 Downlink

6.10.2.4.1.38j.2.1 Transport channel parameters

6.10.2.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38j.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),

	(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)

6.10.2.4.1.38j.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.38k Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing)

6.10.2.4.1.38k.1 Uplink

6.10.2.4.1.38k.1.1 Transport channel parameters

6.10.2.4.1.38k.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38k.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38k.1.1.3 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38k.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38k.1.1.5 TFCS

TFCS size	54 (alt. 24)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB, 32kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0), (TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF2, TF0, TF0), (TF1, TF0, TF0, TF2, TF0, TF0), (TF2, TF1, TF1, TF2, TF0, TF0), (TF0, TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF0, TF1, TF0), (TF0, TF0, TF0, TF1, TF1, TF0), (TF1, TF0, TF0, TF1, TF1, TF0), (TF2, TF1, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF1, TF0),

	(TF1, TF0, TF0, TF2, TF1, TF0), (TF2, TF1, TF1, TF2, TF1, TF0), (TF0, TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF0, TF2, TF0), (TF0, TF0, TF0, TF1, TF2, TF0), (TF1, TF0, TF0, TF1, TF2, TF0), (TF2, TF1, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF2, TF2, TF0), (TF1, TF0, TF0, TF2, TF2, TF0), (TF2, TF1, TF1, TF2, TF2, TF0), (TF0, TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF0, TF1), (TF0, TF0, TF0, TF1, TF0, TF1), (TF1, TF0, TF0, TF1, TF0, TF1), (TF2, TF1, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF2, TF0, TF1), (TF1, TF0, TF0, TF2, TF0, TF1), (TF2, TF1, TF1, TF2, TF0, TF1), (TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1), (TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1, TF1), (TF1, TF0, TF0, TF2, TF1, TF1), (TF2, TF1, TF1, TF2, TF1, TF1), (TF0, TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF0, TF2, TF1), (TF0, TF0, TF0, TF1, TF2, TF1), (TF1, TF0, TF0, TF1, TF2, TF1), (TF2, TF1, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF2, TF2, TF1), (TF1, TF0, TF0, TF2, TF2, TF1), (TF2, TF1, TF1, TF2, TF2, TF1) (alt. (TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0), (TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF0, TF1, TF0), (TF0, TF0, TF0, TF1, TF1, TF0), (TF1, TF0, TF0, TF1, TF1, TF0), (TF2, TF1, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF0, TF1), (TF0, TF0, TF0, TF1, TF0, TF1), (TF1, TF0, TF0, TF1, TF0, TF1), (TF2, TF1, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1), (TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1))

6.10.2.4.1.38k.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.8

6.10.2.4.1.38k.2 Downlink

6.10.2.4.1.38k.2.1 Transport channel parameters

6.10.2.4.1.38k.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38k.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23d.2.1.1.

6.10.2.4.1.38k.2.1.3 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23d.2.1.1.

6.10.2.4.1.38k.2.1.4 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.

6.10.2.4.1.38k.2.1.5 TFCS

TFCS size	54
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB, 32kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0), (TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF2, TF0, TF0), (TF1, TF0, TF0, TF2, TF0, TF0),

	(TF2, TF1, TF1, TF2, TF0, TF0), (TF0, TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF0, TF1, TF0), (TF0, TF0, TF0, TF1, TF1, TF0), (TF1, TF0, TF0, TF1, TF1, TF0), (TF2, TF1, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF1, TF0), (TF1, TF0, TF0, TF2, TF1, TF0), (TF2, TF1, TF1, TF2, TF1, TF0), (TF0, TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF0, TF2, TF0), (TF0, TF0, TF0, TF1, TF2, TF0), (TF1, TF0, TF0, TF1, TF2, TF0), (TF2, TF1, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF2, TF2, TF0), (TF1, TF0, TF0, TF2, TF2, TF0), (TF2, TF1, TF1, TF2, TF2, TF0), (TF0, TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF0, TF1), (TF0, TF0, TF0, TF1, TF0, TF1), (TF1, TF0, TF0, TF1, TF0, TF1), (TF2, TF1, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF2, TF0, TF1), (TF1, TF0, TF0, TF2, TF0, TF1), (TF2, TF1, TF1, TF2, TF0, TF1), (TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1), (TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1, TF1), (TF1, TF0, TF0, TF2, TF1, TF1), (TF2, TF1, TF1, TF2, TF1, TF1), (TF0, TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF0, TF2, TF1), (TF0, TF0, TF0, TF1, TF2, TF1), (TF1, TF0, TF0, TF1, TF2, TF1), (TF2, TF1, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF2, TF2, TF1), (TF1, TF0, TF0, TF2, TF2, TF1), (TF2, TF1, TF1, TF2, TF2, TF1)

6.10.2.4.1.38k.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.39.1 Uplink

See clause 6.10.2.4.1.38.1.

6.10.2.4.1.39.2 Downlink

6.10.2.4.1.39.2.1 Transport channel parameters

6.10.2.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.39.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),

	(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.39.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.40.1 Uplink

6.10.2.4.1.40.1.1 Transport channel parameters

6.10.2.4.1.40.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.40.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.40.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.40.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.40.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.76

6.10.2.4.1.40.2 Downlink

See clause 6.10.2.4.1.39.2.

6.10.2.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.41.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.41.2 Downlink

6.10.2.4.1.41.2.1 Transport channel parameters

6.10.2.4.1.41.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.41.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.41.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.41.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.41.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.42.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.42.2 Downlink

6.10.2.4.1.42.2.1 Transport channel parameters

6.10.2.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.2.4.1.31.2.1.1.

6.10.2.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.42.2.1.4 TFCS

TFCS size	30 (alt. 42)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 256 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1))

6.10.2.4.1.42.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	Number of DPDCH		1
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.43.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.43.2 Downlink

6.10.2.4.1.43.2.1 Transport channel parameters

6.10.2.4.1.43.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.43.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.2.4.1.32.2.1.1.

6.10.2.4.1.43.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.43.2.1.4 TFCS

TFCS size	36 (alt. 54)

TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1) (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1) (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1))

6.10.2.4.1.43.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	Number of DPDCH		1
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.44.1 Uplink

6.10.2.4.1.44.1.1 Transport channel parameters

6.10.2.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.44.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=

	(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.44.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	8
	Max number of DPDCH data bits/radio frame	4 800
	Puncturing Limit	0.92

6.10.2.4.1.44.2 Downlink

6.10.2.4.1.44.2.1 Transport channel parameters

6.10.2.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.2.4.1.35.2.1.1.

6.10.2.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.44.2.1.4 TFCS

TFCS size	66 (alt. 114)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1) (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF11, TF0), (TF1, TF0, TF0, TF11, TF0), (TF2, TF1, TF1, TF11, TF0), (TF0, TF0, TF0, TF12, TF0), (TF1, TF0, TF0, TF12, TF0), (TF2, TF1, TF1, TF12, TF0), (TF0, TF0, TF0, TF13, TF0), (TF1, TF0, TF0, TF13, TF0), (TF2, TF1, TF1, TF13, TF0), (TF0, TF0, TF0, TF14, TF0), (TF1, TF0, TF0, TF14, TF0), (TF2, TF1, TF1, TF14, TF0), (TF0, TF0, TF0, TF15, TF0), (TF1, TF0, TF0, TF15, TF0), (TF2, TF1, TF1, TF15, TF0), (TF0, TF0, TF0, TF16, TF0), (TF1, TF0, TF0, TF16, TF0), (TF2, TF1, TF1, TF16, TF0), (TF0, TF0, TF0, TF17, TF0), (TF1, TF0, TF0, TF17, TF0), (TF2, TF1, TF1, TF17, TF0), (TF0, TF0, TF0, TF18, TF0), (TF1, TF0, TF0, TF18, TF0), (TF2, TF1, TF1, TF18, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1), (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1), (TF0, TF0, TF0, TF11, TF1), (TF1, TF0, TF0, TF11, TF1), (TF2, TF1, TF1, TF11, TF1), (TF0, TF0, TF0, TF12, TF1), (TF1, TF0, TF0, TF12, TF1), (TF2, TF1, TF1, TF12, TF1), (TF0, TF0, TF0, TF13, TF1), (TF1, TF0, TF0, TF13, TF1), (TF2, TF1, TF1, TF13, TF1), (TF0, TF0, TF0, TF14, TF1), (TF1, TF0, TF0, TF14, TF1), (TF2, TF1, TF1, TF14, TF1), (TF0, TF0, TF0, TF15, TF1), (TF1, TF0, TF0, TF15, TF1), (TF2, TF1, TF1, TF15, TF1), (TF0, TF0, TF0, TF16, TF1), (TF1, TF0, TF0, TF16, TF1), (TF2, TF1, TF1, TF16, TF1), (TF0, TF0, TF0, TF17, TF1), (TF1, TF0, TF0, TF17, TF1), (TF2, TF1, TF1, TF17, TF1), (TF0, TF0, TF0, TF18, TF1), (TF1, TF0, TF0, TF18, TF1), (TF2, TF1, TF1, TF18, TF1))

6.10.2.4.1.44.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		4
	Number of DPDCH		3
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	1 248
		Number of data bits/frame	18 720

6.10.2.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.45.1 Uplink

6.10.2.4.1.45.1.1 Transport channel parameters

6.10.2.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.2.4.1.17.1.1.1.

6.10.2.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.45.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.45.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.88

6.10.2.4.1.45.2 Downlink

6.10.2.4.1.45.2.1 Transport channel parameters

6.10.2.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.2.4.1.17.2.1.1.

6.10.2.4.1.45.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.45.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

6.10.2.4.1.45.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.46 Void

6.10.2.4.1.47 Void

6.10.2.4.1.48 Void

6.10.2.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.49.1 Uplink

6.10.2.4.1.49.1.1 Transport channel parameters

6.10.2.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.49.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)

6.10.2.4.1.49.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400

	Puncturing Limit	0.72

6.10.2.4.1.49.2 Downlink

6.10.2.4.1.49.2.1 Transport channel parameters

6.10.2.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.11.

6.10.2.4.1.49.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)

6.10.2.4.1.49.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.49a.1 Uplink

6.10.2.4.1.49a.1.1 Transport channel parameters

6.10.2.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.49a.1.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),

	(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)

6.10.2.4.1.49a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.72

6.10.2.4.1.49a.2 Downlink

6.10.2.4.1.49a.2.1 Transport channel parameters

6.10.2.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.49a.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)

6.10.2.4.1.49a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.50.1 Uplink

6.10.2.4.1.50.1.1 Transport channel parameters

6.10.2.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.50.1.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)

6.10.2.4.1.50.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	8
	Max number of DPDCH data bits/radio frame	4 800
	Puncturing Limit	0.92

6.10.2.4.1.50.2 Downlink

6.10.2.4.1.50.2.1 Transport channel parameters

6.10.2.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.50.2.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)

6.10.2.4.1.50.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51.1 Uplink

6.10.2.4.1.51.1.1 Transport channel parameters

6.10.2.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)

6.10.2.4.1.51.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	8
	Max number of DPDCH data bits/radio frame	4 800
	Puncturing Limit	0.88

6.10.2.4.1.51.2 Downlink

6.10.2.4.1.51.2.1 Transport channel parameters

6.10.2.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.51.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)

6.10.2.4.1.51.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51a.1 Uplink

6.10.2.4.1.51a.1.1 Transport channel parameters

6.10.2.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51a.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51a.1.1.4 TFCS

TFCS size	8
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)

6.10.2.4.1.51a.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.72

6.10.2.4.1.51a.2 Downlink

6.10.2.4.1.51a.2.1 Transport channel parameters

6.10.2.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.51a.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.51a.2.1.4 TFCS

TFCS size	8
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)

6.10.2.4.1.51a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51b.1 Uplink

6.10.2.4.1.51b.1.1 Transport channel parameters

6.10.2.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51b.1.1.2 Transport channel parameters for Interactive or Background / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB

RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Uplink: Max number of bits/radio frame before rate matching		531
	RM attribute		135 to 175

6.10.2.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51b.1.1.4 TFCS

TFCS size	12
TFCS	(64 kbps Conversational RAB, 16 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1)

6.10.2.4.1.51b.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.64

6.10.2.4.1.51b.2 Downlink

See clause 6.10.2.4.1.51.2.

6.10.2.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.52.1 Uplink

See clause 6.10.2.4.1.51.1.

6.10.2.4.1.52.2 Downlink

6.10.2.4.1.52.2.1 Transport channel parameters

6.10.2.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.52.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)

6.10.2.4.1.52.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.53.1 Uplink

6.10.2.4.1.53.1.1 Transport channel parameters

6.10.2.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.53.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)

6.10.2.4.1.53.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Puncturing Limit	0.96

6.10.2.4.1.53.2 Downlink

See clause 6.10.2.4.1.52.2.

6.10.2.4.1.54 Void

6.10.2.4.1.55 Void

6.10.2.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.56.1 Uplink

6.10.2.4.1.56.1.1 Transport channel parameters

6.10.2.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 080	
	Uplink: Max number of bits/radio frame before rate matching		270	
	RM attribute		135 to 175	

6.10.2.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.56.1.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1)

6.10.2.4.1.56.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.10.2.4.1.56.2 Downlink

6.10.2.4.1.56.2.1 Transport channel parameters

6.10.2.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	

	Max number of bits/TTI after channel coding	1 080
	RM attribute	135 to 175

6.10.2.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.56.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1)

6.10.2.4.1.56.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.57.1 Uplink

6.10.2.4.1.57.1.1 Transport channel parameters

6.10.2.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	Uplink: Max number of bits/radio frame before rate matching		2 142	
	RM attribute		130 to 170	

6.10.2.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.57.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)=

	(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)

6.10.2.4.1.57.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.92

6.10.2.4.1.57.2 Downlink

6.10.2.4.1.57.2.1 Transport channel parameters

6.10.2.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	RM attribute		130 to 170	

6.10.2.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.57.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)

6.10.2.4.1.57.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.58.1 Uplink

6.10.2.4.1.58.1.1 Transport channel parameters

6.10.2.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Uplink: Max number of bits/radio frame before rate matching		534
	RM attribute		135 to 175

6.10.2.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.58.1.1.4 TFCS

TFCS size	8
TFCS	(16 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1)

6.10.2.4.1.58.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	1.0

6.10.2.4.1.58.2 Downlink

6.10.2.4.1.58.2.1 Transport channel parameters

6.10.2.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	640
	Max data rate, bps	64 000
	AM PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH

	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 076
	RM attribute		125 to 165

6.10.2.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.58.2.1.4 TFCS

TFCS size	16
TFCS	(64 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1)

6.10.2.4.1.58.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.58a Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.58a.1 Uplink

6.10.2.4.1.58a.1.1 Transport channel parameters

6.10.2.4.1.58a.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

See clause 6.10.2.4.1.58.1.1.1.

6.10.2.4.1.58a.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.58a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.58a.1.1.4 TFCS

See clause 6.10.2.4.1.58.1.1.4.

6.10.2.4.1.58a.1.2 Physical channel parameters

See clause 6.10.2.4.1.58.1.2.

6.10.2.4.1.58a.2 Downlink

6.10.2.4.1.58a.2.1 Transport channel parameters

6.10.2.4.1.58a.2.1.1 Transport channel parameters for Streaming / unknown / DL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		128 000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	3x656
		TF4, bits	4x656
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 076
	RM attribute		125 to 165

6.10.2.4.1.58a.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.58a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.58a.2.1.4 TFCS

TFCS size	20
TFCS	(128 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0), (TF4,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0), (TF4,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1), (TF4,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1), (TF4,TF1,TF1),

6.10.2.4.1.58a.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		16
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	288
		Number of data bits/frame	4 320

6.10.2.4.1.59 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.59.1 Uplink

6.10.2.4.1.59.1.1 Transport channel parameters

6.10.2.4.1.59.1.1.1 Transport channel parameters for Conversational / speech / UL:42.8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
PDCP	PDCP header size, bit		8
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		920, 304, 96
	Max data rate, bps		46 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		928, 312, 104
	TFS	TF0, bits	0x928
		TF1, bits	1x104
		TF2, bits	1x312
		TF3, bits	1x928
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 844
	Uplink: Max number of bits/radio frame before rate matching		1 422
	RM attribute		180 to 220

6.10.2.4.1.59.1.1.2 Transport channel parameters for Interactive / UL:16kbps / PS RAB + UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		16 000	16 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2X340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		2 148	
	Uplink: Max number of bits/radio frame before rate matching		537	
	RM attribute		135 to 175	

6.10.2.4.1.59.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.59.1.1.4 TFCS

TFCS size	24
TFCS	(42.8 kbps Conversational RAB, Interactive 16kbps+16kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1) (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1) (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1) (TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1)

6.10.2.4.1.59.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.76

6.10.2.4.1.59.2 Downlink

6.10.2.4.1.59.2.1 Transport channel parameters

6.10.2.4.1.59.2.1.1 Transport channel parameters for Conversational / speech / DL:42.8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
PDCP	PDCP header size, bit		8
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		920, 304, 96
	Max data rate, bps		46 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		928, 312, 104
	TFS	TF0, bits	0x928
		TF1, bits	1x104
		TF2, bits	1x312
		TF3, bits	1x928
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 844
	RM attribute		180 to 220

6.10.2.4.1.59.2.1.2 Transport channel parameters for Interactive / DL:16kbps / PS RAB + DL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		16 000	16 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		2 148	
	RM attribute		135 to 175	

6.10.2.4.1.59.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.59.2.1.4 TFCS

TFCS size	24
TFCS	(42.8 kbps Conversational RAB, Interactive 16kbps+16kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1) (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1) (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1) (TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1)

6.10.2.4.1.59.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.60 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.60.1 Uplink

6.10.2.4.1.60.1.1 Transport channel parameters

6.10.2.4.1.60.1.1.1 Transport channel parameters for Conversational / speech / UL:42.8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
PDCP	PDCP header size, bit		8
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		920, 304, 96
	Max data rate, bps		46 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		928, 312, 104
	TFS	TF0, bits	0x928
		TF1, bits	1x104
		TF2, bits	1x312
		TF3, bits	1x928
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 844
	Uplink: Max number of bits/radio frame before rate matching		1 422
	RM attribute		180 to 220

6.10.2.4.1.60.1.1.2 Transport channel parameters for Interactive / UL:16kbps / PS RAB

See clause 6.10.2.4.1.23b.1.1.1

6.10.2.4.1.60.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.60.1.1.4 TFCS

TFCS size	24
TFCS	(42.8 kbps Conversational RAB, Interactive 16kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1) (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1) (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1) (TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1)

6.10.2.4.1.60.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.76

6.10.2.4.1.60.2 Downlink

6.10.2.4.1.60.2.1 Transport channel parameters

6.10.2.4.1.60.2.1.1 Transport channel parameters for Conversational / speech / DL:42.8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
PDCP	PDCP header size, bit		8
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		920, 304, 96
	Max data rate, bps		46 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		928, 312, 104
	TFS	TF0, bits	0x928
		TF1, bits	1x104
		TF2, bits	1x312
		TF3, bits	1x928
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 844
	RM attribute		180 to 220

6.10.2.4.1.60.2.1.2 Transport channel parameters for Interactive / DL:16kbps PS RAB

See clause 6.10.2.4.1.23b.2.1.1

6.10.2.4.1.60.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.60.2.1.4 TFCS

TFCS size	24
TFCS	(42.8 kbps Conversational RAB, Interactive 16kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1) (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1) (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1) (TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1)

6.10.2.4.1.60.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.10.2.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.61.1 Uplink

6.10.2.4.1.61.1.1 Transport channel parameters

6.10.2.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB

RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328) (note)
	TFS	TF0, bits	0x328 (alt 1x0) (note)
		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	Uplink: Max number of bits/radio frame before rate matching		261
	RM attribute		135 to 175
NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).			

6.10.2.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2

6.10.2.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.61.1.1.4 TFCS

TFCS size	8
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)

6.10.2.4.1.61.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	32
	Max number of DPDCH data bits/radio frame	1 200
	Puncturing Limit	1.0

6.10.2.4.1.61.2 Downlink

6.10.2.4.1.61.2.1 Transport channel parameters

6.10.2.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328) (note)
	TFS	TF0, bits	0x328 (alt 1x0) (note)
		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	RM attribute		135 to 175

NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.61.2.1.4 TFCS

TFCS size	8
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)

6.10.2.4.1.61.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		64
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	60
		Number of data bits/frame	900

6.10.2.4.1.62 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.1.62.1 Uplink

6.10.2.4.1.62.1.1 Transport channel parameters

6.10.2.4.1.62.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.65 8.85 6.6) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3 (note 2)
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		40, 54, 64, 72 (alt. 0, 40, 54, 64, 72)	78, 113, 181	60
	Max data rate, bps		12 650		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		40, 54, 64, 72 (alt. 0, 40, 54, 64, 72)	78, 113, 181	60
	TFS	TF0, bits	0x72(alt. 1x0) (note 1)	0x181	0x60
		TF1, bits	1x40	1x78	N/A
		TF2 bits	1x54	1x113	N/A
		TF3, bits	1x64	1x181	N/A
		TF4, bits	1x72	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/3
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		276	567	0
	Uplink: Max number of bits/radio frame before rate matching		138	284	0
	RM attribute		180 to 220	170 to 210	256

NOTE 1: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).
NOTE 2: RAB subflow #3 does not exist in Iu interface. UTRAN establishes this additional "dummy" subflow when the RAB for Wideband AMR is assigned.

6.10.2.4.1.62.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.62.1.1.3 TFCS

TFCS size	10
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow #3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1)

6.10.2.4.1.62.1.1.4 TFC subset list

TFC subset list size	3
TFC subset list	0 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1)}, 1 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1)}, 2 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1)}

6.10.2.4.1.62.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.84

6.10.2.4.1.62.2 Downlink

6.10.2.4.1.62.2.1 Transport channel parameters

6.10.2.4.1.62.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.65 8.85 6.6) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3 (note 3)
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		0, 40, 54, 64, 72	78, 113, 181	60
	Max data rate, bps		12 650		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		0, 40, 54, 64, 72	78, 113, 181	60
	TFS (note 1)	TF0, bits	1x0 (note 2)	0x181	0x60
		TF1, bits	1x40	1x78	N/A
		TF2, bits	1x54	1x113	N/A
		TF3, bits	1x64	1x181	N/A
		TF4, bits	1x72	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/3
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		276	567	0
	RM attribute		180 to 220	170 to 210	256

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).
NOTE 2: RAB subflow #3 does not exist in Iu interface. UTRAN establishes this additional "dummy" subflow when the RAB for Wideband AMR is assigned

6.10.2.4.1.62.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.62.2.1.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

Higher layer	RAB/signalling RB	SRB#5
	User of Radio Bearer	RRC
RLC	Logical channel type	DCCH
	RLC mode	TM
	Payload sizes, bit	3

	Max data rate, bps		150
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		3 (alt 0, 3) (note)
	TFS	TF0, bits	0x3 (alt 1x0) (note)
		TF1, bits	1x3
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		8
	Max number of bits/TTI before rate matching		57
	RM attribute		155 to 256
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.			

6.10.2.4.1.62.2.1.4 TFCS

TFCS size	20
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH 3.4, DCCH 0.15)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1)

6.10.2.4.1.62.2.2 Physical channel parameters

DPCH Downlink	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	0
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.10.2.4.1.63 Interactive or background / UL:64 DL:768 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.63.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.63.2 Downlink

6.10.2.4.1.63.2.1 Transport channel parameters

6.10.2.4.1.63.2.1.1 Transport channel parameters for Interactive or background / DL:768 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	768 000
	AMD PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A

Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	16x336
		TF7, bits	20x336
		TF8, bits	24x336
		TF9, bits	N/A (alt 28x336)
		TF10, bits	N/A (alt 32x336)
		TF11, bits	N/A (alt 36x336)
		TF12, bits	N/A (alt 40x336)
		TF13, bits	N/A (alt 44x336)
		TF14, bits	N/A (alt 48x336)
	TTI, ms		10 (alt 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		25 368 (alt 50 736)
	RM attribute		110 to 150

6.10.2.4.1.63.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.63.2.1.3 TFCS

TFCS size	18 (alt. 30)
TFCS	(768 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1) (alt . (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1) (TF9, TF1), (TF10, TF1), (TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1))

6.10.2.4.1.63.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		8
	Number of DPCH		2
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	8
		Number of Pilot bits/slot	16
	DPDCH	Number of data bits/slot	608
		Number of data bits/frame	9 120

6.10.2.4.2 Combinations on PDSCH and DPCH

6.10.2.4.2.1 Void

6.10.2.4.2.2 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.2.2.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.2.2.2 Downlink

6.10.2.4.2.2.1 Transport channel parameters

6.10.2.4.2.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		18
	MAC multiplexing		Logical channel multiplexing on a frame by frame basis
Layer 1	TrCH type		DSCH
	TB sizes, bit		354
	TFS	TF0, bits	0x354
		TF1, bits	1x354
		TF2, bits	2x354
		TF3, bits	4 x354
		TF4, bits	8 x354
		TF5, bits	12 x354
		TF6, bits	N/A (alt. 16x354)
		TF7, bits	N/A (alt. 20x354)
		TF8, bits	N/A (alt. 24x354)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		13 332 (alt. 26 664)
	RM attribute		110 to 150

6.10.2.4.2.2.1.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.2.1.3 TFCS

PDSCH	TFCS size	6 (alt.9)
	TFCS	384 kbps RAB = TF0, TF1, TF2, TF3, TF4, TF5 (alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8)
DPCH Downlink associated with PDSCH	TFCS size	2
	TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.2.2.2 Physical channel parameters

PDSCH	RAB or SRB, TrCh		**Interactive or background / 384 kbps / PS RAB, DSCH**
	DTX position		N/A (SingleTrCH)
	Minimum spreading factor		8
DPCH Downlink associated with PDSCH	RAB or SRB, TrCh		**3.4 kbps SRB for DCCH, DCH**
	DTX position		N/A (SingleTrCH)
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	12
		Number of data bits/frame	180

6.10.2.4.2.3 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.2.3.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.2.3.2 Downlink

6.10.2.4.2.3.2.1 Transport channel parameters

6.10.2.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		2 048 000
	AMD PDU header, bit		16
MAC	MAC header, bit		18
	MAC multiplexing		Logical channel multiplexing on a frame by frame basis
Layer 1	TrCH type		DSCH
	TB sizes, bit		674
	TFS	TF0, bits	0x674
		TF1, bits	1x674
		TF2, bits	2x674
		TF3, bits	4 x674
		TF4, bits	8 x674
		TF5, bits	12x674
		TF6, bits	16x674
		TF7, bits	20x674
		TF8, bits	24x674
		TF9, bits	28x674
		TF10, bits	32x674
		TF11, bits	N/A (alt. 36x674)
		TF12, bits	N/A (alt. 40x674)
		TF13, bits	N/A (alt. 44x674)
		TF14, bits	N/A (alt. 48x674)
		TF15, bits	N/A (alt. 52x674)
		TF16, bits	N/A (alt. 56x674)
		TF17, bits	N/A (alt. 60x674)
		TF18, bits	N/A (alt. 64x674)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		66 300 (alt. 132 588)
	RM attribute		130 to 170

6.10.2.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.2.3.2.1.3 TFCS

PDSCH	TFCS size	11 (alt.19)
	TFCS	2 048 kbps RAB = TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10 (alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16, TF17, TF18)
DPCH Downlink associated with PDSCH	TFCS size	2
	TFCS	SRBs for DCCH = TF0, TF1

6.10.2.4.2.3.2.2 Physical channel parameters

PDSCH	RAB or SRB, TrCh		**Interactive or background / 2 048 kbps / PS RAB, DSCH**
	DTX position		N/A (SingleTrCH)
	Minimum spreading factor		4
DPCH Downlink associated with PDSCH	RAB or SRB, TrCh		3.4 kbps SRB for DCCH, DCH
	DTX position		N/A (SingleTrCH)
	Spreading factor		256
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2

		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	12
		Number of data bits/frame	180

6.10.2.4.2.4 Void

6.10.2.4.2.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.2.5.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.2.5.2 Downlink

6.10.2.4.2.5.2.1 Transport channel parameters

6.10.2.4.2.5.2.1.1 Transport channel parameters for Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.2.5.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.2.4.2.2.1.1.

6.10.2.4.2.5.2.1.3 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.5.2.1.4 TFCS

PDSCH	TFCS size	6 (alt.9)
	TFCS	384 kbps RAB = TF0, TF1, TF2, TF3, TF4, TF5 (alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8)
DPCH Downlink associated with PDSCH	TFCS size	6
	TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH) = (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.2.5.2.2 Physical channel parameters

PDSCH	RAB or SRB, TrCh		**Interactive or background / 384 kbps / PS RAB, DSCH**
	DTX position		N/A (SingleTrCH)
	Minimum spreading factor		8
DPCH Downlink associated with PDSCH	RAB or SRB, TrCh		Conversational / speech / 12.2 kbps / CS RAB, DCH **+ 3.4 kbps SRBs for DCCH. DCH**
	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.2.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.2.6.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.2.6.2 Downlink

6.10.2.4.2.6.2.1 Transport channel parameters

6.10.2.4.2.6.2.1.1 Transport channel parameters for Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.2.6.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.2.4.2.3.2.1.1.

6.10.2.4.2.6.2.1.3 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.6.2.1.4 TFCS

PDSCH	TFCS size	11 (alt.19)
	TFCS	2 048 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10 (alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16, TF17, TF18)
DPCH Downlink associated with PDSCH	TFCS size	6
	TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH) = (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)

6.10.2.4.2.6.2.2 Physical channel parameters

PDSCH	RAB or SRB, TrCh		**Interactive or background / 2 048 kbps / PS RAB, DSCH**
	DTX position		N/A (SingleTrCH)
	Minimum spreading factor		4
DPCH Downlink associated with PDSCH	RAB or SRB, TrCh		Conversational / speech / 12.2 kbps / CS RAB, DCH **+ 3.4 kbps SRBs for DCCH. DCH**
	DTX position		Fixed
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	32
		Number of data bits/frame	480

6.10.2.4.3 Combinations on SCCPCH

6.10.2.4.3.1 Stand-alone signalling RB for PCCH

6.10.2.4.3.1.1 Transport channel parameters

6.10.2.4.3.1.1.1 Transport channel parameter of SRB for PCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		RRC
RLC	Logical channel type		PCCH
	RLC mode		TM
	Payload sizes, bit		240 (alt. 80)
	Max data rate, bps		24 000 (alt. 8 000)
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		PCH
	TB sizes, bit		240 (alt. 80)
	TFS	TF0, bts	0x240 (alt. 0x80)
		TF1, bits	1x240 (alt. 1x80)
	TTI, ms		10
	Coding type		CC 1/2
	CRC, bit		16

	Max number of bits/TTI before rate matching	528 (alt. 208)
	RM attribute	210 to 250

6.10.2.4.3.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for PCCH = TF0, TF1

6.10.2.4.3.1.2 Physical channel parameters

SCCPCH	TFCS size	2
	DTX position	Fixed
	Spreading factor	128 (alt. 256)
	Number of TFCI bits/slot	0
	Number of Pilot bits/slot	0
	Number of data bits/slot	40 (alt. 20)
	Number of data bits/frame	600 (alt. 300)

6.10.2.4.3.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.2.1 Transport channel parameters

6.10.2.4.3.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		Interactive/ Background RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		24
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		360
	TFS	TF0, bits	0x360
		TF1, bits	1x360
	TTI, ms		10
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		1 140
	RM attribute		110 to 150

6.10.2.4.3.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH	BCCH
	RLC mode	UM	UM	AM	AM	AM	TM
	Payload sizes, bit	152	136 or 120 (note)	128	128	128	166
	Max data rate, bps	30 400 (alt. 45 600)	27 200 or 24 000 (alt. 40 800 or 36 000)	25 600 (alt. 38 400)	25 600 (alt. 38 400)	25 600 (alt. 38 400)	33 200 (alt. 49 800)
	AMD/UMD/TrD PDU header, bit	8	8	16	16	16	0
MAC	MAC header, bit	8	24 or 40	24	24	24	2

Higher layer	RAB/signalling RB		SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer		RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
	MAC multiplexing		6 logical channel multiplexing					
Layer 1	TrCH type		FACH					
	TB sizes, bit		168					
	TFS	TF0, bits	0x168					
		TF1, bits	1x168					
		TF2, bits	2x168					
		TF3, bits	N/A (alt. 3x168)					
	TTI, ms		10					
	Coding type		CC 1/2					
	CRC, bit		16					
	Max number of bits/TTI before rate matching		752 (alt. 1 136)					
	RM attribute		200 to 240					
NOTE: MAC header size and PLC payload size depend on use of U-RNTI or C-RNTI.								

6.10.2.4.3.2.1.3 TFCS

TFCS size	4 or 5, (alt. 4, 5 or 6)
TFCS	(SRBs for CCCH/DCCH/BCCH, 32kbps RAB) = (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), [TF1, TF1] (note) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), [TF3, TF0] (note), (TF0, TF1), [TF1, TF1] (note))
NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF2, TF0).	

6.10.2.4.3.2.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	64
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	72
	Number of data bits/frame	1 080

6.10.2.4.3.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.2a.1 Transport channel parameters

6.10.2.4.3.2a.1.1 Transport channel parameters for Interactive or background / 32 kbps / PS RAB + 32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		32 000	32 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		24	24
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		FACH	
	TB sizes, bit		360	
	TFS	TF0, bits	0x360	
		TF1, bits	1x360	
	TTI, ms		10	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 140	
	RM attribute		110 to 150	

6.10.2.4.3.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.2.4.3.2.1.2

6.10.2.4.3.2a.1.3 TFCS

TFCS size	4 or 5 (alt. 4, 5 or 6)
TFCS	(SRBs for CCCH/DCCH/BCCH, 32kbps RAB + 32kbps RAB) = (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), [TF1, TF1] (note) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), [TF3, TF0] (note), (TF0, TF1), [TF1, TF1] (note))
NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF2, TF0).	

6.10.2.4.3.2a.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	64
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	72
	Number of data bits/frame	1 080

6.10.2.4.3.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.3.1 Transport channel parameters

6.10.2.4.3.3.1.1 Transport channel parameters of SRB for Interactive/Background 32 kbps RAB

See clause 6.10.2.4.3.2.1.

6.10.2.4.3.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.2.4.3.1.1.

6.10.2.4.3.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.2.4.3.2.1.2

6.10.2.4.3.3.1.4 TFCS

TFCS size	6, 7 or 8 for 240 bits PCH TrBlk size and TF3 not used (alt 6, 7, 8 or 9 for 80 bits PCH TrBlk size and TF3 not used) (alt 6, 7, 8 or 9 for 240 bits PCH TrBlk size and TF3 used) (alt. 6, 7, 8, 9, 10, or 11 for 80 bits PCH TrBlk size and TF3 used)
TFCS	(SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH, 32 kbps RAB) = (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), (TF0, TF0, TF1), [TF0, TF1, TF1] (see note) for 240 bits PCH TrBlk size and TF3 not used (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), (TF0, TF0, TF1), [TF1, TF0, TF1] (see note), [TF0, TF1, TF1] (see note) for 80 bits PCH TrBlk size and TF3 not used) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), [TF0, TF3, TF0] (see note), (TF0, TF0, TF1), [TF0, TF1, TF1] (see note) for 240 bits PCH TrBlk size and TF3 used) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), [TF0, TF3, TF0] (see note), [TF1, TF3, TF0] (see note), (TF0, TF0, TF1), [TF1, TF0, TF1] (see note), [TF0, TF1, TF1] (see note) for 80 bits PCH TrBlk size and TF3 used)
NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF0, TF2, TF0).	

6.10.2.4.3.3.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	64

	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	72
	Number of data bits/frame	1 080

6.10.2.4.3.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.10.2.4.3.4.1 Transport channel parameters

6.10.2.4.3.4.1.1 Transport channel parameters of RB for CTCH

Higher layer	RAB/signalling RB		N/A
	User of Radio Bearer		BMC
RLC	Logical channel type		CTCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		15 200
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		168
	TFS	TF0, bts	0x168
		TF1, bits	1x168
	TTI, ms		10
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		576
	RM attribute		200 to 240

6.10.2.4.3.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	BCCH
	RLC mode		UM	TM
	Payload sizes, bit		152	166
	Max data rate, bps		15 200	16 600
	AMD/UMD/TrD PDU header, bit		8	0
MAC	MAC header, bit		8	2
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		FACH	
	TB sizes, bit		168	
	TFS	TF0, bits	0x168	
		TF1, bits	1x168	
	TTI, ms		10	
	Coding type		CC 1/3	
	CRC, bit		16	
	Max number of bits/TTI before rate matching		576	
	RM attribute		200 to 240	

6.10.2.4.3.4.1.3 TFCS

TFCS size	3
TFCS	(SRBs for CCCH/ BCCH, RB for CTCH) = (TF0, TF0), (TF1, TF0), (TF0, TF1)

6.10.2.4.3.4.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	128
	Number of TFCI bits/slot	2
	Number of Pilot bits/slot	0
	Number of data bits/slot	38
	Number of data bits/frame	570

6.10.2.4.3.5 64.8kbps RB for MTCH with 80 ms TTI

6.10.2.4.3.5.1 Transport channel parameters

6.10.2.4.3.5.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		64800
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		16344
	RM attribute		160

6.10.2.4.3.5.1.2 TFCS

TFCS size	9
TFCS	64 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8

6.10.2.4.3.5.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	32
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	152
	Number of data bits/frame	2280

6.10.2.4.3.6 129.6 kbps RB for MTCH with 80 ms TTI

6.10.2.4.3.6.1 Transport channel parameters

6.10.2.4.3.6.1.1 Transport channel parameters for 128 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		32679
	RM attribute		160

6.10.2.4.3.6.1.2 TFCS

TFCS size	17
TFCS	128 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.10.2.4.3.6.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	16
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	312
	Number of data bits/frame	4680

6.10.2.4.3.7 259.2 kbps RB for MTCH with 40 ms TTI

6.10.2.4.3.7.1 Transport channel parameters

6.10.2.4.3.7.1.1 Transport channel parameters for 256 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		259200
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		32679
	RM attribute		160

6.10.2.4.3.7.1.2 TFCS

TFCS size	17
TFCS	256 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.10.2.4.3.7.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	8
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	632
	Number of data bits/frame	9480

6.10.2.4.3.8 7.6 kbps signalling RB for MCCH

6.10.2.4.3.8.1 Transport channel parameters

6.10.2.4.3.8.1.1 Transport channel parameters for 7.6 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		7600
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		160
	TFS	TF0, bits	0x160
		TF1, bits	1x160
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		552
	RM attribute		160

6.10.2.4.3.8.1.2 TFCS

TFCS size	2
TFCS	MBMS SRB =TF0, TF1

6.10.2.4.3.8.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	256
	Number of TFCI bits/slot	2
	Number of Pilot bits/slot	0
	Number of data bits/slot	18
	Number of data bits/frame	270

6.10.2.4.3.9 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH + SRB for MCCH

6.10.2.4.3.9.1 Transport channel parameters

6.10.2.4.3.9.1.1 Transport channel parameters of SRB for Interactive/Background 32 kbps RAB

See clause 6.10.2.4.3.2.1.1

6.10.2.4.3.9.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.2.4.3.1.1.1

6.10.2.4.3.9.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.2.4.3.2.1.2

6.10.2.4.3.9.1.4 Transport channel parameters of SRBs for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		7600
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		160
	TFS	TF0, bits	0x160
		TF1, bits	1x160
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		552
	RM attribute		215 to 235

6.10.2.4.3.9.1.4 TFCS

TFCS size	12 or 13 for 240 bits PCH TrBlk size and TF3 not used
TFCS	(SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH, 32 kbps RAB, SRB for MCCH) = (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF0, TF1, TF0, TF0), (TF1, TF1, TF0, TF0), (TF0, TF2, TF0, TF0), (TF0, TF0, TF1, TF0), (TF0, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF0, TF1, TF0, TF1), [TF1, TF1, TF0, TF1] (see note), (TF0, TF2, TF0, TF1), (TF0, TF0, TF1, TF1) for 240 bits PCH TrBlk size and TF3 not used
NOTE: Some TFC's are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC (TF0, TF2, TF0, TF0).	

6.10.2.4.3.9.2 Physical channel parameters

SCCPCH	DTX position	Flexible
	Spreading factor	64
	Number of TFCI bits/slot	8
	Number of Pilot bits/slot	0
	Number of data bits/slot	72
	Number of data bits/frame	1080

6.10.2.4.4 Combinations on PRACH

6.10.2.4.4.1 Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.2.4.4.1.1 Transport channel parameters

6.10.2.4.4.1.1.1 Transport channel parameter for Interactive/Background 32 kbps PS RAB, SRB for CCCH, SRB for DCCH

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	320	166/238 (Rel6, see Note)	136	128	128	128

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
	Max data rate, bps	16 000 (alt. 32 000)	8 300 /11900 Rel 6 (alt.16 600/23800 Rel6, see Note)	6 800 (alt.13 600)	6 400 (alt.12 800)	6 400 (alt.12 800)	6 400 (alt.12 800)
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16
MAC	MAC header, bit	24	2	24	24	24	24
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	360	168/240 (Rel6, see Note)	168	168	168	168
	TFS TF0, bits	1x168					
	TF1, bits	1x360					
	TF2, bits (Rel 6, see Note)	1x240					
	TTI, ms	20 (alt. 10)					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	768	384/528 (Rel 6, see Note)	384	384	384	384
	Max number of bits/ Radio frame before rate matching	384 (alt. 768)	192/264 Rel 6 (alt. 384/528 Rel 6, see Note)	192 (alt. 384)	192 (alt. 384)	192 (alt. 384)	192 (alt. 384)

6.10.2.4.4.1.1.2 TFCS

TFCS size	2, 3 (in Rel 6 , see Note)
TFCS	32 kbps + SRBs for CCCH/ DCCH = TF0, TF1, TF2 (in Rel 6 , see Note)

6.10.2.4.4.1.2 Physical channel parameters

PRACH	Minimum Spreading factor	64 (alt. 32)
	Max number of data bits/radio frame	600 (alt. 1 200)
	Puncturing Limit	1

NOTE: In Release 6 UEs shall use the TF/TFC as indicated in the IE "Additional Dynamic Transport Format Information for CCCH" and the IE "Additional RACH TFCS for CCCH" for CCCH if available. In this configuration the indicated TF / TFC will be transmitted in these IEs.

6.10.2.4.4.2 Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.2.4.4.2.1 Transport channel parameters

6.10.2.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB, Interactive/Background 32 kbps PS RAB, SRB for CCCH, SRB for DCCH

Higher layer	RAB/signalling RB	RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type	DTCH	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH

Higher layer	RAB/signalling RB		RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
	RLC mode		AM	AM	TM	UM	AM	AM	AM
	Payload sizes, bit		320	320	166/238 (Rel6, see Note)	136	128	128	128
	Max data rate, bps		16 000 (alt. 32 000)	16 000 (alt. 32 000)	83 00 /11900 Rel 6 (alt. 16 600/23800 Rel6, see Note)	6 800 (alt. 13 600)	6 400 (alt. 12 800)	6 400 (alt. 12 800)	6 400 (alt.12 800)
	AMD/UMD/TrD PDU header, bit		16	16	0	8	16	16	16
MAC	MAC header, bit		24	24	2	24	24	24	24
	MAC multiplexing		7 logical channel multiplexing						
Layer 1	TrCH type		RACH						
	TB sizes, bit		360	360	168/240 (Rel6, see Note)	168	168	168	168
	TFS	TF0, bits	1x168						
		TF1, bits	1x360						
		TF2, bits	1x240						
	TTI, ms		20 (alt. 10)						
	Coding type		CC 1/2						
	CRC, bit		16						
	Max number of bits/TTI after channel coding		768	768	384/528 (Rel 6, see Note)	384	384	384	384
	Max number of bits/ Radio frame before rate matching		384 (alt. 768)	384 (alt 768)	192/264 (alt. 384/528 Rel 6, see Note)	192 (alt. 384)	192 (alt. 384)	192 (alt. 384)	192 (alt. 384)

6.10.2.4.4.2.1.2 TFCS

TFCS size	2, 3 (in Rel 6 , see Note)
TFCS	32 kbps RAB+ 32 kbps RAB + SRBs for CCCH/ DCCH = TF0, TF1, TF2 (in Rel 6, see Note)

6.10.2.4.4.2.2 Physical channel parameters

PRACH	Minimum Spreading factor	64 (alt. 32)
	Max number of data bits/radio frame	600 (alt. 1 200)
	Puncturing Limit	1

NOTE: In Release 6 UEs shall use the TF/TFC as indicated in the IE "Additional Dynamic Transport Format Information for CCCH" and the IE "Additional RACH TFCS for CCCH" for CCCH if available. In this configuration the indicated TF / TFC will be transmitted in these IEs.

6.10.2.4.4.3 Interactive/Background / UL:32 DL: [max bit rate depending on UE category] with fixed RLC and MAC-ehs / PS RAB + SRBs for DCCH on RACH and SRB with fixed RLC and MAC-ehs on HS-DSCH / DL:QPSK

6.10.2.4.4.3.1 Uplink

6.10.2.4.4.3.1.1 Transport channel parameters

6.10.2.4.4.3.1.1.1 Transport channel for Interactive/Background / UL: 32 kbps / PS RAB + SRBs for DCCH

See clause 6.10.2.4.4.1.1.1

6.10.2.4.4.3.1.1.2 TFCS

See clause 6.10.2.4.4.1.1.2

6.10.2.4.4.3.1.2 Physical channel parameters

See clause 6.10.2.4.4.1.2

6.10.2.4.4.3.2 Downlink

6.10.2.4.4.3.2.1 Transport channel parameters

6.10.2.4.4.3.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.4.3.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

		Fixed RLC + MAC-ehs (Rel-7 and later releases)
Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	depends on UE category (NOTE 1)
	AMD PDU header, bit	16
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-hs Type	MAC-ehs
	MAC-hs/MAC-ehs header fixed part, bit	24
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24
	Applicable modulation schemes	QPSK, 16QAM
NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-ehs PDU (see 3GPP TS 25.321 [38]).		

6.10.2.4.4.3.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2, alt 2.

6.10.2.4.4.3.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.4.3.2.2.1 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5 Combinations on DPCH and HS-PDSCH

6.10.2.4.5.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.1.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.5.1.2 Downlink

6.10.2.4.5.1.2.1 Transport channel parameters

6.10.2.4.5.1.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.1.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2	**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	AM		
	Payload sizes, bit	320 (alt. 640)	320 (alt. 640)	Flexible up to 12000 (NOTE 3)
	Max data rate, bps	depends on UE category NOTE 1		
	AMD PDU header, bit	16	16	16
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	336 (alt. 656)	336 (alt. 656)	Flexible
	MAC-hs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs/MAC-ehs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms	2 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes

NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see 3GPP TS 25.321 [38]).
NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter “Max SDU size”, ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.5.1.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.1.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.1.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.1.2.2 Physical channel parameters

6.10.2.4.5.1.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.1.2.2.2 Physical channel parameters on HS-PDSCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE HS-DSCH Physical Layer category 1 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	2, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.2 Mbps, (alt. 400 kbps)

UE HS-DSCH Physical Layer category 2 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	2, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.2 Mbps, (alt. 600 kbps)]

UE HS-DSCH Physical Layer category 3 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	3, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.8 Mbps, (alt. 900 kbps)

UE HS-DSCH Physical Layer category 4 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	3, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.8 Mbps, (alt. 1.2 Mbps)

UE HS-DSCH Physical Layer category 5 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	3.65 Mbps, (alt. 3.6 Mbps)

UE HS-DSCH Physical Layer category 6 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	3.65 Mbps, (alt. 3.65 Mbps)

UE HS-DSCH Physical Layer category 7 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	7.2 Mbps, (alt. 7.2 Mbps)

UE HS-DSCH Physical Layer category 8 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	7.2 Mbps, (alt. 7.2 Mbps)

UE HS-DSCH Physical Layer category 9 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	10.1 Mbps, (alt. 10.1 Mbps)

UE HS-DSCH Physical Layer category 10 (Rel-5 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	14.0 Mbps, (alt. 10.8 Mbps)

UE HS-DSCH Physical Layer category 11 (Rel-5 and later releases; QPSK):

HS-PDSCH	Number of processes	3, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	900 kbps, (alt. 450 kbps)

UE HS-DSCH Physical Layer category 12 (Rel-5 and later releases; QPSK):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.8 Mbps, (alt. 1.8 Mbps)

UE HS-DSCH Physical Layer category 13 (Rel-7 and later releases; QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	17.6 Mbps, (alt. 16.2 Mbps)

UE HS-DSCH Physical Layer category 14 (Rel-7 and later releases; QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	21.1 Mbps, (alt. 16.2 Mbps)

UE HS-DSCH Physical Layer category 15 (Rel-7 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	11.7 Mbps, (alt. 11.7Mbps)

UE HS-DSCH Physical Layer category 15 (Rel-7 and later releases; MIMO + QPSK or 16QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes
	Max Data Rate	23.4 Mbps, (alt. 21.6 Mbps)

UE HS-DSCH Physical Layer category 16 (Rel-7 and later releases; QPSK or 16QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	14.0. Mbps, (alt. 14.0 Mbps)

UE HS-DSCH Physical Layer category 16 (Rel-7 and later releases; MIMO + QPSK or 16QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes

	Max Data Rate	28,0 Mbps, (alt. 21.6 Mbps)

UE HS-DSCH Physical Layer category 17 (Rel-7 and later releases; QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	17.6 Mbps, (alt. 16.2 Mbps)

UE HS-DSCH Physical Layer category 17 (Rel-7 and later releases; MIMO + QPSK or 16QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes
	Max Data Rate	23,4 Mbps, (alt. 21.6 Mbps)

UE HS-DSCH Physical Layer category 18 (Rel-7 and later releases; QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	21.1 Mbps, (alt. 16.2 Mbps)

UE HS-DSCH Physical Layer category 18 (Rel-7 and later releases; MIMO + QPSK or 16QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes
	Max Data Rate	28.0 Mbps, (alt. 21.6 Mbps)

UE HS-DSCH Physical Layer category 19 (Rel-8 and later releases; MIMO + QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes
	Max Data Rate	35.3 Mbps, (alt. 32.4 Mbps)

UE HS-DSCH Physical Layer category 20 (Rel-8 and later releases; MIMO + QPSK, 16QAM or 64QAM):

HS-PDSCH	Number of processes	12, (alt. 16)
	Process memory size	Split equally among all processes
	Max Data Rate	42.2 Mbps, (alt. 32.4 Mbps)

6.10.2.4.5.1a Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.1a.1 Uplink

See clause 6.10.2.4.1.28.1.

6.10.2.4.5.1a.2 Downlink

6.10.2.4.5.1a.2.1 Transport channel parameters

6.10.2.4.5.1a.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.1a.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.1a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.1a.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.1a.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.1a.2.2 Physical channel parameters

6.10.2.4.5.1a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.1a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.2 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.2.1 Uplink

See clause 6.10.2.4.1.34.1.

6.10.2.4.5.2.2 Downlink

6.10.2.4.5.2.2.1 Transport channel parameters

6.10.2.4.5.2.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.2.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.2.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.2.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.2.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.2.2.2 Physical channel parameters

6.10.2.4.5.2.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.2.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.3.1 Uplink

6.10.2.4.5.3.1.1 Transport channel parameters

6.10.2.4.5.3.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.5.3.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.2.4.1.34.1.1.1.

6.10.2.4.5.3.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.3.1.1.4 TFCS

TFCS size	54 (alt. 36)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1))

6.10.2.4.5.3.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Number of DPDCH	1
	Puncturing Limit	0.60

6.10.2.4.5.3.2 Downlink

6.10.2.4.5.3.2.1 Transport channel parameters

6.10.2.4.5.3.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.3.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.3.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.3.2.1.2.2 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.3.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.3.2.2 Physical channel parameters

6.10.2.4.5.3.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.4.2.2.

6.10.2.4.5.3.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.3a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.3a.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.5.3a.2 Downlink

6.10.2.4.5.3a.2.1 Transport channel parameters

6.10.2.4.5.3a.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.

6.10.2.4.5.3a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.3a.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.3a.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.3a.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.3a.2.2 Physical channel parameters

6.10.2.4.5.3a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.4.2.2.

6.10.2.4.5.3a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.4 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.4.1 Uplink

6.10.2.4.5.4.1.1 Transport channel parameters

6.10.2.4.5.4.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.5.4.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.2.4.1.34.1.1.1.

6.10.2.4.5.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.4.1.1.4 TFCS

TFCS size	36 (alt. 24)
TFCS	(64 kbps RAB, 384 kbps RAB , DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF6, TF0), (TF1, TF6, TF0), (TF0, TF7, TF0), (TF1, TF7, TF0), (TF0, TF8, TF0), (TF1, TF8, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1), (TF0, TF6, TF1), (TF1, TF6, TF1), (TF0, TF7, TF1), (TF1, TF7, TF1), (TF0, TF8, TF1), (TF1, TF8, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1))

6.10.2.4.5.4.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Number of DPDCH	1
	Puncturing Limit	0.52

6.10.2.4.5.4.2 Downlink

6.10.2.4.5.4.2.1 Transport channel parameters

6.10.2.4.5.4.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.4.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.4.2.1.2.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.5.4.2.1.2.2 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.4.2.1.2.3 TFCS

See clause 6.10.2.4.1.13.2.1.3.

6.10.2.4.5.4.2.2 Physical channel parameters

6.10.2.4.5.4.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.13.2.2.

6.10.2.4.5.4.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.4a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.4a.1 Uplink

See clause 6.10.2.4.1.51.1.

6.10.2.4.5.4a.2 Downlink

6.10.2.4.5.4a.2.1 Transport channel parameters

6.10.2.4.5.4a.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.

6.10.2.4.5.4a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.4a.2.1.2.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.5.4a.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.4a.2.1.2.3 TFCS

See clause 6.10.2.4.1.13.2.1.3.

6.10.2.4.5.4a.2.2 Physical channel parameters

6.10.2.4.5.4a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.13.2.2.

6.10.2.4.5.4a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.5 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH6.10.2.4.5.5.1 Uplink

6.10.2.4.5.5.1.1 Transport channel parameters

6.10.2.4.5.5.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB + UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		384 000	384 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	4x340	
		TF4, bits	8x340	
		TF5, bits	12x340	
	TTI, ms		10	
	Coding type		TC	

	CRC, bit	16
	Max number of bits/TTI after channel coding	12 828
	Uplink: Max number of bits/radio frame before rate matching	12 828
	RM attribute	110-180

6.10.2.4.5.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.5.1.1.3 TFCS

TFCS size	12
TFCS	(384 kbps RAB + 384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)

6.10.2.4.5.5.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Number of DPDCH	1
	Puncturing Limit	0.64

6.10.2.4.5.5.2 Downlink

6.10.2.4.5.5.2.1 Transport channel parameters

6.10.2.4.5.5.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.5.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.5.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.5.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.5.2.2 Physical channel parameters

6.10.2.4.5.5.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.5.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.5a Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.5a.1 Uplink

See clause 6.10.2.4.1.57.1.

6.10.2.4.5.5a.2 Downlink

6.10.2.4.5.5a.2.1 Transport channel parameters

6.10.2.4.5.5a.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.5a.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5a.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.5a.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.5a.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.5a.2.2 Physical channel parameters

6.10.2.4.5.5a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.5a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.6 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.6.1 Uplink

6.10.2.4.5.6.1.1 Transport channel parameters

6.10.2.4.5.6.1.1.1 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

Higher Layer	RAB/Signalling RB		**RAB**
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		128000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8076
	Uplink: Max number of bits/radio frame before rate matching		4038
	RM attribute		125-165

6.10.2.4.1.6.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.6.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.6.1.1.4 TFCS

TFCS size	40
TFCS	(128 kbps RAB, 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF2, TF2, TF0), (TF3, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF2, TF3, TF0), (TF3, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF2, TF4, TF0), (TF3, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF2, TF2, TF1), (TF3, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF2, TF3, TF1), (TF3, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF2, TF4, TF1), (TF3, TF4, TF1)

6.10.2.4.1.6.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9600
	Puncturing Limit	0.96

6.10.2.4.5.6.2 Downlink

6.10.2.4.5.6.2.1 Transport channel parameters

6.10.2.4.5.6.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.6.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2	**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	AM		
	Payload sizes, bit	640 (alt. 320)	640 (alt. 320)	Flexible up to 12000 (NOTE 3)
	Max data rate, bps	depends on UE category NOTE1		
	AMD PDU header, bit	16	16	16
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	656 (alt. 336)	656 (alt. 336)	Flexible
	MAC-hs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs/MAC-ehs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms	2 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes

NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see 3GPP TS 25.321 [38]).

NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitly stated in the test case.

NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter “Max SDU size”, ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.5.6.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.6.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.6.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.6.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.6.2.2 Physical channel parameters

6.10.2.4.5.6.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.7.1 Uplink

6.10.2.4.5.7.1.1 Transport channel parameters

6.10.2.4.5.7.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.5.7.1.1.2 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

See clause 6.10.2.4.5.6.1.1.1.

6.10.2.4.1.7.1.1.3 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.7.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7.1.1.5 TFCS

TFCS size	62
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1,TF0), (TF0,TF0,TF0,TF2,TF1,TF0), (TF1,TF0,TF0,TF2,TF1,TF0), (TF2,TF1,TF1,TF2,TF1,TF0), (TF0,TF0,TF0,TF3,TF1,TF0), (TF1,TF0,TF0,TF3,TF1,TF0), (TF2,TF1,TF1,TF3,TF1,TF0), (TF0,TF0,TF0,TF2,TF2,TF0), (TF1,TF0,TF0,TF2,TF2,TF0), (TF2,TF1,TF1,TF2,TF2,TF0), (TF0,TF0,TF0,TF3,TF2,TF0), (TF1,TF0,TF0,TF3,TF2,TF0), (TF2,TF1,TF1,TF3,TF2,TF0), (TF0,TF0,TF0,TF1,TF3,TF0), (TF1,TF0,TF0,TF1,TF3,TF0), (TF2,TF1,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF2,TF3,TF0), (TF1,TF0,TF0,TF2,TF3,TF0), (TF2,TF1,TF1,TF2,TF3,TF0), (TF0,TF0,TF0,TF3,TF3,TF0), (TF1,TF0,TF0,TF3,TF3,TF0), (TF2,TF1,TF1,TF3,TF3,TF0), (TF0,TF0,TF0,TF2,TF4,TF0), (TF1,TF0,TF0,TF2,TF4,TF0), (TF2,TF1,TF1,TF2,TF4,TF0), (TF0,TF0,TF0,TF3,TF4,TF0), (TF1,TF0,TF0,TF3,TF4,TF0), (TF2,TF1,TF1,TF3,TF4,TF0), (TF0,TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF0,TF1), (TF0,TF0,TF0,TF2,TF1,TF1), (TF1,TF0,TF0,TF2,TF1,TF1), (TF2,TF1,TF1,TF2,TF1,TF1), (TF0,TF0,TF0,TF3,TF1,TF1), (TF1,TF0,TF0,TF3,TF1,TF1), (TF2,TF1,TF1,TF3,TF1,TF1), (TF0,TF0,TF0,TF2,TF2,TF1), (TF1,TF0,TF0,TF2,TF2,TF1), (TF2,TF1,TF1,TF2,TF2,TF1), (TF0,TF0,TF0,TF3,TF2,TF1), (TF1,TF0,TF0,TF3,TF2,TF1), (TF2,TF1,TF1,TF3,TF2,TF1), (TF0,TF0,TF0,TF1,TF3,TF1), (TF1,TF0,TF0,TF1,TF3,TF1), (TF2,TF1,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF2,TF3,TF1), (TF1,TF0,TF0,TF2,TF3,TF1), (TF2,TF1,TF1,TF2,TF3,TF1), (TF0,TF0,TF0,TF3,TF3,TF1), (TF1,TF0,TF0,TF3,TF3,TF1), (TF2,TF1,TF1,TF3,TF3,TF1), (TF0,TF0,TF0,TF2,TF4,TF1), (TF1,TF0,TF0,TF2,TF4,TF1), (TF2,TF1,TF1,TF2,TF4,TF1), (TF0,TF0,TF0,TF3,TF4,TF1), (TF1,TF0,TF0,TF3,TF4,TF1), (TF2,TF1,TF1,TF3,TF4,TF1)

6.10.2.4.1.7.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9600
	Puncturing Limit	0.88

6.10.2.4.5.7.2 Downlink

6.10.2.4.5.7.2.1 Transport channel parameters

6.10.2.4.5.7.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.7.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

See clause 6.10.2.4.5.6.2.1.1.1.

6.10.2.4.5.7.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.7.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.7.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.7.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.7.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.7.2.2 Physical channel parameters

6.10.2.4.5.7.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.7.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Interactive or Background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.5.8.1 Uplink

6.10.2.4.5.8.1.1 Transport channel parameters

6.10.2.4.5.8.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.65 8.85 6.6) kbps / CS RAB

See clause 6.10.2.4.1.62.1.1.1

6.10.2.4.5.8.1.1.2 Transport Channel parameters for Interactive or background / UL: 384 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	384 000

	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
	TTI, ms		10
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684
	Uplink: Max number of bits/radio frame before rate matching		12 684
	RM attribute		110 to 180

6.10.2.4.5.8.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.8.1.1.4 TFCS

TFCS size	60
TFCS	((RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0), (TF3,TF2,TF0,TF5,TF0), (TF4,TF3,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1), (TF4,TF3,TF0,TF5,TF1)

6.10.2.4.5.8.1.1.5 TFC subset list

TFC subset list size	3
TFC subset list	0 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1),}, 1 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0), (TF3,TF2,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1)}, 2 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0),(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0), (TF3,TF2,TF0,TF5,TF0), (TF4,TF3,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1), (TF4,TF3,TF0,TF5,TF1)}

6.10.2.4.5.8.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	4
	Max number of DPDCH data bits/radio frame	9 600
	Number of DPDCH	1
	Puncturing Limit	0.60

6.10.2.4.5.8.2 Downlink

6.10.2.4.5.8.2.1 Transport channel parameters

6.10.2.4.5.8.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1

6.10.2.4.5.8.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.8.2.1.2.1 Transport channel parameters for Conversational / speech / DL: (12.65 8.85 6.6) kbps / CS RAB

See clause 6.10.2.4.1.62.2.1.1

6.10.2.4.5.8.2.1.2.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.5.8.2.1.2.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

See clause 6.10.2.4.1.62.2.1.3

6.10.2.4.5.8.2.1.2.4 TFCS

See clause 6.10.2.4.1.62.2.1.4

6.10.2.4.5.8.2.2 Physical channel parameters

6.10.2.4.5.8.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.62.2.2

6.10.2.4.5.8.2.2.2 Physical Channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.1

6.10.2.4.5.9 Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.9.1 Uplink

6.10.2.4.5.9.1.1 Transport channel parameters

6.10.2.4.5.9.1.1.1 Transport channel parameters for Streaming MBMS PTP / unknown / UL:16 kbps / PS RAB

Higher Layer	RAB/Signalling RB		**RAB**
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1068
	Uplink: Max number of bits/radio frame before rate matching		534
	RM attribute		135-175

6.10.2.4.5.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.9.1.1.3 TFCS

TFCS size	4
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.10.2.4.5.9.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	0.80

6.10.2.4.5.9.2 Downlink

6.10.2.4.5.9.2.1 Transport channel parameters

6.10.2.4.5.9.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.9.2.1.1.1 MAC-d flow parameters for Streaming MBMS PTP / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2	**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	AM		
	Payload sizes, bit	640	640	Flexible up to 12000 (NOTE 3)
	Max data rate, bps	depends on UE category NOTE 1		
	AMD PDU header, bit	16	16	16
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	656	656	Flexible
	MAC-hs/MAC-ehs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms	2 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes

NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see 3GPP TS 25.321 [38]).
NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter "Max SDU size", ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.5.9.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.9.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.9.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.9.2.2 Physical channel parameters

6.10.2.4.5.9.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.9.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.10 Streaming MBMS PTP / unknown / UL:16 DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.10.1 Uplink

6.10.2.4.5.10.1.1 Transport channel parameters

6.10.2.4.5.10.1.1.1 Transport channel parameters for Streaming MBMS PTP / unknown / UL:16 kbps / PS RAB

See clause 6.10.2.4.5.9.1.1.1.

6.10.2.4.5.10.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.10.2.4.1.57.1.1.1.

6.10.2.4.5.10.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.10.1.1.4 TFCS

TFCS size	20
TFCS	(16 kbps RAB, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1)

6.10.2.4.5.10.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2400
	Puncturing Limit	0.68

6.10.2.4.5.10.2 Downlink

6.10.2.4.5.10.2.1 Transport channel parameters

6.10.2.4.5.10.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.10.2.1.1.1 MAC-d flow parameters for Streaming MBMS PTP / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

See clause 6.10.2.4.5.9.2.1.1.1.

6.10.2.4.5.10.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.10.2.1.1.3 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.10.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.10.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.10.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.10.2.2 Physical channel parameters

6.10.2.4.5.10.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.10.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6 Combinations on HS-PDSCH and E-DPDCH

6.10.2.4.6.1a Stand-alone UL: [max bit rate depending on UE category and TTI] DL:[max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.2.4.6.1a.1 Uplink

6.10.2.4.6.1a.1.1 Transport channel parameters

6.10.2.4.6.1a.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.1a.1.1.1.1 MAC-d flow parameters for Stand-alone UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2

6.10.2.4.6.1a.1.2 Physical channel parameters

6.10.2.4.6.1a.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.1a.2 Downlink

6.10.2.4.6.1a.2.1 Transport channel parameters

6.10.2.4.6.1a.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.1a.2.1.1.1 MAC-d flow parameters for Stand-alone DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2

6.10.2.4.6.1a.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.1a.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.10.2.4.6.1.1 Uplink

6.10.2.4.6.1.1.1 Transport channel parameters

6.10.2.4.6.1.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later) NOTE 2	Alt 2 Fixed RLC + MAC-i/is (Rel-8 and later releases) NOTE 2	Alt 3 Flexible RLC + MAC-i/is (Rel-8 and later releases) NOTE 2
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	AM		
	Payload sizes, bit	320	320	Flexible from 80 up to 12000 (NOTE 3)
	Max data rate, bps	Depends on UE category and TTI		
	AMD PDU header, bit	16		
MAC	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	336	336	Flexible from 96 up to 12016
	MAC type	MAC-e/es	MAC-i/is	MAC-i/is
	MAC-e/es / MAC-i/is header fixed part, bit	18	24	24
Layer 1	TrCH type	E-DCH		
	TTI	10ms (alt. 2ms) (NOTE 1)		
	Coding type	TC		
	CRC, bit	24		

NOTE 1: The support of 2ms TTI depends on the UE category
NOTE 2: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-i/is) or 3 (Flexible RLC + MAC-i/is) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter "Max SDU size", ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.1.1.1.2 Transport channel parameters for DCH

6.10.2.4.6.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.6.1.1.2 Physical channel parameters

6.10.2.4.6.1.1.2.1 Physical channel parameters on E-DPDCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE E-DPDCH Physical Layer category 1:

E-DPDCH	Number of processes	4
	TTI	10 ms
	Max Data Rate	0.711 Mbps

UE E-DPDCH Physical Layer category 2:

E-DPDCH	Number of processes	4	8
	TTI	10 ms	2 ms
	Max Data Rate	1.4484 Mbps	1.399 Mbps

UE E-DPDCH Physical Layer category 3:

E-DPDCH	Number of processes	4

	TTI	10 ms
	Max Data Rate	1.4484 Mbps

UE E-DPDCH Physical Layer category 4:

E-DPDCH	Number of processes	4	8
	TTI	10 ms	2 ms
	Max Data Rate	2.0 Mbps	2.2886 Mbps

UE E-DPDCH Physical Layer category 5:

E-DPDCH	Number of processes	4
	TTI	10 ms
	Max Data Rate	2.0 Mbps

UE E-DPDCH Physical Layer category 6:

E-DPDCH	Number of processes	4	8
	TTI	10 ms	2 ms
	Max Data Rate	2.0 Mbps	5.742Mbps

UE E-DPDCH Physical Layer category 7:

E-DPDCH	Number of processes	4	8
	TTI	10 ms	2 ms
	Max Data Rate	2.0 Mbps	11.498 Mbps

6.10.2.4.6.1.1.2.2 Physical channel parameters for DPCH

See clause 6.10.2.4.1.2.1.2

6.10.2.4.6.1.2 Downlink

See clause 6.10.2.4.5.1.2.

6.10.2.4.6.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.10.2.4.6.2.1 Uplink

6.10.2.4.6.2.1.1 Transport channel parameters

6.10.2.4.6.2.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.2.1.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.2.1.1.1.2 MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later releases) NOTE 2				**Alt 2 Fixed RLC + MAC-i/is (Rel-8 and later releases) NOTE 2**			
Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH

	RLC mode	UM	AM	AM	AM	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128	136	128	128	128
	Max data rate, bps	Depends on UE category and TTI							
	AMD PDU header, bit	8	16	16	16	8	16	16	16
MAC	MAC-es multiplexing	4 logical channel multiplexing				4 logical channel multiplexing			
	MAC-d PDU size, bit	144				144			
	MAC type	MAC-e/es				MAC-i/is			
	MAC-e/es / MAC-i/is header fixed part, bit	18				24			
Layer 1	TrCH type	E-DCH				E-DCH			
	TTI	10ms (alt. 2ms) (NOTE 1)				10ms (alt. 2ms) (NOTE 1)			
	Coding type	TC				TC			
	CRC, bit	24				24			
NOTE 1: The support of 2ms TTI depends on the UE category. NOTE 2: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-i/is) then this shall be explicitly stated in the test case.									

6.10.2.4.6.2.1.2 Physical channel parameters

6.10.2.4.6.2.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.2.2 Downlink

See clause 6.10.2.4.5.1.2.

6.10.2.4.6.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.2.4.6.3.1 Uplink

See clause 6.10.2.4.6.1.1.

6.10.2.4.6.3.1.2 Physical channel parameters

6.10.2.4.6.3.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.3.2 Downlink

6.10.2.4.6.3.2.1 Transport channel parameters

6.10.2.4.6.3.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.3.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.6.3.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2				**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**			
Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128	136	128	128	128
	Max data rate, bps	Depends on UE category (NOTE 1)							
	AMD PDU header, bit	8	16	16	16	8	16	16	16
MAC	MAC-d header, bit	4	4	4	4	0			
	MAC multiplexing	4 logical channel multiplexing				N/A			
	MAC-d PDU size, bit	148				144			
	MAC-hs Type	MAC-hs				MAC-ehs			
	MAC-hs header fixed part, bit	21				24			
Layer 1	TrCH type	HS-DSCH				HS-DSCH			
	TTI	2 ms				2 ms			
	Coding type	TC				TC			
	CRC, bit	24				24			
	Applicable modulation schemes	QPSK, 16QAM				QPSK, 16QAM, 64QAM			
	Applicable with MIMO	No				Yes			

NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).

NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) then this shall be explicitly stated in the test case.

6.10.2.4.6.3.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.3.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.6.4.1 Uplink

6.10.2.4.6.4.1.1 Transport channel parameters

6.10.2.4.6.4.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.4.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.4.1.1.2 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.6.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.6.4.1.1.4 TFCS

See clause 6.10.2.4.1.4.1.1.3.

6.10.2.4.6.4.1.2 Physical channel parameters

6.10.2.4.6.4.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.4.1.2.2 Physical channel parameters on DCH

See clause 6.10.2.4.1.4.1.2.

6.10.2.4.6.4.2 Downlink

See clause 6.10.2.4.5.3.2.

6.10.2.4.6.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.10.2.4.6.5.1 Uplink

6.10.2.4.6.5.1.1 Transport channel parameters

6.10.2.4.6.5.1.1.1 Transport channel parameters for E-DCH

MAC-e multiplexing between all MAC-d flows in the same MAC-e PDU shall be configured.

6.10.2.4.6.5.1.1.1.1 MAC-d flow #1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.5.1.1.1.2 MAC-d flow #2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.5.1.1.1.3 MAC-d flow #3 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.5.1.2 Physical channel parameters

6.10.2.4.6.5.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.5.2 Downlink

See clause 6.10.2.4.5.6.2.

6.10.2.4.6.6 Conversational / unknown or speech / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

6.10.2.4.6.6.1 Uplink

6.10.2.4.6.6.1.1 Transport channel parameters

6.10.2.4.6.6.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.6.1.1.1.1 MAC-d flow #1 parameters for Conversational / unknown or speech / UL: [max bit rate depending on UE category and TTI] / PS RAB

Higher layer	RAB/Signalling RB	RAB
PDCP	PDCP header size, bit	0
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	88, 104, 136, 152, 168, 184, 200, 216, 280, 288, 304, 336 (alt 328)
	Max data rate, bps	Depends on UE category and TTI
	UMD PDU header, bit	8
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	96, 112 , 144, 160, 176, 192, 208, 224, 288, 296, 312, 344 (alt 336)
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	10ms (alt. 2ms) (NOTE)
	Coding type	TC
	CRC, bit	24
NOTE: The support of 2ms TTI depends on the UE category		

6.10.2.4.6.6.1.1.1.2 MAC-d flow #2 parameters for Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.6.1.1.1.3 MAC-d flow #3 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.6.1.2 Physical channel parameters

6.10.2.4.6.6.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.6.2 Downlink

6.10.2.4.6.6.2.1 Transport channel parameters

6.10.2.4.6.6.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.6.2.1.1.1 MAC-d flow#1 parameters for Conversational / unknown or speech / DL: [max bit rate depending on UE category] kbps / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2	**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**
Higher Layer	RAB/Signalling RB	**RAB**		
RLC	Logical channel type	DTCH		
	RLC mode	UM		
	Payload sizes, bit	104, 136, 152, 168, 184, 216, 288, 336 (alt 328)	104, 136, 152, 168, 184, 216, 288, 336 (alt 328)	Flexible up to 12000 (NOTE 3)
	Max data rate, bps	depends on UE category NOTE1		
	UMD PDU header, bit	8	8	8
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	112 , 144, 160, 176, 192, 224, 296, 344 (alt 336)	112 , 144, 160, 176, 192, 224, 296, 344 (alt 336)	Flexible
	MAC-hs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs/ehs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms	2 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes

NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).
NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter "Max SDU size", ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.6.2.1.1.2 MAC-d flow#2 parameters for Streaming or Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.6.6.2.1.1.3 MAC-d flow#3 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2.

6.10.2.4.6.6.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.6.2.2.1 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.7 Conversational / unknown or speech / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

6.10.2.4.6.7.1 Uplink

6.10.2.4.6.7.1.1 Transport channel parameters

6.10.2.4.6.7.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.7.1.1.1.1 MAC-d flow #1 parameters for Conversational / unknown or speech / UL: [max bit rate depending on UE category and TTI] / PS RAB

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later) NOTE 2	**Alt 2 Flexible RLC + MAC-i/is (Rel-8 and later releases) NOTE 2**
Higher layer	RAB/Signalling RB	RAB	
PDCP	PDCP header size, bit	0	
RLC	Logical channel type	DTCH	
	RLC mode	UM	
	Payload sizes, bit	88, 104, 136, 152, 168, 184, 200, 216, 280, 288, 304, 328 (alt 336)	Flexible from 88 up to 12000 (NOTE 3)
	Max data rate, bps	Depends on UE category and TTI	
	UMD PDU header, bit	8	
MAC	MAC multiplexing	N/A	
	MAC-d PDU size, bit	96, 112 , 144, 160, 176, 192, 208, 224, 288, 296, 312, 336 (alt 344)	Flexible from 96 up to 12008
	MAC type	MAC-e/es	MAC-i/is
	MAC-e/es / MAC-i/is header fixed part, bit	18	24
Layer 1	TrCH type	E-DCH	
	TTI	10ms (alt. 2ms) (NOTE 1)	
	Coding type	TC	
	CRC, bit	24	

NOTE 1: The support of 2ms TTI depends on the UE category
NOTE 2: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternative 2 (Flexible RLC + MAC-i/is) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter “Max SDU size”, ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.7.1.1.1.2 MAC-d flow #2 parameters for Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.7.1.1.1.3 MAC-d flow #3 parameters for Streaming or Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.

6.10.2.4.6.7.1.1.1.4 MAC-d flow #4 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.7.1.2 Physical channel parameters

6.10.2.4.6.7.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.7.2 Downlink

6.10.2.4.6.7.2.1 Transport channel parameters

6.10.2.4.6.7.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.7.2.1.1.1 MAC-d flow#1 parameters for Conversational / unknown or speech / DL: [max bit rate depending on UE category] kbps / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE 2	**Alt 2 Fixed RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-7 and later releases) NOTE 2**
Higher Layer	RAB/Signalling RB	**RAB**		
RLC	Logical channel type	DTCH		
	RLC mode	UM		
	Payload sizes, bit	104, 136, 152, 168, 184, 216, 288, 328 (alt 336)	104, 136, 152, 168, 184, 216, 288, 328 (alt 336)	Flexible up to 12000 (NOTE 3)
	Max data rate, bps	depends on UE category NOTE1		
	UMD PDU header, bit	8	8	8
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	112 , 144, 160, 176, 192, 224, 296, 336 (alt 344)	112 , 144, 160, 176, 192, 224, 296, 336 (alt 344)	Flexible
	MAC-hs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs/ehs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms	2 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes

NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).
NOTE 2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitly stated in the test case.
NOTE 3: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter "Max SDU size", ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.7.2.1.1.2 MAC-d flow#2 parameters for Streaming or Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.6.7.2.1.1.3 MAC-d flow#3 parameters for Streaming or Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.6.7.2.1.1.4 MAC-d flow#4 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2.

6.10.2.4.6.7.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.7.2.2.1 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.6.8.1 Uplink

6.10.2.4.6.8.1.1 Transport channel parameters

6.10.2.4.6.8.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.8.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1

6.10.2.4.6.8.1.1.2 Transport channel parameters for Conversational / speech / UL: (12.65 8.85 6.6) kbps / CS RAB

See clause 6.10.2.4.1.62.1.1.1

6.10.2.4.6.8.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.6.8.1.1.4 TFCS

See clause 6.10.2.4.1.62.1.1.3

6.10.2.4.6.8.1.1.5 TFC subset list

See clause 6.10.2.4.1.62.1.1.4

6.10.2.4.6.8.1.2 Physical channel parameters

6.10.2.4.6.8.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1

6.10.2.4.6.8.1.2.2 Physical channel parameters on DCH

See clause 6.10.2.4.1.62.1.2

6.10.2.4.6.8.2 Downlink

See clause 6.10.2.4.5.8.2

6.10.2.4.6.9 Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) kbps DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB on E-DCH and HS-DSCH + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.2.4.6.9.1 Uplink

6.10.2.4.6.9.1.1 Transport channel parameters

6.10.2.4.6.9.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.9.1.1.1.1 MAC-d flow #1 parameters for Conversational / speech / UL:(12.2, 7,75, 5.9, 4.75) kbps / CS RAB (non-scheduled)

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later) NOTE 3	**Alt 2 Flexible RLC + MAC-i/is (Rel-8 and later releases) NOTE 3**
Higher layer	RAB/Signalling RB	RAB	
PDCP	Header size, bit	8	
RLC	Logical channel type	DTCH	
	RLC mode	UM	
	Payload sizes, bit	48, 104, 128, 168, 256	Flexible from 48 up to up to 12000 (NOTE 4)
	Max data rate, bps	Depends on UE category and TTI	
	UMD PDU header, bit	8	
MAC	MAC multiplexing	N/A	
	MAC-d PDU size, bit	56, 112, 136, 176, 264 (non-scheduled) NOTE1	Flexible
	MAC type	MAC-e/es	MAC-i/is
	MAC-e/es / MAC-i/is header fixed part, bit	18	24
Layer 1	TrCH type	E-DCH	
	TTI	10ms (alt. 2ms) (NOTE 2)	
	Coding type	TC	
	CRC, bit	24	

NOTE1: Max MAC-e PDU content sizes depends on non-scheduled grant given by SRNC
NOTE 2: The support of 2ms TTI depends on the UE category
NOTE 3: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternative 2 (Flexible RLC + MAC-i/is) then this shall be explicitly stated in the test case.
NOTE 4: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter "Max SDU size", ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.9.1.1.1.2 MAC-d flow #2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.9.1.2 Physical channel parameters

6.10.2.4.6.9.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.9.2 Downlink

6.10.2.4.6.9.2.1 Transport channel parameters

6.10.2.4.6.9.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.9.2.1.1.1 MAC-d flow#1 parameters for Conversational / speech / DL:(12.2, 7,75, 5.9, 4.75) kbps / CS RAB

		Alt 1 RLC + MAC-hs (Rel-5 and later releases) NOTE2	**Alt 2 RLC + MAC-ehs (Rel-7 and later releases) NOTE2**
Higher Layer	RAB/Signalling RB	RAB	
PDCP	Header size, bit	8	
RLC	Logical channel type	DTCH	
	RLC mode	UM	
	Payload sizes, bit	48, 104, 128, 168, 256	48, 104, 128, 168, 256
	Max data rate, bps	depends on UE category NOTE1	
	UMD PDU header, bit	8	8
MAC	MAC-d header, bit	0	0
	MAC multiplexing	N/A	N/A
	MAC-d PDU size, bit	56, 112, 136, 176, 264	56, 112, 136, 176, 264
	MAC-hs Type	MAC-hs	MAC-ehs
	MAC-hs/ehs header fixed part, bit	21	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms
	Coding type	TC	TC
	CRC, bit	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes

NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see [25.321]).

NOTE2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternative 2 (Fixed RLC + MAC-ehs) then this shall be explicitly stated in the test case.

6.10.2.4.6.9.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2.

6.10.2.4.6.9.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.9.2.2.1 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.10 Conversational / speech / UL:(12.65, 8.85, 6.6) kbps DL: (12.65, 8.85, 6.6) kbps / CS RAB on E-DCH and HS-DSCH + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.2.4.6.10.1 Uplink

6.10.2.4.6.10.1.1 Transport channel parameters

6.10.2.4.6.10.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.10.1.1.1.1 MAC-d flow#1 parameters for Conversational / speech / UL:(12.65, 8.85, 6.6) kbps / CS RAB (non-scheduled)

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later) NOTE 3	**Alt 2 Flexible RLC + MAC-i/is (Rel-8 and later releases) NOTE 3**
Higher layer	RAB/Signalling RB	RAB	
PDCP	Header size, bit	8	
RLC	Logical channel type	DTCH	
	RLC mode	UM	
	Payload sizes, bit	48, 144, 192, 264	Flexible from 48 up to up to12000 (NOTE 4)
	Max data rate, bps	Depends on UE category and TTI	
	UMD PDU header, bit	8	
MAC	MAC multiplexing	N/A	
	MAC-d PDU size, bit	56, 152, 200, 272 (non-scheduled) NOTE1	Flexible
	MAC type	MAC-e/es	MAC-i/is
	MAC-e/es / MAC-i/is header fixed part, bit	18	24
Layer 1	TrCH type	E-DCH	
	TTI	10ms (alt. 2ms) (NOTE 2)	
	Coding type	TC	
	CRC, bit	24	

NOTE 1: Max MAC-e PDU content sizes depends on non-scheduled grant given by SRNC
NOTE 2: The support of 2ms TTI depends on the UE category
NOTE 3: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternative 2 (Flexible RLC + MAC-i/is) then this shall be explicitly stated in the test case.
NOTE 4: The Maximum RLC payload size for Flexible RLC is 12024 bits (1503 octets, ref: TS 25.322 clause 9.2.2.9). The maximum SDU size above PDCP layer is limited to 12000 bits (1500 octets limit in QoS parameter “Max SDU size”, ref: TS 24.008 clause 10.5.6.5). As no PDCP header is used in this radio bearer configuration then the RLC payload size has been limited to 12000 bits.

6.10.2.4.6.10.1.1.1.2. MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.10.1.2 Physical channel parameters

6.10.2.4.6.10.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.10.2 Downlink

6.10.2.4.6.10.2.1 Transport channel parameters

6.10.2.4.6.10.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.10.2.1.1.1 MAC-d flow#1 parameters for Conversational / speech / DL:(12.65, 8.85, 6.6) kbps / CS RAB

		Alt 1 RLC + MAC-hs (Rel-5 and later releases) NOTE2	**Alt 2 RLC + MAC-ehs (Rel-7 and later releases) NOTE2**
Higher Layer	RAB/Signalling RB	RAB	
PDCP	Header size, bit	8	
RLC	Logical channel type	DTCH	
	RLC mode	UM	
	Payload sizes, bit	48, 144, 192, 264	48, 144, 192, 264
	Max data rate, bps	depends on UE category NOTE1	
	UMD PDU header, bit	8	8
MAC	MAC-d header, bit	0	0
	MAC multiplexing	N/A	N/A
	MAC-d PDU size, bit	56, 152, 200, 272	56, 152, 200, 272
	MAC-hs Type	MAC-hs	MAC-ehs
	MAC-hs/ehs header fixed part, bit	21	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH
	TTI	2 ms	2 ms
	Coding type	TC	TC
	CRC, bit	24	24
	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes
NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see [25.321]).			
NOTE2: Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternative 2 (Fixed RLC + MAC-ehs) then this shall be explicitly stated in the test case.			

6.10.2.4.6.10.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2.

6.10.2.4.6.10.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.10.2.2.1 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.3 RAB and signalling RB for TDD

6.10.3.1 RABs and signalling RBs

In the following clauses, the typical parameter sets are presented for reference RABs, signalling RBs and important combinations of them. The data rate given for each RAB is the maximum data rate that can be supported by that RAB.

NOTE: The granularity for each RAB needs to be clarified.

Table 6.10.3.1.1: Prioritized RABs

#	Traffic class[3]	SSD[3]	Max. rate, kbps	CS/PS
1	Conversational	Speech	UL:12.2 DL:12.2	CS
1a	Conversational	Speech	UL: (12.2 7.95 5.9 4.75) DL(12.2 7.95 5.9 4.75)	CS
2	Conversational	Speech	UL:10.2 DL:10.2	CS
2a	Conversational	Speech	UL:(10.2 , 6.7, 5.9, 4.75) DL:10.2, 6.7, 5.9, 4.75)	CS
3	Conversational	Speech	UL:7.95 DL:7.95	CS
4	Conversational	Speech	UL:7.4 DL:7.4	CS
4a	Conversational	Speech	UL:(12.2 7.95 5.9 4.75, DL:(12.2 7.95 5.9 4.75)	CS
5	Conversational	Speech	UL:6.7 DL:6.7	CS
6	Conversational	Speech	UL:5.9 DL:5.9	CS
7	Conversational	Speech	UL:5.15 DL:5.15	CS
8	Conversational	Speech	UL:4.75 DL:4.75	CS
9	Conversational	Unknown	UL:28.8 DL:28.8	CS
10	Conversational	Unknown	UL:64 DL:64	CS
11	Conversational	Unknown	UL:32 DL:32	CS
11a	Conversational	Unknown	UL:8 DL:8	CS
12	Streaming	Unknown	UL:14.4 DL:14.4	CS
13	Streaming	Unknown	UL:28.8 DL:28.8	CS
14	Streaming	Unknown	UL:57.6 DL:57.6	CS
15	Void			
15a	Streaming	Unknown	UL:16 DL:64	PS
16	Void			
17	Void			
18	Void			
19	Void			
20	Interactive or Background	N/A	UL:32 DL:8	PS
20a	Interactive or Background	N/A	UL:8 DL:8	PS
20b	Interactive or Background	N/A	UL:16 DL:16	PS
20c	Interactive or Background	N/A	UL:32 DL:32	PS
21	Void			
22	Interactive or Background	N/A	UL:32 DL:64	PS
23	Interactive or Background	N/A	UL:64 DL:64	PS
24	Interactive or Background	N/A	UL:64 DL:128	PS
25	Interactive or Background	N/A	UL:128 DL:128	PS
26	Interactive or Background	N/A	UL:64 DL:384	PS
27	Interactive or Background	N/A	UL:128 DL:384	PS
28	Interactive or Background	N/A	UL:384 DL:384	PS
29	Interactive or Background	N/A	UL:64 DL:2048	PS
30	Interactive or Background	N/A	UL:128 DL:2048	PS
31	Void			
32	Interactive or Background	N/A	UL:64 DL:256	PS
33	Interactive or Background	N/A	UL:0 DL:32	PS
34	Interactive or Background	N/A	UL:32 DL:0	PS
35	Interactive or Background	N/A	UL:64 DL:144	PS
36	Interactive or Background	N/A	UL:144 DL:144	PS

Table 6.10.3.1.2: Signalling RBs

#	Maximum rate, kbps	Logical channel	PhyCh onto which SRBs are mapped
1	UL:1.7 DL:1.7	DCCH	DPCH
2	UL:3.4 DL:3.4	DCCH	DPCH
3	UL:13.6 DL:13.6	DCCH	DPCH
4	DL:27.2 (alt. 13.6)	DCCH	SCCPCH
5	UL:16.8	CCCH	PRACH
6	DL:32 (alt. 16)	CCCH	SCCPCH
7	DL:33.6 (alt. 16.8)	BCCH	SCCPCH
8	DL:12 (alt. 8)	PCCH	SCCPCH
9	UL:16.8	SHCCH	PRACH
10	UL:16.8	SHCCH	PRACH or PUSCH
11	DL:32 (alt. 16)	SHCCH	SCCPCH
12	DL:16	SHCCH	SCCPCH or PDSCH

6.10.3.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH.

1a) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void.

19) Void.

20) Void.

21) Void.

22) Void..

23) Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

24) Void..

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

37) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / 12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38e) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38f) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38g) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38h) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38i) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38j) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void

47) Void

48) Void

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void.

55) Void

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

57) Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

58) Streaming / unknown / UL:16 DL:64 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

59) Reserved for future use

60) Reserved for future use

61) Conversational / unknown / UL:8 DL:8 kbps / PS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

Combinations on PDSCH, SCCPCH, PUSCH and PRACH

1) Interactive or background / UL:64 DL:256 kbps / PS RAB
   + UL: 3.4/16.8 DL:3.4/ 33.6 kbps SRBs for DCCH, CCCH and BCCH
   + UL:16.8 DL: 16 kbps SRBs for SHCCH.

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
   + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
   + UL: 16.8 DL: 16 kbps SRBs for SHCCH.

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
   + UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
   + UL: 16.8 DL: 16 kbps SRBs for SHCCH.

4) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
   + UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
   + UL: 16.8 DL: 16 kbps SRBs for SHCCH.

Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

1) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
   + UL:3.4 DL:3.4 kbps SRBs for DCCH
   + Interactive or background / UL:64 DL:256 kbps / PS RAB
   + UL:16.8 kbps SRBs for CCCH and SHCCH
   + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

2) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
   + UL:3.4 DL:3.4 kbps SRBs for DCCH
   + Interactive or background / UL:64 DL:384 kbps / PS RAB
   + UL:16.8 kbps SRBs for CCCH and SHCCH
   + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
   + UL:3.4 DL:3.4 kbps SRBs for DCCH
   + Interactive or background / UL:64 DL:2 048 kbps / PS RAB
   + UL:16.8 kbps SRBs for CCCH and SHCCH
   + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

Combinations on SCCPCH

1) Stand-alone 12 kbps SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
   + SRB for CCCH
   + SRBs for DCCH
   + SRB for BCCH.

2a) Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB
   + SRBs for CCCH
   + SRB for DCCH
   + SRB for BCCH

2b) SRBs for CCCH
   + SRB for DCCH
   + SRB for BCCH

3) Interactive or background / DL:32 kbps / PS RAB
   + SRB for PCCH
   + SRB for CCCH
   + SRBs for DCCH
   + SRB for BCCH.

3a) SRB for PCCH
   + SRB for CCCH

+ SRB for DCCH
+ SRB for BCCH

4) RB for CTCH
+ SRB for CCCH
+ SRB for BCCH

Combinations on PRACH

1) Interactive or background / UL:12.8 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

Combinations on DPCH and HS-PDSCH

1) Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

2) Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3) Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

7) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

8) Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

9) Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

10) Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

11) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.10.3.3 Example of linkage between RABs and services

RABs, which are included in the present document, can provide the services as shown in table 6.10.1.1: Traffic classes. Furthermore, the required BER for each RAB, which is assumed in the present document, is shown in table 6.10.3.3.1.

Table 6.10.3.3.1: Example of linkage between RABs and services

RAB				Residual BER[3]	Services
Traffic class[3]	SSD[3]	Max. rate, kbps	CS/PS		
Conversational	Speech	UL:4.75-12.2 DL:4.75-12.2	CS	$5x10^{-4}$, $1x10^{-3}$, $5x10^{-3}$	AMR speech

Conversational	Unknown	UL:64 DL:64	CS	$1x10^{-4}$ or $1x10^{-6}$	UDI 1B, 64k 3G-324M[4]
Conversational	Unknown	UL:32 DL:32	CS	$1x10^{-4}$ or $1x10^{-6}$	32k 3G-324M[4]
Conversational	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	Transparent modem
Streaming	Unknown	UL:14.4 DL:14.4	CS	$1x10^{-3}$	FAX[6]
Streaming	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	FAX[6] PIAFS 32 kbps
Streaming	Unknown	UL:57.6 DL:57.6	CS	$1x10^{-3}$	Modem[6], FTM[5], PIAFS 64 kbps
Streaming	Unknown	UL:64-128 or DL:64-384	CS	$1x10^{-3}$ or $1x10^{-4}$	Streaming video, uni-directional
Interactive or Background	N/A	UL:32-384 DL:8-2048	PS	$1x10^{-3}$ or $1x10^{-4}$	Packet

NOTE 1: SMS can be provided via the signalling RB (DCCH) on DPCH or SCCPCH.

NOTE 2: CBS can be provided via the signalling RB (CTCH) on SCCPCH

NOTE 3: UDI *n*B can be provided via *n* RABs of conversational 64 kbps.

6.10.3.4 Typical radio parameter sets

NOTE The order of tables and MAC-d flow numbering in this section may be different than the RB IDs and MAC-d flow IDs as defined in default messages in section 9.

6.10.3.4.1 Combinations on DPCH

6.10.3.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.1.1 Uplink

6.10.3.4.1.1.1.1 Transport channel parameters

6.10.3.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		80			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		65			
	RM attribute		155 to 185			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

6.10.3.4.1.1.1.2 TFCS

TFCS size	2

TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.1.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	234
	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.1.2 Downlink

6.10.3.4.1.1.2.1 Transport channel parameters

6.10.3.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0 x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		80			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		65			
	RM attribute		155 to 185			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.10.3.4.1.1.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.1.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	236 bits
	TFCI code word	8 bits
	Puncturing limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)

6.10.3.4.1.1a.1 Uplink

6.10.3.4.1.1a.1.1 Transport channel parameters

6.10.3.4.1.1a.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148			
	TFS	TF0, bits	0x148			
		TF1, bits	1x148			
	TTI, ms		20			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		258			

6.10.3.4.1.1a.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)

6.10.3.4.1.1a.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	266
	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	1
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.1a.2 Downlink

6.10.3.4.1.1a.2.1 Transport channel parameters

6.10.3.4.1.1a.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148			
	TFS	TF0, bits	0 x148			
		TF1, bits	1x148			

	TTI, ms	20
	Coding type	CC 1/3
	CRC, bit	16
	Max number of bits/TTI before rate matching	516
	Max number of bits/radio frame before rate matching	258

6.10.3.4.1.1a.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is optional.	

6.10.3.4.1.1a.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	268 bits
	TFCI code word	8 bits
	Puncturing limit	1
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.2.1 Uplink

6.10.3.4.1.2.1.1 Transport channel parameters

6.10.3.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		129			
	RM attribute		155 to 165			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

6.10.3.4.1.2.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.2.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	234 bits
	TFCI code word	8 bits
	TPC	2 bit
	Puncturing Limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.2.2 Downlink

6.10.3.4.1.2.2.1 Transport channel parameters

6.10.3.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0, 148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		129			
	RM attribute		155 to 165			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.10.3.4.1.2.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.2.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	236
	TFCI code word	8 bits
	Puncturing limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.10.3.4.1.3.1 Uplink

6.10.3.4.1.3.1.1 Transport channel parameters

6.10.3.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		516			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

6.10.3.4.1.3.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.3.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	468 bits
	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	0.88
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.3.2 Downlink

6.10.3.4.1.3.2.1 Transport channel parameters

6.10.3.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit	8	16	16	16
MAC	MAC header, bit	4	4	4	4
	MAC multiplexing	4 logical channel multiplexing			
Layer 1	TrCH type	DCH			

	TB sizes, bit		148 (alt. 0,148) (note)
	TFS	TF0, bits	0x148 (alt. 1x0) (note)
		TF1, bits	1x148
	TTI, ms		10
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		516
	Max number of bits/radio frame before rate matching		516
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.			

6.10.3.4.1.3.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.10.3.4.1.3.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	480 bits
	TFCI code word	8 bits
	Puncturing limit	0.92
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.10.3.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.4.1 Uplink

6.10.3.4.1.4.1.1 Transport channel parameters

6.10.3.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.4.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.4.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.72

6.10.3.4.1.4.2 Downlink

6.10.3.4.1.4.2.1 Transport channel parameters

6.10.3.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39,81 (alt. 0, 39, 81)	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39,81 (alt. 0,39,81)	103	60
	TFS	TF0, bits	0x81 (alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.4.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.4.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.10.3.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.4a.1 Uplink

6.10.3.4.1.4a.1.1 Transport channel parameters

6.10.3.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2 bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.4a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.4a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.72

6.10.3.4.1.4a.2 Downlink

6.10.3.4.1.4a.2.1 Transport channel parameters

6.10.3.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.4a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.4a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits

	TFCI code word	16 bits
	Puncturing limit	0.76

6.10.3.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.5.1 Uplink

6.10.3.4.1.5.1.1 Transport channel parameters

6.10.3.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.5.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.5.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.40

6.10.3.4.1.5.2 Downlink

6.10.3.4.1.5.2.1 Transport channel parameters

6.10.3.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39,65 (alt. 0, 39, 65)	99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 65 (alt.0,39,65)	99	40
	TFS	TF0, bits	0x65 (alt,1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.5.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.5.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.40

6.10.3.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.5a.1 Uplink

6.10.3.4.1.5a.1.1 Transport channel parameters

6.10.3.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type	DTCH		
	RLC mode	TM	TM	TM

	Payload sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in TS 25.222).					

6.10.3.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.5a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.5a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.40

6.10.3.4.1.5a.2 Downlink

6.10.3.4.1.5a.2.1 Transport channel parameters

6.10.3.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type	DTCH		
	RLC mode	TM	TM	TM
	Payload sizes, bit	39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	0, 53, 63, 76, 99	40
	Max data rate, bps	10 200		
	TrD PDU header, bit	0		
MAC	MAC header, bit	0		
	MAC multiplexing	N/A		

Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	0, 53, 63, 76, 99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.10.3.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.5a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.5a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.40

6.10.3.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.6.1 Uplink

6.10.3.4.1.6.1.1 Transport channel parameters

6.10.3.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 75 (alt. 0, 39, 75)	84
	Max data rate, bps		7 950	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 75 (alt. 0, 39, 75)	84
	TFS	TF0, bits	0x75 (alt. 1x0) (note)	0x84
		TF1, bits	1x39	1x84
		TF2, bits	1x75	N/A

	TTI, ms	20	20
	Coding type	CC 1/3	CC 1/3
	CRC, bit	12	N/A
	Max number of bits/TTI after channel coding	285	276
	Max number of bits/radio frame before rate matching	143	138
	RM attribute	180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.10.3.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.6.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.6.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44

6.10.3.4.1.6.2 Downlink

6.10.3.4.1.6.2.1 Transport channel parameters

6.10.3.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 75 (alt. 0, 39, 75)	84
	Max data rate, bps		7 950	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 75 (alt. 0, 39, 75)	84
	TFS	TF0, bits	0x75 (alt. 1x0) (note)	0x84
		TF1, bits	1x39	1x84
		TF2, bits	1x75	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		285	276
	Max number of bits/radio frame before rate matching		143	138
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.6.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.6.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.7.1 Uplink

6.10.3.4.1.7.1.1 Transport channel parameters

6.10.3.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 61 (alt. 0, 39, 61)	87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 61 (alt. 0, 39, 61)	87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x87
		TF2, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.7.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.7.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips

	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.10.3.4.1.7.2 Downlink

6.10.3.4.1.7.2.1 Transport channel parameters

6.10.3.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 61 (alt. 0, 39, 61)	87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 61 (alt. 0, 39, 61)	87
	TFS	TF0, bits	0x61(alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x87
		TF2, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB #1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.7.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; optional otherwise.	

6.10.3.4.1.7.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.7a.1 Uplink

6.10.3.4.1.7a.1.1 Transport channel parameters

6.10.3.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.7a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.7a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.10.3.4.1.7a.2 Downlink

6.10.3.4.1.7a.2.1 Transport channel parameters

6.10.3.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.7a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.7a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.8.1 Uplink

6.10.3.4.1.8.1.1 Transport channel parameters

6.10.3.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2
RLC	Logical channel type	DTCH	
	RLC mode	TM	TM

	Payload sizes, bit		39, 58 (alt. 0, 39, 58)	76
	Max data rate, bps		6 700	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 58 (alt. 0, 39, 58)	76
	TFS	TF0, bits	0x58 (alt. 1x0) (note)	0x76
		TF1, bits	1x39	1x76
		TF2, bits	1x58	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		234	252
	Max number of bits/radio frame before rate matching		117	126
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.8.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.8.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52

6.10.3.4.1.8.2 Downlink

6.10.3.4.1.8.2.1 Transport channel parameters

6.10.3.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 58 (alt. 0, 39, 58)	76
	Max data rate, bps		6 700	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 58 (alt. 0,39,58)	76
	TFS	TF0, bits	0x58 (alt.1x0) (note)	0x76
		TF1, bits	1x39	1x76
		TF2, bits	1x58	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A

	Max number of bits/TTI after channel coding	234	252
	Max number of bits/radio frame before rate matching	117	126
	RM attribute	180 to 220	170 to 210
NOTE :	CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).		

6.10.3.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.8.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE:	In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.

6.10.3.4.1.8.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.10.3.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.9.1 Uplink

6.10.3.4.1.9.1.1 Transport channel parameters

6.10.3.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 55 (alt. 0, 39, 55)	63
	Max data rate, bps		5 900	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 55 (alt. 0, 39, 55)	63
	TFS	TF0, bits	0x55 (alt. 1x0) (note)	0x63
		TF1, bits	1x39	1x63
		TF2, bits	1x55	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		225	213
	Max number of bits/radio frame before rate matching		113	107
	RM attribute		180 to 220	170 to 210
NOTE:	In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.10.3.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.9.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.9.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.56

6.10.3.4.1.9.2 Downlink

6.10.3.4.1.9.2.1 Transport channel parameters

6.10.3.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 55 (alt. 0, 39, 55)	63
	Max data rate, bps		5 900	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 55 (alt. 0, 39, 55)	63
	TFS	TF0, bits	0x55 (alt. 1x0) (note)	0x63
		TF1, bits	1x39	1x63
		TF2, bits	1x55	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		225	213
	Max number of bits/radio frame before rate matching		113	107
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.9.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.9.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits

	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.10.1 Uplink

6.10.3.4.1.10.1.1 Transport channel parameters

6.10.3.4.1.10.1.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 49 (alt. 0, 39, 49)	54
	Max data rate, bps		5 150	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 49 (alt. 0, 39, 49)	54
	TFS	TF0, bits	0x49 (alt. 1x0) (note)	0x54
		TF1, bits	1x39	1x54
		TF2, bits	1x49	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		207	186
	Max number of bits/radio frame before rate matching		104	93
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.10.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.

6.10.3.4.1.10.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.10.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72

6.10.3.4.1.10.2 Downlink

6.10.3.4.1.10.2.1 Transport channel parameters

6.10.3.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2

RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 49 (alt. 0, 39, 49)	54
	Max data rate, bps		5 150	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 49 (alt. 0, 39, 49)	54
	TFS	TF0, bits	0x49 (alt. 1x0) (note)	0x54
		TF1, bits	1x39	1x54
		TF2, bits	1x49	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		207	186
	Max number of bits/radio frame before rate matching		104	93
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.10.2.1.2 Transport channel parameters for DL: 1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.10.3.4.1.10.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.10.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.72

6.10.3.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.11.1 Uplink

6.10.3.4.1.11.1.1 Transport channel parameters

6.10.3.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42 (alt. 0, 39, 42)	53
	Max data rate, bps		4 750	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42 (alt. 0, 39, 42)	53
	TFS	TF0, bits	0x42 (alt. 1x0) (note)	0x53
		TF1, bits	1x39	1x53
		TF2, bits	1x42	N/A

	TTI, ms	20	20
	Coding type	CC 1/3	CC 1/3
	CRC, bit	12	N/A
	Max number of bits/TTI after channel coding	186	183
	Max number of bits/radio frame before rate matching	93	92
	RM attribute	180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.10.3.4.1.11.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.10.3.4.1.11.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.11.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76

6.10.3.4.1.11.2 Downlink

6.10.3.4.1.11.2.1 Transport channel parameters

6.10.3.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42 (alt. 0, 39, 42)	53
	Max data rate, bps		4 750	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42 (alt. 0, 39, 42)	53
	TFS	TF0, bits	0X42 (alt.1x0)(note)	0x53
		TF1, bits	1x39	1x53
		TF2, bits	1x42	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		186	183
	Max number of bits/radio frame before rate matching		93	92
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.10.3.4.1.11.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.10.3.4.1.11.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.11.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.10.3.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.12.1 Uplink

6.10.3.4.1.12.1.1 Transport channel parameters

6.10.3.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		160 to 200

6.10.3.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.12.1.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.12.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76

NOTE: In case the first TFC in a TFCS is not configured, the TFCI code word will be 8 bits.

6.10.3.4.1.12.2 Downlink

6.10.3.4.1.12.2.1 Transport channel parameters

6.10.3.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		160 to 200

6.10.3.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.12.2.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.12.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.40
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.13.1 Uplink

6.10.3.4.1.13.1.1 Transport channel parameters

6.10.3.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	640
	Max data rate, bps	64 000
	TrD PDU header, bit	0
MAC	MAC header, bit	0

	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948
	Max number of bits/radio frame before rate matching		1 974
	RM attribute		150 to 195

6.10.3.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.13.1.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.13.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1148 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.13.2 Downlink

6.10.3.4.1.13.2.1 Transport channel parameters

6.10.3.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948
	Max number of bits/radio frame before rate matching		1 974
	RM attribute		150 to 195

6.10.3.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.13.2.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.13.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 204 bits
	TFCI code word	16 bits
	Puncturing limit	0.52
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.14.1 Uplink

6.10.3.4.1.14.1.1 Transport channel parameters

6.10.3.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		32 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	1x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 980
	Max number of bits/radio frame before rate matching		990
	RM attribute		165 to 210

6.10.3.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.14.1.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.14.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.68
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.14.2 Downlink

6.10.3.4.1.14.2.1 Transport channel parameters

6.10.3.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		32 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	1x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 980
	Max number of bits/radio frame before rate matching		990
	RM attribute		165 to 210

6.10.3.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.14.2.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.14.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716 bits
	TFCI code word	16 bits
	Puncturing limit	0.52
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.15.1 Uplink

6.10.3.4.1.15.1.1 Transport channel parameters

6.10.3.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576

		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	Max number of bits/radio frame before rate matching		447
	RM attribute		145 to 185

6.10.3.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.15.1.1.3 TFCS

TFCS size	4
TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.15.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.15.2 Downlink

6.10.3.4.1.15.2.1 Transport channel parameters

6.10.3.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	Max number of bits/radio frame before rate matching		447
	RM attribute		145 to 185

6.10.3.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.15.2.1.3 TFCS

TFCS size	4
TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.15.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.80
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.16.1 Uplink

6.10.3.4.1.16.1.1 Transport channel parameters

6.10.3.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		135 to 175

6.10.3.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.16.1.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.16.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.16.2 Downlink

6.10.3.4.1.16.2.1 Transport channel parameters

6.10.3.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		135 to 175

6.10.3.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.16.2.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.16.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.44
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.17.1 Uplink

6.10.3.4.1.17.1.1 Transport channel parameters

6.10.3.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	576
	Max data rate, bps	57 600
	TrD PDU header, bit	0
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH
	TB sizes, bit	576

	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	Max number of bits/radio frame before rate matching		1 779
	RM attribute		125 to 165

6.10.3.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.17.1.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.17.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44

6.10.3.4.1.17.2 Downlink

6.10.3.4.1.17.2.1 Transport channel parameters

6.10.3.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		57 600
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	Max number of bits/radio frame before rate matching		1 779
	RM attribute		125 to 165

6.10.3.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.17.2.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.17.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	960 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.18 Void

6.10.3.4.1.19 Void

6.10.3.4.1.20 Void

6.10.3.4.1.21 Void

6.10.3.4.1.22 Void

6.10.3.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23.1 Uplink

6.10.3.4.1.23.1.1 Transport channel parameters

6.10.3.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt.144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching		1 062 (alt. 1 206)
	RM attribute		135 to 175

6.10.3.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23.1.1.3 TFCS

TFCS size	6

TFCS	(32 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.10.3.4.1.23.2 Downlink

6.10.3.4.1.23.2.1 Transport channel parameters

6.10.3.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Max number of bits/radio frame before rate matching		267
	RM attribute		135 to 175

6.10.3.4.1.23.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.56
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23a.1 Uplink

6.10.3.4.1.23a.1.1 Transport channel parameters

6.10.3.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	N/A (alt. 5x144)
	TTI, ms		40 (alt. 80)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068 (alt. 2 412)
	Max number of bits/radio frame before rate matching		267 (alt. 302)
	RM attribute		135 to 175

6.10.3.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23a.1.1.3 TFCS

TFCS size	4 (alt. 6)
TFCS	(8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.56 (alt. 0.48)

6.10.3.4.1.23a.2 Downlink

See clause 6.10.3.4.1.23.2.

6.10.3.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23b.1 Uplink

6.10.3.4.1.23b.1.1 Transport channel parameters

6.10.3.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB

RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching		531 (alt. 603)
	RM attribute		135 to 175

6.10.3.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23b.1.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23b.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68 (alt. 0.60)

6.10.3.4.1.23b.2 Downlink

6.10.3.4.1.23b.2.1 Transport channel parameters

6.10.3.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Max number of bits/radio frame before rate matching		531

	RM attribute	135 to 175

6.10.3.4.1.23b.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23b.2.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23b.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23c.1 Uplink

6.10.3.4.1.23c.1.1 Transport channel parameters

6.10.3.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
		TF3, bits	3x336 (alt. 7x144)
		TF4, bits	4x336 (alt. 10x144)
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236 (alt. 4 812)
	Max number of bits/radio frame before rate matching		1 059 (alt. 1 203)
	RM attribute		135 to 175

6.10.3.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23c.1.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23c.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.10.3.4.1.23c.2 Downlink

6.10.3.4.1.23c.2.1 Transport channel parameters

6.10.3.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Max number of bits/radio frame before rate matching		1 059
	RM attribute		135 to 175

6.10.3.4.1.23c.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23c.2.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23c.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716
	TFCI code word	16 bits
	Puncturing limit	0.60

6.10.3.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23d.1 Uplink

6.10.3.4.1.23d.1.1 Transport channel parameters

6.10.3.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching		1 062 (alt. 1 206)
	RM attribute		135 to 175

6.10.3.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23d.1.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23d.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.10.3.4.1.23d.2 Downlink

6.10.3.4.1.23d.2.1 Transport channel parameters

6.10.3.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	32 000
	AMD PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH

	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Max number of bits/radio frame before rate matching		1 062
	RM attribute		135 to 175

6.10.3.4.1.23d.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23d.2.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.23d.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.24 Void

6.10.3.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.25.1 Uplink

See clause 6.10.3.4.1.23.1.

6.10.3.4.1.25.2 Downlink

6.10.3.4.1.25.2.1 Transport channel parameters

6.10.3.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236

	Max number of bits/radio frame before rate matching	2 118
	RM attribute	130 to 170

6.10.3.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.25.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.25.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF16 x 3 codes x 1 time slot + SF16 x 2 codes x 1 time slot	SF16 x 9 codes x 1 time slot
	Max. Number of data bits/radio frame	1 204 bits	2 180 bits
	TFCI code word	16 bits	16 bits
	Puncturing limit	0.52	0.96

6.10.3.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.26.1 Uplink

6.10.3.4.1.26.1.1 Transport channel parameters

6.10.3.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 3x144)
		TF3, bits	3x336 (alt. 7x144)
		TF4, bits	4x336 (alt. 10x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236 (alt. 4 812)
	Max number of bits/radio frame before rate matching		2 118 (alt. 2 406)
	RM attribute		130 to 170

6.10.3.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.26.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.3.4.1.26.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF4 x 1 code x 1 time slot	SF2 x 1 code x 1 time slot + SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1148 bits	2 784 bits
	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.48 (alt. 0.44)	1

6.10.3.4.1.26.2 Downlink

See clause 6.10.3.4.1.25.2.

6.10.3.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.27.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.27.2 Downlink

6.10.3.4.1.27.2.1 Transport channel parameters

6.10.3.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		128 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460
	Max number of bits/radio frame before rate matching		4 230
	RM attribute		120 to 160

6.10.3.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.27.2.1.3 TFCS

TFCS size	10
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),

	(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE:	In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.

6.10.3.4.1.27.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	256 chips	256 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot	SF16 x 4 codes x 2 time slots + SF16 x 3 codes x 2 time slots
	Max. Number of data bits/radio frame	2 192 bits	3848 bits
	TFCI code word	16 bits	16 bits
	Puncturing limit	0.48	0.84

6.10.3.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.28.1 Uplink

6.10.3.4.1.28.1.1 Transport channel parameters

6.10.3.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		128 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 7x144)
		TF3, bits	4x336 (alt 14x144)
		TF4, bits	8x336 (alt. 20x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 9 612)
	Max number of bits/radio frame before rate matching		4 230 (alt. 4 806)
	RM attribute		120 to 160

6.10.3.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.28.1.1.3 TFCS

TFCS size	9 (alt.10)
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1))
NOTE:	In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.

6.10.3.4.1.28.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	256 chips	256 chips

	Codes and time slots	SF2 x 1 code x 1 timeslot	SF2 x 1 code x 2 timeslots + SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits	5 376 bits
	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.44 (alt. 0.40)	1

6.10.3.4.1.28.2 Downlink

See clause 6.10.3.4.1.27.2.

6.10.3.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.29.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.29.2 Downlink

6.10.3.4.1.29.2.1 Transport channel parameters

6.10.3.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		144 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	9x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516
	Max number of bits/radio frame before rate matching		4 758
	RM attribute		140 to 180

6.10.3.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.29.2.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.29.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 9 codes x 1 time slot
	Max. Number of data bits/radio frame	2468 bits

	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.30.1 Uplink

6.10.3.4.1.30.1.1 Transport channel parameters

6.10.3.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		144 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 10x144)
		TF3, bits	4x336 (alt. 20x144)
		TF4, bits	8x336 (alt. 30x144)
		TF5, bits	9x336 (alt. 45x144)
	TTI, ms		20 (alt. 40)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516 (alt. 21 624)
	Max number of bits/radio frame before rate matching		4 758 (alt. 5 406)
	RM attribute		140 to 180

6.10.3.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.30.1.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.30.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF2 x 1 codex 1 time slot
	Max. Number of data bits/radio frame	2340 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44 (alt. 0.40)

6.10.3.4.1.30.2 Downlink

See clause 6.10.3.4.1.29.2.

6.10.3.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.31.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.31.2 Downlink

6.10.3.4.1.31.2.1 Transport channel parameters

6.10.3.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	N/A (alt. 12x336)
		TF6, bits	N/A (alt. 16x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 16 920)
	Max number of bits/radio frame before rate matching		8 460 (alt. 8 460)
	RM attribute		135 to 175

6.10.3.4.1.31.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.31.2.1.3 TFCS

TFCS size	10 (alt.14)
TFCS	(256 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.31.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.32.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.32.2 Downlink

6.10.3.4.1.32.2.1 Transport channel parameters

6.10.3.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	N/A (alt. 16x336)
		TF7, bits	N/A (alt. 20x336)
		TF8, bits	N/A (alt. 24x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684 (alt. 25 368)
	Max number of bits/radio frame before rate matching		12 684 (alt. 12 684)
	RM attribute		110 to 150

6.10.3.4.1.32.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.32.2.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.32.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	256 chips	256 chips
	Codes and time slots	SF16 x 8 codes x 3 time slots	SF16 x 6 codes x 4 time slots + SF16 x 4 codes x 1 time slot (alt. SF1 x 1 code x 3 time slots)
	Max. Number of data bits/radio frame	6 608 bits	7 712 bits (alt. 13232 bits)
	TFCI code word	16 bits	16 bits
	Puncturing Limit	0.48	0.60 (alt. 1)

6.10.3.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.33.1 Uplink

See clause 6.10.3.4.1.28.1.

6.10.3.4.1.33.2 Downlink

See clause 6.10.3.4.1.32.2.

6.10.3.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.34.1 Uplink

6.10.3.4.1.34.1.1 Transport channel parameters

6.10.3.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	N/A (alt. 16x336)
		TF7, bits	N/A (alt. 20x336)
		TF8, bits	N/A (alt. 24x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684 (alt. 25 368)
	Max number of bits/radio frame before rate matching		12 684
	RM attribute		110 to 150

6.10.3.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.34.1.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.34.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	256 chips	256 chips
	Codes and time slots	SF2 x 1 code x 3 time slots	SF2 x 1 code x 5 timeslots + SF4 x 1 code x 2 timeslots (alt. {SF2 x 1 code + SF4 x 1 code} x 4 timeslots)
	Max. Number of data bits/radio frame	6 480 bits	13 104 bits

	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.48	1

6.10.3.4.1.34.2 Downlink

See clause 6.10.3.4.1.32.2.

6.10.3.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.35.1 Uplink

6.10.3.4.1.35.1.1 Transport channel parameters

See clause 6.10.3.4.1.26.1.1.

6.10.3.4.1.35.1.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.88 (alt. 0.80)

6.10.3.4.1.35.2 Downlink

6.10.3.4.1.35.2.1 Transport channel parameters

6.10.3.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		2 048 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
		TF4, bits	8x656
		TF5, bits	12x656
		TF6, bits	16x656
		TF7, bits	20x656
		TF8, bits	24x656
		TF9, bits	28x656
		TF10, bits	31x656 (alt. 32x656)
		TF11, bits	N/A (alt. 36x656)
		TF12, bits	N/A (alt. 40x656)
		TF13, bits	N/A (alt. 44x656)
		TF14, bits	N/A (alt. 48x656)
		TF15, bits	N/A (alt. 52x656)
		TF16, bits	N/A (alt. 56x656)
		TF17, bits	N/A (alt. 60x656)
		TF18, bits	N/A (alt. 64x656)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16

Higher layer	RAB/Signalling RB	RAB
	Max number of bits/TTI after channel coding	62 565 (alt. 129 141)
	Max number of bits/radio frame before rate matching	62 565 (alt. 64 571)
	RM attribute	130 to 170

6.10.3.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.35.2.1.3 TFCS

TFCS size	21 (alt.38)
TFCS	(2 048 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1) (alt. TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1), (TF16, TF1), (TF17, TF1(TF18, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.35.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	256 chips	256 chips
	Codes and time slots	SF1 x 1 code x 11 time slots	SF16 x 13 codes x 4 time slots + SF16 x 12 codes x 7 time slot
	Max. Number of data bits/radio frame	48 560 bits (alt. 48 544)	37 520 bits (alt. 37 504)
	TFCI code word	16 bits (alt. 32 bits)	16 bits (alt. 32 bits)
	Puncturing limit	0.76 (alt.0.72)	0.56

6.10.3.4.1.36 Void

6.10.3.4.1.37 Void

6.10.3.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38.1 Uplink

6.10.3.4.1.38.1.1 Transport channel parameters

6.10.3.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.10.3.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38.1.1.4 TFCS

TFCS size	18
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),

	(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)

6.10.3.4.1.38.2 Downlink

6.10.3.4.1.38.2.1 Transport channel parameters

6.10.3.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.

6.10.3.4.1.38.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,8kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.10.3.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38a.1 Uplink

6.10.3.4.1.38a.1.1 Transport channel parameters

6.10.3.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH

	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt 0x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.10.3.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38a.1.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.10.3.4.1.38a.2 Downlink

6.10.3.4.1.38a.2.1 Transport channel parameters

6.10.3.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	0
	AMD PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A

Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.10.3.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1

6.10.3.4.1.38a.2.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38b.1 Uplink

6.10.3.4.1.38b.1.1 Transport channel parameters

6.10.3.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38b.1.1.4 TFCS

TFCS size	12 (alt. 17)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38b.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48 (alt. 0.56)

6.10.3.4.1.38b.2 Downlink

6.10.3.4.1.38b.2.1 Transport channel parameters

6.10.3.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38b.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38b.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.10.3.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38c.1 Uplink

6.10.3.4.1.38c.1.1 Transport channel parameters

6.10.3.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.10.3.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38c.1.1.4 TFCS

TFCS size	18 (alt. 17)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38c.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.52)

6.10.3.4.1.38c.2 Downlink

6.10.3.4.1.38c.2.1 Transport channel parameters

6.10.3.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.10.3.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38c.2.1.4 TFCS

TFCS size	18
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38c.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	960
	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38d.1 Uplink

6.10.3.4.1.38d.1.1 Transport channel parameters

6.10.3.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320 (alt. 128)	320 (alt. 128)
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340 (alt. 148)	
	TFS	TF0, bits	0x340 (alt 0x148)	
		TF1, bits	1x340 (alt 1x148)	
		TF2, bits	2x340 (alt 3x148)	
		TF3, bits	3x340 (alt 7x148)	
		TF4, bits	4x340 (alt 10x148)	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284 (alt. 4 932)	
	Max number of bits/radio frame before rate matching		2 142 (alt. 2 466)	
	RM attribute		130 to 170	

6.10.3.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38d.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38d.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.10.3.4.1.38d.2 Downlink

6.10.3.4.1.38d.2.1 Transport channel parameters

6.10.3.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	Max number of bits/radio frame before rate matching		2 142	
	RM attribute		130 to 170	

6.10.3.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38d.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38d.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 916 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38e.1 Uplink

6.10.3.4.1.38e.1.1 Transport channel parameters

6.10.3.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.10.3.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38e.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38e.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.10.3.4.1.38e.2 Downlink

6.10.3.4.1.38e.2.1 Transport channel parameters

6.10.3.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.10.3.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38e.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38e.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38f.1 Uplink

6.10.3.4.1.38f.1.1 Transport channel parameters

6.10.3.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38f.1.1.4 TFCS

TFCS size	24 (alt. 32)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) (TF0,TF0,TF0,TF2,TF1), (TF1,TF0, TF0, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38f.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48 (alt.0.56)

6.10.3.4.1.38f.2 Downlink

6.10.3.4.1.38f.2.1 Transport channel parameters

6.10.3.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38f.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38f.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.10.3.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38g.1 Uplink

6.10.3.4.1.38g.1.1 Transport channel parameters

6.10.3.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.10.3.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38g.1.1.4 TFCS

TFCS size	32 (alt. 31)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1))
NOTE 1: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. NOTE 2: The alt. TFCS is used when the 16Kbps RAB alt. is used.	

6.10.3.4.1.38g.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot + SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	664 bits (alt. 696 bits)
	TFCI code word	32 bits (alt. 16 bits)
	TPC	2 bits
	Puncturing Limit	0.56 (alt. 0.60)

6.10.3.4.1.38g.2 Downlink

6.10.3.4.1.38g.2.1 Transport channel parameters

6.10.3.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.10.3.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1

6.10.3.4.1.38g.2.1.4 TFCS

TFCS size	36
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38g.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	700 bits
	TFCI code word	32 bits
	Puncturing limit	0.56

6.10.3.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38h.1 Uplink

6.10.3.4.1.38h.1.1 Transport channel parameters

6.10.3.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.10.3.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38h.1.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38h.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot + SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 084 bits
	TFCI code word	32 bits
	TPC	2 bits
	Puncturing Limit	0.68 (alt.0.60)

6.10.3.4.1.38h.2 Downlink

6.10.3.4.1.38h.2.1 Transport channel parameters

6.10.3.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.10.3.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38h.2.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF1,TF0,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38h.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	944
	TFCI code word	32 bits
	Puncturing limit	0.60

6.10.3.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38i.1 Uplink

6.10.3.4.1.38i.1.1 Transport channel parameters

6.10.3.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38i.1.1.4 TFCS

TFCS size	48
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38i.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 936 bits
	TFCI code word	32 bit
	TPC	2 bits
	Puncturing Limit	0.68 (alt.0.60)

6.10.3.4.1.38i.2 Downlink

6.10.3.4.1.38i.2.1 Transport channel parameters

6.10.3.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38i.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),

	(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38i.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 900 bits
	TFCI code word	32 bits
	Puncturing limit	0.68

6.10.3.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38j.1 Uplink

See clause 6.10.3.4.1.38i.1

6.10.3.4.1.38j.2 Downlink

6.10.3.4.1.38j.2.1 Transport channel parameters

6.10.3.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38j.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),

	(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.38j.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 6 codes x 2 time slots
	Max. Number of data bits/radio frame	3 280 bits
	TFCI code word	32 bits
	Puncturing limit	0.64

6.10.3.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.39.1 Uplink

See clause 6.10.3.4.1.38.1.

6.10.3.4.1.39.2 Downlink

6.10.3.4.1.39.2.1 Transport channel parameters

6.10.3.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.39.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.39.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	1 936 bits
	TFCI code word	16 bits

	Puncturing limit	0.68

6.10.3.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.40.1 Uplink

6.10.3.4.1.40.1.1 Transport channel parameters

6.10.3.4.1.40.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.40.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.40.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.40.1.1.4 TFCS

6.10.3.4.1.40.1.1.4.1 TFCS (one CCTrCH case)

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.40.1.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.40.1.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.40.1.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.40.1.2 Physical channel parameters

6.10.3.4.1.40.1.2.1 Physical channel (one CCTrCH case)

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 1 time slot

	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.64 (alt. 0.56)

6.10.3.4.1.40.1.2.2 Physical channel (two CCTrCH case)

6.10.3.4.1.40.1.2.2.1 Physical channel (conversational + SRB)

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.10.3.4.1.40.1.2.2.2 Physical channel (Interactive or background)

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.84 (alt. 0.72)

6.10.3.4.1.40.2 Downlink

Transport channel parameters

6.10.3.4.1.40.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.40.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.40.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.40.2.1.4 TFCS

6.10.3.4.1.40.2.1.4.1 TFCS (one CCTrCH case)

See Clause 6.10.3.4.1.39.2.1.4.

6.10.3.4.1.40.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.40.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.40.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.40.2.2 Physical channel parameters

6.10.3.4.1.40.2.2.1 Physical channel parameters (one CCTrCH)

See clause 6.10.3.4.1.39.2.2.

6.10.3.4.1.40.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.40.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.40.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 204 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.41.1 Uplink

See clause 6.10.3.4.1.40.1.

6.10.3.4.1.41.2 Downlink

6.10.3.4.1.41.2.1 Transport channel parameters

6.10.3.4.1.41.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.41.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.41.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.41.2.1.4 TFCS

6.10.3.4.1.41.2.1.4.1 TFCS (one CCTrCH case)

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)

NOTE:	In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.

6.10.3.4.1.41.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.41.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.41.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.41.2.2 Physical channel parameters

6.10.3.4.1.41.2.2.1 Physical channel parameters (one CCTrCH)

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 5codes x 2time slots
	Max. Number of data bits/radio frame	2 744 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.10.3.4.1.41.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.41.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.10.3.4.1.41.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0,48

6.10.3.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.42.1 Uplink

6.10.3.4.1.42.1.1 Transport channel parameters

6.10.3.4.1.42.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.42.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.42.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.42.1.1.4 TFCS

See clause 6.10.3.4.1.40.1.1.4.1.

6.10.3.4.1.42.1.2 Physical channel parameters

See clause 6.10.3.4.1.40.1.2.1.

6.10.3.4.1.42.2 Downlink

6.10.3.4.1.42.2.1 Transport channel parameters

6.10.3.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.10.3.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.42.2.1.4 TFCS

TFCS size	30 (alt. 42)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 256 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.42.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 2 time slots + SF16 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	5 504 bits (alt. 5 488)

	TFCI code word	16 bits (alt. 32)
	Puncturing limit	0.60

6.10.3.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.43.1 Uplink

See clause 6.10.3.4.1.40.1.

6.10.3.4.1.43.2 Downlink

6.10.3.4.1.43.2.1 Transport channel parameters

6.10.3.4.1.43.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.43.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.3.4.1.32.2.1.1.

6.10.3.4.1.43.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.43.2.1.4 TFCS

6.10.3.4.1.43.2.1.4.1 TFCS (one CCTrCH case)

TFCS size	36 (alt. 54)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1) (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1) (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.43.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.43.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.43.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	6 (alt. 9)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0) (alt. (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF0, TF0, TF0, TF8, TF0))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.43.2.2 Physical channel parameters

6.10.3.4.1.43.2.2.1 Physical channel parameters (one CCTrCH)

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 592 bits
	TFCI code word	32 bits
	Puncturing limit	0.48

6.10.3.4.1.43.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.43.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.60

6.10.3.4.1.43.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits
	TFCI code word	16 bits
	Puncturing limit	0,52

6.10.3.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.44.1 Uplink

6.10.3.4.1.44.1.1 Transport channel parameters

6.10.3.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.44.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.44.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	{SF8 x 1 code + SF2 x 1 code} x 1 time slot
	Max. Number of data bits/radio frame	2 616 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.44)

6.10.3.4.1.44.2 Downlink

6.10.3.4.1.44.2.1 Transport channel parameters

6.10.3.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.3.4.1.35.2.1.1.

6.10.3.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.44.2.1.4 TFCS

TFCS size	66 (alt. 114)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1) (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF11, TF0), (TF1, TF0, TF0, TF11, TF0), (TF2, TF1, TF1, TF11, TF0), (TF0, TF0, TF0, TF12, TF0), (TF1, TF0, TF0, TF12, TF0), (TF2, TF1, TF1, TF12, TF0), (TF0, TF0, TF0, TF13, TF0), (TF1, TF0, TF0, TF13, TF0), (TF2, TF1, TF1, TF13, TF0), (TF0, TF0, TF0, TF14, TF0), (TF1, TF0, TF0, TF14, TF0), (TF2, TF1, TF1, TF14, TF0), (TF0, TF0, TF0, TF15, TF0), (TF1, TF0, TF0, TF15, TF0), (TF2, TF1, TF1, TF15, TF0), (TF0, TF0, TF0, TF16, TF0), (TF1, TF0, TF0, TF16, TF0), (TF2, TF1, TF1, TF16, TF0), (TF0, TF0, TF0, TF17, TF0), (TF1, TF0, TF0, TF17, TF0), (TF2, TF1, TF1, TF17, TF0), (TF0, TF0, TF0, TF18, TF0), (TF1, TF0, TF0, TF18, TF0), (TF2, TF1, TF1, TF18, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1), (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1), (TF0, TF0, TF0, TF11, TF1), (TF1, TF0, TF0, TF11, TF1), (TF2, TF1, TF1, TF11, TF1), (TF0, TF0, TF0, TF12, TF1), (TF1, TF0, TF0, TF12, TF1), (TF2, TF1, TF1, TF12, TF1), (TF0, TF0, TF0, TF13, TF1), (TF1, TF0, TF0, TF13, TF1), (TF2, TF1, TF1, TF13, TF1), (TF0, TF0, TF0, TF14, TF1), (TF1, TF0, TF0, TF14, TF1), (TF2, TF1, TF1, TF14, TF1), (TF0, TF0, TF0, TF15, TF1), (TF1, TF0, TF0, TF15, TF1), (TF2, TF1, TF1, TF15, TF1), (TF0, TF0, TF0, TF16, TF1), (TF1, TF0, TF0, TF16, TF1), (TF2, TF1, TF1, TF16, TF1), (TF0, TF0, TF0, TF17, TF1), (TF1, TF0, TF0, TF17, TF1), (TF2, TF1, TF1, TF17, TF1), (TF0, TF0, TF0, TF18, TF1), (TF1, TF0, TF0, TF18, TF1), (TF2, TF1, TF1, TF18, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.44.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 12 codes x 11 time slots
	Max. Number of data bits/radio frame	36 400 bits
	TFCI code word	32 bits
	Puncturing limit	0.52

6.10.3.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.45.1 Uplink

6.10.3.4.1.45.1.1 Transport channel parameters

6.10.3.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.10.3.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.45.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.45.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot + SF4 x 1 codex 1 time slot
	Max. Number of data bits/radio frame	1 392 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.56

6.10.3.4.1.45.2 Downlink

6.10.3.4.1.45.2.1 Transport channel parameters

6.10.3.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.10.3.4.1.45.2.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.45.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.45.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 448 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.46 Void

6.10.3.4.1.47 Void

6.10.3.4.1.48 Void

6.10.3.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.49.1 Uplink

6.10.3.4.1.49.1.1 Transport channel parameters

6.10.3.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.49.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.49.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72

6.10.3.4.1.49.2 Downlink

6.10.3.4.1.49.2.1 Transport channel parameters

6.10.3.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.49.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.49.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.10.3.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.49a.1 Uplink

6.10.3.4.1.49a.1.1 Transport channel parameters

6.10.3.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.49a.1.1.4 TFCS

TFCS size	24

TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.49a.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.72

6.10.3.4.1.49a.2 Downlink

6.10.3.4.1.49a.2.1 Transport channel parameters

6.10.3.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.49a.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.49a.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 916 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.50.1 Uplink

6.10.3.4.1.50.1.1 Transport channel parameters

6.10.3.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.50.1.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.50.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 1time slot + SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 784 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.60

6.10.3.4.1.50.2 Downlink

6.10.3.4.1.50.2.1 Transport channel parameters

6.10.3.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.50.2.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.50.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 6codes x 2 time slots
	Max. Number of data bits/radio frame	2 912 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.10.3.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51.1 Uplink

6.10.3.4.1.51.1.1 Transport channel parameters

6.10.3.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.51.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44 (alt. 0.40)

6.10.3.4.1.51.2 Downlink

6.10.3.4.1.51.2.1 Transport channel parameters

6.10.3.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.51.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.51.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51a.1 Uplink

6.10.3.4.1.51a.1.1 Transport channel parameters

6.10.3.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51a.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51a.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) (alt. (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.51a.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.76

6.10.3.4.1.51a.2 Downlink

6.10.3.4.1.51a.2.1 Transport channel parameters

6.10.3.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.51a.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.51a.2.1.4 TFCS

TFCS size	8
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.51a.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 640 bits
	TFCI code word	16 bits
	Puncturing limit	0.60

6.10.3.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51b.1 Uplink

6.10.3.4.1.51b.1.1 Transport channel parameters

6.10.3.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51b.1.1.2 Transport channel parameters for Interactive or Background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.10.3.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51b.1.1.4 TFCS

TFCS size	12
TFCS	(64 kbps Conversational RAB, 16 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.51b.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.68

6.10.3.4.1.51b.2 Downlink

See clause 6.10.3.4.1.51.2.

6.10.3.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.52.1 Uplink

See clause 6.10.3.4.1.51.1.

6.10.3.4.1.52.2 Downlink

6.10.3.4.1.52.2.1 Transport channel parameters

6.10.3.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.52.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.52.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	{SF16 x 8 codes x 1 time slot} + {SF16 x 5 codes x 1 time slot}
	Max. Number of data bits/radio frame	3 156 bits
	TFCI code word	16 bits
	Puncturing limit	0.44

6.10.3.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.53.1 Uplink

6.10.3.4.1.53.1.1 Transport channel parameters

6.10.3.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.53.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.10.3.4.1.53.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 2 timeslots
	Max. Number of data bits/radio frame	3 760 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)

6.10.3.4.1.53.2 Downlink

See clause 6.10.3.4.1.52.2.

6.10.3.4.1.54 Void

6.10.3.4.1.55 Void

6.10.3.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.56.1 Uplink

6.10.3.4.1.56.1.1 Transport channel parameters

6.10.3.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320 (alt. 128)	320 (alt.128)
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340 (alt. 148)	
	TFS	TF0, bits	0x340 (alt. 0x148)	
		TF1, bits	1x340 (alt. 1x148)	
		TF2, bits	N/A (alt. 5x148)	
	TTI, ms		40 (alt. 80)	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 080 (alt. 2 472)	
	Max number of bits/radio frame before rate matching		270 (alt. 309)	
	RM attribute		135 to 175	

6.10.3.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.56.1.1.3 TFCS

TFCS size	4 (alt. 6)
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1) (alt. (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.56.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits (alt. 16 bits).		

6.10.3.4.1.56.2 Downlink

6.10.3.4.1.56.2.1 Transport channel parameters

6.10.3.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 080	
	Max number of bits/radio frame before rate matching		270	
	RM attribute		135 to 175	

6.10.3.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.56.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.56.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.56
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.10.3.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.57.1 Uplink

6.10.3.4.1.57.1.1 Transport channel parameters

6.10.3.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.10.3.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.57.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.57.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.88 (alt. 0.76)

6.10.3.4.1.57.2 Downlink

6.10.3.4.1.57.2.1 Transport channel parameters

6.10.3.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	Max number of bits/radio frame before rate matching		2 142	
	RM attribute		130 to 170	

6.10.3.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.57.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.57.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 364 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.10.3.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.58.1 Uplink

6.10.3.4.1.58.1.1 Transport channel parameters

6.10.3.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Max number of bits/radio frame before rate matching		534
	RM attribute		135 to 175

6.10.3.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.58.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(16 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1) (alt. (TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF2,TF0), (TF1,TF2,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF0,TF2,TF1), (TF1,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.58.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot + SF16 x 1code x 1 time slot
	Max. Number of data bits/radio frame	696 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.68)

6.10.3.4.1.58.2 Downlink

6.10.3.4.1.58.2.1 Transport channel parameters

6.10.3.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 076
	Max number of bits/radio frame before rate matching		2 019
	RM attribute		125 to 165

6.10.3.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.58.2.1.4 TFCS

TFCS size	16
TFCS	(64 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.58.2.2 Physical channel parameters

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF16 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 640 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.10.3.4.1.59 Reserved for future use

6.10.3.4.1.60 Reserved for future use

6.10.3.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.61.1 Uplink

6.10.3.4.1.61.1.1 Transport channel parameters

6.10.3.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	320

	Max data rate, bps		8 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328)
	TFS	TF0, bits	0x328 (alt 1x0) (note)
		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	Max number of bits/radio frame before rate matching		261
	RM attribute		135 to 175
NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.10.3.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.61.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) (alt. (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.61.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.68 (alt. 0.64)

6.10.3.4.1.61.2 Downlink

6.10.3.4.1.61.2.1 Transport channel parameters

6.10.3.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328)
	TFS	TF0, bits	0x328 (alt 1x0) (note)

		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	Max number of bits/radio frame before rate matching		261
	RM attribute		135 to 175
NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.10.3.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.61.2.1.4 TFCS

TFCS size	8
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.10.3.4.1.61.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.10.3.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH

6.10.3.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.1.1 Uplink

6.10.3.4.2.1.1.1 Transport channel parameters

6.10.3.4.2.1.1.1.1 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	TM
	Payload sizes, bit		320 (alt. 128)	168
	Max data rate, bps		64 000	16 800
	AMD/TrD PDU header, bit		16	0
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		USCH	USCH
	TB sizes, bit		337 (alt. 145)	169
	TFS	TF0, bits	0x337 (alt. 0x145)	0x169
		TF1, bits	1x337 (alt. 1x145)	1x169
		TF2, bits	2x337 (alt. 3x145)	N/A
		TF3, bits	3x337 (alt. 7x145)	N/A
		TF4, bits	4x337 (alt. 10x145)	N/A
	TTI, ms		20	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16

	Max number of bits/TTI after channel coding	4 248 (alt. 4 842)	386
	Max number of bits/radio frame before rate matching	2 124 (alt. 2 421)	386
	RM attribute	135 to 175	230 to 250

6.10.3.4.2.1.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		5	5	5	5
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		USCH			
	TB sizes, bit		149			
	TFS	TF0, bits	0x149			
		TF1, bits	1x149			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		519			
	Max number of bits/radio frame before rate matching		130			
	RM attribute		190 to 210			

6.10.3.4.2.1.1.1.3 TFCS for USCH

TFCS size	20
TFCS	(64 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1)

6.10.3.4.2.1.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

6.10.3.4.2.1.1.1.4.1 RACH transport channel configuration without DTCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH
	RLC mode	TM	UM	AM	AM	AM	TM
	Payload sizes, bit	168	136	128	128	128	168
	Max data rate, bps	16 800	13 600	12 800	12 800	12 800	16 800
	AMD/UMD/TrD PDU header, bit	0	8	16	16	16	0
MAC	MAC header, bit	2	26	26	26	26	2
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	170					
	TFS | TF0, bits	1x170					
	TTI, ms	10					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	388					
	Max number of bits/radio frame before rate matching	388					

6.10.3.4.2.1.1.1.4.2 RACH transport channel configuration with DTCH

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH
	RLC mode	AM	TM	UM	AM	AM	AM	TM
	Payload sizes, bit	128	168	136	128	128	128	168
	Max data rate, bps	12 800	16 800	13 600	12 800	12 800	12 800	16 800
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16	0
MAC	MAC header, bit	26	2	26	26	26	26	2
	MAC multiplexing	7 logical channel multiplexing						
Layer 1	TrCH type	RACH						
	TB sizes, bit	170						
	TFS TF0, bits	1x170						
	TTI, ms	10						
	Coding type	CC 1/2						
	CRC, bit	16						
	Max number of bits/TTI after channel coding	388						
	Max number of bits/radio frame before rate matching	388						

6.10.3.4.2.1.1.2 Physical channel parameters

6.10.3.4.2.1.1.2.1 Physical channel parameters for PUSCH

PUSCH	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.60 (alt. 0.56)

6.10.3.4.2.1.1.2.2 Physical channel parameters for PRACH

PRACH	Midamble	512 chips
	Codes and time slots	SF8 (alt. SF16) x 1 code x 1 time slot
	Max. Number of data bits/radio frame	464 (alt. 232)
	Puncturing Limit	1 (alt. 0.56)

6.10.3.4.2.1.2 Downlink

6.10.3.4.2.1.2.1 Transport channel parameters

6.10.3.4.2.1.2.1.1 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB	RAB	SRB#5
RLC	Logical channel type	DTCH	SHCCH
	RLC mode	AM	UM
	Payload sizes, bit	320	160
	Max data rate, bps	256 000	16 000
	AMD/UMD PDU header, bit	16	8
MAC	MAC header, bit	1	1
	MAC multiplexing	N/A	N/A
Layer 1	TrCH type	DSCH	DSCH
	TB sizes, bit	337	169

	TFS	TF0, bits	0x337	0x169
		TF1, bits	1x337	1x169
		TF2, bits	2x337	N/A
		TF3, bits	4x337	N/A
		TF4, bits	8x337	N/A
		TF5, bits	N/A (alt. 12x337)	N/A
		TF6, bits	N/A (alt. 16x337)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		8 484 (alt. 16 968)	386
	Downlink: Max number of bits/radio frame before rate matching		8 484 (alt. 8 484)	386
	RM attribute		135 to 175	230 to 250

6.10.3.4.2.1.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		5	5	5	5
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DSCH			
	TB sizes, bit		149			
	TFS	TF0, bits	0x149			
		TF1, bits	1x149			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		519			
	Max number of bits/radio frame before rate matching		130			
	RM attribute		155 to 165			

6.10.3.4.2.1.2.1.3 TFCS for DSCH

TFCS size	20 (alt. 28)
TFCS	(256 kbps RAB, SHCCH, SRB for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1))

6.10.3.4.2.1.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

6.10.3.4.2.1.2.1.4.1 FACH transport channel configuration without DTCH

Higher layer	RAB/signalling RB		SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		160	136 or 120 (note)	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		8	8	16	16	16	8	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3	3
	MAC multiplexing		7 logical channel multiplexing						
Layer 1	TrCH type		FACH						
	TB sizes, bit		171						
	TFS	TF0, bits	0x171						
		TF1, bits	1x171						
		TF2, bits	2x171						
		TF3, bits	3x171(alt. N/A)						
		TF4, bits	4x171(alt. N/A)						
	TTI, ms		20						
	Coding type		TC						
	CRC, bit		16						
	Max number of bits/TTI after channel coding		2 256 (alt. 1 134)						
	Max number of bits/radio frame before rate matching		1 128 (alt. 567)						
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.									

6.10.3.4.2.1.2.1.4.2 FACH transport channel configuration with DTCH

Higher layer	RAB/signalling RB		RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		DTCH	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		AM	UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		320	160	136 or 120 (note)	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		16	8	8	16	16	16	8	0
MAC	MAC header, bit		27	3	27 or 43	27	27	27	3	3
	MAC multiplexing		8 logical channel multiplexing							
Layer 1	TrCH type		FACH							
	TB sizes, bit		171, 363							
	TFS	TF0, bits	0x171							
		TF1, bits	1x171							
		TF2, bits	2x171							
		TF3, bits	1x363							
		TF4, bits	3x171 (alt N/A)							
		TF5, bits	4x171 (alt. N/A)							
		TF6, bits	2x363 (alt. N/A)							
	TTI, ms		20							
	Coding type		TC							
	CRC, bit		16							
	Max number of bits/TTI after channel coding		2 286 (alt. 1 149)							
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)							
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.										

6.10.3.4.2.1.2.1.5 TFCS for FACH

6.10.3.4.2.1.2.1.5.1 TFCS for FACH transport channel configuration without DTCH

TFCS size	5 (alt. 3)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2))

6.10.3.4.2.1.2.1.5.2 TFCS for FACH transport channel configuration with DTCH

TFCS size	7 (alt. 4)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4), (TF5), (TF6) (alt. FACH = (TF0), (TF1), (TF2), (TF3))

6.10.3.4.2.1.2.2 Physical channel parameters

6.10.3.4.2.1.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing Limit	0.44

6.10.3.4.2.1.2.2.2 Physical channel parameters for SCCPCH

6.10.3.4.2.1.2.2.2.1 Physical channel parameters for SCCPCH without DTCH

SCCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)
	Puncturing Limit	1 (alt. 0.84)

6.10.3.4.2.1.2.2.2.2 Physical channel parameters for SCCPCH with DTCH

SCCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472 bits)
	TFCI code word	16 bits
	Puncturing Limit	1 (alt. 0.80)

6.10.3.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.2.1 Uplink

See clause 6.10.3.4.2.1.1.

6.10.3.4.2.2.2 Downlink

6.10.3.4.2.2.2.1 Transport channel parameters

6.10.3.4.2.2.2.1.1 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB	RAB	SRB#5
RLC	Logical channel type	DTCH	SHCCH
	RLC mode	AM	UM
	Payload sizes, bit	320	160
	Max data rate, bps	384 000	16 000
	AMD/UMD PDU header, bit	16	8

MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DSCH	DSCH
	TB sizes, bit		337	169
	TFS	TF0, bits	0x337	0x169
		TF1, bits	1x337	1x169
		TF2, bits	2x337	N/A
		TF3, bits	4x337	N/A
		TF4, bits	8x337	N/A
		TF5, bits	12x337	N/A
		TF6, bits	N/A (alt. 16x337)	N/A
		TF7, bits	N/A (alt. 20x337)	N/A
		TF8, bits	N/A (alt. 24x337)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		12 720 (alt. 25 440)	386
	Downlink: Max number of bits/radio frame before rate matching		12 720 (alt. 12 720)	386
	RM attribute		135 to 175	230 to 250

6.10.3.4.2.2.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.10.3.4.2.2.2.1.3 TFCS for DSCH

TFCS size	24 (alt. 36)
TFCS	(384 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1))

6.10.3.4.2.2.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH (with & without DTCH)

See clause 6.10.3.4.2.1.2.1.4.

6.10.3.4.2.2.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.5.

6.10.3.4.2.2.2.2 Physical channel parameters

6.10.3.4.2.2.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits (alt. 6 592 bits)
	TFCI code word	16 bits (alt. 32 bits)
	Puncturing Limit	0.48

6.10.3.4.2.2.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.

6.10.3.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.3.1 Uplink

See clause 6.10.3.4.2.1.1.

6.10.3.4.2.3.2 Downlink

6.10.3.4.2.3.2.1 Transport channel parameters

6.10.3.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	UM
	Payload sizes, bit		640	160
	Max data rate, bps		2 048 000	16 000
	AMD/UMD PDU header, bit		16	8
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DSCH	DSCH
	TB sizes, bit		657	169
	TFS	TF0, bits	0x657	0x169
		TF1, bits	1x657	1x169
		TF2, bits	2x657	N/A
		TF3, bits	4x657	N/A
		TF4, bits	8x657	N/A
		TF5, bits	12x657	N/A
		TF6, bits	16x657	N/A
		TF7, bits	20x657	N/A
		TF8, bits	24x657	N/A
		TF9, bits	28x657	N/A
		TF10, bits	30x657 (alt. 32x657)	N/A
		TF11, bits	N/A (alt. 36x657)	N/A
		TF12, bits	N/A (alt. 40x657)	N/A
		TF13, bits	N/A (alt. 44x657)	N/A
		TF14, bits	N/A (alt. 48x657)	N/A
		TF15, bits	N/A (alt. 52x657)	N/A
		TF16, bits	N/A (alt. 56x657)	N/A
		TF17, bits	N/A (alt. 60x657)	N/A
		TF18, bits	N/A (alt. 64x657)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		60 624 (alt. 129 330)	386
	Downlink: Max number of bits/radio frame before rate matching		60 624 (alt. 64 665)	386
	RM attribute		135 to 175	230 to 250

6.10.3.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.10.3.4.2.3.2.1.3 TFCS for DSCH

TFCS size	41 (alt.76)
TFCS	(2 048 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0), (TF10, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1),

	(TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0), (TF10, TF0, TF0),(TF11, TF0, TF0), (TF12, TF0, TF0), (TF13, TF0, TF0), (TF14, TF0, TF0), (TF15, TF0, TF0), (TF16, TF0, TF0), (TF17, TF0, TF0), (TF18, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0), (TF10, TF1, TF0),(TF11, TF1, TF0), (TF12, TF1, TF0), (TF13, TF1, TF0), (TF14, TF1, TF0), (TF15, TF1, TF0), (TF16, TF1, TF0), (TF17, TF1, TF0), (TF18, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1), (TF10, TF0, TF1), (TF11, TF0, TF1), (TF12, TF0, TF1), (TF13, TF0, TF1), (TF14, TF0, TF1), (TF15, TF0, TF1), (TF16, TF0, TF1), (TF17, TF0, TF1), (TF18, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1), (TF10, TF1, TF1),(TF11, TF1, TF1), (TF12, TF1, TF1), (TF13, TF1, TF1), (TF14, TF1, TF1), (TF15, TF1, TF1), (TF16, TF1, TF1), (TF17, TF1, TF1), (TF18, TF1, TF1))

6.10.3.4.2.3.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.2.1.2.1.4.1.

6.10.3.4.2.3.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.45.1.

6.10.3.4.2.3.2.2 Physical channel parameters

6.10.3.4.2.3.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips
	Codes and time slots	SF16 x 12 codes x 11 time slots
	Max. Number of data bits/radio frame	36 400 bits
	TFCI code word	32 bits
	Puncturing Limit	0.56 (alt. 0.52)

6.10.3.4.2.3.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.4.1 Uplink

6.10.3.4.2.4.1.1 Transport channel parameters

6.10.3.4.2.4.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	TM
	Payload sizes, bit		320 (alt. 128)	168
	Max data rate, bps		384 000	16 800
	AMD/TrD PDU header, bit		16	0
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		USCH	USCH
	TB sizes, bit		337 (alt. 145)	169
	TFS	TF0, bits	0x337 (alt. 0x145)	0x169
		TF1, bits	1x337 (alt. 1x145)	1x169

		TF2, bits	2x337 (alt. 5x145)	N/A
		TF3, bits	4x337 (alt. 10x145)	N/A
		TF4, bits	8x337 (alt. 20x145)	N/A
		TF5, bits	12x337 (alt. 30x145)	N/A
		TF6, bits	16x337 (alt. 40x145)	N/A
		TF7, bits	20x337 (alt. 50x145)	N/A
		TF8, bits	24x337 (alt. 60x145)	N/A
	TTI, ms		20	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		25 440 (alt. 29 004)	386
	Max number of bits/radio frame before rate matching		12 720 (alt. 14 502)	386
	RM attribute		135 to 175	230 to 250

6.10.3.4.2.4.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.2.

6.10.3.4.2.4.1.1.3 TFCS for USCH

TFCS size	36
TFCS	(384 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1) (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1)

6.10.3.4.2.4.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.2.1.1.1.4.

6.10.3.4.2.4.1.2 Physical channel parameters

6.10.3.4.2.4.1.2.1 Physical channel parameters for PUSCH

PUSCH	Midamble	512 chips
	Codes and time slots	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	7 264 bits
	TFCI code word	32 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.44)

6.10.3.4.2.4.1.2.2 Physical channel parameters for PRACH

See clause 6.10.3.4.2.1.1.2.2.

6.10.3.4.2.4.2 Downlink

6.10.3.4.2.4.2.1 Transport channel parameters

See clause 6.10.3.4.2.3.2.1.

6.10.3.4.2.4.2.2 Physical channel parameters

6.10.3.4.2.4.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips

	Codes and time slots	SF1 x 1 codes x 9 time slots
	Max. Number of data bits/radio frame	39 712 bits
	TFCI code word	32 bits
	Puncturing Limit	0.64 (alt. 0.60)

6.10.3.4.2.4.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

6.10.3.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH

6.10.3.4.3.1.1 Uplink

6.10.3.4.3.1.1.1 Transport channel parameters

6.10.3.4.3.1.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.3.1.1.1.2 Transport channel parameters for UL SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.3.1.1.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.1.1.3.

6.10.3.4.3.1.1.1.4 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.10.3.4.3.1.1.1.5 TFCS for USCH

TFCS size	10
TFCS	(64 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.10.3.4.3.1.1.1.6 Transport channel parameters for SRB for CCCH and UL SRB for SHCCH mapped on RACH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	SHCCH
	RLC mode		TM	TM
	Payload sizes, bit		168	168
	Max data rate, bps		16 800	16 800
	TrD PDU header, bit		0	0
MAC	MAC header, bit		2	2
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		RACH	
	TB sizes, bit		170	
	TFS	TF0, bits	1x170	
	TTI, ms		10	
	Coding type		CC 1/2	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		388	
	Max number of bits/radio frame before rate matching		388	

6.10.3.4.3.1.1.2 Physical channel parameters

6.10.3.4.3.1.1.2.1 Physical channel parameters for DPCH

See clause 6.10.3.4.1.4.1.2.

6.10.3.4.3.1.1.2.2 Physical channel parameters for PUSCH

PUSCH	Midamble	512 chips
	Codes and time slots	SF2 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76 (alt. 0.68)

6.10.3.4.3.1.1.2.3 Physical channel parameters for PRACH

See clause 6.10.3.4.2.1.1.2.2.

6.10.3.4.3.1.2 Downlink

6.10.3.4.3.1.2.1 Transport channel parameters

6.10.3.4.3.1.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.1.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.1.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.1.2.1.4 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.10.3.4.3.1.2.1.5 TFCS for DSCH

TFCS size	10 (alt. 14)
TFCS	(256 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1))

6.10.3.4.3.1.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

Higher layer	RAB/Signalling RB		SRB#0	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC
RLC	Logical channel type		CCCH	SHCCH	BCCH
	RLC mode		UM	UM	TM
	Payload sizes, bit		160	160	168
	Max data rate, bps		32 000 (alt. 16 000)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	UMD/TrD PDU header, bit		8	8	0
MAC	MAC header, bit		3		
	MAC multiplexing		3 logical channel multiplexing		
Layer 1	TrCH type		FACH		
	TB sizes, bit		171		
	TFS	TF0, bits	0x171		
		TF1, bits	1x171		

		TF2, bits	2x171
		TF3, bits	3x171 (alt. N/A)
		TF4, bits	4x171 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 256 (alt. 1 134)
	Max number of bits/radio frame before rate matching		1 128 (alt 567)

6.10.3.4.3.1.2.1.7 TFCS for FACH

TFCS size	5 (alt. 3)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2))

6.10.3.4.3.1.2.2 Physical channel parameters

6.10.3.4.3.1.2.2.1 Physical channel parameters for DPCH

Seeclause 6.10.3.4.1.4.2.2.

6.10.3.4.3.1.2.2.2 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing Limit	0.48

6.10.3.4.3.1.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.10.3.4.3.2.1 Uplink

See clause 6.10.3.4.3.1.1.

6.10.3.4.3.2.2 Downlink

6.10.3.4.3.2.2.1 Transport channel parameters

6.10.3.4.3.2.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.2.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.2.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.2.2.1.4 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.10.3.4.3.2.2.1.5 TFCS for DSCH

TFCS size	12 (alt. 18)
TFCS	(384 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))

6.10.3.4.3.2.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.3.1.2.1.6.

6.10.3.4.3.2.2.1.7 TFCS for FACH

See clause 6.10.3.4.3.1.2.1.7.

6.10.3.4.3.2.2.2 Physical channel parameters

6.10.3.4.3.2.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.3.2.2.2.2 Physical channel parameters for PDSCH

PDSCH	Midamble	256 chips
	Codes and time slots	SF16 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits
	TFCI code word	16 bits
	Puncturing Limit	0.48

6.10.3.4.3.2.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.10.3.4.3.3.1 Uplink

See clause 6.10.3.4.3.1.1.

6.10.3.4.3.3.2 Downlink

6.10.3.4.3.3.2.1 Transport channel parameters

6.10.3.4.3.3.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.3.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.3.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.3.2.1.4 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.3.2.1.1.

6.10.3.4.3.3.2.1.5 TFCS for DSCH

TFCS size	22 (alt. 38)
TFCS	(2 048 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1), (TF16, TF1), (TF17, TF1), (TF18, TF1))

6.10.3.4.3.3.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.3.1.2.1.6.

6.10.3.4.3.3.2.1.7 TFCS for FACH

See clause 6.10.3.4.3.1.2.1.7.

6.10.3.4.3.3.2.2 Physical channel parameters

6.10.3.4.3.3.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.3.3.2.2.2 Physical channel parameters for PDSCH

DPCH Downlink	Midamble	256 chips
	Codes and time slots	SF1 x 1 code x 7 time slot
	Max. Number of data bits/radio frame	30 896 bits (alt. 30 880)
	TFCI code word	16 bits (alt. 32 bits)
	Puncturing limit	0.48 (alt. 0.44)

6.10.3.4.3.3.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.4 Combinations on SCCPCH

6.10.3.4.4.1 Stand-alone signalling RB for PCCH

6.10.3.4.4.1.1 Transport channel parameters

6.10.3.4.4.1.1.1 Transport channel parameter of SRB for PCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		RRC
RLC	Logical channel type		PCCH
	RLC mode		TM
	Payload sizes, bit		240 (alt. 80)
	Max data rate, bps		12 000 (alt. 8 000)
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		PCH
	TB sizes, bit		240 (alt. 80)
	TFS	TF0, bts	0x240 (alt. 0x80)
		TF1, bits	1x240 (alt. 1x80)
		TF2, bits	N/A (alt.2x80)
	TTI, ms		20

	Coding type	CC 1/2
	CRC, bit	16
	Max number of bits/TTI before rate matching	528 (alt. 400)
	Max number of bits/radio frame before rate matching	264 (alt. 200)
	RM attribute	210 to 250

6.10.3.4.4.1.1.2 TFCS

TFCS size	2 (alt. 3)
TFCS	SRBs for PCCH = (TF0), (TF1) (alt. (TF0), (TF1), (TF2))

6.10.3.4.4.1.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot (alt. SF16 x 1 code x 1 time slot)
	Max. Number of data bits/radio frame	480 bits (alt. 236 bits)
	TFCI code word	8 bits
	Puncturing limit	1

6.10.3.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2.1 Transport channel parameters

6.10.3.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		Interactive/ Background RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000 (alt. 16 000)
	AMD PDU header, bit		16
MAC	MAC header, bit		27
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		363
	TFS	TF0, bits	0 x363
		TF1, bits	1x363
		TF2, bits	2x363 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		2 286 (alt. 1 149)
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)
	RM attribute		110 to 150

6.10.3.4.4.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH	BCCH
	RLC mode	UM	UM	AM	AM	AM	TM
	Payload sizes, bit	160	136 or 120 (note)	128	128	128	168

	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 24 000 (alt. 24 000 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		8	8	16	16	16	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3
	MAC multiplexing		6 logical channel multiplexing					
Layer 1	TrCH type		FACH					
	TB sizes, bit		171					
	TFS	TF0, bits	0x171					
		TF1, bits	1x171					
		TF2, bits	2x171					
		TF3, bits	3x171 (alt. N/A)					
		TF4, bits	4x171 (alt. N/A)					
	TTI, ms		20					
	Coding type		TC					
	CRC, bit		16					
	Max number of bits/TTI before rate matching		2 256 (alt. 1 134)					
	Max number of bits/radio frame before rate matching		1 128 (alt. 567)					
	RM attribute		200 to 240					
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.								

6.10.3.4.4.2.1.3 TFCS

TFCS size	9 (alt. 4)
TFCS	(32kbps RAB, SRBs for CCCH/DCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4),(TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0))
NOTE: First TFCS applies when the alternative for the 3 2kbps RAB and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RAB and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid.	

6.10.3.4.4.2.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472)
	TFCI code word	16 bits
	Puncturing limit	0.60 (alt. 0.48)

6.10.3.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2a.1 Transport channel parameters

6.10.3.4.4.2a.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB

Higher layer	RAB/Signalling RB	RAB	RAB
	User of Radio Bearer	Interactive/Background RAB	Interactive/Background RAB
RLC	Logical channel type	DTCH	DTCH
	RLC mode	AM	AM
	Payload sizes, bit	320	320
	Max data rate, bps	32 000 (alt. 16 000)	32 000 (alt. 16 000)
	AMD PDU header, bit	16	16
MAC	MAC header, bit	27	27

	MAC multiplexing		2 logical channel multiplexing
Layer 1	TrCH type		FACH
	TB sizes, bit		363
	TFS	TF0, bits	0x363
		TF1, bits	1x363
		TF2, bits	2x363 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		2 286 (alt. 1 149)
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)
	RM attribute		110 to 150

6.10.3.4.4.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.2a.1.3 TFCS

TFCS size	9 (alt. 4)
TFCS	(32kbps RAB + 32kbps RAB, SRBs for CCCH/DCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0))
NOTE: First TFCS applies when the alternative for the 32 kbps RABs and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RABs and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid.	

6.10.3.4.4.2a.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472)
	TFCI code word	16 bits
	Puncturing limit	0.60 (alt. 0.48)

6.10.3.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2b.1 Transport channel parameters

6.10.3.4.4.2b.1.1 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.2b.1.2 TFCS

TFCS size	5 (alt. 3)
TFCS	(SRBs for CCCH/DCCH/BCCH) = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. (TF0), (TF1), (TF2))

6.10.3.4.4.2b.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)

	Puncturing limit	1 (alt. 0.84)

6.10.3.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.3.1 Transport channel parameters

6.10.3.4.4.3.1.1 Transport channel parameters for Interactive/Background 32 kbps RAB

See clause 6.10.3.4.4.2.1.1.

6.10.3.4.4.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.10.3.4.4.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.3.1.4 TFCS

TFCS size	30 (alt. 8)
TFCS	(32 kbps RAB, SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) = (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF0, TF3), (TF0, TF0, TF4), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF1, TF2), (TF0, TF1, TF3), (TF0, TF1, TF4), (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1, TF0, TF2), (TF1, TF0, TF3), (TF1, TF0, TF4),(TF1, TF1, TF0), (TF1, TF1, TF1), (TF1, TF1, TF2), (TF1, TF1, TF3), (TF1, TF1, TF4), (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2, TF0, TF2), (TF2, TF0, TF3), (TF2, TF0, TF4),(TF2, TF1, TF0), (TF2, TF1, TF1), (TF2, TF1, TF2), (TF2, TF1, TF3), (TF2, TF1, TF4) (alt. (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF2, TF0), (TF0, TF2, TF1), (TF1, TF0, TF0))
NOTE: Alt. TFCS applies when alts for 32 kbps RAB, SRB for PCCH, and SRBs for CCCH/ DCCH/ BCCH are all configured.	

6.10.3.4.4.3.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 936 bits (alt. 472 bits)
	TFCI code word	16 bits
	Puncturing limit	0.52 (alt. 0.56)
NOTE: Alt. applies when alts for 32 kbps RAB and SRBs for CCCH/ DCCH/ BCCH are both configured.		

6.10.3.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.3a.1 Transport channel parameters

6.10.3.4.4.3a.1.1 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.10.3.4.4.3a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.3a.1.3 TFCS

TFCS size	10 (alt.7)
TFCS	(SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF1, TF3), (TF1, TF4) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0), (TF1, TF1), (TF2, TF0), (TF2, TF1))
NOTE: Alt. TFCS applies when alts for SRB for PCCH and SRBs for CCCH/ DCCH/ BCCH are both configured.	

6.10.3.4.4.3a.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 5 codes x 1 time slot (alt. SF16 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)
	Puncturing limit	0.84 (alt. 0.84)
NOTE: Alt. applies when alt for SRBs for CCCH/ DCCH/ BCCH is configured.		

6.10.3.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.10.3.4.4.4.1 Transport channel parameters

6.10.3.4.4.4.1.1 Transport channel parameters of RB for CTCH

Higher layer	RAB/signalling RB		N/A
	User of Radio Bearer		BMC
RLC	Logical channel type		CTCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		15 200
	UMD PDU header, bit		8
MAC	MAC header, bit		3
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		163
	TFS	TF0, bits	0x163
		TF1, bits	1x163
		TF2, bits	2x163
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		1 098
	Max number of bits/radio frame before rate matching		549
	RM attribute		200 to 240

6.10.3.4.4.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	BCCH
	RLC mode		UM	TM
	Payload sizes, bit		160	168
	Max data rate, bps		16 000	16 800
	AMD/UMD/TrD PDU header, bit		8	0
MAC	MAC header, bit		3	3
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		FACH	
	TB sizes, bit		171	
	TFS	TF0, bits	0x171	
		TF1, bits	1x171	
		TF2, bits	2x171	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI before rate matching		1 134	
	Max number of bits/radio frame before rate matching		567	
	RM attribute		200 to 240	

6.10.3.4.4.4.1.3 TFCS

TFCS size	4
TFCS	(RB for CTCH, SRBs for CCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0) , (TF1, TF1), (TF2, TF0)

6.10.3.4.4.4.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.80

6.10.3.4.4.5 64.8kbps RB for MTCH with 80 ms TTI

6.10.3.4.4.5.1 Transport channel parameters

6.10.3.4.4.5.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		64800
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		16344
	Max number of bits/radio frame before rate matching		2043
	RM attribute		160

6.10.3.4.4.5.1.2 TFCS

TFCS size	9
TFCS	64 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8

6.10.3.4.4.5.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	1936 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.10.3.4.4.6 129.6kbps RB for MTCH with 80 ms TTI

6.10.3.4.4.6.1 Transport channel parameters

6.10.3.4.4.6.1.1 Transport channel parameters for 128 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		32679
	Max number of bits/radio frame before rate matching		4085
	RM attribute		160

6.10.3.4.4.6.1.2 TFCS

TFCS size	17
TFCS	128 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.10.3.4.4.6.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF1 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	3888 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.10.3.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI

6.10.3.4.4.7.1 Transport channel parameters

6.10.3.4.4.7.1.1 Transport channel parameters for 256 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		32679
	Max number of bits/radio frame before rate matching		8170
	RM attribute		160

6.10.3.4.4.7.1.2 TFCS

TFCS size	17
TFCS	256 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.10.3.4.4.7.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF1 x 1 codes x 2 time slot
	Max. Number of data bits/radio frame	7792 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.10.3.4.4.8 7.6 kbps signalling RB for MCCH

6.10.3.4.4.8.1 Transport channel parameters

6.10.3.4.4.8.1.1 Transport channel parameters for 7.6 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		7600
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		160
	TFS	TF0, bits	0x160
		TF1, bits	1x160
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		552
	Max number of bits/radio frame before rate matching		276
	RM attribute		160

6.10.3.4.4.8.1.2 TFCS

TFCS size	2
TFCS	MBMS SRB =TF0, TF1

6.10.3.4.4.8.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF16 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.80

6.10.3.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI

6.10.3.4.4.9.1 Transport channel parameters

6.10.3.4.4.9.1.1 Transport channel parameters for 124 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		124400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	2x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		30078
	Max number of bits/radio frame before rate matching		3760
	RM attribute		128

6.10.3.4.4.9.1.2 TFCS

TFCS size	3
TFCS	124 kbps RAB =TF0, TF1, TF2

6.10.3.4.4.9.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF1 x 1 code x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	4208 bits
	TFCI code word	(16,5)
	Puncturing limit	1

6.10.3.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI

6.10.3.4.4.10.1 Transport channel parameters

6.10.3.4.4.10.1.1 Transport channel parameters for 320 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4272
	Max data rate, bps		320400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4289
	TFS	TF0, bits	0x4289
		TF1, bits	1x4289
		TF2, bits	6x4289
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		77562
	Max number of bits/radio frame before rate matching		9696
	RM attribute		128

6.10.3.4.4.10.1.2 TFCS

TFCS size	3
TFCS	320 kbps RAB =TF0, TF1, TF2

6.10.3.4.4.10.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF1 x 1 code x 1 time slot
	Modulation	16QAM
	Max. Number of data bits/radio frame	8432 bits
	TFCI code word	(16,5)
	Puncturing limit	0.84

6.10.3.4.4.11 497.6kbps RB for MBSFN MTCH with 80 ms TTI

6.10.3.4.4.11.1 Transport channel parameters

6.10.3.4.4.11.1.1 Transport channel parameters for 496 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		497600
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	8x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		120312
	Max number of bits/radio frame before rate matching		15039
	RM attribute		128

6.10.3.4.4.11.1.2 TFCS

TFCS size	3
TFCS	496 kbps RAB =TF0, TF1, TF2

6.10.3.4.4.11.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF1 x 1 code x 2 time slots
	Modulation	QPSK
	Max. Number of data bits/radio frame	8432 bits
	TFCI code word	(16,5) in first slot only
	Puncturing limit	0.56

6.10.3.4.4.12 7.2 kbps signalling RB for MBSFN MCCH

6.10.3.4.4.12.1 Transport channel parameters

6.10.3.4.4.12.1.1 Transport channel parameters for 7.2 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		72
	Max data rate, bps		7200
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		80
	TFS	TF0, bits	0x80
		TF1, bits	1x80
		TF2, bits	2x80
		TF3, bits	4x80
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1164
	Max number of bits/radio frame before rate matching		291
	RM attribute		128

6.10.3.4.4.12.1.2 TFCS

TFCS size	4
TFCS	MBMS SRB =TF0, TF1, TF2, TF3

6.10.3.4.4.12.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	248 bits
	Modulation	QPSK
	TFCI code word	(16,5)
	Puncturing limit	0.84

6.10.3.4.5 Combinations on PRACH

6.10.3.4.5.1 SRB for CCCH + SRB for DCCH

6.10.3.4.5.1.1 Transport channel parameters

6.10.3.4.5.1.1.1 Transport channel parameter for SRB for CCCH, SRB for DCCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	TM	UM	AM	AM	AM
	Payload sizes, bit	168	136	128	128	128
	Max data rate, bps	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	0	8	16	16	16
MAC	MAC header, bit	2	26	26	26	26

	MAC multiplexing		5 logical channel multiplexing				
Layer 1	TrCH type		RACH				
	TB sizes, bit		170				
	TFS	TF0, bits	1x170				
	TTI, ms		10				
	Coding type		CC 1/2				
	CRC, bit		16				
	Max number of bits/TTI after channel coding		388				
	Max number of bits/Radio frame before rate matching		388				

6.10.3.4.5.1.1.2 TFCS

TFCS size	1
TFCS	SRBs for CCCH/ DCCH = (TF0)

6.10.3.4.5.1.2 Physical channel parameters

PRACH	Midamble	512 chips
	Codes and time slots	SF8 (alt. SF16) x 1 code x 1 time slot
	Max. Number of data bits/radio frame	488 bits (alt. 244 bits)
	Puncturing Limit	1.0 (alt. 0.60)

6.10.3.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.3.4.5.2.1 Transport channel parameters

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	168	136	128	128	128
	Max data rate, bps	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16
MAC	MAC header, bit	26	2	26	26	26	26
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	170					
	TFS TF0, bits	1x170					
	TTI, ms	10					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	388					
	Max number of bits/ Radio frame before rate matching	388					

6.10.3.4.5.2.2 Physical channel parameters

See clause 6.10.3.4.5.1.2.

6.10.3.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.3.4.5.3.1 Transport channel parameters

Higher	RAB/signalling RB	RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4

layer	User of Radio Bearer	Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type	DTCH	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	128	168	136	128	128	128
	Max data rate, bps	12 800	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	16	0	8	16	16	16
MAC	MAC header, bit	26	26	2	26	26	26	26
	MAC multiplexing	7 logical channel multiplexing						
Layer 1	TrCH type	RACH						
	TB sizes, bit	170						
	TFS TF0, bits	1x170						
	TTI, ms	10						
	Coding type	CC 1/2						
	CRC, bit	16						
	Max number of bits/TTI after channel coding	388						
	Max number of bits/ Radio frame before rate matching	388						

6.10.3.4.5.3.2 Physical channel parameters

See clause 6.10.3.4.5.1.2.

6.10.3.4.6 Combinations on DPCH and HS-PDSCH

6.10.3.4.6.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.1.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.6.1.2 Downlink

6.10.3.4.6.1.2.1 Transport channel parameters

6.10.3.4.6.1.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.1.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	depends on UE category NOTE1
	AMD PDU header, bit	16
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	10 ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).		

6.10.3.4.6.1.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.1.2.1.2.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1

6.10.3.4.6.1.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.1.2.2 Physical channel parameters

6.10.3.4.6.1.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.10.3.4.6.1.2.2.2 Physical channel parameters on HS-PDSCH

Physical parameters common for all UE physical layer categories

HS-PDSCH	Midamble	512 chips

UE HS-DSCH Physical Layer category 1:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	2
	Max Data Rate	1.2 Mbps

UE HS-DSCH Physical Layer category 2:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	1.2 Mbps

UE HS-DSCH Physical Layer category 3:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	4
	Max Data Rate	2.4 Mbps

UE HS-DSCH Physical Layer category 4:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	2.4 Mbps

UE HS-DSCH Physical Layer category 5:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	6

	Max Data Rate	3.6 Mbps

UE HS-DSCH Physical Layer category 6:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	3.6 Mbps

UE HS-DSCH Physical Layer category 7:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	5.3 Mbps

UE HS-DSCH Physical Layer category 8:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	7.3 Mbps

UE HS-DSCH Physical Layer category 9:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	10.2 Mbps

6.10.3.4.6.2 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.2.1 Uplink

See clause 6.10.3.4.1.28.1.

6.10.3.4.6.2.2 Downlink

6.10.3.4.6.2.2.1 Transport channel parameters

6.10.3.4.6.2.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.2.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.2.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.2.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.2.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.2.2.2 Physical channel parameters

6.10.3.4.6.2.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2..

6.10.3.4.6.2.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.3 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.3.1 Uplink

See clause 6.10.3.4.1.34.1.

6.10.3.4.6.3.2 Downlink

6.10.3.4.6.3.2.1 Transport channel parameters

6.10.3.4.6.3.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.3.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.3.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.3.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.3.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.3.2.2 Physical channel parameters

6.10.3.4.6.3.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2..

6.10.3.4.6.3.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.4.1 Uplink

6.10.3.4.6.4.1.1 Transport channel parameters

6.10.3.4.6.4.1.1.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.6.4.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.3.4.1.34.1.1.1.

6.10.3.4.6.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.6.4.1.1.4 TFCS

TFCS size	36 (alt. 54)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1))

6.10.3.4.6.4.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 3 time slots
	Max. Number of data bits/radio frame	6 480 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.10.3.4.6.4.2 Downlink

6.10.3.4.6.4.2.1 Transport channel parameters

6.10.3.4.6.4.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.4.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.4.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.4.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.6.4.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.4.2.1.2.3 TFCS

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.6.4.2.2 Physical channel parameters

6.10.3.4.6.4.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.6.4.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.5.1 Uplink

See clause 6.10.3.4.1.40.1.

6.10.3.4.6.5.2 Downlink

6.10.3.4.6.5.2.1 Transport channel parameters

6.10.3.4.6.5.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.5.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.5.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.5.2.1.2.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.6.5.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.5.2.1.2.3 TFCS

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.6.5.2.2 Physical channel parameters

6.10.3.4.6.5.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.6.5.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.6 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.6.1 Uplink

6.10.3.4.6.6.1.1 Transport channel parameters

6.10.3.4.6.6.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.6.6.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.3.4.1.34.1.1.1.

6.10.3.4.6.6.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.6.6.1.1.4 TFCS

TFCS size	24 (alt. 36)
TFCS	(64 kbps RAB, 384 kbps RAB , DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF6, TF0), (TF1, TF6, TF0), (TF0, TF7, TF0), (TF1, TF7, TF0), (TF0, TF8, TF0), (TF1, TF8, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1), (TF0, TF6, TF1), (TF1, TF6, TF1), (TF0, TF7, TF1), (TF1, TF7, TF1), (TF0, TF8, TF1), (TF1, TF8, TF1))

6.10.3.4.6.6.1.2 Physical channel parameters

DPCH Uplink	Midamble	256 chips
	Codes and time slots	SF2 x 1 code x 3 time slots
	Max. Number of data bits/radio frame	6 480 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.40

6.10.3.4.6.6.1 Downlink

6.10.3.4.6.6.2.1 Transport channel parameters

6.10.3.4.6.6.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.6.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.6.2.1.1 Transport channel parameters for DCH

6.10.3.4.6.6.2.1.2.1 Transport channel parameters for Conversational / unkown/ DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.6.6.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.6.2.1.2.3 TFCS

See clause 6.10.3.4.1.13.2.1.3.

6.10.3.4.6.6.2.2 Physical channel parameters

6.10.3.4.6.6.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.6.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.7 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.7.1 Uplink

See clause 6.10.3.4.1.57.1.

6.10.3.4.6.7.2 Downlink

6.10.3.4.6.7.2.1 Transport channel parameters

6.10.3.4.6.7.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.7.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.7.2.1.1 Transport channel parameters for DCH

6.10.3.4.6.7.2.1.2.1 Transport channel parameters for Conversational / unkown/ DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.6.7.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.7.2.1.2.3 TFCS

See clause 6.10.3.4.1.13.2.1.3.

6.10.3.4.6.7.2.2 Physical channel parameters

6.10.3.4.6.7.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.6.7.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.8 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.8.1 Uplink

6.10.3.4.6.8.1.1 Transport channel parameters

6.10.3.4.6.8.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB + UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB	RAB
RLC	Logical channel type	DTCH	DTCH
	RLC mode	AM	AM
	Payload sizes, bit	320	320

	Max data rate, bps		384 000	384 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	4x340	
		TF4, bits	8x340	
		TF5, bits	12x340	
	TTI, ms		10	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		12 828	
	Uplink: Max number of bits/radio frame before rate matching		12 828	
	RM attribute		110-180	

6.10.3.4.6.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.6.8.1.1.3 TFCS

TFCS size	12
TFCS	(384 kbps RAB + 384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)

6.10.3.4.6.8.1.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF2 x 1 codes x 3 time slot
	Max. Number of data bits/radio frame	6480 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.6.8.2 Downlink

6.10.3.4.6.8.2.1 Transport channel parameters

6.10.3.4.6.8.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.8.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.8.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.8.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.8.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.8.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.8.2.2 Physical channel parameters

6.10.3.4.6.8.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.10.3.4.6.8.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.9 Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.9.1 Uplink

See clause 6.10.3.4.1.57.1.

6.10.3.4.6.9.2 Downlink

6.10.3.4.6.9.2.1 Transport channel parameters

6.10.3.4.6.9.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.9.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.9.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.9.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.9.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.9.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.9.2.2 Physical channel parameters

6.10.3.4.6.9.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.10.3.4.6.9.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.10 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.10.1 Uplink

6.10.3.4.6.10.1.1 Transport channel parameters

6.10.3.4.6.10.1.1.1 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

Higher Layer	RAB/Signalling RB		**RAB**
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		128000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8076
	Uplink: Max number of bits/radio frame before rate matching		4038
	RM attribute		125-165

6.10.3.4.6.10.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.6.10.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.6.10.1.1.3 TFCS

TFCS size	40
TFCS	(128 kbps RAB, 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF2, TF2, TF0), (TF3, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF2, TF3, TF0), (TF3, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF2, TF4, TF0), (TF3, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF2, TF2, TF1), (TF3, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF2, TF3, TF1), (TF3, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF2, TF4, TF1), (TF3, TF4, TF1)

6.10.3.4.6.10.1.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF2 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	4272 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.10.3.4.6.10.2 Downlink

6.10.3.4.6.10.2.1 Transport channel parameters

6.10.3.4.6.10.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.10.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB

Higher Layer	RAB/Signalling RB	**RAB**
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	640
	Max data rate, bps	depends on UE category NOTE1
	AMD PDU header, bit	16
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	656
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	10 ms
	Coding type	TC
	CRC, bit	24
NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).		

6.10.3.4.6.10.2.1.1.2 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.10.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.10.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.10.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.10.3.4.6.10.2.2 Physical channel parameters

6.10.3.4.6.10.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.10.3.4.6.10.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.6.11 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.6.11.1 Uplink

6.10.3.4.6.11.1.1 Transport channel parameters

6.10.3.4.6.11.1.1.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.6.11.1.1.2 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

See clause 6.10.3.4.6.10.1.1.1.

6.10.3.4.6.11.1.1.3 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.6.11.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.6.11.1.1.5 TFCS

TFCS size	120
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0,TF0), (TF1,TF0,TF0,TF1,TF0,TF0), (TF2,TF1,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF2,TF0,TF0), (TF1,TF0,TF0,TF2,TF0,TF0), (TF2,TF1,TF1,TF2,TF0,TF0), (TF0,TF0,TF0,TF3,TF0,TF0), (TF1,TF0,TF0,TF3,TF0,TF0), (TF2,TF1,TF1,TF3,TF0,TF0), (TF0,TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1,TF0), (TF1,TF0,TF0,TF1,TF1,TF0), (TF2,TF1,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF1,TF0), (TF1,TF0,TF0,TF2,TF1,TF0), (TF2,TF1,TF1,TF2,TF1,TF0), (TF0,TF0,TF0,TF3,TF1,TF0), (TF1,TF0,TF0,TF3,TF1,TF0), (TF2,TF1,TF1,TF3,TF1,TF0), (TF0,TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF0,TF2,TF0), (TF0,TF0,TF0,TF1,TF2,TF0), (TF1,TF0,TF0,TF1,TF2,TF0), (TF2,TF1,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF2,TF0), (TF1,TF0,TF0,TF2,TF2,TF0), (TF2,TF1,TF1,TF2,TF2,TF0), (TF0,TF0,TF0,TF3,TF2,TF0), (TF1,TF0,TF0,TF3,TF2,TF0), (TF2,TF1,TF1,TF3,TF2,TF0), (TF0,TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF0,TF3,TF0), (TF0,TF0,TF0,TF1,TF3,TF0), (TF1,TF0,TF0,TF1,TF3,TF0), (TF2,TF1,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF2,TF3,TF0), (TF1,TF0,TF0,TF2,TF3,TF0), (TF2,TF1,TF1,TF2,TF3,TF0), (TF0,TF0,TF0,TF3,TF3,TF0), (TF1,TF0,TF0,TF3,TF3,TF0), (TF2,TF1,TF1,TF3,TF3,TF0), (TF0,TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF0,TF4,TF0), (TF0,TF0,TF0,TF1,TF4,TF0), (TF1,TF0,TF0,TF1,TF4,TF0), (TF2,TF1,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF2,TF4,TF0), (TF1,TF0,TF0,TF2,TF4,TF0), (TF2,TF1,TF1,TF2,TF4,TF0), (TF0,TF0,TF0,TF3,TF4,TF0), (TF1,TF0,TF0,TF3,TF4,TF0), (TF2,TF1,TF1,TF3,TF4,TF0), (TF0,TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0,TF1), (TF1,TF0,TF0,TF1,TF0,TF1), (TF2,TF1,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF2,TF0,TF1), (TF1,TF0,TF0,TF2,TF0,TF1), (TF2,TF1,TF1,TF2,TF0,TF1), (TF0,TF0,TF0,TF3,TF0,TF1), (TF1,TF0,TF0,TF3,TF0,TF1), (TF2,TF1,TF1,TF3,TF0,TF1), (TF0,TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF0,TF1,TF1), (TF0,TF0,TF0,TF1,TF1,TF1), (TF1,TF0,TF0,TF1,TF1,TF1), (TF2,TF1,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1,TF1), (TF1,TF0,TF0,TF2,TF1,TF1), (TF2,TF1,TF1,TF2,TF1,TF1), (TF0,TF0,TF0,TF3,TF1,TF1), (TF1,TF0,TF0,TF3,TF1,TF1), (TF2,TF1,TF1,TF3,TF1,TF1), (TF0,TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF0,TF2,TF1), (TF0,TF0,TF0,TF1,TF2,TF1), (TF1,TF0,TF0,TF1,TF2,TF1), (TF2,TF1,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF2,TF2,TF1), (TF1,TF0,TF0,TF2,TF2,TF1), (TF2,TF1,TF1,TF2,TF2,TF1), (TF0,TF0,TF0,TF3,TF2,TF1), (TF1,TF0,TF0,TF3,TF2,TF1), (TF2,TF1,TF1,TF3,TF2,TF1), (TF0,TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF0,TF3,TF1), (TF0,TF0,TF0,TF1,TF3,TF1), (TF1,TF0,TF0,TF1,TF3,TF1), (TF2,TF1,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF2,TF3,TF1), (TF1,TF0,TF0,TF2,TF3,TF1), (TF2,TF1,TF1,TF2,TF3,TF1), (TF0,TF0,TF0,TF3,TF3,TF1), (TF1,TF0,TF0,TF3,TF3,TF1), (TF2,TF1,TF1,TF3,TF3,TF1), (TF0,TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF0,TF4,TF1), (TF0,TF0,TF0,TF1,TF4,TF1), (TF1,TF0,TF0,TF1,TF4,TF1), (TF2,TF1,TF1,TF1,TF4,TF1), (TF0,TF0,TF0,TF2,TF4,TF1), (TF1,TF0,TF0,TF2,TF4,TF1), (TF2,TF1,TF1,TF2,TF4,TF1), (TF0,TF0,TF0,TF3,TF4,TF1), (TF1,TF0,TF0,TF3,TF4,TF1), (TF2,TF1,TF1,TF3,TF4,TF1)

6.10.3.4.6.11.1.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF2 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	4272 bits
	TFCI code word	16 bits
	Puncturing limit	0.44

6.10.3.4.6.11.2 Downlink

6.10.3.4.6.11.2.1 Transport channel parameters

6.10.3.4.6.11.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.6.11.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB

See clause 6.10.3.4.6.10.2.1.1.1.

6.10.3.4.6.11.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.6.11.2.1.2 Transport channel parameters for DCH

6.10.3.4.6.11.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.6.11.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.6.11.2.1.2.3 TFCS

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.6.11.2.2 Physical channel parameters

6.10.3.4.6.11.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.6.11.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.7 Combinations on HS-PDSCH and E-PUCH

6.10.3.4.7.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.10.3.4.7.1.1 Uplink

6.10.3.4.7.1.1.1 Transport channel parameters

6.10.3.4.7.1.1.1.1 Transport channel parameters for E-DCH

6.10.3.4.7.1.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	Depends on UE category and TTI
	AMD PDU header, bit	16
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	10ms
	Coding type	TC
	CRC, bit	24

6.10.3.4.7.1.1.1.2 Transport channel parameters for DCH

6.10.3.4.7.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.7.1.1.2 Physical channel parameters

6.10.3.4.7.1.1.2.1 Physical channel parameters on E-PUCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE E-DCH Physical Layer category 1:

E-PUCH	Number of processes	4
	Max Data Rate	1.7360Mbps

UE E-DCH Physical Layer category 2:

E-PUCH	Number of processes	4
	Max Data Rate	3.4752Mbps

UE E-DCH Physical Layer category 3:

E-PUCH	Number of processes	4
	Max Data Rate	5.2416Mbps

UE E-DCH Physical Layer category 4:

E-PUCH	Number of processes	4
	Max Data Rate	6.9536Mbps

UE E-DCH Physical Layer category 5:

E-PUCH	Number of processes	4
	Max Data Rate	10.4864Mbps

6.10.3.4.7.1.1.2.2 Physical channel parameters for DPCH

See clause 6.10.3.4.1.2.1.2

6.10.3.4.7.1.2 Downlink

See clause 6.10.3.4.6.1.2.

6.10.3.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.3.4.7.3.1 Uplink

See clause 6.10.3.4.7.1.1.

6.10.3.4.7.3.1.2 Physical channel parameters

6.10.3.4.7.3.1.2.1 Physical channel parameters on E-PUCH

See clause 6.10.3.4.7.1.1.2.1.

6.10.3.4.7.3.2 Downlink

6.10.3.4.7.3.2.1 Transport channel parameters

6.10.3.4.7.3.2.1.1 Transport channel parameters for HS-DSCH

6.10.3.4.7.3.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.3.4.6.1.2.1.1.1.

6.10.3.4.7.3.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	Depends on UE category (NOTE)			
	AMD PDU header, bit	8	16	16	16
MAC	MAC-d header, bit	4	4	4	4
	MAC multiplexing	4 logical channel multiplexing			
	MAC-d PDU size, bit	148			
	MAC-hs header fixed part, bit	21			
Layer 1	TrCH type	HS-DSCH			
	TTI	10 ms			
	Coding type	TC			
	CRC, bit	24			
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).					

6.10.3.4.7.3.2.2 Physical channel parameters

6.10.3.4.7.3.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.3.4.6.1.2.2.2.

6.10.3.4.7.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.7.4.1 Uplink

6.10.3.4.7.4.1.1 Transport channel parameters

6.10.3.4.7.4.1.1.1 Transport channel parameters for E-DCH

6.10.3.4.7.4.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.3.4.7.1.1.1.1.1.

6.10.3.4.7.4.1.1.2 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.7.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.7.4.1.1.4 TFCS

See clause 6.10.3.4.1.4.1.1.3.

6.10.3.4.7.4.1.2 Physical channel parameters

6.10.3.4.7.4.1.2.1 Physical channel parameters on E-PUCH

See clause 6.10.3.4.7.1.1.2.1.

6.10.3.4.7.4.1.2.2 Physical channel parameters on DCH

See clause 6.10.3.4.1.4.1.2.

6.10.3.4.7.4.2 Downlink

See clause 6.10.3.4.6.3.2.

6.10.3.4.7.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.10.3.4.7.5.1 Uplink

6.10.3.4.7.5.1.1 Transport channel parameters

6.10.3.4.7.5.1.1.1 Transport channel parameters for E-DCH

MAC-e multiplexing between all MAC-d flows in the same MAC-e PDU shall be configured.

6.10.3.4.7.5.1.1.1.1 MAC-d flow #1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.3.4.7.1.1.1.1.1.

6.10.3.4.7.5.1.1.1.2 MAC-d flow #2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.3.4.7.1.1.1.1.1.

6.10.3.4.7.5.1.1.1.3 MAC-d flow #3 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.3.4.7.2.1.1.1.2.

6.10.3.4.7.5.1.2 Physical channel parameters

6.10.3.4.7.5.1.2.1 Physical channel parameters on E-PUCH

See clause 6.10.3.4.7.1.1.2.1.

6.10.3.4.7.5.2 Downlink

See clause 6.10.3.4.6.3.2.

6.11 Common Radio Bearer configurations for other test purposes

The common radio bearer configurations are used for functional testing of various UE functions. Only common configurations that are used by multiple test cases and are not covered by the reference radio bearer configurations in clause 6.10 are specified in the present clause. Radio bearer configurations only used by a single test case are specified in the actual test case itself.

NOTE If not specifically specified then the mid-value of the RM attribute value range as specified by the actual reference radio bearer configuration shall be applied for testing.

NOTE The order of tables and MAC-d flow numbering in this section may be different than the RB IDs and MAC-d flow IDs as defined in default messages in section 9.

6.11.1 Unacknowledged Mode Radio Bearer configuration (7 bit Length Indicator)

This configuration is based on the Interactive or background / UL:8 DL 8 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.23a) with the transport channels parameters of the RAB and TFCS defined as follows.

6.11.1.1 Transport channel parameters for the Uplink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		8 200
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 080
	Uplink: Max number of bits/radio frame before rate matching		270
	RM attribute		135 to 175

6.11.1.2 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.1.3 Transport channel parameters for the Downlink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		8 200
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336

		TF1, bits	1x336
	TTI, ms		40
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 080
	RM attribute		135 to 175

6.11.1.4 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.1a Streaming / unknown / DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6. 11.1a.1 Uplink

6.11.1a.1.1 Transport channel parameters

6.11.1a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.11.1a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.1a.1.1.3 TFCS

See clause 6.10.2.4.1.23a.1.1.3

6.11.1a.1.2 Physical channel parameters

See clause 6.10.2.4.1.23a.1.2

6.11.1a.2 Downlink

6.11.1a.2.1 Transport channel parameters

6.11.1a.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		64 800
	UM PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 076
	RM attribute		125 to 165

6.11.1a.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.11.1a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.1a.2.1.4 TFCS

See clause 6.10.2.4.1.58.2.1.4

6.11.1a.2.2 Physical channel parameters

See clause 6.10.2.4.1.58.2.2

6.11.1b Streaming / unknown / DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.1b.1 Uplink

6.11.1b.1.1 Transport channel parameters

6.11.1b.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.1b.1.1.2 TFCS

See clause 6.10.2.4.1.2.1.1.3

6.11.1b.1.2 Physical channel parameters

See clause 6.10.2.4.1.2.1.2

6.11.1b.2 Downlink

6.11.1b.2.1 Transport channel parameters

6.11.1b.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

See Clause 6.10.2.4.xx.2.1.1

6.11.1b.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.1b.2.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, DCCH)= (TF0,TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1)

6.11.1b.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.11.1c 8kbps RB for MBSFN MTCH (3.84 Mcps TDD)

6.11.1c.1 Transport channel parameters

6.11.1c.1.1 Transport channel parameters for 8 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		8200
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		345
	TFS	TF0, bits	0x345
		TF1, bits	1x345
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1095
	Max number of bits/radio frame before rate matching		274
	RM attribute		128

6.11.1c.1.2 TFCS

TFCS size	2
TFCS	8 kbps RAB =TF0, TF1

6.11.1c.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF16 x 1 code x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	248 bits
	TFCI code word	(16,5)
	Puncturing limit	0.84

6.11.1d 8kbps RB for MBSFN MTCH (7.68 Mcps TDD)

6.11.1d.1 Transport channel parameters

6.11.1d.1.1 Transport channel parameters for 8 kbps PS RAB

See clause 6.11.1c.1.1

6.11.1d.1.2 TFCS

TFCS size	2
TFCS	8 kbps RAB =TF0, TF1

6.11.1d.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF32 x 1 code x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	248 bits
	TFCI code word	(16,5)
	Puncturing limit	0.84

6.11.2 Unacknowledged Mode Radio Bearer configuration (15 bit Length Indicator)

This configuration is based on the Interactive or background / UL:64 DL 64 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.26) with the transport channels parameters of the RAB defined as followed.

6.11.2.1 Transport channel parameters for the Uplink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		1 336
	Max data rate, bps		66 800
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		1 344
	TFS	TF0, bits	0x1344
		TF1, bits	1x1344
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 092
	Uplink: Max number of bits/radio frame before rate matching		2 046
	RM attribute		130 to 170

6.11.2.2 Transport channel parameters for the Downlink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		1 336
	Max data rate, bps		66 800
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		1 344
	TFS	TF0, bits	0x1344
		TF1, bits	1x1344
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 092
	RM attribute		130 to 170

6.11.2a 64kbps RB for MBSFN MTCH (3.84 Mcps TDD)

6.11.2a.1 Transport channel parameters

6.11.2a.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		1336
	Max data rate, bps		66800
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		1353
	TFS	TF0, bits	0x1353
		TF1, bits	1x1353
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4119
	Max number of bits/radio frame before rate matching		2060
	RM attribute		128

6.11.2a.1.2 TFCS

TFCS size	2
TFCS	64 kbps RAB =TF0, TF1

6.11.2a.2 Physical channel parameters

S-CCPCH	Midamble	320 chips (burst type 4)
	Codes and time slots	SF16 x 8 codes x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	2096 bits
	TFCI code word	(16,5)
	Puncturing limit	1

6.11.2b 64kbps RB for MBSFN MTCH (7.68 Mcps TDD)

6.11.2b.1 Transport channel parameters

6.11.2b.1.1 Transport channel parameters for 64 kbps PS RAB

See clause 6.11.2a.1.1

6.11.2b.1.2 TFCS

TFCS size	2
TFCS	64 kbps RAB =TF0, TF1

6.11.2b.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	2096 bits
	TFCI code word	(16,5)
	Puncturing limit	1

6.11.3 Acknowledged Mode Radio Bearer configuration (7 bit Length Indicator)

This configuration is based on the Interactive or background / UL:8 DL 8 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.23a) with the transport channels parameters of the RAB and TFCS defined as follows.

6.11.3.1 Transport channel parameters for the Uplink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		128
	Max data rate, bps		6 400
	UMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		144
	TFS	0x144	0x144
		1x144	1x144
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		504
	Uplink: Max number of bits/radio frame before rate matching		252
	RM attribute		135 to 175

6.11.3.2 TFCS

TFCS size	4
TFCS	(RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.3.3 Transport channel parameters for the Downlink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		128
	Max data rate, bps		6 400
	UMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		144
	TFS	0x144	0x144
		1x144	1x144
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		504
	RM attribute		135 to 175

6.11.3.4 TFCS

TFCS size	4
TFCS	(RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.4 Acknowledged Mode Radio Bearer configuration (15 bit Length Indicator)

This configuration is based on the Interactive or background / UL:64 DL 64 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.26) with the transport channels parameters of the RAB defined as followed.

6.11.4.1 Transport channel parameters for the Uplink RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		1 328
	Max data rate, bps		66 400
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		1 344
	TFS	TF0, bits	0x1344
		TF1, bits	1x1344
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 092
	Uplink: Max number of bits/radio frame before rate matching		2 046
	RM attribute		130 to 170

6.11.4.2 Transport channel parameters for the Downlink RAB

Higher layer	RAB/Signalling RB		**RAB**
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		1 328
	Max data rate, bps		66 400
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		1 344
	TFS	TF0, bits	0x1344
		TF1, bits	1x1344
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 092
	RM attribute		130 to 170

6.11.4a Reference Radio Bearer configurations used in MAC-hs testing

6.11.4a.1 5 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB

This reference radio bearer configuration is used by the MAC-hs test case 7.1.5.2 in 3GPP TS 34.123-1 [1].

6.11.4a.1.1 Uplink

6.11.4a.1.1.1 Uplink Transport channel parameters for DCH

6.11.4a.1.1.1.1 Transport channel parameters for 5 x Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		**RB5**	**RB6**	**RB7**	**RB8**	**RB9**
RLC	Logical channel type		DTCH	DTCH	DTCH	DTCH	DTCH
	RLC mode		UM	UM	AM	AM	AM
	Payload sizes, bit		328	328	320	320	320
	Max data rate, bps		8 200	8 200	8 000	8 000	8 000
	UMD/AMD PDU header, bit		8	8	16	16	16
MAC	MAC header, bit		4	4	4	4	4
	MAC multiplexing		5 logical channel multiplexing				
Layer 1	TrCH type		DCH				
	TB sizes, bit		340				
	TFS	TF0, bits	0x340				
		TF1, bits	1x340				
	TTI, ms		40				
	Coding type		TC				
	CRC, bit		16				
	Max number of bits/TTI after channel coding		1 080				
	Uplink: Max number of bits/radio frame before rate matching		270				
	RM attribute		135 to 175				

6.11.4a.1.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4a.1.1.1.3 Uplink TFCS

TFCS size	4
TFCS	(5x8 kbps PS RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.4a.1.1.2 Uplink physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.11.4a.1.2 Downlink

6.11.4a.1.2.1 Transport channel parameters for HS-DSCH

6.11.4a.1.2.1.1 MAC-d flow #1 parameters for 2 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	**RB5**	**RB6**
RLC	Logical channel type	DTCH	DTCH
	RLC mode	UM	UM
	Payload sizes, bit	328	328
	Max data rate, bps	depends on UE category	depends on UE category
	UMD PDU header, bit	8	8
MAC-d	MAC-d header, bit	4	4
	MAC multiplexing	2 logical channel multiplexing	
	MAC-d PDU size, bit	340	

6.11.4a.1.2.1.2 MAC-d flow #2 parameters for 2 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	**RB7**	**RB8**
RLC	Logical channel type	DTCH	DTCH
	RLC mode	AM	AM
	Payload sizes, bit	320	320
	Max data rate, bps	depends on UE category	depends on UE category
	AMD PDU header, bit	16	16
MAC-d	MAC-d header, bit	4	4
	MAC multiplexing	2 logical channel multiplexing	
	MAC-d PDU size, bit	340	

6.11.4a.1.2.1.3 MAC-d flow#3 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	**RB9**
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	depends on UE category
	AMD PDU header, bit	16
MAC-d	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336

6.11.4a.1.2.1.4 MAC-hs and Layer 1 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

MAC-hs	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24

6.11.4a.1.2.2 Downlink Transport channel parameters for DCH

6.11.4a.1.2.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4a.1.2.2.2 Downlink TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.11.4a.1.2.3 Downlink physical channel parameters

6.11.4a.1.2.3.1 Downlink physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4a.1.2.3.2 Physical channel parameters on HS-PDSCH

UE HS-DSCH Physical Layer:

HS-PDSCH	Number of processes	2
	Process memory size	Split equally among all processes
	Max Data Rate	Depending on UE category

6.11.4b Interactive or background / UL: 0 kbps DL: 0 kbps PS RAB

This reference radio bearer configuration is used by the RRC test case 8.4.1.43 in 3GPP TS 34.123-1 [1].

6.11.4b.1 Uplink

6.11.4b.1.1 Uplink Transport channel parameters for DCH

6.11.4b.1.1.1 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		CC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Uplink: Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.11.4b.1.1.2 Uplink TFCS

TFCS size	2
TFCS	(0 kbps RAB, DCCH)= (TF0, TF0), (TF0, TF1)

6.11.4b.1.2 Uplink physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.11.4b.2 Downlink

6.11.4b.2.1 Downlink Transport channel parameters for DCH

6.11.4b.2.1.1 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		CC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	RM attribute		130 to 170

6.11.4b.2.1.1.2 Downlink TFCS

TFCS size	2
TFCS	(0 kbps RAB, DCCH)= (TF0, TF0), (TF0, TF1)

6.11.4b.2.2 Downlink physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		128
	DPCCH	Number of TFCI bits/slot	2
		Number of TPC bits/slot	2
		Number of Pilot bits/slot	4
	DPDCH	Number of data bits/slot	34
		Number of data bits/frame	510

6.11.4c Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.4c.1 Uplink

6.11.4c.1.1 Transport channel parameters

6.11.4c.1.1.1 Transport channel parameters for E-DCH

6.11.4c.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

		Alt 1 Fixed RLC + MAC-e/es (Rel-6 and later) NOTE 2	**Alt 2 Fixed RLC + MAC-i/is (Rel-8 and later releases) NOTE 2**	**Alt 3 Flexible RLC + MAC-i/is (Rel-8 and later releases) NOTE 2**
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	UM		
	Payload sizes, bit	328	328	Flexible up to 12000
	Max data rate, bps	Depends on UE category and TTI		
	UMD PDU header, bit	8		
MAC	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	336	336	Flexible
	MAC type	MAC-e/es	MAC-i/is	MAC-i/is
	MAC-e/es / MAC-i/is header fixed part, bit	18	24	24
Layer 1	TrCH type	E-DCH		
	TTI	10ms (alt. 2ms) (NOTE 1)		
	Coding type	TC		
	CRC, bit	24		

NOTE 1: The support of 2ms TTI depends on the UE category
NOTE 2: Alternative 1 with Fixed RLC + MAC-e/es is the default configuration. For test cases that use alternatives 2 (Fixed RLC + MAC-i/is) or 3 (Flexible RLC + MAC-i/is) then this shall be explicitely stated in the test case.

6.11.4c.1.1.1.2 Transport channel parameters for DCH

6.11.4c.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4c.1.2 Physical channel parameters

6.11.4c.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1

6.11.4c.1.2.2 Physical channel parameters for DPCH

See clause 6.10.2.4.1.2.1.2

6.11.4c.2 Downlink

6.11.4c.2.1 Transport channel parameters

6.11.4c.2.1.1 Transport channel parameters for HS-DSCH

6.11.4c.2.1.1.1 MAC-d flow parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	depends on UE category NOTE1
	UMD PDU header, bit	8
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).		

6.11.4c.2.1.1.2 Transport channel parameters for DCH

6.11.4c.2.1.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4c.2.1.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.11.4c.2.2 Physical channel parameters

6.11.4c.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4c.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4d Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + DL: 3.4 kbps SRBs for DCCH on E-DCH and DCH

6.11.4d.1 Uplink

6.11.4d.1.1 Transport channel parameters

6.11.4d.1.1.1 Transport channel parameters for E-DCH

6.11.4d.1.1.1.1 MAC-d flow#1 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2

6.11.4d.1.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1.

6.11.4d.1.1.1.3 MAC-d flow#3 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1.

6.11.4d.1.2 Physical channel parameters

6.11.4d.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.11.4d.2 Downlink

6.11.4d.2.1 Transport channel parameters

6.11.4d.2.1.1 Transport channel parameters for HS-DSCH

6.11.4d.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4c.2.1.1.1.

6.11.4d.2.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4c.2.1.1.1.

6.11.4d.2.1.2 Transport channel parameters for DCH

6.11.4d.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4d.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.11.4d.2.2 Physical channel parameters

6.11.4d.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4d.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4e Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.4e.1 Uplink

6.11.4e.1.1 Transport channel parameters

6.11.4e.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		65 800
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Uplink: Max number of bits/radio frame before rate matching		2 118
	RM attribute		130 to 170

6.11.4e.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4e.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.11.4e.1.2 Physical channel parameters

DPCH Uplink	Min spreading factor	16
	Max number of DPDCH data bits/radio frame	2 400
	Puncturing Limit	0.96

6.11.4e.2 Downlink

6.11.4e.2.1 Transport channel parameters

6.11.4e.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		65 800
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	RM attribute		130 to 170

6.11.4e.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4e.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.11.4e.2.2 Physical channel parameters

DPCH Downlink	DTX position		Flexible
	Spreading factor		32
	DPCCH	Number of TFCI bits/slot	8
		Number of TPC bits/slot	4
		Number of Pilot bits/slot	8
	DPDCH	Number of data bits/slot	140
		Number of data bits/frame	2 100

6.11.4f Reference Radio Bearer configurations used in MAC-ehs testing

6.11.4f.1 3 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB

This reference radio bearer configuration is used by the MAC-ehs test case 7.1.x.x in 3GPP TS 34.123-1 [1].

6.11.4f.1.1 Uplink

6.11.4f.1.1.1 Uplink Transport channel parameters for DCH

6.11.4f.1.1.1.1 Transport channel parameters for 3 x Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		**RB5**	**RB6**	**RB7**
RLC	Logical channel type		DTCH	DTCH	DTCH
	RLC mode		UM	UM	UM
	Payload sizes, bit		328	328	320
	Max data rate, bps		8 200	8 200	8 000
	UMD/AMD PDU header, bit		8	8	8
MAC	MAC header, bit		4	4	4
	MAC multiplexing		3 logical channel multiplexing		
Layer 1	TrCH type		DCH		
	TB sizes, bit		340		
	TFS	TF0, bits	0x340		
		TF1, bits	1x340		
	TTI, ms		40		
	Coding type		TC		
	CRC, bit		16		
	Max number of bits/TTI after channel coding		1 080		
	Uplink: Max number of bits/radio frame before rate matching		270		
	RM attribute		135 to 175		

6.11.4f.1.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4f.1.1.1.3 Uplink TFCS

TFCS size	4
TFCS	(5x8 kbps PS RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.4f.1.1.2 Uplink physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.11.4f.1.2 Downlink

6.11.4f.1.2.1 Transport channel parameters for HS-DSCH

6.11.4f.1.2.1.1 parameters for 3 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	**RB5**	**RB6**	**RB7**
RLC	Logical channel type	DTCH	DTCH	DTCH
	RLC mode	UM	UM	UM
	Payload sizes, bit	328	328	328
	Max data rate, bps	depends on UE category		
	UMD PDU header, bit	8	8	8
MAC-d	MAC-d header, bit	None		
	MAC multiplexing	None		
	MAC-d PDU size, bit	336		

6.11.4f.1.2.1.2 MAC-ehs and Layer 1 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

MAC-ehs	MAC-ehs header fixed part, bit	FFS
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC

	CRC, bit	24

6.11.4f.1.2.2 Downlink Transport channel parameters for DCH

6.11.4f.1.2.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4f.1.2.2.2 Downlink TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.11.4f.1.2.3 Downlink physical channel parameters

6.11.4f.1.2.3.1 Downlink physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4f.1.2.3.2 Physical channel parameters on HS-PDSCH

UE HS-DSCH Physical Layer:

HS-PDSCH	Number of processes	2
	Process memory size	Split equally among all processes
	Max Data Rate	Depending on UE category

6.11.4f.2 1 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB

This reference radio bearer configuration is used by the MAC-ehs test case 7.1.x.x in 3GPP TS 34.123-1 [1].

6.11.4f.2.1 Uplink

6.11.4f.2.1.1 Uplink Transport channel parameters for DCH

6.11.4f.2.1.1.1 Transport channel parameters for 1 x Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		**RB5**
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		8 200
	UMD/AMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 080
	Uplink: Max number of bits/radio frame before rate matching		270
	RM attribute		135 to 175

6.11.4f.2.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4f.2.1.1.3 Uplink TFCS

TFCS size	4
TFCS	(8 kbps PS RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)

6.11.4f.2.1.2 Uplink physical channel parameters

DPCH Uplink	Min spreading factor	64
	Max number of DPDCH data bits/radio frame	600
	Puncturing Limit	1.0

6.11.4f.2.2 Downlink

6.11.4f.2.2.1 Transport channel parameters for HS-DSCH

6.11.4f.2.2.1.1 parameters for 1 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	**RB5**
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	depends on UE category
	UMD PDU header, bit	8
MAC-d	MAC-d header, bit	None
	MAC multiplexing	None
	MAC-d PDU size, bit	336
MAC-ehs	MAC-ehs header fixed part, bit	24
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24

6.11.4f.2.2.2 Downlink Transport channel parameters for DCH

6.11.4f.2.2.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4f.2.2.2.2 Downlink TFCS

TFCS size	2
TFCS	SRBs for DCCH = TF0, TF1

6.11.4f.2.2.3 Downlink physical channel parameters

6.11.4f.2.2.3.1 Downlink physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4f.2.2.3.2 Physical channel parameters on HS-PDSCH

UE HS-DSCH Physical Layer:

HS-PDSCH	Number of processes	2
	Process memory size	Split equally among all processes
	Max Data Rate	Depending on UE category

6.11.4g Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.4g.1 Uplink

6.11.4g.1.1 Transport channel parameters

6.11.4g.1.1.1 Transport channel parameters for E-DCH

6.11.4g.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	Depends on UE category and TTI
	UMD PDU header, bit	8
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	10ms
	Coding type	TC
	CRC, bit	24

6.11.4g.1.1.1.2 Transport channel parameters for DCH

6.11.4g.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.4g.1.2 Physical channel parameters

6.11.4g.1.2.1 Physical channel parameters on E-PUCH

See clause 6.10.3.4.7.1.1.2.1

6.11.4g.1.2.2 Physical channel parameters for DPCH

See clause 6.10.3.4.1.2.1.2

6.11.4g.2 Downlink

6.11.4g.2.1 Transport channel parameters

6.11.4g.2.1.1 Transport channel parameters for HS-DSCH

6.11.4g.2.1.1.1 MAC-d flow parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	depends on UE category NOTE1
	UMD PDU header, bit	8
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH

	TTI	10 ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).		

6.11.4g.2.1.1.2 Transport channel parameters for DCH

6.11.4g.2.1.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.4g.2.1.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.4g.2.2 Physical channel parameters

6.11.4g.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.11.4g.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.11.4h Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + DL: 3.4 kbps SRBs for DCCH on E-DCH and DCH for 3.84Mcps TDD

6.11.4h.1 Uplink

6.11.4h.1.1 Transport channel parameters

6.11.4h.1.1.1 Transport channel parameters for E-DCH

6.11.4h.1.1.1.1 MAC-d flow#1 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.3.4.7.2.1.1.1.2

6.11.4h.1.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4g.1.1.1.1.

6.11.4h.1.1.1.3 MAC-d flow#3 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4g.1.1.1.1.

6.11.4h.1.2 Physical channel parameters

6.11.4h.1.2.1 Physical channel parameters on E-PUCH

See clause 6.10.3.4.7.1.1.2.1.

6.11.4h.2 Downlink

6.11.4h.2.1 Transport channel parameters

6.11.4h.2.1.1 Transport channel parameters for HS-DSCH

6.11.4h.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4g.2.1.1.1.

6.11.4h.2.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4g.2.1.1.1.

6.11.4h.2.1.2 Transport channel parameters for DCH

6.11.4h.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.4h.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.4h.2.2 Physical channel parameters

6.11.4h.2.2.1 Physical channel parameters on DPCH

See clause 6.10.3.4.1.2.2.2.

6.11.4h.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.3.4.6.1.2.2.2.

6.11.4i Reference Radio Bearer configurations used in CPC testing

6.11.4i.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / UM PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.11. 4i.1.1 Uplink

6.11. 4i.1.1.1 Transport channel parameters

6.11. 4i.1.1.1.1 Transport channel parameters for E-DCH

6.11. 4i.1.1.1.1.1 MAC-d flow#1parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1

6.11. 4i.1.1.1.1.2 MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2

6.11. 4i.1.1.2 Physical channel parameters

6.11. 4i.1.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1

6.11. 4i.1.2 Downlink

6.11. 4i.1.2.1 Transport channel parameters

6.11. 4i.1.2.1.1 Transport channel parameters for HS-DSCH

6.11. 4i.1.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4c.2.1.1.1

6.11. 4i.1.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2.

6.11. 4i.1.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.11. 4i.1.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4j Reference Radio Bearer configurations used in Improved L2 testing

6.11.4j.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / UM PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.4j.1.1 Uplink

See clause 6.11.4e.1

6.11.4j.1.2 Downlink

6.11. 4j.1.2.1 Transport channel parameters

6.11. 4j.1.2.1.1 Transport channel parameters for HS-DSCH

6.11. 4j.1.2.1.1.1 MAC-d flow parameters for DL: [max bit rate depending on UE category] kbps / PS RAB

		Flexible RLC + MAC-ehs (Rel-7 and later releases)
Higher Layer	RAB/Signalling RB	**RAB**
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	Flexible up to 12000
	Max data rate, bps	depends on UE category NOTE1
	UMD PDU header, bit	8
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	Flexible
	MAC-hs Type	MAC-ehs
	MAC-hs/ehs header fixed part, bit	24
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24
	Applicable modulation schemes	QPSK, 16QAM, 64QAM
	Applicable with MIMO	Yes
NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).		

6.11.4j.1.2.1.1.2 Transport channel parameters for DCH

6.11.4j.1.2.1.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4j.1.2.1.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.11.4j.1.2.2 Physical channel parameters

6.11.4j.1.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4j.1.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4j.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] with Flexible RLC and MAC-ehs SRBs for DCCH on E-DCH and HS-DSCH

6.11.4j.2.1 Uplink

See clause 6.10.2.4.6.1.1.

6.11.4j.2.1.2 Physical channel parameters

6.11.4j.2.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.11.4j.2.2 Downlink

6.11.4j.2.2.1 Transport channel parameters

6.11.4j.2.2.1.1 Transport channel parameters for HS-DSCH

6.11.4j.2.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.11.4j.2.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

Higher layer	RAB/signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	Flexible up to 12000			
	Max data rate, bps	Depends on UE category (NOTE 1)			
	AMD/UMD PDU header, bit	8	16	16	16
MAC	MAC header, bit	4	4	4	4
	MAC multiplexing	N/A			
	MAC-d PDU size, bit	Flexible			
	MAC-hs Type	MAC-ehs			
	MAC-ehs header fixed part, bit	24			
Layer 1	TrCH type	HS-DSCH			
	TTI, ms	2 ms			
	Coding type	TC			
	CRC, bit	24			
	Applicable modulation schemes	QPSK, 16QAM, 64QAM			
	Applicable with MIMO	Yes			

6.11.4j.2.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.11.4j.2.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.2.4.5.1.2.2.2.

6.11.4k Reference Radio Bearer configurations used in Improved L2 UL testing

6.11.4k.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.4k.1.1 Uplink

6.11.4k.1.1.1 Transport channel parameters

6.11.4k.1.1.1.1 Transport channel parameters for E-DCH

6.11.4k.1.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3.

6.11.4k.1.1.1.1.2 Transport channel parameters for DCH

6.11.4k.1.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.11.4k.1.1.2 Physical channel parameters

6.11.4k.1.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1

6.11.4k.1.1.2.2 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.1.2

6.11.4k.1.2 Downlink

6.11.4k.1.2.1 Transport channel parameters

6.11.4k.1.2.1.1 Transport channel parameters for HS-DSCH

6.11.4k.1.2.1.1.1 MAC-d flow parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	Flexible up to 12000
	Max data rate, bps	depends on UE category NOTE1
	UMD PDU header, bit	8
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	Flexible
	MAC-ehs header fixed part, bit	24
Layer 1	TrCH type	HS-DSCH
	TTI	2 ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-ehs PDU (see 3GPP TS 25.321 [38]).		

6.11.4k.1.2.2 Physical channel parameters

6.11.4k.1.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.5.1.2.

6.11.4k.1.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4k.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.11.4k.2.1 Uplink

6.11.4k.2.1.1 Transport channel parameters

6.11.4k.2.1.1.1 Transport channel parameters for E-DCH

6.11.4k.2.1.1.1.1 MAC-d flow#1 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2, alt 2

6.11.4k.2.1.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3

6.11.4k.2.1.2 Physical channel parameters

6.11.4k.2.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1

6.11.4k.2.2 Downlink

6.11.4k.2.2.1 Transport channel parameters

6.11.4k.2.2.1.1 Transport channel parameters for HS-DSCH

6.11.4k.2.2.1.1.1 MAC-d flow#0 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4k.1.2.1.1.1

6.11.4k.2.2.1.1.2 MAC-d flow#1 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2, alt 2.

6.11.4k.2.2.2 Physical channel parameters

6.11.4k.2.2.2.1 Physical channel parameters on DPCH

The physical channel configuration shall use F-DPCH.

6.11.4k.2.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.4k.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] with Flexible RLC and MAC-ehs SRBs for DCCH on E-DCH and HS-DSCH

6.11.4k.3.1 Uplink

6.11.4k.3.1.1 Transport channel parameters

6.11.4k.3.1.1.1 Transport channel parameters for E-DCH

6.11.4k.3.1.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3.

6.11.4k.3.1.1.1.2 MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2, alt 2.

6.11.4k.3.2.1.2 Physical channel parameters

6.11.4k.3.2.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.11.4k.3.2.2 Downlink

6.11.4k.3.2.2.1 Transport channel parameters

6.11.4k.3.2.2.1.1 Transport channel parameters for HS-DSCH

6.11.4k.3.2.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1, alt 3

6.11.4k.3.2.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.10.2.4.6.3.2.1.1.2, alt 2.

6.11.4k.3.2.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.11.4k.3.2.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.2.4.5.1.2.2.2.

6.11.4k.4 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + 3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.11.4k.4.1 Uplink

6.11.4k.4.1.1 Transport channel parameters

6.11.4k.4.1.1.1 Transport channel parameters for E-DCH

6.11.4k.4.1.1.1.1 MAC-d flow#0 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2, alt 2.

6.11.4k.4.1.1.1.2 MAC-d flow#1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3.

6.11.4k.4.1.1.1.3 MAC-d flow#2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3.

6.11.4k.4.1.1.1.4 MAC-d flow#3 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.4c.1.1.1.1, alt 3.

6.11.4k.4.1.2 Physical channel parameters

6.11.4k.4.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.11.4k.4.2 Downlink

6.11.4k.4.2.1 Transport channel parameters

6.11.4k.4.2.1.1 Transport channel parameters for HS-DSCH

6.11.4k.4.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4k.1.2.1.1.1.

6.11.4k.4.2.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4k.1.2.1.1.1.

6.11.4k.4.2.1.1.2 MAC-d flow#3 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.4k.2.1.1.1.

6.11.4k.4.2.1.2 Transport channel parameters for DCH

6.11.4k.4.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4k.4.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2

6.11.4k.4.2.2 Physical channel parameters

6.11.4k.4.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4k.4.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.11.5 Reference Radio Bearer configurations used in Radio Bearer testing for 1.28 Mcps TDD

6.11.5.1 RABs and signalling RBs

See clause 6.10.3.1.

6.11.5.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH.

1a) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (Multiframe).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

2a) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH (Multiframe).

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75)kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void.

19) Void.

20) Void.

21) Void.

22) Void.

23) Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.(20 msTTI)

24) Void.

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Void.

37) Void.

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38e) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38f) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38g) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38h) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38i) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38j) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void.

47) Void.

48) Void.

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void.

55) Void.

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

57) Interactive or Background / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58) Streaming / Unknown / UL:16 DL:64 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

59) Reserved for future use

60) Reserved for future use

61) Conversational / Unknown / UL:8 DL:8 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

Combinations on PDSCH, SCCPCH, PUSCH and PRACH

1) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL:16.8 DL: 16 kbps SRBs for SHCCH.

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

1) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

2) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

Combinations on SCCPCH

1) Stand-alone SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2a) Interactive/Background 32 kbps PS RAB
+ Interactive/Background 32 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2b) SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH.

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3a) SRB for PCCH
+ SRB for CCCH
+ SRB for DCCH
+ SRB for BCCH.

4) RB for CTCH
+ SRB for CCCH
+ SRB for BCCH.

Combinations on PRACH

1) SRB for CCCH
+ SRBs for DCCH.

2) Interactive/Background 12.8 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

3) Interactive/Background 12.8 kbps PS RAB
+ Interactive/Background 12.8 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

Combinations on DPCH and HS-PDSCH

1) Interactive or background / UL:8 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

1a) Interactive or background / UL:8 (multiframe) DL: [max bit rate depending on UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH (multiframe) (REL-5)

2) Interactive or background / UL:16 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

2a) Interactive or background / UL:16(multiframe) DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)

3) Interactive or background / UL:32 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3a) Interactive or background / UL:32(multiframe) DL: [max bit rate depending on UE category] / PS RAB +UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)

4) Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5) Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

7) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

8) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.3 Example of linkage between RABs and services

See clause 6.10.3.3.

6.11.5.4 Typical radio parameter sets

6.11.5.4.1 Combinations on DPCH

6.11.5.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.5.4.1.1.1 Uplink

6.11.5.4.1.1.1.1 Transport channel parameters

6.11.5.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.1.1.1.2 TFCS

See clause 6.10.3.4.1.1.1.1.2.

6.11.5.4.1.1.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.1.2 Downlink

6.11.5.4.1.1.2.1 Transport channel parameters

6.11.5.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.1.2.1.2 TFCS

See clause 6.10.3.4.1.1.2.1.2.

6.11.5.4.1.1.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits

	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)

6.11.5.4.1.1a.1 Uplink

6.11.5.4.1.1a.1.1 Transport channel parameters

6.11.5.4.1.1a.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1a.1.1.1.

6.11.5.4.1.1a.1.1.2 TFCS

See clause 6.10.3.4.1.1a.1.1.2.

6.11.5.4.1.1a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.1a.2 Downlink

6.11.5.4.1.1a.2.1 Transport channel parameters

6.11.5.4.1.1a.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1a.2.1.1.

6.11.5.4.1.1a.2.1.2 TFCS

See clause 6.10.3.4.1.1a.2.1.2.

6.11.5.4.1.1a.2.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.2.1 Uplink

6.11.5.4.1.2.1.1 Transport channel parameters

6.11.5.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.2.1.1.2 TFCS

See clause 6.10.3.4.1.2.1.1.2.

6.11.5.4.1.2.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.2.2 Downlink

6.11.5.4.1.2.2.1 Transport channel parameters

6.11.5.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.2.2.1.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.1.2.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	160 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.2a Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH (multiframe)

6.11.5.4.1.2a.1 Uplink

6.11.5.4.1.2a.1.1 Transport channel parameters

6.11.5.4.1.2a.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH (multiframe)

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			

	Max number of bits/radio frame before rate matching	516
	RM attribute	155 to 165
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.		

6.11.5.4.1.2a.1.1.2 TFCS

See clause 6.10.3.4.1.2.1.1.2.

6.11.5.4.1.2a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	336 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bit
	SS / radio frame	2x2 bit
	Puncturing Limit	0.64
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.2a.2 Downlink

6.11.5.4.1.2a.2.1 Transport channel parameters

6.11.5.4.1.2a.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH (multiframe)

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0, 148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		516			
	RM attribute		155 to 165			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.11.5.4.1.2a.2.1.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.1.2a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	336 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits

	Puncturing Limit	0.64
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.11.5.4.1.3.1 Uplink

6.11.5.4.1.3.1.1 Transport channel parameters

6.11.5.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

See clause 6.10.3.4.1.3.1.1.1.

6.11.5.4.1.3.1.1.2 TFCS

See clause 6.10.3.4.1.3.1.1.2.

6.11.5.4.1.3.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	336 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bit
	SS / radio frame	2x2 bit
	Puncturing Limit	0.64
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.3.2 Downlink

6.11.5.4.1.3.2.1 Transport channel parameters

6.11.5.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

See clause 6.10.3.4.1.3.2.1.1.

6.11.5.4.1.3.2.1.2 TFCS

See clause 6.10.3.4.1.3.2.1.2.

6.11.5.4.1.3.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	336 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64
NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit.		

6.11.5.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.4.1 Uplink

6.11.5.4.1.4.1.1 Transport channel parameters

6.11.5.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.4.1.1.3 TFCS

See clause 6.10.3.4.1.4.1.1.3.

6.11.5.4.1.4.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.4.2 Downlink

6.11.5.4.1.4.2.1 Transport channel parameters

6.11.5.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.4.2.1.3 TFCS

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.1.4.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.4a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.4a.1 Uplink

6.11.5.4.1.4a.1.1 Transport channel parameters

6.11.5.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.4a.1.1.3 TFCS

See clause 6.10.3.4.1.4a.1.1.3.

6.11.5.4.1.4a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots

	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.4a.2 Downlink

6.11.5.4.1.4a.2.1 Transport channel parameters

6.11.5.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.4a.2.1.3 TFCS

See clause 6.10.3.4.1.4a.1.2.1.3.

6.11.5.4.1.4a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.5.1 Uplink

6.11.5.4.1.5.1.1 Transport channel parameters

6.11.5.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

See clause 6.10.3.4.1.5.1.1.1.

6.11.5.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.5.1.1.3 TFCS

See clause 6.10.3.4.1.5.1.1.3.

6.11.5.4.1.5.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.5.2 Downlink

6.11.5.4.1.5.2.1 Transport channel parameters

6.11.5.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

See clause 6.10.3.4.1.5.2.1.1.

6.11.5.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.5.2.1.3 TFCS

See clause 6.10.3.4.1.5.2.1.3.

6.11.5.4.1.5.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.5a Conversational / speech / UL:10.2 6.7 5.9 4.75 DL:10.2 6.7 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.5a.1 Uplink

6.11.5.4.1.5a.1.1 Transport channel parameters

6.11.5.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 6.7 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.5a.1.1.1.

6.11.5.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.5a.1.1.3 TFCS

See clause 6.10.3.4.1.5a.1.1.3.

6.11.5.4.1.5a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.5a.2 Downlink

6.11.5.4.1.5a.2.1 Transport channel parameters

6.11.5.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: 10.2 6.7 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.5a.2.1.1.

6.11.5.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.5a.2.1.3 TFCS

See clause 6.10.3.4.1.5a.2.1.3.

6.11.5.4.1.5a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.6.1 Uplink

6.11.5.4.1.6.1.1 Transport channel parameters

6.11.5.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

See clause 6.10.3.4.1.6.1.1.1.

6.11.5.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.6.1.1.3 TFCS

See clause 6.10.3.4.1.6.1.1.3.

6.11.5.4.1.6.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.6.2 Downlink

6.11.5.4.1.6.2.1 Transport channel parameters

6.11.5.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

See clause 6.10.3.4.1.6.2.1.1.

6.11.5.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.6.2.1.3 TFCS

See clause 6.10.3.4.1.6.2.1.3.

6.11.5.4.1.6.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.7.1 Uplink

6.11.5.4.1.7.1.1 Transport channel parameters

6.11.5.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

See clause 6.10.3.4.1.7.1.1.1.

6.11.5.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.7.1.1.3 TFCS

See clause 6.10.3.4.1.7.1.1.3.

6.11.5.4.1.7.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.7.2 Downlink

6.11.5.4.1.7.2.1 Transport channel parameters

6.11.5.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

See clause 6.10.3.4.1.7.2.1.1.

6.11.5.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.7.2.1.3 TFCS

See clause 6.10.3.4.1.7.2.1.3.

6.11.5.4.1.7.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.7a Conversational / speech / UL:7.4 6.7 5.9 4.75 DL:7.4 6.7 5.9 4.75 / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.7a.1 Uplink

6.11.5.4.1.7a.1.1 Transport channel parameters

6.11.5.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 6.7 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.7a.1.1.1.

6.11.5.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.7a.1.1.3 TFCS

See clause 6.10.3.4.1.7a.1.1.3.

6.11.5.4.1.7a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.7a.2 Downlink

6.11.5.4.1.7a.2.1 Transport channel parameters

6.11.5.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 6.7 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.7a.2.1.1.

6.11.5.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.7a.2.1.3 TFCS

See clause 6.10.3.4.1.7a.2.1.3.

6.11.5.4.1.7a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.8.1 Uplink

6.11.5.4.1.8.1.1 Transport channel parameters

6.11.5.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

See clause 6.10.3.4.1.8.1.1.1.

6.11.5.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.8.1.1.3 TFCS

See clause 6.10.3.4.1.8.1.1.3.

6.11.5.4.1.8.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.8.2 Downlink

6.11.5.4.1.8.2.1 Transport channel parameters

6.11.5.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

See clause 6.10.3.4.1.8.2.1.1.

6.11.5.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.8.2.1.3 TFCS

See clause 6.10.3.4.1.8.2.1.3.

6.11.5.4.1.8.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.9.1 Uplink

6.11.5.4.1.9.1.1 Transport channel parameters

6.11.5.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

See clause 6.10.3.4.1.9.1.1.1.

6.11.5.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.9.1.1.3 TFCS

See clause 6.10.3.4.1.9.1.1.3.

6.11.5.4.1.9.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72

6.11.5.4.1.9.2 Downlink

6.11.5.4.1.9.2.1 Transport channel parameters

6.11.5.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

See clause 6.10.3.4.1.9.2.1.1.

6.11.5.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.9.2.1.3 TFCS

See clause 6.10.3.4.1.9.2.1.3.

6.11.5.4.1.9.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72

6.11.5.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.5.4.1.10.1 Uplink

6.11.5.4.1.10.1.1 Transport channel parameters

6.11.5.4.1.10.1.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

See clause 6.10.3.4.1.10.1.1.1.

6.11.5.4.1.10.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.10.1.1.3 TFCS

See clause 6.10.3.4.1.10.1.1.3.

6.11.5.4.1.10.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.96

6.11.5.4.1.10.2 Downlink

6.11.5.4.1.10.2.1 Transport channel parameters

6.11.5.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

See clause 6.10.3.4.1.10.2.1.1.

6.11.5.4.1.10.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.10.2.1.3 TFCS

See clause 6.10.3.4.1.10.2.1.3.

6.11.5.4.1.10.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.96

6.11.5.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.5.4.1.11.1 Uplink

6.11.5.4.1.11.1.1 Transport channel parameters

6.11.5.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

See clause 6.10.3.4.1.11.1.1.1.

6.11.5.4.1.11.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.11.1.1.3 TFCS

See clause 6.10.3.4.1.11.1.1.3.

6.11.5.4.1.11.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.11.2 Downlink

6.11.5.4.1.11.2.1 Transport channel parameters

6.11.5.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

See clause 6.10.3.4.1.11.2.1.1.

6.11.5.4.1.11.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.11.2.1.3 TFCS

See clause 6.10.3.4.1.11.2.1.3.

6.11.5.4.1.11.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits

	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.12.1 Uplink

6.11.5.4.1.12.1.1 Transport channel parameters

6.11.5.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.12.1.1.1.

6.11.5.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.12.1.1.3 TFCS

See clause 6.10.3.4.1.12.1.1.3.

6.11.5.4.1.12.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.12.2 Downlink

6.11.5.4.1.12.2.1 Transport channel parameters

6.11.5.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.12.2.1.1.

6.11.5.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.12.2.1.3 TFCS

See clause 6.10.3.4.1.12.2.1.3.

6.11.5.4.1.12.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.13.1 Uplink

6.11.5.4.1.13.1.1 Transport channel parameters

6.11.5.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.13.1.1.3 TFCS

See clause 6.10.3.4.1.13.1.1.3.

6.11.5.4.1.13.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.13.2 Downlink

6.11.5.4.1.13.2.1 Transport channel parameters

6.11.5.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.13.2.1.3 TFCS

See clause 6.10.3.4.1.13.2.1.3.

6.11.5.4.1.13.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.14.1 Uplink

6.11.5.4.1.14.1.1 Transport channel parameters

6.11.5.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

See clause 6.10.3.4.1.14.1.1.1.

6.11.5.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.14.1.1.3 TFCS

See clause 6.10.3.4.1.14.1.1.3.

6.11.5.4.1.14.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.44

6.11.5.4.1.14.2 Downlink

6.11.5.4.1.14.2.1 Transport channel parameters

6.11.5.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

See clause 6.10.3.4.1.14.2.1.1.

6.11.5.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.14.2.1.3 TFCS

See clause 6.10.3.4.1.14.2.1.3.

6.11.5.4.1.14.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.44

6.11.5.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.15.1 Uplink

6.11.5.4.1.15.1.1 Transport channel parameters

6.11.5.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

See clause 6.10.3.4.1.15.1.1.1.

6.11.5.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.15.1.1.3 TFCS

See clause 6.10.3.4.1.15.1.1.3.

6.11.5.4.1.15.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.15.2 Downlink

6.11.5.4.1.15.2.1 Transport channel parameters

6.11.5.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

See clause 6.10.3.4.1.15.2.1.1.

6.11.5.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.15.2.1.3 TFCS

See clause 6.10.3.4.1.15.2.1.3.

6.11.5.4.1.15.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 3 code x 2 time slots
	Max. Number of data bits / radio rame	504 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.76

6.11.5.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.16.1 Uplink

6.11.5.4.1.16.1.1 Transport channel parameters

6.11.5.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.16.1.1.1.

6.11.5.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.16.1.1.3 TFCS

See clause 6.10.3.4.1.16.1.1.3.

6.11.5.4.1.16.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.56

6.11.5.4.1.16.2 Downlink

6.11.5.4.1.16.2.1 Transport channel parameters

6.11.5.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.16.2.1.1.

6.11.5.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.16.2.1.3 TFCS

See clause 6.10.3.4.1.16.2.1.3.

6.11.5.4.1.16.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.56

6.11.5.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.17.1 Uplink

6.11.5.4.1.17.1.1 Transport channel parameters

6.11.5.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.11.5.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.17.1.1.3 TFCS

See clause 6.10.3.4.1.17.1.1.3.

6.11.5.4.1.17.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.17.2 Downlink

6.11.5.4.1.17.2.1 Transport channel parameters

6.11.5.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.11.5.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.17.2.1.3 TFCS

See clause 6.10.3.4.1.17.2.1.3.

6.11.5.4.1.17.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits

	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.18 Void

6.11.5.4.1.19 Void

6.11.5.4.1.20 Void

6.11.5.4.1.21 Void

6.11.5.4.1.22 Void

6.11.5.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23.1 Uplink

6.11.5.4.1.23.1.1 Transport channel parameters

6.11.5.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.11.5.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23.1.1.3 TFCS

See clause 6.10.3.4.1.23.1.1.3.

6.11.5.4.1.23.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 4 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	680bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48 (alt. 0.44)

6.11.5.4.1.23.2 Downlink

6.11.5.4.1.23.2.1 Transport channel parameters

6.11.5.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.23.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23.2.1.3 TFCS

See clause 6.10.3.4.1.23.2.1.3.

6.11.5.4.1.23.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 codes x 2 time slots
	Max. Number of data bits/radio frame	336 bits
	TFCI code word / radio frame	8 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits

	Puncturing Limit	0.76

6.11.5.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23a.1 Uplink

6.11.5.4.1.23a.1.1 Transport channel parameters

6.11.5.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23a.1.1.3 TFCS

See clause 6.10.3.4.1.23a.1.1.3.

6.11.5.4.1.23a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72 (alt 0.68)

6.11.5.4.1.23a.2 Downlink

See clause 6.11.5.4.1.23.2.

6.11.5.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23b.1 Uplink

6.11.5.4.1.23b.1.1 Transport channel parameters

6.11.5.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23b.1.1.3 TFCS

See clause 6.10.3.4.1.23b.1.1.3.

6.11.5.4.1.23b.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	688 bits
	TFCI code word / radio frame	16bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.92 alt (0.84)

6.11.5.4.1.23b.2 Downlink

6.11.5.4.1.23b.2.1 Transport channel parameters

6.11.5.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.11.5.4.1.23b.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23b.2.1.3 TFCS

See clause 6.10.3.4.1.23b.2.1.3.

6.11.5.4.1.23b.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 3 code x 2 time slots
	Max. Number of data bits / radio rame	512 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.23c Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23c.1 Uplink

6.11.5.4.1.23c.1.1 Transport channel parameters

6.11.5.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23c.1.1.1.

6.11.5.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23c.1.1.3 TFCS

See clause 6.10.3.4.1.23c.1.1.3.

6.11.5.4.1.23c.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 4 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	680bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48 (alt 0.44)

6.11.5.4.1.23c.2 Downlink

6.11.5.4.1.23c.2.1 Transport channel parameters

6.11.5.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23c.2.1.1.

6.11.5.4.1.23c.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23c.2.1.3 TFCS

See clause 6.10.3.4.1.23c.2.1.3.

6.11.5.4.1.23c.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits/radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.23d Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23d.1 Uplink

6.11.5.4.1.23d.1.1 Transport channel parameters

6.11.5.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23d.1.1.3 TFCS

See clause 6.10.3.4.1.23d.1.1.3.

6.11.5.4.1.23d.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 4 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48 (alt. 0.44)

6.11.5.4.1.23d.2 Downlink

6.11.5.4.1.23d.2.1 Transport channel parameters

6.11.5.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.23d.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23d.2.1.3 TFCS

See clause 6.10.3.4.1.23d.2.1.3.

6.11.5.4.1.23d.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits/radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.24 Void.

6.11.5.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.25.1 Uplink

See clause 6.11.5.4.1.23.1.

6.11.5.4.1.25.2 Downlink

6.11.5.4.1.25.2.1 Transport channel parameters

6.11.5.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.25.2.1.3 TFCS

See clause 6.10.3.4.1.25.2.1.3.

6.11.5.4.1.25.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit/ radio frame	0.56

6.11.5.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.26.1 Uplink

6.11.5.4.1.26.1.1 Transport channel parameters

6.11.5.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.26.1.1.3 TFCS

See clause 6.10.3.4.1.26.1.1.3.

6.11.5.4.1.26.1.2 Physical channel parameters

DPCH Uplink		Physical 1	Physical 2
	Modulation	QPSK	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits	2 792 bits
	TFCI code word / radio frame	16 bits	16 bits
	TPC / radio frame	2x2 bits	2x2 bits
	SS / radio frame	2x2 bits	2x2 bits
	Puncturing Limit	0.56 (alt 0.48)	1

6.11.5.4.1.26.2 Downlink

See clause 6.11.5.4.1.25.2.

6.11.5.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.27.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.27.2 Downlink

6.11.5.4.1.27.2.1 Transport channel parameters

6.11.5.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.27.2.1.3 TFCS

See clause 6.10.3.4.1.27.2.1.3.

6.11.5.4.1.27.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots
	Max. Number of data bits/radio frame	3 144 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.28.1 Uplink

6.11.5.4.1.28.1.1 Transport channel parameters

6.11.5.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.28.1.1.3 TFCS

See clause 6.10.3.4.1.28.1.1.3.

6.11.5.4.1.28.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 codes x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.28.2 Downlink

See clause 6.11.5.4.1.27.2.

6.11.5.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.29.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.29.2 Downlink

6.11.5.4.1.29.2.1 Transport channel parameters

6.11.5.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

See clause 6.10.3.4.1.29.2.1.1.

6.11.5.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.29.2.1.3 TFCS

See clause 6.10.3.4.1.29.2.1.3.

6.11.5.4.1.29.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots
	Max. Number of data bits/radio frame	3 144 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.30.1 Uplink

6.11.5.4.1.30.1.1 Transport channel parameters

6.11.5.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

See clause 6.10.3.4.1.30.1.1.1.

6.11.5.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.30.1.1.3 TFCS

See clause 6.10.3.4.1.30.1.1.3.

6.11.5.4.1.30.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	(SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots)	SF1 x 1code x 2 time slots
	Max. Number of data bits/radio frame	4 200 bits	4 188 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.72 (alt 0.64)	0.72 (alt 0.64)

6.11.5.4.1.30.2 Downlink

See clause 6.11.5.4.1.29.2.

6.11.5.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.31.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.31.2 Downlink

6.11.5.4.1.31.2.1 Transport channel parameters

6.11.5.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.11.5.4.1.31.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.31.2.1.3 TFCS

See clause 6.10.3.4.1.31.2.1.3.

6.11.5.4.1.31.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	5 608 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.56

6.11.5.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.32.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.32.2 Downlink

6.11.5.4.1.32.2.1 Transport channel parameters

6.11.5.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.3.4.1.32.2.1.1.

6.11.5.4.1.32.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.32.2.1.3 TFCS

See clause 6.10.3.4.1.32.2.1.3.

6.11.5.4.1.32.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF1 x 1 code x 6 time slots	SF1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	8 424 bits	8 412 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits

	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.64	0.64

6.11.5.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.33.1 Uplink

See clause 6.11.5.4.1.28.1.

6.11.5.4.1.33.2 Downlink

See clause 6.11.5.4.1.32.2.

6.11.5.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.34.1 Uplink

6.11.5.4.1.34.1.1 Transport channel parameters

6.11.5.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.3.4.1.34.1.1.1.

6.11.5.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.34.1.1.3 TFCS

See clause 6.10.3.4.1.34.1.1.3.

6.11.5.4.1.34.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF 1 x 1 code x 6 time slots	SF 1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	8 424 bits	8 412 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.64	0.64

6.11.5.4.1.34.2 Downlink

See clause 6.11.5.4.1.32.2.

6.11.5.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.35.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.35.2 Downlink

6.11.5.4.1.35.2.1 Transport channel parameters

6.11.5.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	1 704
	Max data rate, bps	2 048 000
	RLC header, bit	16

MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		1720
	TFS	TF0, bits	0x1720
		TF1, bits	1x1720
		TF2, bits	2x1720
		TF3, bits	4x1720
		TF4, bits	8x1720
		TF5, bits	12x1720
		TF6, bits	N/A (alt. 16x1720)
		TF7, bits	N/A (alt. 20x1720)
		TF8, bits	N/A (alt. 24x1720)
	TTI, ms		10 (alt. 20)
	Coding type		No coding
	CRC, bit		24
	Max number of bits/TTI after channel coding		20 928 (alt. 41 856)
	Max number of bits/radio frame before rate matching		20 928 (alt. 20 928)
	RM attribute		130 to 170

6.11.5.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.35.2.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(2 048 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))

6.11.5.4.1.35.2.2 Physical channel parameters

DPCH Downlink	Modulation	8PSK
	Codes and time slots / radio frame	SF1 x 1 code x 10 time slots
	Max. Number of data bits/radio frame	21 084 bits
	TFCI code word / radio frame	24 bits
	TPC / radio frame	2x3 bits
	SS / radio frame	2x3 bits
	Puncturing Limit	1

6.11.5.4.1.36 Void

6.11.5.4.1.37 Void

6.11.5.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38.1 Uplink

6.11.5.4.1.38.1.1 Transport channel parameters

6.11.5.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.11.5.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38.1.1.4 TFCS

See clause 6.10.3.4.1.38.1.1.4.

6.11.5.4.1.38.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72 (alt 0.68)

6.11.5.4.1.38.2 Downlink

6.11.5.4.1.38.2.1 Transport channel parameters

6.11.5.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38.2.1.4 TFCS

See clause 6.10.3.4.1.38.2.1.4.

6.11.5.4.1.38.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 3 codes x 2 time slots
	Max. Number of data bits/radio frame	504 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.44

6.11.5.4.1.38a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38a.1 Uplink

6.11.5.4.1.38a.1.1 Transport channel parameters

6.11.5.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.11.5.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38a.1.1.4 TFCS

See clause 6.10.3.4.1.38a.1.1.4.

6.11.5.4.1.38a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.38a.2 Downlink

6.11.5.4.1.38a.2.1 Transport channel parameters

6.11.5.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.11.5.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38a.2.1.4 TFCS

See clause 6.10.3.4.1.38a.2.1.4.

6.11.5.4.1.38a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.2.5.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38b.1 Uplink

6.11.5.4.1.38b.1.1 Transport channel parameters

6.11.5.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38b.1.1.4 TFCS

See clause 6.10.3.4.1.38b.1.1.4.

6.11.5.4.1.38b.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64 (alt. 0.60)

6.11.5.4.1.38b.2 Downlink

6.11.5.4.1.38b.2.1 Transport channel parameters

6.11.5.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38b.2.1.4 TFCS

See clause 6.10.3.4.1.38b.2.1.4.

6.11.5.4.1.38b.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38c.1 Uplink

6.11.5.4.1.38c.1.1 Transport channel parameters

6.11.5.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38c.1.1.4 TFCS

See clause 6.10.3.4.1.38c.1.1.4.

6.11.5.4.1.38c.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK

	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72 (alt 0.64) for TFCS size=18 0.80 (alt 0.72) for TFCS size=17

6.11.5.4.1.38c.2 Downlink

6.11.5.4.1.38c.2.1 Transport channel parameters

6.11.5.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38c.2.1.4 TFCS

See clause 6.10.3.4.1.38c.2.1.4.

6.11.5.4.1.38c.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72 (alt 0.64)

6.11.5.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38d.1 Uplink

6.11.5.4.1.38d.1.1 Transport channel parameters

6.11.5.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.11.5.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38d.1.1.4 TFCS

See clause 6.10.3.4.1.38d.1.1.4.

6.11.5.4.1.38d.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK	8PSK

	Codes and time slots / radio frame	(SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots)	SF1 x 1code x 2 time slots
	Max. Number of data bits/radio frame	4 200 bits	4 188 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.72 (alt 0.64)	0.72 (alt 0.64)

6.11.5.4.1.38d.2 Downlink

6.11.5.4.1.38d.2.1 Transport channel parameters

6.11.5.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.2.1.2.

6.11.5.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38d.2.1.4 TFCS

See clause 6.10.3.4.1.38d.2.1.4.

6.11.5.4.1.38d.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots
	Max. Number of data bits/radio frame	3 144 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.38e Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38e.1 Uplink

6.11.5.4.1.38e.1.1 Transport channel parameters

6.11.5.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.11.5.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38e.1.1.4 TFCS

See clause 6.10.3.4.1.38e.1.1.4.

6.11.5.4.1.38e.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF8 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.38e.2 Downlink

6.11.5.4.1.38e.2.1 Transport channel parameters

6.11.5.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.11.5.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38e.2.1.4 TFCS

See clause 6.10.3.4.1.38e.2.1.4.

6.11.5.4.1.38e.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 2 code x 2 time slots
	Max. Number of data bits / radio frame	328 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.38f Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38f.1 Uplink

6.11.5.4.1.38f.1.1 Transport channel parameters

6.11.5.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38f.1.1.4 TFCS

See clause 6.10.3.4.1.38f.1.1.4.

6.11.5.4.1.38f.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64 (alt 0.60)

6.11.5.4.1.38f.2 Downlink

6.11.5.4.1.38f.2.1 Transport channel parameters

6.11.5.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38f.2.1.4 TFCS

See clause 6.10.3.4.1.38f.2.1.4.

6.11.5.4.1.38f.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.38g Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38g.1 Uplink

6.11.5.4.1.38g.1.1 Transport channel parameters

6.11.5.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38g.1.1.4 TFCS

See clause 6.10.3.4.1.38g.1.1.4.

6.11.5.4.1.38g.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	1 368 bits (alt. 1 384 bits)
	TFCI code word / radio frame	32 bits (alt. 16 bits)
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.96 (alt 1.0)
NOTE: There are 32 bit and 16 bit TFCIs for the two cases.		

6.11.5.4.1.38g.2 Downlink

6.11.5.4.1.38g.2.1 Transport channel parameters

6.11.5.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.11.5.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38g.2.1.4 TFCS

See clause 6.10.3.4.1.38g.2.1.4.

6.11.5.4.1.38g.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 368 bits
	TFCI code word / radio frame	32 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1.0

6.11.5.4.1.38h Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38h.1 Uplink

6.11.5.4.1.38h.1.1 Transport channel parameters

6.11.5.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38h.1.1.4 TFCS

See clause 6.10.3.4.1.38h.1.1.4.

6.11.5.4.1.38h.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 368 bits
	TFCI code word / radio frame	32 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72 (alt 0.64)

6.11.5.4.1.38h.2 Downlink

6.11.5.4.1.38h.2.1 Transport channel parameters

6.11.5.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38h.2.1.4 TFCS

See clause 6.10.3.4.1.38h.2.1.4.

6.11.5.4.1.38h.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 368 bits
	TFCI code word / radio frame	32 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.72

6.11.5.4.1.38i Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38i.1 Uplink

6.11.5.4.1.38i.1.1 Transport channel parameters

6.11.5.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38i.1.1.4 TFCS

See clause 6.10.3.4.1.38i.1.1.4.

6.11.5.4.1.38i.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	(SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots)	SF1 x 1code x 2 time slots
	Max. Number of data bits/radio frame	4 184 bits	4 164 bits
	TFCI code word / radio frame	32 bits	48 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	1	1

6.11.5.4.1.38i.2 Downlink

6.11.5.4.1.38i.2.1 Transport channel parameters

6.11.5.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38i.2.1.4 TFCS

See clause 6.10.3.4.1.38i.2.1.4.

6.11.5.4.1.38i.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots
	Max. Number of data bits/radio frame	3 128 bits
	TFCI code word / radio frame	32 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	1

6.11.5.4.1.38j Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38j.1 Uplink

See clause 6.11.5.4.1.38i.1.

6.11.5.4.1.38j.2 Downlink

6.11.5.4.1.38j.2.1 Transport channel parameters

6.11.5.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38j.2.1.4 TFCS

See clause 6.10.3.4.1.38j.2.1.4.

6.11.5.4.1.38j.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots
	Max. Number of data bits/radio frame	3 128 bits
	TFCI code word / radio frame	32 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.39.1 Uplink

See clause 6.11.5.4.1.38.1.

6.11.5.4.1.39.2 Downlink

6.11.5.4.1.39.2.1 Transport channel parameters

6.11.5.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.39.2.1.4 TFCS

See clause 6.10.3.4.1.39.2.1.4.

6.11.5.4.1.39.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 10 codes x 2 time slots
	Max. Number of data bits/radio frame	1 736 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.56

6.11.5.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.40.1 Uplink

6.11.5.4.1.40.1.1 Transport channel parameters

See clause 6.10.3.4.1.40.1.1.

6.11.5.4.1.40.1.2 Physical channel parameters

6.11.5.4.1.40.1.2.1 Physical channel parameters (one CCTrCH case)

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots

	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.92 (alt. 0.84)

6.11.5.4.1.40.1.2.2 Physical channel parameters (two CCTrCH case)

6.11.5.4.1.40.1.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.1.2.

6.11.5.4.1.40.1.2.2.2 Physical channel parameters (Interactive or background)

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64 (alt. 0.56)

6.11.5.4.1.40.2 Downlink

6.11.5.4.1.40.2.1 Transport channel parameters

See clause 6.10.3.4.1.40.2.1.

6.11.5.4.1.40.2.2 Physical channel parameters

6.11.5.4.1.40.2.2.1 Physical channel parameters (one CCTrCH)

See clause 6.11.5.4.1.39.2.2.

6.11.5.4.1.40.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.40.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.40.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.64

6.11.5.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.41.1 Uplink

See clause 6.11.5.4.1.40.1.

6.11.5.4.1.41.2 Downlink

6.11.5.4.1.41.2.1 Transport channel parameters

See clause 6.10.3.4.1.41.2.1.

6.11.5.4.1.41.2.2 Physical channel parameters

6.11.5.4.1.41.2.2.1 Physical channel parameters (one CCTrCH case)

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 4 time slots	SF 16 x 12 codes x 2 time slots
	Max. Number of data bits/radio frame	3 144 bits	3 132 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.60	0.60

6.11.5.4.1.41.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.41.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.41.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots	SF 16 x 11 codes x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits	2868 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.64	0.64

6.11.5.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.42.1 Uplink

6.11.5.4.1.42.1.1 Transport channel parameters

See clause 6.10.3.4.1.42.1.1.

6.11.5.4.1.42.1.2 Physical channel parameters

See clause 6.10.3.4.1.40.1.2.1.

6.11.5.4.1.42.2 Downlink

6.11.5.4.1.42.2.1 Transport channel parameters

6.11.5.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.11.5.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.42.2.1.4 TFCS

See clause 6.10.3.4.1.42.2.1.4.

6.11.5.4.1.42.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF1 x 1code x 6 time slots	SF1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	8 408 bits	8 388 bits
	TFCI code word / radio frame	32 bits	48 bits
	TPC / radio frame	2x2 bits	2x3 bits

	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.80	0.80

6.11.5.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.43.1 Uplink

See clause 6.11.5.4.1.40.1.

6.11.5.4.1.43.2 Downlink

6.11.5.4.1.43.2.1 Transport channel parameters

See clause 6.10.3.4.1.43.2.1.

6.11.5.4.1.43.2.2 Physical channel parameters

6.11.5.4.1.43.2.2.1 Physical channel parameters (one CCTrCH)

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF 1 x 1 code x 6 time slots	SF 1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	8 408 bits	8 388 bits
	TFCI code word / radio frame	32 bits	48 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.60	0.60

6.11.5.4.1.43.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.43.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.43.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	(SF 1 x 1 code x 4 time slots) + (SF 16 x 10 codes x 2 time slots)	SF 1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	7 368 bits	8 412 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.56	0.64

6.11.5.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.44.1 Uplink

6.11.5.4.1.44.1.1 Transport channel parameters

6.11.5.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.44.1.1.4 TFCS

See clause 6.10.3.4.1.44.1.1.4.

6.11.5.4.1.44.1.2 Physical channel parameters

DPCH Uplink	Modulation	8PSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	4 188 bits
	TFCI code word / radio frame	24 bits
	TPC / radio frame	2x3 bits
	SS / radio frame	2x3 bits
	Puncturing Limit	0.80 (alt 0.72)

6.11.5.4.1.44.2 Downlink

6.11.5.4.1.44.2.1 Transport channel parameters

6.11.5.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.11.5.4.1.35.2.1.1.

6.11.5.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.44.2.1.4 TFCS

TFCS size	32 (alt. 50)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)= ((TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1)) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1))

For better understanding of the TFCS please note that the following combinations are not included in the table above:

- (TF2, TF1, TF1, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF2, TF1, TF1, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1).

6.11.5.4.1.44.2.2 Physical channel parameters

DPCH Downlink	Modulation	8PSK
	Codes and time slots / radio frame	SF 1 x 1 code x 10 time slots
	Max. Number of data bits/radio frame	21 060 bits
	TFCI code word / radio frame	48 bits
	TPC / radio frame	2x3 bits
	SS / radio frame	2x3 bits
	Puncturing Limit	1

6.11.5.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.45.1 Uplink

6.11.5.4.1.45.1.1 Transport channel parameters

6.11.5.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.11.5.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.45.1.1.4 TFCS

See clause 6.10.3.4.1.45.1.1.4.

6.11.5.4.1.45.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.45.2 Downlink

6.11.5.4.1.45.2.1 Transport channel parameters

6.11.5.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.11.5.4.1.45.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.45.2.1.4 TFCS

See clause 6.10.3.4.1.45.2.1.4.

6.11.5.4.1.45.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 2 time slots
	Max. Number of data bits/radio frame	1 560 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.56

6.11.5.4.1.46 Void

6.11.5.4.1.47 Void

6.11.5.4.1.48 Void

6.11.5.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.49.1 Uplink

6.11.5.4.1.49.1.1 Transport channel parameters

6.11.5.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.49.1.1.4 TFCS

See clause 6.10.3.4.1.49.1.1.4.

6.11.5.4.1.49.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.88

6.11.5.4.1.49.2 Downlink

6.11.5.4.1.49.2.1 Transport channel parameters

6.11.5.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.49.2.1.4 TFCS

See clause 6.10.3.4.1.49.2.1.4.

6.11.5.4.1.49.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 11 codes x 2 time slots
	Max. Number of data bits/radio frame	1 912 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.49a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL: 12.2 7.95 5.9 4.75 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.49a.1 Uplink

6.11.5.4.1.49a.1.1 Transport channel parameters

6.11.5.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.49a.1.1.4 TFCS

See clause 6.10.3.4.1.49a.1.1.4.

6.11.5.4.1.49a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.88

6.11.5.4.1.49a.2 Downlink

6.11.5.4.1.49a.2.1 Transport channel parameters

6.11.5.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.49.2.1.4 TFCS

See clause 6.10.3.4.1.49a.2.1.4.

6.11.5.4.1.49a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 11 codes x 2 time slots
	Max. Number of data bits/radio frame	1 912 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.50.1 Uplink

6.11.5.4.1.50.1.1 Transport channel parameters

6.11.5.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.50.1.1.3 TFCS

See clause 6.10.3.4.1.50.1.1.3.

6.11.5.4.1.50.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.50.2 Downlink

6.11.5.4.1.50.2.1 Transport channel parameters

6.11.5.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.50.2.1.3 TFCS

See clause 6.10.3.4.1.50.2.1.3.

6.11.5.4.1.50.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 15 codes x 2 time slots
	Max. Number of data bits/radio frame	2 616 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.48

6.11.5.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51.1 Uplink

6.11.5.4.1.51.1.1 Transport channel parameters

6.11.5.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51.1.1.4 TFCS

See clause 6.10.3.4.1.51.1.1.4.

6.11.5.4.1.51.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52 (alt. 0.48)

6.11.5.4.1.51.2 Downlink

6.11.5.4.1.51.2.1 Transport channel parameters

6.11.5.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.51.2.1.4 TFCS

See clause 6.10.3.4.1.51.2.1.4.

6.11.5.4.1.51.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51a.1 Uplink

6.11.5.4.1.51a.1.1 Transport channel parameters

6.11.5.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51a.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51a.1.1.4 TFCS

See clause 6.10.3.4.1.51a.1.1.4.

6.11.5.4.1.51a.1.2 Physical channel parameters

DPCH Uplink		Physical 1	Physical 2
	Modulation	QPSK	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits	2 792 bits
	TFCI code word / radio frame	16 bits	16 bits
	TPC / radio frame	2x2 bits	2x2 bits
	SS / radio frame	2x2 bits	2x2 bits
	Puncturing Limit	0.40	0.84

6.11.5.4.1.51a.2 Downlink

6.11.5.4.1.51a.2.1 Transport channel parameters

6.11.5.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.51a.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.51a.2.1.4 TFCS

See clause 6.10.3.4.1.51.2.1.4.

6.11.5.4.1.51a.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.84

6.11.5.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51b.1 Uplink

6.11.5.4.1.51b.1.1 Transport channel parameters

6.11.5.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51b.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51b.1.1.4 TFCS

See clause 6.10.3.4.1.51b.1.1.4.

6.11.5.4.1.51b.1.2 Physical channel parameters

DPCH Uplink		Physical 1	Physical 2
	Modulation	QPSK	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits	2 792 bits
	TFCI code word / radio frame	16 bits	16 bits
	TPC / radio frame	2x2 bits	2x2 bits
	SS / radio frame	2x2 bits	2x2 bits
	Puncturing Limit	0.40	0.76

6.11.5.4.1.51b.2 Downlink

See clause 6.11.5.4.1.51.2.

6.11.5.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.52.1 Uplink

See clause 6.11.5.4.1.51.1.

6.11.5.4.1.52.2 Downlink

6.11.5.4.1.52.2.1 Transport channel parameters

6.11.5.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.52.2.1.4 TFCS

See clause 6.10.3.4.1.52.2.1.4.

6.11.5.4.1.52.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 12 codes x 4 time slots
	Max. Number of data bits/radio frame	4 200 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.53.1 Uplink

6.11.5.4.1.53.1.1 Transport channel parameters

6.11.5.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.53.1.1.4 TFCS

See clause 6.10.3.4.1.53.1.1.4.

6.11.5.4.1.53.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF1 x 1 code x 4 time slots	SF1 x 1code x 2 time slots
	Max. Number of data bits/radio frame	5 608 bits	4 188 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.72 (alt 0.68)	0.52 (alt 0.48)

6.11.5.4.1.53.2 Downlink

See clause 6.11.5.4.1.52.2.

6.11.5.4.1.54 Void

6.11.5.4.1.55 Void

6.11.5.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.56.1 Uplink

6.11.5.4.1.56.1.1 Transport channel parameters

6.11.5.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

See clause 6.10.3.4.1.56.1.1.1.

6.11.5.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.56.1.1.3 TFCS

See clause 6.10.3.4.1.56.1.1.3.

6.11.5.4.1.56.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.84 (alt 0.76)

6.11.5.4.1.56.2 Downlink

6.11.5.4.1.56.2.1 Transport channel parameters

6.11.5.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

See clause 6.10.3.4.1.56.2.1.1.

6.11.5.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.56.2.1.3 TFCS

See clause 6.10.3.4.1.56.2.1.3.

6.11.5.4.1.56.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.84

6.11.5.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.57.1 Uplink

6.11.5.4.1.57.1.1 Transport channel parameters

6.11.5.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.11.5.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.57.1.1.3 TFCS

See clause 6.11.5.4.1.57.1.1.3.

6.11.5.4.1.57.1.2 Physical channel parameters

DPCH Uplink		Physical 1
	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots

	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52 (alt. 0.44)

6.11.5.4.1.57.2 Downlink

6.11.5.4.1.57.2.1 Transport channel parameters

6.11.5.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

See clause 6.10.3.4.1.57.2.1.1.

6.11.5.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.57.2.1.3 TFCS

See clause 6.10.3.4.1.57.2.1.3.

6.11.5.4.1.57.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.52

6.11.5.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.58.1 Uplink

6.11.5.4.1.58.1.1 Transport channel parameters

6.11.5.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.58.1.1.1.

6.11.5.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.58.1.1.4 TFCS

See clause 6.10.3.4.1.58.1.1.4.

6.11.5.4.1.58.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / frame	SF4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60 (alt 0.56)

6.11.5.4.1.58.2 Downlink

6.11.5.4.1.58.2.1 Transport channel parameters

6.11.5.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.58.2.1.1.

6.10.5.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.58.2.1.4 TFCS

See clause 6.10.3.4.1.58.2.1.4.

6.11.5.4.1.58.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 8 code x 2 time slots
	Max. Number of data bits / radio frame	1 384 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.44

6.11.5.4.1.59 Reserved for future use

6.11.5.4.1.60 Reserved for future use

6.11.5.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.61.1 Uplink

6.11.5.4.1.61.1.1 Transport channel parameters

6.11.5.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.61.1.1.1.

6.10.5.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.61.1.1.4 TFCS

See clause 6.10.3.4.1.61.1.1.4.

6.11.5.4.1.61.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 4 x 1 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.84 (alt 0.80)

6.11.5.4.1.61.2 Downlink

6.11.5.4.1.61.2.1 Transport channel parameters

6.11.5.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.61.2.1.1.

6.11.5.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.61.2.1.4 TFCS

See clause 6.10.3.4.1.61.2.1.4.

6.11.5.4.1.61.2.2 Physical channel parameters

DPCH Downlink	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 4 code x 2 time slots
	Max. Number of data bits / radio frame	680 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.84

6.11.5.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH

6.11.5.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.1.1 Uplink

6.11.5.4.2.1.1.1 Transport channel parameters

6.11.5.4.2.1.1.1.1 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.11.5.4.2.1.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.2.

6.11.5.4.2.1.1.1.3 TFCS for USCH

See clause 6.10.3.4.2.1.1.1.3.

6.11.5.4.2.1.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.2.1.1.1.4.

6.11.5.4.2.1.1.2 Physical channel parameters

6.11.5.4.2.1.1.2.1 Physical channel parameters for PUSCH

PUSCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits

	SS / radio frame	2x2 bits
	Puncturing Limit	0.88

6.11.5.4.2.1.1.2.2 Physical channel parameter for PRACH.

See clause 6.11.5.4.5.1.2.

6.11.5.4.2.1.2 Downlink

6.11.5.4.2.1.2.1 Transport channel parameters

6.11.5.4.2.1.2.1.1 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.11.5.4.2.1.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.1.2.1.3 TFCS for DSCH

See clause 6.10.3.4.2.1.2.1.3.

6.11.5.4.2.1.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

6.11.5.4.2.1.2.1.4.1 FACH transport channel configuration without DTCH

Higher layer	RAB/signalling RB		SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		160	136 or 120*	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 000)
	RLC header, bit		8	8	16	16	16	8	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3	3
	MAC multiplexing		7 logical channel multiplexing						
Layer 1	TrCH type		FACH						
	TB sizes, bit		171	171	171	171	171	171	171
	TFS	TF0, bits	0x171						
		TF1, bits	1x171						
		TF2, bits	2x171						
		TF3, bits	3x171(alt. N/A)						
		TF4, bits	4x171(alt. N/A)						
		TF5, bits							
		TF6, bits							
	TTI, ms		20						
	Coding type		CC 1/2						
	CRC, bit		16						
	Max number of bits/TTI after channel coding		1 528 (alt. 764)	1 528 (alt. 764)	1 528 (alt. 764)	1 528 (alt. 764)	1 528 (alt. 764)	1 528 (alt. 764)	1 528 (alt. 764)
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.									

6.11.5.4.2.1.2.1.4.2 FACH transport channel configuration with DTCH

Higher layer	RAB/signalling RB		RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		DTCH	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		AM	UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		320	160	136 or 120 (note)	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		16	8	8	16	16	16	8	0
MAC	MAC header, bit		27	3	27 or 43	27	27	27	3	3
	MAC multiplexing		8 logical channel multiplexing							
Layer 1	TrCH type		FACH							
	TB sizes, bit		171, 363							
	TFS	TF0, bits	0x171							
		TF1, bits	1x171							
		TF2, bits	2x171							
		TF3, bits	1x363							
		TF4, bits	3x171 (alt N/A)							
		TF5, bits	4x171 (alt. N/A)							
		TF6, bits	2x363 (alt. N/A)							
	TTI, ms		20							
	Coding type		CC ½							
	CRC, bit		16							
	Max number of bits/TTI after channel coding		1 532 (alt. 766)							
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.										

6.11.5.4.2.1.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.5.

6.11.5.4.2.1.2.2 Physical channel parameters

6.11.5.4.2.1.2.2.1 Physical channel parameters for PDSCH

PDSCH	Modulation	QPSK	8PSK
	Codes and time slots / radio frame	SF16 x 11 codes x 6 time slots	SF1 x 1 code x 4 time slots
	Max. Number of data bits/radio frame	5 784 bits	6 511 bits
	TFCI code word / radio frame	16 bits	24 bits
	TPC / radio frame	2x2 bits	2x3 bits
	SS / radio frame	2x2 bits	2x3 bits
	Puncturing Limit	0.60	0.68

6.11.5.4.2.1.2.2.2 Physical channel parameters for SCCPCH

6.11.5.4.2.1.2.2.2.1 Physical channel parameters for SCCPCH without DTCH

S-CCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 5 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slot)
	Max. Number of data bits/radio frame	864 bits (alt. 344 bits)
	TFCI code word / radio frame	16 bits
	TP(alt. 8 bits)C/ radio frame	0 bits
	SS / radio frame	0 bits

	Puncturing Limit	1 (alt. 0.88)

6.11.5.4.2.1.2.2.2.2 Physical channel parameters for SCCPCH with DTCH

S-CCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 5 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slot)
	Max. Number of data bits/radio frame	864 bits (alt. 336 bits)
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	1 (alt. 0.84)

6.11.5.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.2.1 Uplink

See clause 6.11.5.4.2.1.1.

6.11.5.4.2.2.2 Downlink

6.11.5.4.2.2.2.1 Transport channel parameters

6.11.5.4.2.2.2.1.1 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.11.5.4.2.2.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.2.2.1.3 TFCS for DSCH

See clause 6.10.3.4.2.2.2.1.3.

6.11.5.4.2.2.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.2.1.2.1.4.

6.11.5.4.2.2.2.1.5 TFCS for FACH

See clause 6.11.5.4.2.1.2.1.5.

6.11.5.4.2.2.2.2 Physical channel parameters

6.11.5.4.2.2.2.2.1 Physical channel parameters for PDSCH

PDSCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 1 x 1 code x 6 time slots
	Max. Number of data bits/radio frame	8 424 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.60

6.11.5.4.2.2.2.2.2 Physical channel parameters for SCCPCH

See clause 6.11.5.4.2.1.2.2.2.

6.11.5.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.3.1 Uplink

See clause 6.11.5.4.2.1.1.

6.11.5.4.2.3.2 Downlink

6.11.5.4.2.3.2.1 Transport channel parameters

6.11.5.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	UM
	Payload sizes, bit		1 704	160
	Max data rate, bps		2 048 000	16 000
	RLC header, bit		16	8
MAC	MAC header, bit		0	0
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DSCH	DSCH
	TB sizes, bit		1720	168
	TFS	TF0, bits	0x1720	0x168
		TF1, bits	1x1720	1x168
		TF2, bits	2x1720	N/A
		TF3, bits	4x1720	N/A
		TF4, bits	8x1720	N/A
		TF5, bits	12x1720	N/A
		TF6, bits	N/A (alt. 16x1720)	N/A
		TF7, bits	N/A (alt. 20x1720)	N/A
		TF8, bits	N/A (alt. 24x1720)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		No Coding	CC 1/2
	CRC, bit		24	16
	Max number of bits/TTI after channel coding		20 928 (alt. 41 856)	384
	Downlink: Max number of bits/radio frame before rate matching		20 928 (alt. 20 928)	384
	RM attribute		135 to 175	180 to 220

6.11.5.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.3.2.1.3 TFCS for DSCH

TFCS size	22 (alt.34)
TFCS	(2 048 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1))

For better understanding of the TFCS please note that the following combinations are not included in the table above:

- (TF5, TF1, TF0), (TF5, TF1, TF1), (TF8, TF1, TF0), (TF8, TF1, TF1).

6.11.5.4.2.3.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.2.1.2.1.4.

6.11.5.4.2.3.2.1.5 TFCS for FACH

See clause 6.11.5.4.2.1.2.1.5.

6.11.5.4.2.3.2.2 Physical channel parameters

6.11.5.4.2.3.2.2.1 Physical channel parameters for PDSCH

PDSCH	Modulation	8PSK
	Codes and time slots / radio frame	SF1 x 1 code x 10 time slots
	Max. Number of data bits/radio frame	21 084 bits
	TFCI code word / radio frame	24 bits
	TPC / radio frame	2x3 bits
	SS / radio frame	2x3 bits
	Puncturing Limit	1

6.11.5.4.2.3.2.2.2 Physical channel parameters for S-CCPCH

See clause 6.11.5.4.2.1.2.2.2.

6.11.5.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

6.11.5.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.1.1 Uplink

6.11.5.4.3.1.1.1 Transport channel parameters

6.11.5.4.3.1.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.3.1.1.1.2 Transport channel parameters for UL SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.3.1.1.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.1.1.3.

6.11.5.4.3.1.1.1.4 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.11.5.4.3.1.1.1.5 TFCS for USCH

See clause 6.10.3.4.3.1.1.1.5.

6.11.5.4.3.1.1.1.6 Transport channel parameters for SRB for CCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.3.1.1.1.6.

6.11.5.4.3.1.1.2 Physical channel parameters

Physical channel parameters for uplink DPCH see clause 6.11.5.4.1.4.1.2.

Physical channel parameters for PUSCH see clause 6.11.5.4.2.1.1.2.

Physical channel parameters for PRACH see clause 6.11.5.4.2.1.1.2.

6.11.5.4.3.1.2 Downlink

6.11.5.4.3.1.2.1 Transport channel parameters

6.11.5.4.3.1.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.3.1.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.1.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.1.2.1.4 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.11.5.4.3.1.2.1.5 TFCS for DSCH

See clause 6.10.3.4.3.1.2.1.5.

6.11.5.4.3.1.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

Higher layer	RAB/Signalling RB		SRB#0	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC
RLC	Logical channel type		CCCH	SHCCH	BCCH
	RLC mode		UM	UM	TM
	Payload sizes, bit		160	160	168
	Max data rate, bps		32 000	32 000	33 600
	RLC header, bit		8	8	0
MAC	MAC header, bit		3		
	MAC multiplexing		3 logical channel multiplexing		
Layer 1	TrCH type		FACH		
	TB sizes, bit		171		
	TFS	TF0, bits	0x171		
		TF1, bits	1x171		
		TF2, bits	2x171		
		TF3, bits	3x171		
		TF4, bits	4x171		
	TTI, ms		20		
	Coding type		CC 1/2		
	CRC, bit		16		
	Max number of bits/TTI after channel coding		1 528		
	Max number of bits/radio frame before rate matching		764		

6.11.5.4.3.1.2.1.7 TFCS for FACH

TFCS size	5
TFCS	FACH = TF0, TF1,TF2,TF3,TF4

6.11.5.4.3.1.2.2 Physical channel parameters

Physical channel parameters for downlink for DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for downlink for PDSCH see clause 6.11.5.4.2.1.2.2.

Physical channel parameters for SCCPCH see clause 6.11.5.4.2.1.2.2.

6.11.5.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.2.1 Uplink

See clause 6.11.5.4.3.1.1.

6.11.5.4.3.2.2 Downlink

6.11.5.4.3.2.2.1 Transport channel parameters

6.11.5.4.3.2.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.1.4.1.4.2.1.1.

6.11.5.4.3.2.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.2.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.2.2.1.4 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.11.5.4.3.2.2.1.5 TFCS for DSCH

See clause 6.10.3.4.3.2.2.1.5.

6.11.5.4.3.2.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.3.1.2.1.6.

6.11.5.4.3.2.2.1.7 TFCS for FACH

See clause 6.11.5.4.3.1.2.1.7.

6.11.5.4.3.2.2.2 Physical channel parameters

Physical channel parameters for downlink for DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for downlink for PDSCH see clause 6.11.5.4.2.2.2.2.

Physical channel parameters for downlink for SCCPCH see clause 6.11.5.4.2.1.2.2.

6.11.5.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.3.1 Uplink

See clause 6.11.5.4.3.1.1.

6.11.5.4.3.3.2 Downlink

6.11.5.4.3.3.2.1 Transport channel parameters

6.11.5.4.3.3.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.3.3.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.3.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.3.2.1.4 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.11.5.4.2.3.2.1.2.

6.11.5.4.3.3.2.1.5 TFCS for DSCH

See clause 6.11.5.4.2.3.2.1.4.

6.11.5.4.3.3.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.3.1.2.1.6.

6.11.5.4.3.3.2.1.7 TFCS for FACH

See clause 6.11.5.4.3.1.2.1.7.

6.11.5.4.3.3.2.2 Physical channel parameters

Physical channel parameters for downlink DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for PDSCH see clause 6.11.5.4.2.3.2.2.

Physical channel parameters for SCCPCH see clause 6.11.5.4.2.1.2.2.

6.11.5.4.4 Combinations on SCCPCH

6.11.5.4.4.1 Stand-alone signalling RB for PCCH

6.11.5.4.4.1.1 Transport channel parameters

6.11.5.4.4.1.1.1 Transport channel parameter of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.1.1.2 TFCS

See clause 6.10.3.4.4.1.1.2.

6.11.5.4.4.1.2 Physical channel parameters

S-CCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 2 codes x 2 time slots (alt. SF16 x 1 codes x 2 time slots)
	Max. Number of data bits/radio frame	344 bits (alt. 168 bits)
	TFCI code word / radio frame	8 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	1 (alt. 0.84)
NOTE: Alt. Puncturing Limit applies when alt. payload sizes and alt. codes and time slots / radio frame are both configured.		

6.11.5.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2.1 Transport channel parameters

6.11.5.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

See clause 6.10.3.4.4.2.1.1.

6.11.5.4.4.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

Higher layer	RAB/signalling RB		SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer		RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type		CCCH	DCCH	DCCH	DCCH	DCCH	BCCH
	RLC mode		UM	UM	AM	AM	AM	TM
	Payload sizes, bit		160	136 or 120	128	128	128	168
	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 2400 (alt. 24 000 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	33 600 (alt. 16 800)
	RLC header, bit		8	8	16	16	16	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3
	MAC multiplexing		6 logical channel multiplexing					
Layer 1	TrCH type		FACH					
	TB sizes, bit		171					
	TFS	TF0, bits	0x171					
		TF1, bits	1x171					
		TF2, bits	2x171					
		TF3, bits	3x171 (alt. N/A)					
		TF4, bits	4x171 (alt. N/A)					
	TTI, ms		20					
	Coding type		CC 1/2					
	CRC, bit		16					
	Max number of bits/TTI before rate matching		1 528 (alt. 764)					
	Max number of bits/radio frame before rate matching		764 (alt. 382)					
	RM attribute		200 to 240					
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.								

6.11.5.4.4.2.1.3 TFCS

See clause 6.10.3.4.4.2.1.3.

6.11.5.4.4.2.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 9 codes x 2 time slots (alt. SF16 x 4 codes x 2 time slots)
	Max. Number of data bits/radio frame	1 568 bits (alt. 688 bits)
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.52 (alt. 0.48)
NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured.		

6.11.5.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2a.1 Transport channel parameters

6.11.5.4.4.2a.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB

See clause 6.10.3.4.2a.1.1.

6.11.5.4.4.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.2a.1.3 TFCS

See clause 6.10.3.4.4.2a.1.3.

6.11.5.4.4.2a.2 Physical channel parameters

See clause 6.11.5.4.4.2.2.

6.11.5.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2b.1 Transport channel parameters

6.11.5.4.4.2b.1.1 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.2b.1.2 TFCS

See clause 6.10.3.4.4.2b.1.2.

6.11.5.4.4.2b.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 4 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slots)
	Max. Number of data bits/radio frame	688 bits (alt. 344 bits)
	TFCI code word / radio frame	16 bits (alt. 8 bits)
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.88
NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured.		

6.11.5.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.3.1 Transport channel parameters

6.11.5.4.4.3.1.1 Transport channel parameters of SRB for Interactive/Background 32 kbps RAB

See clause 6.10.3.4.4.2.1.1.

6.11.5.4.4.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.3.1.4 TFCS

See clause 6.10.3.4.4.3.1.4.

6.11.5.4.4.3.2 Physical channel parameters

S-CCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 10 codes x 2 time slots (alt. SF16 x 6 codes x 2 time slots)
	Max. Number of data bits/radio frame	1 744 bits (alt. 1 040 bits)

	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.48 (alt. 0.52)
NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured.		

6.11.5.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.3a.1 Transport channel parameters

6.11.5.4.4.3a.1.1 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.3a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.3a.1.3 TFCS

See clause 6.10.3.4.4.3a.1.3.

6.11.5.4.4.3a.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 4 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slots)
	Max. Number of data bits/radio frame	688 bits (alt. 336 bits)
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.60 (alt. 0.52)
NOTE: Alt. applies when alts for SRB for PCCH and SRBs for CCCH/ DCCH/ BCCH are both configured.		

6.11.5.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.11.5.4.4.4.1 Transport channel parameters

6.11.5.4.4.4.1.1 Transport channel parameters of RB for CTCH

See clause 6.10.3.4.4.4.1.1.

6.11.5.4.4.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	BCCH
	RLC mode		UM	TM
	Payload sizes, bit		160	168
	Max data rate, bps		16 000	16 800
	AMD/UMD/TrD PDU header, bit		8	0
MAC	MAC header, bit		3	3
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		FACH	
	TB sizes, bit		171	
	TFS	TF0, bits	0x171	
		TF1, bits	1x171	
		TF2, bits	2x171	
	TTI, ms		20	
	Coding type		CC 1/3	
	CRC, bit		16	
	Max number of bits/TTI before rate matching		1 146	

Higher layer	RAB/signalling RB	SRB#0	SRB#5
	User of Radio Bearer	RRC	RRC
	Max number of bits/radio frame before rate matching	573	
	RM attribute	200 to 240	

6.11.5.4.4.4.1.3 TFCS

See clause 6.10.3.4.4.4.1.3.

6.11.5.4.4.4.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 4 codes x 2 time slots
	Max. Number of data bits/radio frame	688 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.52

6.11.5.4.4.5 64.8kbps RB for MTCH with 40 ms TTI

6.11.5.4.4.5.1 Transport channel parameters

6.11.5.4.4.5.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		64800
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		665
	TFS	TF0, bits	0x665
		TF1, bits	1x665
		TF2, bits	2x665
		TF3, bits	3x665
		TF4, bits	4x665
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		8184
	RM attribute		160

6.11.5.4.4.5.1.2 TFCS

TFCS size	5
TFCS	64 kbps RAB =TF0, TF1, TF2, TF3, TF4

6.11.5.4.4.5.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	1392 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.68

6.11.5.4.4.6 129.6 kbps RB for MTCH with 40 ms TTI

6.11.5.4.4.6.1 Transport channel parameters

6.11.5.4.4.6.1.1 Transport channel parameters for 128 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		665
	TFS	TF0, bits	0x665
		TF1, bits	1x665
		TF2, bits	2x665
		TF3, bits	3x665
		TF4, bits	4x665
		TF5, bits	5x665
		TF6, bits	6x665
		TF7, bits	7x665
		TF8, bits	8x665
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		16368
	RM attribute		160

6.11.5.4.4.6.1.2 TFCS

TFCS size	9
TFCS	128 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8

6.11.5.4.4.6.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 16 codes x 2 time slots
	Max. Number of data bits/radio frame	2800 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.68

6.11.5.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI

6.11.5.4.4.7.1 Transport channel parameters

6.11.5.4.4.7.1.1 Transport channel parameters for 256 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		259200
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		665
	TFS	TF0, bits	0x665
		TF1, bits	1x665
		TF2, bits	2x665
		TF3, bits	3x665
		TF4, bits	4x665
		TF5, bits	5x665
		TF6, bits	6x665
		TF7, bits	7x665
		TF8, bits	8x665
		TF9, bits	9x665
		TF10, bits	10x665
		TF11, bits	11x665
		TF12, bits	12x665
		TF13, bits	13x665
		TF14, bits	14x665
		TF15, bits	15x665
		TF16, bits	16x665
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		32724
	RM attribute		160

6.11.5.4.4.7.1.2 TFCS

TFCS size	17
TFCS	256 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.11.5.4.4.7.2 Physical channel parameters

SCCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF 16 x 16 codes x 4 time slots
	Max. Number of data bits/radio frame	5616 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.68

6.11.5.4.4.8 7.6 kbps signalling RB for MCCH

6.11.5.4.4.8.1 Transport channel parameters

See clause 6.10.2.4.3.8.1.1.

6.11.5.4.4.8.1.2 TFCS

See clause 6.10.2.4.3.8.1.2.

6.11.5.4.4.8.2 Physical channel parameters

S-CCPCH	Modulation	QPSK
	Codes and time slots / radio frame	SF16 x 1 codes x 2 time slots
	Max. Number of data bits/radio frame	270
	TFCI code word / radio frame	2 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	1

6.11.5.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI

6.11.5.4.4.9.1 Transport channel parameters

6.11.5.4.4.9.1.1 Transport channel parameters for 124.4 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		124400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	2x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		30078
	Max number of bits/radio frame before rate matching		3760
	RM attribute		128

6.11.5.4.4.9.1.2 TFCS

TFCS size	3
TFCS	124 kbps RAB =TF0, TF1, TF2

6.11.5.4.4.9.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF16 x 8 codes x 2 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	2096 bits
	TFCI code word / radio frame	16
	Puncturing limit	0.54

6.11.5.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI

6.11.5.4.4.10.1 Transport channel parameters

6.11.5.4.4.10.1.1 Transport channel parameters for 320 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4272
	Max data rate, bps		320400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4289
	TFS	TF0, bits	0x4289
		TF1, bits	1x4289
		TF2, bits	6x4289
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		77562
	Max number of bits/radio frame before rate matching		9696
	RM attribute		128

6.11.5.4.4.10.1.1 TFCS

TFCS size	3
TFCS	320 kbps RAB =TF0, TF1, TF2

6.11.5.4.4.10.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF16 x 16 codes x 2 time slot
	Modulation	16QAM
	Max. Number of data bits/radio frame	8432 bits
	TFCI code word / radio frame	16
	Puncturing limit	0.86

6.11.5.4.4.11 497.6 kbps RB for MBSFN MTCH with 80 ms TTI

6.11.5.4.4.11.1 Transport channel parameters

6.11.5.4.4.11.1.1 Transport channel parameters for 496 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		497600
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	8x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		120312
	Max number of bits/radio frame before rate matching		15039
	RM attribute		128

6.11.5.4.4.11.1.2 TFCS

TFCS size	3
TFCS	496 kbps RAB =TF0, TF1, TF2

6.11.5.4.4.11.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF1 x 1 codes x 4 time slots
	Modulation	QPSK
	Max. Number of data bits/radio frame	16880 bits
	TFCI code word / radio frame	16
	Puncturing limit	1

6.11.5.4.4.12 7.2 kbps signalling RB for MBSFN MCCH

6.11.5.4.4.12.1 Transport channel parameters

6.11.5.4.4.12.1.1 Transport channel parameters for 7.2 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		72
	Max data rate, bps		7200
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		80
	TFS	TF0, bits	0x80
		TF1, bits	1x80
		TF2, bits	2x80
		TF3, bits	4x80
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1164
	Max number of bits/radio frame before rate matching		291
	RM attribute		128

6.11.5.4.4.12.1.2 TFCS

TFCS size	4
TFCS	MBMS SRB =TF0, TF1, TF2, TF3

6.11.5.4.4.12.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF16 x 1 code x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	248 bits
	TFCI code word / radio frame	16
	Puncturing limit	0.84

6.11.5.4.4.13 8kbps RB for MBSFN MTCH with 40 ms TTI

6.11.5.4.4.13.1 Transport channel parameters

6.11.5.4.4.13.1.1 Transport channel parameters for 8kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		328
	Max data rate, bps		8200
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		345
	TFS	TF0, bits	0x345
		TF1, bits	1x345
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1095
	Max number of bits/radio frame before rate matching		274
	RM attribute		128

6.11.5.4.4.13.1.2 TFCS

TFCS size	2
TFCS	8 kbps RAB =TF0, TF1

6.11.5.4.4.13.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF16 x 1 codes x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	248 bits
	TFCI code word / radio frame	16
	Puncturing limit	0.84

6.11.5.4.4.10 64kbps RB for MBSFN MTCH with 40 ms TTI

6.11.5.4.4.14.1 Transport channel parameters

6.11.5.4.4.14.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		1336
	Max data rate, bps		66800
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		1353
	TFS	TF0, bits	0x1353
		TF1, bits	1x1353
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4119
	Max number of bits/radio frame before rate matching		2060
	RM attribute		128

6.11.5.4.4.14.1.1 TFCS

TFCS size	2
TFCS	64 kbps RAB =TF0, TF1

6.11.5.4.4.14.2 Physical channel parameters

S-CCPCH	TPC / radio frame	0 bits
	Codes and time slots / radio frame	SF16 x 8 codes x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	2096 bits
	TFCI code word / radio frame	16
	Puncturing limit	1

6.11.5.4.5 Combinations on PRACH

6.11.5.4.5.1 SRB for CCCH + SRBs for DCCH

6.11.5.4.5.1.1 Transport channel parameters

6.11.5.4.5.1.1.1 Transport channel parameter for SRB for CCCH, SRBs for DCCH

See clause 6.10.3.4.5.1.1.1.

6.11.5.4.5.1.1.2 TFCS

See clause 6.10.3.4.5.1.1.2.

6.11.5.4.5.1.2 Physical channel parameters

PRACH	Modulation	QPSK
	Codes and time slots / radio frame	SF 8 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	352 bits
	TPC / radio frame	0 bits
	SS / radio frame	0 bits
	Puncturing Limit	0.88

6.11.5.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH

6.11.5.4.5.2.1 Transport channel parameters

6.11.5.4.5.2.1.1 Transport channel parameters for Interactive or background / 12.8 kbps / PS RAB + SRB for CCCH + SRBs for DCCH

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	168	136	128	128	128
	Max data rate, bps	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16
MAC	MAC header, bit	26	2	26	26	26	26
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	170					
	TFS TF0, bits	1x170					
	TTI, ms	10					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	388					
	Max number of bits/ Radio frame before rate matching	388					

6.11.5.4.5.2.1.2 TFCS

TFCS size	1
TFCS	12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH = (TF0)

6.11.5.4.5.2.2 Physical channel parameters

See clause 6.11.5.4.5.1.2.

6.11.5.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.11.5.4.5.3.1 Transport channel parameters

6.11.5.4.5.3.1.1 Transport channel parameters for Interactive or background / 12.8 kbps / PS RAB + Interactive or background / 12.8 kbps / PS RAB + SRB for CCCH + SRBs for DCCH

Higher layer	RAB/signalling RB	RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type	DTCH	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	128	168	136	128	128	128
	Max data rate, bps	12 800	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	16	0	8	16	16	16
MAC	MAC header, bit	26	26	2	26	26	26	26
	MAC multiplexing	7 logical channel multiplexing						
Layer 1	TrCH type	RACH						
	TB sizes, bit	170						
	TFS TF0, bits	1x170						
	TTI, ms	10						
	Coding type	CC ½						
	CRC, bit	16						

Higher layer	RAB/signalling RB	RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio
	Max number of bits/TTI after channel coding	388						
	Max number of bits/ Radio frame before rate matching	388						

6.11.5.4.5.3.1.2 TFCS

TFCS size	1
TFCS	12.8 kbps PS RAB + 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH = (TF0)

6.11.5.4.5.3.2 Physical channel parameters

See clause 6.11.5.4.5.1.2.

6.11.5.4.6 Combinations on DPCH and HS-PDSCH

6.11.5.4.6.1 Interactive or background / UL:8 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.6.1.1 Uplink

See clause 6.11.5.4.1.23a.1.

6.11.5.4.6.1.2 Downlink

6.11.5.4.6.1.2.1 Transport channel parameters

6.11.5.4.6.1.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.6.1.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

		Alt 1 Fixed RLC + MAC-hs (Rel-5 and later releases) NOTE2	**Alt 2 Fixed RLC + MAC-ehs (Rel-8 and later releases) NOTE2**	**Alt 3 Flexible RLC + MAC-ehs (Rel-8 and later releases) NOTE2**
Higher layer	RAB/Signalling RB	RAB		
RLC	Logical channel type	DTCH		
	RLC mode	AM		
	Payload sizes, bit	320 (alt. 640)	320 (alt. 640)	Flexible up to 12000
	Max data rate, bps	depends on UE category NOTE 1		
	AMD PDU header, bit	16	16	16
MAC	MAC-d header, bit	0	0	0
	MAC multiplexing	N/A	N/A	N/A
	MAC-d PDU size, bit	336 (alt. 656)	336 (alt. 656)	Flexible
	MAC-hs Type	MAC-hs	MAC-ehs	MAC-ehs
	MAC-hs/MAC-ehs header fixed part, bit	21	24	24
Layer 1	TrCH type	HS-DSCH	HS-DSCH	HS-DSCH
	TTI	5 ms	5 ms	5 ms
	Coding type	TC	TC	TC
	CRC, bit	24	24	24

	Applicable modulation schemes	QPSK, 16QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
	Applicable with MIMO	No	Yes	Yes
NOTE 1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs or MAC-ehs PDU (see 3GPP TS 25.321 [38]). NOTE 2: Alternative 1 is for Rel-5 and later releases.Alternative 2 or 3 is for Rel-8 and later releases.Alternative 1 with Fixed RLC + MAC-hs is the default configuration. For test cases that use alternative 2 (Fixed RLC + MAC-ehs) or 3 (Flexible RLC + MAC-ehs) then this shall be explicitely stated in the test case.				

6.11.5.4.6.1.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.1.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.11.5.4.6.1.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.11.5.4.6.1.2.2 Physical channel parameters

6.11.5.4.6.1.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.2.2.2.

6.11.5.4.6.1.2.2.2 Physical channel parameters on HS-PDSCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE HS-DSCH Physical Layer category 1:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	560k bps, (alt. 280 kbps)

UE HS-DSCH Physical Layer category 2:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	560kbps

UE HS-DSCH Physical Layer category 3:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	560k bps

UE HS-DSCH Physical Layer category 4:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.1 Mbps, (alt. 600 kbps)

UE HS-DSCH Physical Layer category 5:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.1 Mbps

UE HS-DSCH Physical Layer category 6:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	1.1 Mbps

UE HS-DSCH Physical Layer category 7:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.6 Mbps, (alt. 800kbps)

UE HS-DSCH Physical Layer category 8:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	1.6 Mbps

UE HS-DSCH Physical Layer category 9:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	1.6 Mbps

UE HS-DSCH Physical Layer category 10:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	2.2 Mbps, (alt. 1.0 Mbps)

UE HS-DSCH Physical Layer category 11:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	2.2 Mbps

UE HS-DSCH Physical Layer category 12:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	2.2 Mbps

UE HS-DSCH Physical Layer category 13:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	2.8 Mbps, (alt. 1.4 Mbps)

UE HS-DSCH Physical Layer category 14:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	2.8 Mbps

UE HS-DSCH Physical Layer category 15:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes

	Max Data Rate	2.8 Mbps

UE HS-DSCH Physical Layer category 16:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	3.2 Mbps, (alt. 1.6 Mbps)

UE HS-DSCH Physical Layer category 17:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	3.2 Mbps

UE HS-DSCH Physical Layer category 18:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	3.2 Mbps

UE HS-DSCH Physical Layer category 19:

HS-PDSCH	Number of processes	4, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	4.2 Mbps, (alt. 2.1 Mbps)

UE HS-DSCH Physical Layer category 20:

HS-PDSCH	Number of processes	6, (alt. 8)
	Process memory size	Split equally among all processes
	Max Data Rate	4.2Mbps

UE HS-DSCH Physical Layer category 21:

HS-PDSCH	Number of processes	8
	Process memory size	Split equally among all processes
	Max Data Rate	4.2Mbps

6.11.5.4.6.1a Interactive or background / UL:8 (multiframe) DL: [max bit rate depending on UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH (multiframe) (REL-5)

6.11.5.4.6.1a.1 Uplink

6.11.5.4.6.1a.1.1 Transport channel parameters

6.11.5.4.6.1a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps (multiframe) / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		20

	Coding type	TC
	CRC, bit	16
	Max number of bits/TTI after channel coding	2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching	1 062 (alt. 1 206)
	RM attribute	135 to 175

6.11.5.4.6.1a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH(multiframe)

See clause 6.11.5.4.1.2a.1.1.1.

6.11.5.4.6.1a.1.1.3 TFCS

See clause 6.10.2.4.1.23d.1.1.3.

6.11.5.4.6.1a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF 2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	680bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Repetition period	8
	Repetition length	2
	Puncturing limit	0.80 (alt. 0.72)

6.11.5.4.6.1a.2 Downlink

6.11.5.4.6.1a.2.1 Transport channel parameters

6.11.5.4.6.1a.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.6.1a.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.6.1a.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.1a.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH (multiframe)

See clause 6.10.3.4.1.2a.2.1.1.

6.11.5.4.6.1a.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.6.1a.2.2 Physical channel parameters

6.11.5.4.6.1a.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.2a.2.2.

6.11.5.4.6.1a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.6.2 Interactive or background / UL:16 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.2.1 Uplink

See clause 6.11.5.4.1.23b.1.

6.11.5.4.6.2.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.6.2a Interactive or background / UL:16(multiframe) DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)

6.11.5.4.6.2a.1 Uplink

6.11.5.4.6.2a.1.1 Transport channel parameters

6.11.5.4.6.2a.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB(multiframe)

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 3x144)
		TF3, bits	3x336 (alt. 7x144)
		TF4, bits	4x336 (alt. 10x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236 (alt. 4 812)
	Max number of bits/radio frame before rate matching		2 118 (alt. 2 406)
	RM attribute		130 to 170

6.11.5.4.6.2a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH(multiframe)

See clause 6.11.5.4.1.2a.1.1.1.

6.11.5.4.6.2a.1.1.3 TFCS

See clause 6.10.2.4.1.26.1.1.3.

6.11.5.4.6.2a.1.2 Physical channel parameters

DPCH Uplink		Physical 1	Physical 2
	Modulation	QPSK	QPSK
	Codes and time slots / radio frame	SF2 x 1 code x 2 time slots	SF1 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	1 384 bits	2 792 bits
	TFCI code word / radio frame	16 bits	16 bits
	TPC / radio frame	2x2 bits	2x2 bits
	SS / radio frame	2x2 bits	2x2 bits
	Repetition period	4	8
	Repetition length	1	2
	Puncturing limit	0.48 (alt. 0.40)	0.96 (alt. 0.84

6.11.5.4.6.2a.2 Downlink

See clause 6.11.5.4.6.1a.2.

6.11.5.4.6.3 Interactive or background / UL:32 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.3.1 Uplink

See clause 6.10.3.4.1.23c.1.

6.11.5.4.6.3.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.6.3a Interactive or background / UL:32(multiframe) DL: [max bit rate depending on UE category] / PS RAB +UL:3.4 DL:3.4 kbps SRBs for DCCH(multiframe) (REL-5)

6.11.5.4.6.3a.1 Uplink

6.11.5.4.6.3a.1.1 Transport channel parameters

6.11.5.4.6.3a.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB(multiframe)

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 7x144)
		TF3, bits	4x336 (alt 14x144)
		TF4, bits	8x336 (alt. 20x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 9 612)
	Max number of bits/radio frame before rate matching		4 230 (alt. 4 806)
	RM attribute		120 to 160

6.11.5.4.6.3a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH(multiframe)

See clause 6.11.5.4.1.2a.1.1.1.

6.11.5.4.6.3a.1.1.3 TFCS

See clause 6.10.2.4.1.28.1.1.3.

6.11.5.4.6.3a.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF1 x 1 codes x 2 time slots
	Max. Number of data bits/radio frame	2 792 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Repetition period	8
	Repetition length	2
	Puncturing limit	0.56 (alt. (0.48)

6.11.5.4.6.3a.2 Downlink

See clause 6.11.5.4.6.1a.2.

6.11.5.4.6.4 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.4.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.6.4.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.6.5 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.5.1 Uplink

See clause 6.11.5.4.1.28.1.

6.11.5.4.6.5.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.6.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.6.1 Uplink

See clause 6.11.5.4.1.38c.1.

6.11.5.4.6.6.2 Downlink

6.11.5.4.6.6.2.1 Transport channel parameters

6.11.5.4.6.6.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.11.5.4.6.1.2.1.1.

6.11.5.4.6.6.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.6.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.11.5.4.6.6.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.5.4.6.6.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.11.5.4.6.6.2.2 Physical channel parameters

6.11.5.4.6.6.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.4.2.2.

6.11.5.4.6.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.6.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.7.1 Uplink

See clause 6.11.5.4.1.40.1.

6.11.5.4.6.7.2 Downlink

See clause 6.11.5.4.6.6.2.

6.11.5.4.6.8 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.5.4.6.8.1 Uplink

See clause 6.11.5.4.1.51.1.

6.11.5.4.6.8.2 Downlink

6.11.5.4.6.8.2.1 Transport channel parameters

6.11.5.4.6.8.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.11.5.4.6.1.2.1.1.

6.11.5.4.6.8.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.8.2.1.2.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.6.8.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.5.4.6.8.2.1.2.3 TFCS

See clause 6.10.3.4.1.13.2.1.3.

6.11.5.4.6.8.2.2 Physical channel parameters

6.11.5.4.6.8.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.13.2.2.

6.11.5.4.6.8.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.6.9 Interactive or background / UL:64 DL: [bit rate depending on UE category] / PS RAB + Interactive or background / UL:64 DL: [bit rate depending on UE category] / PS RAB+UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.6.9.1 Uplink

See clause 6.11.5.4.1.57.1.

6.11.5.4.6.9.2 Downlink

6.11.5.4.6.9.2.1 Transport channel parameters

6.11.5.4.6.9.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.6.9.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.6.9.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.6.9.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.9.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.6.9.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.6.9.2.2 Physical channel parameters

6.11.5.4.6.9.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.2.2.2.

6.11.5.4.6.9.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.6.10 Conversational/Speech/UL:12.2 DL:12.2kbps/CS RAB + interactive or Background / UL:64 kbps DL: [bit rate depending on UE category]/PS RAB + interactive or Background / UL:64 kbps DL: [bit rate depending on UE category]/PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH

6.11.5.4.6.10.1 Uplink

See clause 6.11.5.4.1.38d.

6.11.5.4.6.10.2 Downlink

6.11.5.4.6.10.2.1 Transport channel parameters

6.11.5.4.6.10.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.6.10.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.6.10.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.6.10.2.1.2 Transport channel parameters for DCH

6.11.5.4.6.10.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.6.10.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.6.10.2.1.2.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.6.10.2.2 Physical channel parameters

6.11.5.4.6.10.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.6.10.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.6.11 Streaming/ UL:32 DL:[Rate depending on UE category]/ PS RAB + interactive or Background/ UL:8 kbps DL:[Rate depending on UE category]/ PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH

6.11.5.4.6.11.1 Uplink

6.11.5.4.6.11.1.1 Transport channel parameters

6.11.5.4.6.11.1.1.1 Transport channel parameters for Streaming/ UL:32 kbps / PS RAB + UL:8 kbps / PS RAB + UL:3.4 kbps SRBs for DCCH

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320 (alt. 128)	320 (alt.128)
	Max data rate, bps		8 000	32 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		0	0
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		336	336(alt. 144)
	TFS	TF0, bits	0x336	0x336(alt. 0x144)
		TF1, bits	1x336	1x336(alt. 1x144)
		TF2, bits	2x336	N/A(alt. 5x144)
	TTI, ms		20	40(alt. 80)
	Coding type		TC	TC
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		2124	1068(alt. 2412)
	Max number of bits/radio frame before rate matching		1062	267(alt. 302)
	RM attribute		135 to 175	135 to 175

6.11.5.4.6.11.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.6.11.1.1.3 TFCS

TFCS size	12 (alt. 18)
TFCS	(32kbps RAB, 8kbps RAB, DCCH)== (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF2, TF2, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF2, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.5.4.6.11.1.2 Physical channel parameters

DPCH Uplink	Modulation	QPSK
	Codes and time slots / radio frame	SF4 x 1 code x 2 timeslots +SF8 x 1 code x 2 timeslot
	Max. Number of data bits / radio frame	1032 bits
	TFCI code word / radio frame	16 bits
	TPC / radio frame	2x2 bits
	SS / radio frame	2x2 bits
	Puncturing Limit	0.68

6.11.5.4.6.11.2 Downlink

See clause 6.11.5.4.6.9.2.

6.11.5.4.6.12 Streaming/ UL:16 DL:[Rate depending on UE category]/ PS RAB + interactive or Background/ UL:8 kbps DL:[Rate depending on UE category]/ PS RAB + UL:3.4 DL 3.4 kbps SRB for DCCH

6.11.5.4.6.12.1 Uplink

See clause 6.11.5.4.1.58.1.

6.11.5.4.6.12.2 Downlink

See clause 6.11.5.4.6.9.2.

6.11.5.4.7 Combinations on HS-PDSCH and E-PUCH

6.11.5.4.7.1 Stand-alone UL: [max bit rate depending on UE category and TTI] DL:[max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.11.5.4.7.1.1 Uplink

6.11.5.4.7.1.1.1 Transport channel parameters

6.11.5.4.7.1.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.1.1.1.1.1 MAC-d flow parameters for Stand-alone UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	Depends on UE category and TTI
	AMD PDU header, bit	16
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	5ms
	Coding type	TC
	CRC, bit	24

6.11.5.4.7.1.1.2 Physical channel parameters

6.11.5.4.7.1.1.2.1 Physical channel parameters on E-PUCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE E-PUCH Physical Layer category 1:

E-PUCH	Number of processes	4
	TTI	5 ms
	Max Data Rate	0.5508 Mbps

UE E-PUCH Physical Layer category 2:

E-PUCH	Number of processes	4
	TTI	5ms
	Max Data Rate	0.8324 Mbps

UE E-PUCH Physical Layer category 3:

E-PUCH	Number of processes	4
	TTI	5 ms

	Max Data Rate	1.1064 Mbps

UE E-PUCH Physical Layer category 4:

E-PUCH	Number of processes	4
	TTI	5 ms
	Max Data Rate	1.6696 Mbps

UE E-PUCH Physical Layer category 5:

E-PUCH	Number of processes	4
	TTI	5 ms
	Max Data Rate	2.232 Mbps

UE E-PUCH Physical Layer category 6:

E-PUCH	Number of processes	4
	TTI	5 ms
	Max Data Rate	2.232 Mbps

6.11.5.4.7.1.2 Downlink

6.11.5.4.7.1.2.1 Transport channel parameters

6.11.5.4.7.1.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.7.1.2.1.1.1 MAC-d flow parameters for Stand-alone DL: [max bit rate depending on UE category] SRBs for HS-DSCH

Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	Depends on UE category and TTI			
	AMD PDU header, bit	8	16	16	16
MAC	MAC-d header, bit	4	4	4	4
	MAC-es multiplexing	4 logical channel multiplexing			
	MAC-d PDU size, bit	148			
	MAC-e/es header fixed part, bit	18			
Layer 1	TrCH type	HS-DSCH			
	TTI	5ms			
	Coding type	TC			
	CRC, bit	24			

6.11.5.4.7.1.2.2 Physical channel parameters

6.11.5.4.7.1.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.7.2 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.5.4.7.2.1 Uplink

6.11.5.4.7.2.1.1 Transport channel parameters

6.11.5.4.7.2.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.2.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.5.4.7.1.1.1.1.1.

6.11.5.4.7.2.1.1.2 Transport channel parameters for DCH

6.11.5.4.7.2.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.5.4.1.2.1.1.1.

6.11.5.4.7.2.1.2 Physical channel parameters

6.11.5.4.7.2.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.2.1.2.2 Physical channel parameters for DPCH

See clause 6.11.5.4.1.2.1.2

6.11.5.4.7.2.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.11.5.4.7.3.1 Uplink

6.11.5.4.7.3.1.1 Transport channel parameters

6.11.5.4.7.3.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.3.1.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.5.4.7.1.1.1.1.1.

6.11.5.4.7.3.1.1.1.2 MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	Depends on UE category and TTI			
	AMD PDU header, bit	8	16	16	16
MAC	MAC-es multiplexing	4 logical channel multiplexing			
	MAC-d PDU size, bit	144			
	MAC-e/es header fixed part, bit	18			
Layer 1	TrCH type	E-DCH			
	TTI	5ms			
	Coding type	TC			
	CRC, bit	24			

6.11.5.4.7.3.1.2 Physical channel parameters

6.11.5.4.7.3.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.3.2 Downlink

See clause 6.11.5.4.6.1.2.

6.11.5.4.7.4 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.11.5.4.7.4.1 Uplink

See clause 6.11.5.4.7.1.1.

6.11.5.4.7.4.1.2 Physical channel parameters

6.11.5.4.7.4.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.4.2 Downlink

6.11.5.4.7.4.2.1 Transport channel parameters

6.11.5.4.7.4.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.7.4.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.6.1.2.1.1.1.

6.11.5.4.7.4.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

See clause 6.11.5.4.7.1.2.1.1.1

6.11.5.4.7.4.2.2 Physical channel parameters

6.11.5.4.7.4.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.11.5.4.6.1.2.2.2.

6.11.5.4.7.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.7.5.1 Uplink

6.11.5.4.7.5.1.1 Transport channel parameters

6.11.5.4.7.5.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.5.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.5.4.7.1.1.1.1.1.

6.11.5.4.7.5.1.1.2 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.5.4.1.4.1.1.1.

6.11.5.4.7.5.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.5.4.1.2.1.1.1.

6.11.5.4.7.5.1.1.4 TFCS

See clause 6.11.5.4.1.4.1.1.3.

6.11.5.4.7.5.1.2 Physical channel parameters

6.11.5.4.7.5.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.5.1.2.2 Physical channel parameters on DCH

See clause 6.11.5.4.1.4.1.2.

6.11.5.4.7.5.2 Downlink

See clause 6.11.5.4.6.6.2.

6.11.5.4.7.6 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.5.4.7.6.1 Uplink

6.11.5.4.7.6.1.1 Transport channel parameters

6.11.5.4.7.6.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.6.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	Depends on UE category and TTI
	UMD PDU header, bit	8
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	5ms
	Coding type	TC
	CRC, bit	24

6.11.5.4.7.6.1.1.2 Transport channel parameters for DCH

6.11.5.4.7.6.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.5.4.1.2.1.1.1.

6.11.5.4.7.6.1.2 Physical channel parameters

6.11.5.4.7.6.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.6.2 Downlink

6.11.5.4.7.6.2.1 Transport channel parameters

6.11.5.4.7.6.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.7.6.2.1.1.1 MAC-d flow parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	328
	Max data rate, bps	Depends on UE category and TTI
	UMD PDU header, bit	8
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	5ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).		

6.11.5.4.7.6.2.1.2 Transport channel parameters for DCH

6.11.5.4.7.6.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.5.4.1.2.2.1.1.

6.11.5.4.7.6.2.1.2.2 TFCS

See clause 6.11.5.4.1.2.2.1.2.

6.11.5.4.7.6.2.2 Physical channel parameters

6.11.5.4.7.6.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.2.2.2.

6.11.5.4.7.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause6.11.5.4.6.1.2.2.2..

6.11.5.4.7.7 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.11.5.4.7.7.1 Uplink

6.11.5.4.7.7.1.1 Transport channel parameters

6.11.5.4.7.7.1.1.1 Transport channel parameters for E-DCH

6.11.5.4.7.7.1.1.1.1 MAC-d flow#1 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.11.5.4.7.3.1.1.1.2.

6.11.5.4.7.7.1.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.5.4.7.6.1.1.1.1

6.11.5.4.7.7.1.1.1.3 MAC-d flow#3 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.5.4.7.6.1.1.1.1

6.11.5.4.7.7.1.2 Physical channel parameters

6.11.5.4.7.7.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.5.4.7.1.1.2.1.

6.11.5.4.7.7.2 Downlink

6.11.5.4.7.7.2.1 Transport channel parameters

6.11.5.4.7.7.2.1.1 Transport channel parameters for HS-DSCH

6.11.5.4.7.7.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.7.6.2.1.1.1.

6.11.5.4.7.7.2.1.1.2 MAC-d flow#2 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.5.4.7.6.2.1.1.1.

6.11.5.4.7.7.2.1.2 Transport channel parameters for DCH

6.11.5.4.7.7.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.5.4.1.2.2.1.1.

6.11.5.4.7.7.2.1.2.2 TFCS

See clause 6.11.5.4.1.2.2.1.2.

6.11.5.4.7.7.2.2 Physical channel parameters

6.11.5.4.7.7.2.2.1 Physical channel parameters on DPCH

See clause 6.11.5.4.1.2.2.2.

6.11.5.4.7.7.2.2.2 Physical channel parameters on HS-PDSCH

See clause6.11.5.4.6.1.2.2.2..

6.11.6 Reference Radio Bearer configurations used in Radio Bearer testing for 7.68 Mcps TDD

6.11.6.1 RABs and signalling RBs

In the following clauses, the typical parameter sets are presented for reference RABs, signalling RBs and important combinations of them. The data rate given for each RAB is the maximum data rate that can be supported by that RAB.

NOTE: The granularity for each RAB needs to be clarified.

Table 6.11.6.1.1: Prioritized RABs

#	Traffic class[3]	SSD[3]	Max. rate, kbps	CS/PS
1	Conversational	Speech	UL:12.2 DL:12.2	CS
1a	Conversational	Speech	UL: (12.2 7.95 5.9 4.75) DL(12.2 7.95 5.9 4.75)	CS
2	Conversational	Speech	UL:10.2 DL:10.2	CS
2a	Conversational	Speech	UL:(10.2 , 6.7, 5.9, 4.75) DL:10.2, 6.7, 5.9, 4.75)	CS
3	Conversational	Speech	UL:7.95 DL:7.95	CS
4	Conversational	Speech	UL:7.4 DL:7.4	CS
4a	Conversational	Speech	UL:(12.2 7.95 5.9 4.75, DL:(12.2 7.95 5.9 4.75)	CS
5	Conversational	Speech	UL:6.7 DL:6.7	CS
6	Conversational	Speech	UL:5.9 DL:5.9	CS
7	Conversational	Speech	UL:5.15 DL:5.15	CS

#	Traffic class[3]	SSD[3]	Max. rate, kbps	CS/PS
8	Conversational	Speech	UL:4.75 DL:4.75	CS
9	Conversational	Unknown	UL:28.8 DL:28.8	CS
10	Conversational	Unknown	UL:64 DL:64	CS
11	Conversational	Unknown	UL:32 DL:32	CS
11a	Conversational	Unknown	UL:8 DL:8	CS
12	Streaming	Unknown	UL:14.4 DL:14.4	CS
13	Streaming	Unknown	UL:28.8 DL:28.8	CS
14	Streaming	Unknown	UL:57.6 DL:57.6	CS
15	Void			
15a	Streaming	Unknown	UL:16 DL:64	PS
16	Void			
17	Void			
18	Void			
19	Void			
20	Interactive or Background	N/A	UL:32 DL:8	PS
20a	Interactive or Background	N/A	UL:8 DL:8	PS
20b	Interactive or Background	N/A	UL:16 DL:16	PS
20c	Interactive or Background	N/A	UL:32 DL:32	PS
21	Void			
22	Interactive or Background	N/A	UL:32 DL:64	PS
23	Interactive or Background	N/A	UL:64 DL:64	PS
24	Interactive or Background	N/A	UL:64 DL:128	PS
25	Interactive or Background	N/A	UL:128 DL:128	PS
26	Interactive or Background	N/A	UL:64 DL:384	PS
27	Interactive or Background	N/A	UL:128 DL:384	PS
28	Interactive or Background	N/A	UL:384 DL:384	PS
29	Interactive or Background	N/A	UL:64 DL:2048	PS
30	Interactive or Background	N/A	UL:128 DL:2048	PS
31	Void			
32	Interactive or Background	N/A	UL:64 DL:256	PS
33	Interactive or Background	N/A	UL:0 DL:32	PS
34	Interactive or Background	N/A	UL:32 DL:0	PS
35	Interactive or Background	N/A	UL:64 DL:144	PS
36	Interactive or Background	N/A	UL:144 DL:144	PS

Table 6.11.6.1.2: Signalling RBs

#	Maximum rate, kbps	Logical channel	PhyCh onto which SRBs are mapped
1	UL:1.7 DL:1.7	DCCH	DPCH
2	UL:3.4 DL:3.4	DCCH	DPCH
3	UL:13.6 DL:13.6	DCCH	DPCH
4	DL:27.2 (alt. 13.6)	DCCH	SCCPCH
5	UL:16.8	CCCH	PRACH
6	DL:32 (alt. 16)	CCCH	SCCPCH
7	DL:33.6 (alt. 16.8)	BCCH	SCCPCH
8	DL:12 (alt. 8)	PCCH	SCCPCH
9	UL:16.8	SHCCH	PRACH
10	UL:16.8	SHCCH	PRACH or PUSCH
11	DL:32 (alt. 16)	SHCCH	SCCPCH
12	DL:16	SHCCH	SCCPCH or PDSCH

6.11.6.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH.

1a) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void.

19) Void.

20) Void.

21) Void.

22) Void..

23) Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

24) Void..

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

37) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / 12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38e) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38f) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38g) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38h) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38i) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38j) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void

47) Void

48) Void

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void.

55) Void

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

57) Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

58) Streaming / unknown / UL:16 DL:64 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

59) Reserved for future use

60) Reserved for future use

61) Conversational / unknown / UL:8 DL:8 kbps / PS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

Combinations on PDSCH, SCCPCH, PUSCH and PRACH

1) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL: 3.4/16.8 DL:3.4/ 33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL:16.8 DL: 16 kbps SRBs for SHCCH.

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

4) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

1) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

2) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

Combinations on SCCPCH

1) Stand-alone 12 kbps SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2a) Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB
+ SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH

2b) SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3a) SRB for PCCH
+ SRB for CCCH
+ SRB for DCCH
+ SRB for BCCH

4) RB for CTCH
+ SRB for CCCH
+ SRB for BCCH

Combinations on PRACH

1) Interactive or background / UL:12.8 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

Combinations on DPCH and HS-PDSCH

1) Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

2) Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3) Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

7) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

8) Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

9) Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

10) Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

11) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6.11.6.3 Example of linkage between RABs and services

RABs, which are included in the present document, can provide the services as shown in table 6.10.1.1: Traffic classes. Furthermore, the required BER for each RAB, which is assumed in the present document, is shown in table 6.11.6.3.1.

Table 6.11.6.3.1: Example of linkage between RABs and services

RAB				Residual BER[3]	Services
Traffic class[3]	SSD[3]	Max. rate, kbps	CS/PS		
Conversational	Speech	UL:4.75-12.2 DL:4.75-12.2	CS	$5x10^{-4}$, $1x10^{-3}$, $5x10^{-3}$	AMR speech
Conversational	Unknown	UL:64 DL:64	CS	$1x10^{-4}$ or $1x10^{-6}$	UDI 1B, 64k 3G-324M[4]
Conversational	Unknown	UL:32 DL:32	CS	$1x10^{-4}$ or $1x10^{-6}$	32k 3G-324M[4]
Conversational	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	Transparent modem
Streaming	Unknown	UL:14.4 DL:14.4	CS	$1x10^{-3}$	FAX[6]
Streaming	Unknown	UL:28.8 DL:28.8	CS	$1x10^{-3}$	FAX[6] PIAFS 32 kbps

Streaming	Unknown	UL:57.6 DL:57.6	CS	$1x10^{-3}$	Modem[6], FTM[5], PIAFS 64 kbps
Streaming	Unknown	UL:64-128 or DL:64-384	CS	$1x10^{-3}$ or $1x10^{-4}$	Streaming video, uni-directional
Interactive or Background	N/A	UL:32-384 DL:8-2048	PS	$1x10^{-3}$ or $1x10^{-4}$	Packet

NOTE 1: SMS can be provided via the signalling RB (DCCH) on DPCH or SCCPCH.

NOTE 2: CBS can be provided via the signalling RB (CTCH) on SCCPCH

NOTE 3: UDI *n*B can be provided via *n* RABs of conversational 64 kbps.

6.11.6.4 Typical radio parameter sets

NOTE The order of tables and MAC-d flow numbering in this section may be different than the RB IDs and MAC-d flow IDs as defined in default messages in section 9.

6.11.6.4.1 Combinations on DPCH

6.11.6.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.6.4.1.1.1 Uplink

####### 6.11.6.4.1.1.1.1 Transport channel parameters

######## 6.11.6.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		80			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		65			
	RM attribute		155 to 185			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

######## 6.11.6.4.1.1.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

####### 6.11.6.4.1.1.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	234

	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.1.2 Downlink

6.11.6.4.1.1.2.1 Transport channel parameters

6.11.6.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0 x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		80			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		65			
	RM attribute		155 to 185			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.11.6.4.1.1.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.11.6.4.1.1.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	236 bits
	TFCI code word	8 bits
	Puncturing limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)

6.11.6.4.1.1a.1 Uplink

6.11.6.4.1.1a.1.1 Transport channel parameters

6.11.6.4.1.1a.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM

	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148			
	TFS	TF0, bits	0x148			
		TF1, bits	1x148			
	TTI, ms		20			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		258			

6.11.6.4.1.1a.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)

6.11.6.4.1.1a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	266
	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	1
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.1a.2 Downlink

6.11.6.4.1.1a.2.1 Transport channel parameters

6.11.6.4.1.1a.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		1 700	1 600	1 600	1 600
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148			
	TFS	TF0, bits	0 x148			
		TF1, bits	1x148			
	TTI, ms		20			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		258			

6.11.6.4.1.1a.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is optional.	

6.11.6.4.1.1a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	268 bits
	TFCI code word	8 bits
	Puncturing limit	1
	Repetition period	8
	Repetition length	2
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.2.1 Uplink

6.11.6.4.1.2.1.1 Transport channel parameters

6.11.6.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		129			
	RM attribute		155 to 165			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

6.11.6.4.1.2.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.11.6.4.1.2.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	234 bits
	TFCI code word	8 bits
	TPC	2 bit
	Puncturing Limit	1

NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.

6.11.6.4.1.2.2 Downlink

6.11.6.4.1.2.2.1 Transport channel parameters

6.11.6.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0, 148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		129			
	RM attribute		155 to 165			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.11.6.4.1.2.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.11.6.4.1.2.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	236
	TFCI code word	8 bits
	Puncturing limit	1
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.11.6.4.1.3.1 Uplink

6.11.6.4.1.3.1.1 Transport channel parameters

6.11.6.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit	8	16	16	16

MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		516			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN.						

6.11.6.4.1.3.1.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.11.6.4.1.3.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	468 bits
	TFCI code word	8 bits
	TPC	2 bits
	Puncturing Limit	0.88
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.3.2 Downlink

6.11.6.4.1.3.2.1 Transport channel parameters

6.11.6.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		13 600	12 800	12 800	12 800
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		4	4	4	4
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DCH			
	TB sizes, bit		148 (alt. 0,148) (note)			
	TFS	TF0, bits	0x148 (alt. 1x0) (note)			
		TF1, bits	1x148			
	TTI, ms		10			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		516			
	Max number of bits/radio frame before rate matching		516			
NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE.						

6.11.6.4.1.3.2.1.2 TFCS

TFCS size	2
TFCS	SRBs for DCCH = (TF0), (TF1)
NOTE: The first TFC is required for the alt. case, optional otherwise.	

6.11.6.4.1.3.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	480 bits
	TFCI code word	8 bits
	Puncturing limit	0.92
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits.		

6.11.6.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.4.1 Uplink

6.11.6.4.1.4.1.1 Transport channel parameters

6.11.6.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 81 (alt. 0, 39, 81)	103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.4.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.4.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.72

6.11.6.4.1.4.2 Downlink

6.11.6.4.1.4.2.1 Transport channel parameters

6.11.6.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39,81 (alt. 0, 39, 81)	103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39,81 (alt. 0,39,81)	103	60
	TFS	TF0, bits	0x81 (alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x103	1x60
		TF2, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.4.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.4.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.11.6.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.4a.1 Uplink

6.11.6.4.1.4a.1.1 Transport channel parameters

6.11.6.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2 bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.4a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.4a.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.72

6.11.6.4.1.4a.2 Downlink

6.11.6.4.1.4a.2.1 Transport channel parameters

6.11.6.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	Max data rate, bps		12 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81)	53, 63, 84, 103	60
	TFS	TF0, bits	0x81(alt. 1x0) (note)	0x103	0x60
		TF1, bits	1x39	1x53	1x60
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x84	N/A
		TF4, bits	1x75	1x103	N/A
		TF5, bits	1x81	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		303	333	136
	Max number of bits/radio frame before rate matching		152	167	68
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.4a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.4a.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.11.6.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.5.1 Uplink

6.11.6.4.1.5.1.1 Transport channel parameters

6.11.6.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 65 (alt. 0, 39, 65)	99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.5.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.5.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.40

6.11.6.4.1.5.2 Downlink

6.11.6.4.1.5.2.1 Transport channel parameters

6.11.6.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type	DTCH		
	RLC mode	TM	TM	TM
	Payload sizes, bit	39,65 (alt. 0, 39, 65)	99	40
	Max data rate, bps	10 200		
	TrD PDU header, bit	0		
MAC	MAC header, bit	0		
	MAC multiplexing	N/A		
Layer 1	TrCH type	DCH	DCH	DCH

	TB sizes, bit		39, 65 (alt.0,39,65)	99	40
	TFS	TF0, bits	0x65 (alt,1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x99	1x40
		TF2, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48
	RM attribute		180 to 220	170 to 210	215 to 256
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).					

6.11.6.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.5.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0), (TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.5.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.40

6.11.6.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.5a.1 Uplink

6.11.6.4.1.5a.1.1 Transport channel parameters

6.11.6.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	53, 63, 76, 99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20

	Coding type	CC 1/3	CC 1/3	CC 1/2
	CRC, bit	12	N/A	N/A
	Max number of bits/TTI after channel coding	255	321	96
	Max number of bits/radio frame before rate matching	128	161	48
	RM attribute	180 to 220	170 to 210	215 to 256
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in TS 25.222).				

6.11.6.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.5a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.5a.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.40

6.11.6.4.1.5a.2 Downlink

6.11.6.4.1.5a.2.1 Transport channel parameters

6.11.6.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2	RAB subflow #3
RLC	Logical channel type		DTCH		
	RLC mode		TM	TM	TM
	Payload sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	0, 53, 63, 76, 99	40
	Max data rate, bps		10 200		
	TrD PDU header, bit		0		
MAC	MAC header, bit		0		
	MAC multiplexing		N/A		
Layer 1	TrCH type		DCH	DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65)	0, 53, 63, 76, 99	40
	TFS	TF0, bits	0x65 (alt. 1x0) (note)	0x99	0x40
		TF1, bits	1x39	1x53	1x40
		TF2, bits	1x42	1x63	N/A
		TF3, bits	1x55	1x76	N/A
		TF4, bits	1x58	1x99	N/A
		TF5, bits	1x65	N/A	N/A
	TTI, ms		20	20	20
	Coding type		CC 1/3	CC 1/3	CC 1/2
	CRC, bit		12	N/A	N/A
	Max number of bits/TTI after channel coding		255	321	96
	Max number of bits/radio frame before rate matching		128	161	48

	RM attribute	180 to 220	170 to 210	215 to 256
NOTE:	In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.11.6.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.5a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.5a.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.40

6.11.6.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.6.1 Uplink

6.11.6.4.1.6.1.1 Transport channel parameters

6.11.6.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 75 (alt. 0, 39, 75)	84
	Max data rate, bps		7 950	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 75 (alt. 0, 39, 75)	84
	TFS	TF0, bits	0x75 (alt. 1x0) (note)	0x84
		TF1, bits	1x39	1x84
		TF2, bits	1x75	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		285	276
	Max number of bits/radio frame before rate matching		143	138
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.6.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.6.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44

6.11.6.4.1.6.2 Downlink

6.11.6.4.1.6.2.1 Transport channel parameters

6.11.6.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 75 (alt. 0, 39, 75)	84
	Max data rate, bps		7 950	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 75 (alt. 0, 39, 75)	84
	TFS	TF0, bits	0x75 (alt. 1x0) (note)	0x84
		TF1, bits	1x39	1x84
		TF2, bits	1x75	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		285	276
	Max number of bits/radio frame before rate matching		143	138
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.6.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.6.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.7.1 Uplink

6.11.6.4.1.7.1.1 Transport channel parameters

6.11.6.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 61 (alt. 0, 39, 61)	87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 61 (alt. 0, 39, 61)	87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x87
		TF2, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.7.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.7.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.11.6.4.1.7.2 Downlink

6.11.6.4.1.7.2.1 Transport channel parameters

6.11.6.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2
RLC	Logical channel type	DTCH	
	RLC mode	TM	TM

	Payload sizes, bit		39, 61 (alt. 0, 39, 61)	87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 61 (alt. 0, 39, 61)	87
	TFS	TF0, bits	0x61(alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x87
		TF2, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB #1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.7.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; optional otherwise.	

6.11.6.4.1.7.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.7a.1 Uplink

6.11.6.4.1.7a.1.1 Transport channel parameters

6.11.6.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87

		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		243	285
	Max number of bits/radio frame before rate matching		122	143
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.7a.1.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.7a.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.11.6.4.1.7a.2 Downlink

6.11.6.4.1.7a.2.1 Transport channel parameters

6.11.6.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	Max data rate, bps		7 400	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61)	53, 63, 76, 87
	TFS	TF0, bits	0x61 (alt. 1x0) (note)	0x87
		TF1, bits	1x39	1x53
		TF2, bits	1x42	1x63
		TF3, bits	1x55	1x76
		TF4, bits	1x58	1x87
		TF5, bits	1x61	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3

	CRC, bit	12	N/A
	Max number of bits/TTI after channel coding	243	285
	Max number of bits/radio frame before rate matching	122	143
	RM attribute	180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.11.6.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.7a.2.1.3 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.7a.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.8.1 Uplink

6.11.6.4.1.8.1.1 Transport channel parameters

6.11.6.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 58 (alt. 0, 39, 58)	76
	Max data rate, bps		6 700	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 58 (alt. 0, 39, 58)	76
	TFS	TF0, bits	0x58 (alt. 1x0) (note)	0x76
		TF1, bits	1x39	1x76
		TF2, bits	1x58	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		234	252
	Max number of bits/radio frame before rate matching		117	126
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.8.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.8.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52

6.11.6.4.1.8.2 Downlink

6.11.6.4.1.8.2.1 Transport channel parameters

6.11.6.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 58 (alt. 0, 39, 58)	76
	Max data rate, bps		6 700	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 58 (alt. 0,39,58)	76
	TFS	TF0, bits	0x58 (alt.1x0) (note)	0x76
		TF1, bits	1x39	1x76
		TF2, bits	1x58	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		234	252
	Max number of bits/radio frame before rate matching		117	126
	RM attribute		180 to 220	170 to 210
NOTE : CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.8.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.8.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot

	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.11.6.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.9.1 Uplink

6.11.6.4.1.9.1.1 Transport channel parameters

6.11.6.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 55 (alt. 0, 39, 55)	63
	Max data rate, bps		5 900	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 55 (alt. 0, 39, 55)	63
	TFS	TF0, bits	0x55 (alt. 1x0) (note)	0x63
		TF1, bits	1x39	1x63
		TF2, bits	1x55	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		225	213
	Max number of bits/radio frame before rate matching		113	107
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.9.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.9.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.56

6.11.6.4.1.9.2 Downlink

6.11.6.4.1.9.2.1 Transport channel parameters

6.11.6.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 55 (alt. 0, 39, 55)	63
	Max data rate, bps		5 900	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 55 (alt. 0, 39, 55)	63
	TFS	TF0, bits	0x55 (alt. 1x0) (note)	0x63
		TF1, bits	1x39	1x63
		TF2, bits	1x55	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		225	213
	Max number of bits/radio frame before rate matching		113	107
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.9.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.9.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.6.4.1.10.1 Uplink

6.11.6.4.1.10.1.1 Transport channel parameters

6.11.6.4.1.10.1.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB subflow #1	RAB subflow #2
RLC	Logical channel type	DTCH	
	RLC mode	TM	TM
	Payload sizes, bit	39, 49 (alt. 0, 39, 49)	54
	Max data rate, bps	5 150	
	TrD PDU header, bit	0	

MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 49 (alt. 0, 39, 49)	54
	TFS	TF0, bits	0x49 (alt. 1x0) (note)	0x54
		TF1, bits	1x39	1x54
		TF2, bits	1x49	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		207	186
	Max number of bits/radio frame before rate matching		104	93
	RM attribute		180 to 220	170 to 210
NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.10.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.11.6.4.1.1.1.1.

6.11.6.4.1.10.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.10.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72

6.11.6.4.1.10.2 Downlink

6.11.6.4.1.10.2.1 Transport channel parameters

6.11.6.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 49 (alt. 0, 39, 49)	54
	Max data rate, bps		5 150	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 49 (alt. 0, 39, 49)	54
	TFS	TF0, bits	0x49 (alt. 1x0) (note)	0x54
		TF1, bits	1x39	1x54
		TF2, bits	1x49	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		207	186
	Max number of bits/radio frame before rate matching		104	93
	RM attribute		180 to 220	170 to 210

NOTE:	CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).

6.11.6.4.1.10.2.1.2 Transport channel parameters for DL: 1.7 kbps SRBs for DCCH

See clause 6.11.6.4.1.1.2.1.1.

6.11.6.4.1.10.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE:	In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.

6.11.6.4.1.10.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.72

6.11.6.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.6.4.1.11.1 Uplink

6.11.6.4.1.11.1.1 Transport channel parameters

6.11.6.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42 (alt. 0, 39, 42)	53
	Max data rate, bps		4 750	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42 (alt. 0, 39, 42)	53
	TFS	TF0, bits	0x42 (alt. 1x0) (note)	0x53
		TF1, bits	1x39	1x53
		TF2, bits	1x42	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		186	183
	Max number of bits/radio frame before rate matching		93	92
	RM attribute		180 to 220	170 to 210
NOTE:	In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.11.6.4.1.11.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.11.6.4.1.1.1.1.1.

6.11.6.4.1.11.1.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)=

	(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.11.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76

6.11.6.4.1.11.2 Downlink

6.11.6.4.1.11.2.1 Transport channel parameters

6.11.6.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB subflow #1	RAB subflow #2
RLC	Logical channel type		DTCH	
	RLC mode		TM	TM
	Payload sizes, bit		39, 42 (alt. 0, 39, 42)	53
	Max data rate, bps		4 750	
	TrD PDU header, bit		0	
MAC	MAC header, bit		0	
	MAC multiplexing		N/A	
Layer 1	TrCH type		DCH	DCH
	TB sizes, bit		39, 42 (alt. 0, 39, 42)	53
	TFS	TF0, bits	0X42 (alt.1x0)(note)	0x53
		TF1, bits	1x39	1x53
		TF2, bits	1x42	N/A
	TTI, ms		20	20
	Coding type		CC 1/3	CC 1/3
	CRC, bit		12	N/A
	Max number of bits/TTI after channel coding		186	183
	Max number of bits/radio frame before rate matching		93	92
	RM attribute		180 to 220	170 to 210
NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).				

6.11.6.4.1.11.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.11.6.4.1.1.2.1.1.

6.11.6.4.1.11.2.1.3 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.11.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.11.6.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.12.1 Uplink

6.11.6.4.1.12.1.1 Transport channel parameters

6.11.6.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		160 to 200

6.11.6.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.12.1.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.12.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76
NOTE: In case the first TFC in a TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.12.2 Downlink

6.11.6.4.1.12.2.1 Transport channel parameters

6.11.6.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	576
	Max data rate, bps	28 800
	TrD PDU header, bit	0
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH

	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		160 to 200

6.11.6.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.12.2.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.12.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.40
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.13.1 Uplink

6.11.6.4.1.13.1.1 Transport channel parameters

6.11.6.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948
	Max number of bits/radio frame before rate matching		1 974
	RM attribute		150 to 195

6.11.6.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.13.1.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.13.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot + SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1148 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.13.2 Downlink

6.11.6.4.1.13.2.1 Transport channel parameters

6.11.6.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	2x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 948
	Max number of bits/radio frame before rate matching		1 974
	RM attribute		150 to 195

6.11.6.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.13.2.1.3 TFCS

TFCS size	4
TFCS	(64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.13.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 204 bits
	TFCI code word	16 bits
	Puncturing limit	0.52
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.14.1 Uplink

6.11.6.4.1.14.1.1 Transport channel parameters

6.11.6.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		640
	Max data rate, bps		32 000
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		640
	TFS	TF0, bits	0x640
		TF1, bits	1x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 980
	Max number of bits/radio frame before rate matching		990
	RM attribute		165 to 210

6.11.6.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.14.1.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.14.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.68
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.14.2 Downlink

6.11.6.4.1.14.2.1 Transport channel parameters

6.11.6.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	640
	Max data rate, bps	32 000
	TrD PDU header, bit	0
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH
	TB sizes, bit	640

	TFS	TF0, bits	0x640
		TF1, bits	1x640
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 980
	Max number of bits/radio frame before rate matching		990
	RM attribute		165 to 210

6.11.6.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.14.2.1.3 TFCS

TFCS size	4
TFCS	(32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.14.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716 bits
	TFCI code word	16 bits
	Puncturing limit	0.52
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.15.1 Uplink

6.11.6.4.1.15.1.1 Transport channel parameters

6.11.6.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	Max number of bits/radio frame before rate matching		447
	RM attribute		145 to 185

6.11.6.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.15.1.1.3 TFCS

TFCS size	4

TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.15.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.15.2 Downlink

6.11.6.4.1.15.2.1 Transport channel parameters

6.11.6.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		14 400
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 788
	Max number of bits/radio frame before rate matching		447
	RM attribute		145 to 185

6.11.6.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.15.2.1.3 TFCS

TFCS size	4
TFCS	(14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.15.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.80
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.16.1 Uplink

6.11.6.4.1.16.1.1 Transport channel parameters

6.11.6.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		28 800
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		135 to 175

6.11.6.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.16.1.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.16.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.16.2 Downlink

6.11.6.4.1.16.2.1 Transport channel parameters

6.11.6.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	TM
	Payload sizes, bit	576
	Max data rate, bps	28 800
	TrD PDU header, bit	0
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH

	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		3 564
	Max number of bits/radio frame before rate matching		891
	RM attribute		135 to 175

6.11.6.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.16.2.1.3 TFCS

TFCS size	4
TFCS	(28.8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.16.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.44
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.17.1 Uplink

6.11.6.4.1.17.1.1 Transport channel parameters

6.11.6.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		57 600
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	Max number of bits/radio frame before rate matching		1 779
	RM attribute		125 to 165

6.11.6.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.17.1.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.17.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44

6.11.6.4.1.17.2 Downlink

6.11.6.4.1.17.2.1 Transport channel parameters

6.11.6.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		TM
	Payload sizes, bit		576
	Max data rate, bps		57 600
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		576
	TFS	TF0, bits	0x576
		TF1, bits	1x576
		TF2, bits	2x576
		TF3, bits	3x576
		TF4, bits	4x576
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		7 116
	Max number of bits/radio frame before rate matching		1 779
	RM attribute		125 to 165

6.11.6.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.17.2.1.3 TFCS

TFCS size	10
TFCS	(57.6 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.17.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 4 codes x 1 time slot

	Max. Number of data bits/radio frame	960 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.18 Void

6.11.6.4.1.19 Void

6.11.6.4.1.20 Void

6.11.6.4.1.21 Void

6.11.6.4.1.22 Void

6.11.6.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.23.1 Uplink

6.11.6.4.1.23.1.1 Transport channel parameters

6.11.6.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt.144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching		1 062 (alt. 1 206)
	RM attribute		135 to 175

6.11.6.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.23.1.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.11.6.4.1.23.2 Downlink

6.11.6.4.1.23.2.1 Transport channel parameters

6.11.6.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Max number of bits/radio frame before rate matching		267
	RM attribute		135 to 175

6.11.6.4.1.23.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.23.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.56
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.23a.1 Uplink

6.11.6.4.1.23a.1.1 Transport channel parameters

6.11.6.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		8 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)

		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	N/A (alt. 5x144)
	TTI, ms		40 (alt. 80)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068 (alt. 2 412)
	Max number of bits/radio frame before rate matching		267 (alt. 302)
	RM attribute		135 to 175

6.11.6.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.23a.1.1.3 TFCS

TFCS size	4 (alt. 6)
TFCS	(8 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23a.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.56 (alt. 0.48)

6.11.6.4.1.23a.2 Downlink

See clause 6.11.6.4.1.23.2.

6.11.6.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.23b.1 Uplink

6.11.6.4.1.23b.1.1 Transport channel parameters

6.11.6.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching		531 (alt. 603)
	RM attribute		135 to 175

6.11.6.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.23b.1.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23b.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68 (alt. 0.60)

6.11.6.4.1.23b.2 Downlink

6.11.6.4.1.23b.2.1 Transport channel parameters

6.11.6.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Max number of bits/radio frame before rate matching		531
	RM attribute		135 to 175

6.11.6.4.1.23b.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.23b.2.1.3 TFCS

TFCS size	6
TFCS	(16 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23b.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits

	Puncturing limit	0.68

6.11.6.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.23c.1 Uplink

6.11.6.4.1.23c.1.1 Transport channel parameters

6.11.6.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 5x144)
		TF3, bits	3x336 (alt. 7x144)
		TF4, bits	4x336 (alt. 10x144)
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236 (alt. 4 812)
	Max number of bits/radio frame before rate matching		1 059 (alt. 1 203)
	RM attribute		135 to 175

6.11.6.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.23c.1.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23c.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.11.6.4.1.23c.2 Downlink

6.11.6.4.1.23c.2.1 Transport channel parameters

6.11.6.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM

	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Max number of bits/radio frame before rate matching		1 059
	RM attribute		135 to 175

6.11.6.4.1.23c.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.23c.2.1.3 TFCS

TFCS size	10
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23c.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716
	TFCI code word	16 bits
	Puncturing limit	0.60

6.11.6.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.23d.1 Uplink

6.11.6.4.1.23d.1.1 Transport channel parameters

6.11.6.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt 1x144)
		TF2, bits	2x336 (alt. 5x144)
	TTI, ms		20
	Coding type		TC

	CRC, bit	16
	Max number of bits/TTI after channel coding	2 124 (alt. 2 412)
	Max number of bits/radio frame before rate matching	1 062 (alt. 1 206)
	RM attribute	135 to 175

6.11.6.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.23d.1.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23d.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.11.6.4.1.23d.2 Downlink

6.11.6.4.1.23d.2.1 Transport channel parameters

6.11.6.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 124
	Max number of bits/radio frame before rate matching		1 062
	RM attribute		135 to 175

6.11.6.4.1.23d.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.23d.2.1.3 TFCS

TFCS size	6
TFCS	(32 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.23d.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	716 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.24 Void

6.11.6.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.25.1 Uplink

See clause 6.11.6.4.1.23.1.

6.11.6.4.1.25.2 Downlink

6.11.6.4.1.25.2.1 Transport channel parameters

6.11.6.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	3x336
		TF4, bits	4x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236
	Max number of bits/radio frame before rate matching		2 118
	RM attribute		130 to 170

6.11.6.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.25.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.25.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	1024 chips	1024 chips
	Codes and time slots	SF32 x 3 codes x 1 time slot + SF32 x 2 codes x 1 time slot	SF32 x 9 codes x 1 time slot

	Max. Number of data bits/radio frame	1 204 bits	2 180 bits
	TFCI code word	16 bits	16 bits
	Puncturing limit	0.52	0.96

6.11.6.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.26.1 Uplink

6.11.6.4.1.26.1.1 Transport channel parameters

6.11.6.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		64 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 3x144)
		TF3, bits	3x336 (alt. 7x144)
		TF4, bits	4x336 (alt. 10x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 236 (alt. 4 812)
	Max number of bits/radio frame before rate matching		2 118 (alt. 2 406)
	RM attribute		130 to 170

6.11.6.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.26.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.11.6.4.1.26.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	1024 chips	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot + SF8 x 1 code x 1 time slot	SF4 x 1 code x 1 time slot + SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1148 bits	2 784 bits
	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.48 (alt. 0.44)	1

6.11.6.4.1.26.2 Downlink

See clause 6.11.6.4.1.25.2.

6.11.6.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.27.1 Uplink

See clause 6.11.6.4.1.26.1.

6.11.6.4.1.27.2 Downlink

6.11.6.4.1.27.2.1 Transport channel parameters

6.11.6.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		128 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460
	Max number of bits/radio frame before rate matching		4 230
	RM attribute		120 to 160

6.11.6.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.27.2.1.3 TFCS

TFCS size	10
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.27.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot	SF32 x 4 codes x 2 time slots + SF32 x 3 codes x 2 time slots
	Max. Number of data bits/radio frame	2 192 bits	3848 bits
	TFCI code word	16 bits	16 bits
	Puncturing limit	0.48	0.84

6.11.6.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.28.1 Uplink

6.11.6.4.1.28.1.1 Transport channel parameters

6.11.6.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		128 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 7x144)
		TF3, bits	4x336 (alt 14x144)
		TF4, bits	8x336 (alt. 20x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 9 612)
	Max number of bits/radio frame before rate matching		4 230 (alt. 4 806)
	RM attribute		120 to 160

6.11.6.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.28.1.1.3 TFCS

TFCS size	9 (alt.10)
TFCS	(128 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.28.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF4 x 1 code x 1 timeslot	SF4 x 1 code x 2 timeslots + SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits	5 376 bits
	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.44 (alt. 0.40)	1

6.11.6.4.1.28.2 Downlink

See clause 6.11.6.4.1.27.2.

6.11.6.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.29.1 Uplink

See clause 6.11.6.4.1.26.1.

6.11.6.4.1.29.2 Downlink

6.11.6.4.1.29.2.1 Transport channel parameters

6.11.6.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		144 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	9x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516
	Max number of bits/radio frame before rate matching		4 758
	RM attribute		140 to 180

6.11.6.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.29.2.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.29.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 9 codes x 1 time slot
	Max. Number of data bits/radio frame	2468 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.30.1 Uplink

6.11.6.4.1.30.1.1 Transport channel parameters

6.11.6.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320 (alt. 128)
	Max data rate, bps	144 000
	AMD PDU header, bit	16
MAC	MAC header, bit	0
	MAC multiplexing	N/A
Layer 1	TrCH type	DCH
	TB sizes, bit	336 (alt. 144)

	TFS	TF0, bits	0x336 (alt. 0x144)
		TF1, bits	1x336 (alt. 1x144)
		TF2, bits	2x336 (alt. 10x144)
		TF3, bits	4x336 (alt. 20x144)
		TF4, bits	8x336 (alt. 30x144)
		TF5, bits	9x336 (alt. 45x144)
	TTI, ms		20 (alt. 40)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		9 516 (alt. 21 624)
	Max number of bits/radio frame before rate matching		4 758 (alt. 5 406)
	RM attribute		140 to 180

6.11.6.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.30.1.1.3 TFCS

TFCS size	12
TFCS	(144 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.30.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF32 x 1 code x 1 time slot + SF4 x 1 codex 1 time slot
	Max. Number of data bits/radio frame	2340 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44 (alt. 0.40)

6.11.6.4.1.30.2 Downlink

See clause 6.11.6.4.1.29.2.

6.11.6.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.31.1 Uplink

See clause 6.11.6.4.1.26.1.

6.11.6.4.1.31.2 Downlink

6.11.6.4.1.31.2.1 Transport channel parameters

6.11.6.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336

Higher layer	RAB/Signalling RB		RAB
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	N/A (alt. 12x336)
		TF6, bits	N/A (alt. 16x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 460 (alt. 16 920)
	Max number of bits/radio frame before rate matching		8 460 (alt. 8 460)
	RM attribute		135 to 175

6.11.6.4.1.31.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.31.2.1.3 TFCS

TFCS size	10 (alt.14)
TFCS	(256 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.31.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.32.1 Uplink

See clause 6.11.6.4.1.26.1.

6.11.6.4.1.32.2 Downlink

6.11.6.4.1.32.2.1 Transport channel parameters

6.11.6.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		384 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336

Higher layer	RAB/Signalling RB		RAB
		TF6, bits	N/A (alt. 16x336)
		TF7, bits	N/A (alt. 20x336)
		TF8, bits	N/A (alt. 24x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684 (alt. 25 368)
	Max number of bits/radio frame before rate matching		12 684 (alt. 12 684)
	RM attribute		110 to 150

6.11.6.4.1.32.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.32.2.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0) (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.32.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF32 x 8 codes x 3 time slots	SF32 x 6 codes x 4 time slots + SF32 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	6 608 bits	7 712 bits
	TFCI code word	16 bits	16 bits
	Puncturing Limit	0.48	0.60

6.11.6.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.33.1 Uplink

See clause 6.11.6.4.1.28.1.

6.11.6.4.1.33.2 Downlink

See clause 6.11.6.4.1.32.2.

6.11.6.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.34.1 Uplink

6.11.6.4.1.34.1.1 Transport channel parameters

6.11.6.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	384 000
	AMD PDU header, bit	16

MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
		TF2, bits	2x336
		TF3, bits	4x336
		TF4, bits	8x336
		TF5, bits	12x336
		TF6, bits	N/A (alt. 16x336)
		TF7, bits	N/A (alt. 20x336)
		TF8, bits	N/A (alt. 24x336)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		12 684 (alt. 25 368)
	Max number of bits/radio frame before rate matching		12 684
	RM attribute		110 to 150

6.11.6.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.34.1.1.3 TFCS

TFCS size	12 (alt.18)
TFCS	(384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.34.1.2 Physical channel parameters

DPCH Uplink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF4 x 1 code x 3 time slots	SF4 x 1 code x 5 timeslots + SF8 x 1 code x 2 timeslots (alt. {SF4 x 1 code + SF8 x 1 code} x 4 timeslots)
	Max. Number of data bits/radio frame	6 480 bits	13 104 bits
	TFCI code word	16 bits	16 bits
	TPC	2 bits	2 bits
	Puncturing Limit	0.48	1

6.11.6.4.1.34.2 Downlink

See clause 6.11.6.4.1.32.2.

6.11.6.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.35.1 Uplink

6.11.6.4.1.35.1.1 Transport channel parameters

See clause 6.11.6.4.1.26.1.1.

6.11.6.4.1.35.1.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.88 (alt. 0.80)

6.11.6.4.1.35.2 Downlink

6.11.6.4.1.35.2.1 Transport channel parameters

6.11.6.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		2 048 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
		TF4, bits	8x656
		TF5, bits	12x656
		TF6, bits	16x656
		TF7, bits	20x656
		TF8, bits	24x656
		TF9, bits	28x656
		TF10, bits	31x656 (alt. 32x656)
		TF11, bits	N/A (alt. 36x656)
		TF12, bits	N/A (alt. 40x656)
		TF13, bits	N/A (alt. 44x656)
		TF14, bits	N/A (alt. 48x656)
		TF15, bits	N/A (alt. 52x656)
		TF16, bits	N/A (alt. 56x656)
		TF17, bits	N/A (alt. 60x656)
		TF18, bits	N/A (alt. 64x656)
	TTI, ms		10 (alt. 20)
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		62 565 (alt. 129 141)
	Max number of bits/radio frame before rate matching		62 565 (alt. 64 571)
	RM attribute		130 to 170

6.11.6.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.35.2.1.3 TFCS

TFCS size	21 (alt.38)
TFCS	(2 048 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1) (alt. TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0),

	(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1), (TF16, TF1), (TF17, TF1(TF18, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.35.2.2 Physical channel parameters

DPCH Downlink		Physical Configuration 1	Physical Configuration 2
	Midamble	512 chips	512 chips
	Codes and time slots	SF1 x 1 code x 5 time slots	SF8 x 13 codes x 4 time slots + SF8 x 12 codes x 7 time slot
	Max. Number of data bits/radio frame	44 144 bits (alt. 44 128)	37 520 bits (alt. 37 504)
	TFCI code word	16 bits (alt. 32 bits)	16 bits (alt. 32 bits)
	Puncturing limit	0.68 (alt.0.68)	0.56

6.11.6.4.1.36 Void

6.11.6.4.1.37 Void

6.11.6.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38.1 Uplink

6.11.6.4.1.38.1.1 Transport channel parameters

6.11.6.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.11.6.4.1.23.1.1.1.

6.11.6.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38.1.1.4 TFCS

TFCS size	18
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)

6.11.6.4.1.38.2 Downlink

6.11.6.4.1.38.2.1 Transport channel parameters

6.11.6.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.

6.11.6.4.1.38.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3,8kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.11.6.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38a.1 Uplink

6.11.6.4.1.38a.1.1 Transport channel parameters

6.11.6.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320 (alt. 128)
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336 (alt. 144)
	TFS	TF0, bits	0x336 (alt 0x144)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.11.6.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38a.1.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38a.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.11.6.4.1.38a.2 Downlink

6.11.6.4.1.38a.2.1 Transport channel parameters

6.11.6.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		0
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		0
	Max number of bits/radio frame before rate matching		0
	RM attribute		130 to 170

6.11.6.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1

6.11.6.4.1.38a.2.1.4 TFCS

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38a.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38b.1 Uplink

6.11.6.4.1.38b.1.1 Transport channel parameters

6.11.6.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.11.6.4.1.23a.1.1.1.

6.11.6.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38b.1.1.4 TFCS

TFCS size	12 (alt. 17)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38b.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48 (alt. 0.56)

6.11.6.4.1.38b.2 Downlink

6.11.6.4.1.38b.2.1 Transport channel parameters

6.11.6.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38b.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38b.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.11.6.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38c.1 Uplink

6.11.6.4.1.38c.1.1 Transport channel parameters

6.11.6.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.11.6.4.1.23d.1.1.1.

6.11.6.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38c.1.1.4 TFCS

TFCS size	18 (alt. 17)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38c.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	904 bits
	TFCI code word	16 bits
	TPC	2 bits

	Puncturing Limit	0.52 (alt. 0.52)

6.11.6.4.1.38c.2 Downlink

6.11.6.4.1.38c.2.1 Transport channel parameters

6.11.6.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.11.6.4.1.23d.2.1.1.

6.11.6.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38c.2.1.4 TFCS

TFCS size	18
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38c.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	960
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38d.1 Uplink

6.11.6.4.1.38d.1.1 Transport channel parameters

6.11.6.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320 (alt. 128)	320 (alt. 128)
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340 (alt. 148)	
	TFS	TF0, bits	0x340 (alt 0x148)	

		TF1, bits	1x340 (alt 1x148)
		TF2, bits	2x340 (alt 3x148)
		TF3, bits	3x340 (alt 7x148)
		TF4, bits	4x340 (alt 10x148)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 284 (alt. 4 932)
	Max number of bits/radio frame before rate matching		2 142 (alt. 2 466)
	RM attribute		130 to 170

6.11.6.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38d.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38d.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.64)

6.11.6.4.1.38d.2 Downlink

6.11.6.4.1.38d.2.1 Transport channel parameters

6.11.6.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	

		TF3, bits	3x340
		TF4, bits	4x340
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		4 284
	Max number of bits/radio frame before rate matching		2 142
	RM attribute		130 to 170

6.11.6.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38d.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38d.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 916 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38e.1 Uplink

6.11.6.4.1.38e.1.1 Transport channel parameters

6.11.6.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.11.6.4.1.38a.1.1.2.

6.11.6.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38e.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),

	(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38e.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.11.6.4.1.38e.2 Downlink

6.11.6.4.1.38e.2.1 Transport channel parameters

6.11.6.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.11.6.4.1.38a.2.1.2.

6.11.6.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38e.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38e.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38f.1 Uplink

6.11.6.4.1.38f.1.1 Transport channel parameters

6.11.6.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.11.6.4.1.23a.1.1.1.

6.11.6.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38f.1.1.4 TFCS

TFCS size	24 (alt. 32)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) (TF0,TF0,TF0,TF2,TF1), (TF1,TF0, TF0, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38f.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48 (alt.0.56)

6.11.6.4.1.38f.2 Downlink

6.11.6.4.1.38f.2.1 Transport channel parameters

6.11.6.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38f.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),

	(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38f.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.11.6.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38g.1 Uplink

6.11.6.4.1.38g.1.1 Transport channel parameters

6.11.6.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.11.6.4.1.23b.1.1.1.

6.11.6.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38g.1.1.4 TFCS

TFCS size	32 (alt. 31)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1) (alt. (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1))
NOTE 1: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. NOTE 2: The alt. TFCS is used when the 16Kbps RAB alt. is used.	

6.11.6.4.1.38g.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	664 bits (alt. 696 bits)
	TFCI code word	32 bits (alt. 16 bits)
	TPC	2 bits
	Puncturing Limit	0.56 (alt. 0.60)

6.11.6.4.1.38g.2 Downlink

6.11.6.4.1.38g.2.1 Transport channel parameters

6.11.6.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.11.6.4.1.23b.2.1.1.

6.11.6.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1

6.11.6.4.1.38g.2.1.4 TFCS

TFCS size	36
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38g.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 3 codes x 1 time slot
	Max. Number of data bits/radio frame	700 bits
	TFCI code word	32 bits
	Puncturing limit	0.56

6.11.6.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38h.1 Uplink

6.11.6.4.1.38h.1.1 Transport channel parameters

6.11.6.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.11.6.4.1.23d.1.1.1.

6.11.6.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38h.1.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38h.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF8 x 1 code x 1 time slot + SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 084 bits
	TFCI code word	32 bits
	TPC	2 bits
	Puncturing Limit	0.68 (alt.0.60)

6.11.6.4.1.38h.2 Downlink

6.11.6.4.1.38h.2.1 Transport channel parameters

6.11.6.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.11.6.4.1.23d.2.1.1.

6.11.6.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38h.2.1.4 TFCS

TFCS size	32
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF1,TF0,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)

NOTE:	In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.

6.11.6.4.1.38h.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	944
	TFCI code word	32 bits
	Puncturing limit	0.60

6.11.6.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38i.1 Uplink

6.11.6.4.1.38i.1.1 Transport channel parameters

6.11.6.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.11.6.4.1.26.1.1.1.

6.11.6.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.38i.1.1.4 TFCS

TFCS size	48
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE:	In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.

6.11.6.4.1.38i.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 936 bits
	TFCI code word	32 bit
	TPC	2 bits
	Puncturing Limit	0.68 (alt.0.60)

6.11.6.4.1.38i.2 Downlink

6.11.6.4.1.38i.2.1 Transport channel parameters

6.11.6.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.11.6.4.1.25.2.1.1.

6.11.6.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38i.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38i.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 900 bits
	TFCI code word	32 bits
	Puncturing limit	0.68

6.11.6.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.38j.1 Uplink

See clause 6.11.6.4.1.38i.1

6.11.6.4.1.38j.2 Downlink

6.11.6.4.1.38j.2.1 Transport channel parameters

6.11.6.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.11.6.4.1.27.2.1.1.

6.11.6.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.38j.2.1.4 TFCS

TFCS size	60
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.38j.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 6 codes x 2 time slots
	Max. Number of data bits/radio frame	3 280 bits
	TFCI code word	32 bits
	Puncturing limit	0.64

6.11.6.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.39.1 Uplink

See clause 6.11.6.4.1.38.1.

6.11.6.4.1.39.2 Downlink

6.11.6.4.1.39.2.1 Transport channel parameters

6.11.6.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.11.6.4.1.25.2.1.1.

6.11.6.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.39.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.39.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	1 936 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.6.4.1.40.1 Uplink

6.11.6.4.1.40.1.1 Transport channel parameters

6.11.6.4.1.40.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.

6.11.6.4.1.40.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.11.6.4.1.26.1.1.

6.11.6.4.1.40.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.40.1.1.4 TFCS

6.11.6.4.1.40.1.1.4.1 TFCS (one CCTrCH case)

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.40.1.1.4.2 TFCS (two CCTrCH case)

6.11.6.4.1.40.1.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.40.1.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.40.1.2 Physical channel parameters

6.11.6.4.1.40.1.2.1 Physical channel (one CCTrCH case)

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.64 (alt. 0.56)

6.11.6.4.1.40.1.2.2 Physical channel (two CCTrCH case)

6.11.6.4.1.40.1.2.2.1 Physical channel (conversational + SRB)

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bit
	Puncturing Limit	0.68

6.11.6.4.1.40.1.2.2.2 Physical channel (Interactive or background)

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.84 (alt. 0.72)

6.11.6.4.1.40.2 Downlink

Transport channel parameters

6.11.6.4.1.40.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.40.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.11.6.4.1.25.2.1.1.

6.11.6.4.1.40.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.40.2.1.4 TFCS

6.11.6.4.1.40.2.1.4.1 TFCS (one CCTrCH case)

See Clause 6.11.6.4.1.39.2.1.4.

6.11.6.4.1.40.2.1.4.2 TFCS (two CCTrCH case)

6.11.6.4.1.40.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.40.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.40.2.2 Physical channel parameters

6.11.6.4.1.40.2.2.1 Physical channel parameters (one CCTrCH)

See clause 6.11.6.4.1.39.2.2.

6.11.6.4.1.40.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.6.4.1.40.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.40.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 204 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.41.1 Uplink

See clause 6.11.6.4.1.40.1.

6.11.6.4.1.41.2 Downlink

6.11.6.4.1.41.2.1 Transport channel parameters

6.11.6.4.1.41.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.41.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.11.6.4.1.27.2.1.1.

6.11.6.4.1.41.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.41.2.1.4 TFCS

6.11.6.4.1.41.2.1.4.1 TFCS (one CCTrCH case)

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.41.2.1.4.2 TFCS (two CCTrCH case)

6.11.6.4.1.41.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.41.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	5
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.41.2.2 Physical channel parameters

6.11.6.4.1.41.2.2.1 Physical channel parameters (one CCTrCH)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 5codes x 2time slots
	Max. Number of data bits/radio frame	2 744 bits
	TFCI code word	16 bits
	Puncturing limit	0.52

6.11.6.4.1.41.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.6.4.1.41.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.11.6.4.1.41.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0,48

6.11.6.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.42.1 Uplink

6.11.6.4.1.42.1.1 Transport channel parameters

6.11.6.4.1.42.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.42.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.11.6.4.1.26.1.1.1.

6.11.6.4.1.42.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.42.1.1.4 TFCS

See clause 6.11.6.4.1.40.1.1.4.1.

6.11.6.4.1.42.1.2 Physical channel parameters

See clause 6.11.6.4.1.40.1.2.1.

6.11.6.4.1.42.2 Downlink

6.11.6.4.1.42.2.1 Transport channel parameters

6.11.6.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.11.6.4.1.31.2.1.1.

6.11.6.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.42.2.1.4 TFCS

TFCS size	30 (alt. 42)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 256 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),

	(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.42.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 2 time slots + SF32 x 4 codes x 1 time slot
	Max. Number of data bits/radio frame	5 504 bits (alt. 5 488)
	TFCI code word	16 bits (alt. 32)
	Puncturing limit	0.60

6.11.6.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.43.1 Uplink

See clause 6.11.6.4.1.40.1.

6.11.6.4.1.43.2 Downlink

6.11.6.4.1.43.2.1 Transport channel parameters

6.11.6.4.1.43.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.43.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.11.6.4.1.32.2.1.1.

6.11.6.4.1.43.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.43.2.1.4 TFCS

6.11.6.4.1.43.2.1.4.1 TFCS (one CCTrCH case)

TFCS size	36 (alt. 54)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1) (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1) (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.43.2.1.4.2 TFCS (two CCTrCH case)

6.11.6.4.1.43.2.1.4.2.1 TFCS (conversational + SRB)

TFCS size	6
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.43.2.1.4.2.2 TFCS (Interactive or background)

TFCS size	6 (alt. 9)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0) (alt. (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF0, TF0, TF0, TF8, TF0))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.43.2.2 Physical channel parameters

6.11.6.4.1.43.2.2.1 Physical channel parameters (one CCTrCH)

DPCH Downlink	Midamble	512 chips

	Codes and time slots	SF32 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 592 bits
	TFCI code word	32 bits
	Puncturing limit	0.48

6.11.6.4.1.43.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.6.4.1.43.2.2.2.1 Physical channel parameters (conversational + SRB)

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.60

6.11.6.4.1.43.2.2.2.2 Physical channel parameters (Interactive or background)

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits
	TFCI code word	16 bits
	Puncturing limit	0,52

6.11.6.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.44.1 Uplink

6.11.6.4.1.44.1.1 Transport channel parameters

6.11.6.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.11.6.4.1.28.1.1.1.

6.11.6.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.44.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.44.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	{SF16 x 1 code + SF4 x 1 code} x 1 time slot
	Max. Number of data bits/radio frame	2 616 bits

	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.44)

6.11.6.4.1.44.2 Downlink

6.11.6.4.1.44.2.1 Transport channel parameters

6.11.6.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.11.6.4.1.35.2.1.1.

6.11.6.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.44.2.1.4 TFCS

TFCS size	66 (alt. 114)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1) (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0), (TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0), (TF0, TF0, TF0, TF11, TF0), (TF1, TF0, TF0, TF11, TF0), (TF2, TF1, TF1, TF11, TF0), (TF0, TF0, TF0, TF12, TF0), (TF1, TF0, TF0, TF12, TF0), (TF2, TF1, TF1, TF12, TF0), (TF0, TF0, TF0, TF13, TF0), (TF1, TF0, TF0, TF13, TF0), (TF2, TF1, TF1, TF13, TF0), (TF0, TF0, TF0, TF14, TF0), (TF1, TF0, TF0, TF14, TF0), (TF2, TF1, TF1, TF14, TF0), (TF0, TF0, TF0, TF15, TF0), (TF1, TF0, TF0, TF15, TF0), (TF2, TF1, TF1, TF15, TF0), (TF0, TF0, TF0, TF16, TF0), (TF1, TF0, TF0, TF16, TF0), (TF2, TF1, TF1, TF16, TF0), (TF0, TF0, TF0, TF17, TF0), (TF1, TF0, TF0, TF17, TF0), (TF2, TF1, TF1, TF17, TF0), (TF0, TF0, TF0, TF18, TF0), (TF1, TF0, TF0, TF18, TF0), (TF2, TF1, TF1, TF18, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1), (TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1), (TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1), (TF0, TF0, TF0, TF11, TF1), (TF1, TF0, TF0, TF11, TF1), (TF2, TF1, TF1, TF11, TF1), (TF0, TF0, TF0, TF12, TF1), (TF1, TF0, TF0, TF12, TF1), (TF2, TF1, TF1, TF12, TF1), (TF0, TF0, TF0, TF13, TF1), (TF1, TF0, TF0, TF13, TF1), (TF2, TF1, TF1, TF13, TF1), (TF0, TF0, TF0, TF14, TF1), (TF1, TF0, TF0, TF14, TF1), (TF2, TF1, TF1, TF14, TF1), (TF0, TF0, TF0, TF15, TF1), (TF1, TF0, TF0, TF15, TF1), (TF2, TF1, TF1, TF15, TF1), (TF0, TF0, TF0, TF16, TF1), (TF1, TF0, TF0, TF16, TF1), (TF2, TF1, TF1, TF16, TF1), (TF0, TF0, TF0, TF17, TF1), (TF1, TF0, TF0, TF17, TF1), (TF2, TF1, TF1, TF17, TF1), (TF0, TF0, TF0, TF18, TF1), (TF1, TF0, TF0, TF18, TF1), (TF2, TF1, TF1, TF18, TF1))
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.44.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 12 codes x 11 time slots
	Max. Number of data bits/radio frame	36 400 bits
	TFCI code word	32 bits
	Puncturing limit	0.52

6.11.6.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.45.1 Uplink

6.11.6.4.1.45.1.1 Transport channel parameters

6.11.6.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.11.6.4.1.17.1.1.1.

6.11.6.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.45.1.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.45.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF8 x 1 codex 1 time slot
	Max. Number of data bits/radio frame	1 392 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.56

6.11.6.4.1.45.2 Downlink

6.11.6.4.1.45.2.1 Transport channel parameters

6.11.6.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.11.6.4.1.17.2.1.1.

6.11.6.4.1.45.2.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.45.2.1.4 TFCS

TFCS size	30
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.45.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 448 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.46 Void

6.11.6.4.1.47 Void

6.11.6.4.1.48 Void

6.11.6.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.49.1 Uplink

6.11.6.4.1.49.1.1 Transport channel parameters

6.11.6.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.49.1.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.49.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72

6.11.6.4.1.49.2 Downlink

6.11.6.4.1.49.2.1 Transport channel parameters

6.11.6.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.49.2.1.4 TFCS

TFCS size	12
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.49.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0.76

6.11.6.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.49a.1 Uplink

6.11.6.4.1.49a.1.1 Transport channel parameters

6.11.6.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.1.1.1.

6.11.6.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.49a.1.1.4 TFCS

TFCS size	24

TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.49a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.72

6.11.6.4.1.49a.2 Downlink

6.11.6.4.1.49a.2.1 Transport channel parameters

6.11.6.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.11.6.4.1.4a.2.1.1.

6.11.6.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.49a.2.1.4 TFCS

TFCS size	24
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.49a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 7 codes x 1 time slot
	Max. Number of data bits/radio frame	1 916 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.50.1 Uplink

6.11.6.4.1.50.1.1 Transport channel parameters

6.11.6.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.50.1.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.50.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 1time slot + SF8 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 784 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.60

6.11.6.4.1.50.2 Downlink

6.11.6.4.1.50.2.1 Transport channel parameters

6.11.6.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.50.2.1.3 TFCS

TFCS size	8
TFCS	(64 kbps RAB, 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0) (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.50.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 6codes x 2 time slots
	Max. Number of data bits/radio frame	2 912 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.11.6.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.51.1 Uplink

6.11.6.4.1.51.1.1 Transport channel parameters

6.11.6.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.11.6.4.1.26.1.1.1.

6.11.6.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.51.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.51.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.44 (alt. 0.40)

6.11.6.4.1.51.2 Downlink

6.11.6.4.1.51.2.1 Transport channel parameters

6.11.6.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.11.6.4.1.25.2.1.1.

6.11.6.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.51.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.51.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	2 192 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.51a.1 Uplink

6.11.6.4.1.51a.1.1 Transport channel parameters

6.11.6.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.51a.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.11.6.4.1.23a.1.1.1.

6.11.6.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.51a.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) (alt. (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.51a.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.76

6.11.6.4.1.51a.2 Downlink

6.11.6.4.1.51a.2.1 Transport channel parameters

6.11.6.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / PS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.51a.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.51a.2.1.4 TFCS

TFCS size	8
TFCS	(64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.51a.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 640 bits
	TFCI code word	16 bits
	Puncturing limit	0.60

6.11.6.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.51b.1 Uplink

6.11.6.4.1.51b.1.1 Transport channel parameters

6.11.6.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.51b.1.1.2 Transport channel parameters for Interactive or Background / UL:16 kbps / PS RAB

See clause 6.11.6.4.1.23b.1.1.1.

6.11.6.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.51b.1.1.4 TFCS

TFCS size	12
TFCS	(64 kbps Conversational RAB, 16 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.51b.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bit
	TPC	2 bits
	Puncturing Limit	0.68

6.11.6.4.1.51b.2 Downlink

See clause 6.11.6.4.1.51.2.

6.11.6.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.52.1 Uplink

See clause 6.11.6.4.1.51.1.

6.11.6.4.1.52.2 Downlink

6.11.6.4.1.52.2.1 Transport channel parameters

6.11.6.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.11.6.4.1.27.2.1.1.

6.11.6.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.52.2.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.52.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	{SF32 x 8 codes x 1 time slot} + {SF32 x 5 codes x 1 time slot}
	Max. Number of data bits/radio frame	3 156 bits
	TFCI code word	16 bits
	Puncturing limit	0.44

6.11.6.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.53.1 Uplink

6.11.6.4.1.53.1.1 Transport channel parameters

6.11.6.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.11.6.4.1.28.1.1.1.

6.11.6.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.53.1.1.4 TFCS

TFCS size	20
TFCS	(Conv. 64 kbps RAB, I/B 128kbps RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1)
NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise.	

6.11.6.4.1.53.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 2 timeslots
	Max. Number of data bits/radio frame	3 760 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)

6.11.6.4.1.53.2 Downlink

See clause 6.11.6.4.1.52.2.

6.11.6.4.1.54 Void

6.11.6.4.1.55 Void

6.11.6.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.56.1 Uplink

6.11.6.4.1.56.1.1 Transport channel parameters

6.11.6.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320 (alt. 128)	320 (alt.128)
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340 (alt. 148)	
	TFS	TF0, bits	0x340 (alt. 0x148)	
		TF1, bits	1x340 (alt. 1x148)	
		TF2, bits	N/A (alt. 5x148)	
	TTI, ms		40 (alt. 80)	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 080 (alt. 2 472)	
	Max number of bits/radio frame before rate matching		270 (alt. 309)	
	RM attribute		135 to 175	

6.11.6.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.56.1.1.3 TFCS

TFCS size	4 (alt. 6)
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1) (alt. (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.56.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	226 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.48)
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits (alt. 16 bits).		

6.11.6.4.1.56.2 Downlink

6.11.6.4.1.56.2.1 Transport channel parameters

6.11.6.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		8 000	8 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
	TTI, ms		40	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		1 080	
	Max number of bits/radio frame before rate matching		270	
	RM attribute		135 to 175	

6.11.6.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.56.2.1.3 TFCS

TFCS size	4
TFCS	(8 kbps RAB + 8 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.56.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.56
NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits.		

6.11.6.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.57.1 Uplink

6.11.6.4.1.57.1.1 Transport channel parameters

6.11.6.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.11.6.4.1.38d.1.1.2.

6.11.6.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.57.1.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.57.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	2 064 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.88 (alt. 0.76)

6.11.6.4.1.57.2 Downlink

6.11.6.4.1.57.2.1 Transport channel parameters

6.11.6.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	RAB
RLC	Logical channel type		DTCH	DTCH
	RLC mode		AM	AM
	Payload sizes, bit		320	320
	Max data rate, bps		64 000	64 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	3x340	
		TF4, bits	4x340	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		4 284	
	Max number of bits/radio frame before rate matching		2 142	
	RM attribute		130 to 170	

6.11.6.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.57.2.1.3 TFCS

TFCS size	10
TFCS	(64 kbps RAB + 64 kbps RAB, DCCH)= (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.57.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot
	Max. Number of data bits/radio frame	1 364 bits
	TFCI code word	16 bits
	Puncturing limit	0.56

6.11.6.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.58.1 Uplink

6.11.6.4.1.58.1.1 Transport channel parameters

6.11.6.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		16 000
	AMD PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		336
	TFS	TF0, bits	0x336
		TF1, bits	1x336
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 068
	Max number of bits/radio frame before rate matching		534
	RM attribute		135 to 175

6.11.6.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.11.6.4.1.23a.1.1.1.

6.11.6.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.58.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(16 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1) (alt. (TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF2,TF0), (TF1,TF2,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF0,TF2,TF1), (TF1,TF2,TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.58.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot + SF32 x 1code x 1 time slot
	Max. Number of data bits/radio frame	696 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.72 (alt. 0.68)

6.11.6.4.1.58.2 Downlink

6.11.6.4.1.58.2.1 Transport channel parameters

6.11.6.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		64 000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8 076
	Max number of bits/radio frame before rate matching		2 019
	RM attribute		125 to 165

6.11.6.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.58.2.1.4 TFCS

TFCS size	16
TFCS	(64 kbps RAB, 8 kbps RAB, DCCH)= (TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0), (TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.58.2.2 Physical channel parameters

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF32 x 6 codes x 1 time slot
	Max. Number of data bits/radio frame	1 640 bits
	TFCI code word	16 bits
	Puncturing limit	0.64

6.11.6.4.1.59 Reserved for future use

6.11.6.4.1.60 Reserved for future use

6.11.6.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.1.61.1 Uplink

6.11.6.4.1.61.1.1 Transport channel parameters

6.11.6.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	UM
	Payload sizes, bit	320

	Max data rate, bps		8 000
	UMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328)
	TFS	TF0, bits	0x328 (alt 1x0) (note)
		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	Max number of bits/radio frame before rate matching		261
	RM attribute		135 to 175
NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.11.6.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.11.6.4.1.23a.1.1.1.

6.11.6.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.1.61.1.1.4 TFCS

TFCS size	8 (alt. 12)
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) (alt. (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1))
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.61.1.2 Physical channel parameters

DPCH Uplink	Midamble	1024 chips
	Codes and time slots	SF16 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	452 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.68 (alt. 0.64)

6.11.6.4.1.61.2 Downlink

6.11.6.4.1.61.2.1 Transport channel parameters

6.11.6.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		UM
	Payload sizes, bit		320
	Max data rate, bps		8 000
	AMD PDU header, bit		8
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		328 (alt 0, 328)
	TFS	TF0, bits	0x328 (alt 1x0) (note)

		TF1, bits	1x328
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1 044
	Max number of bits/radio frame before rate matching		261
	RM attribute		135 to 175
NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]).			

6.11.6.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.11.6.4.1.23.2.1.1.

6.11.6.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.1.61.2.1.4 TFCS

TFCS size	8
TFCS	(8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)= (TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)
NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.	

6.11.6.4.1.61.2.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.68

6.11.6.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH

6.11.6.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.6.4.2.1.1 Uplink

6.11.6.4.2.1.1.1 Transport channel parameters

6.11.6.4.2.1.1.1.1 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	TM
	Payload sizes, bit		320 (alt. 128)	168
	Max data rate, bps		64 000	16 800
	AMD/TrD PDU header, bit		16	0
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		USCH	USCH
	TB sizes, bit		337 (alt. 145)	169
	TFS	TF0, bits	0x337 (alt. 0x145)	0x169
		TF1, bits	1x337 (alt. 1x145)	1x169
		TF2, bits	2x337 (alt. 3x145)	N/A
		TF3, bits	3x337 (alt. 7x145)	N/A
		TF4, bits	4x337 (alt. 10x145)	N/A
	TTI, ms		20	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16

	Max number of bits/TTI after channel coding	4 248 (alt. 4 842)	386
	Max number of bits/radio frame before rate matching	2 124 (alt. 2 421)	386
	RM attribute	135 to 175	230 to 250

6.11.6.4.2.1.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		5	5	5	5
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		USCH			
	TB sizes, bit		149			
	TFS	TF0, bits	0x149			
		TF1, bits	1x149			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		519			
	Max number of bits/radio frame before rate matching		130			
	RM attribute		190 to 210			

6.11.6.4.2.1.1.1.3 TFCS for USCH

TFCS size	20
TFCS	(64 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1)

6.11.6.4.2.1.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

6.11.6.4.2.1.1.1.4.1 RACH transport channel configuration without DTCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH
	RLC mode	TM	UM	AM	AM	AM	TM
	Payload sizes, bit	168	136	128	128	128	168
	Max data rate, bps	16 800	13 600	12 800	12 800	12 800	16 800
	AMD/UMD/TrD PDU header, bit	0	8	16	16	16	0
MAC	MAC header, bit	2	26	26	26	26	2
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	170					
	TFS | TF0, bits	1x170					
	TTI, ms	10					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	388					
	Max number of bits/radio frame before rate matching	388					

6.11.6.4.2.1.1.1.4.2 RACH transport channel configuration with DTCH

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH
	RLC mode	AM	TM	UM	AM	AM	AM	TM
	Payload sizes, bit	128	168	136	128	128	128	168
	Max data rate, bps	12 800	16 800	13 600	12 800	12 800	12 800	16 800
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16	0
MAC	MAC header, bit	26	2	26	26	26	26	2
	MAC multiplexing	7 logical channel multiplexing						
Layer 1	TrCH type	RACH						
	TB sizes, bit	170						
	TFS TF0, bits	1x170						
	TTI, ms	10						
	Coding type	CC 1/2						
	CRC, bit	16						
	Max number of bits/TTI after channel coding	388						
	Max number of bits/radio frame before rate matching	388						

6.11.6.4.2.1.1.2 Physical channel parameters

6.11.6.4.2.1.1.2.1 Physical channel parameters for PUSCH

PUSCH	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.60 (alt. 0.56)

6.11.6.4.2.1.1.2.2 Physical channel parameters for PRACH

PRACH	Midamble	1024 chips
	Codes and time slots	SF16 (alt. SF32) x 1 code x 1 time slot
	Max. Number of data bits/radio frame	464 (alt. 232)
	Puncturing Limit	1 (alt. 0.56)

6.11.6.4.2.1.2 Downlink

6.11.6.4.2.1.2.1 Transport channel parameters

6.11.6.4.2.1.2.1.1 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB	RAB	SRB#5
RLC	Logical channel type	DTCH	SHCCH
	RLC mode	AM	UM
	Payload sizes, bit	320	160
	Max data rate, bps	256 000	16 000
	AMD/UMD PDU header, bit	16	8
MAC	MAC header, bit	1	1
	MAC multiplexing	N/A	N/A
Layer 1	TrCH type	DSCH	DSCH
	TB sizes, bit	337	169

	TFS	TF0, bits	0x337	0x169
		TF1, bits	1x337	1x169
		TF2, bits	2x337	N/A
		TF3, bits	4x337	N/A
		TF4, bits	8x337	N/A
		TF5, bits	N/A (alt. 12x337)	N/A
		TF6, bits	N/A (alt. 16x337)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		8 484 (alt. 16 968)	386
	Downlink: Max number of bits/radio frame before rate matching		8 484 (alt. 8 484)	386
	RM attribute		135 to 175	230 to 250

6.11.6.4.2.1.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

Higher layer	RAB/signalling RB		SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer		RRC	RRC	NAS_DT High prio	NAS_DT Low prio
RLC	Logical channel type		DCCH	DCCH	DCCH	DCCH
	RLC mode		UM	AM	AM	AM
	Payload sizes, bit		136	128	128	128
	Max data rate, bps		3 400	3 200	3 200	3 200
	AMD/UMD PDU header, bit		8	16	16	16
MAC	MAC header, bit		5	5	5	5
	MAC multiplexing		4 logical channel multiplexing			
Layer 1	TrCH type		DSCH			
	TB sizes, bit		149			
	TFS	TF0, bits	0x149			
		TF1, bits	1x149			
	TTI, ms		40			
	Coding type		CC 1/3			
	CRC, bit		16			
	Max number of bits/TTI before rate matching		519			
	Max number of bits/radio frame before rate matching		130			
	RM attribute		155 to 165			

6.11.6.4.2.1.2.1.3 TFCS for DSCH

TFCS size	20 (alt. 28)
TFCS	(256 kbps RAB, SHCCH, SRB for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1))

6.11.6.4.2.1.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

6.11.6.4.2.1.2.1.4.1 FACH transport channel configuration without DTCH

Higher layer	RAB/signalling RB		SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		160	136 or 120 (note)	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		8	8	16	16	16	8	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3	3
	MAC multiplexing		7 logical channel multiplexing						
Layer 1	TrCH type		FACH						
	TB sizes, bit		171						
	TFS	TF0, bits	0x171						
		TF1, bits	1x171						
		TF2, bits	2x171						
		TF3, bits	3x171(alt. N/A)						
		TF4, bits	4x171(alt. N/A)						
	TTI, ms		20						
	Coding type		TC						
	CRC, bit		16						
	Max number of bits/TTI after channel coding		2 256 (alt. 1 134)						
	Max number of bits/radio frame before rate matching		1 128 (alt. 567)						
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.									

6.11.6.4.2.1.2.1.4.2 FACH transport channel configuration with DTCH

Higher layer	RAB/signalling RB		RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5	SRB#6
	User of Radio Bearer		Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC	RRC
RLC	Logical channel type		DTCH	CCCH	DCCH	DCCH	DCCH	DCCH	SHCCH	BCCH
	RLC mode		AM	UM	UM	AM	AM	AM	UM	TM
	Payload sizes, bit		320	160	136 or 120 (note)	128	128	128	160	168
	Max data rate, bps		32 000 (alt. 16 000)	32 000 (alt. 16 000)	27 200 or 24 000 (alt. 13 600 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		16	8	8	16	16	16	8	0
MAC	MAC header, bit		27	3	27 or 43	27	27	27	3	3
	MAC multiplexing		8 logical channel multiplexing							
Layer 1	TrCH type		FACH							
	TB sizes, bit		171, 363							
	TFS	TF0, bits	0x171							
		TF1, bits	1x171							
		TF2, bits	2x171							
		TF3, bits	1x363							
		TF4, bits	3x171 (alt N/A)							
		TF5, bits	4x171 (alt. N/A)							
		TF6, bits	2x363 (alt. N/A)							
	TTI, ms		20							
	Coding type		TC							
	CRC, bit		16							
	Max number of bits/TTI after channel coding		2 286 (alt. 1 149)							
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)							
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.										

6.11.6.4.2.1.2.1.5 TFCS for FACH

6.11.6.4.2.1.2.1.5.1 TFCS for FACH transport channel configuration without DTCH

TFCS size	5 (alt. 3)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2))

6.11.6.4.2.1.2.1.5.2 TFCS for FACH transport channel configuration with DTCH

TFCS size	7 (alt. 4)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4), (TF5), (TF6) (alt. FACH = (TF0), (TF1), (TF2), (TF3))

6.11.6.4.2.1.2.2 Physical channel parameters

6.11.6.4.2.1.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing Limit	0.44

6.11.6.4.2.1.2.2.2 Physical channel parameters for SCCPCH

6.11.6.4.2.1.2.2.2.1 Physical channel parameters for SCCPCH without DTCH

SCCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)
	Puncturing Limit	1 (alt. 0.84)

6.11.6.4.2.1.2.2.2.2 Physical channel parameters for SCCPCH with DTCH

SCCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472 bits)
	TFCI code word	16 bits
	Puncturing Limit	1 (alt. 0.80)

6.11.6.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.6.4.2.2.1 Uplink

See clause 6.11.6.4.2.1.1.

6.11.6.4.2.2.2 Downlink

6.11.6.4.2.2.2.1 Transport channel parameters

6.11.6.4.2.2.2.1.1 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB	RAB	SRB#5
RLC	Logical channel type	DTCH	SHCCH
	RLC mode	AM	UM
	Payload sizes, bit	320	160
	Max data rate, bps	384 000	16 000
	AMD/UMD PDU header, bit	16	8

MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DSCH	DSCH
	TB sizes, bit		337	169
	TFS	TF0, bits	0x337	0x169
		TF1, bits	1x337	1x169
		TF2, bits	2x337	N/A
		TF3, bits	4x337	N/A
		TF4, bits	8x337	N/A
		TF5, bits	12x337	N/A
		TF6, bits	N/A (alt. 16x337)	N/A
		TF7, bits	N/A (alt. 20x337)	N/A
		TF8, bits	N/A (alt. 24x337)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		12 720 (alt. 25 440)	386
	Downlink: Max number of bits/radio frame before rate matching		12 720 (alt. 12 720)	386
	RM attribute		135 to 175	230 to 250

6.11.6.4.2.2.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.11.6.4.2.1.2.1.2.

6.11.6.4.2.2.2.1.3 TFCS for DSCH

TFCS size	24 (alt. 36)
TFCS	(384 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1))

6.11.6.4.2.2.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH (with & without DTCH)

See clause 6.11.6.4.2.1.2.1.4.

6.11.6.4.2.2.2.1.5 TFCS for FACH

See clause 6.11.6.4.2.1.2.1.5.

6.11.6.4.2.2.2.2 Physical channel parameters

6.11.6.4.2.2.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits (alt. 6 592 bits)
	TFCI code word	16 bits (alt. 32 bits)
	Puncturing Limit	0.48

6.11.6.4.2.2.2.2.2 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.

6.11.6.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.6.4.2.3.1 Uplink

See clause 6.11.6.4.2.1.1.

6.11.6.4.2.3.2 Downlink

6.11.6.4.2.3.2.1 Transport channel parameters

6.11.6.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	UM
	Payload sizes, bit		640	160
	Max data rate, bps		2 048 000	16 000
	AMD/UMD PDU header, bit		16	8
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		DSCH	DSCH
	TB sizes, bit		657	169
	TFS	TF0, bits	0x657	0x169
		TF1, bits	1x657	1x169
		TF2, bits	2x657	N/A
		TF3, bits	4x657	N/A
		TF4, bits	8x657	N/A
		TF5, bits	12x657	N/A
		TF6, bits	16x657	N/A
		TF7, bits	20x657	N/A
		TF8, bits	24x657	N/A
		TF9, bits	28x657	N/A
		TF10, bits	30x657 (alt. 32x657)	N/A
		TF11, bits	N/A (alt. 36x657)	N/A
		TF12, bits	N/A (alt. 40x657)	N/A
		TF13, bits	N/A (alt. 44x657)	N/A
		TF14, bits	N/A (alt. 48x657)	N/A
		TF15, bits	N/A (alt. 52x657)	N/A
		TF16, bits	N/A (alt. 56x657)	N/A
		TF17, bits	N/A (alt. 60x657)	N/A
		TF18, bits	N/A (alt. 64x657)	N/A
	TTI, ms		10 (alt. 20)	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		60 624 (alt. 129 330)	386
	Downlink: Max number of bits/radio frame before rate matching		60 624 (alt. 64 665)	386
	RM attribute		135 to 175	230 to 250

6.11.6.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.11.6.4.2.1.2.1.2.

6.11.6.4.2.3.2.1.3 TFCS for DSCH

TFCS size	41 (alt.76)
TFCS	(2 048 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0), (TF10, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1),

	(TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0), (TF10, TF0, TF0),(TF11, TF0, TF0), (TF12, TF0, TF0), (TF13, TF0, TF0), (TF14, TF0, TF0), (TF15, TF0, TF0), (TF16, TF0, TF0), (TF17, TF0, TF0), (TF18, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0), (TF10, TF1, TF0),(TF11, TF1, TF0), (TF12, TF1, TF0), (TF13, TF1, TF0), (TF14, TF1, TF0), (TF15, TF1, TF0), (TF16, TF1, TF0), (TF17, TF1, TF0), (TF18, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1), (TF10, TF0, TF1), (TF11, TF0, TF1), (TF12, TF0, TF1), (TF13, TF0, TF1), (TF14, TF0, TF1), (TF15, TF0, TF1), (TF16, TF0, TF1), (TF17, TF0, TF1), (TF18, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1), (TF10, TF1, TF1),(TF11, TF1, TF1), (TF12, TF1, TF1), (TF13, TF1, TF1), (TF14, TF1, TF1), (TF15, TF1, TF1), (TF16, TF1, TF1), (TF17, TF1, TF1), (TF18, TF1, TF1))

6.11.6.4.2.3.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.6.4.2.1.2.1.4.1.

6.11.6.4.2.3.2.1.5 TFCS for FACH

See clause 6.11.6.4.2.1.2.1.45.1.

6.11.6.4.2.3.2.2 Physical channel parameters

6.11.6.4.2.3.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips
	Codes and time slots	SF32 x 12 codes x 11 time slots
	Max. Number of data bits/radio frame	36 400 bits
	TFCI code word	32 bits
	Puncturing Limit	0.56 (alt. 0.52)

6.11.6.4.2.3.2.2.2 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.1.

6.11.6.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.6.4.2.4.1 Uplink

6.11.6.4.2.4.1.1 Transport channel parameters

6.11.6.4.2.4.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB		RAB	SRB#5
RLC	Logical channel type		DTCH	SHCCH
	RLC mode		AM	TM
	Payload sizes, bit		320 (alt. 128)	168
	Max data rate, bps		384 000	16 800
	AMD/TrD PDU header, bit		16	0
MAC	MAC header, bit		1	1
	MAC multiplexing		N/A	N/A
Layer 1	TrCH type		USCH	USCH
	TB sizes, bit		337 (alt. 145)	169
	TFS	TF0, bits	0x337 (alt. 0x145)	0x169
		TF1, bits	1x337 (alt. 1x145)	1x169

		TF2, bits	2x337 (alt. 5x145)	N/A
		TF3, bits	4x337 (alt. 10x145)	N/A
		TF4, bits	8x337 (alt. 20x145)	N/A
		TF5, bits	12x337 (alt. 30x145)	N/A
		TF6, bits	16x337 (alt. 40x145)	N/A
		TF7, bits	20x337 (alt. 50x145)	N/A
		TF8, bits	24x337 (alt. 60x145)	N/A
	TTI, ms		20	10
	Coding type		TC	CC 1/2
	CRC, bit		16	16
	Max number of bits/TTI after channel coding		25 440 (alt. 29 004)	386
	Max number of bits/radio frame before rate matching		12 720 (alt. 14 502)	386
	RM attribute		135 to 175	230 to 250

6.11.6.4.2.4.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

See clause 6.11.6.4.2.1.1.1.2.

6.11.6.4.2.4.1.1.3 TFCS for USCH

TFCS size	36
TFCS	(384 kbps RAB, SHCCH, SRBs for DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1) (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1)

6.11.6.4.2.4.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

See clause 6.11.6.4.2.1.1.1.4.

6.11.6.4.2.4.1.2 Physical channel parameters

6.11.6.4.2.4.1.2.1 Physical channel parameters for PUSCH

PUSCH	Midamble	1024 chips
	Codes and time slots	SF2 x 1 code x 2 time slots
	Max. Number of data bits/radio frame	7 264 bits
	TFCI code word	32 bits
	TPC	2 bits
	Puncturing Limit	0.52 (alt. 0.44)

6.11.6.4.2.4.1.2.2 Physical channel parameters for PRACH

See clause 6.11.6.4.2.1.1.2.2.

6.11.6.4.2.4.2 Downlink

6.11.6.4.2.4.2.1 Transport channel parameters

See clause 6.11.6.4.2.3.2.1.

6.11.6.4.2.4.2.2 Physical channel parameters

6.11.6.4.2.4.2.2.1 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips

	Codes and time slots	SF1 x 1 codes x 4 time slots
	Max. Number of data bits/radio frame	35 296 bits
	TFCI code word	32 bits
	Puncturing Limit	0.56 (alt. 0.52)

6.11.6.4.2.4.2.2.2 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.1.

6.11.6.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

6.11.6.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH

6.11.6.4.3.1.1 Uplink

6.11.6.4.3.1.1.1 Transport channel parameters

6.11.6.4.3.1.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.3.1.1.1.2 Transport channel parameters for UL SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.3.1.1.1.3 TFCS for DCH

See clause 6.11.6.4.1.4.1.1.3.

6.11.6.4.3.1.1.1.4 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.11.6.4.2.1.1.1.1.

6.11.6.4.3.1.1.1.5 TFCS for USCH

TFCS size	10
TFCS	(64 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)

6.11.6.4.3.1.1.1.6 Transport channel parameters for SRB for CCCH and UL SRB for SHCCH mapped on RACH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	SHCCH
	RLC mode		TM	TM
	Payload sizes, bit		168	168
	Max data rate, bps		16 800	16 800
	TrD PDU header, bit		0	0
MAC	MAC header, bit		2	2
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		RACH	
	TB sizes, bit		170	
	TFS	TF0, bits	1x170	
	TTI, ms		10	
	Coding type		CC 1/2	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		388	
	Max number of bits/radio frame before rate matching		388	

6.11.6.4.3.1.1.2 Physical channel parameters

6.11.6.4.3.1.1.2.1 Physical channel parameters for DPCH

See clause 6.11.6.4.1.4.1.2.

6.11.6.4.3.1.1.2.2 Physical channel parameters for PUSCH

PUSCH	Midamble	1024 chips
	Codes and time slots	SF4 x 1 code x 1 time slot
	Max. Number of data bits/radio frame	1 808 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.76 (alt. 0.68)

6.11.6.4.3.1.1.2.3 Physical channel parameters for PRACH

See clause 6.11.6.4.2.1.1.2.2.

6.11.6.4.3.1.2 Downlink

6.11.6.4.3.1.2.1 Transport channel parameters

6.11.6.4.3.1.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.3.1.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.3.1.2.1.3 TFCS for DCH

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.3.1.2.1.4 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.11.6.4.2.1.2.1.1.

6.11.6.4.3.1.2.1.5 TFCS for DSCH

TFCS size	10 (alt. 14)
TFCS	(256 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1))

6.11.6.4.3.1.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

Higher layer	RAB/Signalling RB		SRB#0	SRB#5	SRB#6
	User of Radio Bearer		RRC	RRC	RRC
RLC	Logical channel type		CCCH	SHCCH	BCCH
	RLC mode		UM	UM	TM
	Payload sizes, bit		160	160	168
	Max data rate, bps		32 000 (alt. 16 000)	32 000 (alt. 16 000)	33 600 (alt. 16 800)
	UMD/TrD PDU header, bit		8	8	0
MAC	MAC header, bit		3		
	MAC multiplexing		3 logical channel multiplexing		
Layer 1	TrCH type		FACH		
	TB sizes, bit		171		
	TFS	TF0, bits	0x171		
		TF1, bits	1x171		

		TF2, bits	2x171
		TF3, bits	3x171 (alt. N/A)
		TF4, bits	4x171 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		2 256 (alt. 1 134)
	Max number of bits/radio frame before rate matching		1 128 (alt 567)

6.11.6.4.3.1.2.1.7 TFCS for FACH

TFCS size	5 (alt. 3)
TFCS	FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2))

6.11.6.4.3.1.2.2 Physical channel parameters

6.11.6.4.3.1.2.2.1 Physical channel parameters for DPCH

Seeclause 6.11.6.4.1.4.2.2.

6.11.6.4.3.1.2.2.2 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 2 time slots
	Max. Number of data bits/radio frame	4 400 bits
	TFCI code word	16 bits
	Puncturing Limit	0.48

6.11.6.4.3.1.2.2.3 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.1.

6.11.6.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.6.4.3.2.1 Uplink

See clause 6.11.6.4.3.1.1.

6.11.6.4.3.2.2 Downlink

6.11.6.4.3.2.2.1 Transport channel parameters

6.11.6.4.3.2.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.3.2.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.3.2.2.1.3 TFCS for DCH

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.3.2.2.1.4 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.11.6.4.2.2.2.1.1.

6.11.6.4.3.2.2.1.5 TFCS for DSCH

TFCS size	12 (alt. 18)
TFCS	(384 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1))

6.11.6.4.3.2.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.6.4.3.1.2.1.6.

6.11.6.4.3.2.2.1.7 TFCS for FACH

See clause 6.11.6.4.3.1.2.1.7.

6.11.6.4.3.2.2.2 Physical channel parameters

6.11.6.4.3.2.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.3.2.2.2.2 Physical channel parameters for PDSCH

PDSCH	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 3 time slots
	Max. Number of data bits/radio frame	6 608 bits
	TFCI code word	16 bits
	Puncturing Limit	0.48

6.11.6.4.3.2.2.2.3 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.1.

6.11.6.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.6.4.3.3.1 Uplink

See clause 6.11.6.4.3.1.1.

6.11.6.4.3.3.2 Downlink

6.11.6.4.3.3.2.1 Transport channel parameters

6.11.6.4.3.3.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.3.3.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.3.3.2.1.3 TFCS for DCH

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.3.3.2.1.4 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.11.6.4.2.3.2.1.1.

6.11.6.4.3.3.2.1.5 TFCS for DSCH

TFCS size	22 (alt. 38)
TFCS	(2 048 kbps RAB, SHCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1) (alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1), (TF16, TF1), (TF17, TF1), (TF18, TF1))

6.11.6.4.3.3.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.6.4.3.1.2.1.6.

6.11.6.4.3.3.2.1.7 TFCS for FACH

See clause 6.11.6.4.3.1.2.1.7.

6.11.6.4.3.3.2.2 Physical channel parameters

6.11.6.4.3.3.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.3.3.2.2.2 Physical channel parameters for PDSCH

DPCH Downlink	Midamble	512 chips
	Codes and time slots	SF1 x 1 code x 4 time slot
	Max. Number of data bits/radio frame	35 312 bits (alt. 35 296)
	TFCI code word	16 bits (alt. 32 bits)
	Puncturing limit	0.56 (alt. 0.52)

6.11.6.4.3.3.2.2.3 Physical channel parameters for SCCPCH

See clause 6.11.6.4.2.1.2.2.2.1.

6.11.6.4.4 Combinations on SCCPCH

6.11.6.4.4.1 Stand-alone signalling RB for PCCH

6.11.6.4.4.1.1 Transport channel parameters

6.11.6.4.4.1.1.1 Transport channel parameter of SRB for PCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		RRC
RLC	Logical channel type		PCCH
	RLC mode		TM
	Payload sizes, bit		240 (alt. 80)
	Max data rate, bps		12 000 (alt. 8 000)
	TrD PDU header, bit		0
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		PCH
	TB sizes, bit		240 (alt. 80)
	TFS	TF0, bts	0x240 (alt. 0x80)
		TF1, bits	1x240 (alt. 1x80)
		TF2, bits	N/A (alt.2x80)
	TTI, ms		20

	Coding type	CC 1/2
	CRC, bit	16
	Max number of bits/TTI before rate matching	528 (alt. 400)
	Max number of bits/radio frame before rate matching	264 (alt. 200)
	RM attribute	210 to 250

6.11.6.4.4.1.1.2 TFCS

TFCS size	2 (alt. 3)
TFCS	SRBs for PCCH = (TF0), (TF1) (alt. (TF0), (TF1), (TF2))

6.11.6.4.4.1.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot (alt. SF32 x 1 code x 1 time slot)
	Max. Number of data bits/radio frame	480 bits (alt. 236 bits)
	TFCI code word	8 bits
	Puncturing limit	1

6.11.6.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.6.4.4.2.1 Transport channel parameters

6.11.6.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		Interactive/ Background RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		320
	Max data rate, bps		32 000 (alt. 16 000)
	AMD PDU header, bit		16
MAC	MAC header, bit		27
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		363
	TFS	TF0, bits	0 x363
		TF1, bits	1x363
		TF2, bits	2x363 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		2 286 (alt. 1 149)
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)
	RM attribute		110 to 150

6.11.6.4.4.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4	SRB#5
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio	RRC
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH	BCCH
	RLC mode	UM	UM	AM	AM	AM	TM
	Payload sizes, bit	160	136 or 120 (note)	128	128	128	168

	Max data rate, bps		32 000 (alt. 16 000)	27 200 or 24 000 (alt. 24 000 or 12 000)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	25 600 (alt. 12 800)	33 600 (alt. 16 800)
	AMD/UMD/TrD PDU header, bit		8	8	16	16	16	0
MAC	MAC header, bit		3	27 or 43	27	27	27	3
	MAC multiplexing		6 logical channel multiplexing					
Layer 1	TrCH type		FACH					
	TB sizes, bit		171					
	TFS	TF0, bits	0x171					
		TF1, bits	1x171					
		TF2, bits	2x171					
		TF3, bits	3x171 (alt. N/A)					
		TF4, bits	4x171 (alt. N/A)					
	TTI, ms		20					
	Coding type		TC					
	CRC, bit		16					
	Max number of bits/TTI before rate matching		2 256 (alt. 1 134)					
	Max number of bits/radio frame before rate matching		1 128 (alt. 567)					
	RM attribute		200 to 240					
NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI.								

6.11.6.4.4.2.1.3 TFCS

TFCS size	9 (alt. 4)
TFCS	(32kbps RAB, SRBs for CCCH/DCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4),(TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0))
NOTE: First TFCS applies when the alternative for the 3 2kbps RAB and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RAB and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid.	

6.11.6.4.4.2.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472)
	TFCI code word	16 bits
	Puncturing limit	0.60 (alt. 0.48)

6.11.6.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.6.4.4.2a.1 Transport channel parameters

6.11.6.4.4.2a.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB

Higher layer	RAB/Signalling RB	RAB	RAB
	User of Radio Bearer	Interactive/Background RAB	Interactive/Background RAB
RLC	Logical channel type	DTCH	DTCH
	RLC mode	AM	AM
	Payload sizes, bit	320	320
	Max data rate, bps	32 000 (alt. 16 000)	32 000 (alt. 16 000)
	AMD PDU header, bit	16	16
MAC	MAC header, bit	27	27

	MAC multiplexing		2 logical channel multiplexing
Layer 1	TrCH type		FACH
	TB sizes, bit		363
	TFS	TF0, bits	0x363
		TF1, bits	1x363
		TF2, bits	2x363 (alt. N/A)
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI before rate matching		2 286 (alt. 1 149)
	Max number of bits/radio frame before rate matching		1 143 (alt. 575)
	RM attribute		110 to 150

6.11.6.4.4.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.6.4.4.2.1.2.

6.11.6.4.4.2a.1.3 TFCS

TFCS size	9 (alt. 4)
TFCS	(32kbps RAB + 32kbps RAB, SRBs for CCCH/DCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0))
NOTE: First TFCS applies when the alternative for the 32 kbps RABs and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RABs and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid.	

6.11.6.4.4.2a.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 472)
	TFCI code word	16 bits
	Puncturing limit	0.60 (alt. 0.48)

6.11.6.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.6.4.4.2b.1 Transport channel parameters

6.11.6.4.4.2b.1.1 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.6.4.4.2.1.2.

6.11.6.4.4.2b.1.2 TFCS

TFCS size	5 (alt. 3)
TFCS	(SRBs for CCCH/DCCH/BCCH) = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. (TF0), (TF1), (TF2))

6.11.6.4.4.2b.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)

	Puncturing limit	1 (alt. 0.84)

6.11.6.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.6.4.4.3.1 Transport channel parameters

6.11.6.4.4.3.1.1 Transport channel parameters for Interactive/Background 32 kbps RAB

See clause 6.11.6.4.4.2.1.1.

6.11.6.4.4.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.11.6.4.4.1.1.1.

6.11.6.4.4.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.6.4.4.2.1.2.

6.11.6.4.4.3.1.4 TFCS

TFCS size	30 (alt. 8)
TFCS	(32 kbps RAB, SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) = (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF0, TF3), (TF0, TF0, TF4), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF1, TF2), (TF0, TF1, TF3), (TF0, TF1, TF4), (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1, TF0, TF2), (TF1, TF0, TF3), (TF1, TF0, TF4),(TF1, TF1, TF0), (TF1, TF1, TF1), (TF1, TF1, TF2), (TF1, TF1, TF3), (TF1, TF1, TF4), (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2, TF0, TF2), (TF2, TF0, TF3), (TF2, TF0, TF4),(TF2, TF1, TF0), (TF2, TF1, TF1), (TF2, TF1, TF2), (TF2, TF1, TF3), (TF2, TF1, TF4) (alt. (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF2, TF0), (TF0, TF2, TF1), (TF1, TF0, TF0))
NOTE: Alt. TFCS applies when alts for 32 kbps RAB, SRB for PCCH, and SRBs for CCCH/ DCCH/ BCCH are all configured.	

6.11.6.4.4.3.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 936 bits (alt. 472 bits)
	TFCI code word	16 bits
	Puncturing limit	0.52 (alt. 0.56)
NOTE: Alt. applies when alts for 32 kbps RAB and SRBs for CCCH/ DCCH/ BCCH are both configured.		

6.11.6.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.6.4.4.3a.1 Transport channel parameters

6.11.6.4.4.3a.1.1 Transport channel parameters of SRB for PCCH

See clause 6.11.6.4.4.1.1.1.

6.11.6.4.4.3a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.6.4.4.2.1.2.

6.11.6.4.4.3a.1.3 TFCS

TFCS size	10 (alt.7)
TFCS	(SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF1, TF3), (TF1, TF4) (alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0), (TF1, TF1), (TF2, TF0), (TF2, TF1))
NOTE: Alt. TFCS applies when alts for SRB for PCCH and SRBs for CCCH/ DCCH/ BCCH are both configured.	

6.11.6.4.4.3a.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 5 codes x 1 time slot (alt. SF32 x 2 codes x 1 time slot)
	Max. Number of data bits/radio frame	1 204 bits (alt. 480 bits)
	TFCI code word	16 bits (alt. 8 bits)
	Puncturing limit	0.84 (alt. 0.84)
NOTE: Alt. applies when alt for SRBs for CCCH/ DCCH/ BCCH is configured.		

6.11.6.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.11.6.4.4.4.1 Transport channel parameters

6.11.6.4.4.4.1.1 Transport channel parameters of RB for CTCH

Higher layer	RAB/signalling RB		N/A
	User of Radio Bearer		BMC
RLC	Logical channel type		CTCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		15 200
	UMD PDU header, bit		8
MAC	MAC header, bit		3
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		163
	TFS	TF0, bits	0x163
		TF1, bits	1x163
		TF2, bits	2x163
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI before rate matching		1 098
	Max number of bits/radio frame before rate matching		549
	RM attribute		200 to 240

6.11.6.4.4.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

Higher layer	RAB/signalling RB		SRB#0	SRB#5
	User of Radio Bearer		RRC	RRC
RLC	Logical channel type		CCCH	BCCH
	RLC mode		UM	TM
	Payload sizes, bit		160	168
	Max data rate, bps		16 000	16 800
	AMD/UMD/TrD PDU header, bit		8	0
MAC	MAC header, bit		3	3
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		FACH	
	TB sizes, bit		171	
	TFS	TF0, bits	0x171	
		TF1, bits	1x171	
		TF2, bits	2x171	
	TTI, ms		20	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI before rate matching		1 134	
	Max number of bits/radio frame before rate matching		567	
	RM attribute		200 to 240	

6.11.6.4.4.4.1.3 TFCS

TFCS size	4
TFCS	(RB for CTCH, SRBs for CCCH/BCCH) = (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0) , (TF1, TF1), (TF2, TF0)

6.11.6.4.4.4.2 Physical channel parameters

S-CCPCH	Midamble	1024 chips
	Codes and time slots	SF32 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	472 bits
	TFCI code word	16 bits
	Puncturing limit	0.80

6.11.6.4.4.5 64.8kbps RB for MTCH with 80 ms TTI

6.11.6.4.4.5.1 Transport channel parameters

6.11.6.4.4.5.1.1 Transport channel parameters for 64 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		64800
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		16344
	Max number of bits/radio frame before rate matching		2043
	RM attribute		160

6.11.6.4.4.5.1.2 TFCS

TFCS size	9
TFCS	64 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8

6.11.6.4.4.5.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Max. Number of data bits/radio frame	1936 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.11.6.4.4.6 129.6kbps RB for MTCH with 80 ms TTI

6.11.6.4.4.6.1 Transport channel parameters

6.11.6.4.4.6.1.1 Transport channel parameters for 128 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		32679
	Max number of bits/radio frame before rate matching		4085
	RM attribute		160

6.11.6.4.4.6.1.1 TFCS

TFCS size	17
TFCS	128 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.11.6.4.4.6.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF32 x 16 codes x 1 time slot
	Max. Number of data bits/radio frame	3888 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.11.6.4.4.7 259.2 kbps RB for MTCH with 40 ms TTI

6.11.6.4.4.7.1 Transport channel parameters

6.11.6.4.4.7.1.1 Transport channel parameters for 256 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		648
	Max data rate, bps		129600
	UMD PDU header, bit		8
MAC	MAC header, bit		8
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		664
	TFS	TF0, bits	0x664
		TF1, bits	1x664
		TF2, bits	2x664
		TF3, bits	3x664
		TF4, bits	4x664
		TF5, bits	5x664
		TF6, bits	6x664
		TF7, bits	7x664
		TF8, bits	8x664
		TF9, bits	9x664
		TF10, bits	10x664
		TF11, bits	11x664
		TF12, bits	12x664
		TF13, bits	13x664
		TF14, bits	14x664
		TF15, bits	15x664
		TF16, bits	16x664
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		32679
	Max number of bits/radio frame before rate matching		8170
	RM attribute		160

6.11.6.4.4.7.1.2 TFCS

TFCS size	17
TFCS	256 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16

6.11.6.4.4.7.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF1 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	7792 bits
	TFCI code word	16 bits
	Puncturing limit	0.92

6.11.6.4.4.8 7.6 kbps signalling RB for MCCH

6.11.6.4.4.8.1 Transport channel parameters

6.11.6.4.4.8.1.1 Transport channel parameters for 7.6 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		152
	Max data rate, bps		7600
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		160
	TFS	TF0, bits	0x160
		TF1, bits	1x160
	TTI, ms		20
	Coding type		CC 1/3
	CRC, bit		16
	Max number of bits/TTI after channel coding		552
	Max number of bits/radio frame before rate matching		276
	RM attribute		160

6.11.6.4.4.8.1.2 TFCS

TFCS size	2
TFCS	MBMS SRB =TF0, TF1

6.11.6.4.4.8.2 Physical channel parameters

S-CCPCH	Midamble	512 chips
	Codes and time slots	SF32 x 1 codes x 1 time slot
	Max. Number of data bits/radio frame	228 bits
	TFCI code word	16 bits
	Puncturing limit	0.80

6.11.6.4.4.9 124.4kbps RB for MBSFN MTCH with 80 ms TTI

6.11.6.4.4.9.1 Transport channel parameters

6.11.6.4.4.9.1.1 Transport channel parameters for 124 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		124400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	2x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		30078
	Max number of bits/radio frame before rate matching		3760
	RM attribute		128

6.11.6.4.4.9.1.2 TFCS

TFCS size	3
TFCS	124 kbps RAB =TF0, TF1, TF2

6.11.6.4.4.9.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF32 x 8 codes x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	2096 bits
	TFCI code word	(16,5)
	Puncturing limit	0.54

6.11.6.4.4.10 320.4kbps RB for MBSFN MTCH with 80 ms TTI

6.11.6.4.4.10.1 Transport channel parameters

6.11.6.4.4.10.1.1 Transport channel parameters for 320 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4272
	Max data rate, bps		320400
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4289
	TFS	TF0, bits	0x4289
		TF1, bits	1x4289
		TF2, bits	6x4289
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		77562
	Max number of bits/radio frame before rate matching		9696
	RM attribute		128

6.11.6.4.4.10.1.1 TFCS

TFCS size	3
TFCS	320 kbps RAB =TF0, TF1, TF2

6.11.6.4.4.10.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF32 x 16 codes x 1 time slot
	Modulation	16QAM
	Max. Number of data bits/radio frame	8432 bits
	TFCI code word	(16,5)
	Puncturing limit	0.86

6.11.6.4.4.11 497.6 kbps RB for MBSFN MTCH with 80 ms TTI

6.11.6.4.4.11.1 Transport channel parameters

6.11.6.4.4.11.1.1 Transport channel parameters for 496 kbps PS RAB

Higher layer	RAB/signalling RB		RAB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MTCH
	RLC mode		UM
	Payload sizes, bit		4976
	Max data rate, bps		497600
	UMD PDU header, bit		8
MAC	MAC header, bit		9
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		4993
	TFS	TF0, bits	0x4993
		TF1, bits	1x4993
		TF2, bits	8x4993
	TTI, ms		80
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		120312
	Max number of bits/radio frame before rate matching		15039
	RM attribute		128

6.11.6.4.4.11.1.2 TFCS

TFCS size	3
TFCS	496 kbps RAB =TF0, TF1, TF2

6.11.6.4.4.11.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF1 x 1 codes x 2 time slots
	Modulation	QPSK
	Max. Number of data bits/radio frame	16880 bits
	TFCI code word	(16,5) in first slot only
	Puncturing limit	1

6.11.6.4.4.12 7.2 kbps signalling RB for MBSFN MCCH

6.11.6.4.4.12.1 Transport channel parameters

6.11.6.4.4.12.1.1 Transport channel parameters for 7.2 kbps signalling RB for MCCH

Higher layer	RAB/signalling RB		SRB
	User of Radio Bearer		MBMS
RLC	Logical channel type		MCCH
	RLC mode		UM
	Payload sizes, bit		72
	Max data rate, bps		7200
	UMD PDU header, bit		8
MAC	MAC header, bit		-
	MAC multiplexing		N/A
Layer 1	TrCH type		FACH
	TB sizes, bit		80
	TFS	TF0, bits	0x80
		TF1, bits	1x80
		TF2, bits	2x80
		TF3, bits	4x80
	TTI, ms		40
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		1164
	Max number of bits/radio frame before rate matching		291
	RM attribute		128

6.11.6.4.4.12.1.2 TFCS

TFCS size	4
TFCS	MBMS SRB =TF0, TF1, TF2, TF3

6.11.6.4.4.12.2 Physical channel parameters

S-CCPCH	Midamble	640 chips (burst type 4)
	Codes and time slots	SF32 x 1 code x 1 time slot
	Modulation	QPSK
	Max. Number of data bits/radio frame	248 bits
	TFCI code word	(16,5)
	Puncturing limit	0.84

6.11.6.4.5 Combinations on PRACH

6.11.6.4.5.1 SRB for CCCH + SRB for DCCH

6.11.6.4.5.1.1 Transport channel parameters

6.11.6.4.5.1.1.1 Transport channel parameter for SRB for CCCH, SRB for DCCH

Higher layer	RAB/signalling RB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	RRC	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	TM	UM	AM	AM	AM
	Payload sizes, bit	168	136	128	128	128
	Max data rate, bps	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	0	8	16	16	16
MAC	MAC header, bit	2	26	26	26	26
	MAC multiplexing	5 logical channel multiplexing				

Layer 1	TrCH type		RACH				
	TB sizes, bit		170				
	TFS	TF0, bits	1x170				
	TTI, ms		10				
	Coding type		CC 1/2				
	CRC, bit		16				
	Max number of bits/TTI after channel coding		388				
	Max number of bits/Radio frame before rate matching		388				

6.11.6.4.5.1.1.2 TFCS

TFCS size	1
TFCS	SRBs for CCCH/ DCCH = (TF0)

6.11.6.4.5.1.2 Physical channel parameters

PRACH	Midamble	1024 chips
	Codes and time slots	SF16 (alt. SF32) x 1 code x 1 time slot
	Max. Number of data bits/radio frame	488 bits (alt. 244 bits)
	Puncturing Limit	1.0 (alt. 0.60)

6.11.6.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.11.6.4.5.2.1 Transport channel parameters

Higher layer	RAB/signalling RB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High priority	NAS_DT Low priority
RLC	Logical channel type	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	168	136	128	128	128
	Max data rate, bps	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	0	8	16	16	16
MAC	MAC header, bit	26	2	26	26	26	26
	MAC multiplexing	6 logical channel multiplexing					
Layer 1	TrCH type	RACH					
	TB sizes, bit	170					
	TFS / TF0, bits	1x170					
	TTI, ms	10					
	Coding type	CC 1/2					
	CRC, bit	16					
	Max number of bits/TTI after channel coding	388					
	Max number of bits/ Radio frame before rate matching	388					

6.11.6.4.5.2.2 Physical channel parameters

See clause 6.11.6.4.5.1.2.

6.11.6.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.11.6.4.5.3.1 Transport channel parameters

Higher layer	RAB/signalling RB	RAB	RAB	SRB#0	SRB#1	SRB#2	SRB#3	SRB#4
	User of Radio Bearer	Interactive/ Background RAB	Interactive/ Background RAB	RRC	RRC	RRC	NAS_DT High prio	NAS_DT Low prio

RLC	Logical channel type	DTCH	DTCH	CCCH	DCCH	DCCH	DCCH	DCCH
	RLC mode	AM	AM	TM	UM	AM	AM	AM
	Payload sizes, bit	128	128	168	136	128	128	128
	Max data rate, bps	12 800	12 800	16 800	13 600	12 800	12 800	12 800
	AMD/UMD/TrD PDU header, bit	16	16	0	8	16	16	16
MAC	MAC header, bit	26	26	2	26	26	26	26
	MAC multiplexing	7 logical channel multiplexing						
Layer 1	TrCH type	RACH						
	TB sizes, bit	170						
	TFS TF0, bits	1x170						
	TTI, ms	10						
	Coding type	CC 1/2						
	CRC, bit	16						
	Max number of bits/TTI after channel coding	388						
	Max number of bits/ Radio frame before rate matching	388						

6.11.6.4.5.3.2 Physical channel parameters

See clause 6.11.6.4.5.1.2.

6.11.6.4.6 Combinations on DPCH and HS-PDSCH

6.11.6.4.6.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.1.1 Uplink

See clause 6.11.6.4.1.26.1.

6.11.6.4.6.1.2 Downlink

6.11.6.4.6.1.2.1 Transport channel parameters

6.11.6.4.6.1.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.1.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	depends on UE category NOTE1
	AMD PDU header, bit	16
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	10 ms
	Coding type	TC
	CRC, bit	24
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).		

6.11.6.4.6.1.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.1.2.1.2.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1

6.11.6.4.6.1.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.1.2.2 Physical channel parameters

6.11.6.4.6.1.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2.

6.10.3.4.6.1.2.2.2 Physical channel parameters on HS-PDSCH

Physical parameters common for all UE physical layer categories

HS-PDSCH	Midamble	1024 chips

UE HS-DSCH Physical Layer category 1:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	2
	Max Data Rate	1.2 Mbps

UE HS-DSCH Physical Layer category 2:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	1.2 Mbps

UE HS-DSCH Physical Layer category 3:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	4
	Max Data Rate	2.4 Mbps

UE HS-DSCH Physical Layer category 4:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	2.4 Mbps

UE HS-DSCH Physical Layer category 5:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	6
	Max Data Rate	3.6 Mbps

UE HS-DSCH Physical Layer category 6:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	3.6 Mbps

UE HS-DSCH Physical Layer category 7:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	5.3 Mbps

UE HS-DSCH Physical Layer category 8:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	7.3 Mbps

UE HS-DSCH Physical Layer category 9:

HS-PDSCH	Number of processes	3
	Process memory size	Split equally among all processes
	Maximum number of HS-DSCH timeslots per TTI	12
	Max Data Rate	10.2 Mbps

6.11.6.4.6.2 Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.2.1 Uplink

See clause 6.11.6.4.1.28.1.

6.11.6.4.6.2.2 Downlink

6.11.6.4.6.2.2.1 Transport channel parameters

6.11.6.4.6.2.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.2.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.2.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.2.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.2.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.2.2.2 Physical channel parameters

6.11.6.4.6.2.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2..

6.11.6.4.6.2.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.3 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.3.1 Uplink

See clause 6.11.6.4.1.34.1.

6.11.6.4.6.3.2 Downlink

6.11.6.4.6.3.2.1 Transport channel parameters

6.11.6.4.6.3.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.3.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.3.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.3.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.3.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.3.2.2 Physical channel parameters

6.11.6.4.6.3.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2..

6.11.6.4.6.3.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.4.1 Uplink

6.11.6.4.6.4.1.1 Transport channel parameters

6.11.6.4.6.4.1.1.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.6.4.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.11.6.4.1.34.1.1.1.

6.11.6.4.6.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.6.4.1.1.4 TFCS

TFCS size	36 (alt. 54)
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1) (alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0), (TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0), (TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1), (TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1), (TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1), (TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1))

6.11.6.4.6.4.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 3 time slots
	Max. Number of data bits/radio frame	6 480 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.48

6.11.6.4.6.4.2 Downlink

6.11.6.4.6.4.2.1 Transport channel parameters

6.11.6.4.6.4.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.4.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.4.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.4.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.6.4.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.4.2.1.2.3 TFCS

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.6.4.2.2 Physical channel parameters

6.11.6.4.6.4.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.6.4.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.5.1 Uplink

See clause 6.11.6.4.1.40.1.

6.11.6.4.6.5.2 Downlink

6.11.6.4.6.5.2.1 Transport channel parameters

6.11.6.4.6.5.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.5.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.5.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.5.2.1.2.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.6.5.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.5.2.1.2.3 TFCS

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.6.5.2.2 Physical channel parameters

6.11.6.4.6.5.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.6.5.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.6 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.6.1 Uplink

6.11.6.4.6.6.1.1 Transport channel parameters

6.11.6.4.6.6.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.1.1.1.

6.11.6.4.6.6.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.11.6.4.1.34.1.1.1.

6.11.6.4.6.6.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.6.6.1.1.4 TFCS

TFCS size	24 (alt. 36)
TFCS	(64 kbps RAB, 384 kbps RAB , DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1) (alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0), (TF0, TF6, TF0), (TF1, TF6, TF0), (TF0, TF7, TF0), (TF1, TF7, TF0), (TF0, TF8, TF0), (TF1, TF8, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1), (TF0, TF6, TF1), (TF1, TF6, TF1), (TF0, TF7, TF1), (TF1, TF7, TF1), (TF0, TF8, TF1), (TF1, TF8, TF1))

6.11.6.4.6.6.1.2 Physical channel parameters

DPCH Uplink	Midamble	512 chips
	Codes and time slots	SF4 x 1 code x 3 time slots
	Max. Number of data bits/radio frame	6 480 bits
	TFCI code word	16 bits
	TPC	2 bits
	Puncturing Limit	0.40

6.11.6.4.6.6.1 Downlink

6.11.6.4.6.6.2.1 Transport channel parameters

6.11.6.4.6.6.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.6.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.6.2.1.1 Transport channel parameters for DCH

6.11.6.4.6.6.2.1.2.1 Transport channel parameters for Conversational / unkown/ DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.6.6.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.6.2.1.2.3 TFCS

See clause 6.11.6.4.1.13.2.1.3.

6.11.6.4.6.6.2.2 Physical channel parameters

6.11.6.4.6.6.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.6.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.7 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.7.1 Uplink

See clause 6.11.6.4.1.57.1.

6.11.6.4.6.7.2 Downlink

6.11.6.4.6.7.2.1 Transport channel parameters

6.11.6.4.6.7.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.7.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.7.2.1.1 Transport channel parameters for DCH

6.11.6.4.6.7.2.1.2.1 Transport channel parameters for Conversational / unkown/ DL:64 kbps / CS RAB

See clause 6.11.6.4.1.13.2.1.1.

6.11.6.4.6.7.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.7.2.1.2.3 TFCS

See clause 6.11.6.4.1.13.2.1.3.

6.11.6.4.6.7.2.2 Physical channel parameters

6.11.6.4.6.7.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.6.7.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.8 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.8.1 Uplink

6.11.6.4.6.8.1.1 Transport channel parameters

6.11.6.4.6.8.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB + UL:384 kbps / PS RAB

Higher layer	RAB/Signalling RB	RAB	RAB
RLC	Logical channel type	DTCH	DTCH
	RLC mode	AM	AM
	Payload sizes, bit	320	320

	Max data rate, bps		384 000	384 000
	AMD PDU header, bit		16	16
MAC	MAC header, bit		4	4
	MAC multiplexing		2 logical channel multiplexing	
Layer 1	TrCH type		DCH	
	TB sizes, bit		340	
	TFS	TF0, bits	0x340	
		TF1, bits	1x340	
		TF2, bits	2x340	
		TF3, bits	4x340	
		TF4, bits	8x340	
		TF5, bits	12x340	
	TTI, ms		10	
	Coding type		TC	
	CRC, bit		16	
	Max number of bits/TTI after channel coding		12 828	
	Uplink: Max number of bits/radio frame before rate matching		12 828	
	RM attribute		110-180	

6.11.6.4.6.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.6.8.1.1.3 TFCS

TFCS size	12
TFCS	(384 kbps RAB + 384 kbps RAB, DCCH)= (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)

6.11.6.4.6.8.1.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF4 x 1 codes x 3 time slot
	Max. Number of data bits/radio frame	6480 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.6.8.2 Downlink

6.11.6.4.6.8.2.1 Transport channel parameters

6.11.6.4.6.8.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.8.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.8.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.8.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.8.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.8.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.8.2.2 Physical channel parameters

6.11.6.4.6.8.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2.

6.11.6.4.6.8.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.9 Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.9.1 Uplink

See clause 6.11.6.4.1.57.1.

6.11.6.4.6.9.2 Downlink

6.11.6.4.6.9.2.1 Transport channel parameters

6.11.6.4.6.9.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.9.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.9.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.9.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.9.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.9.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.9.2.2 Physical channel parameters

6.11.6.4.6.9.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2.

6.11.6.4.6.9.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.10 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.10.1 Uplink

6.11.6.4.6.10.1.1 Transport channel parameters

6.11.6.4.6.10.1.1.1 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

Higher Layer	RAB/Signalling RB		RAB
RLC	Logical channel type		DTCH
	RLC mode		AM
	Payload sizes, bit		640
	Max data rate, bps		128000
	AM PDU header, bit		16
MAC	MAC header, bit		0
	MAC multiplexing		N/A
Layer 1	TrCH type		DCH
	TB sizes, bit		656
	TFS	TF0, bits	0x656
		TF1, bits	1x656
		TF2, bits	2x656
		TF3, bits	4x656
	TTI, ms		20
	Coding type		TC
	CRC, bit		16
	Max number of bits/TTI after channel coding		8076
	Uplink: Max number of bits/radio frame before rate matching		4038
	RM attribute		125-165

6.11.6.4.6.10.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.11.6.4.1.28.1.1.1.

6.11.6.4.6.10.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.6.10.1.1.3 TFCS

TFCS size	40
TFCS	(128 kbps RAB, 128 kbps RAB, DCCH)= (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF0, TF2, TF0), (TF1, TF2, TF0), (TF2, TF2, TF0), (TF3, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0), (TF2, TF3, TF0), (TF3, TF3, TF0), (TF0, TF4, TF0), (TF1, TF4, TF0), (TF2, TF4, TF0), (TF3, TF4, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1), (TF2, TF2, TF1), (TF3, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1), (TF2, TF3, TF1), (TF3, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1), (TF2, TF4, TF1), (TF3, TF4, TF1)

6.11.6.4.6.10.1.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF4 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	4272 bits
	TFCI code word	16 bits
	Puncturing limit	0.48

6.11.6.4.6.10.2 Downlink

6.11.6.4.6.10.2.1 Transport channel parameters

6.11.6.4.6.10.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.10.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB

Higher Layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	640
	Max data rate, bps	depends on UE category NOTE1
	AMD PDU header, bit	16
MAC	MAC-d header, bit	0
	MAC multiplexing	N/A
	MAC-d PDU size, bit	656
	MAC-hs header fixed part, bit	21
Layer 1	TrCH type	HS-DSCH
	TTI	10 ms
	Coding type	TC
	CRC, bit	24
NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).		

6.11.6.4.6.10.2.1.1.2 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.10.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.10.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.10.2.1.2.2 TFCS

See clause 6.11.6.4.1.2.2.1.2.

6.11.6.4.6.10.2.2 Physical channel parameters

6.11.6.4.6.10.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.2.2.2.

6.11.6.4.6.10.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.6.11 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.6.11.1 Uplink

6.11.6.4.6.11.1.1 Transport channel parameters

6.11.6.4.6.11.1.1.1 Transport channel parameters for Conversational / Speech / UL12.2kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.6.11.1.1.2 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

See clause 6.11.6.4.6.10.1.1.1.

6.11.6.4.6.11.1.1.3 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.11.6.4.1.28.1.1.1.

6.11.6.4.6.11.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.6.11.1.1.5 TFCS

TFCS size	120
TFCS	(RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0,TF0), (TF1,TF0,TF0,TF1,TF0,TF0), (TF2,TF1,TF1,TF1,TF0,TF0), (TF0,TF0,TF0,TF2,TF0,TF0), (TF1,TF0,TF0,TF2,TF0,TF0), (TF2,TF1,TF1,TF2,TF0,TF0), (TF0,TF0,TF0,TF3,TF0,TF0), (TF1,TF0,TF0,TF3,TF0,TF0), (TF2,TF1,TF1,TF3,TF0,TF0), (TF0,TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1,TF0), (TF1,TF0,TF0,TF1,TF1,TF0), (TF2,TF1,TF1,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF1,TF0), (TF1,TF0,TF0,TF2,TF1,TF0), (TF2,TF1,TF1,TF2,TF1,TF0), (TF0,TF0,TF0,TF3,TF1,TF0), (TF1,TF0,TF0,TF3,TF1,TF0), (TF2,TF1,TF1,TF3,TF1,TF0), (TF0,TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF0,TF2,TF0), (TF0,TF0,TF0,TF1,TF2,TF0), (TF1,TF0,TF0,TF1,TF2,TF0), (TF2,TF1,TF1,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF2,TF0), (TF1,TF0,TF0,TF2,TF2,TF0), (TF2,TF1,TF1,TF2,TF2,TF0), (TF0,TF0,TF0,TF3,TF2,TF0), (TF1,TF0,TF0,TF3,TF2,TF0), (TF2,TF1,TF1,TF3,TF2,TF0), (TF0,TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF0,TF3,TF0), (TF0,TF0,TF0,TF1,TF3,TF0), (TF1,TF0,TF0,TF1,TF3,TF0), (TF2,TF1,TF1,TF1,TF3,TF0), (TF0,TF0,TF0,TF2,TF3,TF0), (TF1,TF0,TF0,TF2,TF3,TF0), (TF2,TF1,TF1,TF2,TF3,TF0), (TF0,TF0,TF0,TF3,TF3,TF0), (TF1,TF0,TF0,TF3,TF3,TF0), (TF2,TF1,TF1,TF3,TF3,TF0), (TF0,TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF0,TF4,TF0), (TF0,TF0,TF0,TF1,TF4,TF0), (TF1,TF0,TF0,TF1,TF4,TF0), (TF2,TF1,TF1,TF1,TF4,TF0), (TF0,TF0,TF0,TF2,TF4,TF0), (TF1,TF0,TF0,TF2,TF4,TF0), (TF2,TF1,TF1,TF2,TF4,TF0), (TF0,TF0,TF0,TF3,TF4,TF0), (TF1,TF0,TF0,TF3,TF4,TF0), (TF2,TF1,TF1,TF3,TF4,TF0), (TF0,TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0,TF1), (TF1,TF0,TF0,TF1,TF0,TF1), (TF2,TF1,TF1,TF1,TF0,TF1), (TF0,TF0,TF0,TF2,TF0,TF1), (TF1,TF0,TF0,TF2,TF0,TF1), (TF2,TF1,TF1,TF2,TF0,TF1), (TF0,TF0,TF0,TF3,TF0,TF1), (TF1,TF0,TF0,TF3,TF0,TF1), (TF2,TF1,TF1,TF3,TF0,TF1), (TF0,TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF0,TF1,TF1), (TF0,TF0,TF0,TF1,TF1,TF1), (TF1,TF0,TF0,TF1,TF1,TF1), (TF2,TF1,TF1,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1,TF1), (TF1,TF0,TF0,TF2,TF1,TF1), (TF2,TF1,TF1,TF2,TF1,TF1), (TF0,TF0,TF0,TF3,TF1,TF1), (TF1,TF0,TF0,TF3,TF1,TF1), (TF2,TF1,TF1,TF3,TF1,TF1), (TF0,TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF0,TF2,TF1), (TF0,TF0,TF0,TF1,TF2,TF1), (TF1,TF0,TF0,TF1,TF2,TF1), (TF2,TF1,TF1,TF1,TF2,TF1), (TF0,TF0,TF0,TF2,TF2,TF1), (TF1,TF0,TF0,TF2,TF2,TF1), (TF2,TF1,TF1,TF2,TF2,TF1), (TF0,TF0,TF0,TF3,TF2,TF1), (TF1,TF0,TF0,TF3,TF2,TF1), (TF2,TF1,TF1,TF3,TF2,TF1), (TF0,TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF0,TF3,TF1), (TF0,TF0,TF0,TF1,TF3,TF1), (TF1,TF0,TF0,TF1,TF3,TF1), (TF2,TF1,TF1,TF1,TF3,TF1), (TF0,TF0,TF0,TF2,TF3,TF1), (TF1,TF0,TF0,TF2,TF3,TF1), (TF2,TF1,TF1,TF2,TF3,TF1), (TF0,TF0,TF0,TF3,TF3,TF1), (TF1,TF0,TF0,TF3,TF3,TF1), (TF2,TF1,TF1,TF3,TF3,TF1), (TF0,TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF0,TF4,TF1), (TF0,TF0,TF0,TF1,TF4,TF1), (TF1,TF0,TF0,TF1,TF4,TF1), (TF2,TF1,TF1,TF1,TF4,TF1), (TF0,TF0,TF0,TF2,TF4,TF1), (TF1,TF0,TF0,TF2,TF4,TF1), (TF2,TF1,TF1,TF2,TF4,TF1), (TF0,TF0,TF0,TF3,TF4,TF1), (TF1,TF0,TF0,TF3,TF4,TF1), (TF2,TF1,TF1,TF3,TF4,TF1)

6.11.6.4.6.11.1.2 Physical channel parameters

DPCH Downlink	Midamble	1024 chips
	Codes and time slots	SF4 x 2 codes x 1 time slot
	Max. Number of data bits/radio frame	4272 bits
	TFCI code word	16 bits
	Puncturing limit	0.44

6.11.6.4.6.11.2 Downlink

6.11.6.4.6.11.2.1 Transport channel parameters

6.11.6.4.6.11.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.6.11.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB

See clause 6.11.6.4.6.10.2.1.1.1.

6.11.6.4.6.11.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.6.11.2.1.2 Transport channel parameters for DCH

6.11.6.4.6.11.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.2.1.1.

6.11.6.4.6.11.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.2.1.1.

6.11.6.4.6.11.2.1.2.3 TFCS

See clause 6.11.6.4.1.4.2.1.3.

6.11.6.4.6.11.2.2 Physical channel parameters

6.11.6.4.6.11.2.2.1 Physical channel parameters on DPCH

See clause 6.11.6.4.1.4.2.2.

6.11.6.4.6.11.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.7 Combinations on HS-PDSCH and E-PUCH

6.11.6.4.7.1 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.11.6.4.7.1.1 Uplink

6.11.6.4.7.1.1.1 Transport channel parameters

6.11.6.4.7.1.1.1.1 Transport channel parameters for E-DCH

6.11.6.4.7.1.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

Higher layer	RAB/Signalling RB	RAB
RLC	Logical channel type	DTCH
	RLC mode	AM
	Payload sizes, bit	320
	Max data rate, bps	Depends on UE category and TTI
	AMD PDU header, bit	16
MAC	MAC multiplexing	N/A
	MAC-d PDU size, bit	336
	MAC-e/es header fixed part, bit	18
Layer 1	TrCH type	E-DCH
	TTI	10ms
	Coding type	TC
	CRC, bit	24

6.11.6.4.7.1.1.1.2 Transport channel parameters for DCH

6.11.6.4.7.1.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.7.1.1.2 Physical channel parameters

6.11.6.4.7.1.1.2.1 Physical channel parameters on E-PUCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE E-DCH Physical Layer category 1:

E-PUCH	Number of processes	4
	Max Data Rate	1.7360Mbps

UE E-DCH Physical Layer category 2:

E-PUCH	Number of processes	4
	Max Data Rate	3.4752Mbps

UE E-DCH Physical Layer category 3:

E-PUCH	Number of processes	4
	Max Data Rate	5.2416Mbps

UE E-DCH Physical Layer category 4:

E-PUCH	Number of processes	4
	Max Data Rate	6.9536Mbps

UE E-DCH Physical Layer category 5:

E-PUCH	Number of processes	4
	Max Data Rate	8.7200Mbps

UE E-DCH Physical Layer category 6:

E-PUCH	Number of processes	4
	Max Data Rate	13.9104Mbps

UE E-DCH Physical Layer category 7:

E-PUCH	Number of processes	4
	Max Data Rate	20.9760Mbps

6.11.6.4.7.1.1.2.2 Physical channel parameters for DPCH

See clause 6.11.6.4.1.2.1.2

6.11.6.4.7.1.2 Downlink

See clause 6.11.6.4.6.1.2.

6.11.6.4.7.3 Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.11.6.4.7.3.1 Uplink

See clause 6.11.6.4.7.1.1.

6.11.6.4.7.3.1.2 Physical channel parameters

6.11.6.4.7.3.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.6.4.7.1.1.2.1.

6.11.6.4.7.3.2 Downlink

6.11.6.4.7.3.2.1 Transport channel parameters

6.11.6.4.7.3.2.1.1 Transport channel parameters for HS-DSCH

6.11.6.4.7.3.2.1.1.1 MAC-d flow#1 parameters for Streaming or interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.11.6.4.6.1.2.1.1.1.

6.11.6.4.7.3.2.1.1.2 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

Higher layer	RAB/Signalling RB	SRB#1	SRB#2	SRB#3	SRB#4
RLC	Logical channel type	DCCH	DCCH	DCCH	DCCH
	RLC mode	UM	AM	AM	AM
	Payload sizes, bit	136	128	128	128
	Max data rate, bps	Depends on UE category (NOTE)			
	AMD PDU header, bit	8	16	16	16
MAC	MAC-d header, bit	4	4	4	4
	MAC multiplexing	4 logical channel multiplexing			
	MAC-d PDU size, bit	148			
	MAC-hs header fixed part, bit	21			
Layer 1	TrCH type	HS-DSCH			
	TTI	10 ms			
	Coding type	TC			
	CRC, bit	24			
NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]).					

6.11.6.4.7.3.2.2 Physical channel parameters

6.11.6.4.7.3.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.11.6.4.6.1.2.2.2.

6.11.6.4.7.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.6.4.7.4.1 Uplink

6.11.6.4.7.4.1.1 Transport channel parameters

6.11.6.4.7.4.1.1.1 Transport channel parameters for E-DCH

6.11.6.4.7.4.1.1.1.1 MAC-d flow parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.6.4.7.1.1.1.1.1.

6.11.6.4.7.4.1.1.2 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.11.6.4.1.4.1.1.1.

6.11.6.4.7.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.11.6.4.1.2.1.1.1.

6.11.6.4.7.4.1.1.4 TFCS

See clause 6.11.6.4.1.4.1.1.3.

6.11.6.4.7.4.1.2 Physical channel parameters

6.11.6.4.7.4.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.6.4.7.1.1.2.1.

6.11.6.4.7.4.1.2.2 Physical channel parameters on DCH

See clause 6.11.6.4.1.4.1.2.

6.11.6.4.7.4.2 Downlink

See clause 6.11.6.4.6.3.2.

6.11.6.4.7.5 Streaming or interactive or background / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.11.6.4.7.5.1 Uplink

6.11.6.4.7.5.1.1 Transport channel parameters

6.11.6.4.7.5.1.1.1 Transport channel parameters for E-DCH

MAC-e multiplexing between all MAC-d flows in the same MAC-e PDU shall be configured.

6.11.6.4.7.5.1.1.1.1 MAC-d flow #1 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.6.4.7.1.1.1.1.1.

6.11.6.4.7.5.1.1.1.2 MAC-d flow #2 parameters for Streaming or interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.11.6.4.7.1.1.1.1.1.

6.11.6.4.7.5.1.1.1.3 MAC-d flow #3 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.11.6.4.7.2.1.1.1.2.

6.11.6.4.7.5.1.2 Physical channel parameters

6.11.6.4.7.5.1.2.1 Physical channel parameters on E-PUCH

See clause 6.11.6.4.7.1.1.2.1.

6.11.6.4.7.5.2 Downlink

See clause 6.11.6.4.6.3.2.

7 Generic setup procedures

7.1 Basic Generic Procedures

7.1.1 UE Test States for Basic Generic Procedures

This clause describes a set of procedures for use by test cases in 3GPP TS 34.123-1 [1]. Describing these procedures in a generic manner allows their use in many test cases. By using these procedures, test case descriptions need not detail signalling that is not relevant to its purpose or understanding.

The procedures are based upon default values that are adapted to the most common usage. Test cases that require values different from the default will, when specifying the Basic Generic Procedure, also specify those parameters that are modified.

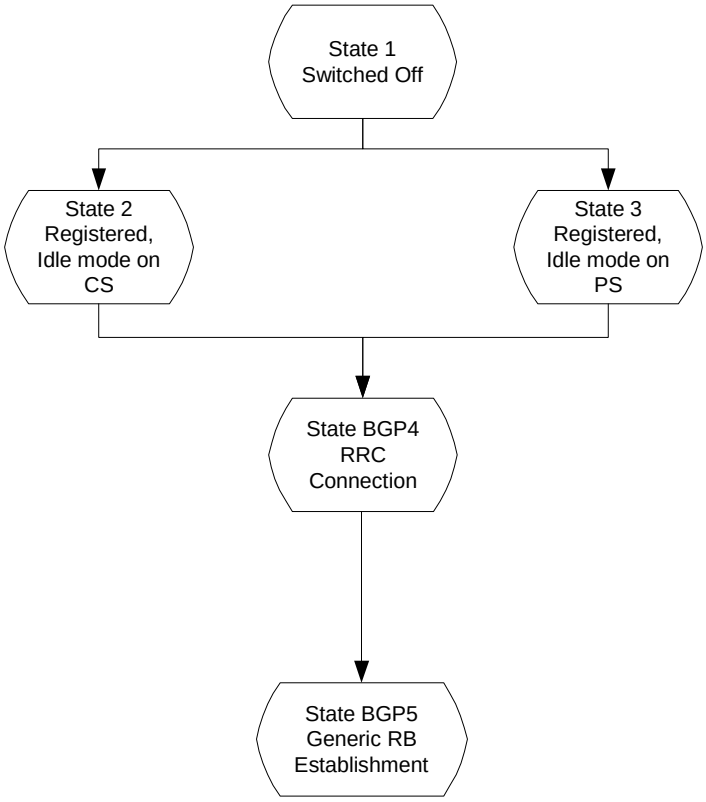


Figure 7.1.1: UE Test States for Basic Generic Procedures

In order that the UE can set up a call in UTRAN, there are a number of procedures to be undertaken in a hierachical sequence to move between known states. The sequences are shown in figure 7.1.1 and the status of the relevant protocols in the UE in the different states are given in table 7.1.1.

Table 7.1.1: The UE states

		RRC	CC	MM	SM	GMM
State 1	Power OFF	-----	null	null	pdp-inactive	GMM-null
State 2	CS Registered Idle Mode	idle	null	idle	pdp-inactive	GMM-deregistered
State 3	PS Registered Idle Mode	idle	null	null	pdp-inactive	GMM-registered
State BGP4	RRC Connection	connected	null	as previous	pdp-inactive	as previous
State BGP5	Generic RB Establishment	connected	null	as previous	pdp-inactive	as previous

7.1.2 Mobile terminated establishment of Radio Resource Connection

7.1.2.1 Initial conditions

System Simulator:

The system simulator will start from the default idle state. Parameters will the default parameters for a single cell, unless otherwise specified in the test case.

User Equipment:

Unless otherwise specified in the test case, the UE will be in the following state:

- Default test operating conditions.
- The UE shall have followed the generic registration procedure for CS or PS operations, and will be in Idle Mode, Camped-on (State 2 or State 3).

7.1.2.2 Definition of system information messages

The default system information messages are used.

7.1.2.3 Procedure

- The SS sends a PAGING TYPE 1 message to the UE on the appropriate paging block, and with the IE "Paging record" containing the TMSI or P-TMSI of the UUT.
- The SS receives an RRC CONNECTION REQUEST message from the UE.
- On receipt of the RRC CONNECTION REQUEST the SS shall transmit a RRC CONNECTION SETUP message to the UE. The SS shall wait for the receipt of an RRC CONNECTION SETUP COMPLETE message from the UE.
- On receipt of an RRC CONNECTION SETUP COMPLETE message, the procedure is complete.

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Default SI messages
2	←		PAGING TYPE 1 (PCCH)	Sent on appropriate cycle
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC

7.1.2.4 Specific message contents

7.1.2.4.1 PAGING TYPE 1

This message is sent from the SS to the UE, using the TM RLC SAP, on the PCCH logical channel.

Information Element				Value/Remark
Message Type				PAGING TYPE 1
UE Information elements				
Paging record list	Paging record	CN originator	Paging cause	Terminating Speech Call (note)
			CN domain identity	CS domain (note)
			UE Identity	TMSI (GSM-MAP) As specified during Registration procedure
Other information elements				
BCCH modification info				omit
NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the Paging cause, CN domain identity and UE Identity are selected in accordance with the requirements of the following procedure.				

7.1.2.4.2 RRC CONNECTION REQUEST

This message is sent by the UE to the SS using the TM-RLC SAP. It is sent on the CCCH Logical channel.

Information Element			Value/Remark
Message Type			RRC CONNECTION REQUEST
UE information elements			
Initial UE identity	TMSI and LAI	TMSI (GSM-MAP)	As specified during Registration procedure
		LAI (GSM-MAP)	As specified by default 1 cell environment
Initial UE capability	Maximum number of AM entities		As declared in UE ICS
Establishment cause			As appropriate
Protocol error indicator			FALSE
>UE Specific Behaviour Information 1 idle			This IE will not be checked by default behaviour, but in specific test case.
Measurement information elements			
Measured results on RACH			Not checked
NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the UE Identity is selected in accordance with the requirements of the following procedure.			

7.1.2.4.3 RRC CONNECTION SETUP

This message is sent from the SS to the UE using the UM-RLC SAP. The message is sent on the CCCH Logical channel.

The default RRC CONNECTION SETUP message for the transition to connected mode CELL_DCH is used except for the IE fields specified below.

Information Element			Value/Remark
Message Type			RRC CONNECTION SETUP
UE Information Elements			
Initial UE identity	TMSI and LAI	TMSI (GSM-MAP)	As specified during Registration procedure
		LAI (GSM-MAP)	As specified by default 1 cell environment
RB Information Elements			
Use default			
TrCH Information Elements			
Use default			
PhyCH Information Elements			
Frequency info			As specified by default 1 cell environment
Uplink radio resources			
Use default			
Downlink radio resources			
Use default			
NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the UE Identity is selected in accordance with the requirements of the following procedure.			

7.1.2.4.4 RRC CONNECTION SETUP COMPLETE

This message is sent by the UE to the SS using AM-RLC SAP. The message is sent on the DCCH Logical channel.

Information Element			Value/Remark
Message Type			RRC CONNECTION SETUP COMPLETE
UE Information Elements			
Hyper frame number			Not checked
UE radio access capability	Conformance test compliance		R99
	PDCP capability	Support for lossless SRNS relocation	Not checked

Information Element			Value/Remark
		Supported algorithm types	Not checked
	RLC capability	Total RLC AM buffer size	Not checked
		Maximum number of AM entities	Not checked
	Transport channel capability	**Downlink**	
		Max no of bits received	Not checked
		Max convolutionally coded bits received	Not checked
		Max turbo coded bits received	Not checked
		Maximum number of simultaneous transport channels	Not checked
		Max no of received transport blocks	Not checked
		Maximum number of TFC in the TFCS	Not checked
		Maximum number of TF	Not checked
		Support for turbo decoding	Not checked
		Uplink	
		Max no of bits transmitted	Not checked
		Max convolutionally coded bits received	Not checked
		Max turbo coded bits received	Not checked
		Maximum number of simultaneous transport channels	Not checked
		Max no of transmitted transport blocks	Not checked
		Maximum number of TFC in the TFCS	Not checked
		Maximum number of TF	Not checked
		Support for turbo encoding	Not checked
	RF capability	UE power class	As declared for UE
		Tx/Rx frequency separation	Not checked
	Physical channel capability	**Downlink**	
		Maximum number of simultaneous CCTrCH	Not checked
		Max no DPCH/PDSCH codes	Not checked (PDSCH: R99 and Rel-4 only)
		Max no physical channel bits received	Not checked
		Support for SF 512	Not checked
		Support of PDSCH	Not checked (R99 and Rel-4 only)
		Simultaneous reception of SCCPCH and DPCH	Not checked
		Max no of S-CCPCH RL	Not checked
		Uplink	
		Maximum number of DPDCH bits transmitted per 10 ms	Not checked
		Support of PCPCH (R99 and Rel-4 only)	Not checked
	UE multi-mode/multi-RAT capability	Multi-RAT capability	
		Multi-mode capability	FDD or FDD/TDD
	Security capability	Ciphering algorithm capability	Not checked
		Integrity protection algorithm capability	Not checked
	LCS capability	Standalone location method(s) supported	Not checked

Information Element			Value/Remark
		UE based OTDOA supported	Not checked
		Network Assisted GPS support	Not checked
		GPS reference time capable	Not checked
		Support for IPDL	Not checked
	Measurement capability	Need for downlink compressed mode	Not checked
		FDD measurements DL	Not checked
		TDD measurements DL	Not checke
		GSM 900 DL	Not checked
		DCS 1800 DL	Not checked
		GSM 1900 DL	Not checked
		Multi-carrier measurement DL	Not checked
		Need for uplink compressed mode	Not checked
		FDD measurements UL	Not checked
		TDD measurements UL	Not checked
		GSM 900 UL	Not checked
		DCS 1800 UL	Not checked
		GSM 1900 UL	Not checked
		Multi-carrier measurement UL	Not checked
UE system specific capability			Not checked

7.1.3 Radio Bearer Setup Procedure

7.1.3.1 Initial conditions

The procedure specified in clause 7.1.2 will be run. This procedure starts from the successful completion of clause 7.1.2.

7.1.3.2 Definition of system information messages

The default system information messages are used.

7.1.3.3 Procedure

- The SS sends a RADIO BEARER SETUP message to the UE on the DCCH established by the RRC Connection Establishment procedure.
- The SS receives a RADIO BEARER SETUP COMPLETE message from the UE in RLC Acknowledged mode on the DCCH.

On receiption of the RADIO BEARER SETUP COMPLETE the procedure is complete.

Step	Direction		Message	Comments
	UE	SS		
1	←		RADIO BEARER SETUP (DCCH)	RRC
2	→		RADIO BEARER SETUP COMPLETE (DCCH)	RRC

7.1.3.4 Specific message contents

7.1.3.4.1 RADIO BEARER SETUP

The RADIO BEARER SETUP message is sent from the System Simulator to the UE, using AM-RLC on the DCCH logical channel.

The default RRC CONNECTION SETUP message for the setup of a speech radio access bearer is used except for the IE fields specified below.

Information Element		Value/Remark
Message Type		RADIO BEARER SETUP
UE Information Elements		
CN Information Elements		
RB Information Elements		
RAB information for setup	Default parameters for 12.2 kbps speech RAB + 3.4 kbps signalling radio bearer according to clause 6.10.2.4.1.4 for FDD, clause 6.10.3.4.1.4 for 3.84 Mcps TDD and 6.11.5.4.1.4 for 1.28 Mcps TDD	

7.1.3.4.2 RADIO BEARER SETUP COMPLETE

The RADIO BEARER SETUP COMPLETE message is sent from the UE to the System Simulator, using AM-RLC on the DCCH logical channel.

The default RADIO BEARER SETUP COMPLETE message is used.

Information Element	Value/Remark
Message Type	RADIO BEARER SETUP COMPLETE
Use default	

7.2 Generic setup procedures

7.2.1 UE Test States for Generic setup procedures

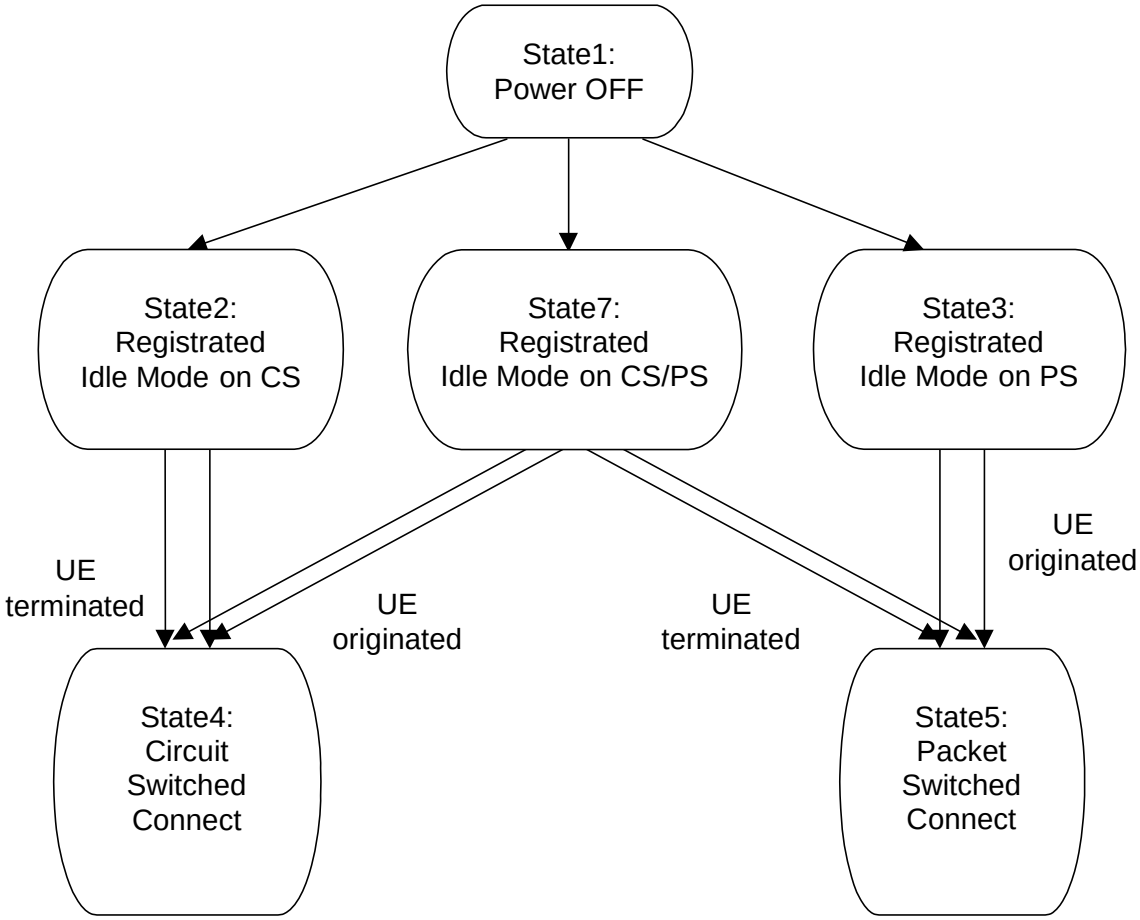


Figure 7.2.1.1: UE Test States for Generic setup procedures

In order that the UE can set up a call in UTRAN, there are a number of procedures to be undertaken in a hierarchical sequence to move between known states. The sequences are shown in figure 7.2.1.1 and the status of the relevant protocols in the UE in the different states are given in table 7.2.1.1.

Table 7.2.1.1: The UE states

		RRC	CC	MM	SM	GMM
State1	Power OFF	-----	null	null	pdp-inactive	GMM-null
State2	Registered Idle Mode on CS	idle	null	MM idle	pdp-inactive	GMM-deregistered
State3	Registered Idle Mode on PS	idle	null	null	pdp-inactive	GMM-registered
State4	Circuit Switched Connect	connected	active	MM connection active	pdp-inactive	same as previous state
State5	Packet Switched Connect	connected	null	same as previous state	pdp-active	GMM-registered
State7	Registered Idle Mode on CS/PS	idle	null	MM idle	pdp-inactive	GMM-registered

7.2.2 Registration of UE

The default procedures required to achieve the changes of state between State 1, in clause 7.2.1, and States 2, 3 and 7 are illustrated in the following clauses.

The choice of which procedure to use given a UE supporting packet services is influenced by the Network Mode of Operation being simulated by the SS and by the Operation Mode of the UE, as described in 3GPP TS 24.008 [32] clause 1.7.2.2. Table 7.2.2 shows the appropriate clause number for each combination of these two modes of operation.

Table 7.2.2: Registration Procedures for UEs Supporting Packet Services

Network Mode		NMO I	NMO II
UE Mode	**PS/CS**	7.2.2.3	7.2.2.4
	PS	7.2.2.2	7.2.2.2

7.2.2.1 Registration on CS

7.2.2.1.1 Initial condition

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.2.1.2 Definition of system information messages

The default system information messages are used.

7.2.2.1.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	NW Broadcast
2	→		RRC CONNECTION REQUEST (CCCH)	RRC
3	←		RRC CONNECTION SETUP (CCCH)	RRC
4	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5	→		LOCATION UPDATING REQUEST	MM
6	←		AUTHENTICATION REQUEST	MM
7	→		AUTHENTICATION RESPONSE	MM
8	←		SECURITY MODE COMMAND	RRC
9	→		SECURITY MODE COMPLETE	RRC
10	←		LOCATION UPDATING ACCEPT	MM
11	→		TMSI REALLOCATION COMPLETE	MM
12	←		RRC CONNECTION RELEASE	RRC
13	→		RRC CONNECTION RELEASE COMPLETE	RRC

7.2.2.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.2.2 Registration on PS

7.2.2.2.1 Initial condition

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.2.2.2 Definition of system information messages

The default system information messages are used.

7.2.2.2.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	NW Broadcast
2	→		RRC CONNECTION REQUEST (CCCH)	RRC
3	←		RRC CONNECTION SETUP (CCCH)	RRC
4	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5	→		ATTACH REQUEST	GMM
6	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
7	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
8	←		SECURITY MODE COMMAND	RRC
9	→		SECURITY MODE COMPLETE	RRC
10	←		ATTACH ACCEPT	GMM
11	→		ATTACH COMPLETE	GMM
12	←		RRC CONNECTION RELEASE	RRC
13	→		RRC CONNECTION RELEASE COMPLETE	RRC

7.2.2.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.2.3 Registration on CS / PS combined environment

7.2.2.3.1 Initial condition

System Simulator:

- 1 cell operating in network operation mode I, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.2.3.2 Definition of system information messages

The default system information messages are used.

7.2.2.3.3 Procedure UE establish PS registration immediately after the UE has been switched on

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	NW Broadcast
2		→	RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5		→	ATTACH REQUEST	GMM
6		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
7		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
8		←	SECURITY MODE COMMAND	RRC
9		→	SECURITY MODE COMPLETE	RRC
10		←	ATTACH ACCEPT	GMM
11		→	ATTACH COMPLETE	GMM
12		←	RRC CONNECTION RELEASE	RRC
13		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.2.2.3.3a Procedure UE establish PS registration later the user decides to use the PS services

CS registration has been successfully completed and RRC connection is released, cee clause 7.2.2.1. Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	NW Broadcast
1a				The UE initiates an attach by MMI or by AT command.
2		→	RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5		→	ATTACH REQUEST	GMM
6		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
7		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
8		←	SECURITY MODE COMMAND	RRC
9		→	SECURITY MODE COMPLETE	RRC
10		←	ATTACH ACCEPT	GMM
11		→	ATTACH COMPLETE	GMM
12		←	RRC CONNECTION RELEASE	RRC
13		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.2.2.3.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.2.4 Registration on CS / PS non-combined environment

7.2.2.4.1 Initial condition

System Simulator:

- 1 cell operating in network operation mode II, default parameters.

User Equipment:

- The UE set to Operation mode A
- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.2.4.2 Definition of system information messages

The default system information messages are used.

7.2.2.4.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Registrations in the CS domain and in the PS domain shall execute independently. The separate registration procedures may occur sequentially or in parallel. If the procedures occur sequentially PS domain registration can be started immediately after power on or the UE can initiate PS registration by MMI or by AT command. If MMI or AT commands are used, registrations are done with two separate RRC connections. The procedures for CS and PS registration shall be as defined in clauses 7.2.2.1 and 7.2.2.2.

7.2.2.4.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.3 Call setup

7.2.3.1 Generic call set up procedure for mobile terminating circuit switched calls

7.2.3.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.3.1.2 Definition of system information messages

The default system information messages are used.

7.2.3.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING (PCCH)	Paging

Step	Direction		Message	Comments
	UE	SS		
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		PAGING RESPONSE	RR
7	←		AUTHENTICATION REQUEST	MM
8	→		AUTHENTICATION RESPONSE	MM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		SET UP	CC (see note)
12	→		CALL CONFIRMED	CC
13	←		RADIO BEARER SETUP	RRC RAB SETUP
14	→		RADIO BEARER SETUP COMPLETE	RRC
15	→		ALERTING	CC (this message is optional)
16	→		CONNECT	CC
17	←		CONNECT ACKNOWLEDGE	CC
NOTE: The "Signal" information element is not included in the SETUP message.				

7.2.3.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.3.2 Generic call set-up procedure for mobile originating circuit switched calls

7.2.3.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.3.2.2 Definition of system information messages

The default system information messages are used.

7.2.3.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	→		RRC CONNECTION REQUEST (CCCH)	RRC
3	←		RRC CONNECTION SETUP (CCCH)	RRC
4	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5	→		CM SERVICE REQUEST	MM
6	←		AUTHENTICATION REQUEST	MM
7	→		AUTHENTICATION RESPONSE	MM
8	←		SECURITY MODE COMMAND	RRC
9	→		SECURITY MODE COMPLETE	RRC
10	→		SET UP	CC
11	←		CALL PROCEEDING	CC
12	←		RADIO BEARER SETUP	RRC RAB SETUP
13	→		RADIO BEARER SETUP COMPLETE	RRC
14	←		ALERTING	CC
15	←		CONNECT	CC
16	→		CONNECT ACKOWLEDGE	CC

7.2.3.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.4 Session setup

7.2.4.1 Generic session set up procedure for mobile terminating packet switched sessions

7.2.4.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

7.2.4.1.2 Definition of system information messages

The default system information messages are used.

7.2.4.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	SERVICE REQUEST	GMM
7		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
8		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
9		←	SECURITY MODE COMMAND	RRC
10		→	SECURITY MODE COMPLETE	RRC
11		←	REQUEST PDP CONTEXT ACTIVATION	SM
12		→	ACTIVATE PDP CONTEXT REQUEST	SM
13		←	RADIO BEARER SETUP	RRC RAB SETUP
14		→	RADIO BEARER SETUP COMPLETE	RRC
15		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.2.4.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.2.4.2 Generic session set up procedure for mobile originating packet switched sessions

7.2.4.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.

- The Test-USIM shall be inserted.

7.2.4.2.2 Definition of system information messages

The default system information messages are used.

7.2.4.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		→	RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5		→	SERVICE REQUEST	GMM
6		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
7		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
8		←	SECURITY MODE COMMAND	RRC
9		→	SECURITY MODE COMPLETE	RRC
10		→	ACTIVATE PDP CONTEXT REQUEST	SM
11		←	RADIO BEARER SETUP	RRC RAB SETUP
12		→	RADIO BEARER SETUP COMPLETE	RRC
13		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.2.4.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

7.3 Test procedures for RF test

NOTE: In general parameters defined for specific test cases in 3GPP TS 34.121 [2] take priority over the default parameters defined in the present document.

7.3.1 UE Test States for RF testing

In this clause, the states of the UE for the test are defined. For RF testing the same UE test states as specified in section 7.2.1 apply plus an additional RB Test Mode State. The RB Test Mode State can be reached from the UE States 2, 3 and 7 according to section 7.2.1. For this RB Test Mode State the different protocols shall be in the following states:

	RRC	CC	MM	SM	GMM
RB Test Mode State	connected	null	see Note	pdp-inactive	same as previous state
Note: The MM state is "MM connection active" if an RRC connection exists for the CS domain otherwise it is "same as previous state".					

7.3.2 Test procedure for TX, RX and Performance Requirement (without handover)

7.3.2.1 Initial conditions

System Simulator

- test cases using 1 cell:
  - 1cell, default parameters.
- other test cases using this test procedure:
  - Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.2.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.2.3 Procedure

7.3.2.3.1 For UE supporting CS

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (CS domain, TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC

6	→	PAGING RESPONSE	RR
7	←	AUTHENTICATION REQUEST	MM
8	→	AUTHENTICATION RESPONSE	MM
9	←	SECURITY MODE COMMAND	RRC
10	→	SECURITY MODE COMPLETE	RRC
11	←	ACTIVATE RB TEST MODE	TC
12	→	ACTIVATE RB TEST MODE COMPLETE	TC
13	←	RADIO BEARER SETUP	RRC (RAB SETUP)
14	→	RADIO BEARER SETUP COMPLETE	RRC
15	←	CLOSE UE TEST LOOP (DCCH)	TC (UE test loop mode set up)
16	→	CLOSE UE TEST LOOP COMPLETE	TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated)
17	←	OPEN UE TEST LOOP	TC
18	→	OPEN UE TEST LOOP COMPLETE	TC
19	←	RRC CONNECTION RELEASE	RRC
20	→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.2.3.2 For UE supporting PS only

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		SERVICE REQUEST	GMM
7	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
8	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		ACTIVATE RB TEST MODE	TC
12	→		ACTIVATE RB TEST MODE COMPLETE	TC
13	←		RADIO BEARER SETUP	RRC (RAB SETUP)
14	→		RADIO BEARER SETUP COMPLETE	RRC
15	←		CLOSE UE TEST LOOP (DCCH)	TC (UE test loop mode set up)
16	→		CLOSE UE TEST LOOP COMPLETE	TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated)
17	←		OPEN UE TEST LOOP	TC
18	→		OPEN UE TEST LOOP COMPLETE	TC
19	←		RRC CONNECTION RELEASE	RRC
20	→		RRC CONNECTION RELEASE COMPLETE	RRC

7.3.2.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

7.3.2.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE, used for the UE supporting PS only.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.2.4.2 Reference measurement channels

The configurations of the reference measurement channels for RF tests are described in 3GPP TS 34.121 [2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD.

7.3.2.4.3 Void

7.3.2.4.4 Compressed mode

[T.B.D]

7.3.2.4.5 Transmit diversity mode

[T.B.D]

7.3.3 Test procedure for test cases using Cell_PCH or URA_PCH state

7.3.3.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.3.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2

- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.3.3 Procedure

7.3.3.3.1 For UE supporting PS

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		SERVICE REQUEST	GMM
7	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
8	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		ACTIVATE RB TEST MODE	TC
12	→		ACTIVATE RB TEST MODE COMPLETE	TC
13	←		RADIO BEARER SETUP	RRC - RAB SETUP using Reference Radio Bearer Configuration
14	→		RADIO BEARER SETUP COMPLETE	RRC
15	←		PHYSICAL CHANNEL RECONFIGURATION	RRC - RRC state indicator is set to "Cell_PCH" or "URA_PCH" depending on the test case
16	→		PHYSICAL CHANNEL RECONFIGURATION COMPLETE	RRC The UE sends this message before it completes state transition.
17			Void	SS sends the L2 ack on the PHYSICAL CHANNEL RECONFIGURATION COMPLETE message. NOTE: The SS should continue to keep the dedicated channel configuration during the time when the L2 ack is sent to the UE.

7.3.3.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

The RADIO BEARER SETUP message is defined in clause 9.2.1, "Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only)".

The PHYSICAL CHANNEL RECONFIGURATION message is defined in clause 9.1.1, "Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM" using condition A8 for URA_PCH and condition A10 for Cell_PCH.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.4 Test procedure for Handover

NOTE: This test procedure is also used for some other test cases involving more than 1 cell.

7.3.4.1 Initial conditions

System Simulator:

- Intra-frequency hard handover and soft handover (for FDD) case:
  - 2 cells, default parameters according to Cell 1 and Cell 2 in clause 6.1.4.
- Inter-frequency hard handover case:
  - 2 cells, default parameters according to Cell 1 and Cell 4 in clause 6.1.4.
- Inter-system handover UTRAN to GSM case:
  - 2 cells, default parameters according to Cell 1 and Cell 9 in clause 6.1.4.
- other test cases using this test procedure:
  - Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] for FDD and TS 34.122 [5] for TDD and take priority over the default parameters.

UserEquipment:

- The UE shall be initially operated under the normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.4.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE

- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

For the intra-frequency hard handover and soft handover (for FDD) case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 2 in clause 6.1.4 are used.

For the inter-frequency hard handover case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 4 in clause 6.1.4 are used.

For the inter-system handover from UTRAN to GSM case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 9 in clause 6.1.4 are used.

7.3.4.3 Procedure

7.3.4.3.1 For UE supporting CS

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (CS domain, TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		PAGING RESPONSE	RR
7	←		AUTHENTICATION REQUEST	MM
8	→		AUTHENTICATION RESPONSE	MM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		ACTIVATE RB TEST MODE	TC
12	→		ACTIVATE RB TEST MODE COMPLETE	TC
13	←		RADIO BEARER SETUP	RRC - RAB SETUP using Reference Radio Bearer Configuration - RRC state indicator is set to "CELL_DCH"
14	→		RADIO BEARER SETUP COMPLETE	RRC
15	←		RRC CONNECTION RELEASE	RRC
16	→		RRC CONNECTION RELEASE COMPLETE	RRC

7.3.4.3.2 For UE supporting PS only

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		SERVICE REQUEST	GMM
7	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
8	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		ACTIVATE RB TEST MODE	TC
12	→		ACTIVATE RB TEST MODE COMPLETE	TC

13	←	RADIO BEARER SETUP	RRC - RAB SETUP using Reference Radio Bearer Configuration - RRC state indicator is set to "CELL_DCH"
14	→	RADIO BEARER SETUP COMPLETE	RRC
15	←	RRC CONNECTION RELEASE	RRC
16	→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.4.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.5 Test procedure for test cases using CELL_FACH state

7.3.5.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.5.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.5.3 Procedure

7.3.5.3.1 For UE supporting CS

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging (CS domain, TMSI)
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	PAGING RESPONSE	RR
7		←	AUTHENTICATION REQUEST	MM
8		→	AUTHENTICATION RESPONSE	MM
9		←	SECURITY MODE COMMAND	RRC
10		→	SECURITY MODE COMPLETE	RRC
11		←	ACTIVATE RB TEST MODE	TC
12		→	ACTIVATE RB TEST MODE COMPLETE	TC
13		←	DEACTIVATE RB TEST MODE	TC
14		→	DEACTIVATE RB TEST MODE COMPLETE	TC
15		←	RRC CONNECTION RELEASE	RRC
16		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.5.3.2 For UE supporting PS only

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	SERVICE REQUEST	GMM
7		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
8		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
9		←	SECURITY MODE COMMAND	RRC
10		→	SECURITY MODE COMPLETE	RRC
11		←	ACTIVATE RB TEST MODE	TC
12		→	ACTIVATE RB TEST MODE COMPLETE	TC
13		←	DEACTIVATE RB TEST MODE	TC
14		→	DEACTIVATE RB TEST MODE COMPLETE	TC
15		←	RRC CONNECTION RELEASE	RRC
16		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.5.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

The RRC connection setup is defined in clause 9.1.1, "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)".

7.3.6 Test procedure for HSDPA RF Performance Requirement

7.3.6.1 Initial conditions

System Simulator:

- 1 HS-DSCH cell, default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.6.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64

- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.6.3 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging (CS domain, TMSI)
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	PAGING RESPONSE	RR
7		←	AUTHENTICATION REQUEST	MM
8		→	AUTHENTICATION RESPONSE	MM
9		←	SECURITY MODE COMMAND	RRC (CS domain)
10		→	SECURITY MODE COMPLETE	RRC
11		←	PAGING TYPE2 (DCCH)	Paging (TMSI (GSM-MAP)/ P-TMSI)
12		→	SERVICE REQUEST	GMM
13		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
14		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
15		←	SECURITY MODE COMMAND	RRC (PS domain, IE Integrity protection mode command set to "modify")
16		→	SECURITY MODE COMPLETE	RRC
17		←	ACTIVATE RB TEST MODE	TC
18		→	ACTIVATE RB TEST MODE COMPLETE	TC
19		←	RADIO BEARER SETUP	RRC (RAB SETUP RMC 12.2 CS)
20		→	RADIO BEARER SETUP COMPLETE	RRC
21		←	RADIO BEARER SETUP	RRC (RAB SETUP HSDPA PS)
22		→	RADIO BEARER SETUP COMPLETE	RRC
A23		←	CLOSE UE TEST LOOP (DCCH)	TC (UE test loop mode set up to loop the RMC 12.2 to UL RMC 12.2) . Test steps A23, A24, A26 and A27 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used.
A24		→	CLOSE UE TEST LOOP COMPLETE	TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated)
25	<-->			Perform test
A26		←	OPEN UE TEST LOOP	TC
A27		→	OPEN UE TEST LOOP COMPLETE	TC
28		←	RRC CONNECTION RELEASE	RRC
29		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.6.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

7.3.6.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.6.4.2 RADIO BEARER SETUP

For step 19, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)" is used with condition A1. For step 21, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)" is used.

The configurations of the fixed reference channels for HSDPA RF tests are described in 3GPP TS 34.121[2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD.

7.3.6.4.3 RRC CONNECTION SETUP

For step 4 , the message in clause 9.2, "Contents of RRC CONNECTION SETUP message: UM" is used with the following exceptions:

Contents of RRC CONNECTION SETUP message: UM

Information Element	Value/remark
- Default DPCH Offset Value	Arbitrary set to value 1536..306176 by step of 2560 (this corresponds to a 0.5 slot timing offset between the DPCCH and the HS-DPCCH)

7.3.7 Test procedure for inter-RAT handover used in RRM testing

7.3.7.1 Initial conditions

System Simulator:

- 2 cells, default parameters according to Cell 1 and Cell 9 in clause 6.1.4.

UserEquipment:

- The UE shall be initially operated under the normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.7.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	

Information Element	Value/remark
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

For the inter-system handover from UTRAN FDD to GSM case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 9 in clause 6.1.4 are used.

7.3.7.3 Procedure

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING (PCCH)	Paging
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC (Transition to cell DCH)
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		PAGING RESPONSE	RR
7	←		AUTHENTICATION REQUEST	MM
8	→		AUTHENTICATION RESPONSE	MM
9	←		SECURITY MODE COMMAND	RRC
10	→		SECURITY MODE COMPLETE	RRC
11	←		SET UP	CC (see note)
12	→		CALL CONFIRMED	CC
13	←		RADIO BEARER SETUP	RRC RAB SETUP
14	→		RADIO BEARER SETUP COMPLETE	RRC
15	→		ALERTING	CC (this message is optional)
16	→		CONNECT	CC
17	←		CONNECT ACKNOWLEDGE	CC
NOTE: The "Signal" information element is not included in the SETUP message.				

7.3.7.4 Specific message contents

The default message contents specified in clause 9.1 are used with the following exceptions.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.8 Test procedure for inter-RAT cell FACH reselection used in RRM testing

7.3.8.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.8.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.8.3 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2	→		RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5	→		SERVICE REQUEST	GMM
6		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
7	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
8		←	SECURITY MODE COMMAND	RRC
9	→		SECURITY MODE COMPLETE	RRC
10	→		ACTIVATE PDP CONTEXT REQUEST	SM
11		←	RADIO BEARER SETUP	RRC RAB SETUP
12	→		RADIO BEARER SETUP COMPLETE	RRC
13		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.3.8.4 Specific message contents

The default message contents specified in clause 9.1 are used with the following exceptions.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

The RRC connection setup is defined in clause 9.1.1, "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)".

7.3.9 Test procedure for E-DCH RF test cases

7.3.9.1 Initial conditions

System Simulator:

- 1 HS-DSCH plus E-DCH cell, default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.9.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP

Information Element	Value/remark
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.9.3 Procedure

7.3.9.3.1 For UE not transmitting E-DPDCH with SF4x2 and SF2x2 configuration

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging (CS domain, TMSI)
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	PAGING RESPONSE	RR
7		←	AUTHENTICATION REQUEST	MM
8		→	AUTHENTICATION RESPONSE	MM
9		←	SECURITY MODE COMMAND	RRC (CS domain)
10		→	SECURITY MODE COMPLETE	RRC
11		←	PAGING TYPE2 (DCCH)	Paging (TMSI (GSM-MAP)/ P-TMSI)
12		→	SERVICE REQUEST	GMM
13		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
14		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
15		←	SECURITY MODE COMMAND	RRC (PS domain, IE Integrity protection mode command set to "modify")
16		→	SECURITY MODE COMPLETE	RRC
17		←	ACTIVATE RB TEST MODE	TC
18		→	ACTIVATE RB TEST MODE COMPLETE	TC
19		←	RADIO BEARER SETUP	RRC (RAB SETUP RMC 12.2 CS)
20		→	RADIO BEARER SETUP COMPLETE	RRC
21		←	RADIO BEARER SETUP	RRC (RAB SETUP HSDPA and E-DCH PS)
22		→	RADIO BEARER SETUP COMPLETE	RRC

A23	←	CLOSE UE TEST LOOP (DCCH)	TC (UE test loop mode set up) . Test steps A23, A24, A26 and A27 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used.
A24	→	CLOSE UE TEST LOOP COMPLETE	TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated)
25	<-->		Perform test
A26	←	OPEN UE TEST LOOP	TC
A27	→	OPEN UE TEST LOOP COMPLETE	TC
28	←	RRC CONNECTION RELEASE	RRC
29	→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.9.3.2 For UE transmitting E-DPDCH with SF4x2 and SF2x2 configuration

Step	Direction		Message	Comments
	UE	SS		
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3	→		RRC CONNECTION REQUEST (CCCH)	RRC
4	←		RRC CONNECTION SETUP (CCCH)	RRC
5	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6	→		SERVICE REQUEST	GMM
7	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
8	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
9	←		SECURITY MODE COMMAND	RRC (PS domain)
10	→		SECURITY MODE COMPLETE	RRC
11	←		ACTIVATE RB TEST MODE	TC
12	→		ACTIVATE RB TEST MODE COMPLETE	TC
13	←		RADIO BEARER SETUP	RRC (RAB SETUP HSDPA and E-DCH PS)
14	→		RADIO BEARER SETUP COMPLETE	RRC
A15	←		CLOSE UE TEST LOOP (DCCH)	TC (UE test loop mode set up) . Test steps A15, A16, A18 and A19 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used.
A16	→		CLOSE UE TEST LOOP COMPLETE	TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated)
17	<-->			Perform test
A18	←		OPEN UE TEST LOOP	TC
A19	→		OPEN UE TEST LOOP COMPLETE	TC
20	←		RRC CONNECTION RELEASE	RRC
21	→		RRC CONNECTION RELEASE COMPLETE	RRC

7.3.9.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

7.3.9.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.9.4.2 RADIO BEARER SETUP

For step 19, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)" is used with condition A1. For step 21, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)" is used.

The configurations of the fixed reference channels for HSDPA RF tests are described in 3GPP TS 34.121[2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD. The configurations of the reference channels for E-DCH RF tests are described in 3GPP TS 34.121[2].

7.3.9.4.3 RRC CONNECTION SETUP

For step 4 , the messages in clause 9.2, "Contents of RRC CONNECTION SETUP message: UM" is used with the following exceptions:

Contents of RRC CONNECTION SETUP message: UM

Information Element	Value/remark
- Default DPCH Offset Value	Arbitrary set to value 1536..306176 by step of 2560 (this corresponds to a 0.5 slot timing offset between the DPCCH and the HS-DPCCH)

7.3.10 Test procedure for MBMS RF/RRM test cases

7.3.10.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.10.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark

- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

As specified in 34.108 clause 6.1.0b with the following exceptions and using condition M2.

Information Element	Value/remark
- Secondary CCPCH system information - Secondary CCPCH info - CHOICE mode - Secondary scrambling code - STTD indicator - Spreading factor - Code number - Pilot symbol existence - TFCI existence - Fixed or Flexible position - Timing offset	 FDD Not Present FALSE 64 2 FALSE TRUE (default value) Flexible (default value) Not Present Absence of this IE is equivalent to default value 0

7.3.10.3 Procedure

Step	Direction		Message	Comments
	UE	SS		
0				UE selects the required MBMS broadcast service
1	←		SYSTEM INFORMATION (BCCH)	Broadcast
2	←		MBMS MODIFIED SERVICES INFORMATION (MCCH)	No Service in Modified Service list
3	←		MBMS UNMODIFIED SERVICES INFORMATION (MCCH)	MBMS required UE action " set to acquire PTM RB info".
4	←		MBMS GENERAL INFORMATION (MCCH)	
5	←		MBMS COMMON P-T-M RB INFORMATION (MCCH)	Contains configuration of the MTCH radio bearer.
6	←		MBMS CURRENT CELL P-T-M RB INFORMATION (MCCH)	Indicates the radio bearer configuration to be used for reception of the service.
7	←		MBMS NEIGHBOURING CELL P-T-M RB INFORMATION (MCCH)	Optional depending on whether neighbour cells are required.
8	←		PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
9	→		RRC CONNECTION REQUEST (CCCH)	RRC
10	←		RRC CONNECTION SETUP (CCCH)	RRC
11	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
12	→		SERVICE REQUEST	GMM
13	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
14	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
15	←		SECURITY MODE COMMAND	RRC
16	→		SECURITY MODE COMPLETE	RRC
17	←		ACTIVATE RB TEST MODE	TC
18	→		ACTIVATE RB TEST MODE COMPLETE	TC
19			void	
20	←		CLOSE UE TEST LOOP	TC (UE test loop mode 3 set up) The RLC SDU counting shall be performed by the UE
21	→		CLOSE UE TEST LOOP COMPLETE	TC (test loop mode 3 on MTCH is activated)
A22	←		PHYSICAL CHANNEL RECONFIGURATION	RRC - RRC state indicator set to "Cell_PCH" Test steps A22 and A23 are only executed when the test method in TS 34.121 [2] specifies that transition to CELL_PCH state is required.

A23	→	PHYSICAL CHANNEL RECONFIGURATION COMPLETE	RRC The UE sends this message before it completes state transition.
	<-->		Perform test.
A24	←	PAGING TYPE 1	RRC - RRC state indicator set to "Cell_FACH" Test steps A24, A25 and A26 are only executed when the test method in TS 34.121 [2] specifies that the test is performed in CELL_PCH state.
A25	→	CELL UPDATE	RRC
A26	←	CELL UPDATE CONFIRM	RRC
27	←	OPEN UE TEST LOOP	TC
28	→	OPEN UE TEST LOOP COMPLETE	TC
29	←	RRC CONNECTION RELEASE	RRC
30	→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.10.4 Specific message contents

The default message contents specified in clause 9.1.1 are used with the following exceptions.

Contents of MBMS GENERAL INFORMATION message: UM (Step 4)

Information Element	Value/remark	Version
MICH configuration information		Rel-6
- MICH Power offset	0dB	Rel-6
- CHOICE *Mode*	FDD	Rel-6
- Channelisation code	7	Rel-6
- Number of NI per frame	18	Rel-6
- STTD indicator	FALSE	Rel-6

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (Step 5)

For step 5, the message in clause 9.2.1 “Contents of MBMS COMMON P-T-M RB INFORMATION message: UM” is used

Contents of MBMS CURRENT P-T-M RB INFORMATION message: UM (Step 6)

For step 6, the message in clause 9.1.1 “Contents of MBMS CURRENT P-T-M RB INFORMATION message: UM” is used with condition A2.

PHYSICAL CHANNEL RECONFIGURATION (Step A22)

For step A22, the message in clause 9.1.1 "Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM" is used with condition A9 for Cell_PCH.

PAGING TYPE 1 (Step A24)

Information Element	Value/remark
Message Type	
Paging record list	
-Paging record	
- CHOICE Used paging identity	Utran-Identity
-U-RNTI	
-SRNC-Identity	'000000000001'B
-S-RNTI	'00000000000000000001'B
BCCH modification info	Not Present

Contents of ATTACH ACCEPT message: GMM

This message is sent from the SS to the UE.

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.11 Test procedure for HSDPA with F-DPCH RF Performance Requirement

7.3.11.1 Initial conditions

System Simulator:

- 1 HS-DSCH with F-DPCH cell, default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

7.3.11.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

Information Element	Value/remark
- CN domain system information	
- CN domain identity	PS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00 00
- CN domain specific DRX cycle length coefficient	7
- CN domain identity	CS
- CHOICE CN Type	GSM-MAP
- CN domain specific NAS system information	
- GSM-MAP NAS system information	00(T3212 is set to infinity) 01
- CN domain specific DRX cycle length coefficient	7
- UE Timers and constants in connected mode	
- T305	Infinity

Contents of System information block type 3 and 4: RRC

Information Element	Value/remark
- Qrxlevmin	-115

Contents of System Information Block type 5 (FDD)

Information Element	Value/remark
- Secondary CCPCH system information	
- Secondary CCPCH info	
- CHOICE mode	FDD
- Secondary scrambling code	Not Present
- STTD indicator	FALSE
- Spreading factor	64
- Code number	2
- Pilot symbol existence	FALSE
- TFCI existence	TRUE (default value)
- Fixed or Flexible position	Flexible (default value)
- Timing offset	Not Present Absence of this IE is equivalent to default value 0

7.3.11.3 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	SYSTEM INFORMATION (BCCH)	Broadcast
2		←	PAGING TYPE1 (PCCH)	Paging (PS domain, P-TMSI)
3		→	RRC CONNECTION REQUEST (CCCH)	RRC
4		←	RRC CONNECTION SETUP (CCCH)	RRC
5		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
6		→	SERVICE REQUEST	GMM
7		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
8		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
9		←	SECURITY MODE COMMAND	RRC (PS domain)
10		→	SECURITY MODE COMPLETE	RRC
11		←	ACTIVATE RB TEST MODE	TC
12		→	ACTIVATE RB TEST MODE COMPLETE	TC
13		←	RADIO BEARER SETUP	RRC (RAB SETUP HSDPA with F-DPCH)
14		→	RADIO BEARER SETUP COMPLETE	RRC
15	<-->			Perform test
16		←	RRC CONNECTION RELEASE	RRC
17		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.3.11.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

7.3.11.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE.

Contents of Attach Accept message: GMM

Information Element	Value/remark
Periodic RA update timer	E0 (timer is deactivated)

7.3.11.4.2 RADIO BEARER SETUP

For step 13, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (HSDPA with F-DPCH)" is used.

The configurations of the fixed reference channels for HSDPA RF tests are described in 3GPP TS 34.121[2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD.

7.3.11.4.3 RRC CONNECTION SETUP

For step 4 , the message in clause 9.2, "Contents of RRC CONNECTION SETUP message: UM" is used with the following exceptions:

Contents of RRC CONNECTION SETUP message: UM

Information Element	Value/remark
- Default DPCH Offset Value	Arbitrary set to value 1536..306176 by step of 2560 (this corresponds to a 0.5 slot timing offset between the DPCCH and the HS-DPCCH)

7.4 Common generic procedures for AS testing

7.4.1 UE RRC Test States for common procedures

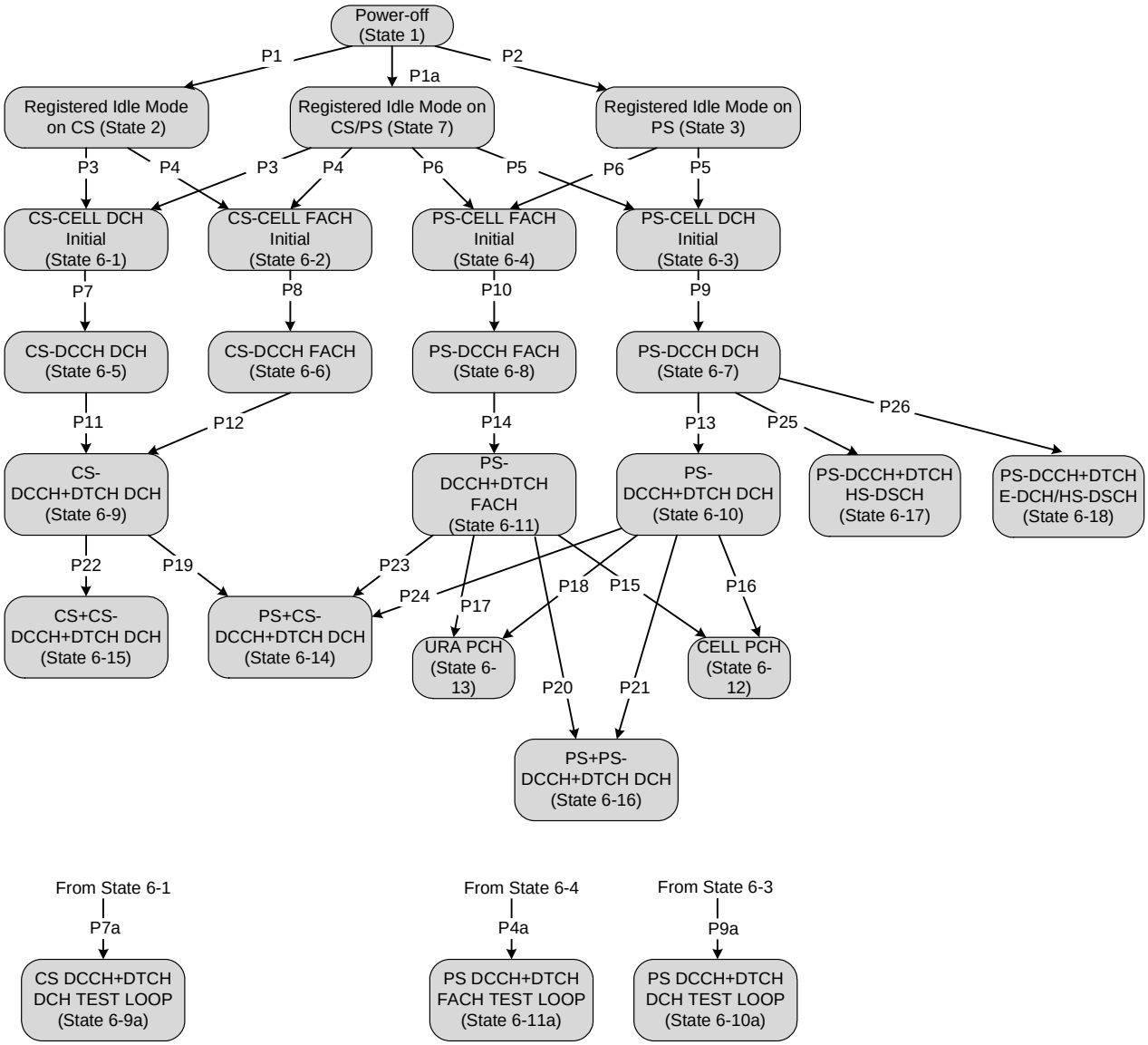


Figure 7.4.1.1: UE RRC test initial states and common procedures

For UE to set up a call in UTRAN, there are a number of procedures to be undertaken in a hierarchical sequence to move between known states. The sequences are shown in figure 7.4.1.1, the operating states for various protocols in the UE are given in table 7.4.1.1.

It is noted that figure 7.4.1.1 should not be construed as a formal state transition diagram, in any manner. The intention here is to define the starting state of UE following the execution of the procedures indicated above.

Table 7.4.1.1: The UE states

		RRC	CC	MM	SM	GMM
State 1	Power OFF	-----	Null	Null	Pdp-Inactive	GMM-null
State 2	Registered Idle Mode on CS	Idle	Null	MM Idle	Pdp-Inactive	GMM-deregistered
State 3	Registered Idle Mode on PS	Idle	Null	Null	Pdp-Inactive	GMM-registered
State 7	Registered Idle Mode on CS/PS	Idle	Null	MM Idle	Pdp-Inactive	GMM-registered
State BGP6-1	CS-CELL_DCH_Initial	Connected	Null	MM Idle	Pdp-Inactive	As previous
State BGP6-2	CS-CELL_FACH_Initial	Connected	Null	MM Idle	Pdp-nactive	As previous
State BGP6-3	PS-CELL_DCH_Initial	Connected	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-4	PS-CELL_FACH_Initial	Connected	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-5	CS-DCCH_DCH	Connected (CELL_DCH)	Null	MM Idle	Pdp-Inactive	As previous
State BGP6-6	CS-DCCH_FACH	Connected (CELL_FACH)	Null	MM Idle	Pdp-Inactive	As previous
State BGP6-7	PS-DCCH_DCH	Connected (CELL_DCH)	Null	As previous	Pdp-Active pending	GMM registered
State BGP6-8	PS-DCCH_FACH	Connected (CELL_FACH)	Null	As previous	Pdp-Active pending	GMM registered
State BGP6-9	CS-DCCH+DTCH_DCH	Connected (CELL_DCH)	Active	MM connection active	Pdp-Inactive	As previous
State BGP6-9a	CS-DCCH+DTCH_DCH_TEST_LOOP	Connected (CELL_DCH)	Null	MM Idle	Pdp-Inactive	As previous
State BGP6-10	PS-DCCH+DTCH_DCH	Connected (CELL_DCH)	Null	As previous	Pdp-Active	GMM registered
State BGP6-10a	PS-DCCH+DTCH_DCH_TEST_LOOP	Connected (CELL_DCH)	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-11	PS-DCCH+DTCH_FACH	Connected (CELL_FACH)	Null	As previous	Pdp-Active	GMM registered
State BGP6-11a	PS-DCCH+DTCH_FACH_TEST_LOOP	Connected (CELL_FACH)	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-12	CELL_PCH	Connected (CELL_PCH)	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-13	URA_PCH	Connected (URA_PCH)	Null	As previous	Pdp-Inactive	GMM registered
State BGP6-14	PS+CS-DCCH+DTCH_DCH	Connected (CELL_DCH)	Active	MM connection active	Pdp-Active	GMM registered
State BGP6-15	CS+CS-DCCH+DTCH_DCH	Connected (CELL_DCH)	Active	MM connection active	Pdp-Inactive	As previous
State BGP6-16	PS+PS-DCCH+DTCH_DCH	Connected (CELL_DCH)	Null	As previous	Pdp-Active	GMM registered
State BGP6-17	PS-DCCH+DTCH_HS-DSCH	Connected (CELL_DCH)	Null	As previous	Pdp-Active	GMM registered
State BGP6-18	PS-DCCH+DTCH_E-DCH/HS-DSCH	Connected (CELL_DCH)	Null	As previous	Pdp-Active	GMM registered

State 1, state 2, state 3, P1, P2 and P1a are described in clause 7.2. States 6-X (for X=1 to 17) are described below.

7.4.2 Generic Setup Procedure for RRC test cases

7.4.2.1 RRC connection establishment procedure for circuit-switched calls (procedure P3 and P4)

7.4.2.1.1 Mobile terminating call

7.4.2.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

7.4.2.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.1.1.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE 1 (PCCH)	RRC
2		→	RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5		→	PAGING RESPONSE	RR

7.4.2.1.1.4 Specific message contents

To execute procedure P3, all specific message contents shall be referred to clause 9.

To execute procedure P4, all specific message contents with the exception of step 3 shall be referred to clause 9. For step 3, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

7.4.2.1.2 Mobile originating calls

7.4.2.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

7.4.2.1.2.2 Definition of system information messages

The default system information messages specified in clause 6.1 are used.

7.4.2.1.2.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		

1	→	RRC CONNECTION REQUEST (CCCH)	RRC
2	←	RRC CONNECTION SETUP (CCCH)	RRC
3	→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
4	→	CM SERVICE REQUEST	MM

7.4.2.1.2.4 Specific message contents

To execute procedure P3, all specific message contents shall be referred to clause 9.

To execute procedure P4, all specific message contents with the exception of step 2 shall be referred to clause 9. For step 2, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

7.4.2.2 RRC connection establishment procedure for packet switched sessions (procedure P5 and P6)

7.4.2.2.1 Mobile terminating session

7.4.2.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

7.4.2.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.2.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE1 (PCCH)	Paging
2		→	RRC CONNECTION REQUEST (CCCH)	RRC
3		←	RRC CONNECTION SETUP (CCCH)	RRC
4		→	RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
5		→	SERVICE REQUEST	GMM

7.4.2.2.1.4 Specific message contents

To execute procedure P5, all specific message contents shall be referred to clause 9.

To execute procedure P6, all specific message contents with the exception of step 3 shall be referred to clause 9. For step 3, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

7.4.2.2.2 Mobile originating sessions

7.4.2.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

7.4.2.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.2.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	→		RRC CONNECTION REQUEST (CCCH)	RRC
2	←		RRC CONNECTION SETUP (CCCH)	RRC
3	→		RRC CONNECTION SETUP COMPLETE (DCCH)	RRC
4	→		SERVICE REQUEST	GMM

7.4.2.2.2.4 Specific message contents

To execute procedure P5, all specific message contents shall be referred to clause 9.

To execute procedure P6, all specific message contents with the exception of step 2 shall be referred to clause 9. For step 2, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

7.4.2.3 NAS call set up procedure for circuit switched calls (procedure P7 and P8)

7.4.2.3.1 Mobile terminating call

7.4.2.3.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1 or state 6-2.
- The Test USIM shall be inserted.

7.4.2.3.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.3.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		AUTHENTICATION REQUEST	MM
2	→		AUTHENTICATION RESPONSE	MM
3	←		SECURITY MODE COMMAND	RRC
4	→		SECURITY MODE COMPLETE	RRC
5	←		SET UP	CC
6	→		CALL CONFIRMED	CC

7.4.2.3.1.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

7.4.2.3.2 Mobile originating calls

7.4.2.3.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1or state 6-2.
- The Test USIM shall be inserted.

7.4.2.3.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.3.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		AUTHENTICATION REQUEST	MM
2	→		AUTHENTICATION RESPONSE	MM
3	←		SECURITY MODE COMMAND	RRC
4	→		SECURITY MODE COMPLETE	RRC
5	→		SET UP	CC
6	←		CALL PROCEEDING	CC

7.4.2.3.2.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

7.4.2.4 NAS session activation procedure for packet switched sessions (procedure P9 and P10)

7.4.2.4.1 Mobile terminating session

7.4.2.4.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

7.4.2.4.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.4.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
2	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
3	←		SECURITY MODE COMMAND	RRC
4	→		SECURITY MODE COMPLETE	RRC

5	←	REQUEST PDP CONTEXT ACTIVATION	SM
6	→	ACTIVATE PDP CONTEXT REQUEST	SM

7.4.2.4.1.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

7.4.2.4.2 Mobile originating sessions

7.4.2.4.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

7.4.2.4.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.4.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
2	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
3	←		SECURITY MODE COMMAND	RRC
4	→		SECURITY MODE COMPLETE	RRC
5	→		ACTIVATE PDP CONTEXT REQUEST	SM

7.4.2.4.2.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

7.4.2.5 Radio access bearer establishment procedure for circuit switched calls (procedure P11 and P12)

7.4.2.5.1 Mobile terminating call

7.4.2.5.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-5 or state 6-6.
- The Test USIM shall be inserted.

7.4.2.5.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.5.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		RADIO BEARER SETUP	RRC RAB SETUP
2	→		RADIO BEARER SETUP COMPLETE	RRC
3	→		ALERTING	CC (This message is optional)
4	→		CONNECT	CC
5	←		CONNECT ACKNOWLEDGE	CC

7.4.2.5.1.4 Specific message contents

To execute procedure P11, use the message titled "CS speech" (defined in clause 9) for the message in step 1. To execute procedure 12, use the message "The others of speech in CS" (defined in clause 9) for the message in step 1.

7.4.2.5.2 Mobile originating calls

7.4.2.5.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-5 or state 6-6.
- The Test USIM shall be inserted.

7.4.2.5.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.5.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		RADIO BEARER SETUP	RRC RAB SETUP
2	→		RADIO BEARER SETUP COMPLETE	RRC
3	←		ALERTING	CC
4	←		CONNECT	CC
5	→		CONNECT ACKOWLEDGE	CC

7.4.2.5.2.4 Specific message contents

To execute procedure P11, use the message titled "CS speech" (defined in clause 9) for the message in step 1. To execute procedure 12, use the message "The others of speech in CS" (defined in clause 9) for the message in step 1.

7.4.2.5a Test loop activation and radio access bearer establishment procedure for circuit switched calls (procedure P7a)

7.4.2.5a.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1.
- The Test USIM shall be inserted.

7.4.2.5a.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.5a.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	AUTHENTICATION REQUEST	MM
2		→	AUTHENTICATION RESPONSE	MM
3		←	SECURITY MODE COMMAND	RRC
4		→	SECURITY MODE COMPLETE	RRC
5		←	ACTIVATE RB TEST MODE (DCCH)	TC
6		→	ACTIVATE RB TEST MODE COMPLETE (DCCH)	TC
1		←	RADIO BEARER SETUP	RRC RAB SETUP
2		→	RADIO BEARER SETUP COMPLETE	RRC
14		←	CLOSE UE TEST LOOP (DCCH)	TC UE test mode 1 RLC SDU size set as specified for the actual test case.
15		→	CLOSE UE TEST LOOP COMPLETE (DCCH)	TC

7.4.2.5a.4 Specific message contents

To execute procedure P7a, use the message titled "CS speech" (defined in clause 9) for the message in step 1.

7.4.2.6 Radio access bearer establishment procedure for packet switched sessions (procedure P13, P14, P25 and P26)

7.4.2.6.1 Mobile terminating session

7.4.2.6.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-7 or state 6-8.
- The Test USIM shall be inserted.

7.4.2.6.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.6.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	RADIO BEARER SETUP	RRC RAB SETUP
2		→	RADIO BEARER SETUP COMPLETE	RRC
3		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.6.1.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P13, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure P14, use the message titled "Packet to CELL_FACH from CELL_FACH in PS". To execute procedure P25, use the message titled "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS". To execute procedure P26, use the RADIO BEARER SETUP message with one of the conditions A12, A13, A14, A15, A16, A19, A20,A21 or A22..

7.4.2.6.2 Mobile originating sessions

7.4.2.6.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-7 or state 6-8.
- The Test USIM shall be inserted.

7.4.2.6.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.6.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	RADIO BEARER SETUP	RRC RAB SETUP
2		→	RADIO BEARER SETUP COMPLETE	RRC
3		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.6.2.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P13, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure P14, use the message titled "Packet to CELL_FACH from CELL_FACH in PS". To execute procedure P25, use the message titled "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS". To execute procedure P26, use the RADIO BEARER SETUP message with one of the conditions A12, A13, A14, A15 or A16.

7.4.2.6a Test loop activation and radio access bearer establishment procedure for packet switched sessions (procedure P4a and P9a)

7.4.2.6a.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

7.4.2.6a.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.6a.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
2		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
3		←	SECURITY MODE COMMAND	RRC
4		→	SECURITY MODE COMPLETE	RRC
5		←	ACTIVATE RB TEST MODE (DCCH)	TC

6	→	ACTIVATE RB TEST MODE COMPLETE (DCCH)	TC
7	←	RADIO BEARER SETUP	RRC RAB SETUP. The 'pdcp info' IE shall be omitted.
8	→	RADIO BEARER SETUP COMPLETE	RRC
14	←	CLOSE UE TEST LOOP (DCCH)	TC UE test mode 1 RLC SDU size set as specified for the actual test case.
15	→	CLOSE UE TEST LOOP COMPLETE (DCCH)	TC

7.4.2.6a.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P9a, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure 4a, use the message titled "Packet to CELL_FACH from CELL_FACH in PS" with the exception that the 'pdcp info' IE shall be omitted.

7.4.2.7 Procedure for transitions to CELL_PCH or URA_PCH state (procedure P15, P16, P17 and P18)

7.4.2.7.1 Transition to CELL_PCH (procedure P15 and P16)

7.4.2.7.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.7.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.7.1.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	←		PHYSICAL CHANNEL RECONFIGURATION	RRC
2	→		PHYSICAL CHANNEL RECONFIGURATION COMPLETE	RRC

7.4.2.7.1.4 Specific message contents

Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1)

Information Element	Value/remark
Message Type	
RRC State Indicator	CELL_PCH

7.4.2.7.2 Transition to URA_PCH (procedure P17 and P18)

7.4.2.7.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.7.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.7.2.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PHYSICAL CHANNEL RECONFIGURATION	RRC
2		→	PHYSICAL CHANNEL RECONFIGURATION COMPLETE	RRC

7.4.2.7.2.4 Specific message contents

Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1)

Information Element	Value/remark
Message Type	
RRC State Indicator	URA_PCH

7.4.2.8 Radio access bearer establishment procedure with packet switched sessions for transitions to Multi Call state (procedure P19, 20 and 21)

7.4.2.8.1 Transition to PS+CS-DCCH+DTCH DCH (procedure P19)

7.4.2.8.1.1 Mobile terminating session

7.4.2.8.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall have registered in CS/PS.
- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

7.4.2.8.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.1.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE2 (DCCH)	Paging
2		→	SERVICE REQUEST	GMM
3		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
4		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
5		←	SECURITY MODE COMMAND	RRC
6		→	SECURITY MODE COMPLETE	RRC
7		←	REQUEST PDP CONTEXT ACTIVATION	SM
8		→	ACTIVATE PDP CONTEXT REQUEST	SM
9		←	RADIO BEARER SETUP	RRC RAB SETUP

Step	Direction		Message	Comments
	UE	SS		
10	→		RADIO BEARER SETUP COMPLETE	RRC
11	←		ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.8.1.1.4 Specific message contents

FFS

7.4.2.8.1.2 Mobile originating sessions

7.4.2.8.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

7.4.2.8.1.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.1.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1	→		SERVICE REQUEST	GMM
2	←		AUTHENTICATION AND CIPHERING REQUEST	GMM
3	→		AUTHENTICATION AND CIPHERING RESPONSE	GMM
4	←		SECURITY MODE COMMAND	RRC
5	→		SECURITY MODE COMPLETE	RRC
6	→		ACTIVATE PDP CONTEXT REQUEST	SM
7	←		RADIO BEARER SETUP	RRC RAB SETUP
8	→		RADIO BEARER SETUP COMPLETE	RRC
9	←		ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.8.1.2.4 Specific message contents

FFS

7.4.2.8.2 Transition to PS+PS-DCCH+DTCH DCH (procedure P20 and P21)

7.4.2.8.2.1 Mobile terminating session

7.4.2.8.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.8.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.2.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE2 (DCCH)	Paging
2		→	SERVICE REQUEST	GMM
3		←	SERVICE ACCEPT	GMM
4		←	REQUEST PDP CONTEXT ACTIVATION	SM
5		→	ACTIVATE PDP CONTEXT REQUEST	SM
6		←	RADIO BEARER SETUP	RRC RAB SETUP
7		→	RADIO BEARER SETUP COMPLETE	RRC
8		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.8.2.1.4 Specific message contents

FFS

7.4.2.8.2.2 Mobile originating sessions

7.4.2.8.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.8.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.2.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		→	SERVICE REQUEST	GMM
2		←	SERVICE ACCEPT	GMM
3		→	ACTIVATE PDP CONTEXT REQUEST	SM
4		←	RADIO BEARER SETUP	RRC RAB SETUP
5		→	RADIO BEARER SETUP COMPLETE	RRC
6		←	ACTIVATE PDP CONTEXT ACCEPT	SM

7.4.2.8.2.2.4 Specific message contents

FFS

7.4.2.9 Radio access bearer establishment procedure with circuit switched calls for transitions to Multi Call state (procedure P22, P23 and P24)

7.4.2.9.1 Transition to CS+CS-DCCH+DTCH DCH (procedure P22)

7.4.2.9.1.1 Mobile terminating call

7.4.2.9.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

7.4.2.9.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.9.1.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE2 (DCCH)	Paging
2		→	PAGING RESPONSE	RR
3		←	SET UP	CC
4		→	CALL CONFIRMED	CC
5		←	RADIO BEARER SETUP	RRC RAB SETUP
6		→	RADIO BEARER SETUP COMPLETE	RRC
7		→	ALERTING	CC (this message is optional)
8		→	CONNECT	CC
9		←	CONNECT ACKNOWLEDGE	CC

7.4.2.9.1.1.4 Specific message contents

FFS

7.4.2.9.1.2 Mobile originating calls

7.4.2.9.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

7.4.2.9.1.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.9.1.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		→	CM SERVICE REQUEST	MM
2		←	CM SERVICE ACCEPT	MM
3		→	SET UP	CC
4		←	CALL PROCEEDING	CC
5		←	RADIO BEARER SETUP	RRC RAB SETUP
6		→	RADIO BEARER SETUP COMPLETE	RRC
7		←	ALERTING	CC
8		←	CONNECT	CC
9		→	CONNECT ACKNOWLEDGE	CC

7.4.2.9.1.2.4 Specific message contents

FFS

7.4.2.9.2 Transition to PS+CS-DCCH+DTCH DCH (procedure P23 and 24)

7.4.2.9.2.1 Mobile terminating call

7.4.2.9.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall have registered in CS/PS.
- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.9.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.9.2.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		←	PAGING TYPE2 (DCCH)	Paging
2		→	PAGING RESPONSE	RR
3		←	AUTHENTICATION REQUEST	MM
4		→	AUTHENTICATION RESPONSE	MM
5		←	SECURITY MODE COMMAND	RRC
6		→	SECURITY MODE COMPLETE	RRC
7		←	SET UP	CC
8		→	CALL CONFIRMED	CC
9		←	RADIO BEARER SETUP	RRC RAB SETUP
10		→	RADIO BEARER SETUP COMPLETE	RRC
11		→	ALERTING	CC (this message is optional)
12		→	CONNECT	CC
13		←	CONNECT ACKNOWLEDGE	CC

7.4.2.9.2.1.4 Specific message contents

FFS

7.4.2.9.2.2 Mobile originating calls

7.4.2.9.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.9.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.9.2.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

Step	Direction		Message	Comments
	UE	SS		
1		→	CM SERVICE REQUEST	MM
2		←	AUTHENTICATION REQUEST	MM
3		→	AUTHENTICATION RESPONSE	MM
4		←	SECURITY MODE COMMAND	RRC
5		→	SECURITY MODE COMPLETE	RRC
6		→	SET UP	CC
7		←	CALL PROCEEDING	CC
8		←	RADIO BEARER SETUP	RRC RAB SETUP
9		→	RADIO BEARER SETUP COMPLETE	RRC
10		←	ALERTING	CC
11		←	CONNECT	CC
12		→	CONNECT ACKOWLEDGE	CC

7.4.2.9.2.2.4 Specific message contents

FFS

7.5 Test procedures for A-GPS Performance requirements testing

This clause specifies the procedures that shall be used for testing of A-GPS Performance requirements in TS 34.171 [41].

7.5.1 Normal UE based A-GPS procedure

The procedure in this clause shall be used for all UE-based A-GPS TTFF test cases in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

7.5.1.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

7.5.1.2 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	RESET UE POSITIONING STORED INFORMATION	TC
2		←	RRC MEASUREMENT CONTROL	RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4, 5 (1))
3		←	RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9 (1), Iono Model)
4		←	RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criterion, GPS Ref time (1), ReferencePosition (1))
5		→	RRC MEASUREMENT REPORT	RRC (Position Estimate), 1st test instance
6		←	RESET UE POSITIONING STORED INFORMATION	TC
7		←	RRC MEASUREMENT CONTROL	RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4,5 (2))
8		←	RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9 (2), Iono Model)
9		←	RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criterion, GPS Ref time (2), ReferencePosition (2))

Step	Direction		Message	Comments
	UE	SS		
10	→		RRC MEASUREMENT REPORT	RRC (Position Estimate), 2nd test instance
11	←		RESET UE POSITIONING STORED INFORMATION	TC
….	……		…….	
n	→		RRC MEASUREMENT REPORT	RRC (Position Estimate), nth test instance

7.5.1.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

Information Element	Value/remark
UE Positioning Technology	AGPS

Contents of MEASUREMENT CONTROL messages: RRC

MEASUREMENT CONTROL (Steps 2 + (n-1)*5):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Setup	
Measurement Reporting Mode - Measurement report transfer mode - Periodical reporting / Event trigger reporting mode	 Acknowledged mode RLC Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type* - UE positioning measurement - UE positioning reporting quantity - Method type - Positioning methods - Response time - Horizontal accuracy - Vertical accuracy - GPS timing of cell wanted - Multiple sets - Additional assistance data request - Environmental characterization - Measurement validity - UE state - CHOICE *Reporting criteria* - No reporting - UE pos OTDOA assistance data for UE-assisted - UE pos OTDOA assistance data for UE-based - UE positioning GPS assistance data - UE positioning GPS navigation model	UE positioning measurement UE based GPS 128 19 (51 m) 48 (102 m) FALSE FALSE FALSE Not present All states Not present Not present Satellites 1-5 as specified in clause 10	 R99 only R99 only
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

MEASUREMENT CONTROL (Steps 3 + (n-1)*5):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Modify	
Measurement Reporting Mode		
- Measurement report transfer mode	Acknowledged mode RLC	
- Periodical reporting / Event trigger reporting mode	Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type*	UE positioning measurement	
- UE positioning measurement		
- UE positioning reporting quantity		
- Method type	UE based	
- Positioning methods	GPS	
- Response time	128	R99 only
- Horizontal accuracy	19 (51 m)	
- Vertical accuracy	48 (102 m)	
- GPS timing of cell wanted	FALSE	
- Multiple sets	FALSE	R99 only
- Additional assistance data request	FALSE	
- Environmental characterization	Not present	
- Measurement validity		
- UE state	All states	
- CHOICE *Reporting criteria*		
- No reporting		
- UE pos OTDOA assistance data for UE-assisted	Not present	
- UE pos OTDOA assistance data for UE-based	Not present	
- UE positioning GPS assistance data		
- UE positioning GPS navigation model	Satellites 6-9 as specified in clause 10	
- UE positioning GPS ionospheric model	As specified in clause 10	
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

MEASUREMENT CONTROL (Steps 4 + (n-1)*5):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Modify	
Measurement Reporting Mode - Measurement report transfer mode - Periodical reporting / Event trigger reporting mode	 Acknowledged mode RLC Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type* - UE positioning measurement - UE positioning reporting quantity - Method type - Positioning methods - Response time - Horizontal accuracy - Vertical accuracy - GPS timing of cell wanted - Multiple sets - Additional assistance data request - Environmental characterization - Measurement validity - UE state - CHOICE *Reporting criteria* - Amount of reporting - Reporting Interval - UE pos OTDOA assistance data for UE-assisted - UE pos OTDOA assistance data for UE-based - UE positioning GPS assistance data - UE positioning GPS reference time - UE positioning GPS reference UE position	UE positioning measurement UE based GPS 128 19 (51 m) 48 (102 m) FALSE FALSE FALSE Not present All states Periodical Reporting Criteria 1 20000 Not present Not present As specified in clause 10 As specified in clause 10	 R99 only R99 only
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

7.5.2 UE based A-GPS procedure for moving scenario and periodic update test case

The procedure in this clause shall be used for the UE-based A-GPS moving scenario and periodic update test case in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

7.5.2.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

7.5.2.2 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	RESET UE POSITIONING STORED INFORMATION	TC
2		←	RRC MEASUREMENT CONTROL	RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4, 5)
3		←	RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9, Iono Model)
4		←	RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criterion, GPS Ref time, ReferencePosition)
5		→	RRC MEASUREMENT REPORT	RRC (Position Estimate)
6		→	RRC MEASUREMENT REPORT	RRC (Position Estimate)
…..		→	……	
n		→	RRC MEASUREMENT REPORT	RRC (Position Estimate)
NOTE: In the actual testing the UE may report error messages at step 5 until it has been able to acquire a position estimate.				

7.5.2.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

The contents of the Reset UE Positioning Stored Information message in Step 1 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1.

Contents of MEASUREMENT CONTROL message: RRC

The contents of the Measurement Control message in steps 2 and 3 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1.

The contents of the Measurement Control message in step 4 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1 with the following exceptions:

Information Element	Value/remark
Amount of reporting Reporting interval	Infinite (see note) 2 000 ms
NOTE: Infinite means during the complete test time.	

7.5.3 Void

7.5.4 Normal UE assisted GPS procedure

The procedure in this clause shall be used for all UE-assisted A-GPS TTFF test cases in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

7.5.4.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

7.5.4.2 Procedure

Step	Direction		Message	Comments
	UE	SS		
1		←	RESET UE POSITIONING STORED INFORMATION	TC
2		←	RRC MEASUREMENT CONTROL	RRC (Setup, Periodical Reporting Criteria, GPS Ref time)
3		→	RRC MEASUREMENT REPORT	RRC (Additional Assistance Data Request)
4		←	RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9)
5		←	RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criteria)
6		→	RRC MEASUREMENT REPORT	RRC (GPS Measured Results), 1st test instance
7		←	RESET UE POSITIONING STORED INFORMATION	TC
8		←	RRC MEASUREMENT CONTROL	RRC (Setup, Periodical Reporting Criteria, GPS Ref time)
9		→	RRC MEASUREMENT REPORT	RRC (Additional Assistance Data Request)
10		←	RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9)
11		←	RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criteria)
12		→	RRC MEASUREMENT REPORT	RRC (GPS Measured Results), 2nd test instance
13		←	RESET UE POSITIONING STORED INFORMATION	TC
….		……	…….	
n		→	RRC MEASUREMENT REPORT	RRC (GPS Measured Results), nth test instance

7.5.4.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

Information Element	Value/remark
UE Positioning Technology	AGPS

Contents of MEASUREMENT CONTROL messages: RRC

MEASUREMENT CONTROL (Steps 2 + (n-1)*6):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Setup	
Measurement Reporting Mode - Measurement report transfer mode - Periodical reporting / Event trigger reporting mode	 Acknowledged mode RLC Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type* - UE positioning measurement - UE positioning reporting quantity - Method type - Positioning methods - Response time - Horizontal accuracy - Vertical accuracy - GPS timing of cell wanted - Multiple sets - Additional assistance data request - Environmental characterization - Measurement validity - UE state - CHOICE *Reporting criteria* - Amount of reporting - Reporting Interval - UE pos OTDOA assistance data for UE-assisted - UE pos OTDOA assistance data for UE-based - UE positioning GPS assistance data - UE positioning GPS reference time	UE positioning measurement UE assisted GPS 128 19 (51 m) 48 (102 m) FALSE FALSE TRUE Not present All states Periodical Reporting Criteria 1 20000 Not present Not present As specified in clause 10	 R99 only R99 only
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

MEASUREMENT REPORT (Steps 3 + (n-1)*6):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measured Results - CHOICE *Measurement* - UE positioning measured results - UE positioning OTDOA measured results - UE positioning position estimate info - UE positioning GPS measured results - UE positioning error - Error reason - GPS additional assistance data request	 Not present Not present Not present Assistance Data Missing Defines assistance data requested by the UE	
Measured Results on RACH	Not present	
Additional Measured Results	Not present	
Event Results	Not present	

MEASUREMENT CONTROL (Steps 4 + (n-1)*6):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Modify	
Measurement Reporting Mode - Measurement report transfer mode - Periodical reporting / Event trigger reporting mode	 Acknowledged mode RLC Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type* - UE positioning measurement - UE positioning reporting quantity - Method type - Positioning methods - Response time - Horizontal accuracy - Vertical accuracy - GPS timing of cell wanted - Multiple sets - Additional assistance data request - Environmental characterization - Measurement validity - UE state - CHOICE *Reporting criteria* - No reporting - UE pos OTDOA assistance data for UE-assisted - UE pos OTDOA assistance data for UE-based - UE positioning GPS assistance data	UE positioning measurement UE assisted GPS 128 19 (51 m) 48 (102 m) FALSE FALSE FALSE Not present All states Not present Not present As specified in clause 10 and requested by the UE in Step 3+(n-1)*6	 R99 only R99 only
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

MEASUREMENT CONTROL (Steps 5 + (n-1)*6):

Information element	Value/remark	Version
Measurement Information Elements		
Measurement Identity	10	
Measurement Command	Modify	
Measurement Reporting Mode - Measurement report transfer mode - Periodical reporting / Event trigger reporting mode	 Acknowledged mode RLC Periodical reporting	
Additional Measurements List	Not present	
CHOICE *Measurement type* - UE positioning measurement - UE positioning reporting quantity - Method type - Positioning methods - Response time - Horizontal accuracy - Vertical accuracy - GPS timing of cell wanted - Multiple sets - Additional assistance data request - Environmental characterization - Measurement validity - UE state - CHOICE *Reporting criteria* - Amount of reporting - Reporting Interval - UE pos OTDOA assistance data for UE-assisted - UE pos OTDOA assistance data for UE-based - UE positioning GPS assistance data	UE positioning measurement UE assisted GPS 128 19 (51 m) 48 (102 m) FALSE FALSE FALSE Not present All states Periodical Reporting Criteria 1 20000 Not present Not present Not present	 R99 only R99 only
Physical Channel Information Elements		
DPCH compressed mode status info	Not present	

7.5.5 UE assisted A-GPS procedure for moving scenario and periodic update test case

The procedure in this clause shall be used for the UE-assisted A-GPS moving scenario and periodic update test case in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

7.5.5.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

7.5.5.2 Procedure

Step	Direction		Message	Comments
	UE	SS		
1	←		RESET UE POSITIONING STORED INFORMATION	TC
2	←		RRC MEASUREMENT CONTROL	RRC (Setup, Periodical Reporting Criteria, GPS Ref time)
3	→		RRC MEASUREMENT REPORT	RRC (Additional Assistance Data Request)
4	←		RRC MEASUREMENT CONTROL	RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9)
5	←		RRC MEASUREMENT CONTROL	RRC (Modify, Periodical Reporting Criteria)
6	→		RRC MEASUREMENT REPORT	RRC (GPS Measured Results), 1st test instance
7	→		RRC MEASUREMENT REPORT	RRC (GPS Measured Results), 2nd test instance
…..	→		……	
n	→		RRC MEASUREMENT REPORT	RRC (GPS Measured Results), nth test instance
NOTE: In the actual testing the UE may report error messages at step 6 until it has been able to acquire GPS measured results.				

7.5.5.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

The contents of the Reset UE Positioning Stored Information message in Step 1 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4.

Contents of MEASUREMENT CONTROL message: RRC

The contents of the Measurement Control message in steps 2 and 4 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4.

The contents of the Measurement Control message in step 5 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4 with the following exceptions:

Information Element	Value/remark
Amount of reporting	Infinite (see note)
Reporting interval	2 000 ms
NOTE: Infinite means during the complete test time.	

7.6 Test procedures for MBMS testing

This clause specifies the procedures that shall be used for MBMS testing.

7.6.1 GMM-REGISTERED with 1 MBMS Service Activated

7.6.1.1 Initial conditions

User Equipment:

For MBMS broadcast test cases:

- The UE is in registered Idle Mode on PS state (state 3) if the UE only supports PS domain or registered Idle Mode on CS/PS (state 7) if the UE supports both CS and PS domain. The UE states are specified in clause 7.4.For MBMS multicast test cases:

  - The UE is in CELL_DCH (6-10 PS-DCCH+DTCH_DCH) or CELL_FACH (6-11 PS-DCCH+DTCH_FACH) state after executing the procedure P13 or P14 defined in clause 7.4.2.6.

For MBMS broadcast and multicast test cases:

- Subsequent to the broadcasting of System Information, MCCH messages are transmitted by the SS using MBMS configuration C1 and Default1 MCCH scheduling (No ongoing service). See subclause 11.1.

- For MBMS the RRC messages used throughout the generic setup procedures make use of specific parameter settings as specified in clause 9.1.

7.6.1.2 Procedure

For MBMS broadcast test cases:

a) The SS request the UE to active the requested MBMS broadcast service.

b) The SS waits until the MBMS broadcast service has been activated in the UE

c) The SS change UE state to CELL_DCH (6-10 PS-DCCH+DTCH_DCH) or CELL_FACH (6-11 PS-DCCH+DTCH_FACH) state by executing the procedure P13 or P14 defined in clause 7.4.2.6 except for the reception of MBMS MODIFICATION REQUEST message after RRC CONNECTION SETUP COMPLETE message during P5 or P6 procedure in case of MBMS Selected Service.

Specific Message Content for MBMS MODIFICATION REQUEST:

MBMS MODIFICATION REQUEST

Information Element	Value/remark
MBMS preferred frequency request	Check that the IE is not present
MBMS RB list requested to be released	Check that the IE is not present
MBMS Selected Service Info	
- CHOICE Status	Some
- MBMS Selected Services Full	
- MBMS Selected Service ID	
- MBMS Service ID	MBMS service ID of the activated MBMS service
- CHOICE PLMN identity	Check to see that one of the below choice element is present
- SameAs-MIB	(no data)
- explicitPLMN_Id	Check to see if it is set to the same value as "PLMN ID" in the Master Information block transmitted for the current serving cell.

For MBMS multicast test cases:

Step	Direction		Message	Comments
	UE	SS		
1				Make UE join a multicast service
2		→	IGMP/MLD JOIN	IGMP/MLD
3		←	REQUEST MBMS CONTEXT ACTIVATION	SM
4		→	ACTIVATE MBMS CONTEXT REQUEST	SM
5		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
6		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
7		←	ACTIVATE MBMS CONTEXT ACCEPT	SM

If required, transition from CELL_FACH (6-11) to CELL_PCH (6-12) or URA_PCH (6-13) state using the procedures P15 or P17 respectively will be performed.

7.6.1.3 Specific message contents

All specific message contents shall be referred to clause 9 with the following exceptions:

PHYSICAL CHANNEL RECONFIGURATION (procedures 6-12 and 6-13)

Information Element	Value/remark
UTRAN DRX cycle length coefficient	7

Attach Accept message: GMM

Information Element	Value/remark
NetworkFeatureSupport	MBMS supported

Service Request message: GMM

Information Element	Value/remark
ServiceType	MBMS Service Reception

REQUEST MBMS CONTEXT ACTIVATION

Information Element	Value/remark
Linked NSAPI	5
Offered Multicast address	Present
Access point name	Present
MBMS protocol configuration options	Not present

ACTIVATE MBMS CONTEXT REQUEST

Information Element	Value/remark
Requested MBMS NSAPI	128
Requested LLC SAPI	Present
Supported MBMS bearer capabilities	Present
Requested Multicast address	Present
Access point name	Present
MBMS protocol configuration options	Not present

ACTIVATE MBMS CONTEXT ACCEPT

Information Element	Value/remark
Temporary Mobile Group Identity	
- MBMS Service Id	Present
- MCC	Present
- MNC	Present
Negotiated LLC SAPI	Present
MBMS protocol configuration options	Not present

IPv4_Datagram

Information Element	Value/remark
Version	0x4
HeaderLength	Present
TypeOfService	Present
TotalLength	Present
Identification	Present
ReservedFlag	Present
MoreFragments	Present
FragmentationOffset	Present
TimeToLive	Present
Protocol	Present
HeaderCheckum	Present
SourceAddress	Present
DestinationAddress	Present
OptionsList	Router alert option with value 0 (0x94040000)
Data	IGMP PDU

IGMP/JOIN (IPv4)

Information Element	Value/remark

Type	0x16 (Version 2 Membership Report)
Max Resp Time	Present
Checksum	Present
Group Address	Multicast IP address

IGMP/Leave (IPv4)

Information Element	Value/remark
Type	0x17 (Version 2 Membership Report)
Max Resp Time	Present
Checksum	Present
Group Address	Multicast IP address

IPv6_Datagram

Information Element	Value/remark
Version	0x6
TrafficClass	Present
FlowLabel	Present
PayloadLength	Present
NextHeader	Present
HopLimit	Present
SourceAddress	Present
DestinationAddress	Present
ExtensionHeaders	Router alert option with value 0 (0x05020000)
Data	MLD PDU

MLD/JOIN (IPv6)

Information Element	Value/remark
Type	0x83
Code	Present
Checksum	Present
MaximumResponseDelay	Present
Reserved	Present
Group Address	Multicast IP address

MLD/LEAVE (IPv6)

Information Element	Value/remark
Type	0x84
Code	Present
Checksum	Present
MaximumResponseDelay	Present
Reserved	Present
Group Address	Multicast IP address

7.6.2 IDLE with 1 MBMS Service Activated

7.6.2.1 Initial conditions

User Equipment:

For MBMS broadcast test cases:

- The UE is in registered Idle Mode on PS state (state 3) if the UE only supports PS domain or registered Idle Mode on CS/PS (state 7) if the UE supports both CS and PS domain. The UE states are specified in clause 7.4.

For MBMS multicast test cases:

- The UE is in CELL_FACH (6-11 PS-DCCH+DTCH_FACH) state after executing the procedure P14 defined in clause 7.4.2.6.

For MBMS broadcast and multicast test cases:

- Subsequent to the broadcasting of System Information, MCCH messages are transmitted by the SS using MBMS configuration C1 (No ongoing service) and Default1 MCCH scheduling. See subclause 11.1.

- For MBMS the RRC messages used throughout the generic setup procedures make use of specific parameter settings as specified in clause 9.1.

7.6.2.2 Procedure

For MBMS broadcast test cases:

a) The SS request the UE to active the requested MBMS broadcast service.

b) The SS waits until the MBMS broadcast service has been activated in the UE

For MBMS multicast test cases:

Step	Direction		Message	Comments
	UE	SS		
1				Make UE join a multicast service
2		→	IGMP/MLD JOIN	IGMP/MLD
3		←	REQUEST MBMS CONTEXT ACTIVATION	SM
4		→	ACTIVATE MBMS CONTEXT REQUEST	SM
5		←	AUTHENTICATION AND CIPHERING REQUEST	GMM
6		→	AUTHENTICATION AND CIPHERING RESPONSE	GMM
7		←	ACTIVATE MBMS CONTEXT ACCEPT	SM
8		←	RRC CONNECTION RELEASE	RRC
9		→	RRC CONNECTION RELEASE COMPLETE	RRC

7.6.2.3 Specific message contents

All specific message contents shall be referred to clause 9 with the following exceptions:

Attach Accept message: GMM

Information Element	Value/remark
NetworkFeatureSupport	MBMS supported

Service Request message: GMM

Information Element	Value/remark
ServiceType	MBMS Service Reception

REQUEST MBMS CONTEXT ACTIVATION

Information Element	Value/remark
Linked NSAPI	5
Offered Multicast address	Present
Access point name	Present
MBMS protocol configuration options	Not present

ACTIVATE MBMS CONTEXT REQUEST

Information Element	Value/remark
Requested MBMS NSAPI	128
Requested LLC SAPI	Present
Supported MBMS bearer capabilities	Present

Requested Multicast address	Present
Access point name	Present
MBMS protocol configuration options	Not present

ACTIVATE MBMS CONTEXT ACCEPT

Information Element	Value/remark
Temporary Mobile Group Identity	
- MBMS Service Id	Present
- MCC	Present
- MNC	Present
Negotiated LLC SAPI	Present
MBMS protocol configuration options	Not present

IPv4_Datagram

Information Element	Value/remark
Version	0x4
HeaderLength	Present
TypeOfService	Present
TotalLength	Present
Identification	Present
ReservedFlag	Present
MoreFragments	Present
FragmentationOffset	Present
TimeToLive	Present
Protocol	Present
HeaderCheckum	Present
SourceAddress	Present
DestinationAddress	Present
OptionsList	Router alert option with value 0 (0x94040000)
Data	IGMP PDU

IGMP/JOIN (IPv4)

Information Element	Value/remark
Type	0x16 (Version 2 Membership Report)
Max Resp Time	Present
Checksum	Present
Group Address	Multicast IP address

IGMP/Leave (IPv4)

Information Element	Value/remark
Type	0x17 (Version 2 Membership Report)
Max Resp Time	Present
Checksum	Present
Group Address	Multicast IP address

IPv6_Datagram

Information Element	Value/remark
Version	0x6
TrafficClass	Present
FlowLabel	Present
PayloadLength	Present
NextHeader	Present
HopLimit	Present
SourceAddress	Present
DestinationAddress	Present
ExtensionHeaders	Router alert option with value 0 (0x05020000)

Data	MLD PDU

MLD/JOIN (IPv6)

Information Element	Value/remark
Type	0x83
Code	Present
Checksum	Present
MaximumResponseDelay	Present
Reserved	Present
Group Address	Multicast IP address

MLD/LEAVE (IPv6)

Information Element	Value/remark
Type	0x84
Code	Present
Checksum	Present
MaximumResponseDelay	Present
Reserved	Present
Group Address	Multicast IP address

7.6.3 MBSFN IDLE (3.84/FDD, 7.68 Mcps TDD)

7.6.3.1 Initial conditions

System Simulator:

- 1 MBMS MBSFN Cell 31 with default parameters.

  In addition to broadcasting System Information, MCCH messages are transmitted by the SS using MBMS configuration C1 and Default1 MCCH scheduling (No ongoing service). See subclause 11.1.

- 1 unicast carrier Cell 1 with default parameters.

User Equipment:

- The UE is in MBSFN Idle mode with one service activated on the MBSFN cell as specified in subclause 7.6.4.

- On the unicast carrier cell the UE is in registered Idle Mode on PS (state 3) if the UE only supports PS domain or registered Idle Mode on CS/PS (state 7) if the UE supports both CS and PS domains. See subclause 7.6.4. The UE states are specified in subclause 7.4.

7.6.3.2 Procedure

a) The SS requests the UE to de-activate the requested MBMS broadcast service.

b) The SS waits until the MBMS broadcast service has been de-activated in the UE

Expected Sequence:

Step	Direction		Carrier	Message	Comment
	UE	SS			
1	←		M	SYSTEM INFORMATION (BCCH)	
2	←		M	MBMS MCCH Message Configuration C1	MBMS configuration C1 and Default1 MCCH scheduling. No session ongoing.
3	SS				SS requests the UE to de-activate the required MBMS broadcast service
4	SS				SS waits until the MBMS broadcast service has been de-activated in the UE.

7.6.3.3 Specific message contents

All message contents shall be as specified in clause 9.1.

7.6.4 MBSFN IDLE with 1 MBMS Service Activated (3.84/FDD, 7.68 Mcps TDD)

7.6.4.1 Initial conditions

System Simulator:

- 1 MBMS MBSFN Cell 31 with default parameters.

  In addition to broadcasting System Information, MCCH messages are transmitted by the SS using MBMS configuration C2 and Default1 MCCH scheduling (No modified services. One ongoing service corresponding to that to be activated at the UE. 124 kbps PS RAB). See subclause 11.1.

- 1 unicast carrier Cell 1 with default parameters.

User Equipment:

- The UE is is switched off.
- The Test-USIM shall be inserted.
- The UE shall be operated under normal test conditions.

7.6.4.2 Procedure

a) The UE shall be switched on and the unicast carrier mobile termination shall be activated.

b) The UE registers on the unicast carrier Cell 1. The UE registers on PS, as specified in clause 7.2.2.2 of TS 34.108 (state 3) if the UE only supports PS domain or registers on CS/PS, as specified in clause 7.2.2.3 of TS 34.108. (state 7) if the UE supports both CS and PS domains.

c) The SS sends ACTIVATE RB TEST MODE on the unicast carrier cell. The UE acknowledges by send ing ACTIVATE RB TEST MODE COMPLETE.

d) The MBMS MBSFN mobile termination shall be activated. (See Note 1)

e) The SS sends CLOSE UE TEST LOOP via the unicast carrier cell, requesting activation of Test Loop Mode 3 specifying the MBSFN MBMS short transmission identity of the MTCH for the activated service (on Cell 31).

f) The SS requests the UE to activate the requested MBMS broadcast service.

g) The UE performs an MBSFN cell search, reads System Information and camps on Cell 31.

h) The UE reads the MCCH messages transmitted by the SS in accordance with Combination C2 and with Default1 MCCH information scheduling. See subclause 11.1. The UE shall continue acquiring the above MBMS messages until it has received a consistent set of MCCH information in the same modification period.

i) The UE shall establish the p-t-m radio bearer for the ongoing activated MBMS service indicated in the MBMS UNMODIFIED SERVICES INFORMATION message according to the configuration defined in the MBMS CURRENT CELL P-T-M INFORMATION (one ongoing session corresponding to the service activated at the UE. The UE closes the test loop and starts counting successfully received RLC PDUs on the MTCH. The UE will send CLOST UE TEST LOOP COMPLETE.

j) The Test Loop is opened and RB Test Mode is deactivated.

Expected Sequence:

Step	Direction		Carrier	Message	Comment
	UE	SS			
1	UE		U		UE switched on and unicast carrier mobile termination is activated.
2			U		UE registers on Unicast carrier Cell 1.
3	←		U	ACTIVATE RB TEST MODE	
4	→		U	ACTIVATE RB TEST MODE COMPLETE	
5	UE				MBMS MBSFN mobile termination is activated. UE starts MBSFN cell search. Note 1.
6	←		U	CLOSE UE TEST LOOP	Loop back mode 3 on MTCH on Cell 31 is requested.
7	SS				SS requests the UE to activate the required MBMS broadcast service
8	←		M	SYSTEM INFORMATION (BCCH)	
9	←		M	MBMS MCCH Message Configuration C2	No modified services. One ongoing service corresponding to that activated at the UE. 124 kbps PS RAB
10	UE		M		The UE shall continue acquiring the above MBMS messages until it has received a consistent set of MCCH information in the same modification period.
11	→		M	CLOSE UE TEST LOOP COMPLETE	The UE shall establish the indicated p-t-m radio bearer and close the test loop.
12	←		U	OPEN UE TEST LOOP	
13	→		U	OPEN UE TEST LOOP COMPLETE	
14	←		U	DEACTIVATE RB TEST MODE	
15	→		U	DEACTIVATE RB TEST MODE COMPLETE	

Note 1: If possible, activation of the MBMS Mobile Termination shall be delayed until registration on the unicast carrier is complete.

7.6.4.3 Specific message contents

All message contents shall be as specified in clause 9.1.

8 Test USIM Parameters

8.1 Introduction

This clause defines default parameters for programming the elementary files of the test USIM. The requirements of this clause do not apply to the USIM/ME tests of 3GPP TS 31.120 [39] and 3GPP TS 31.121 [40].

8.1.1 Definitions

"Test USIM card":

A USIM card supporting the test algorithm for authentication, programmed with the parameters defined in this clause. The electrical, mechanical and environmental requirements of the test USIM card are specified in 3GPP TS 31.101 [22] and 3GPP TS 31.102 [23].

"Test USIM":

Either a test USIM card or the USIM simulator programmed with the parameters defined in this clause.

8.1.2 Definition of the test algorithm for authentication

In order to be able to easily test the UMTS authentication and key agreement procedure as specified in 3GPP TS 33.102 [24] and 3GPP TS 33.105 [26] along the whole system, the availability of a test algorithm for generation of authentication vector based on quintets is needed (in GSM triplets was used). Additionally, calculation of the parameters for re-synchronization requests is needed. The definition of the test algorithm are the functions f1, f2, f3, f4, f5 and the corresponding functions for re-synchronization are f1* and f5*.

For test USIM intended to be used for inter-RAT test cases then the test USIM shall support the conversion function c3 according to 3GPP TS 33.102 [24], clause 6.8.1.2 to derive the GSM ciphering key Kc from the UMTS cipher/integrity keys CK and IK.

The test algorithm defined in the present clause shall be implemented in test USIM cards as well in test USIM simulators and SS. The test algorithm may also, for test purposes, be implemented in AUC.

The following procedure employs bit wise modulo 2 addition ("XOR").

The following convention applies:

All data variables in the specification of this test algorithm are presented with the most significant substring on the left hand side and the least significant substring on the right hand side. A substring may be a bit, byte or other arbitrary length bitstring. Where a variable is broken down into a number of substrings, the leftmost (most significant) substring is numbered 0, the next most significant is numbered 1, and so on through to the least significant.

8.1.2.1 Authentication and key derivation in the test USIM and SS

The following steps describe sequence of operations for the functions f1, f2, f3, f4 and f5 to perform in the test USIM and SS, in order to obtain the XMAC/MAC, RES/XRES, CK, IK, Kc and AK respectively, to be used in the authentication and key agreement procedure.

Step 1:

XOR to the challenge **RAND**, a predefined number **K** (in which at least one bit is not zero, see clause 8.2), having the same bit length (128 bits) as **RAND**.

The result **XDOUT** of this is:

XDOUT[bits 0,1, . . .126,127] = **K** [bits 0,1, . . .126,127] XOR **RAND**[bits 0,1, . . .126,127]

Step 2:

RES (test USIM)**, XRES** (SS), **CK, IK** and **AK** are extracted from **XDOUT** this way:

RES[bits 0,1, . . .n-1,n] = **f2**(**XDOUT**,n) = **XDOUT**[bits 0,1, . . .n-1,n] (with 30 < n < 128)

NOTE: Suggested length for RES is 128 bits (i.e. n = 127).
In SS and AUC, the XRES calculation is identical to RES.

CK[bits 0,1, . . .126,127] = **f3**(**XDOUT**) = **XDOUT**[bits 8,9, . . .126,127,0,1, . . .6,7]

IK[bits 0,1, . . . 126,127] = **f4**(**XDOUT**) = **XDOUT**[bits 16,17, . . .126,127,0,1, . . .14,15]

AK[bits 0,1, . . . 46,47] = **f4**(**XDOUT**) = **XDOUT**[bits 24,25, . . .70,71]

For test USIM intended for inter-RAT testing the GSM ciphering key Kc shall be derived from the UMTS cipher/integrity keys:

Kc[bits 0,1, . . . 62,63] **=** **c3(CK,IK)**, see 3GPP TS 33.102 [24], clause 6.8.1.2.

Step 3:

Concatenate **SQN** with **AMF** to obtain **CDOUT** like this:

CDOUT[bits 0,1,. . .62,63] = **SQN**[bits 0,1,. . .46,47] || **AMF**[bits 0,1,. . .14,15]

NOTE: For test USIM the **SQN** = $\mathbf{SQN_{MS}}$ = $\mathbf{SQN_{SS}}$[bits 0,1,. . .46,47] = **AUTN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTN is the received authentication token.

Step 4:

XMAC (test USIM) and **MAC** (SS) are calculated from **XDOUT** and **CDOUT** this way:

XMAC[bits 0,1, . . .62, 63] = **f1**(**XDOUT, CDOUT**) = **XDOUT**[bits 0,1. . .62,63] XOR **CDOUT**[bits 0,1,. . .62,63]

NOTE: In SS and AUC, the MAC calculation is identical to XMAC.

Step 5:

The SS calculates the authentication token **AUTN**:

AUTN[bits 0,1,..126,127] = **SQN** ⊕ **AK**[bits 0,1,. . .46,47] || **AMF**[bits 0,1,. . .14,15] || **MAC**[bits 0,1, . . .62, 63]

Where **SQN** ⊕ **AK**[bits 0,1,. . .46,47] = **SQN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47]

8.1.2.2 Generation of re-synchronization parameters in the USIM

For SS to be able to initiate an authentication re-synchronization procedure a specific AMF value has been defined.

$AMF_{RESYNCH}$ = AMF[bits 0,1,..14,15] = "1111 1111 1111 1111"

When the test USIM receives an authentication token (AUTN) having the value of AMF field equal to the $AMF_{RESYNCH}$ value then the test USIM shall initiate the re-synchronization procedure.

When the test USIM starts the re-synchronization procedure, the MAC-S and AK have to be calculated using the functions f1* and f5*, which in the test algorithm are identical to f1 and f5, respectively.

Step 1:

XOR to the challenge **RAND**, a predefined number **K** (in which at least one bit is not zero, see 8.2), having the same bit length (128 bits) as **RAND**.

The result **XDOUT** of this is:

XDOUT[bits 0,1, . . .126,127] = **K**[bits 0,1, . . .126,127] XOR **RAND**[bits 0,1, . . .126,127]

Step 2:

AK is extracted from **XDOUT** this way:

AK[bits 0,1, . . . 46,47] = **f5***(**XDOUT**) = **XDOUT**[bits 24,25, . . .70,71]

Step 3:

Concatenate $\mathbf{SQN_{MS}}$ with **AMF*** to obtain **CDOUT** like this:

CDOUT[bits 0,1,. . .62,63] = $\mathbf{SQN_{MS}}$[bits 0,1,. . .46,47] || **AMF***[bits 0,1,. . .14,15]

Where **AMF*** assumes a dummy value of all zeros.

NOTE 1: For test USIM the $\mathbf{SQN_{MS}}$ = $\mathbf{SQN_{SS}}$[bits 0,1,. . .46,47] = **AUTN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTN is the received authentication token.

NOTE 2: For SS and AUC the $\mathbf{SQN_{MS}}$ = **AUTS**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTS is the received re-synchronization parameter.

Step 4:

MAC-S is calculated from **XDOUT** and **CDOUT** this way:

MAC-S[bits 0,1, . . .62, 63] = **f1***(**XDOUT, CDOUT**) = **XDOUT**[bits 0,1. . .62,63] XOR **CDOUT**[bits 0,1,. . .62,63]

NOTE: In SS and AUC, the XMAC-S calculation is identical to MAC-S.

Step 5:

The test USIM calculates the re-synchronization parameter **AUTS**:

AUTS[bits 0,1,..110,111] = $\mathbf{SQN_{MS}} \oplus$ **AK**[bits 0,1,. . .46,47] || **MAC-S**[bits 0,1, . . .62, 63]

Where $\mathbf{SQN_{MS}} \oplus$ **AK**[bits 0,1,. . .46,47] = $\mathbf{SQN_{MS}}$ [bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47]

8.1.2.3 Using the authentication test algorithm for UE conformance testing

8.1.2.3.1 Authentication accept case

The authentication accept case is illustrated in figures 8.1.2.3.1 and 8.1.2.3.2.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value different from the $AMF_{RESYNCH}$ value.

The SS sends an authentication request, including RAND and AUTN parameters, to the ME/USIM.

Based on the received RAND parameter the test USIM calculates the RES, CK, IK, Kc and XMAC parameters according to clause 8.1.2.1 (step 1 to step 4). The test USIM extracts the $SQN_{MS} = SQN_{SS}$, AMF and MAC parameters from the received authentication token AUTN.

The test USIM checks that XMAC = MAC and then return the RES, CK and IK parameters to the ME.

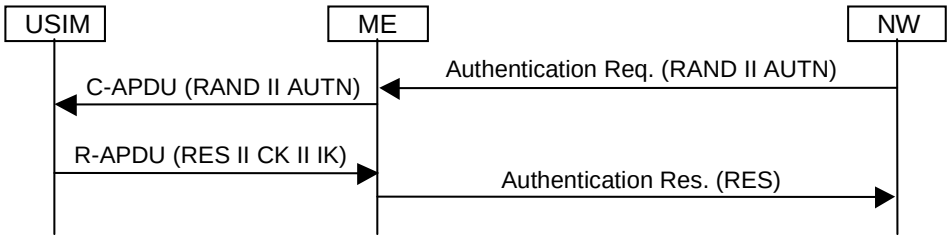


Figure 8.1.2.3.1: Network accepted by UE (USIM not supporting derivation of GSM cipher key Kc)

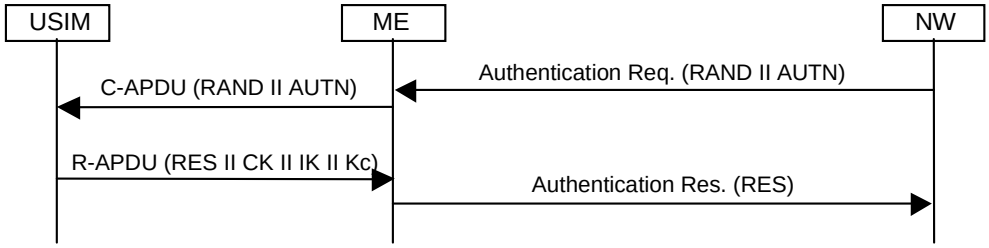


Figure 8.1.2.3.2: Network accepted by UE (USIM supporting derivation of GSM cipher key Kc)

8.1.2.3.2 MAC failure case

The MAC failure case is illustrated in figure 8.1.2.3.2.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value different from the $AMF_{RESYNCH}$ value and a MAC value different from what is calculated in clause 8.1.2.1 step 4.

The SS sends an authentication request, including RAND and AUTN parameters, to the ME/USIM.

Based on the received RAND parameter The test USIM calculates the RES, CK, IK, Kc and XMAC parameters according to clause 8.1.2.1 (step 1 to step 4).

The test USIM extracts the SQN_{MS} = SQN_{SS}, AMF and MAC parameters from the received authentication token AUTN.

When the test USIM identifies that the calculated XMAC value is different from the MAC value received in AUTN then the USIM notifies the ME of the MAC failure and the ME sends an AUTENTICATION FAILURE message to the SS (cause "MAC failure").

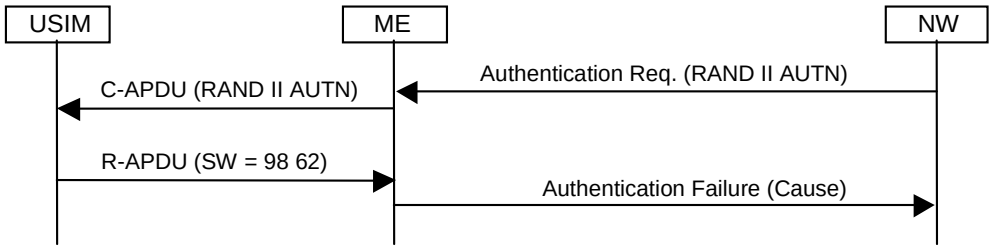


Figure 8.1.2.3.2: MAC failure cases

8.1.2.3.3 SQN failure case

The SQN failure case is illustrated in figure 8.1.2.3.3.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value equal to $AMF_{RESYNCH}$.

The SS sends an authentication request, including RAND and AUTN parameters, to the UE/USIM.

The test USIM extracts the SQN_{MS} = SQN_{SS}, AMF and MAC parameters from the received authentication token AUTN.

When the test USIM identifies that the AMF field is equal to the $AMF_{RESYNCH}$ value it calculates the re-synchronization parameter AUTS as specified in clause 8.1.2.2 (step 1 to step 5) and forward it to the ME.

The ME sends an AUTHENTICATION FAILURE message to the SS including the AUTS parameter.

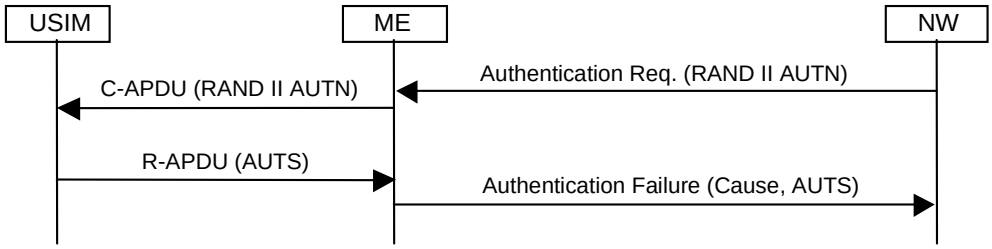


Figure 8.1.2.3.3: SQN failure case

8.1.3 Definition of the test algorithm for VGCS/VBS VSTK generation

In order to be able to easily test the VGCS/VBS key generation and encryption as specified in TS 43.020 [44] and TS 31.102 [23] along the whole system, the availability of a test algorithm for generation of the VSTK-key is needed.

The test algorithm defined in the present clause shall be implemented in test USIM cards as well in test USIM simulators and SS.

The following procedure employs bit wise modulo 2 addition ("XOR").

The following convention applies:

All data variables in the specification of this test algorithm are presented with the most significant substring on the left hand side and the least significant substring on the right hand side. A substring may be a bit, byte or other arbitrary length bitstring. Where a variable is broken down into a number of substrings, the leftmost (most significant) substring is numbered 0, the next most significant is numbered 1, and so on through to the least significant.

8.1.3.1 VSTK generation in the test USIM and SS

The following steps describe the sequence of operations for the function A8_V (TS 43.020 [44]) to be performed in the test USIM and SS, in order to obtain the VSTK, to be used in the subsequent ME/BSS key derivation steps for VGCS/VBS ciphering.

Step 1:

Expand the 36-bit value **VSTK_RAND** to an intermediate 40-bit value **EXPAND**:

FILLER[bits 0,..7] = "11111111"

EXPAND [bits 0,1, . . .39] = **FILLER**[bits 0,..3] || **VSTK_RAND**[bits 0,1, . . .35]

Expand the 40-bit value **EXPAND** to a 128-bit value **EXP_RAND**:

EXP_RAND[bits 0,1, . . .126,127] = **EXPAND**[bits 0,1, . . .39] || **EXPAND**[bits 0,1, . . .39] || **EXPAND**[bits 0,1, . . .39] || **FILLER**[bits 0,..7]

Step 2:

XOR the expanded 128 bit **EXP_RAND** with a stored **V_Ki** i.e. a 128 bit Voice Group or Broadcast Group Key (128 bit) number taken by the USIM from an internal table indexed by VK_Id and Group_Id

The result **VSTK** of this is:

VSTK[bits 0,1, . . .126,127]= **V_Ki** [bits 0,1, . . .126,127] XOR **EXP_RAND**[bits 0,1, . . .126,127]

8.2 Default Parameters for the test USIM

K:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00

Bytes 2 (HEX): 01

Bytes 3 (HEX): 02

Bytes 4 (HEX): 03

Bytes 5 (HEX): 04

Bytes 6 (HEX): 05

Bytes 7 (HEX): 06

Bytes 8 (HEX): 07

Bytes 9 (HEX): 08

Bytes 10 (HEX): 09

Bytes 11 (HEX): 0A

Bytes 12 (HEX): 0B

Bytes 13 (HEX): 0C

Bytes 14 (HEX): 0D

Bytes 15 (HEX): 0E

Bytes 16 (HEX): 0F

PIN Disabling:

The PIN enabled / disabled flag will be set to "PIN Disabled". This ensures that when the Test USIM is inserted into a UE the user will not be prompted for PIN entry.

8.3 Default settings for the Elementary Files (EFs)

The format and coding of elementary files of the USIM are defined in 3GPP TS 31.101 [22] and 3GPP TS 31.102 [23]. The following clauses define the default parameters to be programmed into each elementary file. Some files may be updated by the UE based on information received from the SS. These are identified in the following clauses.

If EFs have an unassigned value, it may not be clear from the main text what this value should be. This clause suggests values in these cases.

8.3.1 Contents of the EFs at the MF level

8.3.1.1 EF_{DIR}

8.3.1.2 EF_{ICCID} (ICC Identity)

The programming of this EF is a test house option.

8.3.1.3 EF_{PL} (Preferred Languages)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.1.4 EF_{ARR} (Access rule reference)

The programming of this EF is a test house option.

8.3.2 Contents of files at the USIM ADF (Application DF) level

8.3.2.1 EF_{LI} (Language Indication)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.2 EF_{IMSI} (IMSI)

The IMSI value will be chosen by the test house. The IMSI used by the SS will align this value.

File size: 9 bytes

Default values: Byte 1 (DEC): 8

Bytes 2 to 9 (HEX): 09 10 10 ** ** ** ** **

49 24 10 ** ** ** ** ** (for Band VI)

"*" indicates any number between 0 and 9 subject to the restriction that IMSI mod 1000 (i.e. bytes 7, 8 and 9) lies in one of the following ranges:

- 063 to 125, 189 to 251, 315 to 377, 441 to 503, 567 to 629, 693 to 755, 819 to 881 or 945 to 999.

NOTE: This ensures that the UE can listen to the second CCCH when more than one basic physical channel is configured for the CCCH. This is necessary for the test of "paging re-organization".

8.3.2.3 EF_{Keys} (Ciphering and Integrity Keys)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.4 EF_{KeysPS} (Ciphering and Integrity Keys for Packet Switched domain)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.5 $EF_{PLMNwAcT}$ (User controlled PLMN selector with Access Technology)

File size:	5n bytes		
Default values (HEX):	Bytes 1 to 3:	32 F4 10	(MCC, MNC) - Translates to 234, 01
	Bytes 4 to 5:	80 00	(Access Technology) - Translates to UTRAN
	Bytes 6 to 8:	32 F4 20	(MCC, MNC)
	Bytes 9 to 10:	80 00	(Access Technology)
	Bytes 11 to 13:	32 F4 30	(MCC, MNC)
	....		
	....		
	....		
	Bytes(5n-4) to (5n-2):	32 F4 43	(MCC, MNC)
	Bytes (5n-1) to 5n:	80 00	(Access Technology)

PLMNs are shown coded above since this is the largest number required for a test. It is necessary to take this into account since the USIM cards must be dimensioned to cope with this number of records.

8.3.2.6 EF_{HPPLMN} (Higher Priority PLMN search period)

File size: 1 byte

Default value (HEX): 00 (No higher priority PLMN search attempts)

8.3.2.7 EF_{ACMmax} (ACM maximum value)

File size:	3 bytes	
Default:	Byte 1:	00
	Byte 2:	00
	Byte 3:	00

The above translates to: "Not valid".

8.3.2.8 EF_{UST} (USIM Service Table)

Services will be allocated and activated as follows.

Services		Activated	Version
Service n°1 :	Local Phone Book	Option	
Service n°2 :	Fixed Dialling Numbers (FDN)	Option	
Service n°3 :	Extension 2	Option	
Service n°4 :	Service Dialling Numbers (SDN)	Option	
Service n°5 :	Extension3	Option	
Service n°6 :	Barred Dialling Numbers (BDN)	Option	
Service n°7 :	Extension4	Option	
Service n°8 :	Outgoing Call Information (OCI and OCT)	Option	
Service n°9 :	Incoming Call Information (ICI and ICT)	Option	
Service n°10:	Short Message Storage (SMS)	Yes	
Service n°11:	Short Message Status Reports (SMSR)	Option	
Service n°12:	Short Message Service Parameters (SMSP)	Yes	
Service n°13:	Advice of Charge (AoC)	Yes	
Service n°14:	Capability Configuration Parameters (CCP)	Yes	

Services		Activated	Version
Service n°15:	Cell Broadcast Message Identifier	Yes	
Service n°16:	Cell Broadcast Message Identifier Ranges	Yes	
Service n°17:	Group Identifier Level 1	Option	
Service n°18:	Group Identifier Level 2	Option	
Service n°19:	Service Provider Name	Option	
Service n°20:	User controlled PLMN selector with Access Technology	Yes	
Service n°21:	MSISDN	Option	
Service n°22:	Image (IMG)	Option	
Service n°23:	Not used (reserved for SoLSA)	No	
Service n°24:	Enhanced Multi-Level Precedence and Pre-emption Service	Option	
Service n°25:	Automatic Answer for Emlpp	Option	
Service n°26:	RFU	No	
Service n°27:	GSM Access	Yes	
Service n°28:	Data download via SMS-PP	Option	
Service n°29:	Data download via SMS-CB	Option	
Service n°30:	Call Control by USIM	Option	
Service n°31:	MO-SMS Control by USIM	Option	
Service n°32:	RUN AT COMMAND command	Option	
Service n°33:	Packet Switched Domain	Yes	
Service n°34:	Enabled Services Table	Yes	
Service n°35:	APN Control List (ACL)	Option	
Service n°36:	Depersonalization Control Keys	Option	
Service n°37:	Co-operative Network List	Option	
Service n°38:	GSM security context	Yes	
Service n°39:	CPBCCH Information	Yes	
Service n°40:	Investigation Scan	Yes	
Service n°41:	MExE	Option	
Service n°42	Operator controlled PLMN selector with Access Technology	Yes	
Service n°43	HPLMN selector with Access Technology	Yes	
Service n°44	Extension 5	Option	
Service n°45	PLMN Network Name	Option	
Service n°46	Operator PLMN List	Option	
Service n°47	Mailbox Dialling Numbers	Option	
Service n°48	Message Waiting Indication Status	Option	
Service n°49	Call Forwarding Indication Status	Option	
Service n°50	Reserved and shall be ignored	Option	
Service n°51	Service Provider Display Information	Option	
Service n°52	Multimedia Messaging Service (MMS)	Option	
Service n°53	Extension 8	Option	
Service n°54	Call control on GPRS by USIM	Option	
Service n°55	MMS User Connectivity Parameters	Option	
Service n°56	Network's indication of alerting in the MS (NIA)	Option	
Service n°57	VGCS Group Identifier List (EF_{VGCS} and EF_{VGCSS})	YES	
Service n°58	VBS Group Identifier List (EF_{VBS} and EF_{VBSS})	YES	
Service n°59	Pseudonym	Option	REL-6 and later
Service n°60	User Controlled PLMN selector for WLAN access	Option	REL-6 and later
Service n°61	Operator Controlled PLMN selector for WLAN access	Option	REL-6 and later
Service n°62	User controlled WSID list	Option	REL-6 and later
Service n°63	Operator controlled WSID list	Option	REL-6 and later
Service n°64	VGCS security	YES	REL-6 and later
Service n°65	VBS security	YES	REL-6 and later
Service n°66	WLAN Reauthentication Identity	Option	REL-6 and later
Service n°67	Multimedia Messages Storage	Option	REL-6 and later
Service n°68	Generic Bootstrapping Architecture (GBA)	Option	REL-6 and later
Service n°69	MBMS security	Option	REL-6 and later
Service n°70	Data download via USSD and USSD application mode	Option	REL-6 and later
Service n°71	Equivalent HPLMN	Option	REL-6 and later
Service n°72	Additional TERMINAL PROFILE after UICC activation	Option	REL-6 and later
Service n°73	Equivalent HPLMN Presentation Indication	Option	REL-6 and later

8.3.2.9 EF_{ACM} (Accumulated Call Meter)

File size: 3 bytes

Default: Byte 1: 00

Byte 2: 00

Byte 3: 00

The above translates to: "Not yet implemented".

8.3.2.10 EF_{GID1} (Group Identifier Level 1)

The programming of this EF is a test house option.

8.3.2.11 EF_{GID2} (Group Identifier Level 2)

The programming of this EF is a test house option.

8.3.2.12 EF_{SPN} (Service Provider Name)

The programming of this EF is a test house option.

8.3.2.13 EF_{PUCT} (Price per Unit and Currency Table)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.14 EF_{CBMI} (Cell Broadcast Message identifier selection)

The programming of this EF is a test house option.

The file size is 2n bytes, where n is the number of Cell broadcast message identifier records - each record defining a type of Cell Broadcast message which may be accessed by the UE. Care should be taken when dimensioning the USIM to take into account the number of Cell Broadcast message identifier records required.

8.3.2.15 EF_{ACC} (Access Control Class)

The EFACC can be selected by a test house in two types.

Type A;

File size: 2 Bytes

Default values (BIN): Byte 1: 000000**

Byte 2: ********

The test house may set any single bit shown by "*" to "1". All remaining bits of byte 2 will be set to "0". This determines the access control class of the USIM.

Type B;

Default values (BIN): Byte 1: 111110**

Byte 2: ********

The test house may set any single bit shown by "*" to "1". All remaining bits of byte 2 will be set to "0". This determines the access control class of the USIM.

Type C;

File size: 2 Bytes

Default values (BIN): Byte 1: 100010**

Byte 2: ********

The test house may set any single bit shown by "*" to "1". This determines the access control class of the USIM.

Type D;

Default values (BIN): Byte 1: 011100**

Byte 2: ********

The test house may set any single bit shown by "*" to "1". This determines the access control class of the USIM.

8.3.2.16 EF_{FPLMN} (Forbidden PLMNs)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.17 EF_{LOCI} (Location Information)

File size: 11 Bytes

Default values: Bytes 1 to 4 (HEX): FF FF FF FF (TMSI)

Bytes 5 to 9 (HEX): 42 F6 18 FF FE (LAI)

Byte 10 (HEX): FF (RFU)

Byte 11 (BIN): 00000001 (Location Update Status = "not updated")

Bytes 5 to 9: LAI-MCC = 246 (bytes 5 to 6) and LAI-MNC = 81 (byte 7) are frequently used. The LAC (bytes 8 to 9) is set to "FF FE" since this, in conjunction with byte 11 setting of "01", is used to ensure that the UE performs a location update at the beginning of a test.

Bytes in this file (e.g. TMSI in bytes 1 to 4) may be updated as a result of a location update attempt by the UE.

8.3.2.18 EF_{AD} (Administrative Data)

File size: 4 bytes

Default values Byte 1: 10000000 - (type approval operations)

Byte 2: 00000000

Byte 3: 00000000

Byte 4: 00000010

8.3.2.19 Void

8.3.2.20 EF_{CBMID} (Cell Broadcast Message Identifier for Data Download)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.21 EF_{ECC} (Emergency Call Codes)

The programming of this EF is a test house option.

8.3.2.22 EF_{CBMIR} (Cell Broadcast Message Identifier Range selection)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.23 EF_{PSLOCI} (Packet Switched location information)

File size: 14 Bytes

Default values: Bytes 1 to 4 (HEX): FF FF FF FF (P-TMSI)

Bytes 5 to 7 (HEX): FF FF FF (P-TMSI signature value)

Bytes 8 to 13 (HEX): 42 F6 18 FF FE FF (RAI)

Byte 14 (BIN): 00000001 (Routing Area update status = "not updated")

Bytes 8 to 13: RAI-MCC = 246 (bytes 8 to 9) and RAI-MNC = 81 (byte 10) are frequently used. The LAC (bytes 11 to 12) is set to "FF FE" since this, in conjunction with byte 14 setting of "01", is used to ensure that the UE performs a location update at the beginning of a test.

Bytes in this file (e.g. P-TMSI in bytes 1 to 4) may be updated as a result of a location update attempt by the UE.

8.3.2.24 EF_{FDN} (Fixed Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.25 EF_{SMS} (Short messages)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.26 EF_{MSISDN} (MSISDN)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.27 EF_{SMSP} (Short message service parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.28 EF_{SMSS} (SMS status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.29 EF_{SDN} (Service Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.30 EF_{EXT2} (Extension2)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.31 EF_{EXT3} (Extension3)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.32 EF_{SMSR} (Short message status reports)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.33 EF_{ICI} (Incoming Call Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.34 EF_{OCI} (Outgoing Call Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.35 EF_{ICT} (Incoming Call Timer)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.36 EF_{OCT} (Outgoing Call Timer)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.37 EF_{EXT5} (Extension5)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.38 EF_{CCP2} (Capability Configuration Parameters 2)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.39 EF_{eMLPP} (enhanced Multi Level Precedence and Pre-emption)

The programming of this EF is a test house option.

8.3.2.40 EF_{AAeM} (Automatic Answer for eMLPP Service)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.41 Void

8.3.2.42 $EF_{Hiddenkey}$ (Key for hidden phone book entries)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.43 Void

8.3.2.44 EF_{BDN} (Barred dialling numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.45 EF_{EXT4} (Extension 4)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.46 EF_{CMI} (Comparison method information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.47 EF_{EST} (Enabled service table)

The programming of this EF is a test house option.

8.3.2.48 EF_{ACL} (Access point name control list)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.49 EF_{DCK} (Depersonalization control keys)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.50 EF_{CNL} (Co-operative network list)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.51 $EF_{START-HFN}$ (Initialisation values for Hyperframe number)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.52 $EF_{THRESHOLD}$ (Maximum value of START)

The programming of this EF is a test house option.

8.3.2.53 $EF_{OPLMNwACT}$ (Operator controlled PLMN selector with Access Technology)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.54 $EF_{HPLMNwAcT}$ (HPLMN selector with Access Technology)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.55 EF_{ARR} (Access rule reference)

The programming of this EF is a test house option.

8.3.2.56 Void

8.3.2.57 EF_{NETPAR} (Network Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.58 EF_{PNN} (PLMN Network Name)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.59 EF_{OPL} (Operator PLMN List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.60 EF_{MBDN} (Mailbox Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.61 EF_{EXT6} (Extension6)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.62 EF_{MBI} (Mailbox Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.63 EF_{MWIS} (Message Waiting Indication Status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.64 EF_{CFIS} (Call Forwarding Indication Status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.65 EF_{EXT7} (Extension7)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.66 EF_{SPDI} (Service Provider Display Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.67 EF_{MMSN} (MMS Notification)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.68 EF_{EXT8} (Extension 8)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.69 EF_{MMSICP} (MMS Issuer Connectivity Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.70 EF_{MMSUP} (MMS User Preferences)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.71 EF_{MMSUCP} (MMS User Connectivity Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.72 EF_{NIA} (Network's Indication of Alerting)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

If service n°57 is "available", this file shall be present.

This EF contains a list of those VGCS group identifiers the user has subscribed to. The elementary file is used by the ME for group call establishment and group call reception.

File size: Bytes 200

Default values:

Bytes	Group ID	Value	BCD encoding in the USIM
1-4	1	12	21 FF FF FF
5-8	2	123	21 F3 FF FF
9-12	3	1234	21 43 FF FF
13-16	4	12348	21 43 F8 FF
17-20	5	123491	21 43 19 FF
21-24	6	1235029	21 53 20 F9
25-28	7	12351	21 53 F1 FF
29-32	8	12352	21 53 F2 FF
33-36	9	12353	21 53 F3 FF
37-40	10	12354	21 53 F4 FF
41-44	11	12355	21 53 F5 FF
45-48	12	12356	21 53 F6 FF
49-52	13	12357	21 53 F7 FF
53-56	14	12358	21 53 F8 FF
57-60	15	12359	21 53 F9 FF
61-64	16	20000	02 00 F0 FF
65-68	17	20001	02 00 F1 FF
69-72	18	20002	02 00 F2 FF
73-76	19	20003	02 00 F3 FF
77-80	20	20004	02 00 F4 FF
81-84	21	20005	02 00 F5 FF
85-88	22	20006	02 00 F6 FF
89-92	23	20007	02 00 F7 FF
93-96	24	20008	02 00 F8 FF
97-100	25	20009	02 00 F9 FF
101-104	26	20010	02 10 F0 FF
105-108	27	66660	66 66 F0 FF
109-112	28	66661	66 66 F1 FF
113-116	29	66662	66 66 F2 FF
117-120	30	666638	66 66 83 FF
121-124	31	66664	66 66 F4 FF
125-128	32	66665	66 66 F5 FF
129-132	33	66666	66 66 F6 FF
133-136	34	66667	66 66 F7 FF
137-140	35	66668	66 66 F8 FF
141-144	36	66669	66 66 F9 FF
145-148	37	66670	66 76 F0 FF
149-152	38	80120	08 21 F0 FF
153-156	39	80121	08 21 F1 FF
157-160	40	80122	08 21 F2 FF
161-164	41	80123	08 21 F3 FF
165-168	42	80124	08 21 F4 FF
169-172	43	80125	08 21 F5 FF
173-176	44	80126	08 21 F6 FF
177-180	45	80127	08 21 F7 FF
181-184	46	80128	08 21 F8 FF
185-188	47	80129	08 21 F9 FF

Bytes	Group ID	Value	BCD encoding in the USIM
189-192	48	80130	08 31 F0 FF
193-196	49	99999	99 99 F9 FF
197-200	50	1111119	11 11 11 F9

For Group Id = 1 V_Ki with VK_Id = 0:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00
Bytes 2 (HEX): 01
Bytes 3 (HEX): 02
Bytes 4 (HEX): 03
Bytes 5 (HEX): 04
Bytes 6 (HEX): 05
Bytes 7 (HEX): 06
Bytes 8 (HEX): 07
Bytes 9 (HEX): 08
Bytes 10 (HEX): 09
Bytes 11 (HEX): 0A
Bytes 12 (HEX): 0B
Bytes 13 (HEX): 0C
Bytes 14 (HEX): 0D
Bytes 15 (HEX): 0E
Bytes 16 (HEX): 0F

Group Id= 1: V_Ki with VK_Id = 1:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 01
Bytes 2 (HEX): 02
Bytes 3 (HEX): 03
Bytes 4 (HEX): 04
Bytes 5 (HEX): 05
Bytes 6 (HEX): 06
Bytes 7 (HEX): 07
Bytes 8 (HEX): 08
Bytes 9 (HEX): 09
Bytes 10 (HEX): 0A
Bytes 11 (HEX): 0B
Bytes 12 (HEX): 0C
Bytes 13 (HEX): 0D
Bytes 14 (HEX): 0E
Bytes 15 (HEX): 0F
Bytes 16 (HEX): 00

8.3.2.74 EF_{VGCSS} (Voice Group Call Service Status)

.If service n°57 is "available", this file shall be present.

This EF contains the status of activation for the VGCS group identifiers. The elementary file is directly related to the EF_{VGCS}. This EF shall always be allocated if EF_{VGCS} is allocated. The following list of group ID are activated: 1, 4, 20, 30, 50.

File size: 7 Bytes

Default value(HEX) : Bytes 1-7: '09 00 08 20 00 00 FE'

8.3.2.75 EF_{VBS} (Voice Broadcast Service)

If service n°58 is "available", this file shall be present.

This EF contains a list of those VBS group identifiers the user has subscribed to. The elementary file is used by the ME for broadcast call establishment and broadcast call reception.

File size: Bytes 200

Default values:

Bytes	Group ID	Value	BCD encoding in the USIM
1-4	1	12	21 FF FF FF
5-8	2	123	21 F3 FF FF
9-12	3	1234	21 43 FF FF
13-16	4	12348	21 43 F8 FF
17-20	5	123491	21 43 19 FF
21-24	6	1235029	21 53 20 F9
25-28	7	12351	21 53 F1 FF
29-32	8	12352	21 53 F2 FF
33-36	9	12353	21 53 F3 FF
37-40	10	12354	21 53 F4 FF
41-44	11	12355	21 53 F5 FF
45-48	12	12356	21 53 F6 FF
49-52	13	12357	21 53 F7 FF
53-56	14	12358	21 53 F8 FF
57-60	15	12359	21 53 F9 FF
61-64	16	20000	02 00 F0 FF
65-68	17	20001	02 00 F1 FF
69-72	18	20002	02 00 F2 FF
73-76	19	20003	02 00 F3 FF
77-80	20	20004	02 00 F4 FF
81-84	21	20005	02 00 F5 FF
85-88	22	20006	02 00 F6 FF
89-92	23	20007	02 00 F7 FF
93-96	24	20008	02 00 F8 FF
97-100	25	20009	02 00 F9 FF
101-104	26	20010	02 10 F0 FF
105-108	27	66660	66 66 F0 FF
109-112	28	66661	66 66 F1 FF
113-116	29	66662	66 66 F2 FF
117-120	30	666638	66 66 83 FF
121-124	31	66664	66 66 F4 FF
125-128	32	66665	66 66 F5 FF
129-132	33	66666	66 66 F6 FF
133-136	34	66667	66 66 F7 FF
137-140	35	66668	66 66 F8 FF
141-144	36	66669	66 66 F9 FF
145-148	37	66670	66 76 F0 FF
149-152	38	80120	08 21 F0 FF
153-156	39	80121	08 21 F1 FF
157-160	40	80122	08 21 F2 FF
161-164	41	80123	08 21 F3 FF

Bytes	Group ID	Value	BCD encoding in the USIM
165-168	42	80124	08 21 F4 FF
169-172	43	80125	08 21 F5 FF
173-176	44	80126	08 21 F6 FF
177-180	45	80127	08 21 F7 FF
181-184	46	80128	08 21 F8 FF
185-188	47	80129	08 21 F9 FF
189-192	48	80130	08 31 F0 FF
193-196	49	99999	99 99 F9 FF
197-200	50	1111119	11 11 11 F9

8.3.2.76 EF_{VBSS} (Voice Broadcast Service Status)

If service n°58 is "available", this file shall be present.

This EF contains the status of activation for the VBS group identifiers. The elementary file is directly related to the EF_{VBS}. This EF shall always be allocated if EF_{VBS} is allocated.

The following list of group ID are activated: 1, 4, 20, 30, 50.

File size: 7 Bytes

Default values (HEX): Bytes 1-7: '09 00 08 20 00 00 FE'

For Group ID= 1 V_Ki with VK_Id = 0:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 0F
Bytes 2 (HEX): 0E
Bytes 3 (HEX): 0D
Bytes 4 (HEX): 0C
Bytes 5 (HEX): 0B
Bytes 6 (HEX): 0A
Bytes 7 (HEX): 09
Bytes 8 (HEX): 08
Bytes 9 (HEX): 07
Bytes 10 (HEX): 06
Bytes 11 (HEX): 05
Bytes 12 (HEX): 04
Bytes 13 (HEX): 03
Bytes 14 (HEX): 02
Bytes 15 (HEX): 01
Bytes 16 (HEX): 00

For Group Id=1 V_Ki with VK_Id = 1:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00
Bytes 2 (HEX): 0F
Bytes 3 (HEX): 0E
Bytes 4 (HEX): 0D
Bytes 5 (HEX): 0C

Bytes 6 (HEX): 0B

Bytes 7 (HEX): 0A

Bytes 8 (HEX): 09

Bytes 9 (HEX): 08

Bytes 10 (HEX): 07

Bytes 11 (HEX): 06

Bytes 12 (HEX): 05

Bytes 13 (HEX): 04

Bytes 14 (HEX): 03

Bytes 15 (HEX): 02

Bytes 16 (HEX): 01

8.3.2.77 EF_{VGCSCA} (Voice Group Call Service Ciphering Algorithm)

If service n°64 is "available", this file shall be present.

This EF contains the ciphering algorithm identifiers for each of the Master Group Key (V_Ki) of each VGCS group that the user has subscribed to (defined in EF_{VGCS}).

File size: 2 Bytes

Default value: Byte 1 = '01' (i.e. A5/1) and Byte 2 = '03' (i.e. A5/3)

8.3.2.78 EF_{VBSCA} (Voice Broadcast Service Ciphering Algorithm)

If service n°65 is "available", this file shall be present.

This EF contains the ciphering algorithm identifiers for each of the Master Group Key (V_Ki) of each VBS group that the user has subscribed to (defined in EF_{VBS}).

File size: 2 Bytes

Default value: Byte 1 = '01' (i.e. A5/1) and Byte 2 = '03' (i.e. A5/3)

8.3.2.79 EF_{GBABP} (GBA Bootstrapping parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.80 EF_{MSK} (MBMS Service Keys List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.81 EF_{MUK} (MBMS User Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.82 Void

8.3.2.83 EF_{GBANL} (GBA NAF List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.84 EF_{EHPLMN} (Equivalent HPLMN)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.85 $EF_{EHPLMNPI}$ (Equivalent HPLMN Presentation Indication)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.2.86 $EF_{LRPLMNSI}$ (Last RPLMN Selection Indication)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

<END OF MODIFIED SECTION>

8.3.3 Contents of DFs at the USIM ADF (Application DF) level

8.3.3.1 Contents of files at the USIM SoLSA level

8.3.3.1.1 EF_{SAI} (SoLSA Access Indicator)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.2 EF_{SLL} (SoLSA LSA List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.3 LSA Descriptor files

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.4 Contents of files at the MExE level

8.3.3.1.4.1 $EF_{MExE-ST}$ (MExE Service table)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.4.2 EF_{ORPK} (Operator Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.4.3 EF_{ARPK} (Administrator Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.4.4 EF_{TPRPK} (Third Party Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.1.4.5 EF_{TKCDF} (Trusted Key/Certificates Data Files)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2 Contents of files at the DF PHONEBOOK level

8.3.3.2.1 EF_{PBR} (Phone Book Reference file)

The programming of this EF is a test house option.

8.3.3.2.2 EF_{IAP} (Index Administration Phone book)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.3 EF_{ADN} (Abbreviated dialling numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.4 EF_{EXT1} (Extension1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.5 EF_{PBC} (Phone Book Control)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.6 EF_{GRP} (Grouping file)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.7 EF_{AAS} (Additional number Alpha String)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.8 EF_{GAS} (Grouping information Alpha String)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.9 EF_{ANR} (Additional Number)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.10 EF_{SNE} (Second Name Entry)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.11 EF_{CCP1} (Capability Configuration Parameters 1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.12 Phone Book Synchronization

8.3.3.2.12.1 EF_{UID} (Unique Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.12.2 EF_{PSC} (Phone book Synchronization Counter)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.12.3 EF_{CC} (Change Counter)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.12.4 EF_{PUID} (Previous Unique Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.2.13 EF_{EMAIL} (e-mail address)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.3 Contents of files at the DF GSM-ACCESS level (Files required for GSM Access)

8.3.3.3.1 EF_{Kc} (GSM Ciphering key Kc)

File size: 9 Bytes

Default values (HEX): Bytes 1 to 8: Align with Kc used by SS

Byte 9: 07

Byte 9 is set to 07 to indicate that there is no key available at the start of a test.

The bytes within this elementary file may be updated by the UE as a result of a successful authentication attempt.

8.3.3.3.2 EF_{KcGPRS} (GPRS Ciphering key KcGPRS)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.3.3 Void

8.3.3.3.4 EF_{CPBCCH} (CPBCCH Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.3.3.5 $EF_{InvScan}$ (Investigation Scan)

The programming of this EF follows default parameter.

8.3.4 Contents of EFs at the TELECOM level

8.3.4.1 EF_{ADN} (Abbreviated dialling numbers)

The programming of this EF is a test house option. It should be noted that sufficient space should be provided on the USIM card for 101 records.

8.3.4.2 EF_{EXT1} (Extension1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.4.3 EF_{ECCP} (Extended Capability Configuration Parameter)

The programming of this EF is a test house option.

8.3.4.4 EF_{SUME} (SetUpMenu Elements)

The programming of this EF is a test house option.

8.3.4.5 EF_{ARR} (Access rule reference)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.5 Contents of DFs at the TELECOM level

8.3.5.1 Contents of files at the $DF_{GRAPHICS}$ level

8.3.5.1.1 EF_{IMG} (Image)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.5.1.2 Image Instance Data Files

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

8.3.5.2 Contents of files at the $DF_{PHONEBOOK}$ under the $DF_{TELECOM}$

The programming of this EF is a test house option.

9 Default Message Contents

9.1 Default Message Contents for Signalling

9.1.1 Default RRC Message Contents (FDD)

This clause contains the default values of common messages, which unless indicated otherwise in specific clauses of 3GPP TS 34.123-1 [1], shall be transmitted and checked by the system simulator.

In this clause, decimal values are normally used. However, sometimes a hexadecimal value, indicated by an "H", or a binary value, indicated by a "B" is used.

The necessary L3 messages are listed in alphabetic order, with the exception of the SYSTEM INFORMATION messages, where it is the information elements which are listed in alphabetic order (this is because some information elements occur in several SYSTEM INFORMATION types).

Default SYSTEM INFORMATION:

NOTE: SYSTEM INFORMATION BLOCK TYPE 1 (except for PLMN type "GSM-MAP"), SYSTEM INFORMATION BLOCK TYPE 8, SYSTEM INFORMATION BLOCK TYPE 9, SYSTEM INFORMATION BLOCK TYPE 10, SYSTEM INFORMATION BLOCK TYPE 14, SYSTEM INFORMATION BLOCK TYPE 15 and SYSTEM INFORMATION BLOCK TYPE 16 messages are not used.

Contents of ACTIVE SET UPDATE message: AM

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects one integer between 0 to 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Activation time	Now	
New U-RNTI	Not Present	
New H-RNTI	Not Present	Rel-6
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
CN information info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX Information	Not Present	Rel-7
HS-SCCH less Information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	Not Present - use default value	
Radio link addition information	Not Present	
Radio link removal information	Not Present	
TX Diversity Mode	None	
SSDT information	Not Present	R99 and Rel-4 only
DPC Mode	[FFS]	Rel-5
Serving HS-DSCH cell information	Not Present	Rel-6
E-DCH reconfiguration information	Not Present	Rel-6
UL 16QAM configuration	Not Present	Rel-7
E-DCH reconfiguration information same serving cell	Not Present	Rel-7
E-TFC Boost Info	Not Present	Rel-7
E-DPDCH power interpolation	Not Present	Rel-7

Contents of ACTIVE SET UPDATE COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the same value used in the corresponding downlink ACTIVE SET UPDATE message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.

Contents of ACTIVE SET UPDATE FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the same value used in the corresponding downlink ACTIVE SET UPDATE message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Refer to test requirement

Contents of CELL UPDATE message: TM

Information Element	Value/remark	Version
Message Type		
U-RNTI	Checked to see if it is set to the following values	
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
RRC transaction identifier	Checked to see if it is absent	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
START List	Checked to see if the 'CN domain identity' and 'START' IEs are present for all CN domains supported by the UE .	
	Checked to see if the 'CN domain identity' and 'START' IEs are present for each CN domain for which RABs are established or is the latest configured CN domain.	Rel-6
- CN domain identity	Checked to see if it is one of the supported CN domains	
- START	This IE is checked to see if it is present. The first/ leftmost bit of the bit string contains the most significant bit of the START.	
AM_RLC error indication (RB2, RB3 or RB4)	Checked to see if it is set to 'FALSE'	
AM_RLC error indication (RB>4)	Checked to see if it is set to 'FALSE'	
Cell update cause	See the specific test case	
Traffic volume indicator	Checked to see if it is absent	Rel-6
Failure cause	Checked to see if it is absent	
RB timer indicator		
- T314 expired	Checked to see if it is set to 'FALSE'	
- T315 expired	Checked to see if it is set to 'FALSE'	
Establishment cause	This IE is checked to see if it is absent	Rel-5
CS Call Type	Not Present	Rel-7
HS-PDSCH in CELL_FACH	Not checked	Rel-7
UE Mobility State Indicator	Not Present	Rel-7
Capability change indicator	Not Present	Rel-7
Reconfiguration Status Indicator	Checked to see if it is absent	Rel-6

Measured results on RACH	Not checked	

Contents of CELL UPDATE CONFIRM message: UM

Information Element	Value/remark	Version
Message Type		
U-RNTI	If this message is sent on CCCH, use the following values. Else, this IE is absent.	
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
RRC transaction identifier	Selects an arbitrary integer between 0 to 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present - use default value	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New DSCH-RNTI	Not Present	R99 and Rel-4 only
New H-RNTI	Not Present	Rel-5
New Primary E-RNTI	Not present	Rel-6
New Secondary E-RNTI	Not present	Rel-6
RRC State indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	Not Present	
RLC re-establish indicator (RB2, RB3 and RB4)	FALSE	
RLC re-establish indicator (RB5 and upwards)	FALSE	
CN information info	Not Present	
URA identity	Not Present	
RNC support for change of UE capability	Not Present	Rel-7
RB information to release list	Not Present	
RB information to reconfigure list	Not Present	
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information common for all transport channels	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE Mode	FDD	
- CPCH set ID	Not Present	R99 and Rel-4 only
- Added or Reconfigured TrCH information for DRAC list	Not Present	R99 and Rel-4 only
DL Transport channel information common for all transport channels	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
Frequency info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX Information	Not Present	Rel-7
HS-SCCH less Information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	Not Present	
CHOICE channel requirement	Not Present	
E-DCH Info	Not Present	Rel-6
CHOICE mode	FDD	R99 and Rel-4 only
- Downlink PDSCH information	Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	Not Present	Rel-5
Downlink information common for all radio	Not Present	

Information Element	Value/remark	Version
links		
Downlink information per radio link list	Not Present	
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of DOWNLINK DIRECT TRANSFER message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.
CN domain identity	CS domain or PS domain
NAS message	See Specific Message Content for each test case

Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects one integer between 0 to 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.
Activation time	now
RAB Info	
- RAB identity	0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.
- CN domain identity	CS domain
- NAS Synchronization Indicator	Not present
- Re-establishment timer	Use T314
Inter-system message	
- CHOICE System type	GSM
- Frequency Band	Set to "GSM/ PCS 1900" if GSM/ PCS 1900 is used in this test. Otherwise set to "GSM/DCS 1800 Band"
- CHOICE GSM message	Single GSM message
- Single GSM message	GSM HANDOVER COMMAND formatted and coded according to GSM specifications as BIT STRING (1..512). The first/ l*eftmost/ most significant* bit of the bit string contains bit 8 of the first octet of the GSM message. The contents of the HANDOVER COMMAND is to be defined in the specific test case.

Contents of HANDOVER FROM UTRAN FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the same value used in the corresponding downlink HANDOVER FROM UTRAN COMMAND -GSM message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Inter-RAT handover failure	
-Inter-RAT handover failure cause	physical channel failure
Inter-system message	Not Checked

Contents of INITIAL DIRECT TRANSFER message: AM

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
PLMN identity	This IE is checked to see if it is absent	Rel-6
CN domain identity	Checked to see if set to supported CN domain as specified in the IXIT statements.	
Intra Domain NAS Node Selector		
- CHOICE version	R99	
- CHOICE CN type	GSM-MAP	
- CHOICE Routing basis	Local (P)TMSI	
- Routing parameter	If the IE "CN domain identity" is equal to "CS domain", this bit string is set to to bits b14 through b23 of the TMSI. If the IE "CN domain identity" is equal to "PS domain", this bit string is set to to bits b14 through b23 of the P-TMSI. The TMSI/P-TMSI consists of 4 octets (32bits). This can be represented by a string of bits numbered from b0 to b31, with bit b0 being the least significant The "Routing parameter" bit string consists of bits b14 through b23 of the TMSI/ PTMSI. The first/ leftmost/ most significant bit of the bit string contains bit b23 of the TMSI/ PTMSI.	
- Entered parameter	Not checked	
NAS message	Set according to that indicated in specific message content for each test case	
START	This IE is checked to see if it is present.	
Establishment cause	This IE is checked to see if it is absent	Rel-5
Measured results on RACH	Not checked	
MBMS joined information	This IE is checked to see if it is absent	Rel-6

Contents of MBMS ACCESS INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Service list	1 entry in the list	Rel-6
- MBMS short transmission ID	Index to the MBMS transmission identity in the previous MBMS MODIFIED SERVICES INFORMATION or MBMS UNMODIFIED SERVICES INFORMATION corresponding to the service for which the current counting procedure applies.	Rel-6
- Access probability factor – Idle	0 (corresponding to the actual probability factor value 1)	Rel-6
- Connected mode counting scope		Rel-6
- URA_PCH	FALSE	Rel-6
- CELL_PCH	FALSE	Rel-6
- CELL_FACH	FALSE	Rel-6

Contents of MBMS GENERAL INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
MBMS preferred frequency information	Not Present	Rel-6
MBMS timers and counters		Rel-6
- T318	4000 ms	Rel-6
MICH configuration information		Rel-6
- MICH Power offset	-5dB	Rel-6
- CHOICE *Mode*	FDD	Rel-6
- Channelisation code	Reference to clause 5.5.1.4 “Downlink physical channels code allocation for MBMS test cases”	Rel-6

Information Element	Value/remark	Version
- Number of NI per frame	18	Rel-6
- STTD indicator	FALSE	Rel-6
Cell group identity	'000000000001' (cells with mid range UARFCN) '000000000010' (cells with low range UARFCN) '000000000011' (cells with high range UARFCN)	Rel-6
Default MSCH configuration information	Not Present	Rel-6
Indicate changes in MBMS Selected Services	Not Present	Rel-6

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
RB information list	2 entries in the list	Rel-6
- RB identity	14	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
- RB identity	15	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	

Information Element	Value/remark	Version
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
TrCh information for each TrCh	2 entries in the list	Rel-6
- Transport channel identity	17	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
- Transport channel identity	23	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
TrCh information for each CCTrCh	2 entries in the list	Rel-6
- CCTrCH identity	1	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	

Information Element	Value/remark	Version
- Power offset information	Not Present	
- CCTrCH identity	2	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	
- Power offset information	Not Present	
PhyCh information	2 entries in list	Rel-6
- PhyCh identity	13	Rel-6
- Secondary CCPCH info MBMS		
- CHOICE *mode*	FDD	
- Secondary scrambling code	Not Present	
- STTD indicator	FALSE	
- Spreading factor	Reference to clause 6.10 "Parameter Set"	
- Code number	Reference to clause 5.5.1.4 “Downlink physical channels code allocation for MBMS test cases”	
- Timing Offset	Set to (Cell No. – 21) * 18 for MBMS Cell Nos. 21-28. (actual value = IE value * 256 chips)	
- PhyCh identity	17	Rel-6
- Secondary CCPCH info MBMS		
- CHOICE *mode*	FDD	
- Secondary scrambling code	Not Present	
- STTD indicator	FALSE	
- Spreading factor	Reference to clause 6.10 "Parameter Set"	
- Code number	Reference to clause 5.5.1.4 “Downlink physical channels code allocation for MBMS test cases”	
- Timing Offset	Set to (Cell No. – 21) * 18 for MBMS Cell Nos. 21-28. (actual value = IE value * 256 chips)	

Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM

Information Element	Condition	Value/remark	Version
Message type	A1, A2, A3		Rel-6
S-CCPCH list	A1	Not Present	Rel-6
S-CCPCH list	A2	Contains 1 S-CCPCH	Rel-6
S-CCPCH list	A3	Contains 2 S-CCPCH	Rel-6
- S-CCPCH identity	A2, A3	1 if combining is used in the test (MBMS NEIGHBOURING CELL P-T-M RB INFORMATION is transmitted in the same modification period). Not Present if combining is not used in the test (MBMS NEIGHBOURING CELL P-T-M RB INFORMATION is not transmitted in the same modification period).	Rel-6
- Secondary CCPCH info		13	Rel-6
- MBMS Soft Combining Timing Offset		Not Present	Rel-6
- TrCh information common for all TrCh		1	Rel-6
- TrCH information list			Rel-6
- TrCh information		17	Rel-6
- RB information list			Rel-6
- RB information		14	Rel-6
- MBMS short transmission ID		Refers to the index of the service in the list of services on the cell which is being provided on this RB	Rel-6
- MBMS logical channel identity		1	Rel-6
- MSCH configuration information		Not Present	Rel-6
- S-CCPCH identity	A3	Not Present	Rel-6
- Secondary CCPCH info		17	Rel-6
- MBMS Soft Combining Timing Offset		Not Present	Rel-6
- TrCh information common for all TrCh		2	Rel-6
- TrCH information list			Rel-6
- TrCh information		23	Rel-6
- RB information list			Rel-6
- RB information		15	Rel-6
- MBMS short transmission ID		Refers to the index of the service in the list of services on the cell which is being provided on this RB	Rel-6
- MBMS logical channel identity		2	Rel-6
- MSCH configuration information		Not Present	Rel-6
S-CCPCH in SIB type 5	A1, A2, A3	Not Present	Rel-6

Condition	Explanation
A1	No services ongoing or starting
A2	1 service ongoing or starting
A3	2 services ongoing or starting

Contents of MBMS MODIFIED SERVICES INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Modified services list	1 entry per modified service - maximum 12. If no services are modified in the current modification period this IE is Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	Set to the value of the service ID being modified (e.g. '000001')	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	'01'	Rel-6
- MBMS required UE action	Acquire PTM RB info	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- Continue MCCH reading	FALSE	Rel-6
MBMS re- acquire MCCH	Not Present	Rel-6
MBMS dynamic persistence level	Not Present	Rel-6
End of modified MCCH information	Not Present	Rel-6
MBMS number of neighbour cells	0	Rel-6
MBMS all unmodified p-t-m services	Not Present	Rel-6
MBMS p-t-m activation time	Not Present	Rel-6

Contents of MBMS NEIGHBOURING CELL P-T-M RB INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Neighbouring cell identity	The intra-frequency cell id of the cell of the MBMS neighbouring cell referred to in the test procedure, obtained from the IE 'Intra-frequency Cell Info list' in SIB 11.	Rel-6
Neighbouring cell's S-CCPCH list	1 entry in the list	Rel-6
- Secondary CCPCH info	Refers to the Physical channel identity being used for the service under test in the common RB info of the current cell	Rel-6
- Secondary CCPCH Power Offset Difference	Not Present	Rel-6
- L1 combining	Not Present	Rel-6
- CHOICE L23 configuration	SameAs Current cell	Rel-6
- Current cell's S-CCPCH	1 (same as the S-CCPCH identity in the MBMS CURRENT CELL P-T-M RB INFORMATION)	Rel-6
- MSCH configuration information	Not Present	Rel-6

Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Unmodified services list	12 services by default. See NOTE 1.	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000001'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000002'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000003'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000004'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000005'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000006'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000007'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000008'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000009'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6

Information Element	Value/remark	Version
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000A'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000B'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000C'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6

Information Element	Condition	Value/remark	Explanation
- MBMS Session ID	A1	Not Present	Condition used when the session is currently not being transmitted
- MBMS required UE action		'None'	
- MBMS Session ID	A2	'01'	Condition used when the session is currently ongoing
- MBMS required UE action		'Acquire PTM RB info'	

NOTE 1: Any service ID which is included in MBMS MODIFIED SERVICES INFORMATION in the current modification period shall be Not Present in the list of services in this message.

Contents of MEASUREMENT CONTROL message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an unused integer between 0 to 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
Measurement Identity	1
Measurement Command	Setup
Measurement Reporting Mode	
- Measurement Report Transfer Mode	Acknowledged mode RLC
- Periodical Reporting/Event Trigger Reporting Mode	Periodical reporting
Additional measurement list	Not Present
CHOICE Measurement type	Intra-frequency measurement
- Intra-frequency measurement	
- Intra-frequency cell info list	
- CHOICE intra-frequency cell removal	Not present
- New intra-frequency cell	
- Intra-frequency cell-id	1
- Cell info	
- Cell individual offset	0 (0dB)
- Reference time difference to cell	Not Present
- Read SFN number	FALSE
- CHOICE mode	FDD

Information Element	Value/remark
- Primary CPICH info	
- Primary scrambling code	Different from the Default setting in clause 6.1 (FDD)
- Primary CPICH Tx power	Not Present
- TX Diversity indicator	FALSE
- Cells for measurement	Not present
- Intra-frequency measurement quantity	Not Present
- Intra-frequency reporting quantity	
- Reporting quantities for active set cells	
- Cell synchronization information reporting indicator	FALSE
- Cell Identity reporting indicator	TRUE
- CPICH Ec/N0 reporting indicator	FALSE
- CPICH RSCP reporting indicator	TRUE
- Pathloss reporting indicator	FALSE
- Reporting quantities for monitored set cells	
- Cell synchronization information reporting indicator	FALSE
- Cell Identity reporting indicator	TRUE
- CPICH Ec/N0 reporting indicator	FALSE
- CPICH RSCP reporting indicator	TRUE
- Pathloss reporting indicator	FALSE
- Reporting quantities for detected set cells	Not Present
- Reporting cell status	
- CHOICE reported cell	Report cell within active set and/or monitored cells on used frequency
- Maximum number of reported cells	2
- Measurement validity	Not Present
- CHOICE report criteria	Periodic reporting criteria
- Amount of reporting	Infinity
- Reporting interval	64 s
DPCH Compressed mode status info	Not Present

Contents of MEASUREMENT CONTROL FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it's set to the identical value for the same IE in the downlink MEASUREMENT CONTROL message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	See the test content

Contents of MEASUREMENT REPORT message: AM

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results		
- Intra-frequency measured results		
- Cell measured results		
- Cell Identity	Not present	
- Cell synchronization information	Checked that this IE is absent	
- Primary CPICH info		
- Primary scrambling code	Different from the Default setting in clause 6.1 (FDD)	

- CPICH Ec/N0 - CPICH RSCP - Pathloss Measured results on RACH Additional measured results Event results	Checked that this IE is absent Checked that this IE is present Checked that this IE is absent Checked that this IE is absent Checked that this IE is absent Checked that this IE is absent	
GSM OTD reference cell	Checked that this IE is absent	Rel-4
Inter-RAT cell info indication	Checked that this IE is absent	Rel-5

Contents of PAGING TYPE 1 message: TM (Speech in CS)

Information Element	Value/remark
Message Type Paging record list - Paging record - CHOICE Used paging identity - Paging cause - CN domain identity - CHOICE UE identity - IMSI (GSM-MAP) BCCH modification info	 CN identity Terminating Conversational Call CS domain Set to the same octet string as in the IMSI stored in the USIM card Not Present

Contents of PAGING TYPE 1 message: TM (The others of speech in CS)

Information Element	Value/remark
Message Type Paging record list - Paging record - CHOICE Used paging identity - Paging cause - CN domain identity - CHOICE UE identity - IMSI (GSM-MAP) BCCH modification info	 CN identity Terminating Streaming Call CS domain Set to the same octet string as in the IMSI stored in the USIM card Not Present

Contents of PAGING TYPE 1 message: TM (Packet in PS)

Information Element	Value/remark
Message Type Paging record list - Paging record - CHOICE Used paging identity - Paging cause - CN domain identity - CHOICE UE identity - P-TMSI BCCH modification info	 CN identity Terminating Interactive Call PS domain Use P-TMSI allocated by SS at initial attach. Not Present

Contents of PAGING TYPE 1 message: TM (SMS in CS)

Information Element	Value/remark
Message Type Paging record list - Paging record - CHOICE Used paging identity - Paging cause - CN domain identity - CHOICE UE identity - IMSI (GSM-MAP) BCCH modification info	 CN identity Terminating Low Priority Signalling CS domain Set to the same octet string as in the IMSI stored in the TEST USIM card Not Present

Contents of PAGING TYPE 1 message: TM (SMS in PS)

Information Element	Value/remark
Message Type	
Paging record list	
- Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Low Priority Signalling
- CN domain identity	PS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the TEST USIM card
BCCH modification info	Not Present

Contents of PAGING TYPE 2 message: AM (Speech in CS)

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
Paging cause	Terminating Conversational Call
CN domain identity	CS domain
Paging record type identifier	Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message.

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6, A7, A8, A9, A10	Not Present	
Delay restriction flag	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	R99 and Rel-4 only
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UE Mobility State Indicator		Not Present	Rel-7
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
RNC support for change of UE capability		Not Present	Rel-7
Reconfiguration in response to requested change of UE capability		Not Present	Rel-7
Downlink counter synchronization info		Not Present	
Frequency info	A1, A2, A3, A4, A5		
- UARFCN uplink (Nu)		Not present Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]	
- UARFCN downlink (Nd)		Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX Information		Not Present	Rel-7
HS-SCCH less Information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
HARQ Info		Not Present	Rel-7
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE *channel requirement*	A1, A2, A3, A4	Uplink DPCH info	
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- HARQ_preamble_mode		0	Rel-6
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Number of TPC bits		Not Present	Rel-7
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
E-DCH Info		Not Present	Rel-6
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	FDD	
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information for each radio links	A1, A2,A3		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value : Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		5	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Code number		0	
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-

Information Element	Condition	Value/remark	Version
			4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio links	A4		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value : Default DPCH Offset Value mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		5	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A5		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A6, A7, A8, A9, A10	Not Present	

Information Element	Condition	Value/remark	Version
MBMS PL Service Restriction Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for "Packet to URA_PCH from CELL_FACH in PS"
A8	This IE need for "Packet to URA_PCH from CELL_DCH in PS"
A9	This IE need for "Packet to CELL_PCH from CELL_FACH in PS"
A10	This IE need for "Packet to CELL_PCH from CELL_DCH in PS"

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked
CHOICE mode	FDD
Deferred measurement control reading	Not present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of RADIO BEARER SETUP message: AM or UM

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A11		
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22		Rel-7
	, A23, A24		Rel-7 Rel-8
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	

Information Element	Condition	Value/remark	Version
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A11	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
Activation time	A4, A5, A6, A7, A8 A10	Not Present	Rel-5
New U-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
New C-RNTI	A1, A2, A3, A4, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	R99 and Rel-4 only
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	Rel-5
New H-RNTI	A9, A10	'1010 1010 1010 1010'	Rel-5
	A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
New Primary E-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11	Not Present	Rel-6
	, A17, A17a, A18, A24		Rel-7
New Primary E-RNTI	A12, A13, A14, A15, A16	'1010 1010 1010 1010'	Rel-6
	A19, A20, A21, A22		Rel-7
	, A23		Rel-7 Rel-8
New Secondary E-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16	Not Present	Rel-6

Information Element	Condition	Value/remark	Version
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
RRC State indicator	A1, A2, A3, A4, A7, A8, A11	CELL_DCH	
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22		Rel-7
	, A23		Rel-7 Rel-8
RRC State indicator	A5, A6, A24	CELL_FACH	Rel-7
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
CN information info		Not Present	
URA identity		Not Present	
RNC support for change of UE capability		Not Present	Rel-7
CHOICE Specification mode		Complete specification	Rel-6
- Signalling RB information to setup		Not Present	
- RAB information for setup	A1, A7		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel		DCH	

Information Element	Condition	Value/remark	Version
type			
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RAB information for setup	A2, A8		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		11	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	

Information Element	Condition	Value/remark	Version
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		7	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		12	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		8	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RAB information for setup	A3, A4, A5, A6		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	

Information Element	Condition	Value/remark	Version
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	Rel-5
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A9		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B	

Information Element	Condition	Value/remark	Version
		The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow		Not Present	

Information Element	Condition	Value/remark	Version
identity			
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A10		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	

Information Element	Condition	Value/remark	Version
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A11		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression		Not present	

Information Element	Condition	Value/remark	Version
information			
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	Rel-5
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		4	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		9	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	

Information Element	Condition	Value/remark	Version
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A12		Rel-6
	A19		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			

Information Element	Condition	Value/remark	Version
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- RAB information for setup	A13, A14, A15, A16		Rel-6
- RAB info - RAB identity		(high-speed AM DTCH for PS domain) 0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	

Information Element	Condition	Value/remark	Version
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	

Information Element	Condition	Value/remark	Version
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A15		Rel-6
- RAB info		(second high-speed AM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		17	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		8	
- E-DCH MAC-d flow identity		3	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		6	

Information Element	Condition	Value/remark	Version
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		2	
- Logical channel identity		Not Present	
- RAB information for setup	A16		Rel-6
- RAB info		(Conversational UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		27	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		32	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		9	
- E-DCH MAC-d flow identity		4	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		7	
- RLC PDU size list		12 RLC PDU sizes	
- RLC PDU size		96 bits	
- RLC PDU size		112 bits	
- RLC PDU size		144 bits	
- RLC PDU size		160 bits	
- RLC PDU size		176 bits	

Information Element	Condition	Value/remark	Version
- RLC PDU size		192 bits	
- RLC PDU size		208 bits	
- RLC PDU size		224 bits	
- RLC PDU size		288 bits	
- RLC PDU size		296 bits	
- RLC PDU size		312 bits	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		3	
- Logical channel identity		Not Present	
- RAB information for setup	A17, A17a		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		This IE is present and set to “7” if Downlink RLC PDU Size is set to “Flexible”	

Information Element	Condition	Value/remark	Version
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		0	
- Logical channel identity		7	
- RAB information for setup	A18		Rel-7
- RAB info		(high-speed UM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		Not present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		TRUE	
- Use special value of HE field		Not present	
- RB mapping info			

Information Element	Condition	Value/remark	Version
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		0	
- Logical channel identity		7	
- RAB information for setup	,A20, A21		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	

Information Element	Condition	Value/remark	Version
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-hs	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A21		Rel-7
- RAB info		(Conversational UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		27	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		32	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	

Information Element	Condition	Value/remark	Version
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		9	
- E-DCH MAC-d flow identity		4	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		7	
- RLC PDU size list		12 RLC PDU sizes	
- RLC PDU size		96 bits	
- RLC PDU size		112 bits	
- RLC PDU size		144 bits	
- RLC PDU size		160 bits	
- RLC PDU size		176 bits	
- RLC PDU size		192 bits	
- RLC PDU size		208 bits	
- RLC PDU size		224 bits	
- RLC PDU size		288 bits	
- RLC PDU size		296 bits	
- RLC PDU size		312 bits	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-hs	
- DL HS-DSCH MAC-d flow identity		3	
- Logical channel identity		Not Present	
- RAB information for setup	A22		Rel-7
- RAB info		(second high-speed UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		27	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	

Information Element	Condition	Value/remark	Version
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		15	
- DL Reception Window Size		Not present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		TRUE	
- Use special value of HE field		Not present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		9	
- E-DCH MAC-d flow identity		4	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		6	
- RLC PDU size list		See clause 6.10	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		3	
- Logical channel identity		9	
- RAB information for setup	A22		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			

Information Element	Condition	Value/remark	Version
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		This IE is present and set to “7” if Downlink RLC PDU Size is set to “Flexible”	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		6	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		0	
- Logical channel identity		7	
- RAB information for setup	A23		Rel-7 Rel-8
- RAB info		(high-speed UM DTCH for CS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	

Information Element	Condition	Value/remark	Version
- CN domain identity		CS domain	
- NAS Synchronization Indicator		'1010' if WB-AMR is tested, otherwise '0110'	
- Re-establishment timer		useT314	
- CS-HSPA information			
- UL AMR rate		Not Present	
- Max CS delay		160	
- RB information to setup			
- RB identity		26	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		present	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		Not present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		Not present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- CHOICE RLC PDU size	MAC-I-FIXED	Fixed size	Rel-8
- DDI		6	
- RLC PDU size list		Reference to clause 6.10 Parameter Set	
- CHOICE RLC PDU size	MAC-I-FLEX	Flexible size	Rel-8
- Length indicator size		Not present	
- Minimum UL RLC PDU size		See clause 6.10	
- Largest UL RLC PDU size		See clause 6.10	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		0	
- Logical channel identity		7	
- RAB information for setup	A24		
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	

Information Element	Condition	Value/remark	Version
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		0	

Information Element	Condition	Value/remark	Version
- Logical channel identity		Not Present	
RB information to reconfigure list	A1, A2, A3, A4, A5, A6, A7, A8 A9, A10 A12, A13, A14, A15, A16 , A17, A17a, A18, A19, A20, A21, A22	Not Present	 Rel-5 Rel-6 Rel-7
	, A23		Rel-7 Rel-8
RB information to be affected	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10 A12 , A17, A17a, A18, A19, A20, A21, A24	Not Present	 Rel-5 Rel-6 Rel-7
	, A23		Rel-7 Rel-8
RB information to be affected - RB identity - RB mapping info - Information for each multiplexing option - RLC logical channel mapping indicator - Number of uplink RLC logical channels - Uplink transport channel type - Logical channel identity - E-DCH MAC-d flow identity - CHOICE RLC PDU size - DDI - RLC PDU size list - RLC PDU size - Include in scheduling info - MAC logical channel priority - Downlink RLC logical channel info - Number of RLC logical channels - Downlink transport channel type - DL DCH Transport channel identity - DL DSCH Transport channel identity - Logical channel identity - RB identity - RB mapping info - Information for each multiplexing option - RLC logical channel mapping indicator - Number of uplink RLC logical channels - Uplink transport channel type - Logical channel identity - E-DCH MAC-d flow identity - CHOICE RLC PDU size - DDI - RLC PDU size list - RLC PDU size - Include in scheduling info - MAC logical channel priority - Downlink RLC logical channel info	A13, A15	 1 (UM DCCH for RRC) 1 RBMuxOption Not Present 1 E-DCH 1 1 Fixed size 1 1 RLC PDU size 144 bits FALSE 1 1 DCH 10 Not Present 1 2 (AM DCCH for RRC) 1 RBMuxOption Not Present 1 E-DCH 2 1 Fixed size 2 1 RLC PDU size 144 bits FALSE 2	Rel-6 Rel-8

Information Element	Condition	Value/remark	Version
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
RB information to be affected	A14, A16		Rel-6
- RB identity		1 (UM DCCH for RRC)	

Information Element	Condition	Value/remark	Version
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	

Information Element	Condition	Value/remark	Version
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
RB information to be affected	A22		Rel-7
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		1	
- RLC PDU size list		1 RLC PDU size	

Information Element	Condition	Value/remark	Version
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			

Information Element	Condition	Value/remark	Version
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		4	
Downlink counter synchronization info	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
PDCP ROHC target mode	A9, A10	Not Present	Rel-5
	, A12, A13, A14, A15, A16		Rel-6
	, A17, A17a, A18, A19, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
UL Transport channel information for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8, A11		
	, A9, A10		Rel-5
	, A17, A17a, A18		Rel-7
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	

Information Element	Condition	Value/remark	Version
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set	
- CTFC		Reference to clause 6.10.2.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Gain factor βc		11 (below 64 kbps) 9 (equal or higher than 64 kbps) when HSDPA is not configured 9 (equal or higher than 64 kbps and below 384 kbps) when HSDPA is also configured 6 (equal or higher than 384 kbps) when HSDPA is also configured (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Gain factor βd		15 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P p-m		Not Present	
UL Transport channel information for all transport channels	A12		Rel-6
	A19		Rel-7
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((UL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors	
- CTFC		1 ((UL DCH RAB, DCCH)=(TF0, TF1))	
- Power offset information			
- CHOICE Gain Factors		Signalled Gain Factors	
- Gain factor βc		11 (below 64 kbps) 9 (equal or higher than 64 kbps) when HSDPA is not configured 9 (equal or higher than 64 kbps and below 384 kbps) when HSDPA is also configured 6 (equal or higher than 384 kbps) when HSDPA is also configured (Not Present if the CHOICE Gain Factors is set to	

Information Element	Condition	Value/remark	Version
		Computed Gain Factors)	
- Gain factor βd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P p-m		Not Present	
UL Transport channel information for all transport channels	A13, A14, A15, A16	Not Present	Rel-6
	, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
Deleted UL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10 , A12 , A17, A17a, A18, A19, A20, A21, A24	Not Present	 Rel-5 Rel-6 Rel-7
	, A23		Rel-7 Rel-8
Deleted UL TrCH information	A13, A14, A15, A16		Rel-6
	, A22		Rel-7
- Uplink transport channel type - UL transport channel identity Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information	DCH 5 A1, A3 A4, A5, A6, A7 , A9, A10 , A17, A17a, A18	 1 DCH added, 1 DCH reconfigured (if from cell_DCH) OR 2 DCHs added (if from cell_FACH) DCH 1 Dedicated transport channels	 Rel-5 Rel-7
- RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS	

Information Element	Condition	Value/remark	Version
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A12	1 E-DCH added, 1 DCH added, 1 DCH reconfigured	Rel-6
	A19		Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- UL MAC header type		Not present	Rel-8
- UL MAC header type	MAC-I-FIXED, MAC-I-FLEX	MAC-i/is	Rel-8
- E-DCH Transmission Time Interval		set to 2ms if supported by the UE E-DCH category, or 10ms if the UE E-DCH category does not support 2ms TTI	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow			
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	

Information Element	Condition	Value/remark	Version
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A13, A14	1 E-DCH added with one DCCH MAC-d flow and one DTCH MAC-d flow	Rel-6
	, A20		Rel-7
	, A23		Rel-7 Rel-8
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- UL MAC header type		Not present	Rel-8
- UL MAC header type	MAC-I-FIXED, MAC-I-FLEX	MAC-i/is	Rel-8
- E-DCH Transmission Time Interval		set to 2ms if supported by the UE E-DCH category, or 10ms if the UE E-DCH category does not support 2ms TTI	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- Max MAC-e PDU contents size	MAC-I-FIXED, MAC-I-FLEX	168 bits	Rel-8
- Max MAC-e PDU contents size		162 bits	
- 2 ms non-scheduled transmission grant HARQ process allocation		Not Present	
- Added or reconfigured E-DCH MAC-d flow		(for DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A15	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-6
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	

Information Element	Condition	Value/remark	Version
- UL MAC header type		Not present	Rel-8
- UL MAC header type	MAC-I-FIXED, MAC-I-FLEX	MAC-i/is	Rel-8
- E-DCH Transmission Time Interval		set to 2ms if supported by the UE E-DCH category, or 10ms if the UE E-DCH category does not support 2ms TTI	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- Max MAC-e PDU contents size		162 bits	
- Max MAC-e PDU contents size	MAC-I-FIXED, MAC-I-FLEX	168 bits	Rel-8
- 2 ms non-scheduled transmission grant HARQ process allocation		Not Present	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		3	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A16	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-6
	, A21, A22		Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- UL MAC header type		Not present	Rel-8
- UL MAC header type	MAC-I-FIXED, MAC-I-FLEX	MAC-i/is	Rel-8
- E-DCH Transmission Time Interval		set to 2ms if supported by the UE E-DCH category, or 10ms if the UE E-DCH category does not support 2ms TTI	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	

Information Element	Condition	Value/remark	Version
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- Max MAC-e PDU contents size		162 bits	
- Max MAC-e PDU contents size	MAC-I-FIXED, MAC-I-FLEX	168 bits	Rel-8
- 2 ms non-scheduled transmission grant HARQ process allocation		Not Present	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		4	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured TrCH information list	A24	Not Present	Rel-7
CHOICE *mode*	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10 , A12, A13 , A17, A18, A17a, A19, A22	Not Present	 Rel-5 Rel-6 Rel-7
DL Transport channel information common for all transport channel	A1, A2, A7, A8		
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		SameasUL	
DL Transport channel information common for all transport channel	A3, A4, A5, A6, A11 A10 A12, A13, A15 , A17, A18, A17a, A19		 Rel-5 Rel-6 Rel-7
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Explicit	
- DL DCH TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and	

Information Element	Condition	Value/remark	Version
- CTFC		reference to clause 6.10.2.4 Reference to clause 6.10.2.4 Parameter Set	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel - SCCPCH TFCS - CHOICE mode - CHOICE DL parameters - DL DCH TFCS - CHOICE TFCI Signalling - TFCI Field 1 Information - CHOICE TFCS representation - TFCS complete reconfigure - CHOICE CTFC Size - CTFC information - CTFC - Power offset information - CTFC - Power offset information	A9	 Not Present FDD Explicit Normal Complete reconfiguration ctfc2bit 0 ((DL DCH RAB, DCCH)=(TF0, TF0)) Not Present 1 ((DL DCH RAB, DCCH)=(TF0, TF1)) Not Present	Rel-5
DL Transport channel information common for all transport channel	A14, A16	Not Present	Rel-6
	, A20, A21, A22, A24		Rel-7
	, A23		Rel-7 Rel-8
Deleted DL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10 , A12, A13 , A17, A18, A17a, A19, A20, A21, A24	Not Present	 Rel-5 Rel-6 Rel-7
Deleted DL TrCH information	A14, A16		Rel-6
	A22		Rel-7
- Downlink transport channel type - DL Transport channel identity		DCH 10	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value	A1	1 DCH added, 1 DCH reconfigured DCH 6 Same as UL DCH 1 -20 (-2.0) DCH 10 Same as UL DCH 5 -20 (-2.0)	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters	A3, A4, A5, A6, A7	2 TrCHs(DCH for DCCH and DCH for DTCH) DCH 10 Same as UL DCH 5 -20 (-2.0) DCH 6 Explicit Except for RAB with the symmetric DL and UL rate: Same as UL	

Information Element	Condition	Value/remark	Version
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set only including TF0	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		7	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			

Information Element	Condition	Value/remark	Version
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		8	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
Added or Reconfigured DL TrCH information	A9	3 TrCHs (DCH for DCCH and DCH plus HS-DSCH for DTCH)	Rel-5
	A12		Rel-6
	A19		Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			

Information Element	Condition	Value/remark	Version
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A10	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-5
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A11	1 DCH for DTCH	
- Downlink transport channel type		DCH	
- DL Transport channel identity		9	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A13	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-6
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A14	1 TrCH (HS-DSCH for DTCH and DCCH)	Rel-6
	, A20		Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queues)	
- MAC-hs queue Id		0 (for DTCH)	
- MAC-d Flow Identity		0	
- T1		50	

Information Element	Condition	Value/remark	Version
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A15	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-6
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queues)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		2 (for second DTCH)	
- MAC-d Flow Identity		2	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A16	1 TrCH (HS-DSCH for 2 DTCHs and DCCH)	Rel-6
	, A21		Rel-7

Information Element	Condition	Value/remark	Version
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(three queues)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue Id		3 (for second DTCH)	
- MAC-d Flow Identity		3	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		112	
- MAC-d PDU size index		0	
- MAC-d PDU size		144	
- MAC-d PDU size index		1	
- MAC-d PDU size		160	
- MAC-d PDU size index		2	
- MAC-d PDU size		176	
- MAC-d PDU size index		3	
- MAC-d PDU size		192	
- MAC-d PDU size index		4	
- MAC-d PDU size		224	
- MAC-d PDU size index		5	
- MAC-d PDU size		296	
- MAC-d PDU size index		6	
- MAC-d PDU size		344	
- MAC-d PDU size index		7	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A17, A17a, A18	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- CHOICE *DL MAC header type*		MAC-ehs	

Information Element	Condition	Value/remark	Version
- Added or reconfigured MAC-ehs reordering queue			
- MAC-ehs queue to add or reconfigure list		(one queue)	
- MAC-ehs queue Id		0	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A22	1 TrCH (HS-DSCH for 2 DTCHs and DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- CHOICE *DL MAC header type*		MAC-ehs	
- Added or reconfigured MAC-ehs reordering queue			
- MAC-ehs queue to add or reconfigure list		(three queues)	
- MAC-ehs queue Id		0 (for first DTCH)	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue Id		1 (for DCCH)	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue Id		3 (for second DTCH)	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A23		Rel-7 Rel-8
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- CHOICE *DL MAC header type*		MAC-ehs	
- Added or reconfigured MAC-ehs reordering queue			
- MAC-ehs queue to add or reconfigure list		(one queue)	
- MAC-ehs queue Id		0	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured TrCH information list	A24	Not present	
Frequency info	A1, A2, A3, A4, A5, A7, A8, 11 , A9, A10 , A12, A13, A14, A15, A16 , A17, A17a, A18, A19, A20, A21, A22, A24		 Rel-5 Rel-6 Rel-7

Information Element	Condition	Value/remark	Version
	, A23		Rel-7 Rel-8
- UARFCN uplink (Nu) - UARFCN downlink (Nd) Frequency info	 A6	Reference to clause 5.1 Test frequencies. This IE should be present, if the default duplex distance defined for the operating frequency band is not used and frequency is different from the current frequency, otherwise set to Not Present. Reference to clause 5.1 Test frequencies if frequency is different from the current frequency otherwise set to Not Present. Not Present	
DTX-DRX timing information	A20, A21 , A23		Rel-7 Rel-7 Rel-8
CHOICE *timing* - New timing - Enabling Delay - UE DTX DRX Offset DTX-DRX Information - CHOICE *E-DCH TTI length* - UE DTX cycle 1 - UE DTX cycle 2 - MAC DTX cycle - Inactivity Threshold for UE DTX cycle 2 - UE DTX long preamble length - MAC Inactivity Threshold - CQI DTX Timer - UE DPCCH burst_1 - UE DPCCH burst_2 DRX Information - UE DRX cycle - Inactivity Threshold for UE DRX cycle - Inactivity Threshold for UE Grant Monitoring - UE DRX Grant Monitoring Uplink DPCCH slot format information HS-SCCH less information		 0 1 if 2ms TTI selected, otherwise 0 Unless stated otherwise, this should be set to 2ms if the UE supports 2ms TTI, or 10ms if the UE does not support 2ms TTI. 8 if 2ms TTI selected, otherwise 10 16 if 2ms TTI selected, otherwise 20 8 if 2ms TTI selected, otherwise 10 32 if 2ms TTI selected, otherwise 8 4 1 if 2ms TTI selected, otherwise 8 32 1 1 8 if 2ms TTI selected, otherwise 10 32 32 if 2ms TTI selected, otherwise 8 TRUE 1 Not Present	
Maximum allowed UL TX power	A1, A2, A3, A4, A7, A8, A11 , A9, A10 , A12, A13, A14, A15, A16 , A17, A17a, A18, A19, A20, A21, A22, A24	33dBm	 Rel-5 Rel-6 Rel-7
	, A23		Rel-7 Rel-8
Maximum allowed UL TX power	A5, A6	Not Present	
CHOICE channel requirement - Uplink DPCH power control info - DPCCH power offset - PC Preamble - SRB delay - Power Control Algorithm - TPC step size - Δ_{NACK} - Δ_{NACK} - Ack-Nack repetition factor	A1, A2, A3, A4, A7, A8, A11	Uplink DPCH info -40 (-80dB) 1 frame 7 frames Algorithm1 0 (1dB) Not Present Not Present Not Present	Rel-5 and earlier Rel-5 Rel-5 Rel-5

Information Element	Condition	Value/remark	Version
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Number of TPC bits		Not Present	Rel-7
CHOICE channel requirement	A9, A10	Uplink DPCH info	Rel-5
	, A17, A17a, A18		Rel-7
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		3	
- Δ_{NACK}		3	
- Ack-Nack repetition factor		1	
- HARQ_preamble_mode		0	Rel-6
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
CHOICE channel requirement	A5,A6	Not Present	Rel-5 and earlier
Uplink DPCH info	A12		Rel-6
	A19		Rel-7
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		3	
- Δ_{NACK}		3	
- Ack-Nack repetition factor		1	
- HARQ_preamble_mode		0	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Number of TPC bits		Not Present	Rel-7
Uplink DPCH info	A13, A14, A15, A16		Rel-6
	, A20, A21, A22		Rel-7
	, A23		Rel-7
			Rel-8
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		3	
- Δ_{NACK}		3	
- Ack-Nack repetition factor		1	
- HARQ_preamble_mode		0	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	

Information Element	Condition	Value/remark	Version
- Number of DPDCH - spreading factor - TFCI existence - Number of FBI bit - Puncturing Limit		0 Not Present FALSE Not Present Not Present	
- Number of TPC bits		Not Present	Rel-7
E-DCH info	A12, A13, A14, A15, A16		Rel-6
	, A20, A21, A22		Rel-7
	, A23		Rel-7 Rel-8
- MAC-es/e reset indicator - E-DPCCH info - E-DPCCH/DPCCH power offset - Happy bit delay condition - E-TFC Boost Info - E-DPDCH power interpolation - E-DPDCH info - E-TFCI table index - E-DCH minimum set E-TFCI - Reference E-TFCIs - Reference E-TFCI - Reference E-TFCI PO - Reference E-TFCI - Reference E-TFCI PO - Maximum channelisation codes - PLnon-max - Scheduling Information Configuration - Periodicity for Scheduling Info – no grant - Periodicity for Scheduling Info – grant - Power Offset for Scheduling Info - 3-Index-Step Threshold - 2-Index-Step Threshold - Scheduled Transmission configuration - 2ms scheduled transmission grant HARQ process allocation - Serving Grant -UL 16QAM settings		TRUE 0 100 ms Not Present Not Present 0 9 2 E-TFCIs 11 4 83 16 2sf4 0.84 Not present Not present 0 Not present Not present Not present Not present Not Present	Rel-7 Rel-7
E-DCH info	A19		Rel-7
- MAC-es/e reset indicator - E-DPCCH info - E-DPCCH/DPCCH power offset - Happy bit delay condition - E-TFC Boost Info - E-DPDCH power interpolation - E-DPDCH info - E-TFCI table index - E-DCH minimum set E-TFCI - Reference E-TFCIs - Reference E-TFCI - Reference E-TFCI PO - Reference E-TFCI - Reference E-TFCI PO - Maximum channelisation codes - PLnon-max - Scheduling Information Configuration - Periodicity for Scheduling Info – no grant - Periodicity for Scheduling Info – grant - Power Offset for Scheduling Info - 3-Index-Step Threshold		TRUE 0 100 ms Not Present Not Present 0 9 2 E-TFCIs 11 4 83 16 2sf2and2sf4 0.84 Not present Not present 0 Not present	Rel-7 Rel-7

Information Element	Condition	Value/remark	Version
- 2-Index-Step Threshold - Scheduled Transmission configuration - 2ms scheduled transmission grant HARQ process allocation - Serving Grant -UL 16QAM settings -BetaEd gain E-AGCH table selection		Not present Not present Not present 1	 Rel-7
E-DCH info	A24	Not Present	Rel-7
CHOICE Mode - Downlink PDSCH information	A1, A2, A3, A4, A5, A6, A7, A8, A11 	FDD Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A11 A24	Not Present Not Present	Rel-5 Rel-7
Downlink HS-PDSCH Information	A9, A10 A12, A13, A14, A15, A16 A17, A18, A19, A20, A21, A22		Rel-5 Rel-6 Rel-7
- HS-SCCH Info - CHOICE mode - DL Scrambling Code - HS-SCCH Channelisation Code Information - HS-SCCH Channelisation Code - Measurement Feedback Info - CHOICE mode - POhsdsch - CQI Feedback cycle, k - CQI repetition factor - Δ_{CQI} - CHOICE mode		 FDD Not present 7 FDD 6 dB 4 ms 1 5 (corresponds to 0dB in relative power offset) FDD (no data)	
- Downlink 64QAM configured		Not present	Rel-7
- HS-DSCH TB size table		Not present	Rel-7
Downlink HS-PDSCH Information - HS-SCCH Info - CHOICE mode - DL Scrambling Code - HS-SCCH Channelisation Code Information - HS-SCCH Channelisation Code - Measurement Feedback Info - CHOICE mode - POhsdsch - CQI Feedback cycle, k - CQI repetition factor - Δ_{CQI} - CHOICE mode - Downlink 64QAM configured - HS-DSCH TB size table	A17a	 FDD Not Present 7 FDD 6 dB 4 ms 1 5 (corresponds to 0dB in relative power offset) FDD TRUE Not Present	Rel-7 Rel-7 Rel-7
Downlink HS-PDSCH Information - HS-SCCH Info - CHOICE mode - DL Scrambling Code - HS-SCCH Channelisation Code Information - HS-SCCH Channelisation Code - Measurement Feedback Info - CHOICE mode	A23	 FDD Not present 7 FDD	Rel-7 Rel-8

Information Element	Condition	Value/remark	Version
- POhsdsch		6 dB	
- CQI Feedback cycle, k		4 ms	
- CQI repetition factor		1	
- Δ_{CQI}		5 (corresponds to 0dB in relative power offset)	
- CHOICE mode		FDD (no data)	
- Downlink 64QAM configured		Not present	Rel-7
- HS-DSCH TB size table		Octet Aligned	Rel-7
Downlink information common for all radio links	A1, A2, A3, A11		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A9 A12, A13, A15 A17, A17a, A18. A19		Rel-5 Rel-6 Rel-7
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset PPilot-DPDCH		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	
Downlink information common for all radio links	A4,A7,A8		
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	

Information Element	Condition	Value/remark	Version
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
Downlink information common for all radio links	A10		Rel-5
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset PPilot-DPDCH		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		Not Present	
Downlink information common for all radio links	A14, A16		Rel-6
	, A20, A21, A22		Rel-7
	, A23		Rel-7 Rel-8
- Downlink F-DPCH info common for all RL			
- Timing Indication		Maintain	
- Timing maintained Synchronization indicator		FALSE	
- Downlink F-DPCH power control information			
- DPC mode		0 (single)	
- TPC command error rate target		0.04	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	
Downlink information common for all radio links	A5,A6	Not Present	
	A24	Not Present	Rel-7
Downlink information for each radio link list	A1, A2, A3, A4, A7, A8, A11		
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and

Information Element	Condition	Value/remark	Version
			Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		1	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A5		
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Downlink DPCH info for each RL		Not present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A9, A10		Rel-5
	, A17, A18		Rel-7
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value	

Information Element	Condition	Value/remark	Version
		(as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		1	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A17a		Rel-7
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set 13	
- Code number			
- Scrambling code change		Not Present	
- TPC combination index		0	
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
Downlink information for each radio link list	A12, A13, A15		Rel-6
	A19		Rel-7
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		TRUE	

Information Element	Condition	Value/remark	Version
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		1	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2")	Default1
		Set to value: OMIT (otherwise)	Default2
- TPC combination index		0	
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info			
- E-AGCH Channelisation Code		10	
- CHOICE E-HICH Information			
- E-HICH Information			
- Channelisation code		4	
- Signature sequence		1	
- CHOICE E-RGCH Information			
- E-RGCH Information			
- Signature Sequence		0	
- RG combination index		0	
Downlink information for each radio link list	A14, A16		Rel-6
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		TRUE	
- Downlink DPCH info for each RL		Not Present	
- Downlink F-DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- F-DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- F-DPCH slot format		3 if UE supports enhanced F-DPCH, otherwise Not Present	Rel-7
- Secondary CPICH info		Not Present	
- Secondary scrambling code		Not Present	
- Code number		12	
- TPC combination index		0	
- E-AGCH Info			
- E-AGCH Channelisation Code		10	
- CHOICE E-HICH Information			
- E-HICH Information			
- Channelisation code		4	
- Signature sequence		1	
- CHOICE E-RGCH Information		Not Present	
Downlink information for each radio link list	A20, A21, A22		Rel-7
	, A23		Rel-7 Rel-8
- Downlink information for each radio			

Information Element	Condition	Value/remark	Version
link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		TRUE	
- Downlink DPCH info for each RL		Not Present	
- Downlink F-DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- F-DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- F-DPCH slot format		3 if UE supports enhanced F-DPCH, otherwise Not Present	Rel-7
- Secondary CPICH info		Not Present	
- Secondary scrambling code		Not Present	
- Code number		12	
- TPC combination index		0	
- E-AGCH Info			
- E-AGCH Channelisation Code		10	
- CHOICE E-HICH Information			
- E-HICH Information			
- Channelisation code		4	
- Signature sequence		1	
- CHOICE E-RGCH Information			
- E-RGCH Information			
- Signature Sequence		0	
- RG combination index		0	
Downlink information for each radio link list	A6, A24	Not Present	
MBMS PL Service Restriction Information	A1, A2, A3, A4, A5, A6, A7, A8 A9, A10 A12, A13, A14, A15, A16 A17, A17a, A18, A19, A20, A21, A22, A23, A24	Not Present	 Rel-5 Rel-6 Rel-7 Rel-7 Rel-8

Condition	Explanation	Version
A1	This IE is needed for "Non speech to CELL_DCH from CELL_DCH in CS"	
A2	This IE is needed for "Speech to CELL_DCH from CELL_DCH in CS"	
A3	This IE is needed for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE is needed for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE is needed for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE is needed for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE is needed for "Non speech to CELL_DCH from CELL_FACH in CS"	
A8	This IE is needed for "Speech to CELL_DCH from CELL_FACH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_FACH in PS"	Rel-5
A11	This IE is needed for " Packet RAB Setup after Speech RAB Setup in CELL_DCH"	
A12	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using three multiplexing options (3/3) and SRBs mapped on DCH/DCH"	Rel-6
A13	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-6
A14	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-6

A15	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (two streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-6
A16	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (one conversational and one streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-6
A17	This IE is needed for "Packet to CELL_DCH / HS-DSCH with enhanced data rate and RLC AM"	Rel-7
A17a	This IE is needed for "Packet to CELL_DCH / HS-DSCH [DL : 64QAM] with enhanced data rate and RLC AM"	Rel-7
A18	This IE is needed for "Packet to CELL_DCH / HS-DSCH with enhanced data rate and RLC UM"	Rel-7
A19	This IE is needed for "Packet to CELL_DCH / E-DCH[UL : 16QAM] / HS-DSCH using three multiplexing options (3/3) and SRBs mapped on DCH/DCH"	Rel-7
A20	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with DTX/DRX using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7
A21	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with DTX/DRX and multiple RABs (one conversational and one streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7
A22	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (one conversational and one streaming/interactive/background) with enhanced data rate using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7
A23	This IE is needed for "Speech to CELL_DCH / E-DCH / HS-DSCH CS RAB with DTX/DRX and enhanced data rate using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH" .	Rel-7 Rel-8 (Note 1)
A24	This IE is needed for "Packet to CELL_FACH from CELL_FACH using one multiplexing option (1/1) and SRBs mapped on RACH/HS-DSCH"	Rel-7

NOTE 1: Support depends on the UE capability: Support for CS voice over HSPA. This is supported in Rel-8 and may be supported in Rel-7.

Condition	Explanation	Version
MAC-I-FIXED	Used with other condition when MAC-i/is with Fixed RLC PDU size is configured	Rel-8
MAC-I-FLEX	Used with other condition when MAC-i/is with Flexible RLC PDU size is configured	Rel-8

Contents of RADIO BEARER SETUP message: AM or UM, for MBMS PtP Radio Bearer Setup

Information Element	Condition	Value/remark
- RAB information for setup		
- RAB info		(UM DTCH for PS domain DL only)
- RAB identity		11111111B For Selected Service and Set to same as Enhanced NSAPI received in Service Request (10000000B to 11111110B) for Multicast service. The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.
- CN domain identity		PS domain
- NAS Synchronization Indicator		Not Present
- Re-establishment timer		useT315
- RB information to setup		
- RB identity		21
- MBMS Service Identity		Present for Selected Service only
- MBMS Service ID		MBMS Service ID of the service UE has selected
- MBMS Session identity		
- MBMS Session ID		Ongoing Session ID
- PDCP info		
- Support for lossless SRNS relocation		FALSE
- Max PDCP SN window size		Not present
- PDCP PDU header		Absent
- Header compression information		Not present
- CHOICE RLC info type		RLC info
- CHOICE Uplink RLC mode		Not Present
- CHOICE Downlink RLC mode		UM RLC
- DL UM RLC LI size		7
- DL Reception Window Size		Not Present
- RB mapping info		
- Information for each multiplexing option		1 RBMuxOptions
- RLC logical channel mapping indicator		Not Present
- Number of uplink RLC logical channels		Not Present
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels		1
- Downlink transport channel type		DCH
- DL DCH Transport channel identity		7
- DL DSCH Transport channel identity		Not Present
- Logical channel identity		Not Present
Added or Reconfigured UL TrCH information	B5, B2	5 DCH reconfigured
- Uplink transport channel type		DCH
- UL Transport channel identity		5
- TFS		
- CHOICE Transport channel type		Dedicated transport channels
- Dynamic Transport format information		
- RLC Size		Reference to clause 6. 11.1b Parameter Set
- Number of TBs and TTI List		(This IE is repeated for TFI number.)
- Transmission Time Interval		Not Present
- Number of Transport blocks		Reference to clause 6. 11.1b Parameter Set
- CHOICE Logical channel list		All
- Semi-static Transport Format information		
- Transmission time interval		Reference to clause 6. 11.1b Parameter Set
- Type of channel coding		Reference to clause 6. 11.1b Parameter Set
- Coding Rate		Reference to clause 6. 11.1b Parameter Set
- Rate matching attribute		Reference to clause 6. 11.1b Parameter Set

Information Element	Condition	Value/remark
- CRC size		Reference to clause 6. 11.1b Parameter Set
Added or Reconfigured DL TrCH information	B1, B2	2 TrCHs(DCH for DCCH and 1 DCH for DTCH)
- Downlink transport channel type		DCH
- DL Transport channel identity		10
- CHOICE DL parameters		Same as UL
- Uplink transport channel type		DCH
- UL TrCH identity		5
- DCH quality target		
- BLER Quality value		-20 (-2.0)
- Downlink transport channel type		DCH
- DL Transport channel identity		7
- CHOICE DL parameters		Explicit
- TFS		
- CHOICE Transport channel type		Dedicated transport channel
- Dynamic transport format information		
- RLC Size		Reference to clause 6. 11.1b Parameter Set
- Number of TBs and TTI List		(This IE is repeated for TFI number.)
- Transmission Time Interval		Not Present
- Number of Transport blocks		Reference to clause 6. 11.1b Parameter Set only including TF0
- CHOICE Logical channel list		All
- Semi-static Transport Format information		
- Transmission time interval		Reference to clause 6. 11.1b Parameter Set
- Type of channel coding		Reference to clause 6. 11.1b Parameter Set
- Coding Rate		Reference to clause 6. 11.1b Parameter Set
- Rate matching attribute		Reference to clause 6. 11.1b Parameter Set
- CRC size		Reference to clause 6. 11.1b Parameter Set
- DCH quality target		
- BLER Quality value		-20 (-2.0)
Added or Reconfigured DL TrCH information	B3, B4	3 TrCHs(DCH for DCCH and 2 DCH for DTCH's)
- Downlink transport channel type		DCH
- DL Transport channel identity		10
- CHOICE DL parameters		Same as UL
- Uplink transport channel type		DCH
- UL TrCH identity		5
- DCH quality target		
- BLER Quality value		-20 (-2.0)
- Downlink transport channel type		DCH
- DL Transport channel identity		6
- CHOICE DL parameters		Same as UL
- Uplink transport channel type		DCH
- UL TrCH identity		1
- DCH quality target		
- BLER Quality value		-20 (-2.0)
- Downlink transport channel type		DCH
- DL Transport channel identity		7
- CHOICE DL parameters		Explicit
- TFS		
- CHOICE Transport channel type		Dedicated transport channel
- Dynamic transport format information		
- RLC Size		Reference to clause 6. 11.1a Parameter Set
- Number of TBs and TTI List		(This IE is repeated for TFI number.)
- Transmission Time Interval		Not Present
- Number of Transport blocks		Reference to clause 6. 11.1a Parameter
- CHOICE Logical channel list		All
- Semi-static Transport Format information		
- Transmission time interval		Reference to clause 6. 11.1a Parameter Set
- Type of channel coding		Reference to clause 6. 11.1a Parameter Set
- Coding Rate		Reference to clause 6. 11.1a Parameter Set
- Rate matching attribute		Reference to clause 6. 11.1a Parameter Set
- CRC size		Reference to clause 6. 11.1a Parameter Set
- DCH quality target		
- BLER Quality value		-20 (-2.0)
All other IEs	B1	Use the values defined in the RADIO BEARER SETUP message indicated as "Packet to CELL_DCH from

Information Element	Condition	Value/remark
		CELL_DCH in PS" condition A3 except RB parameter set referred in 6.11.1b instead of 6.10
All other IEs	B2	Use the values defined in the RADIO BEARER SETUP message indicated as "Packet to CELL_DCH from CELL_FACH in PS" condition A4 except RB parameter set referred in 6.11.1b instead of 6.10
All other IEs	B3	Use the values defined in the RADIO BEARER SETUP message indicated as "Packet to CELL_DCH from CELL_DCH in PS" condition A3 except RB parameter set referred in 6.11.1a instead of 6.10
All other IEs	B4	Use the values defined in the RADIO BEARER SETUP message indicated as "Packet to CELL_DCH from CELL_FACH in PS" condition A4 except RB parameter set referred in 6.11.1a instead of 6.10

Condition	Explanation	Version
B1	This IE is needed for " MBMS PtP Radio Bearer Setup when UE state is state 6-7"	
B2	This IE is needed for " MBMS PtP Radio Bearer Setup when UE state is state 6-8"	
B3	This IE is needed for " MBMS PtP Radio Bearer Setup, when UE state is 6-10"	
B4	This IE is needed for " MBMS PtP Radio Bearer Setup, when UE state is 6-11"	

Contents of RADIO BEARER SETUP COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink RADIO BEARER SETUP message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked.
CHOICE mode	FDD
START	Not checked (if ciphering is OFF), check the presence if ciphering is ON.
Deferred measurement control reading	Not present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	The UE shall include this IE if the following two conditions are fulfilled: (a) The RADIO BEARER SETUP message did not contain the IE "Ciphering activation time for DPCH" and (b) The RADIO BEARER SETUP message established the first RB(s) mapped to RLC-TM for a CN domain. Else, this IE is absent.
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of RADIO BEARER SETUP FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER SETUP message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded	Not checked

Contents of RADIO BEARER RECONFIGURATION message: AM or UM

Information Element	Condition	Value/remark	Version
Message Type	A1,A2,A3,A4,A5,A6		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1,A2,A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5,A6	Not Present	
Delay restriction flag	A1,A2,A3,A4,A5,A6	Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4,	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6	Not Present	R99 and Rel-4 only
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
UE Mobility State Indicator		Not Present	Rel-7
UTRAN DRX cycle length coefficient	A1,A2,A3,A4,A5,A6	Not Present	
CN information info		Not Present	
URA identity		Not Present	
CHOICE specification mode		Complete specification	Rel-5
RNC support for change of UE capability		Not Present	Rel-7
Reconfiguration in response to requested change of UE capability		Not Present	Rel-7
RAB information to reconfigure list		Not Present	
RB information to reconfigure list	A1	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	

Information Element	Condition	Value/remark	Version
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A2	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		11	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH) (This IE is needed for 12.2 kbps and 10.2 kbps)	
- RB identity		12	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A3,A4,A5,A6	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	

Information Element	Condition	Value/remark	Version
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DTCH)	
- RB identity		20	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to be affected	A1, A2, A3,A4,A5,A6	Not Present	
RB with PDCP context relocation info list		Not Present	Rel-5
PDCP ROHC target mode		Not Present	Rel-5
UL Transport channel information common for all transport channels	A1, A2, A5,A6	Not Present	
UL Transport channel information common for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set	
- CTFC		Reference to clause 6.10.2.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Gain factor βc		11 (below 64 kbps) 9 (equal or higher than 64 kbps) when HSDPA is not configured 9 (equal or higher than 64 kbps and	

Information Element	Condition	Value/remark	Version
		below 384 kbps) when HSDPA is also configured 6 (equal or higher than 384 kbps) when HSDPA is also configured (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- Gain factor βd		15 (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P p-m		Not Present	
Deleted UL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured UL TrCH information	A1, A2, A5,A6	Not Present	
Added or Reconfigured UL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A3	(DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	

Information Element	Condition	Value/remark	Version
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
CHOICE mode	A1,A2,A3,A4,A5,A6	Not Present	
DL Transport channel information common for all transport channel	A1, A2, A5, A6	Not Present	
DL Transport channel information common for all transport channel	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Explicit	
- DL DCH TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.2.4	
- CTFC		Reference to clause 6.10.2.4 Parameter Set	
- Power offset information		Not Present	
Deleted DL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured DL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured DL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A3		
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	

Information Element	Condition	Value/remark	Version
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Frequency info	A1,A2,A3,A4,A5		
- UARFCN uplink (Nu)		Not present Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]	
- UARFCN downlink (Nd)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX Information		Not Present	Rel-7
HS-SCCH less Information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power	A1,A2,A3,A4,A5,A6	33dBm	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
-Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Number of TPC bits		Not Present	Rel-7
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
CHOICE channel requirement	A5, A6	Not Present	
E-DCH Info		Not Present	Rel-6
CHOICE Mode	A1,A2,A3,A4,A5,A6	FDD	
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	

Information Element	Condition	Value/remark	Version
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Present Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information per radio link list -Downlink information for each radio link	A1, A2, A3		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- Secondary scrambling code			
- channelisation code			
- DL channelisation code			
- Secondary scrambling code		2	

Information Element	Condition	Value/remark	Version
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list -Downlink information for each radio link	A4		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value : Default DPCH Offset Value mod 38 400	
- Secondary CPICH info		Not Present	
- Secondary scrambling code			
- channelisation code			
- DL channelisation code			
- Secondary scrambling code		2	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A5		

Information Element	Condition	Value/remark	Version
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL		Not present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A6		R99
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL		Not present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A6	Not Present	Rel-4 on
MBMS PL Service Restriction Information	A1,A2,A3,A4,A5,A6	Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded List	Not checked

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION COMPLETE message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked
CHOICE mode	FDD
Deferred measurement control reading	Not present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of RADIO BEARER RELEASE message: AM or UM

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A7, A8	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9, A10		Rel-5
Activation time	A4, A5, A6	Not Present	
New U-RNTI		Not Present	
New C-RNTI	A1,A2,A3,A4	Not Present	
	, A9		Rel-5
New C-RNTI	A5, A6, A7, A8	'1010 1010 1010 1010'	
	, A10		Rel-5
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	R99 and Rel-4 only
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10,		Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator	A1,A2, A3, A4	CELL_DCH	
	, A9		Rel-5
RRC State indicator	A5, A6, A7, A8	CELL_FACH	
	, A10		Rel-5
UE Mobility State Indicator		Not Present	Rel-7
UTRAN DRX cycle length coefficient	A1,A2,A3,A4,A5,A6, A7, A8	Not Present	

Information Element	Condition	Value/remark	Version
	, A9, A10		Rel-5
CN information info		Not Present	
Signalling Connection release indication		Not Present	
URA identity		Not Present	
RNC support for change of UE capability		Not Present	Rel-7
RAB information to reconfigure list		Not Present	
RB information to release	A1,A2, A7, A8		
- RB identity		10	
RB information to release	A2, A8		
- RB identity		11	
RB information to release	A2, A8		
- RB identity		12	
RB information to release	A3, A4, A5, A6		
- RB identity		20	
RB information to release	A9, A10		Rel-5
- RB identity		25	
RB information to reconfigure list	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
RB information to be affected	A1,A2, A3,A4,A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
Downlink counter synchronization info	A1,A2,A3,A4,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
UL Transport channel information for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted UL TrCH Information	A1,A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
- Uplink transport channel type		DCH	
- Transport channel identity		1	
Deleted UL TrCH Information	A2, A8		
- Uplink transport channel type		DCH	
- Transport channel identity		2	
Deleted UL TrCH Information	A2, A8		
- Uplink transport channel type		DCH	
- Transport channel identity		3	
Added or Reconfigured UL TrCH information	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Added or Reconfigured UL TrCH information	A1, A2, A3, A4	TrCHs(DCH for DCCH)	
	, A9		Rel-5
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of Transport blocks		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	

Information Element	Condition	Value/remark	Version
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Type of channel coding		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Coding Rate		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Rate matching attribute		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CRC size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
DL Transport channel information for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted DL TrCH Information	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9		Rel-5
- Downlink transport channel type		DCH	
- Transport channel identity		6	
Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		7	
Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		8	
Deleted DL TrCH Information	A9, A10		Rel-5
- Downlink transport channel type		HS-DSCH	
- DL HS-DSCH MAC-d flow identity		0	
Added or Reconfigured DL TrCH information	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Added or Reconfigured DL TrCH information	A1, A2, A3, A4	1 TrCHs(DCH for DCCH)	
	, A9		Rel-5
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
Frequency info	A1,A2,A3,A4,A5, A7, A8		
	, A9, A10		Rel-5
- UARFCN uplink (Nu)		Not present Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]	
- UARFCN downlink (Nd)		Reference to clause 5.1 Test frequencies	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX Information		Not Present	Rel-7
HS-SCCH less Information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power		33dBm	
Frequency info	A6	Not Present	
CHOICE *channel requirement*	A5, A6, A7,	Not Present	

Information Element	Condition	Value/remark	Version
	A8 , A10		Rel-5
CHOICE channel requirement	A1, A2, A3, A4 , A9	Uplink DPCH info	Rel-5
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- TPC step size		0 (1dB)	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	
- Number of TPC bits		Not Present	Rel-7
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
E-DCH Info		Not Present	Rel-6
CHOICE Mode	A1,A2,A3,A4,A5,A6, A7, A8 , A9, A10	FDD	Rel-5
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A5, A6, A7, A8 , A10	Not Present	Rel-5
Downlink information common for all radio links	A1,A2, A3 , A9		Rel-5
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information for each radio link list	A1,A2,A3, A9		Rel-5
-Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- Secondary scrambling code			
- channelisation code			
- DL channelisation code			
- Secondary scrambling code		3	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Code number		0	
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A4		
-Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	

Information Element	Condition	Value/remark	Version
- DPCH frame offset		Set to value : Default DPCH Offset Value mod 38 400	
- Secondary CPICH info		Not Present	
- Secondary scrambling code			
- channelisation code			
- DL channelisation code			
- Secondary scrambling code		3	
- Spreading factor		Reference to clause 6.10 Parameter Set 0	
- Code number			
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A5, A7, A8		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL		Not present	
- E-AGCH Info		Not present	Rel-6
- E-HICH Information		Not present	Rel-6
- E-RGCH Information		Not present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A6, A10	Not Present	
MBMS PL Service Restriction Information	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
MBMS RB list released to change transfer mode		Not Present	Rel-6

Condition	Explanation	Version
A1	This IE need for "Non speech in CS"	
A2	This IE need for "Speech in CS"	
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE need for "Non speech to CELL_FACH from CELL_DCH in CS"	
A8	This IE need for "Speech to CELL_FACH from CELL_DCH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH from CELL_DCH / HS-DSCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_FACH from CELL_DCH / HS-DSCH in PS"	Rel-5

Contents of RADIO BEARER RELEASE COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see the value is identical to the same IE in the downlink RADIO BEARER RELEASE message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked.
CHOICE mode	FDD
Deferred measurement control reading	Not present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of RADIO BEARER RELEASE FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RELEASE message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded	Not checked

Contents of RRC CONNECTION REQUEST message: TM

Information Element	Condition	Value/remark	Version
Message Type			
Predefined configuration status information		To be checked against requirement if specified	Rel-5
Initial UE identity			
- CHOICE UE id type			
- TMSI and LAI (GSM-MAP)		Set to the UE's TMSI and LAI.	
Establishment cause		To be checked against requirement if specified	
Protocol error indicator		FALSE	
UE Specific Behaviour Information 1 idle		This IE will not be checked by default behaviour, but in specific test case.	
UE capability indication		To be checked against requirement if specified	Rel-6
Support for F-DPCH	A1	TRUE	Rel-6
Support for F-DPCH	A2	Not Present	Rel-6
UE Mobility State Indicator		Not Present	Rel-7
Support for Enhanced F-DPCH		To be checked against requirement if specified	Rel-7
HS-PDSCH in CELL_FACH		To be checked against requirement if specified	Rel-7
MAC-ehs support		To be checked against requirement if specified	Rel-7
DPCCH Discontinuous Transmission support		To be checked against requirement if specified	Rel-7
Measured results on RACH		To be checked against requirement if specified	Rel-4
Access stratum release indicator		To be checked against requirement if specified	Rel-4

Condition	Explanation
A1	This IE need to be set to TRUE when F-DPCH is fully supported by the UE.
A2	This IE need to be absent when F-DPCH is not fully supported by the UE.

Contents of RRC CONNECTION REJECT message: UM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Initial UE identity	Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message.
Rejection cause	Unspecified
Wait Time	0
Redirection info	Not Present

Contents of RRC CONNECTION RELEASE message: UM

Information Element	Value/remark	Version
Message Type		
U-RNTI	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	R99, Rel-4
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
CHOICE identity type	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	Rel-5
- U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
- Group identity	[FFS]	
- Group release information	[FFS]	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info	This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted.	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.	
N308	2 (for CELL_DCH state). Not Present (for UE in other connected mode states).	
Release cause	Normal event	
UE Mobility State Indicator	Not Present	Rel-7
Rplmn information	Not Present	

Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM

Information Element	Semantics description
Message Type	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION RELEASE message.
Integrity check info	
- Message authentication code	Checked to see if it's identical to the value of XMAC-I calculated by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	Checked to see if it is present. This number is used by the SS to compute the XMAC-I
Error indication	Not checked

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH or HS-PDSCH in CELL_FACH)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3 , A4, A5		
Initial UE identity		Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	

Information Element	Condition	Value/remark	Version
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Activation time		Not Present(Now)	
New U-RNTI			
- SRNC identity		0000 0000 0001B	
- S-RNTI		0000 0000 0000 0000 0001B	
New C-RNTI	A1, A2, A3	Not present	
	A4	'1010 1010 1010 1010'	Rel-7
New H-RNTI	A1	Not present	Rel-6
	A2	'1010 1010 1010 1010'	Rel-6
	A3, A4		Rel-7
	A5		Rel-8
New Primary E-RNTI	A1	Not present	Rel-6
	A2, A3	'1010 1010 1010 1010'	Rel-7
	A5		Rel-8
New Secondary E-RNTI		Not present	Rel-6
RRC State Indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		9	
Capability update requirement			
- UE radio access FDD capability update requirement		TRUE	
- UE radio access TDD capability update requirement		FALSE	
- UE radio access 3.84 Mcps TDD capability update requirement		FALSE	Rel-4
- UE radio access 1.28 Mcps TDD capability update requirement		FALSE	Rel-4
- System specific capability update requirement list		GSM	
RNC support for change of UE capability		FALSE	Rel-7
CHOICE *specification mode*		Complete specification	Rel-5
- Complete specification			Rel-5
- Signalling RB information to setup	A1	(UM DCCH for RRC)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7 bit	Rel-6
- One sided RLC re-establishment		FALSE	Rel-6
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		1	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	

Information Element	Condition	Value/remark	Version
- UL Transport channel identity		Not Present	
- Logical channel identity		1	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- Signalling RB information to setup	A2	(UM DCCH for RRC)	Rel-6
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7 bit	Rel-6
- One sided RLC re-establishment		FALSE	Rel-6
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		1	
- Signalling RB information to setup	A3 A5	(UM DCCH for RRC)	Rel-7 Rel-8
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7 bit	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		TRUE	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing		1 RBMuxOption	

Information Element	Condition	Value/remark	Version
option			
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		1	
- Signalling RB information to setup	A4	(UM DCCH for RRC)	Rel-7
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7 bit	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		TRUE	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		1	
- CHOICE RLC size list		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		1	
- Signalling RB information to setup	A1	(AM DCCH for RRC)	
- RB identity		Not Present	
- CHOICE RLC info type			

Information Element	Condition	Value/remark	Version
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	Rel-6
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		2	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		2	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- Signalling RB information to setup	A2	(AM DCCH for RRC)	Rel-6
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
- Signalling RB information to setup	A3	(AM DCCH for RRC)	Rel-7
	A5		Rel-8
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	

Information Element	Condition	Value/remark	Version
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- CHOICE RLC PDU size		Fixed size	Rel-8
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		2	
- Signalling RB information to setup	A4	(AM DCCH for RRC)	Rel-7
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	

Information Element	Condition	Value/remark	Version
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		2	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		2	
- Signalling RB information to setup	A1	(AM DCCH for NAS_DT High priority)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		According to clause 6.10.2.4.1.3 (standalone	Rel-6

Information Element	Condition	Value/remark	Version
		13.6 kbps signalling radio bearer)	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		3	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		3	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- Signalling RB information to setup	A2	(AM DCCH for NAS_DT High priority)	Rel-6
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	

Information Element	Condition	Value/remark	Version
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		3	
- Signalling RB information to setup	A3	(AM DCCH for NAS_DT High priority)	Rel-7
	A5		Rel-8
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			

Information Element	Condition	Value/remark	Version
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		3	
- Signalling RB information to setup	A4	(AM DCCH for NAS_DT High priority)	Rel-7
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	

Information Element	Condition	Value/remark	Version
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		3	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		3	
- Signalling RB information to setup	A1	(AM DCCH for NAS_DT Low priority)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	Rel-6
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		4	

Information Element	Condition	Value/remark	Version
- CHOICE RLC size list		Configured	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		4	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
- Signalling RB information to setup	A2	(AM DCCH for NAS_DT Low priority)	Rel-6
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	

Information Element	Condition	Value/remark	Version
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
- Signalling RB information to setup	A3, A5	(AM DCCH for NAS DT Low priority)	Rel-7
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- CHOICE RLC PDU size		Fixed size	Rel-8
- DDI		2	
- RLC PDU size list		1 RLC PDU size	

Information Element	Condition	Value/remark	Version
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		4	
- Signalling RB information to setup	A4	(AM DCCH for NAS DT Low priority)	Rel-7
- RB identity		Not present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		32	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- Length indicator size		7	
- In-sequence delivery		TRUE	
- Receiving window size		32	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- Alternative E-bit interpretation		Not Present	
- Use special value of HE field		TRUE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		4	
- CHOICE RLC size list		Explicit List	
- RLC size index		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	

Information Element	Condition	Value/remark	Version
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- CHOICE *DL MAC header type*		MAC-ehs	
- DL HS-DSCH MAC-ehs Queue Id		1	
- Logical channel identity		4	
UL Transport channel information for all transport channels	A1		
- PRACH TFCS		Not Present	
- CHOICE Mode		FDD	
- TFC subset		Nor Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete	
- TFCS complete reconfigure			
- CHOICE CTFC Size		2bit CTFC	
- CTFC information		This IE is repeated for TFC numbers according to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CTFC		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Gain factor ßc		11 (below 64 kbps) 9 (higher than 64 kbps) (Not Present if the above is set to Computed Gain Factors)	
- Gain factor ßd		15 (Not Present if the above is set to Computed Gain Factors)	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
UL Transport channel information for all transport channels	A2	Not Present	Rel-6
UL Transport channel information for all transport channels	A3, A4	Not Present	Rel-7
Added or Reconfigured UL TrCH information	A1		
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of TBs and TTI lists		(This IE is repeated for TFI number)	
- Transmission Time Interval		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of Transport blocks		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Type of channel coding		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Coding Rate		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	

Information Element	Condition	Value/remark	Version
- Rate matching attribute		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CRC size		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
Added or Reconfigured UL TrCH information	A2	1 E-DCH added with one DCCH MAC-d flow	Rel-6
	A3		Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- E-DCH Transmission Time Interval		set to 2ms if supported by the UE E-DCH category, or 10ms if the UE E-DCH category does not support 2ms TTI	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- Max MAC-e PDU contents size		162 bits	
- 2 ms non-scheduled transmission grant HARQ process allocation		Not Present	
Added or Reconfigured UL TrCH information	A4		Rel-7
- Uplink transport channel type		DCH	
- UL Transport channel identity		15	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information		2 Transport formats	
- RLC size		240	
- Number of TBs and TTI lists			
- Transmission Time Interval		20 ms	
- Number of Transport blocks		1	
- CHOICE Logical channel list		Configured	
- RLC size		360	
- Number of TBs and TTI lists			
- Transmission Time Interval		20 ms	
- Number of Transport blocks		1	
- CHOICE Logical channel list		Configured	
- Semi-static Transport Format information			
- Transmission time interval		20 ms	
- Type of channel coding		Convolutional	
- Coding Rate		1/2	
- Rate matching attribute		150	
- CRC size		16	
Added or Reconfigured UL TrCH information	A5	1 E-DCH added with one DCCH MAC-d flow	Rel-8
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- UL MAC header type		MAC-i/is	
- E-DCH Transmission Time Interval		10 ms	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	

Information Element	Condition	Value/remark	Version
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- Max MAC-e PDU contents size		168 bits	
- 2 ms non-scheduled transmission grant HARQ process allocation		Not Present	
DL Transport channel information common for all transport channel	A1		
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Same as UL	
DL Transport channel information common for all transport channel	A2	Not Present	Rel-6
DL Transport channel information common for all transport channel	A3, A4	Not Present	Rel-7
Added or Reconfigured DL TrCH information	A1		
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH Identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A2	1 TrCH (HS-DSCH for DCCH)	Rel-6
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A3 A5	1 TrCH (HS-DSCH for DCCH)	Rel-7 Rel-8
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- CHOICE *DL MAC header type*		MAC-ehs	
- Added or reconfigured MAC-ehs reordering queue			
- MAC-ehs queue to add or reconfigure list		(one queue)	

Information Element	Condition	Value/remark	Version
- MAC-ehs queue Id		1	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue to delete list		Not present	
- DCH quality target		Not present	
Frequency info		Not Present	
DTX-DRX timing information		Not present	Rel-7
DTX-DRX Information		Not present	Rel-7
HS-SCCH less Information		Not present	Rel-7
Maximum allowed UL TX power		Not Present	
Added or Reconfigured DL TrCH information	A4	1 TrCH (HS-DSCH for DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- CHOICE *DL MAC header type*		MAC-ehs	
- Added or reconfigured MAC-ehs reordering queue			
- MAC-ehs queue to add or reconfigure list		(one queue)	
- MAC-ehs queue Id		1	
- T1		50	
- MAC-ehs window size		16	
- MAC-ehs queue to delete list		Not present	
- DCH quality target		Not present	
Frequency info		Not Present	
DTX-DRX timing information		Not present	Rel-7
DTX-DRX Information		Not present	Rel-7
HS-SCCH less Information		Not present	Rel-7
Maximum allowed UL TX power		Not Present	
Uplink DPCH info	A1		
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- Spreading factor		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- TFCI existence		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of FBI bit		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Puncturing Limit		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Number of TPC bits		Not present	Rel-7
Uplink DPCH info	A2		Rel-6
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		3	
- Δ_{NACK}		3	
- Ack-Nack repetition factor		1	
- HARQ_preamble_mode		0	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		0	

Information Element	Condition	Value/remark	Version
- spreading factor		Not Present	
- TFCI existence		FALSE	
- Number of FBI bit		Not Present	
- Puncturing Limit		Not Present	
Uplink DPCH info	A4	Not Present	Rel-7
E-DCH Info	A1	Not Present	Rel-6
E-DCH info	A2		Rel-6
	A5		Rel-8
- MAC-es/e reset indicator		TRUE	
- E-DPCCH info			
- E-DPCCH/DPCCH power offset		0	
- Happy bit delay condition		100 ms	
- E-TFC Boost Info		Not present	Rel-7
- E-DPDCH power interpolation		Not present	Rel-7
- E-DPDCH info			
- E-TFCI table index		0	
- E-DCH minimum set E-TFCI		9	
- Reference E-TFCIs		2 E-TFCIs	
- Reference E-TFCI		11	
- Reference E-TFCI PO		4	
- Reference E-TFCI		83	
- Reference E-TFCI PO		16	
- Maximum channelisation codes		2sf4	
- PLnon-max		0.84	
- Scheduling Information Configuration			
- Periodicity for Scheduling Info – no grant		Not present	
- Periodicity for Scheduling Info – grant		Not present	
- Power Offset for Scheduling Info		0	
- 3-Index-Step Threshold		Not present	
- 2-Index-Step Threshold		Not present	
- Scheduled Transmission configuration			
- 2ms scheduled transmission grant HARQ process allocation		Not present	
- Serving Grant		Not present	
- UL 16QAM settings		Not present	Rel-7
E-DCH info	A4	Not Present	Rel-7
Downlink HS-PDSCH Information	A1	Not Present	Rel-6
Downlink HS-PDSCH Information	A2		Rel-6
	A5		Rel-8
	A3		Rel-7
- HS-SCCH Info			
- CHOICE mode		FDD	
- DL Scrambling Code		Not present	
- HS-SCCH Channelisation Code Information			
- HS-SCCH Channelisation Code		7	
- Measurement Feedback Info			
- CHOICE mode		FDD	
- POhsdsch		6 dB	
- CQI Feedback cycle, k		4 ms	
- CQI repetition factor		1	
- Δ_{CQI}		5 (corresponds to 0dB in relative power offset)	
- CHOICE mode		FDD (no data)	
- Downlink 64QAM configured		Not present	Rel-7
- HS-DSCH TB size table		Not present	Rel-7
Downlink HS-PDSCH Information	A4		Rel-7
- HS-SCCH Info			
- CHOICE mode		FDD	
- DL Scrambling Code		Not present	
- HS-SCCH Channelisation Code Information		Use 1 HS-SCCH	
- HS-SCCH Channelisation		7	

Information Element	Condition	Value/remark	Version
Code			
- Measurement Feedback Info		Not Present	
- CHOICE mode		FDD (no data)	
- Downlink 64QAM configured		Not present	Rel-7
- HS-DSCH TB size table		Octet aligned	Rel-7
Downlink information common for all radio links	A1		
- Downlink DPCH info common for all RL			
- Timing Indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- CHOICE mode		FDD	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Fixed or Flexible Position		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- TFCI existence		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CHOICE SF		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
Downlink information common for all radio links	A2		Rel-6
	A3		Rel-7
- Downlink F-DPCH info common for all RL			
- Timing Indication		Initialise	
- Downlink F-DPCH power control information			
- DPC mode		0 (single)	
- TPC command error rate target		0.04	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		TRUE	
Downlink information common for all radio links	A4		Rel-7
- Downlink F-DPCH info common for all RL			
- Timing Indication		Initialise	
- Downlink F-DPCH power control information		Not Present	
- TPC command error rate target		0.04	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		TRUE	
Downlink information for each radio links list	A1		
- Downlink information for each radio links			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Reference to clause 6.1 "Default settings (FDD)"	

Information Element	Condition	Value/remark	Version
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-6
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value: Default DPCH Offset Value mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		1	
- Spreading factor		According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Code number		0	
- Scrambling code change		Not Present	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A2		Rel-6
	A3		Rel-7
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		TRUE	
- Downlink DPCH info for each RL		Not Present	
- Downlink F-DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- F-DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- F-DPCH slot format		Not Present	Rel-7
- Secondary CPICH info		Not Present	
- Secondary scrambling code		Not Present	
- Code number		12	
- TPC combination index		0	
- E-AGCH Info			
- E-AGCH Channelisation Code		10	
- CHOICE E-HICH Information			
- E-HICH Information			
- Channelisation code		4	
- Signature sequence		1	
- CHOICE E-RGCH Information			
- E-RGCH Information			
- Signature Sequence		0	

Information Element	Condition	Value/remark	Version
- RG combination index		0	
Downlink information for each radio link list	A4		Rel-7
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- Downlink F-DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- F-DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- F-DPCH slot format		Not Present	Rel-7
- Secondary CPICH info		Not Present	
- Secondary scrambling code		Not Present	
- Code number		12	
- TPC combination index		0	
- E-AGCH Info		Not Present	
- E-HICH Information		Not Present	
- E-RGCH Information		Not Present	

Condition	Explanation	Version
A1	This IE is needed for “Stand-alone SRBs mapped on DCH/DCH”	
A2	This IE is needed for "Stand-alone SRBs mapped on E-DCH and HS-DSCH "	Rel-6
A3	This IE is needed for "Stand-alone SRBs mapped on E-DCH and HS-DSCH using MAC-ehs”	Rel-7
A4	This IE is needed for "Stand-alone SRBs mapped on RACH and HS-DSCH using MAC-ehs” for HS-DSCH reception in CELL_FACH	Rel-7
A5	This IE is needed for "Stand-alone SRBs mapped on E-DCH using MAC-i/is and HS-DSCH using MAC-ehs”	Rel-8
NOTE: If not specified, then A1 will be the default condition		

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present (Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	0000 0000 0000 0001B	
New H-RNTI	Not present	Rel-6
New Primary E-RNTI	Not present	Rel-6
New Secondary E-RNTI	Not present	Rel-6
RRC state indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	TRUE	
- UE radio access TDD capability update requirement	FALSE	
- UE radio access 3.84 Mcps TDD capability update requirement	FALSE	Rel-4
- UE radio access 1.28 Mcps TDD capability update requirement	FALSE	Rel-4

Information Element	Value/remark	Version
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	timerBasedNoExplicit : dt50	
- SDU discard mode	Not present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7 bit	Rel-6
- One sided RLC re-establishment	FALSE	Rel-6
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	1	
- CHOICE RLC size list	Explicit list	
- RLC size index	According to clause 6.10.2.4.4.1	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	

Information Element	Value/remark	Version
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- DL RLC PDU size	According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	Rel-6
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit list	
- RLC size index	According to clause 6.10.2.4.4.1	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	

Information Element	Value/remark	Version
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- DL RLC PDU size	According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	Rel-6
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL DCH Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit list	
- RLC size index	According to clause 6.10.2.4.4.1	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	

Information Element	Value/remark	Version
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- DL RLC PDU size	According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	Rel-6
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit list	
- RLC size index	According to clause 6.10.2.4.4.1	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE Mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	

Information Element	Value/remark	Version
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete	
- TFCS complete reconfigure		
- CHOICE CTFC Size	2bit CTFC	
- CTFC information	This IE is repeated for TFC numbers according to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- CTFC	According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer)	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Gain factor ßc	11 (below 64 kbps) 9 (equal or higher than 64 kbps) when HSDPA is not configured 9 (equal or higher than 64 kbps and below 384 kbps) when HSDPA is also configured 6 (equal or higher than 384 kbps) when HSDPA is also configured (Not Present if the above is set to Computed Gain Factors)	
- Gain factor ßd	15 (Not Present if the above is set to Computed Gain Factors)	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset Pp-m	Not Present	
Added or Reconfigured TrCH information list	TS 25.331 specifies that "Although this IE is not required when the IE "RRC state indicator" is set to "CELL_FACH", need is MP to align with ASN.1"	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC Size	bitMode sizeType2 {part1 2, part2 OMIT} This results in an RLC size of 144 bits	
- Number of TBs and TTI List	List with two entry	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	0	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	1	
- CHOICE Logical channel List	ALL	
- Semi-static Transport Format information		
- Transmission time interval	40 ms	
- Type of channel coding	Convolutional	
- Coding Rate	1/3	
- Rate matching attribute	-170	
- CRC size	16	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	FDD	
- CHOICE DL parameters	Same as UL	
Added or Reconfigured TrCH information list	TS 25.331 specifies that "Although this IE is not required when the IE "RRC state indicator" is set to "CELL_FACH", need is MP to align with ASN.1"	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink Transport channel type	DCH	
- UL TrCH identity	5	
- DCH quality target	Not Present	
Frequency info	Not present	

Information Element	Value/remark	Version
Maximum allowed UL TX power	Not present	
CHOICE channel requirement	Not Present	
E-DCH Info	Not Present	Rel-6
Downlink HS-PDSCH Information	Not Present	Rel-6
Downlink information common for all radio links	Not Present	
Downlink information for each radio link list	Not present	

Contents of RRC CONNECTION SETUP COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION SETUP message.
START list	This IE is checked to see if it is present.
UE radio access capability	Not checked
UE radio access capability extension	Not checked
UE system specific capability	Not checked
Deferred measurement control reading	Not Present for Rel-7 or later, otherwise Not checked

Contents of RRC STATUS message: AM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Identification of received message	Not Checked
Protocol error information	
- Protocol error cause	Refer to test requirement.

Contents of SECURITY MODE COMMAND message: AM

Information Element	Condition	Value/remark
Message Type	A1, A2	
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3
Integrity check info		
- Message authentication code		Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message Sequence Number		Set to an arbitrarily selected integer between 0 and 15
Security capability		
- Ciphering algorithm capability		
- UEA0		If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE.
- UEA1		If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE.
- Spare		Spare 2-15 = FALSE
- Integrity protection algorithm capability		0000000000000010B (UIA1)
- UIA1		TRUE
- Spare		Spare 0 and Spare 2-15 = FALSE
Ciphering mode info		This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the

Information Element	Condition	Value/remark
		values of the sub IEs as stated below. Else, this IE is omitted.
- Ciphering mode command		Start/restart
- Ciphering algorithm		UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message.
- Ciphering activation time for DPCH		Not Present
- Radio bearer downlink ciphering activation time info		
- Radio bearer activation time		
- RB identity		1
- RLC sequence number		Current RLC SN
- RB identity		2
- RLC sequence number		Current RLC SN+2
- RB identity		3
- RLC sequence number		Current RLC SN
- RB identity		4
- RLC sequence number		Current RLC SN
Integrity protection mode info		
- Integrity protection mode command		Start
- Downlink integrity protection activation info		Not Present
- Integrity protection algorithm		UIA1
- Integrity protection initialisation number		SS selects an arbitrary 32 bits number for FRESH
CN domain identity		CS or PS
UE system specific security capability	A1	Not Checked
UE system specific security capability	A2	
- Inter-RAT UE security capability		
- CHOICE *system*		GSM
- GSM security capability		The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message.

Condition	Explanation
A1	UE not supporting GSM
A2	UE supporting GSM

Contents of SECURITY MODE COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked.
Radio bearer uplink ciphering activation time info	If ciphering is not activated in SECURITY MODE COMMAND message, this IE must be absent. Else, SS checks this IE for the presence of activation times for all ciphered uplink RLC-UM and RLC-AM RBs.

Contents of SECURITY MODE FAILURE message: AM

Information Element	Value/remark

Message Type	
RRC transaction identifier	Checked to see if the value is the identical to the same IE in the downlink SECURITY MODE COMMAND message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Refer to test requirement.

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6	Not Present	
Delay restriction flag	A1, A2, A3, A4, A5, A6	Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6	Not Present	R99 and Rel-4 only
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
UE Mobility State Indicator		Not Present	Rel-7
UTRAN DRX cycle length coefficient	A1, A2, A3, A4,A5,A6	Not Present	
CN information info		Not Present	
URA identity		Not Present	
RNC support for change of UE capability		Not Present	Rel-7
Reconfiguration in response to requested change of UE capability		Not Present	Rel-7
Downlink counter synchronization info		Not Present	
UL Transport channel information for all transport channels	A1, A2, A5, A6	Not Present	
UL Transport channel information for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from	

Information Element	Condition	Value/remark	Version
		clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set	
- CTFC		Reference to clause 6.10.2.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Gain factor βc		11 (equal or below 64 kbps) when HSDPA is not configured 9 (equal or higher than 64 kbps and below 384 kbps) when HSDPA is also configured 6 (equal or higher than 384 kbps) when HSDPA is also configured 9 (higher than 64 kbps) (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- Gain factor βd		15 (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P p-m		Not Present	
Added or Reconfigured UL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured UL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A3	(DCH for DTCH)	
- Uplink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
CHOICE *mode*	A1,A2,A3,A4,A5,A6	Not Present	
DL Transport channel information common for all transport channel	A1, A2, A5,A6	Not Present	
DL Transport channel information common for all transport channel	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Explicit	
- DL DCH TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.2.4	
- CTFC		Reference to clause 6.10.2.4 Parameter Set	
- Power offset information		Not Present	
Added or Reconfigured DL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured DL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit Except for RAB with the symmetric DL and UL rate: Same as UL	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A3		
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit Except for RAB with the symmetric DL and UL rate: Same as UL	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Frequency info	A1,A2,A3,A4,A5		
- UARFCN uplink (Nu)		Not present Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11]	
- UARFCN downlink (Nd)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX Information		Not Present	Rel-7
HS-SCCH less Information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power	A1, A2, A3, A4, A5, A6	33dBm	
CHOICE *channel requirement*	A5, A6	Not Present	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
-Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present(1)	
- spreading factor		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- Number of FBI bit		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Number of TPC bits		Not Present	Rel-7
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
E-DCH Info		Not Present	Rel-6
CHOICE Mode	A1, A2, A3, A4, A5, A6	FDD	R99 and Rel-4 only
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indicator		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position		Reference to clause 6.10 Parameter Set	
- TFCI existence		Reference to clause 6.10 Parameter Set	
- CHOICE SF		Reference to clause 6.10 Parameter Set	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information for each radio link list	A1, A2, A3		
- Downlink information for each radio links			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-

Information Element	Condition	Value/remark	Version
			4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Power offset $P_{Pilot-DPDCH}$		0	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		4	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Code number		0	
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
Downlink information for each radio link list	A4		
- Downlink information for each radio link			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value: Default DPCH Offset Value mod 38 400	
- Power offset $P_{Pilot-DPDCH}$		0	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		4	
- Spreading factor		Reference to clause 6.10 Parameter Set	
- Code number		0	
- Scrambling code change		Set to value Default1: No code change (if the UE has a compressed mode pattern sequence configured in variable TGPS_IDENTITY or included in the message including IE "Downlink DPCH info for each RL", which is using compressed mode method "SF/2") Set to value Default2: OMIT (otherwise)	
- TPC combination index		0	

Information Element	Condition	Value/remark	Version
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A5		
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to the Default setting in clause 6.1 (FDD)	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL		Not present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- Downlink information for each radio link	A6	Not Present	
MBMS PL Service Restriction Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked
CHOICE mode	FDD
Deferred measurement control reading	Not present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the

	downlink TRANSPORT CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH)

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.
CHOICE mode	FDD
DPCH/PUSCH TFCS in Uplink	
- CHOICE *Subset representation*	Allowed transport format combination list
- Allowed Transport format combination	0 (The TFC is constructed from ALL TF0)
Activation time for TFC subset	Not Present
TFC Control duration	Not Present

Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of UE CAPABILITY ENQUIRY message: AM or UM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.
Capability update requirement	
- UE radio access FDD capability update requirement	TRUE
- UE radio access TDD capability update requirement	FALSE
- System specific capability update requirement list	Not Present

Contents of UE CAPABILITY INFORMATION message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message.

Information Element	Value/remark
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
UE radio access capability	Value will be checked. Stated capability must be compatible with 3GPP TS 34.123-2 [3] (ICS statements) and the user settings
- Access stratum release indicator	
- PDCP Capability	
- RLC Capability	
- Transport channel capability	
- RF Capability FDD	
- RF Capability TDD	
- Physical channel capability	
- UE multi-mode/multi-RAT capability	
- Security Capability	
- UE positioning Capability	
- Measurement capability	
UE radio access capability extension	Value will be checked. Stated capability must be compatible with 3GPP TS 34.123-2 [3] (ICS statements) and the user settings
UE system specific capability	Not Checked

Contents of UE CAPABILITY INFORMATION CONFIRM message: AM or UM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Set to the same value as received in the UE CAPABILITY INFORMATON message.
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.

Contents of URA UPDATE message: TM

Information Element	Value/remark
Message Type	
U-RNTI	
- SRNC identity	0000 0000 0001B
- S-RNTI	0000 0000 0000 0000 0001B
RRC transaction identifier	Checked to see if it is absent
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
URA update cause	See the test content
Protocol error indicator	Checked to see if it is absent or set to 'FALSE'
HS-PDSCH in CELL_PCH and URA_PCH	Not checked
HS-PDSCH in CELL_FACH	Not checked
Protocol error information	Checked to see if it is absent

Contents of URA UPDATE CONFIRM message: UM

Information Element	Value/remark
Message Type	
U-RNTI	If this message is sent on CCCH, use the following values.

	Else, this IE is absent.
- SRNC identity	0000 0000 0001B
- S-RNTI	0000 0000 0000 0000 0001B
RRC transaction identifier	Arbitrarily selects and integer between 0 and 3
Integrity check info	
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
Integrity protection mode info	Not Present
Ciphering mode info	Not Present
New U-RNTI	Not Present
New C-RNTI	Not Present
RRC state indicator	URA_PCH
UTRAN DRX cycle length coefficient	3
CN information info	Not Present
URA identity	Not Present
RNC support for change of UE capability	Not Present
Downlink counter synchronization info	Not Present

Contents of UPLINK DIRECT TRANSFER message: AM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
CN domain identity	Checked to see if set to a CN domain for which a signalling connection exists
NAS message	Set according to that indicated in specific message content clause
Measured results on RACH	Not checked

Contents of UTRAN MOBILITY INFORMATION message: AM or UM

Information Element	Value/remark
Message Type	
Integrity check info	
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity protection mode info	Not Present
Ciphering mode info	Not Present
New U-RNTI	See the test content
New C-RNTI	See the test content
New H-RNTI	Not Present
UE Timers and constants in connected mode	
- T301	2 000 milliseconds
- N301	2
- T302	4 000 milliseconds
- N302	3
- T304	1 000 milliseconds
- N304	3
- T305	60 minutes
- T307	50 seconds
- T308	320 milliseconds
- T309	8 seconds
- T310	320 milliseconds
- N310	5

Information Element	Value/remark
- T311	500 milliseconds
- T312	5 seconds
- N312	200
- T313	10 seconds
- N313	200
- T314	20 seconds
- T315	30 seconds
- N315	200
- T316	50 seconds
- T317	1 800 seconds
CN information info	Not Present
URA identity	Not present
RNC support for change of UE capability	Not Present
Downlink counter synchronization info	Not Present

Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked
Deferred measurement control reading	Not Present for Rel-7 or later, otherwise Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not present

Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure Cause	Checked to see if it meets test requirement

9.1.2 Default Message Contents for Signalling (TDD)

Contents of RRC STATUS message: AM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Identification of received message	Not checked
Protocol error information	
- Protocol error cause	Refer to test requirement.

Contents of SECURITY MODE FAILURE message: AM

Information Element	Value/remark
Message Type	
UE information elements	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Refer to test requirement.

Contents of URA UPDATE message: TM

Information Element	Value/remark
Message Type	
U-RNTI	Checked to see if it is set to the following values
- SRNC identity	0000 0000 0001B
- S-RNTI	0000 0000 0000 0000 0001B
RRC transaction identifier	Checked to see if it is absent
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
URA update cause	See the test content
Protocol error indicator	Checked to see if it is absent or set to 'FALSE'
Protocol error information	Checked to see if it is absent

Contents of URA UPDATE CONFIRM message: UM

Information Element	Value/remark
Message Type	
U-RNTI	If this message is sent on CCCH, use the following values. Else, this IE is absent.
- SRNC identity	0000 0000 0001B
- S-RNTI	0000 0000 0000 0000 0001B
RRC transaction identifier	Arbitrarily selects and integer between 0 and 3
Integrity check info	
- Message authentication code	Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message Sequence Number	Set to an arbitrarily selected integer between 0 and 15

Information Element	Value/remark
Integrity protection mode info	Not present
Ciphering mode info	Not present
New U-RNTI	Not present
New C-RNTI	Not present
RRC State Indicator	URA_PCH
UTRAN DRX cycle length coefficient	3
CN Information info	Not present
URA identity	See the test content
Downlink counter synchronization info	Not present

Contents of UPLINK DIRECT TRANSFER message: AM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
CN domain identity	Checked to see if set to a CN domain for which a signalling connection exists
NAS message	Set according to that indicated in specific message content for each test case
Measured results on RACH	Not checked

Contents of UTRAN MOBILITY INFORMATION message: AM or UM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message Sequence Number	Set to an arbitrarily selected integer between 0 and 15
RRC transaction identifier	Arbitrarily selects and integer between 0 and 3
Integrity protection mode info	Not present
Ciphering mode info	Not present
New U-RNTI	See the test content
New C-RNTI	See the test content
UE Timers and constants in connected mode	
- T301	2 000 milliseconds
- N301	2
- T302	4 000 milliseconds
- N302	3
- T304	1 000 milliseconds
- N304	3
- T305	60 minutes
- T307	50 seconds
- T308	320 milliseconds
- T309	8 seconds
- T310	320 milliseconds
- N310	5
- T311	500 milliseconds
- T312	5 seconds
- N312	200
- T313	10 seconds
- N313	200
- T314	20 seconds
- T315	30 seconds
- N315	200
- T316	50 seconds
- T317	1 800 seconds
CN Information info	Not present

Information Element	Value/remark
URA identity	Not present
Downlink counter synchronization info	Not present

Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked
COUNT-C activation time	Not checked
Radio bearer uplink ciphering activation time info	Not checked
Uplink counter synchronization info	Not checked

Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure Cause	Checked to see if it meets test requirement

Contents of UE CAPABILITY ENQUIRY message

Information Element	Value/remark
Message Type	UE CAPABILITY ENQUIRY
Integrity check info	Not Present
- Message authentication code	If present, SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	If present, SS provides the value of this IE, from its internal counter.
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Capability update requirement	
- UE radio access FDD capability update requirement	FALSE
- UE radio access 3.84 Mcps TDD capability update requirement	FALSE
- UE radio access 1.28 Mcps TDD capability update requirement	TRUE
- System specific capability update requirement list	Not Present

Contents of UE CAPABILITY INFORMATION message (1.28 Mpcs TDD)

Information Element	Value/remark
Message Type	UE CAPABILITY INFORMATION
Integrity check info	Not Present
- Message authentication code	If present, SS calculates the value of MAC-I for this message

Information Element	Value/remark
	and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	If present, SS provides the value of this IE, from its internal counter.
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message.
UE radio access capability	Present
- Access stratum release indicator	Rel-5
- DL capability with simultaneous HS-DSCH configuration	Not Present
- PDCP capability	
- Support for lossless SRNS relocation	TRUE
- Support for RFC2507	TRUE
- Max HC context space	512
- Support for RFC3095	FALSE
- RLC capability	
- Total RLC AM buffer size	150
- Maximum RLC AM Window Size	2 047
- Maximum number of AM entities	30
- Transport channel capability	
- Downlink transport channel capability information elements	
- Max number of bits received	640
- Max convolutionally coded bits received	6 400
- Max turbo coded bits received	6 400
- Max number of simultaneous transport channels	8
- Maximum number of simultaneous CCtrCH	1
- Max number of received transport blocks	32
- Max number of TFC	128
- Max number of TF	64
- Turbo decoding supported	TRUE
- Uplink transport channel capability information elements	
- Max number of bits transmitted	6 400
- Max convolutionally coded bits transmitted	6 400
- Max turbo coded bits transmitted	6 400
- Max number of simultaneous transport channels	8
- Max number of simultaneous CCTrCH of DCH	1
- Max number of transmitted transport blocks	16
- max number of TFC	64
- Max number of TF	32
- Turbo coding supported	TRUE
- RF capability FDD	Not Present
- RF capability TDD	Present
- UE power class	1
- Radio frequency bands	a
- Chip rate capability	1.28 Mcps
- Physical channel capability	
-Downlink physical channel capability information	
- FDD physical channel capability	Not Present
- 3.84 Mcps TDD downlink physical channel capability	Not Present
- 1.28 Mcps TDD downlink physical channel capability	Present
- maxTS per subFrame	6
- max physical channel per frame	96
- min. SF	16
- Support of PDSCH	FALSE
- Support of HS-PDSCH	Unsupported
- max. physical channel per TS	16
- Support of 8psk	FALSE
-Uplink physical channel capability information	
- FDD physical channel capability	Not Present
- 3.84 Mcps TDD uplink physical channel capability	Not Present
- 1.28 Mcps TDD uplink physical channel capability	Present
- maxTS per subFrame	6
- max physical channel per timeslot	2

Information Element	Value/remark
- min. SF	16
- Support of PDSCH	FALSE
- max. physical channel per TS	16
- Support of 8psk	FALSE
- UE multi-mode/multi-RAT capability	
- MultiRAT capability List	
- Support of GSM	FALSE
- Support of Multicarrier	TRUE
- MultiMode capability	TDD
- Support of UTRAN to GERAN NACC	FALSE
- Security capability	
- Ciphering algorithm capability	
- UEA0	FALSE
- UEA1	FALSE
- Spare	FALSE
- Integrity protection algorithm	
- UIA1	FALSE
- Spare	FALSE
- UE positioning capability	
- Standalone location method(s) supported	FALSE
- UE based OTDOA supported	FASLE
- Network Assisted GPS support	None
- Support for GPS timing of cell frames measurement	FALSE
- Support for IPDL	FALSE
- Support for RX-TX time difference type2 measurement	FALSE
- Support for Up measurement validaity in CELL-PCH and URA-PCH states	FALSE
- Measurement capability	Not Present
UE system specific capability	Not present

Contents of UE CAPABILITY INFORMATION CONFIRM message

Information Element	Value/remark
Message Type	UE CAPABILITY INFORMATION
Integrity check info	Not Present
- Message authentication code	If present, SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	If present, SS provides the value of this IE, from its internal counter.
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message.

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 +	

Information Element	Condition	Value/remark	Version
Activation time	A4, A5, A6, A7, A8, A9, A10	8))MOD 256 Not Present	
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4,A5,A6	Not Present	
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronisation info		Not Present	
UL Transport channel information common for all transport channels	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
UL Transport channel information common for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE *TFCI signalling*		Normal	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.10.3.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 10.3.4 Parameter Set	
- CTFC		Reference to TS34.108 clause 6. 10.3.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- TFC subset			
- CHOICE Subset representation		Full transport format combination set	
- TFC subset list		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.10 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.10 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
Added or Reconfigured TrCH information list	A3	(DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.10 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			

Information Element	Condition	Value/remark	Version
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
CHOICE *mode*	A1,A2,A3, A4,A5,A6	TDD	
Downlink HS-PDSCH Information			Rel-5
DL Transport channel information common for all transport channels	A1, A2, A5,A6	Not Present	
DL Transport channel information common for all transport channels	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Independent	
- DL TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6. 10.3.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 10.3.4	
- CTFC		Reference to TS34.108 clause 6. 10.3.4 Parameter Set	
- Power offset information		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A3		
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
Frequency info	A1, A2, A3,		

Information Element	Condition	Value/remark	Version
- Choice mode - UARFCN (Nt)	A4, A5	TDD Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE *channel requirement* - Uplink DPCH power control info - CHOICE mode - CHOICE TDD option - UL target SIR - CHOICE UL OL PC info - CHOICE TDD option - Individual timeslot interference info - Individual timeslot interference - CHOICE TDD option - Timeslot number - TDD UL interference - Primary CCPCH Tx Power - CHOICE mode - Uplink Timing Advance Control - CHOICE Timing Advance - CHOICE TDD option	A1, A2, A3, A4	Uplink DPCH info TDD 3.84 Mcps TDD 6 Individually Signalled 3.84 Mcps TDD Reference to TS34.108 clause 6.10.3 Parameter Set (for number of TS's) 3.84 Mcps TDD As required by, Reference to TS34.108 clause 6.10.3 Parameter Set As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified - 60 dBm) 18 Integer(6..43) (-70 dBm Received if pathloss not specified) TDD Enabled 3.84 Mcps TDD	 Rel-4 Rel-4 Rel-4 Rel-4
- UL CCTrCH List - TFCS ID		 1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info - Activation time - Duration - Common timeslot info - 2nd interleaving mode - TFCI coding - Puncturing limit - Repetition period - Repetition length - Uplink DPCH timeslots and code - Dynamic SF usage - First individual timeslot info - Timeslot number - CHOICE TDD option - Timeslot number - TFCI existence - Midamble shift and burst type		 (256+CFN-(CFN MOD 8 + 8))MOD 256 Infinite Default value is "Frame" Reference to TS34.108 clause 6 Parameter set Reference to TS34.108 clause 6 Parameter set 1 null FALSE 3.84 Mcps TDD 1 OR 2 OR 3 TRUE	 Rel-4
- CHOICE TDD option - Midamble allocation mode - Midamble configuration		3.84 Mcps TTD Default midamble 16	Rel-4

Information Element	Condition	Value/remark	Version
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	TDD	
- Downlink PDSCH information		No data	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information per radio link list	A1, A2, A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	

Information Element	Condition	Value/remark	Version
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			

Information Element	Condition	Value/remark	Version
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6, A7, A8, A9, A10	Not Present	

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for "Packet to URA_PCH from CELL_FACH in PS"
A8	This IE need for "Packet to URA_PCH from CELL_DCH in PS"
A9	This IE need for "Packet to CELL_PCH from CELL_FACH in PS"
A10	This IE need for "Packet to CELL_PCH from CELL_DCH in PS"

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6, A7, A8, A9, A10	Not Present	
Delay restriction flag		Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
CHOICE *mode*		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4,A5,A6	Not Present	
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronization info		Not Present	
UL Transport channel information for all transport channels	A1, A2, A5, A6	Not Present	
UL Transport channel information for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	

Information Element	Condition	Value/remark	Version
- UL TFCS			
- CHOICE *TFCI signalling*		Normal	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- TFC subset			
- CHOICE Subset representation		Full transport format combination set	
- TFC subset list		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF	

Information Element	Condition	Value/remark	Version
		number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured TrCH information list	A3	(DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
CHOICE *mode*	A1,A2,A3,A4,A5,A6	TDD	
Downlink HS-PDSCH Information			Rel-5
DL Transport channel information common for all transport channels	A1, A2, A5,A6	Not Present	
DL Transport channel information common for all transport channel	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Independent	
- DL TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC	

Information Element	Condition	Value/remark	Version
		numbers and reference to clause 6.11.5.4	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset information		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A3		
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
Frequency info	A1, A2, A3, A4, A5		
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE *channel requirement*	A1, A2, A3, A4	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	Rel-4
- CHOICE TDD option		1.28 Mcps TDD	Rel-4
- PRXPDPCHdes		-80 Integer(-120…-58 by step of 1)	
- CHOICE UL OL PC info		Individually Signalled	
- CHOICE TDD option		1.28 Mcps TDD	
- TPC step size		1	
- Primary CCPCH Tx Power		20 Integer(6..43)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		1.28 Mcps TDD	
- Uplink synchronization parameters			
- Uplink synchronization step size		1	
- Uplink synchronization frequency		1	
- Synchronization parameters			
- SYNC_UL codes bitmap		01010101	
- FPACH info			
- Timeslot number		0	
- Channelisation code		16/15	
- Midamble Shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble Allocation Mode		Default midamble	
- Midamble configuration		16 Integer(2, 4, 6, 8, 10, 12, 14, 16)	
- WT		4 Integer(1..4)	
- PRXUpPCHdes		-80 dBm	
- SYNC_UL procedure			
- Max SYNC_UL Transmissions		2	
- Power Ramp Step		2	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length			
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	

Information Element	Condition	Value/remark	Version
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TTD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info		Not Present	Rel-7
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	TDD	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information per radio link list	A1, A2, A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	

Information Element	Condition	Value/remark	Version
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	

Information Element	Condition	Value/remark	Version
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A6, A7, A8, A9, A10	Not Present	
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"

A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for "Packet to URA_PCH from CELL_FACH in PS"
A8	This IE need for "Packet to URA_PCH from CELL_DCH in PS"
A9	This IE need for "Packet to CELL_PCH from CELL_FACH in PS"
A10	This IE need for "Packet to CELL_PCH from CELL_DCH in PS"

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6, A7, A8, A9, A10	Not Present	
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4,A5,A6	Not Present	
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronisation info		Not Present	
UL Transport channel information common for all transport channels	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
UL Transport channel information common for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE *TFCI signalling*		Normal	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.11 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 11 Parameter Set	
- CTFC		Reference to TS34.108 clause 6. 11 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- TFC subset			
- CHOICE Subset representation		Full transport format combination set	
- TFC subset list		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.11 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.11 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.11 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.11 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.11 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.11 Parameter Set	
- Number of TBs and TTI List		This IE is repeated for maxTF number	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.11 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.11 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.11 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.11 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.11 Parameter Set	
Added or Reconfigured TrCH information list	A3	(DCH for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			

Information Element	Condition	Value/remark	Version
- Transmission time interval		Reference to TS34.108 clause 6.11 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.11 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.11 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.11 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.11 Parameter Set	
CHOICE *mode*	A1,A2,A3, A4,A5,A6	TDD	
Downlink HS-PDSCH Information			Rel-5
DL Transport channel information common for all transport channels	A1, A2, A5,A6	Not Present	
DL Transport channel information common for all transport channels	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Independent	
- DL TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6. 11Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 11	
- CTFC		Reference to TS34.108 clause 6. 11 Parameter Set	
- Power offset information		Not Present	
Added or Reconfigured TrCH information list	A1, A2, A5, A6	Not Present	

Information Element	Condition	Value/remark	Version
Added or Reconfigured TrCH information list	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.11 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.11 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.11 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.11 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A3		
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.11 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.11 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.11 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.11 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Transparent mode signalling info		Not Present	
Frequency info	A1, A2, A3,		

Information Element	Condition	Value/remark	Version
- Choice mode - UARFCN (Nt)	A4, A5	TDD Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE *channel requirement* - Uplink DPCH power control info - CHOICE mode - UL target SIR - CHOICE UL OL PC info - CHOICE TDD option - Individual timeslot interference info - Individual timeslot interference - CHOICE TDD option - Timeslot number - TDD UL interference - Primary CCPCH Tx Power - CHOICE mode - Uplink Timing Advance Control - CHOICE Timing Advance - CHOICE TDD option	A1, A2, A3, A4	Uplink DPCH info TDD 6 Individually Signalled 7.68 Mcps TDD Reference to TS34.108 clause 6.11 Parameter Set (for number of TS's) 7.68 Mcps TDD As required by, Reference to TS34.108 clause 6.11Parameter Set As required by, Reference to TS34.108 clause 6.11 Parameter Set (if not specified -60 dBm) 18 Integer(6..43) (-70 dBm Received if pathloss not specified) TDD Enabled 7.68 Mcps TDD	 Rel-7 Rel-7 Rel-7
- Extended Timning Advance	-	-	
- CHOICE TDD mode		7.68 Mcps TDD	Rel-7
- Extended UL Timing Advance	-	0	Rel-7
- UL CCTrCH List - TFCS ID		 1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info - Activation time - Duration - Common timeslot info - 2nd interleaving mode - TFCI coding - Puncturing limit - Repetition period - Repetition length - Uplink DPCH timeslots and code - Dynamic SF usage - First individual timeslot info - Timeslot number - CHOICE TDD option - Timeslot number - TFCI existence - Midamble shift and burst type		 (256+CFN-(CFN MOD 8 + 8))MOD 256 Infinite Default value is "Frame" Reference to TS34.108 clause 6 .11 Parameter set Reference to TS34.108 clause 6.11 Parameter set 1 null FALSE 7.68 Mcps TDD 1 OR 2 OR 3 TRUE	 Rel-7
- CHOICE TDD option		7.68 Mcps TTD	Rel-7

Information Element	Condition	Value/remark	Version
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6.11 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6.11 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	TDD	
- Downlink PDSCH information		No data	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	Rel-4
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-4
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information per radio link list	A1, A2, A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			

Information Element	Condition	Value/remark	Version
- CHOICE mode		7.68Mcps TDD	Rel-7
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6.11 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes VHCR			Rel-7
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot channelisation codes VHCR		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	Rel-7
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			

Information Element	Condition	Value/remark	Version
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6.11 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot channelisation codes VHCR		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	Rel-7
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6, A7, A8, A9, A10	Not Present	

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for "Packet to URA_PCH from CELL_FACH in PS"
A8	This IE need for "Packet to URA_PCH from CELL_DCH in PS"
A9	This IE need for "Packet to CELL_PCH from CELL_FACH in PS"
A10	This IE need for "Packet to CELL_PCH from CELL_DCH in PS"

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	Rel-4
- Uplink Timing Advance	0	
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	Rel-4
CHOICE TDD option	1.28 Mcps TDD	Rel-4
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronization info	Not checked	

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- Extended Uplink Timing Advance	0	Rel-7
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH)

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.
CHOICE mode	TDD
- TFCS Id	
- TFCS ID	1
- Shared Channel Indicator	FALSE
DPCH/PUSCH TFCS in uplink	
- CHOICE *Subset representation*	Allowed transport format combination list
- Allowed transport format combination list	0 (The TFC is constructed from ALL TF0)
Activation time for TFC subset	Now
TFC Control duration	Not Present

Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message.

Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of RRC CONNECTION REJECT message: UM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Initial UE identity	Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message.
Rejection cause	Unspecified
Wait Time	0
Redirection info	Not Present

Contents of CELL UPDATE message: TM

Information Element	Value/remark
Message Type	
U-RNTI	Checked to see if it is set to the following values
- SRNC identity	0000 0000 0001B
- S-RNTI	0000 0000 0000 0000 0001B
RRC transaction identifier	Checked to see if it is absent
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
START List	Checked to see if the 'CN domain identity' and 'START' IEs are present for all CN domains supported by the UE
- CN domain identity	Checked to see if it is one of the supported CN domains
- START	Checked to see if it is present
AM_RLC error indication (RB2, RB3 or RB4)	Checked to see if it is set to 'FALSE'
AM_RLC error indication (RB>4)	Checked to see if it is set to 'FALSE'
Cell update cause	See the test content
Failure cause	Checked to see if it is absent
RB timer indicator	
- T314 expired	Checked to see if it is set to 'FALSE'
- T315 expired	Checked to see if it is set to 'FALSE'
Measured results on RACH	Not checked

Contents of CELL UPDATE CONFIRM message: UM

Information Element	Value/remark	Version
Message Type		
U-RNTI	If this message is sent on CCCH, use the following values. Else, this IE is absent.	
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
RRC transaction identifier	Selects an arbitrary integer between 0 to 3	
Integrity check info		
- Message authentication code	Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message Sequence Number	Set to an arbitrarily selected integer between 0 and 15	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present - use default value	

New U-RNTI	Not Present	
New C-RNTI	Not Present	
New DSCH-RNTI	Not Present	R99 and Rel-4 only
New H-RNTI	Not Present	Rel-5
CHOICE *mode*	TDD	Rel-7
- New E-RNTI	Not Present	Rel-7
RRC State indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	Not Present	
RLC re-establish indicator (RB2, RB3 and RB4)	FALSE	
RLC re-establish indicator (RB5 and upwards)	FALSE	
CN information info	Not Present	
URA identity		
-URA identity	0000 0000 0000 0001B	
RB information to release list	Not Present	
RB information to reconfigure list	Not Present	
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
UL Transport channel information common for all transport channels	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE Mode	TDD	
DL Transport channel information common for all transport channels	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present	
CHOICE channel requirement	Not Present	
E-DCH Info	Not Present	Rel-6
CHOICE mode	TDD	
Downlink information common for all radio links	Not Present	
Downlink information per radio link list	Not Present	
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Integrity check info	
- Message authentication code	Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I
- RRC Message sequence number	Set to an arbitrarily selected integer between 0 and 15
Activation time	Not Present - use default value "now"
RAB info	For each RAB to be handed over. In this version, the maximum size of the list of 1 shall be applied for all system types.
- RAB identity	0000 0001B
- CN domain identity	CS domain
- NAS Synchronization Indicator	Not present
- Re-establishment time	Use T315
CHOICE *System type*	GSM
- Frequency band	Set to "GSM/ PCS 1900" if GSM/ PCS 1900 is used in this test. Otherwise set to "GSM/DCS 1800 Band"
- CHOIC GSM message	Single GSM message
- Single GSM message	GSM HANDOVER COMMAND formatted and coded according to GSM specifications as BIT STRING (1..512). The first/ l*eftmost/ most significant* bit of the bit string contains bit 8 of the first octet of the GSM message. The contents of the HANDOVER COMMAND is to be defined in the specific test case.

Contents of HANDOVER FROM UTRAN FAILURE message: AM

Information Element/Group name	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it matches the same value used in the corresponding downlink HANDOVER FROM UTRAN COMMAND - GSM message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Inter-RAT handover failure	
- Inter-RAT handover failure cause	physical channel failure
- Protocol error information	Check to see if it is absent
Inter-system message	Not checked

Contents of MEASUREMENT CONTROL Message: AM (Intra-frequence measurement) (1.28 Mcps TDD)

Information Element	Value/remark
Message Type	
UE information elements	
RRC transaction identifier	Arbitrarily selects an unused integer between 0 to 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
Measurement information elements	
Measurement Identity	1
Measurement Command	Setup
Measurement Reporting Mode	
- Measurement Report Transfer Mode	Acknowledged mode RLC
- Periodical Reporting/Event Trigger Reporting Mode	Periodical reporting
Additional measurement list	Not Present
CHOICE Measurement type	Intra-frequency measurement
- Intra-frequency measurement	
- Intra-frequency cell info list	
- CHOICE intra-frequency cell removal	Not present
- New intra-frequency cell	
- Intra-frequency cell-id	1
- Cell info	
- Cell individual offset	0 (0dB)
- Reference time difference to cell	Not Present
- Read SFN number	FALSE
- CHOICE mode	TDD
- Primary CCPCH info	
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
-TSTD indicator	FALSE
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)
- SCTD indicator	FALSE
- Primary CCPCH Tx power	Not present
- Timeslot list	Not present
- Cells for measurement	Not present
- **Intra-frequency measurement quantity**	
- Filter coefficient	Not present (use default 0)
- CHOICE mode	TDD
- Measurement quantity list	
- Measurement quantity	Primary CCPCH RSCP
- Intra-frequency reporting quantity	
- Reporting quantities for active set cells	
- Cell synchronization information reporting indicator	FALSE
- Cell Identity reporting indicator	TRUE
- CHOICE mode	TDD
- Timeslot ISCP reporting indicator	FALSE

Information Element	Value/remark
- Proposed TGSN reporting indicator	FALSE
- Primary CCPCH RSCP reporting indicator	FALSE
- Pathloss reporting indicator	FALSE
- Reporting quantities for monitored set cells	
- Cell synchronization information reporting indicator	FALSE
- Cell Identity reporting indicator	TRUE
- CHOICE mode	TDD
- Timeslot ISCP reporting indicator	FALSE
- Proposed TGSN reporting indicator	FALSE
- Primary CCPCH RSCP reporting indicator	FALSE
- Pathloss reporting indicator	FALSE
- Reporting quantities for detected set cells	Not present
- Reporting cell status	Not present
- Measurement validity	Not present
- CHOICE report criteria	Intra-frequency measurement reporting criteria
- Parameters required for each event	
- Intra-frequency event identity	1g
- Triggering condition 1	Not present (this IE is MP only for event "1b" or "1f", TDD should not present)
- Triggering condition 2	Not present (this IE is MP only for event "1c", TDD should not present)
- Reporting Range Constant	Not present (this IE is MP only for event "1a" or "1b", TDD should not present)
- Cells forbidden to affect Reporting range	Not present (this IE is MP only for event "1a" or "1b", TDD should not present)
- W	Not present (this IE is MP only for event "1a" or "1b", TDD should not present)
- Hysteresis	0 (0 dBm)
- Threshold used frequency	Not present (this IE is MP only for event "1e", "1f", "1h" or "1i")
- Reporting deactivation threshold	Not present (this IE is MP only for event "1a", TDD should not present)
- Replacement activation threshold	Not present (this IE is MP only for event "1c" TDD should not present)
- Time to trigger	0 ms
- Amount of reporting	Not present (this IE is MP only for event "1a" or "1c" TDD should not present)
- Reporting interval	Not present (this IE is MP only for event "1a" or "1c", TDD should not present)
- Reporting cell status	Not present
Physical channel information elements	
DPCH Compressed mode status info	Not Present

Contents of MEASUREMENT CONTROL Message: AM (Inter-frequence measurement) (1.28 Mcps TDD)

Information Element	Value/remark
Message Type	
UE information elements	
RRC transaction identifier	Arbitrarily selects an unused integer between 0 to 3
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
Measurement information elements	
Measurement Identity	2
Measurement Command	Setup

Information Element	Value/remark
Measurement Reporting Mode	
- Measurement Report Transfer Mode	Acknowledged mode RLC
- Periodical Reporting/Event Trigger Reporting Mode	Periodical reporting
Additional measurement list	Not present
CHOICE Measurement type	Inter-frequency measurement
- Inter-frequency measurement	
- Inter-frequency cell info list	
- CHOICE inter-frequency cell removal	Not present
- New inter-frequency cell	
- Inter-frequency cell-id	4
- Frequency info	
- CHOICE mode	TDD
- UARFCN (Nt)	Reference to table 6.1.7 for cell 4
- Cell info	
- Cell individual offset	0 (0dB)
- Reference time difference to cell	Not Present
- Read SFN number	FALSE
- CHOICE mode	TDD
- Primary CCPCH info	
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
-TSTD indicator	FALSE
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 4(TDD)
- SCTD indicator	FALSE
- Primary CCPCH Tx power	Not present
- Timeslot list	Not present
- Cells for measurement	Not present
- **Inter-frequency measurement quantity**	
- CHOICE *reporting criteria*	Inter-frequency reporting criteria
- Inter-frequency reporting criteria	
- Filter coefficient	Not present (use default 0)
- CHOICE *mode*	TDD
- Measurement quantity for frequency quality estimate	Primary CCPCH RSCP
- Inter-frequency reporting quantity	
- UTRA Carrier RSSI	FALSE
- Frequency quality estimate	FALSE This parameters is not used in this release and should be set to FALSE. It shall be ignored by the UE.
- Non frequency related cell reporting quantities	
- Cell synchronization information reporting indicator	FALSE
- Cell Identity reporting indicator	FALSE
- CHOICE mode	TDD
- Timeslot ISCP reporting indicator	FALSE
- Proposed TGSN reporting indicator	FALSE
- Primary CCPCH RSCP reporting indicator	FALSE
- Pathloss reporting indicator	FASLE
- Reporting cell status	Not present
- Measurement validity	Not present
- **Inter-frequency set update**	Not present (this IE only for FDD)
- CHOICE report criteria	Inter-frequency measurement reporting criteria
- Parameters required for each event	
- Inter-frequency event identity	2b
- Threshold used frequency	-70 dBm (this IE is MP for event 2b, 2d, or 2f Ranges used depend on measurement quantity. CPICH Ec/No -24..0dB CPICH/Primary CCPCH RSCP -115..-25dBm)
- W used frequency	0 (0) (this IE is MP for event 2a, 2b, 2d or 2f Real(0, 0.1..2.0 by step of 0.1))
- Hysteresis	2 (1 dBm)
- Time to trigger	5 000 ms
- Reporting cell status	Within active set or within virtual active set or of the other RAT

Information Element	Value/remark
- Maximum number of reporting cells	1
- Parameters required for each non-used frequency	
- Threshold non used frequency	-70 dBm (this IE is MP for event 2a, 2b, 2c or 2e Ranges used depend on measurement quantity. CPICH Ec/No -24..0dB CPICH/Primary CCPCH RSCP -115..-25dBm. This IE is not needed if the IE "Inter-frequency event identity" is set to 2a. However, it is specified to be mandatory to align with the ASN.1)
- W non-used frequency	0 (0) (this IE is MP if 2a, 2b, 2c or 2e Real(0, 0.1..2.0 by step of 0.1))
Physical channel information elements	
DPCH Compressed mode status info	Not Present

Contents of MEASUREMENT CONTROL FAILURE Message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Checked to see if it's set to the identical value for the same IE in the downlink MEASUREMENT CONTROL message
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	See the test content

Contents of MEASUREMENT REPORT message: AM (intra-frequency measurement (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results		
- Intra-frequency measured results		
- Cell measured results		
- Cell Identity	Checked that this IE is present	
- Cell synchronisation information	Checked that this IE is absent	
- CHOICE mode	TDD	
- Cell parameters Id	Different from the Default setting in TS34.108 clause 6.1 (TDD)	
- Proposed TGSN	Checked that this IE is absent	
- Primary CCPCH RSCP	Checked that this IE is absent	
- Pathloss	Checked that this IE is absent	
- Timeslot list	Checked that this IE is absent	
Measured results on RACH	Checked that this IE is absent	
Additional measured results	Checked that this IE is absent	
Event results		
- CHOICE *event result*	Intra-frequency measurement event results	
- Intra-frequency measurement event results		
- Intra-frequency event identity	Ig	
- Cell measurement event results		
- CHOICE *mode*	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0)	
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)	
- SCTD indicator	FALSE	

Contents of MEASUREMENT REPORT message: AM (intra-frequency measurement) (1.28 Mcps TDD)

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Measurement identity	1
Measured Results	
- Intra-frequency measured results	
- Cell measured results	
- Cell Identity	Checked that this IE is present
- Cell synchronization information	Checked that this IE is absent
- CHOICE mode	TDD
- Cell parameters Id	Different from the Default setting in clause 6.1 (TDD)
- Proposed TGSN	Checked that this IE is absent
- Primary CCPCH RSCP	Checked that this IE is absent
- Pathloss	Checked that this IE is absent
- Timeslot list	Checked that this IE is absent
Measured results on RACH	Checked that this IE is absent

Additional measured results	Checked that this IE is absent
Event results	
- CHOICE *event result*	Intra-frequency measurement event results
- Intra-frequency measurement event results	
- Intra-frequency event identity	Ig
- Cell measurement event results	
- CHOICE *mode*	TDD
- Primary CCPCH info	
- CHOICE mode	TDD
- CHOICE TDD option	1.28 Mcps TDD
-TSTD indicator	FALSE
- Cell parameters ID	Reference in clause 6.1.4 Default settings for cell 1(TDD)
- SCTD indicator	FALSE

Contents of MEASUREMENT REPORT message: AM (intra-frequency measurement (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results		
- Intra-frequency measured results		
- Cell measured results		
- Cell Identity	Checked that this IE is present	
- Cell synchronisation information	Checked that this IE is absent	
- CHOICE mode	TDD	
- Cell parameters Id	Different from the Default setting in TS34.108 clause 6.1 (TDD)	
- Proposed TGSN	Checked that this IE is absent	
- Primary CCPCH RSCP	Checked that this IE is absent	
- Pathloss	Checked that this IE is absent	
- Timeslot list	Checked that this IE is absent	
Measured results on RACH	Checked that this IE is absent	
Additional measured results	Checked that this IE is absent	
Event results		
- CHOICE *event result*	Intra-frequency measurement event results	
- Intra-frequency measurement event results		
- Intra-frequency event identity	Ig	
- Cell measurement event results		
- CHOICE *mode*	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0)	
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)	
- SCTD indicator	FALSE	

Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results	Checked that this IE is absent	
Measured results on RACH	Checked that this IE is absent	
Additional measured results	Checked that this IE is absent	
Event results		
- CHOICE *event result*	Inter-frequency measurement event results	
- Inter-frequency measurement event results		
- Inter-frequency event identity	2b	
- Inter-frequency cells		
- Frequency info	Reference to table 6.1.7 for cell 4	
- Non frequency related measurement event results		
- Cell measurement event results		
- CHOICE *mode*	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0)	
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)	
- SCTD indicator	FALSE	
GSM OTD reference cell	Checked that this IE is absent	Rel-4

Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results	Checked that this IE is absent	
Measured results on RACH	Checked that this IE is absent	
Additional measured results	Checked that this IE is absent	
Event results		
- CHOICE *event result*	Inter-frequency measurement event results	
- Inter-frequency measurement event results		
- Inter-frequency event identity	2b	
- Inter-frequency cells		
- Frequency info	Reference to table 6.1.7 for cell 4	
- Non frequency related measurement event results		
- Cell measurement event results		
- CHOICE *mode*	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	

Information Element	Value/remark	Version
- CHOICE TDD option	1.28 Mcps TDD	
-TSTD indicator	FALSE	
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)	
- SCTD indicator	FALSE	
GSM OTD reference cell	Checked that this IE is absent	Rel-4

Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Measurement identity	1	
Measured Results	Checked that this IE is absent	
Measured results on RACH	Checked that this IE is absent	
Additional measured results	Checked that this IE is absent	
Event results		
- CHOICE *event result*	Inter-frequency measurement event results	
- Inter-frequency measurement event results		
- Inter-frequency event identity	2b	
- Inter-frequency cells		
- Frequency info	Reference to table 6.1.7 for cell 4	
- Non frequency related measurement event results		
- Cell measurement event results		
- CHOICE *mode*	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0)	
- Cell parameters ID	Reference clause 6.1.4 Default settings for cell 1(TDD)	
- SCTD indicator	FALSE	
GSM OTD reference cell	Checked that this IE is absent	Rel-4

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6, A7, A8, A9,	Not Present	

Information Element	Condition	Value/remark	Version
New U-RNTI	A10	Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6	Not Present	
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronisation info		Not Present	
Frequency info	A1, A2, A3, A4, A5		
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		6	
- CHOICE *UL OL PC info*		Individually Signalled	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Individual timeslot interference info		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Individual timeslot interference			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		As required by, Reference to TS34.108 clause 6.10.3 Parameter Set	
- TDD UL interference		As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified -60 dBm)	
- Primary CCPCH Tx Power		18 Integer(6..43) (-70 dBm Received if pathloss not specified)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		3.84 Mcps TDD (Default)	Rel-4
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	

Information Element	Condition	Value/remark	Version
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		Null	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD (No Data)	Rel-4
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots (No Data)	
- UL CCTrCH List to Remove		Not present	
Downlink radio resources			
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	TDD	
- Downlink PDSCH information		No data	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD (No Data)	Rel-4
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD (No Data)	Rel-4
- Default DPCH Offset Value			
- CHOICE mode		TDD	

Information Element	Condition	Value/remark	Version
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information per radio link list	A1, A2,A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots (No Data)	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	

Information Element	Condition	Value/remark	Version
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List		Not Present	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6, A7, A8, A9, A10	Not Present	

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for “Packet to URA_PCH from CELL_FACH in PS”
A8	This IE need for “Packet to URA_PCH from CELL_DCH in PS”
A9	This IE need for “Packet to CELL_PCH from CELL_FACH in PS”
A10	This IE need for “Packet to CELL_PCH from CELL_DCH in PS”

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6	Now	
Delay restriction flag	A1,A2,A3,A4,A5,A6	Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3,	Not Present	

Information Element	Condition	Value/remark	Version
	A4, A5, A6		
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
CHOICE *mode*		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6	Not Present	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronization info		Not Present	
Frequency info	A1, A2, A3, A4, A5		
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement	A5, A6	Not Present	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- PRXPDPCHdes		-80 Integer(-120…-58 by step of 1)	
- CHOICE UL OL PC info		Individually Signalled	
- CHOICE TDD option		1.28 Mcps TDD	
- TPC step size		1	
- Primary CCPCH Tx Power		20 Integer(6..43)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		1.28 Mcps TDD	
- Uplink synchronization parameters			
- Uplink synchronization step size		1	
- Uplink synchronization frequency		1	
- Synchronization parameters			
- SYNC_UL codes bitmap		01010101	
- FPACH info			
- Timeslot number		0	
- Channelisation code		16/15	
- Midamble Shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble Allocation Mode		Default midamble	
- Midamble configuration		16 Integer(2, 4, 6, 8, 10, 12, 14, 16)	
- WT		4 Integer(1..4)	
- PRXUpPCHdes		-80 dBm	
- SYNC_UL procedure			
- Max SYNC_UL Transmissions		2	
- Power Ramp Step		2	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5 dB)	
		Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		Null	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			

Information Element	Condition	Value/remark	Version
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info		Not Present	Rel-7
CHOICE Mode	A1, A2, A3, A4, A5, A6	TDD	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information per radio link list	A1, A2,A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	

Information Element	Condition	Value/remark	Version
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List		Not Present	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7

Information Element	Condition	Value/remark	Version
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A6	Not Present	
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (7.68 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6, A7, A8, A9, A10	Not Present	
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4, A7, A8, A9, A10	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
RRC State indicator	A1, A2, A3, A4	CELL_DCH	
RRC State indicator	A5, A6	CELL_FACH	

Information Element	Condition	Value/remark	Version
RRC State indicator	A7, A8	URA_PCH	
RRC State indicator	A9, A10	CELL_PCH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6	Not Present	
UTRAN DRX cycle length coefficient	A7, A8, A9, A10	3	
CN information info		Not Present	
URA identity		Not Present	
Downlink counter synchronisation info		Not Present	
Frequency info	A1, A2, A3, A4, A5		
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6, A7, A8, A9, A10	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement	A5, A6, A7, A8, A9, A10	Not Present	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		6	
- CHOICE *UL OL PC info*		Individually Signalled	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Individual timeslot interference info		Reference to TS34.108 clause 6.11 Parameter Set	
- Individual timeslot interference			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		As required by, Reference to TS34.108 clause 6.11 Parameter Set	
- TDD UL interference		As required by, Reference to TS34.108 clause 6.11 Parameter Set (if not specified -60 dBm)	
- Primary CCPCH Tx Power		18 Integer(6..43) (-70 dBm Received if pathloss not specified)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Extended UL Timing Advance	-	0	Rel-7
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		Null	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Midamble allocation mode		Default midamble	

Information Element	Condition	Value/remark	Version
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD (No Data)	Rel-7
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots (No Data)	
- UL CCTrCH List to Remove		Not present	
Downlink radio resources			
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	TDD	
- Downlink PDSCH information		No data	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD (No Data)	Rel-7
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD (No Data)	Rel-7
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A5, A6, A7, A8, A9, A10	Not Present	
Downlink information per radio link list	A1, A2,A3		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	

Information Element	Condition	Value/remark	Version
- Downlink DPCH info for each RL			
- CHOICE mode		7.68 TDD	Rel-7
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes VHCR			Rel-7
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot channelisation codes VHCR		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6.11 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots (No Data)	
- UL CCTrCH TPC List		Default (is previous list or all defined UL CCTrCHs.)	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List		Not Present	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			

Information Element	Condition	Value/remark	Version
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6, A7, A8, A9, A10	Not Present	

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"
A7	This IE need for “Packet to URA_PCH from CELL_FACH in PS”
A8	This IE need for “Packet to URA_PCH from CELL_DCH in PS”
A9	This IE need for “Packet to CELL_PCH from CELL_FACH in PS”
A10	This IE need for “Packet to CELL_PCH from CELL_DCH in PS”

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	Rel-4
- Uplink Timing Advance	0	
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	

Information Element	Value/remark	Version
CHOICE TDD option	1.28 Mcps TDD	Rel-4
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronization info	Not checked	

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- Extended Uplink Timing Advance	0	Rel-7
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement

Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A2,A3, A4,A5,A6		
UE Information elements			
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1,A2,A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5,A6	Not Present MD Integer(0..255) default is ‘now”	

Information Element	Condition	Value/remark	Version
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4,	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
RRC State indicator	A1, A2, A3, A4	CELL_DCH Indicates to a UE the RRC state to be entered.	
RRC State indicator	A5, A6	CELL_FACH	
UTRAN DRX cycle length coefficient **CN information elements** CN information info **UTRAN mobility information elements** URA identity CHOICE specification mode	A1,A2,A3, A4,A5,A6	Not Present A coefficient in the formula to count the paging occasions to be used by a specific UE Not Present Not Present [FFS]	 Rel-5
RB information elements			
RAB information to reconfigure list		Not Present	
RB information to reconfigure list - RB information to reconfigure - RB identity - PDCP info - PDCP SN info - RLC info - RB mapping info - RB stop/continue - RB information to reconfigure - RB identity - PDCP info - PDCP SN info - RLC info - RB mapping info - RB stop/continue - RB information to reconfigure - RB identity - PDCP info - PDCP SN info - RLC info - RB mapping info - RB stop/continue - RB information to reconfigure - RB identity - PDCP info - PDCP SN info - RLC info - RB mapping info - RB stop/continue - RB information to reconfigure - RB identity - PDCP info - PDCP SN info - RLC info - RB mapping info - RB stop/continue	A1	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". (UM DCCH for RRC) 1 Not Present Not Present Not Present Not Present Not Present (AM DCCH for RRC) 2 Not Present Not Present Not Present Not Present Not Present (AM DCCH for NAS_DT High priority) 3 Not Present Not Present Not Present Not Present Not Present (AM DCCH for NAS_DT Low priority) 4 Not Present Not Present Not Present Not Present Not Present (TM DTCH) 10 Not Present Not Present Not Present Not Present Not Present	
RB information to reconfigure list - RB information to reconfigure - RB identity - PDCP info	A2	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". (UM DCCH for RRC) 1 Not Present	

Information Element	Condition	Value/remark	Version
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		11	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH) (This IE is needed for 12.2 kbps and 10.2 kbps)	
- RB identity		12	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A3,A4,A5, A6	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	

Information Element	Condition	Value/remark	Version
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DTCH)	
- RB identity		20	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to be affected	A1, A2, A3,A4,A5, A6	Not Present	
TrCH Information Elements			
Uplink transport channels			
UL Transport channel information for all transport channels	A1, A2, A5,A6	Not Present	
UL Transport channel information for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE *TFCI signalling*		Normal (another option “split” only for FDD)	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.10.3.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6.10.3.4 Parameter Set	
- CTFC		Reference to TS34.108 clause 6.10.3.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE *mode*		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE *mode*		TDD	
- TFC subset			

Information Element	Condition	Value/remark	Version
- CHOICE *Subset representation*		Minimum allowed Transport format combination index	
- Allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Full transport format combination set		Not present	
- TFC subset list		Not present	
Deleted TrCH information list			
Deleted UL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured UL TrCH information	A1, A2, A5,A6	Not Present	
Added or Reconfigured UL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CHOICE Logical Channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10.3 Parameter Set	
Added or Reconfigured UL TrCH information	A3	(DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	

Information Element	Condition	Value/remark	Version
- TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical Channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size		 Dedicated transport channels Reference to TS34.108 clause 6.10.3 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.10.3 Parameter Set All Reference to TS34.108 clause 6.10.3 Parameter Set Reference to TS34.108 clause 6.10.3 Parameter Set Reference to TS34.108 clause 6.10.3 Parameter Set Reference to TS34.108 clause 6.10.3 Parameter Set Reference to TS34.108 clause 6.10.3 Parameter Set	
CHOICE mode - (no data)	A1,A2,A3, A4,A5,A6	TDD	
Downlink transport channels			
DL Transport channel information common for all transport channel	A1, A2, A5, A6	Not Present	
DL Transport channel information common for all transport channel - SCCPCH TFCS - CHOICE mode - Individual DL CCTrCH information - DL TFCS Identity - TFCS ID - Shared Channel Indicator - CHOICE *DL parameters* - DL TFCS – CHOICE *TFCI signalling* - TFCI Field 1 Information	A3,A4	 Not Present TDD Independent Normal (Normal' : meaning no split in the TFCI field either 'Logical' or 'Hard')	
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information - CHOICE CTFC Size - CTFC information - CTFC - Power offset information		 Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6.10.3.4 Parameter Set. This IE is repeated for TFC numbers and reference to TS34.108 clause 6.10.3.4 Reference to TS34.108 clause 6.10.3.4 Parameter Set Not Present	
Deleted TrCH information list			
Deleted DL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured DL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type	A4	2 TrCHs(DCH for DCCH and DCH for DTCH) DCH 10 Same as UL DCH	

Information Element	Condition	Value/remark	Version
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A3		
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10.3 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10.3 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Preconfiguration	A1,A2,A3, A4,A5,A6	[FFS]	Rel-5
PhyCH information elements			
Frequency info	A1,A2,A3, A4,A5		
- CHOICE mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
Uplink radio resources			
Maximum allowed UL TX power	A1,A2,A3,	33dBm	

Information Element	Condition	Value/remark	Version
	A4,A5,A6		
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
-Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		6	
- CHOICE *UL OL PC info*		Individually Signalled	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Individual timeslot interference info		Reference to TS34.108 clause 6.10.3 Parameter Set	
- Individual timeslot interference			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		As required by, Reference to TS34.108 clause 6.10.3 Parameter Set	
- TDD UL interference		As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified -60 dBm)	
- Primary CCPCH Tx Power		18 Integer(6..43) (-70 dBm Received if pathloss not specified)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE *Timing Advance*		Enabled	
- CHOICE *TDD option*		3.84 Mcps TDD (Default)	Rel-4
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD (No Data)	Rel-4
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots (No Data)	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A5, A6	Not Present	
Downlink radio resources			

Information Element	Condition	Value/remark	Version
CHOICE Mode	A1,A2,A3, A4,A5,A6	TDD	
- Downlink PDSCH information		No date	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indicaton		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		3.84 Mcps TDD (No Data)	Rel-4
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links			
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		3.84 Mcps TDD (no Data)	Rel-4
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0	
Downlink information per radio link list	A1, A2, A3, A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	Rel-4
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID	Integer(1.8)	Identity of this CCTrCh.Default value is 1	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	Rel-4

Information Element	Condition	Value/remark	Version
- CHOICE *Burst Type* - Midamble allocation mode - Midamble configuration - Midamble Shift - CHOICE TDD option - First timeslot channelisation codes - CHOICE codes representation - Channelisation codes bitmap - CHOICE more timeslots - UL CCTrCH TPC List - UL TPC TFCS Identity - TFCS ID - Shared Channel Indicator - DL CCTrCH List to Remove - SCCPCH Information for FACH		Type 1 Default midamble 16 Not Present 3.84 Mcps TDD (No Data) Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. Bitmap Reference to TS34.108 clause 6.10 Parameter Set No more timeslots (No Data) Default (is previous list or all defined UL CCTrCHs.) 1 FALSE Not present Not Present	Rel-4 R99 and Rel-4 only
Downlink information per radio link list - Downlink information for each radio link - Choice mode - Primary CCPCH info - Choice mode - Choice TDD Option - TSTD indicator - Cell parameters ID - SCTD indicator	A5	 TDD TDD 3.84 Mcps TDD FALSE Reference clause 6.1.4 Default settings for cell 1 FALSE	 Rel-4
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6		
- Downlink information for each radio link		Not Present	

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of RADIO BEARER RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A2,A3,A4, A5,A6		
UE Information elements			
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info - message authentication code		 SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	

Information Element	Condition	Value/remark	Version
Activation time	A1,A2,A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5,A6	Not Present MD Integer(0..255) default is 'now"	
Delay restriction flag	A1,A2,A3,A4, A5,A6	Not Present	Rel-6
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4,	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
CHOICE *mode*		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1, A2, A3, A4	CELL_DCH Indicates to a UE the RRC state to be entered.	
RRC State indicator	A5, A6	CELL_FACH	
UTRAN DRX cycle length coefficient	A1,A2,A3,A4, A5,A6	Not Present A coefficient in the formula to count the paging occasions to be used by a specific UE	
CN information elements			
CN information info		Not Present	
UTRAN mobility information elements			
URA identity		Not Present	
CHOICE specification mode		[FFS]	Rel-5
RB information elements			
RAB information to reconfigure list		Not Present	
RB information to reconfigure list	A1	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	

Information Element	Condition	Value/remark	Version
- RB stop/continue		Not Present	
RB information to reconfigure list	A2	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		11	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH) (This IE is needed for 12.2 kbps and 10.2 kbps)	
- RB identity		12	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A3,A4,A5,A6	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	

Information Element	Condition	Value/remark	Version
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DTCH)	
- RB identity		20	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to be affected	A1, A2, A3,A4,A5,A6	Not Present	
TrCH Information Elements			
Uplink transport channels			
UL Transport channel information for all transport channels	A1, A2, A5,A6	Not Present	
UL Transport channel information for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE *TFCI signalling*		Normal (another option "split" only for FDD)	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	

Information Element	Condition	Value/remark	Version
- CHOICE *mode*		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE *mode*		TDD	
- TFC subset			
- CHOICE *Subset representation*		Minimum allowed Transport format combination index	
- Allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Full transport format combination set		Not present	
- TFC subset list		Not present	
Deleted TrCH information list			
Deleted UL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured UL TrCH information	A1, A2, A5,A6	Not Present	
Added or Reconfigured UL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11.5 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11.5 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11.5 Parameter Set	
- Type of channel coding		Reference to clause 6.11.5 Parameter Set	
- Coding Rate		Reference to clause 6.11.5 Parameter Set	
- Rate matching attribute		Reference to clause 6.11.5 Parameter Set	
- CRC size		Reference to clause 6.11.5 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11.5 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11.5 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11.5 Parameter Set	
- Type of channel coding		Reference to clause 6.11.5 Parameter Set	
- Coding Rate		Reference to clause 6.11.5 Parameter Set	
- Rate matching attribute		Reference to clause 6.11.5 Parameter Set	

Information Element	Condition	Value/remark	Version
- CRC size		Reference to clause 6.11.5 Parameter Set	
Added or Reconfigured UL TrCH information	A3	(DCH for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11.5 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11.5 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11.5 Parameter Set	
- Type of channel coding		Reference to clause 6.11.5 Parameter Set	
- Coding Rate		Reference to clause 6.11.5 Parameter Set	
- Rate matching attribute		Reference to clause 6.11.5 Parameter Set	
- CRC size		Reference to clause 6.11.5 Parameter Set	
CHOICE mode	A1,A2,A3,A4, A5,A6	TDD	
- (no data)			
Downlink transport channels			
DL Transport channel information common for all transport channel	A1, A2, A5, A6	Not Present	
DL Transport channel information common for all transport channel	A3,A4		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID			
- Shared Channel Indicator			
- CHOICE *DL parameters*		Independent	
- DL TFCS			
- CHOICE *TFCI signalling*		Normal (Normal' : meaning no split in the TFCI field either 'Logical' or 'Hard')	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset information		Not Present	
Deleted TrCH information list			
Deleted DL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured DL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured DL TrCH information	A4	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11.5 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11.5 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11.5 Parameter Set	
- Type of channel coding		Reference to clause 6.11.5 Parameter Set	
- Coding Rate		Reference to clause 6.11.5 Parameter Set	
- Rate matching attribute		Reference to clause 6.11.5 Parameter Set	
- CRC size		Reference to clause 6.11.5 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A3		
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11.5 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11.5 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11.5 Parameter Set	
- Type of channel coding		Reference to clause 6.11.5 Parameter Set	
- Coding Rate		Reference to clause 6.11.5 Parameter Set	
- Rate matching attribute		Reference to clause 6.11.5 Parameter Set	
- CRC size		Reference to clause 6.11.5 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Preconfiguration	A1,A2,A3,A4, A5,A6	[FFS]	Rel-5
PhyCH information elements			
Frequency info	A1,A2,A3,A4, A5		
- CHOICE mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	

Information Element	Condition	Value/remark	Version
Frequency info	A6	Not Present	
Uplink radio resources			
Maximum allowed UL TX power	A1,A2,A3,A4, A5,A6	33dBm	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
-Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	Rel-4
- $PRX_{PDPCHdes}$		Integer(-120…-58 by step of 1)	
- CHOICE *UL OL PC info*			
- Broadcast UL OL PC info		Null	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE *Timing Advance*		Enabled	
- CHOICE *TDD option*		1.28 Mcps TDD	
- Uplink synchronization parameters			
- Uplink synchronization step size		1	
- Uplink synchronization frequency		1	
- Synchronization parameters		Not Present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A5, A6	Not Present	
E-DCH Info		Not Present	Rel-7
Downlink radio resources			
CHOICE Mode	A1,A2,A3,A4, A5,A6	TDD	
- Downlink PDSCH information		No date	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		

Information Element	Condition	Value/remark	Version
- Downlink DPCH info common for all RL			
- Timing indicaton		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0	
Downlink information per radio link list	A1, A2, A3, A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Reference clause 6.1.4 Default settings for cell 1	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID	Integer(1.8)	Identity of this CCTrCh.Default value is 1	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	

Information Element	Condition	Value/remark	Version
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Reference clause 6.1.4 Default settings for cell 1	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A6		
- Downlink information for each radio link		Not Present	
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of RADIO BEARER RECONFIGURATION message: AM or UM (7.68 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A2,A3, A4,A5,A6		
UE Information elements			
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1,A2,A3	(256+CFN-(CFN MOD 8 + 8))MOD 256	

Information Element	Condition	Value/remark	Version
Activation time	A4, A5,A6	Not Present MD Integer(0..255) default is 'now"	
New U-RNTI		Not Present	
New C-RNTI	A1, A2, A3, A4,	Not Present	
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
CHOICE mode		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1, A2, A3, A4	CELL_DCH Indicates to a UE the RRC state to be entered.	
RRC State indicator	A5, A6	CELL_FACH	
UTRAN DRX cycle length coefficient	A1,A2,A3, A4,A5,A6	Not Present A coefficient in the formula to count the paging occasions to be used by a specific UE	
CN information elements			
CN information info		Not Present	
UTRAN mobility information elements			
URA identity		Not Present	
CHOICE specification mode		[FFS]	Rel-5
RB information elements			
RAB information to reconfigure list		Not Present	
RB information to reconfigure list	A1	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A2	TS25.331 specifies that "Although this IE is not always required, need is MP to	

Information Element	Condition	Value/remark	Version
		align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		10	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH)	
- RB identity		11	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(TM DTCH) (This IE is needed for 12.2 kbps and 10.2 kbps)	
- RB identity		12	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to reconfigure list	A3,A4,A5, A6	TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1".	
- RB information to reconfigure		(UM DCCH for RRC)	
- RB identity		1	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for RRC)	
- RB identity		2	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	

Information Element	Condition	Value/remark	Version
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
- RB information to reconfigure		(AM DTCH)	
- RB identity		20	
- PDCP info		Not Present	
- PDCP SN info		Not Present	
- RLC info		Not Present	
- RB mapping info		Not Present	
- RB stop/continue		Not Present	
RB information to be affected	A1, A2, A3,A4,A5, A6	Not Present	
TrCH Information Elements			
Uplink transport channels			
UL Transport channel information for all transport channels	A1, A2, A5,A6	Not Present	
UL Transport channel information for all transport channels	A3, A4		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE *TFCI signalling*		Normal (another option “split” only for FDD)	
- TFCI Field 1 Information			
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE *CTFC Size*		Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.11 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to TS34.108 clause 6.11 Parameter Set	
- CTFC		Reference to TS34.108 clause 6.11 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors (The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE *mode*		TDD	
- Gain Factor β_d		15	

Information Element	Condition	Value/remark	Version
- Reference TFC ID - CHOICE *mode*		0 Integer(0.. 3) TDD	
- TFC subset			
- CHOICE *Subset representation*		Minimum allowed Transport format combination index	
- Allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Non-allowed transport format combination list		Not present	
- Full transport format combination set		Not present	
- TFC subset list		Not present	
Deleted TrCH information list			
Deleted UL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured UL TrCH information Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical Channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical Channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size	A1, A2, A5,A6 A4	Not Present 2 TrCHs(DCH for DCCH and DCH for DTCH) DCH 5 Dedicated transport channels Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.11 Parameter Set All Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set DCH 1 Dedicated transport channels Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.11 Parameter Set All Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical Channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size	A3	(DCH for DTCH) DCH 1 Dedicated transport channels Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.11 Parameter Set All Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set	
CHOICE mode - (no data)	A1,A2,A3, A4,A5,A6	TDD	
Downlink transport channels			
DL Transport channel information common for all transport channel	A1, A2, A5, A6	Not Present	
DL Transport channel information common for all transport channel - SCCPCH TFCS - CHOICE mode - Individual DL CCTrCH information - DL TFCS Identity - TFCS ID - Shared Channel Indicator - CHOICE *DL parameters* - DL TFCS – CHOICE *TFCI signalling* - TFCI Field 1 Information	A3,A4	 Not Present TDD Independent Normal (Normal' : meaning no split in the TFCI field either 'Logical' or 'Hard')	
- CHOICE *TFCS representation*		Complete reconfiguration	
- TFCS complete reconfiguration information - CHOICE CTFC Size - CTFC information - CTFC - Power offset information		 Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6.11 Parameter Set. This IE is repeated for TFC numbers and reference to TS34.108 clause 6.11 Reference to TS34.108 clause 6.11 Parameter Set Not Present	
Deleted TrCH information list			
Deleted DL TrCH information	A1, A2, A3, A4, A5,A6	Not Present	
Added or Reconfigured TrCH information list			
Added or Reconfigured DL TrCH information	A1, A2, A5, A6	Not Present	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity	A4	2 TrCHs(DCH for DCCH and DCH for DTCH) DCH 10	

Information Element	Condition	Value/remark	Version
- CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - TFS - CHOICE Transport channel type - Dynamic transport format information - RLC Size - Number of TBs and TTI List - Dynamic transport format information - Transmission Time Interval - Number of Transport blocks - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - DCH quality target - BLER Quality value		Same as UL DCH 5 Not Present DCH 6 Explicit Dedicated transport channel Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set -20 (-2.0)	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - TFS - CHOICE Transport channel type - Dynamic transport format information - RLC Size - Number of TBs and TTI List - Dynamic transport format information - Transmission Time Interval - Number of Transport blocks - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - DCH quality target - BLER Quality value	A3	 DCH 6 Explicit Dedicated transport channel Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set -20 (-2.0)	
Preconfiguration	A1,A2,A3, A4,A5,A6	[FFS]	Rel-5
PhyCH information elements			
Frequency info - CHOICE mode - UARFCN (Nt)	A1,A2,A3, A4,A5	 TDD Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	

Information Element	Condition	Value/remark	Version
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX information		Not Present	Rel-7
HS-SCCH less information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Uplink radio resources			
Maximum allowed UL TX power	A1,A2,A3, A4,A5,A6	33dBm	
CHOICE channel requirement	A1, A2, A3, A4	Uplink DPCH info	
-Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		6	
- CHOICE *UL OL PC info*		Individually Signalled	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Individual timeslot interference info		Reference to TS34.108 clause 6.11 Parameter Set	
- Individual timeslot interference			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		As required by, Reference to TS34.108 clause 6.11 Parameter Set	
- TDD UL interference		As required by, Reference to TS34.108 clause 6.11 Parameter Set (if not specified -60 dBm)	
- Primary CCPCH Tx Power		18 Integer(6..43) (-70 dBm Received if pathloss not specified)	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE *Timing Advance*		Enabled	
- CHOICE *TDD option*		7.68 Mcps TDD (Default)	Rel-7
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD (No Data)	Rel-4
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned	

Information Element	Condition	Value/remark	Version
- CHOICE more timeslots		code matching the SF specified in TS34.108 clause 6 Parameter Set. No more timeslots (No Data)	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A5, A6	Not Present	
E-DCH info		Not Present	Rel-6
Downlink radio resources			
CHOICE Mode	A1,A2,A3, A4,A5,A6	TDD	
- Downlink PDSCH information		No date	
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6	Not Present	Rel-5
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indicaton		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		7.68 Mcps TDD (No Data)	Rel-7
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links			
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE *TDD option*		7.68 Mcps TDD (no Data)	Rel-7
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0	
Downlink information per radio link list	A1, A2, A3, A4		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		Reference clause 6.1.4 Default settings for cell 1	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		7.68 Mcps TDD	Rel-7
- DL CCTrCh List			
- TFCS ID		Identity of this CCTrCh.Default value is 1	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		empty	
- Downlink DPCH timeslots and codes			Rel-7

Information Element	Condition	Value/remark	Version
VHCR - First individual timeslot info - Timeslot number - CHOICE TDD option		 7.68 Mcps TDD	 Rel-7
- Timeslot number - TFCI existence - Midamble shift and burst type - CHOICE TDD option - CHOICE *Burst Type* - Midamble allocation mode - Midamble configuration - Midamble Shift - CHOICE TDD option - First timeslot channelisation codes VHCR - CHOICE codes representation - Channelisation codes bitmap - CHOICE more timeslots - UL CCTrCH TPC List - UL TPC TFCS Identity - TFCS ID - Shared Channel Indicator - DL CCTrCH List to Remove - SCCPCH Information for FACH		4 OR 5 OR 6 TRUE 7.68 Mcps TDD Type 1 Default midamble 8 Not Present 7.68 Mcps TDD (No Data) Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6.11 Parameter Set. Bitmap Reference to TS34.108 clause 6.11 Parameter Set No more timeslots (No Data) Default (is previous list or all defined UL CCTrCHs.) 1 FALSE Not present Not Present	 Rel-7 Rel-7 R99 and Rel-4 only
Downlink information per radio link list - Downlink information for each radio link - Choice mode - Primary CCPCH info - Choice mode - Choice TDD Option - TSTD indicator - Cell parameters ID - SCTD indicator	A5	 TDD TDD 7.68 Mcps TDD FALSE Reference clause 6.1.4 Default settings for cell 1 FALSE	 Rel-7
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6		
- Downlink information for each radio link		Not Present	
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	This IE need for "Non speech in CS"
A2	This IE need for "Speech in CS"
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type RRC transaction identifier Integrity check info	 Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION message	

- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE *TDD option*	3.84 Mcps TDD	Rel-4
- UL Timing Advance	0	
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE *TDD option*	1.28 Mcps TDD (No data)	Rel-4
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronization info	Not checked	

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION message	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked	
CHOICE mode	TDD	
- CHOICE *TDD option*	7.68 Mcps TDD	Rel-7
- Extended UL Timing Advance	0	Rel-7
COUNT-C activation time	Not checked	
Radio bearer uplink ciphering activation time info	Not checked	
Uplink counter synchronisation info	Not checked	

Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM

Information Element	Value/remark

Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded List	Not checked

Contents of RADIO BEARER RELEASE message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info - message authentication code - RRC message sequence number		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A7, A8	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6	Not Present	
	, A9, A10		Rel-5
New U-RNTI		Not Present	
New C-RNTI	A1,A2,A3,A4	Not Present	
	, A9		Rel-5
New C-RNTI	A5, A6, A7, A8	'1010 1010 1010 1010'	
	, A10		Rel-5
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10,		Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator	A1,A2, A3, A4	CELL_DCH	
	, A9		Rel-5
RRC State indicator	A5, A6, A7, A8	CELL_FACH	
	, A10		Rel-5
UTRAN DRX cycle length coefficient	A1,A2,A3,A4 ,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
CN information info		Not Present	
Signalling Connection release indication		Not Present	
URA identity		Not Present	

Information Element	Condition	Value/remark	Version
RAB information to reconfigure list		Not Present	
RB information to release list RB information to release - RB identity	A1, A7	 10	
RB information to release list RB information to release - RB identity RB information to release - RB identity RB information to release - RB identity	A2, A8	 10 11 12	
RB information to release list	A3, A4, A5, A6		
RB information to release - RB identity		 20	
RB information to release	A9, A10		Rel-5
- RB identity		25	
RB information to reconfigure list	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
RB information to be affected list	A1,A2, A3,A4,A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
Downlink counter synchronisation info	A1,A2,A3,A4 ,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
UL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted TrCH information list	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
Deleted UL TrCH Information	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
- Uplink transport channel type - Transport channel identity Deleted UL TrCH Information - Uplink transport channel type - Transport channel identity Deleted UL TrCH Information - Uplink transport channel type - Transport channel identity	 A2, A8 A2, A8	DCH 1 DCH 2 DCH 3	
Deleted TrCH information list	A4, A6	Not Present	
Added or Reconfigured TrCH information list	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Added or Reconfigured TrCH information list	A1, A2, A3, A4	TrCHs (DCH for DCCH)	
	, A9		Rel-5
Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks		 DCH 5 Dedicated transport channels Reference to TS34.108 clause 6.10 Parameter Set (This IE is repeated for TFI number.) Not present Reference to TS34.108 clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- CHOICE Logical Channel list		All (NULL)	
- Semi-static Transport Format information			
- Transmission time interval		Reference to TS34.108 clause 6.10 Parameter Set	
- Type of channel coding		Reference to TS34.108 clause 6.10 Parameter Set	
- Coding Rate		Reference to TS34.108 clause 6.10 Parameter Set	
- Rate matching attribute		Reference to TS34.108 clause 6.10 Parameter Set	
- CRC size		Reference to TS34.108 clause 6.10 Parameter Set	
CHOICE *mode*		TDD (No data)	R99 and Rel-4 only
DL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted TrCH information list			
- Deleted DL TrCH Information	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9		Rel-5
- Downlink transport channel type		DCH	
- Transport channel identity		6	
- Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		7	
- Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		8	
- Deleted DL TrCH Information	A9, A10		Rel-5
- Downlink transport channel type		HS-DSCH	
- DL HS-DSCH MAC-d flow identity		0	
Added or Reconfigured TrCH information list			
- Added or Reconfigured DL TrCH information	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
- Added or Reconfigured DL TrCH information	A1, A2, A3, A4	1 TrCHs (DCH for DCCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		Not Present	
Frequency info	A1, A2, A3, A4, A5, A7, A8		
	, A9, A10		Rel-5
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6 , A7, A8	Not Present	R99 and Rel-4 only
CHOICE *channel requirement*	A1, A2, A3, A4	Uplink DPCH info	R99 and Rel-4 only
Uplink DPCH info	A10	Not Present	Rel-5
Uplink DPCH info	A9		Rel-5
- Uplink DPCH power control info		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			

Information Element	Condition	Value/remark	Version
- TFCS ID		1	
- UL Target SIR		+20dB	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length			
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD (no data)	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info	Not Present		Rel-6
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8	TDD	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A5, A6, A7, A8 , A10	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3 , A9		Rel-5
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5

Information Element	Condition	Value/remark	Version
Downlink information common for all radio links	A4		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information per radio link list	A1, A2, A3, A4		
	, A9		Rel-5
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Downlink DPCH info for each RL	Rel-6
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		3.84 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		3.84 Mcps TDD (no data)	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots	

Information Element	Condition	Value/remark	Version
- UL CCTrCH TPC List		Default is all	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not present	Rel-6
- CHOICE E-HICH Information		Not present	Rel-6
- CHOICE E-RGCH Information		Not present	Rel-6
Downlink information per radio link list	A5 ,A7, A8		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		3.84 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- E-AGCH Info		Not present	Rel-6
- CHOICE E-HICH Information		Not present	Rel-6
- CHOICE E-RGCH Information		Not present	Rel-6
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6	Not Present	
	, A10		Rel-5
MBMS PL Service Restriction Information	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
MBMS RB list released to change transfer mode		Not Present	Rel-6

Condition	Explanation	Version
A1	This IE need for "Non speech in CS"	
A2	This IE need for "Speech in CS"	
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE need for "Non speech to CELL_FACH from CELL_DCH in CS"	
A8	This IE need for "Speech to CELL_FACH from CELL_DCH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH from CELL_DCH / HS-DSCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_FACH from CELL_DCH / HS-DSCH in PS"	Rel-5

Contents of RADIO BEARER RELEASE message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	

Information Element	Condition	Value/remark	Version
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A7, A8	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9, A10		Rel-5
Activation time	A4, A5, A6	Not Present	
New U-RNTI		Not Present	
New C-RNTI	A1,A2,A3,A4	Not Present	
	, A9		Rel-5
New C-RNTI	A5, A6, A7, A8	'1010 1010 1010 1010'	
	, A10		Rel-5
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
CHOICE *mode*		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1,A2, A3, A4	CELL_DCH	
	, A9		Rel-5
RRC State indicator	A5, A6, A7, A8	CELL_FACH	
	, A10		Rel-5
UTRAN DRX cycle length coefficient	A1,A2,A3,A4, A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
CN information info		Not Present	
Signalling Connection release indication		Not Present	
URA identity		Not Present	
RAB information to reconfigure list		Not Present	
RB information to release list	A1, A7		
RB information to release			
- RB identity		10	
RB information to release list	A2, A8		
RB information to release			
- RB identity		10	
RB information to release			
- RB identity		11	
RB information to release			
- RB identity		12	
RB information to release list	A3, A4, A5, A6		
RB information to release			
- RB identity		20	
RB information to release	A9, A10		Rel-5
- RB identity		25	
RB information to reconfigure list		Not Present	Rel-6
RB information to be affected list	A1,A2, A3,A4,A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
Downlink counter synchronization info	A1,A2,A3,A4,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
UL Transport channel information common for all transport channels	A1, A2, A3, A4	TFCS reconfigured to fit the new transport channel configuration.	

Information Element	Condition	Value/remark	Version
	, A9		Rel-5
UL Transport channel information common for all transport channels	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Deleted TrCH information list	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
Deleted UL TrCH Information	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
- Uplink transport channel type		DCH	
- Transport channel identity		1	
Deleted UL TrCH Information	A2, A8		
- Uplink transport channel type		DCH	
- Transport channel identity		2	
Deleted UL TrCH Information	A2, A8		
- Uplink transport channel type		DCH	
- Transport channel identity		3	
Deleted TrCH information list	A4, A6	Not Present	
Added or Reconfigured TrCH information list	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Added or Reconfigured TrCH information list	A1, A2, A3, A4	TrCHs (DCH for DCCH)	
	, A9		Rel-5
Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All (NULL)	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
CHOICE *mode*		TDD (No data)	
DL Transport channel information common for all transport channels	A1, A2, A3, A4,	TFCS reconfigured to fit the new transport channel configuration.	
	, A9		Rel-5
DL Transport channel information common for all transport channels	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Deleted TrCH information list			
- Deleted DL TrCH Information	A1, A2, A3, A5,A7, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		6	
- Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		7	
- Deleted DL TrCH Information	A2, A8		
- Downlink transport channel type		DCH	
- Transport channel identity		8	
Deleted TrCH information list	A4, A6	Not Present	
Deleted DL TrCH Information	A9, A10		Rel-5
- Downlink transport channel type		HS-DSCH	
- DL HS-DSCH MAC-d flow identity		0	

Information Element	Condition	Value/remark	Version
Added or Reconfigured TrCH information list			
- Added or Reconfigured DL TrCH information	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
- Added or Reconfigured DL TrCH information	A1, A2, A3, A4	1 TrCHs (DCH for DCCH)	
	, A9		Rel-5
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Frequency info	A1, A2, A3, A4, A5, A7, A8		
	, A9, A10		Rel-5
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
Maximum allowed UL TX power	A1, A2, A3, A4, A7, A8	33dBm	
Maximum allowed UL TX power	A5, A6	using the default value	
CHOICE *channel requirement*	A5, A6 , A7, A8	Not Present	
	, A10		Rel-5
CHOICE *channel requirement*	A1, A2, A3, A4	Uplink DPCH info	
	, A9		Rel-5
- Uplink DPCH power control info		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length			
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	

Information Element	Condition	Value/remark	Version
- Additional TPC-SS Symbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info		Not Present	Rel-7
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8	TDD	
	, A9, A10		Rel-5
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Downlink information common for all radio links	A1, A2, A3		
	, A9		Rel-5
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information per radio link list	A1, A2, A3, A4,		
	, A9		Rel-5
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	

Information Element	Condition	Value/remark	Version
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to clause 6.10 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A5 ,A7, A8		
	, A10		Rel-5
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A6	Not Present	
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation	Version
A1	This IE need for "Non speech in CS"	
A2	This IE need for "Speech in CS"	
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE need for "Non speech to CELL_FACH from CELL_DCH in CS"	
A8	This IE need for "Speech to CELL_FACH from CELL_DCH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH from CELL_DCH / HS-DSCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_FACH from CELL_DCH / HS-DSCH in PS"	Rel-5

Contents of RADIO BEARER RELEASE message: AM or UM (7.68 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info - message authentication code - RRC message sequence number		 SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A7, A8	(256+CFN-(CFN MOD 8 + 8))MOD 256	
Activation time	A4, A5, A6	Not Present	
	, A9, A10		Rel-5
New U-RNTI		Not Present	
New C-RNTI	A1,A2,A3,A4	Not Present	
	, A9		Rel-5
New C-RNTI	A5, A6, A7, A8	'1010 1010 1010 1010'	
	, A10		Rel-5
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10,		Rel-5
CHOICE mode		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator	A1,A2, A3, A4	CELL_DCH	
	, A9		Rel-5
RRC State indicator	A5, A6, A7, A8	CELL_FACH	
	, A10		Rel-5
UTRAN DRX cycle length coefficient	A1,A2,A3,A4 ,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
CN information info		Not Present	
Signalling Connection release indication		Not Present	

Information Element	Condition	Value/remark	Version
URA identity		Not Present	
RAB information to reconfigure list		Not Present	
RB information to release list RB information to release - RB identity	A1, A7	 10	
RB information to release list RB information to release - RB identity RB information to release - RB identity RB information to release - RB identity	A2, A8	 10 11 12	
RB information to release list	A3, A4, A5, A6		
RB information to release - RB identity		 20	
RB information to release	A9, A10		Rel-5
- RB identity		25	
RB information to reconfigure list	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
RB information to be affected list	A1,A2, A3,A4,A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
Downlink counter synchronisation info	A1,A2,A3,A4 ,A5,A6, A7, A8	Not Present	
	, A9, A10		Rel-5
UL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted TrCH information list	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
Deleted UL TrCH Information	A1,A2, A3, A5, A7, A8		
	, A9, A10		Rel-5
- Uplink transport channel type - Transport channel identity Deleted UL TrCH Information - Uplink transport channel type - Transport channel identity Deleted UL TrCH Information - Uplink transport channel type - Transport channel identity	 A2, A8 A2, A8	DCH 1 DCH 2 DCH 3	
Deleted TrCH information list	A4, A6	Not Present	
Added or Reconfigured TrCH information list	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
Added or Reconfigured TrCH information list	A1, A2, A3, A4	TrCHs (DCH for DCCH)	
	, A9		Rel-5
Added or Reconfigured UL TrCH information - Uplink transport channel type - UL Transport channel identity - TFS - CHOICE Transport channel type - Dynamic Transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks		 DCH 5 Dedicated transport channels Reference to TS34.108 clause 6.11 Parameter Set (This IE is repeated for TFI number.) Not present Reference to TS34.108 clause 6.11	

Information Element	Condition	Value/remark	Version
- CHOICE Logical Channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size		Parameter Set All (NULL) Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set Reference to TS34.108 clause 6.11 Parameter Set	
CHOICE *mode*		TDD (No data)	R99 and Rel-4 only
DL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8	TFCS reconfigured to fit the new transport channel configuration.	
	, A9, A10		Rel-5
Deleted TrCH information list			
- Deleted DL TrCH Information	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9		Rel-5
- Downlink transport channel type - Transport channel identity - Deleted DL TrCH Information - Downlink transport channel type - Transport channel identity - Deleted DL TrCH Information - Downlink transport channel type - Transport channel identity	 A2, A8 A2, A8	DCH 6 DCH 7 DCH 8	
- Deleted DL TrCH Information	A9, A10		Rel-5
- Downlink transport channel type		HS-DSCH	
- DL HS-DSCH MAC-d flow identity		0	
Added or Reconfigured TrCH information list			
- Added or Reconfigured DL TrCH information	A5, A6, A7, A8	Not Present	
	, A10		Rel-5
- Added or Reconfigured DL TrCH information	A1, A2, A3, A4	1 TrCHs (DCH for DCCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-Not Present	
Frequency info	A1, A2, A3, A4, A5, A7, A8		
	, A9, A10		Rel-5
- Choice mode - UARFCN (Nt)		TDD Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX information		Not Present	Rel-7
HS-SCCH less information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power		33dBm	
CHOICE *channel requirement*	A5, A6 , A7, A8	Not Present	R99 and Rel-4 only
CHOICE *channel requirement*	A1, A2, A3, A4	Uplink DPCH info	R99 and Rel-4 only
Uplink DPCH info	A10	Not Present	Rel-5

Information Element	Condition	Value/remark	Version
Uplink DPCH info	A9		Rel-5
- Uplink DPCH power control info		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		+20dB	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6.11 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length			
- Choice TDD option		7.68Mcps TDD	Rel-7
- Uplink DPCH timeslots and codes VHCR			Rel-7
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	
- CHOICE *Burst Type*		Type 1	
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD (no data)	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info	Not Present		Rel-6
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8	TDD	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-5
Downlink information common for all radio links	A5, A6, A7, A8 , A10	Not Present	Rel-5
Downlink information common for all radio links	A1, A2, A3 , A9		Rel-5
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	

Information Element	Condition	Value/remark	Version
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information common for all radio links	A4		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Downlink DPCH info common for all RL			
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information per radio link list	A1, A2, A3, A4		
	, A9		Rel-5
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Downlink DPCH info for each RL	Rel-6
- Downlink DPCH info for each RL			
- CHOICE mode		7.68Mcps TDD	Rel-7
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to TS34.108 clause 6.11 Parameter set	
- Puncturing limit		Reference to TS34.108 clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes VHCR			Rel-7
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Midamble allocation mode		Default midamble	
- Midamble configuration		8	
- Midamble Shift		Not Present	
- CHOICE TDD option		7.68 Mcps TDD (no data)	Rel-7
- First timeslot channelisation codes VHCR		Repeated (1,2) for each channelisation code assigned in the	Rel-7

Information Element	Condition	Value/remark	Version
		slot to meet the needs of TS34.108 clause 6 Parameter Set.	
- CHOICE codes representation		Bitmap	
- Channelisation codes bitmap		Reference to TS34.108 clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		Default is all	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not present	Rel-6
- CHOICE E-HICH Information		Not present	Rel-6
- CHOICE E-RGCH Information		Not present	Rel-6
Downlink information per radio link list	A5 ,A7, A8		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		7.68 Mcps TDD	Rel-7
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- E-AGCH Info		Not present	Rel-6
- CHOICE E-HICH Information		Not present	Rel-6
- CHOICE E-RGCH Information		Not present	Rel-6
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
Downlink information per radio link list	A6	Not Present	
	, A10		Rel-5
MBMS PL Service Restriction Information	A1,A2, A3,A4,A5, A6, A7, A8, A9, A10	Not Present	Rel-6
MBMS RB list released to change transfer mode		Not Present	Rel-6

Condition	Explanation	Version
A1	This IE need for "Non speech in CS"	
A2	This IE need for "Speech in CS"	
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE need for "Non speech to CELL_FACH from CELL_DCH in CS"	
A8	This IE need for "Speech to CELL_FACH from CELL_DCH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH from CELL_DCH / HS-DSCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_FACH from CELL_DCH / HS-DSCH in PS"	Rel-5

Contents of DOWNLINK DIRECT TRANSFER message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	0
Integrity check info	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.

CN domain identity	CS domain or PS domain
NAS message	See Specific Message Content for each test case

Contents of INITIAL DIRECT TRANSFER message: AM

Information Element	Value/remark	Version
Message Type		
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
CN domain identity	CS domain or PS domain	
Intra Domain NAS Node Selector	Set to the same octet string as in the IMSI stored in the USIM card	
NAS message	Set according to that indicated in specific message content for each test case	
START	This IE is checked to see if it is present.	
Establishment cause	See the specific test case	Rel-5
Measured results on RACH	Not checked	

Contents of MBMS GENERAL INFORMATION message: UM (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message type		Rel-6
MBMS preferred frequency information	Not Present	Rel-6
MBMS timers and counters		Rel-6
- T318	4000 ms	Rel-6
MICH configuration information		Rel-6
- MICH Power offset	-5dB	Rel-6
- CHOICE *Mode*	TDD	Rel-6
- Timeslot Number	1	Rel-6
- CHOICE TDD option	3.84 Mcps	Rel-6
- CHOICE Burst Type	Type 1	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	3.84 Mcps TDD	Rel-6
- Chanelisation code	16/1	Rel-6
- Repetition period/length	(4,2)	Rel-6
- Offset	0	Rel-6
- MBMS Notification indicator length	4	Rel-6
Cell group identity	'000000000001'	Rel-6
Default MSCH configuration information	Not Present	Rel-6
Indicate changes in MBMS Selected Services	FALSE	Rel-6

Contents of MBMS GENERAL INFORMATION message: UM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message type		Rel-6
MBMS preferred frequency information	Not Present	Rel-6
MBMS timers and counters		Rel-6
- T318	4000 ms	Rel-6
MICH configuration information		Rel-6
- MICH Power offset	-5dB	Rel-6
- CHOICE *Mode*	TDD	Rel-6
- Timeslot Number	1	Rel-6
- Midamble shift and burst type		
- CHOICE TDD option	1.28 Mcps TDD	Rel-6
- Midamble Allocation Mode	Default midamble	Rel-6

Information Element	Value/remark	Version
- Midamble configuration	16	Rel-6
- Midamble Shift	Not Present	Rel-6
- CHOICE TDD option	1.28 Mcps TDD	Rel-6
- Chanelisation code	16/15	Rel-6
- Repetition period/length	(4,2)	Rel-6
- Offset	0	Rel-6
- MBMS Notification indicator length	4	Rel-6
Cell group identity	'000000000001'	Rel-6
Default MSCH configuration information	Not Present	Rel-6
Indicate changes in MBMS Selected Services	FALSE	Rel-6

Contents of MBMS GENERAL INFORMATION message: UM (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message type		Rel-6
MBMS preferred frequency information	Not Present	Rel-6
MBMS timers and counters		Rel-6
- T318	4000 ms	Rel-6
MICH configuration information		Rel-6
- MICH Power offset	-5dB	Rel-6
- CHOICE *Mode*	TDD	Rel-6
- Timeslot Number	1	Rel-6
- CHOICE TDD option	7.68 Mcps	Rel-6
- CHOICE Burst Type	Type 1	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	7.68 Mcps TDD	Rel-6
- Chanelisation code	32/1	Rel-6
- Repetition period/length	(4,2)	Rel-6
- Offset	0	Rel-6
- MBMS Notification indicator length	4	Rel-6
Cell group identity	'000000000001'	Rel-6
Default MSCH configuration information	Not Present	Rel-6
Indicate changes in MBMS Selected Services	FALSE	Rel-6

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (3.84 Mcps)

Information Element	Value/remark	Version
Message type		Rel-6
RB information list	2 entries in the list	Rel-6
- RB identity	14	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
- RB identity	15	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	

Information Element	Value/remark	Version
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
TrCh information for each TrCh	2 entries in the list	Rel-6
- Transport channel identity	1	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
- Transport channel identity	2	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
TrCh information for each CCTrCh	2 entries in the list	Rel-6
- CCTrCH identity	1	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	
- Power offset information	**Not Present**	

Information Element	Value/remark	Version
- CCTrCH identity	2	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	
- Power offset information	**Not Present**	
PhyCh information	2 entries in list	Rel-6
- PhyCh identity	13	Rel-6
- Secondary CCPCH info MBMS		Rel-6
- CHOICE *mode*	1.28/3.84 Mcps TDD	Rel-6
- 2nd interleaving mode	Frame	Rel-6
- TFCI coding	Reference to clause 6.10 "Parameter Set"	Rel-6
- Puncturing limit	Reference to clause 6.10 "Parameter Set"	Rel-6
- Timeslot number	2	Rel-6
- TFCI information	TRUE	Rel-6
- CHOICE Burst Type	Reference to clause 6.10 "Parameter Set"	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	3.84Mcps TDD	Rel-6
- CHOICE codes representation	Consecutive codes	Rel-6
- First channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- Last channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- CHOICE more timeslots	No more timeslots	Rel-6
- Modulation	Reference to clause 6.10 "Parameter Set"	Rel-7
- PhyCh identity	17	Rel-6
- Secondary CCPCH info MBMS		Rel-6
- CHOICE *mode*	1.28/3.84 Mcps TDD	Rel-6
- 2nd interleaving mode	Frame	Rel-6
- TFCI coding	Reference to clause 6.10 "Parameter Set"	Rel-6
- Puncturing limit	Reference to clause 6.10 "Parameter Set"	Rel-6
- Timeslot number	2	Rel-6
- TFCI information	TRUE	Rel-6
- CHOICE Burst Type	Reference to clause 6.10 "Parameter Set"	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	3.84Mcps TDD	Rel-6
- CHOICE codes representation	Consecutive codes	Rel-6
- First channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- Last channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- CHOICE more timeslots	No more timeslots	Rel-6
- Modulation	Reference to clause 6.10 "Parameter Set"	Rel-7

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message type		Rel-6
RB information list	2 entries in the list	Rel-6
- RB identity	14	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
- RB identity	15	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	

Information Element	Value/remark	Version
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
TrCh information for each TrCh	2 entries in the list	Rel-6
- Transport channel identity	1	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.11 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.11 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.11 parameter set	
- Type of channel coding	Reference to clause 6.11 parameter set	
- Coding Rate	Reference to clause 6.11 parameter set	
- Rate matching attribute	Reference to clause 6.11 parameter set	
- CRC size	Reference to clause 6.11 parameter set	
- Transport channel identity	2	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.11 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.11 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.11 parameter set	
- Type of channel coding	Reference to clause 6.11 parameter set	
- Coding Rate	Reference to clause 6.11 parameter set	
- Rate matching attribute	Reference to clause 6.11 parameter set	
- CRC size	Reference to clause 6.11 parameter set	
TrCh information for each CCTrCh	2 entries in the list	Rel-6
- CCTrCH identity	1	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.11 "Parameter Set"	
- CTFC	Reference to clause 6.11 "Parameter Set"	
- Power offset information	Not Present	

Information Element	Value/remark	Version
- CCTrCH identity	2	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.11 "Parameter Set"	
- CTFC	Reference to clause 6.11 "Parameter Set"	
- Power offset information	Not Present	

Information Element	Value/remark	Version
PhyCh information	2 entries in list	Rel-6
- PhyCh identity	13	Rel-6
- Secondary CCPCH info MBMS		
- CHOICE *mode*	1.28/3.84 Mcps TDD	
- Common timeslot info MBMS		
- 2nd interleaving mode	Frame	
- TFCI coding	Reference to clause 6.11 "Parameter Set"	
- Puncturing limit	Reference to clause 6.11 "Parameter Set"	
- Downlink Timeslots and Codes		
- First Individual timeslot info		
- Timeslot number	4	
- TFCI information	TRUE	
- Midamble Shift and burst type		
- Midamble allocation mode	Default midamble	
- Midamble configuration	16	
- Midamble Shift	Not Present	
- CHOICE TDD option	1.28Mcps TDD	
- Modulation	QPSK	
- SS-TPC Symbols	1	
- Additional TPC-SS Symbols	Not Present	
- First timeslot channelisation codes		
- CHOICE codes representation	Consecutive codes	
- First channelisation code	Reference to clause 6.11 "Parameter Set"	
- Last channelisation code	Reference to clause 6.11 "Parameter Set"	
- CHOICE more timeslots	No more timeslots	
- Modulation	Reference to clause 6.11 "Parameter Set"	
- PhyCh identity	17	Rel-6
- Secondary CCPCH info MBMS		
- CHOICE *mode*	1.28/3.84 Mcps TDD	
- Common timeslot info MBMS		
- 2nd interleaving mode	Frame	
- TFCI coding	Reference to clause 6.11 "Parameter Set"	
- Puncturing limit	Reference to clause 6.11 "Parameter Set"	
- Downlink Timeslots and Codes		
- First Individual timeslot info		
- Timeslot number	4	
- TFCI information	TRUE	
- Midamble Shift and burst type		
- Midamble allocation mode	Default midamble	
- Midamble configuration	16	
- Midamble Shift	Not Present	
- CHOICE TDD option	1.28Mcps TDD	
- Modulation	QPSK	
- SS-TPC Symbols	1	
- Additional TPC-SS Symbols	Not Present	
- First timeslot channelisation codes		
- CHOICE codes representation	Consecutive codes	
- First channelisation code	Reference to clause 6.11 "Parameter Set"	
- Last channelisation code	Reference to clause 6.11 "Parameter Set"	
- CHOICE more timeslots	No more timeslots	
- Modulation	Reference to clause 6.11 "Parameter Set"	

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (7.68 Mcps)

Information Element	Value/remark	Version
Message type		Rel-6
RB information list	2 entries in the list	Rel-6
- RB identity	14	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
- RB identity	15	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	absent	
- Header compression information	Not Present	

Information Element	Value/remark	Version
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
TrCh information for each TrCh	2 entries in the list	Rel-6
- Transport channel identity	1	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
- Transport channel identity	2	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 parameter set	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 parameter set	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 parameter set	
- Type of channel coding	Reference to clause 6.10 parameter set	
- Coding Rate	Reference to clause 6.10 parameter set	
- Rate matching attribute	Reference to clause 6.10 parameter set	
- CRC size	Reference to clause 6.10 parameter set	
TrCh information for each CCTrCh	2 entries in the list	Rel-6
- CCTrCH identity	1	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	
- Power offset information	**Not Present**	

Information Element	Value/remark	Version
- CCTrCH identity	2	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.	
- CTFC information	This IE is repeated for number of CTFCs in clause 6.10 "Parameter Set"	
- CTFC	Reference to clause 6.10 "Parameter Set"	
- Power offset information	**Not Present**	
PhyCh information	2 entries in list	Rel-6
- PhyCh identity	13	Rel-6
- Secondary CCPCH info MBMS		Rel-6
- CHOICE *mode*	7.68 Mcps TDD	Rel-6
- 2nd interleaving mode	Frame	Rel-6
- TFCI coding	Reference to clause 6.10 "Parameter Set"	Rel-6
- Puncturing limit	Reference to clause 6.10 "Parameter Set"	Rel-6
- Timeslot number	2	Rel-6
- TFCI information	TRUE	Rel-6
- CHOICE Burst Type	Reference to clause 6.10 "Parameter Set"	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	3.84Mcps TDD	Rel-6
- CHOICE codes representation	Consecutive codes	Rel-6
- First channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- Last channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- CHOICE more timeslots	No more timeslots	Rel-6
- Modulation	Reference to clause 6.10 "Parameter Set"	Rel-7
- PhyCh identity	17	Rel-6
- Secondary CCPCH info MBMS		Rel-6
- CHOICE *mode*	1.28/3.84 Mcps TDD	Rel-6
- 2nd interleaving mode	Frame	Rel-6
- TFCI coding	Reference to clause 6.10 "Parameter Set"	Rel-6
- Puncturing limit	Reference to clause 6.10 "Parameter Set"	Rel-6
- Timeslot number	2	Rel-6
- TFCI information	TRUE	Rel-6
- CHOICE Burst Type	Reference to clause 6.10 "Parameter Set"	Rel-6
- Midamble allocation mode	Default	Rel-6
- Midamble configuration burst type 1 and 3	8	Rel-6
- CHOICE TDD option	3.84Mcps TDD	Rel-6
- CHOICE codes representation	Consecutive codes	Rel-6
- First channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- Last channelisation code	Reference to clause 6.10 "Parameter Set"	Rel-6
- CHOICE more timeslots	No more timeslots	Rel-6
- Modulation	Reference to clause 6.10 "Parameter Set"	Rel-7

Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM

Information Element	Condition	Value/remark	Version
Message type	A1, A2, A3		Rel-6
S-CCPCH list	A1	Not Present	Rel-6
S-CCPCH list	A2	Contains 1 S-CCPCH	Rel-6
S-CCPCH list	A3	Contains 2 S-CCPCH	Rel-6
- S-CCPCH identity	A2, A3	Not Present	Rel-6
- Secondary CCPCH info		1	Rel-6
- MBMS Soft Combining Timing Offset		Not Present	Rel-6
- TrCh information common for all TrCh		1	Rel-6
- TrCH information list			Rel-6
- TrCh information		1	Rel-6
- RB information list			Rel-6
- RB information		1	Rel-6
- MBMS short transmission ID		Refers to the index of the service in the list of services on the cell which is being provided on this RB	Rel-6
- MBMS logical channel identity		1	Rel-6
- MSCH configuration information		Not Present	Rel-6
- S-CCPCH identity	A3	Not Present	Rel-6
- Secondary CCPCH info		2	Rel-6
- MBMS Soft Combining Timing Offset		Not Present	Rel-6
- TrCh information common for all TrCh		2	Rel-6
- TrCH information list			Rel-6
- TrCh information		2	Rel-6
- RB information list			Rel-6
- RB information		2	Rel-6
- MBMS short transmission ID		Refers to the index of the service in the list of services on the cell which is being provided on this RB	Rel-6
- MBMS logical channel identity		2	Rel-6
- MSCH configuration information		Not Present	Rel-6
S-CCPCH in SIB type 5	A1, A2, A3	Not Present	Rel-6

Condition	Explanation
A1	No services ongoing or starting
A2	1 service ongoing or starting
A3	2 services ongoing or starting

Contents of MBMS MODIFIED SERVICES INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Modified services list	1 entry per modified service - maximum 12. If no services are modified in the current modification period this IE is Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	Set to the value of the service ID being modified (e.g. '000001')	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	'01'	Rel-6
- MBMS required UE action	Acquire PTM RB info	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS dispersion indicator	Not Present	Rel-6
- Continue MCCH reading	FALSE	Rel-6
MBMS re- acquire MCCH	Not Present	Rel-6
MBMS dynamic persistence level	Not Present	Rel-6
End of modified MCCH information	Not Present	Rel-6
MBMS number of neighbour cells	0	Rel-6
MBMS all unmodified p-t-m services	Not Present	Rel-6
MBMS p-t-m activation time	Set to the 11 LSB of the first SFN of the next modification period.	Rel-6

Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Unmodified services list	12 services by default. See NOTE 1.	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000001'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000002'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000003'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000004'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000005'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000006'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000007'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000008'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'000009'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6

Information Element	Value/remark	Version
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000A'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000B'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6
- MBMS Transmission identity		Rel-6
- MBMS Service ID		
- MBMS Service ID	'00000C'	Rel-6
- CHOICE PLMN identity	SameAs-MIB	Rel-6
- MBMS Session ID	Value set according to table for condition A1 or A2	Rel-6
- MBMS required UE action	Value set according to table for condition A1 or A2	Rel-6
- MBMS preferred frequency	Not Present	Rel-6

Information Element	Condition	Value/remark	Explanation
- MBMS Session ID	A1	Not Present	Condition used when the session is currently not being transmitted
- MBMS required UE action		'None'	
- MBMS Session ID	A2	'01'	Condition used when the session is currently ongoing
- MBMS required UE action		'Acquire PTM RB info'	

NOTE 1: Any service ID which is included in MBMS MODIFIED SERVICES INFORMATION in the current modification period shall be Not Present in the list of services in this message.

Contents of PAGING TYPE 1 message: TM (Speech in CS)

Information Element	Value/remark
Message Type	
Paging record list	
- Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Conversational Call
- CN domain identity	CS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of PAGING TYPE 1 message: TM (The others of speech in CS)

Information Element	Value/remark
Message Type	
Paging record list	
- Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Streaming Call
- CN domain identity	CS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card

BCCH modification info	Not Present

Contents of PAGING TYPE 1 message: TM (Packet in PS)

Information Element	Value/remark
Message Type	
Paging record list	
- Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Interactive Call
- CN domain identity	PS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A11		
	, A9, A10		Rel-5
			Rel-6
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A11	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9		Rel-5
			Rel-6
Activation time	A4, A5, A6, A7, A8 A10	Not Present	Rel-5
New U-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
			Rel-6
New C-RNTI	A1, A2, A3, A4, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
			Rel-6
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	Rel-5
New H-RNTI	A9, A10 A12, A13, A14, A15	'1010 1010 1010 1010'	Rel-5 Rel-6
Choice mode		TDD	Rel-7
- New E-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11	Not Present	Rel-7
- New E-RNTI	A12, A13, A14, A15, A16	'1010 1010 1010 1010'	Rel-7
RRC State indicator	A1, A2, A3, A4, A7, A8, A11	CELL_DCH	
	, A9, A10		Rel-5
			Rel-6
RRC State indicator	A5, A6	CELL_FACH	

Information Element	Condition	Value/remark	Version
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10 A12, A13, A14, A15, A16	Not Present	 Rel-5 Rel-7
CN information info		Not Present	
URA identity		Not Present	
CHOICE Specification mode		Complete specification	Rel-6
- Signalling RB information to setup		Not Present	
- RAB information for setup	A1, A7		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RAB information for setup	A2, A8		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	

Information Element	Condition	Value/remark	Version
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		11	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		7	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		12	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping		Not Present	

Information Element	Condition	Value/remark	Version
indicator			
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		8	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RAB information for setup	A3, A4, A5, A6		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	

Information Element	Condition	Value/remark	Version
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A9		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	

Information Element	Condition	Value/remark	Version
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping		Not Present	

Information Element	Condition	Value/remark	Version
indicator			
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A10		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	

Information Element	Condition	Value/remark	Version
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A11		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	

Information Element	Condition	Value/remark	Version
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		4	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		9	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- RAB information for setup	A12		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			

Information Element	Condition	Value/remark	Version
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel			

Information Element	Condition	Value/remark	Version
info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- RAB information for setup	A13, A14		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- DDI		5	

Information Element	Condition	Value/remark	Version
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A15		Rel-7
- RAB info		(second high-speed AM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		17	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing		1 RBMuxOption	

Information Element	Condition	Value/remark	Version
option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		8	
- E-DCH MAC-d flow identity		3	
- DDI		6	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		2	
- Logical channel identity		Not Present	
- RAB information for setup	A16		Rel-6
- RAB info		(Conversational UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		27	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		32	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		9	
- E-DCH MAC-d flow identity		4	
- DDI		7	
- RLC PDU size list		12 RLC PDU sizes	
- RLC PDU size		96 bits	

Information Element	Condition	Value/remark	Version
- RLC PDU size		112 bits	
- RLC PDU size		144 bits	
- RLC PDU size		160 bits	
- RLC PDU size		176 bits	
- RLC PDU size		192 bits	
- RLC PDU size		208 bits	
- RLC PDU size		224 bits	
- RLC PDU size		288 bits	
- RLC PDU size		296 bits	
- RLC PDU size		312 bits	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		3	
- Logical channel identity		Not Present	
RB information to reconfigure list	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	A9, A10		Rel-5
	A12, A13, A14, A15, A16		Rel-7
RB information to be affected	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10,		Rel-5
	A12		Rel-7
RB information to be affected	A13, A15		Rel-7
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	

Information Element	Condition	Value/remark	Version
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	

Information Element	Condition	Value/remark	Version
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
RB information to be affected	A14, A16		Rel-7
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
Downlink counter synchronization info	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
	A12, A13, A14, A15		Rel-7
PDCP ROHC target mode	A9, A10	Not Present	Rel-5
	A12, A13, A14, A15		Rel-7

Information Element	Condition	Value/remark	Version
UL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10		Rel-5
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	R99 and Rel-4 only
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.3.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.3.4 Parameter Set	
- CTFC		Reference to clause 6.10.3.4 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain factor βd		8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID		0	
- CHOICE mode		TDD	
- TFC subset		Not Present	
- TFC subset list		Not Present	
UL Transport channel information for all transport channels	A12		Rel-7
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((UL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors	
- CTFC		1 ((UL DCH RAB, DCCH)=(TF0, TF1))	
- Power offset information			

Information Element	Condition	Value/remark	Version
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		TDD	
- Gain factor βd		8	
- Reference TFC ID		0	
- CHOICE mode		TDD	
- TFC subset		Not Present	
- TFC subset list		Not Present	
UL Transport channel information for all transport channels	A13, A14, A15, A16	Not Present	Rel-7
Deleted UL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10 A12	Not Present	 Rel-5 Rel-7
Deleted UL TrCH information	A13, A14, A15, A16		Rel-7
- Uplink transport channel type	DCH		
- UL transport channel identity	5		
Added or Reconfigured UL TrCH information	A1, A3 A4, A5, A6, A7 , A9, A10	1 DCH added, 1 DCH reconfigured (if from cell_DCH) OR 2 DCHs added (if from cell_FACH)	 Rel-5
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A11	1 DCH added for DTCH	
- Uplink transport channel type		DCH	
- UL Transport channel identity		4	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- TFS			

Information Element	Condition	Value/remark	Version
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A12	1 E-DCH added, 1 DCH added, 1 DCH reconfigured	Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE mode		TDD	
- HARQ info for E-DCH			
- CHOICE UL parameters		E-DCH	
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow			
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
Added or Reconfigured UL TrCH information	A13, A14	1 E-DCH added with one DCCH MAC-d flow and one DTCH MAC-d flow	Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE mode		TDD	
- HARQ info for E-DCH			
- CHOICE mode		TDD	
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84/7.68Mcps TDD	
- Timeslot resource related information		Bit map with all TS configured for E-DCH operation set to ‘1’ all others set to ‘0’	
- Power Resource Related Information		32	
- Activation time		Set to the CFN on which the non-scheduled grant becomes active	
- Repetition period and length		Not present	
- Code resource information		2/1	
- Added or reconfigured E-DCH MAC-d flow		(for DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A15	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE mode		TDD	
- HARQ info for E-DCH			
- CHOICE mode		TDD	
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84/7.68Mcps TDD	
- Timeslot resource related information		Bit map with all TS configured for E-DCH operation set to ‘1’ all others set to ‘0’	

Information Element	Condition	Value/remark	Version
- Power Resource Related Information		32	
- Activation time		Set to the CFN on which the non-scheduled grant becomes active	
- Repetition period and length		Not present	
- Code resource information		2/1	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		3	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A16	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE mode		TDD	
- HARQ info for E-DCH			
- CHOICE mode		TDD	
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84/7.68Mcps TDD	
- Timeslot resource related information		Bit map with all TS configured for E-DCH operation set to '1' all others set to '0'	
- Power Resource Related Information		32	
- Activation time		Set to the CFN on which the non-scheduled grant becomes active	
- Repetition period and length		Not present	
- Code resource information		2/1	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow		Not Present	

Information Element	Condition	Value/remark	Version
multiplexing list			
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		4	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
DL Transport channel information common for all transport channel	A1, A2, A7, A8		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		SameasUL	
- UL DCH TFCS Identity		1	
DL Transport channel information common for all transport channel	A3, A4, A5, A6, A11 A10 A12, A13, A15		 Rel-5 Rel-7
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		Independent	
- DL TFCS			
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.3.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.10.3.4	
- CTFC		Reference to clause 6.10.3.4 Parameter Set	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel	A9		Rel-5
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		Independent	
- DL TFCS			
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((DL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information		Not Present	
- CTFC		1 ((DL DCH RAB, DCCH)=(TF0, TF1))	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel	A14, A16	Not Present	Rel-7

Information Element	Condition	Value/remark	Version
Deleted DL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10 , A12, A13	Not Present	 Rel-5 Rel-7
Deleted DL TrCH information - Downlink transport channel type - DL Transport channel identity	A14, A16	 DCH 10	Rel-6
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value	A1	1 DCH added, 1 DCH reconfigured DCH 6 Same as UL DCH 1 -2.0 DCH 10 Same as UL DCH 5 -2.0	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - TFS - CHOICE Transport channel type - Dynamic transport format information - RLC Size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel list - Semi-static Transport Format information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute	A3, A4, A5, A6, A7	2 TrCHs(DCH for DCCH and DCH for DTCH) DCH 10 Same as UL DCH 5 -2.0 DCH 6 Explicit Except for RAB with the symmetric DL and UL rate: Same as UL Dedicated transport channel Reference to clause 6.10 Parameter Set (This IE is repeated for TFI number.) Not Present Reference to clause 6.10 Parameter Set only including TF0 All Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set	
- CRC size - DCH quality target - BLER Quality value		Reference to clause 6.10 Parameter Set -2.0	
Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - Uplink transport channel type - UL TrCH identity - DCH quality target - BLER Quality value - Downlink transport channel type - DL Transport channel identity	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH) DCH 10 Same as UL DCH 5 2.0 DCH 6	

Information Element	Condition	Value/remark	Version
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		7	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		8	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	

Information Element	Condition	Value/remark	Version
Added or Reconfigured DL TrCH information	A9 A12	3 TrCHs (DCH for DCCH and DCH plus HS-DSCH for DTCH)	Rel-5 Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A10	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-5
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10.2.4.5	

Information Element	Condition	Value/remark	Version
		Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A11	1 DCH for DTCH	
- Downlink transport channel type		DCH	
- DL Transport channel identity		9	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-2.0	
Added or Reconfigured DL TrCH information	A13	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	

Information Element	Condition	Value/remark	Version
- HARQ Info			
- Number of Processes		Reference to clause 6.10 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A14	1 TrCH (HS-DSCH for DTCH and DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queues)	
- MAC-hs queue Id		0 (for DTCH)	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A15	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queues)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		2 (for second DTCH)	
- MAC-d Flow Identity		2	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A16	1 TrCH (HS-DSCH for 2 DTCHs and DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.10 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(three queues)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue Id		3 (for second DTCH)	
- MAC-d Flow Identity		3	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		112	
- MAC-d PDU size index		0	
- MAC-d PDU size		144	
- MAC-d PDU size index		1	
- MAC-d PDU size		160	
- MAC-d PDU size index		2	

Information Element	Condition	Value/remark	Version
- MAC-d PDU size		176	
- MAC-d PDU size index		3	
- MAC-d PDU size		192	
- MAC-d PDU size index		4	
- MAC-d PDU size		224	
- MAC-d PDU size index		5	
- MAC-d PDU size		296	
- MAC-d PDU size index		6	
- MAC-d PDU size		344	
- MAC-d PDU size index		7	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Frequency info	A1, A2, A3, A4, A5, A7, A8, 11 , A9, A10 , A12, A13, A14, A15, A16		 Rel-5 Rel-7
- CHOICE mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies if frequency is different from the current frequency otherwise set to Not Present.	
Frequency info	A6	Not Present	
Maximum allowed UL TX power	A1, A2, A3, A4, A7, A8, A11 , A9, A10 , A12, A13, A14, A15, A16	33dBm	 Rel-5 Rel-7
Maximum allowed UL TX power	A5, A6	Not Present	
CHOICE channel requirement	A1, A2, A3, A4, A7, A8, A9, A10, A11	Uplink DPCH info	R99 and Rel-4 only
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		Not Present	
- CHOICE UL OL PC info		Broadcast UL OL PC info	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		3.84 Mcps TDD	
- UL Timing Advance		Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List		1 CCTrCh	
- TFCS Id		1	
- UL target SIR		+20dB	
- Activation time		Not present	
- Duration		Not present	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.10 Parameter Set	
- TFCI coding		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Repetition Period		Reference to clause 6.10 Parameter Set	
- Repetition Length		Reference to clause 6.10 Parameter Set	
- Uplink DPCH timeslots and codes			
- Dynamic SF usage		TRUE	
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps	
- CHOICE Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration		Choose lowest possible Kcell value given burst type	
- CHOICE TDD option		3.84 Mcps TDD	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the	

Information Element	Condition	Value/remark	Version
- Channelisation code		needs of clause 6.10 Parameter Set. (i/SF) where i denotes an unassigned code matching the SF specified in clause 6.10 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6.10 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A5,A6	Not Present	Rel-5 and earlier
Uplink DPCH info	A12, A13, A14, A15, A16		Rel-7
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		Not Present	
- CHOICE UL OL PC info		Broadcast UL OL PC info	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		3.84 Mcps TDD	
- UL Timing Advance		Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List		1 CCTrCh	
- TFCS Id		1	
- UL target SIR		+20dB	
- Activation time		Not present	
- Duration		Not present	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.10 Parameter Set	
- TFCI coding		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Repetition Period		Reference to clause 6.10 Parameter Set	
- Repetition Length		Reference to clause 6.10 Parameter Set	
- Uplink DPCH timeslots and codes			
- Dynamic SF usage		TRUE	
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps	
- CHOICE Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration		Choose lowest possible Kcell value given burst type	
- CHOICE TDD option		3.84 Mcps TDD	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6.10 Parameter Set.	
- Channelisation code		(i/SF) where i denotes an unassigned code matching the SF specified in clause 6.10 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6.10 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove		Not present	
E-DCH info	A12, A13, A14, A15, A16		Rel-7
- MAC-es/e reset indicator		TRUE	
- CHOICE mode		TDD	
- CHOICE TDD mode		3.84/7.68Mcps TDD	

Information Element	Condition	Value/remark	Version
- E-RUCCH info			
- E-RUCCH constant value		0dB	
- E-RUCCH persistance scaling		0.9	
- T-RUCCH		100ms	
- E-RUCCH timeslot number		Not Present	
- E-RUCCH midamble		Not Present	
- T-adv		Not Present	
- T-SCHED		Not Present	
- CHOICE TDD option		3.84Mcps TDD	
- CHOICE SF		Not present	
- E-PUCH info			
- E-TFCS information			
- Reference Beta Information QPSK list		Reference to clause 6.10 Parameter Set	
- Reference Code Rate		Reference to clause 6.10 Parameter Set	
- Reference beta		Reference to clause 6.10 Parameter Set	
- Reference Beta Information 16QAM list		Reference to clause 6.10 Parameter Set	
- Reference Code Rate		Reference to clause 6.10 Parameter Set	
- Reference beta		Reference to clause 6.10 Parameter Set	
- CHOICE TDD mode		3.84/7.68 Mcps TDD	
- N_{E-UCCH}		Not Present	
- E-PUCH constant value		0dB	
- E-PUCH TS configuration list		Reference to clause 6.10 Parameter Set	
- TS number		Reference to clause 6.10 Parameter Set	
- CHOICE *Burst Type*		Reference to clause 6.10 Parameter Set	
- Midmable configuration		Reference to clause 6.10 Parameter Set	
- E-PUCH code hopping		TRUE	
- E-PUCH TPC step size		1dB	
- Minimum allowed code rate		Reference to clause 6.10 Parameter Set	
- Maximum allowed code rate		Reference to clause 6.10 Parameter Set	
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A11	TDD	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	Rel-5
Downlink HS-PDSCH Information	A9, A10		Rel-5
	A12, A13, A14, A15, A16		Rel-7
- HS-SCCH Info			
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps	
- Ack-Nack Power Offset		0dB	
- HS-SICH Power Control Info			
- UL SIR target		0dB	
- HS-SICH Constant Value		-10dB	
- $D_{hs-sync}$		Not present	
- HS-SCCH Set Configuration		4	
- Timeslot number		The timeslot in which HS-SCCH is to be configured	
- Channelisation code		CC16/x where x is a previously unassigned channelisation code in this TS	
- Midamble Allocation mode		Default	
- Midamble configuration		8	
- BLER target		-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	
- HS-SICH configuration			
- Timeslot number		The timeslot in which HS-SICH is to be configured	
- Channelisation code		CC16/x where x is a previously unassigned channelisation code in this TS	
- Midamble Allocation mode		Default	
- Midamble configuration		8	
- Measurement Feedback Info		Not Present	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	

Information Element	Condition	Value/remark	Version
- HS-PDSCH Timeslot Configuration			
- HS-PDSCH Timeslot Configuration List		Reference to clause 6.10 Parameter Set	
- Timeslot Number		The timeslot(s) in which HS-HS-DSCH is to be configured	
- CHOICE Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		8	
Downlink information common for all radio links	A5, A6	Not present	
Downlink information common for all radio links	A1, A2, A3, A9, A11		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	R99 and Rel-4 only
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Default DPCH Offset Value		Not Present	
- Mac-hs reset indicator		Not Present	
Downlink information common for all radio links	A4, A7, A8, A10		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	R99 and Rel-4 only
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	Rel-4
- Default DPCH Offset Value		Not Present	
- Mac-hs reset indicator		Not Present	
Downlink information for each radio link list	A1, A2, A3, A4, A7, A8, A9, A10, A11	1	
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	
- CHOICE SyncCase		Sync Case 2	
- Timeslot		0	
- Cell parameters ID		10	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Downlink DPCH info for each RL	Rel-6
- CHOICE mode		TDD	
- DL CCTrCH List		1 CCTrCh	
- TFCS ID		1	
- Activation time		Not Present	
- Duration		Not Present	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.10 Parameter Set	
- TFCI coding		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Repetition Period		Reference to clause 6.10 Parameter Set	

Information Element	Condition	Value/remark	Version
- Repetition Length		Reference to clause 6.10 Parameter Set	
- Downlink DPCH timeslots and codes			
- Individual timeslot info			
- Timeslot number		The number of a downlink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps	
- CHOICE Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration		Set Kcell to lowest possible value given the number of codes defined in clause 6.10 Parameter Set	
- CHOICE TDD option		3.84 Mcps	
- First timeslot channelisation codes			
- CHOICE codes representation		Consecutive codes	
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6.10 Parameter Set.	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot as specified in clause 6.10 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6.10 Parameter Set t could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List		1	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared channel indicator		False	
- DL CCTrCH List to Remove		Not Present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
Downlink information for each radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	
- CHOICE SyncCase		Sync Case 2	
- Timeslot		0	
- Cell parameters ID		10	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Not present	Rel-6
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
Downlink information for each radio link list	A6	Not present	
Downlink information for each radio link list	A12, A13, A14, A15, A16		Rel-7
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- CHOICE TDD option		3.84 Mcps TDD	
- CHOICE SyncCase		Sync Case 2	

Information Element	Condition	Value/remark	Version
- Timeslot		0	
- Cell parameters ID		10	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Not present	
- E-AGCH Info		Present	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84Mcps	
- Long Term Grant Indicator		FALSE	
- Length of TTRI field		Reference to clause 6.10 Parameter Set	
- E-AGCH set configuration		Reference to clause 6.10 Parameter Set	
- TS number		Reference to clause 6.10 Parameter Set	
- Channelisation code		Reference to clause 6.10 Parameter Set	
- CHOICE Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble allocation		Reference to clause 6.10 Parameter Set	
- E-AGCH BLER Target		-2	
- CHOICE mode		TDD	
- E-HICH info		Present	
- CHOICE mode		TDD	
- CHOICE TDD option		3.84Mcps	
- N_{E-HICH}		4	
- TS Number		Reference to clause 6.10 Parameter Set	
- Channelisation code		Reference to clause 6.10 Parameter Set	
- Burst Type		Reference to clause 6.10 Parameter Set	
- Midamble allocation mode		Reference to clause 6.10 Parameter Set	
MBMS PL Service Restriction Information	A1, A2, A3, A4, A5, A6, A7, A8 A9, A10 A12, A13, A14, A15, A16	Not Present	 Rel-5 Rel-7

Condition	Explanation	Version
A1	This IE is needed for "Non speech to CELL_DCH from CELL_DCH in CS"	
A2	This IE is needed for "Speech to CELL_DCH from CELL_DCH in CS"	
A3	This IE is needed for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE is needed for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE is needed for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE is needed for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE is needed for "Non speech to CELL_DCH from CELL_FACH in CS"	
A8	This IE is needed for "Speech to CELL_DCH from CELL_FACH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_FACH in PS"	Rel-5
A11	This IE is needed for " Packet RAB Setup after Speech RAB Setup in CELL_DCH"	
A12	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using three multiplexing options (3/3) and SRBs mapped on DCH/DCH"	Rel-7
A13	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-7
A14	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7
A15	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (two streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-7
A16	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (one conversational and one streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7

Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8		

Information Element	Condition	Value/remark	Version
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A7, A8	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
Activation time	A4, A5, A6	Now	
	, A10		Rel-5
New U-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
New C-RNTI	A1, A2, A3, A4, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	Rel-5
New H-RNTI	A9, A10	'1010 1010 1010 1010'	Rel-5
	, A11, A12, A13, A14, A15		Rel-7
CHOICE *mode*		TDD	Rel-7
- New E-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10	Not Present	Rel-7
- New E-RNTI	, A11, A12, A13, A14, A15	'1010 1010 1010 1010'	Rel-7
RRC State indicator	A1, A2, A3, A4, A7, A8	CELL_DCH	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
RRC State indicator	A5, A6	CELL_FACH	

Information Element	Condition	Value/remark	Version
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
CN information info		Not Present	
URA identity		Not Present	
- Signalling RB information to setup list		Not Present	
- RAB information for setup list	A1, A7		
- RAB info			
- RAB identity			
- CHOICE RAB identity type		RAB identity (GSM-MAP)	
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE *RLC size list*		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RAB information to setup list	A2, A8		
- RAB info			
- RAB identity			
- CHOICE RAB identity type		RAB identity (GSM-MAP)	
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	

Information Element	Condition	Value/remark	Version
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE *RLC size list*		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		11	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- Logical channel identity		Not Present	
- CHOICE *RLC size list*		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		7	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		12	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- Logical channel identity		Not Present	
- CHOICE *RLC size list*		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			

Information Element	Condition	Value/remark	Version
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		8	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RAB information for setup list	A3, A4, A5, A6		
- RAB info			
- RAB identity			
- CHOICE RAB identity type		RAB identity (GSM-MAP)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup list			
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Not present	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE *SDU Discard Mode*		Max DAT retransmissions	
- MAX_DAT		4	
- Timer_MRW		100	
- MaxMRW		4	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		200	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			

Information Element	Condition	Value/remark	Version
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		8	
- RAB information for setup	A9		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	

Information Element	Condition	Value/remark	Version
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A10		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	

Information Element	Condition	Value/remark	Version
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A11		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	

Information Element	Condition	Value/remark	Version
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	

Information Element	Condition	Value/remark	Version
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- RAB information for setup	A12, A13, A14, A15		Rel-7
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	

Information Element	Condition	Value/remark	Version
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A14		Rel-7
- RAB info		(second high-speed AM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		17	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		256	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		8	
- E-DCH MAC-d flow identity		3	
- DDI		6	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	

Information Element	Condition	Value/remark	Version
- DL HS-DSCH MAC-d flow identity		2	
- Logical channel identity		Not Present	
- RAB information for setup	A15		Rel-7
- RAB info		(Conversational UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		27	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	
- DL Reception Window Size		32	
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		9	
- E-DCH MAC-d flow identity		4	
- DDI		7	
- RLC PDU size list		12 RLC PDU sizes	
- RLC PDU size		96 bits	
- RLC PDU size		112 bits	
- RLC PDU size		144 bits	
- RLC PDU size		160 bits	
- RLC PDU size		176 bits	
- RLC PDU size		192 bits	
- RLC PDU size		208 bits	
- RLC PDU size		224 bits	
- RLC PDU size		288 bits	
- RLC PDU size		296 bits	
- RLC PDU size		312 bits	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		3	
- Logical channel identity		Not Present	
RB information to reconfigure list	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7

Information Element	Condition	Value/remark	Version
RB information to be affected list	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11		Rel-7
RB information to be affected	A12, A14		Rel-7
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	

Information Element	Condition	Value/remark	Version
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
RB information to be affected	A13, A15		Rel-7
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	

Information Element	Condition	Value/remark	Version
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
Downlink counter synchronization info	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
PDCP ROHC target mode	A9, A10	Not Present	Rel-5
	, A11, A12, A13, A14, A15		Rel-7
UL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8		
	, A9, A10		Rel-5
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC		Reference to clause 6.11.5.4	

Information Element	Condition	Value/remark	Version
		Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain Factor β_d		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- TFC subset			
- CHOICE Subset representation		Full transport format combination set	
- TFC subset list		Not Present	
UL Transport channel information common for all transport channels	A11		Rel-7
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((UL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0 Integer(0.. 3)	
- CTFC		1 ((UL DCH RAB, DCCH)=(TF0, TF1))	
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		TDD	
- Gain factor βd		15	
- Reference TFC ID		0 Integer(0.. 3)	
- CHOICE mode		TDD	
- TFC subset			
- CHOICE Subset representation		Full transport format combination set	
- TFC subset list		Not Present	
UL Transport channel information common for all transport channels	A12, A13, A14, A15		Rel-7

Information Element	Condition	Value/remark	Version
Deleted TrCH information list	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11		Rel-7
Deleted TrCH information list	A12, A13, A14, A15		Rel-7
- Uplink transport channel type		DCH	
- UL transport channel identity		5	
Added or Reconfigured UL TrCH information	A1, A3 A4, A5, A6, A7	1 DCH added, 1 DCH reconfigured	
	, A9, A10		Rel-5
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured TrCH information list	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	

Information Element	Condition	Value/remark	Version
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter	

Information Element	Condition	Value/remark	Version
		Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List	1 to maxTF	(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured UL TrCH information	A11	1 E-DCH added, 1 DCH added, 1 DCH reconfigured	Rel-7
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE *mode*		TDD	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow			
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter	

Information Element	Condition	Value/remark	Version
		Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured UL TrCH information	A12, A13	1 E-DCH added with one DCCH MAC-d flow and one DTCH MAC-d flow	Rel-6
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE *mode*		TDD	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE *mode*		TDD(NULL)	
- Added or reconfigured E-DCH MAC-d flow		(for DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A14	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-6
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE *mode*		TDD	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of		3	

Information Element	Condition	Value/remark	Version
retransmissions			
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE mode		TDD(NULL)	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		3	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
Added or Reconfigured UL TrCH information	A15	1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows	Rel-6
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
- CHOICE mode		TDD	
- HARQ info for E-DCH			
- HARQ RV Configuration		rvtable	
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		3	
- E-DCH MAC-d flow retransmission timer		60	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	
- CHOICE mode		TDD(NULL)	
- Added or reconfigured E-DCH MAC-d flow		(for first DTCH)	
- E-DCH MAC-d flow identity		2	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
- Added or reconfigured E-DCH MAC-d flow		(for second DTCH)	
- E-DCH MAC-d flow identity		4	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Scheduled grant info	
CHOICE mode		TDD (no data)	
DL Transport channel information common for all transport channel	A1, A2, A7, A8		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			

Information Element	Condition	Value/remark	Version
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		SameAsUL	
- UL DCH TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
DL Transport channel information common for all transport channel	A3, A4, A5, A6		
	A10		Rel-5
	, A11, A12, A14		Rel-7
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Independent	
- DL TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel	A9		Rel-5
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		2	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Explicit	
- DL DCH TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((DL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information		Not Present	
- CTFC		1 ((DL DCH RAB, DCCH)=(TF0, TF1))	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel	A13, A15	Not Present	Rel-7
Deleted TrCH information list	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	, A9, A10		Rel-5
	, A11, A12		Rel-7
Deleted DL TrCH information	A13, A15		Rel-7
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL Transport channel identity		10	
Added or Reconfigured TrCH information list	A1	1 DCH added, 1 DCH reconfigured	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A3, A4, A5, A6, A7	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- Transparent mode signalling info		Not Present	
Added or Reconfigured TrCH information list	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- Transparent mode signalling info		Not Present	
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		7	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		8	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured DL TrCH information	A9,A10	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-5
	, A11		Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11.5.4.6 Parameter Set	
- CHOICE Memory Partitioning		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A12	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11.5.4.6 Parameter Set	
- CHOICE Memory Partitioning		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A13	1 TrCH (HS-DSCH for DTCH and DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11.5.4.6 Parameter Set	
- CHOICE Memory Partitioning		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queue)	
- MAC-hs queue Id		0(for DTCH)	
- MAC-d Flow Identity		0	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1(for DCCH)	
- MAC-d Flow Identity		1	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A14	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-7
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TF number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter	

Information Element	Condition	Value/remark	Version
		Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11.5.4.6 Parameter Set	
- CHOICE Memory Partitioning		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(two queue)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		2 (for second DTCH)	
- MAC-d Flow Identity		2	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A15	1 TrCH (HS-DSCH for 2 DTCH and DCCH)	Rel-7
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11.5.4.6 Parameter Set	
- CHOICE Memory Partitioning		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(three queue)	
- MAC-hs queue Id		0 (for first DTCH)	
- MAC-d Flow Identity		0	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue Id		3 (for second DTCH)	
- MAC-d Flow Identity		3	

Information Element	Condition	Value/remark	Version
- T1		120	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		112	
- MAC-d PDU size index		0	
- MAC-d PDU size		144	
- MAC-d PDU size index		1	
- MAC-d PDU size		160	
- MAC-d PDU size index		2	
- MAC-d PDU size		176	
- MAC-d PDU size index		3	
- MAC-d PDU size		192	
- MAC-d PDU size index		4	
- MAC-d PDU size		224	
- MAC-d PDU size index		5	
- MAC-d PDU size		296	
- MAC-d PDU size index		6	
- MAC-d PDU size		344	
- MAC-d PDU size index		7	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Frequency info	A1, A2, A3, A4, A5, A7, A8		
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
- Choice mode		TDD	
- UARFCN (Nt)		Reference to clause 5.1 Test frequencies	
Frequency info	A6	Not Present	
Maximum allowed UL TX power	A1, A2, A3, A4, A7, A8	33dBm	
	, A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
Maximum allowed UL TX power	A5, A6	Not Present	
CHOICE *channel requirement*	A5, A6	Not Present	
CHOICE *channel requirement*	A1, A2, A3, A4, A7, A8	Uplink DPCH info	
	, A9, A10		Rel-5
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- $PRX_{PDPCHdes}$		Integer (-120…-58 by step of 1)	
- CHOICE *UL OL PC info*			
- Broadcast UL OL PC info		Null	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	

Information Element	Condition	Value/remark	Version
- Repetition period		1	
- Repetition length			
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	

Information Element	Condition	Value/remark	Version
Uplink DPCH info	A11, A12, A13, A14, A15	Uplink DPCH info	Rel-7
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- PRXPDPCHdes		Integer (-120…-58 by step of 1)	
- CHOICE UL OL PC info			
- Broadcast UL OL PC info		Null	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		1	
- Uplink DPCH timeslots and code			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info	A11, A12, A13, A14, A15		Rel-7
- MAC-es/e reset indicator		TRUE	
- CHOICE *mode*		TDD	
- E-RUCCH info			
- CHOICE TDD mode		1.28 Mcps TDD	
- T-RUCCH		200	
- N-RUCCH		3	
- T-WAIT		40	
- T-SI		40	
- Extended Estimation Window		Not present	
- PRACH Information		Not present	
- E-PUCH info			
- E-TFCS information			

Information Element	Condition	Value/remark	Version
- Reference Beta Information QPSK list			
- Reference Code Rate		2	
- Reference Beta		-10	
- Reference Code Rate		8	
- Reference Beta		-3	
- Reference Beta Information 16QAM list			
- Reference Code Rate		2	
- Reference Beta		-5	
- Reference Code Rate		8	
- Reference Beta		2	
- CHOICE TDD mode		1.28 Mcps TDD	
- SNPL Reporting Type		type1	
- PRXdes_base		-112	
- Beacon PL Est		FALSE	
- TPC step size		1	
- Uplink synchronisation parameters		Not present	
- E-PUCH TS configuration list			
- TS number		3	
- Midamble shift and burst type			
- Midamble Allocation Mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- Minimum allowed code rate		0	
- Maximum allowed code rate		63	
- Power Offset for Scheduling Info		Not Present	
- Non-scheduled transmission grant info	A11	Not Present	
- Non-scheduled transmission grant info	A12, A13, A14, A15		
- CHOICE *TDD Option*		1.28 Mcps TDD	
- $N_{E\text{-}UCCH}$		1	
- $N_{E\text{-}HICH}$		4	
- Timeslot Resource Related InformationI		00100	
- Power Resource Related Information		1	
- Activation Time		0	
- Subframe number		0	
- Repetition period and length			
- Repetition period		1	
- Repetition Length		1	
- Code Resource Information		8/1	
- E-HICH Information			
- Timeslot number		6	
- Channelisation code		16/6	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		16	
- Signature Sequence Group Index		0	
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8	TDD	
	, A9, A10		Rel-5
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	Rel-5
Downlink HS-PDSCH Information	A9, A10		Rel-5
	, A11, A12, A13, A14, A15		Rel-7
- HS-SCCH Info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps	
- HS-SCCH Set Configuration			
- Timeslot number		0	
- First Channelisation code		(16/5)	
- Second Channelisation code		(16/6)	

Information Element	Condition	Value/remark	Version
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/11)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$		-120	
- TPC step size		1dB	
- Timeslot number		0	
- First Channelisation code		(16/7)	
- Second Channelisation code		(16/8)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/12)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$		-120	
- TPC step size		1dB	
- Timeslot number		0	
- First Channelisation code		(16/9)	
- Second Channelisation code		(16/10)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/13)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$		-120	
- TPC step size		1dB	
- Timeslot number		0	
- First Channelisation code		(16/11)	
- Second Channelisation code		(16/12)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/14)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$		-120	
- TPC step size		1dB	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- HS-PDSCH Midamble Configuration			
- Midamble Allocation Mode		Default midamble	
- Midamble Configuration		8	
- Midamble Shift		Not present	
Downlink information common for all radio links	A5, A6	Not Present	
Downlink information common for all radio links	A1, A2, A3		
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	

Information Element	Condition	Value/remark	Version
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A9		Rel-5
	, A11, A12, A14, A15		Rel-7
- Downlink DPCH info common for all RL			
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		TRUE	
Downlink information common for all radio links	A4, A7, A8		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value			
- CHOICE mode		TDD	
- Default DPCH Offset Value		0 Integer(0..7)	
Downlink information common for all radio links	A10		Rel-5
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		TRUE	
Downlink information common for all radio links	A13, A15		Rel-7
- Downlink F-DPCH info common for all RL			
- Timing indication		Maintain	
- Timing maintained Synchronization indicator		FALSE t	
- Downlink F-DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1	
- TPC command error rate target		0.04	

Information Element	Condition	Value/remark	Version
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	
Downlink information per radio link list	A1, A2, A3, A4, A7, A8		
	, A9, A10		Rel-5
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7

Information Element	Condition	Value/remark	Version
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A11, A12, A13, A14, A15		Rel-7
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCh List			
- TFCS ID		2 Integer(1.8)	
- Time info			
- Activation time		Now	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Default value is "Frame"	
- TFCI coding		Reference to clause 6 Parameter set	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		4 OR 5 OR 6	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- CHOICE codes representation			
- Channelisation codes bitmap		Reference to clause 6.11 Parameter Set	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH TPC List		This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- DL CCTrCH List to Remove		Not present	
- E-AGCH Info			
- CHOICE *mode*		TDD	
- CHOICE TDD Option		1.28 Mcps TDD	
- RDI Indicator		FALSE	
- TPC step size		1	
- E-AGCH set configuration			
- Timeslot number		6	
- First Channelisation code		16/3	

Information Element	Condition	Value/remark	Version
- Second Channelisation code		16/4	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- E-AGCH BLER target		-0.05	
- CHOICE *mode*		TDD	
- CHOICE *TDD Option*		1.28 Mcps TDD	
- N_{E-HICH}		15	
- E-HICH set configuration			
- EI		2	
- Timeslot number		6	
- Channelisation code		16/5	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
Downlink information per radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- Choice TDD Option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127)	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL		Not Present	
- SCCPCH Information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE *mode*		TDD	Rel-7
- E-HICH Information		Not Present	Rel-7
Downlink information per radio link list	A6	Not Present	

Condition	Explanation	Version
A1	This IE need for "Non speech to CELL_DCH from CELL_DCH in CS"	
A2	This IE need for "Speech to CELL_DCH from CELL_DCH in CS"	
A3	This IE need for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE need for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE need for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE need for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE need for "Non speech to CELL_DCH from CELL_FACH in CS"	
A8	This IE need for "Speech to CELL_DCH from CELL_FACH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_FACH in PS"	Rel-5
A11	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using three multiplexing options (3/3) and SRBs mapped on DCH/DCH"	Rel-7
A12	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-7
A13	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7
A14	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (two streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH"	Rel-7
A15	This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs (one conversational and one streaming/interactive/background) using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH"	Rel-7

Contents of RADIO BEARER SETUP message: AM or UM (7.68 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2, A3, A4, A5, A6, A7, A8, A11		
	, A9, A10		Rel-5
			Rel-6
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1, A2, A3, A11	(256+CFN-(CFN MOD 8 + 8))MOD 256	
	, A9		Rel-5
			Rel-6
Activation time	A4, A5, A6, A7, A8 A10	Not Present	Rel-5
New U-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
			Rel-6
New C-RNTI	A1, A2, A3, A4, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
			Rel-6
New C-RNTI	A5, A6	'1010 1010 1010 1010'	
New DSCH-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
New H-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	Rel-5
New H-RNTI	A9, A10 A12, A13, A14, A15	'1010 1010 1010 1010'	Rel-5 Rel-6
CHOICE mode	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11	TDD	Rel-7
- New E-RNTI	A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11	Not Present	Rel-7
RRC State indicator	A1, A2, A3, A4, A7, A8, A11	CELL_DCH	
	, A9, A10		Rel-5
			Rel-6
RRC State indicator	A5, A6	CELL_FACH	
UTRAN DRX cycle length coefficient	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	
	, A9, A10		Rel-5
			Rel-6
CN information info		Not Present	
URA identity		Not Present	
CHOICE Specification mode		Complete specification	Rel-6
- Signalling RB information to setup		Not Present	
- RAB information for setup	A1, A7		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			

Information Element	Condition	Value/remark	Version
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RAB information for setup	A2, A8		
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT314	
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		11	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		7	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity		12	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		6	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		8	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- RAB information for setup	A3, A4, A5, A6		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping		Not Present	

Information Element	Condition	Value/remark	Version
indicator			
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
- RAB information for setup	A9		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	

Information Element	Condition	Value/remark	Version
- RB mapping info			
- Information for each multiplexing option		3 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- Logical channel identity		7	
- RAB information for setup	A10		Rel-5
- RAB info		(high-speed AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		100	
- Timer_poll		100	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- CHOICE Downlink RLC PDU Size		Reference to clause 6 Parameter Set	
- In-sequence delivery		TRUE	
- Receiving window size		768	
- Downlink RLC status info			
- Timer_status_prohibit		100	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel		Not present	

Information Element	Condition	Value/remark	Version
identity			
- DL DSCH Transport channel identity		Not present	
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
- RAB information for setup	A11		
- RAB info		(AM DTCH for PS domain)	
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		20	
- PDCP info			
- Support for lossless SRNS relocation		FALSE	
- Max PDCP SN window size		Not present	
- PDCP PDU header		Absent	
- Header compression information		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		4	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL DCH Transport channel identity		9	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
RB information to reconfigure list	A1, A2, A3, A4, A5, A6, A7, A8 A9, A10	Not Present	 Rel-5 Rel-6
RB information to be affected	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10	Not Present	 Rel-5 Rel-6
Downlink counter synchronization info	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10	Not Present	 Rel-5 Rel-6
PDCP ROHC target mode	A9, A10	Not Present	Rel-5 Rel-6
UL Transport channel information common for all transport channels	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10		 Rel-5
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information			
- UL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	R99 and Rel-4 only
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11 Parameter Set	
- CTFC		Reference to clause 6.11 Parameter Set	
- Power offset information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0 Integer(0.. 3)	

Information Element	Condition	Value/remark	Version
- CHOICE mode		TDD	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain factor βd		8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID		0	
- CHOICE mode		TDD	
- TFC subset		Not Present	
- TFC subset list		Not Present	
UL Transport channel information for all transport channels		Not Present	Rel-6
Deleted UL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8, A11 , A9, A10	Not Present	Rel-5 Rel-6
Added or Reconfigured UL TrCH information	A1, A3 A4, A5, A6, A7 , A9, A10	1 DCH added, 1 DCH reconfigured (if from cell_DCH) OR 2 DCHs added (if from cell_FACH)	Rel-5
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured UL TrCH information	A11	1 DCH added for DTCH	
- Uplink transport channel type		DCH	
- UL Transport channel identity		4	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
Added or Reconfigured UL TrCH information	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- Uplink transport channel type		DCH	
- UL Transport channel identity		2	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- Uplink transport channel type		DCH	
- UL Transport channel identity		3	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
DL Transport channel information common for all transport channel	A1, A2, A7, A8		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		SameasUL	
- UL DCH TFCS Identity		1	
DL Transport channel information common for all transport channel	A3, A4, A5, A6, A11 A10		Rel-5 Rel-6
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		Independent	
- DL TFCS			
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		Number of bits used must be enough to cover all combinations of CTFC from clause 6.11 Parameter Set.	
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11	
- CTFC		Reference to clause 6.11 Parameter Set	
- Power offset information		Not Present	
DL Transport channel information common for all transport channel	A9		Rel-5
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- Individual DL CCTrCH information		1 CCTrCh	
- DL TFCS identity		1	
- CHOICE DL parameters		Independent	
- DL TFCS			
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		ctfc2bit	
- CTFC information			
- CTFC		0 ((DL DCH RAB, DCCH)=(TF0, TF0))	
- Power offset information		Not Present	
- CTFC		1 ((DL DCH RAB, DCCH)=(TF0, TF1))	
- Power offset information		Not Present	

Information Element	Condition	Value/remark	Version
Deleted DL TrCH information	A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10	Not Present	Rel-5 Rel-6
Added or Reconfigured DL TrCH information	A1	1 DCH added, 1 DCH reconfigured	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
Added or Reconfigured DL TrCH information	A3, A4, A5, A6, A7	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
		Except for RAB with the symmetric DL and UL rate: Same as UL	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set only including TF0	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-2.0	
Added or Reconfigured DL TrCH information	A2, A8	4 TrCHs(DCH for DCCH and 3DCHs for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		2.0	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	

Information Element	Condition	Value/remark	Version
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		7	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
- Downlink transport channel type		DCH	
- DL Transport channel identity		8	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		Not Present	
Added or Reconfigured DL TrCH information	A9	3 TrCHs (DCH for DCCH and DCH plus HS-DSCH for DTCH)	Rel-5 Rel-6
- Downlink transport channel type		DCH	

Information Element	Condition	Value/remark	Version
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A10	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	Rel-5
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		5	
- DCH quality target			
- BLER Quality value		-2.0	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to clause 6.11 Parameter Set	
- CHOICE *Memory Partitioning*		Implicit	
- Added or reconfigured MAC-d flow			

Information Element	Condition	Value/remark	Version
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		336	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A11	1 DCH for DTCH	
- Downlink transport channel type		DCH	
- DL Transport channel identity		9	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channel	
- Dynamic transport format information			
- RLC Size		Reference to clause 6.11 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Dynamic transport format information			
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- CRC size		Reference to clause 6.11 Parameter Set	
- DCH quality target			
- BLER Quality value		-2.0	
Frequency info	A6	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX information		Not Present	Rel-7
HS-SCCH less information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power	A1, A2, A3, A4, A7, A8, A11	33dBm	
	, A9, A10		Rel-5
			Rel-6
Maximum allowed UL TX power	A5, A6	Not Present	
CHOICE channel requirement	A1, A2, A3, A4, A7, A8, A9, A10, A11	Uplink DPCH info	R99 and Rel-4 only
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL target SIR		Not Present	
- CHOICE UL OL PC info		Broadcast UL OL PC info	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Extended UL Timing Advance		Determined by observed timing deviation of the RACH at the node B	Rel-7
- UL CCTrCH List		1 CCTrCh	
- TFCS Id		1	
- UL target SIR		+20dB	
- Activation time		Not present	
- Duration		Not present	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter Set	
- TFCI coding		Reference to clause 6.11 Parameter Set	
- Puncturing Limit		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- Repetition Period		Reference to clause 6.11 Parameter Set	
- Repetition Length		Reference to clause 6.11 Parameter Set	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Uplink DPCH timeslots and codes VHCR			
- Dynamic SF usage		TRUE	
- First invidual timeslot info			
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps	Rel-7
- CHOICE Burst Type		Reference to clause 6.11 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration		Choose lowest possible Kcell value given burst type	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6.11 Parameter Set.	
- Channelisation code		(i/SF) where i denotes an unassigned code matching the SF specified in clause 6.11 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6.11 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A5,A6	Not Present	Rel-5 and earlier
CHOICE Mode	A1, A2, A3, A4, A5, A6, A7, A8, A11	TDD	R99 and Rel-4 only
Downlink HS-PDSCH Information	A1, A2, A3, A4, A5, A6, A7, A8, A11	Not Present	Rel-5
Downlink HS-PDSCH Information	A9, A10		Rel-5 Rel-6
- HS-SCCH Info			
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps	Rel-7
- Ack-Nack Power Offset		0dB	
- HS-SICH Power Control Info			
- UL SIR target		0dB	
- HS-SICH Constant Value		-10dB	
- $D_{hs\text{-}sync}$		Not present	
- HS-SCCH Set Configuration		4	
- Timeslot number		The timeslot in which HS-SCCH is to be configured	
- Channelisation code		CC32/x where x is a previously unassigned channelisation code in this TS	
- Midamble Allocation mode		Default	
- Midamble configuration		8	
- BLER target		-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	
- HS-SICH configuration			
- Timeslot number		The timeslot in which HS-SICH is to be configured	
- Channelisation code		CC32/x where x is a previously unassigned channelisation code in this TS	
- Midamble Allocation mode		Default	
- Midamble configuration		8	
- Measurement Feedback Info		Not Present	
- CHOICE mode		TDD	

Information Element	Condition	Value/remark	Version
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- HS-PDSCH Timeslot Configuration VHCR			
- HS-PDSCH Timeslot Configuration List		Reference to clause 6.11 Parameter Set	
- Timeslot Number		The timeslot(s) in which HS-HS-DSCH is to be configured	
- CHOICE Burst Type		Reference to clause 6.11 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		8	
Downlink information common for all radio links	A5, A6	Not present	
Downlink information common for all radio links	A1, A2, A3, A9, A11		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Timing indication		Maintain	
- CFN-targetSFN frame offset		Not Present	R99 and Rel-4 only
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Default DPCH Offset Value		Not Present	
- Mac-hs reset indicator		Not Present	
Downlink information common for all radio links	A4, A7, A8, A10		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Timing indication		Initialise	
- CFN-targetSFN frame offset		Not Present	R99 and Rel-4 only
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Default DPCH Offset Value		Not Present	
- Mac-hs reset indicator		Not Present	
Downlink information for each radio link list	A1, A2, A3, A4, A7, A8, A9, A10, A11	1	
- Downlink information for each radio link			
- Choice mode		7.68 Mcps TDD	Rel-7
- Primary CCPCH info			
- Choice mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- CHOICE SyncCase		Sync Case 2	
- Timeslot		0	
- Cell parameters ID		10	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Downlink DPCH info for each RL	Rel-6
- CHOICE mode		TDD	
- DL CCTrCH List		1 CCTrCh	
- TFCS ID		1	
- Activation time		Not Present	
- Duration		Not Present	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter Set	
- TFCI coding		Reference to clause 6.11 Parameter Set	
- Puncturing Limit		Reference to clause 6.11 Parameter Set	

Information Element	Condition	Value/remark	Version
- Repetition Period		Reference to clause 6.11 Parameter Set	
- Repetition Length		Reference to clause 6.11 Parameter Set	
- Downlink DPCH timeslots and codes VHCR			Rel-7
- Individual timeslot info			
- Timeslot number		The number of a downlink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps	Rel-7
- CHOICE Burst Type		Reference to clause 6.11 Parameter Set	
- Midamble Allocation Mode		Default	
- Midamble configuration		Set Kcell to lowest possible value given the number of codes defined in clause 6.11 Parameter Set	
- CHOICE TDD option		7.68 Mcps	Rel-7
- First timeslot channelisation codes VHCR			Rel-7
- CHOICE codes representation		Consecutive codes	
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6.11 Parameter Set.	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot as specified in clause 6.11 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6.11 Parameter Set t could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List		1	
- UL TPC TFCS Identity			
- TFCS ID		1	
- Shared channel indicator		False	
- DL CCTrCH List to Remove		Not Present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
Downlink information for each radio link list	A5		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- Choice mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	
- CHOICE SyncCase		Sync Case 2	
- Timeslot		0	
- Cell parameters ID		10	
- SCTD indicator		FALSE	
- CHOICE DPCH info		Not present	Rel-6
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
Downlink information for each radio link list	A6	Not present	
MBMS PL Service Restriction Information	A1, A2, A3, A4, A5, A6, A7, A8	Not Present	
	A9, A10		Rel-5 Rel-6

Condition	Explanation	Version
A1	This IE is needed for "Non speech to CELL_DCH from CELL_DCH in CS"	

A2	This IE is needed for "Speech to CELL_DCH from CELL_DCH in CS"	
A3	This IE is needed for "Packet to CELL_DCH from CELL_DCH in PS"	
A4	This IE is needed for "Packet to CELL_DCH from CELL_FACH in PS"	
A5	This IE is needed for "Packet to CELL_FACH from CELL_DCH in PS"	
A6	This IE is needed for "Packet to CELL_FACH from CELL_FACH in PS"	
A7	This IE is needed for "Non speech to CELL_DCH from CELL_FACH in CS"	
A8	This IE is needed for "Speech to CELL_DCH from CELL_FACH in CS"	
A9	This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS"	Rel-5
A10	This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_FACH in PS"	Rel-5
A11	This IE is needed for " Packet RAB Setup after Speech RAB Setup in CELL_DCH"	

Contents of RADIO BEARER SETUP COMPLETE message: AM

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see if the value is identical to the same IE in the downlink RADIO BEARER SETUP message.	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked.	
CHOICE mode	TDD	
- CHOICE *TDD option*	Check that this IE is present	Rel-4
START	Not checked (if ciphering is OFF), check the presence if ciphering is ON.	
COUNT-C activation time	The presence of this IE depends on the following 2 factors: (a) There exists RB(s) mapped to RLC-TM and (b) UE is transiting to CELL_DCH state after the RB establishment procedure. Else, this IE is absent.	
Radio bearer uplink ciphering activation time info	If ciphering is not activated in RADIO BEARER SETUP message, this IE must be absent. Else, SS checks this IE for the presence of activation times of all ciphered uplink RLC-UM and RLC-AM RBs.	
Uplink counter synchronization info	Not present	

Contents of RADIO BEARER SETUP FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded	Not Check

Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		

Information Element	Value/remark	Version
RRC transaction identifier	Checked to see the value is identical to the same IE in the downlink RADIO BEARER RELEASE message.	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked.	
CHOICE mode	TDD	
- CHOICE TDD option	1.28 Mcps TDD (no data)	
COUNT-C activation time	The presence of this IE depends on the following 2 factors: (a) There exists RB(s) mapped to RLC-TM and (b) UE is transiting to CELL_DCH state after the RB release procedure. Else, this IE is absent.	
Radio bearer uplink ciphering activation time info	If ciphering is not activated in RADIO BEARER RELEASE message, this IE must be absent. Else, SS checks this IE for the presence of activation times of all ciphered uplink RLC-UM and RLC-AM RBs.	
Uplink counter synchronization info	Not checked	

Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Checked to see the value is identical to the same IE in the downlink RADIO BEARER RELEASE message.	
Integrity check info		
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.	
Uplink integrity protection activation info	Not checked.	
CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD (no data)	
- Extended Uplink Timing advance	Not Present	
COUNT-C activation time	The presence of this IE depends on the following 2 factors: (a) There exists RB(s) mapped to RLC-TM and (b) UE is transiting to CELL_DCH state after the RB release procedure. Else, this IE is absent.	
Radio bearer uplink ciphering activation time info	If ciphering is not activated in RADIO BEARER RELEASE message, this IE must be absent. Else, SS checks this IE for the presence of activation times of all ciphered uplink RLC-UM and RLC-AM RBs.	

Contents of RADIO BEARER RELEASE FAILURE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identitifer	Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RELEASE message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Failure cause	Checked to see if it meets test requirement
Radio bearers for which reconfiguration would have succeeded	Not checked

Contents of RRC CONNECTION REQUEST message: TM

Information Element	Value/remark	Version
Message Type		
Predefined configuration status information	To be checked against requirement if specified	Rel-5
Initial UE identity		
- CHOICE UE id type		
- IMSI (GSM-MAP)	Set to the UE's IMSI (GSM-MAP) or TMSI.	
Establishment cause	To be checked against requirement if specified	
Protocol error indicator	FALSE	
UE Specific Behaviour Information 1 idle	This IE will not be checked by default behaviour, but in specific test case.	
Measured results on RACH	To be checked against requirement if specified	
Access stratum release indicator	To be checked against requirement if specified	Rel-4

Contents of RRC CONNECTION REJECT message: UM

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3
Initial UE identity	Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message.
Rejection cause	Unspecified
Wait Time	0
Redirection info	Not Present

Contents of RRC CONNECTION RELEASE message: UM

Information Element	Value/remark	Version
Message Type		
U-RNTI	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	R99, Rel-4
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
CHOICE identity type	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	Rel-5
- U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
- Group identity	[FFS]	
- Group release information	[FFS]	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info	This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted.	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.	
N308	2 (for CELL_DCH state). Not Present (for UE in other connected mode states).	
Release cause	Normal event	
Rplmn information	Not Present	

Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM

Information Element	Semantics description
Message Type	

RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION RELEASE message.
Integrity check info	
- Message authentication code	Checked to see if it's identical to the value of XMAC-I calculated by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	Checked to see if it is present. This number is used by the SS to compute the XMAC-I
Error indication	Not checked

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (3.84 Mcps TDD option)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	0	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	Not Present	
RRC State Indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	

Information Element	Value/remark	Version
- Logical channel identity	1	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	

Information Element	Value/remark	Version
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	

Information Element	Value/remark	Version
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	

Information Element	Value/remark	Version
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS ID	(This IE is repeated for TFC number.)	
- UL TFCS		
- TFC subset	Default value is the complete existing set of transport format combinations	
- Allowed Transport Format combination	0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS	(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- TFCS complete reconfigure information		
- CHOICE TFCS Size	Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC size	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- Number of TBs and TTI lists	(This IE is repeated for TFI number)	
- CHOICE mode	TDD	
- Transmission Time Interval	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator		
- CHOICE DL parameters	Same as UL	
Added or Reconfigured TrCH information list		
- Added or Reconfigured DL TrCH		

Information Element	Value/remark	Version
information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
-DCH quality target		
- BLER Quality value	-63 (-6.3)	
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present	
CHOICE channel requirement	Uplink DPCH info	
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE *TDD option*	3.84 Mcps	
- UL target SIR	Reference to clause 6.10 Parameter set	
- CHOICE mode	TDD	
- CHOICE *UL OL PC info*	Individually signalled	
- CHOICE *TDD option*	3.84 Mcps	
- Individual timeslot interference	Not Present	
info		
- Individual timeslot interference		
- DPCH Constant Value		
- Primary CCPCH Tx Power	Not Present	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to clause 6.10 Parameter Set	
- TFCI coding	Reference to clause 6.10 Parameter Set	
- Puncturing Limit	Reference to clause 6.10 Parameter Set	
- Repetition Period	Reference to clause 6.10 Parameter Set	
- Repetition Length	Reference to clause 6.10 Parameter Set	
- Uplink DPCH timeslots and	Default is to use the old timeslots and codes	
codes		
- CPCH SET Info	(no data)	R99 and Rel-4 only
Downlink information common for all radio links		
- Downlink DPCH info common for all		
RL		
- Timing indicator	Maintain	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control		
information		
- DPC mode	0 (single)	
- CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps (no data)	
- Default DPCH Offset Value	Not Present	
Downlink information for each radio link list		
- Downlink information for each radio link		
- Choice mode	TDD	
- Primary CCPCH info		
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	PCCPCH timeslot	
- Cell parameters ID	0	
- SCTD indicator		
- Downlink DPCH info for each RL		
- CHOICE mode	TDD	
- DL CCTrCH List		
- TFCS ID	1	
- Time info		
- Activation time	(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration	infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to the present document	
- TFCI coding	TRUE	
- Puncturing limit	Reference to clause 6 Parameter set	

Information Element	Value/remark	Version
- Repetition period	1	
- Repetition length	Empty	
- Downlink DPCH timeslots and codes		
- CHOICE *more timeslots*		
- CHOICE TDD option	3.84 Mcps	
- Timeslot number	The number of a downlink timeslot that has unassigned codes in a frame.	
- Individual timeslot info		
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	3.84 Mcps	
-CHOICE Burst Type		
-Type 1		
-Midamble Allocation Mode	Default	
- Midamble configuration burst type 1 and 3	As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes		
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	Not Present	
-SCCPCH information for FACH	Not Present	R99 and Rel-4 only

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD option)

Information Element	Condition	Value/remark	Version
Message Type	A1,A2		
Initial UE identity		Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier		0	
Activation time		Not Present(Now)	
New U-RNTI			
- SRNC identity		0000 0000 0001B	
- S-RNTI		0000 0000 0000 0000 0001B	
New C-RNTI		Not Present	
RRC State Indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		9, Integer(3...9)	
Capability update requirement			
- UE radio access FDD capability update requirement		FALSE	
- UE radio access 3.84 Mcps TDD capability update requirement		FALSE	
- UE radio access 1.28 Mcps TDD capability update requirement		TRUE	
- System specific capability update requirement list		Not Present	
CHOICE *specification mode*		Complete specification	Rel-5
- Complete specification			Rel-5
- Signalling RB information to setup list			
- Signalling RB information to setup	A1	(UM DCCH for RRC)	
- RB identity		1	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- RB mapping info			

Information Element	Condition	Value/remark	Version
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		1	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		1	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		1	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		1	
- Signalling RB information to setup	A2	(UM DCCH for RRC)	
- RB identity		1	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow		1	

Information Element	Condition	Value/remark	Version
identity			
- Logical channel identity		1	
- Signalling RB information to setup	A1	(AM DCCH for RRC)	
- RB identity		2	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		2	
- CHOICE RLC size list		Configure	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity			
- Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		2	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		2	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		2	
- Signalling RB information to setup	A2	(AM DCCH for RRC)	Rel-6
- RB identity		2	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Window		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
Signalling RB information to setup	A1	(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	

Information Element	Condition	Value/remark	Version
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		3	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity			
- Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		3	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		3	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		3	
- Signalling RB information to setup	A2	(AM DCCH for NAS_DT High priority)	
- RB identity		3	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	

Information Element	Condition	Value/remark	Version
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		2	
- Signalling RB information to setup	A1	(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			

Information Element	Condition	Value/remark	Version
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		4	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity			
- Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		4	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		4	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		Not Present	
- Logical channel identity		4	
- Signalling RB information to setup	A2	(AM DCCH for NAS_DT Low priority)	
- RB identity		4	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	

Information Element	Condition	Value/remark	Version
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		4	
- E-DCH MAC-d flow identity		1	
- DDI		4	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		144 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- DL HS-DSCH MAC-d flow identity		1	
- Logical channel identity		4	
- UL Transport channel information for all transport channels	A1		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
-Individual UL CCTrCH information			
- UL TFCS ID		(This IE is repeated for TFC number.)	
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- UL TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	

Information Element	Condition	Value/remark	Version
- CTFC information		This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC		Reference to clause 6.11.5.4 Parameter Set	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID		0, Integer(0.. 3)	
- CHOICE Gain Factors		Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode		TDD	
- Gain Factor □d		15	
- Reference TFC ID		0, Integer (0..3)	
- CHOICE mode		TDD	
- TFC subset		Default value is the complete existing set of transport format combinations	
- CHOICE *Subset representation*		Allowed transport format combination list	
- Allowed Transport Format combination		0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- Transport format combination		Integer (0.. 1023)	
- TFC subset list		Not present	
- Added or Reconfigured UL TrCH information list			
- UL Transport channel information for all transport channels	A2	Not Present	
- Added or Reconfigured UL TrCH information	A1		
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC size		According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- Number of TBs and TTI lists		(This IE is repeated for TFI number)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.11 Parameter Set	
- CHOICE Logical channel list		All	
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 Parameter Set	
- Type of channel coding		Reference to clause 6.11 Parameter Set	
- Coding Rate		Reference to clause 6.11 Parameter Set	
- Rate matching attribute		Reference to clause 6.11 Parameter Set	
- Added or Reconfigured UL TrCH information	A2	1 E-DCH added with one DCCH MAC-d flow	Rel-6
- Uplink transport channel type		E-DCH	
- CHOICE UL parameters		E-DCH	
-CHOICE mode		TDD	Rel-7
- HARQ info for E-DCH			Rel-7
- HARQ RV Configuration		rvtable	Rel-7
- Added or reconfigured E-DCH MAC-d flow		(for DCCH)	
- E-DCH MAC-d flow identity		1	
- E-DCH MAC-d flow power offset		0	
- E-DCH MAC-d flow maximum number of retransmissions		7	
- E-DCH MAC-d flow multiplexing list		Not Present	
- CHOICE transmission grant type		Non-scheduled grant info	Rel-6

Information Element	Condition	Value/remark	Version
-CHOICE mode		TDD	Rel-7
-CHOICE *TDD Option*		1.28 Mcps TDD	Rel-7
-$N_{E\text{-}UCCH}$		1	Rel-7
-Timeslot Resource Related Information		00001	Rel-7
-Power Resource Related Information		32	Rel-7
-Activation Time		now	Rel-7
-Subframe number		1	Rel-7
-Repetition period		1	Rel-7
-Repetition Length		0	Rel-7
-Code Resource Information		4/1	Rel-7
-E-HICH Information		Not present	Rel-7
-Timeslot number		1	
- Channelisation code		16/1	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not present	
- Signature Sequence Group Index		0	
DL Transport channel information common for all transport channel			
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
-Individual DL CCTrCH information			
- DL TFCS Identity			
- TFCS ID		1	
- Shared Channel Indicator		FALSE	
- CHOICE DL parameters		Same as UL	
DL Transport channel information common for all transport channel	A2	Not Present	
- Added or Reconfigured TrCH information list	A1		
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
-DCH quality target			
- BLER Quality value		-63 (-6.3)	
- Added or Reconfigured TrCH information list	A2		
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		HS-DSCH	
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		1 (for DCCH)	
- MAC-d Flow Identity		1	
- T1		50	
- MAC-hs window size		Integer(32, 64, 96, 128, 160, 192, 256)	Rel-7
- MAC-d PDU size Info			
- MAC-d PDU size		148	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
- Added or reconfigured MAC-d flow			
Frequency info		Not Present	
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement	A1	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE *TDD option*		1.28 Mcps TDD	

Information Element	Condition	Value/remark	Version
- $PRX_{PDPCHdes}$		Reference to clause 6.11 Parameter set	
- CHOICE *UL OL PC info*		Individually signalled	
- CHOICE *TDD option*		1.28 Mcps TDD	
- TPC step size		1 dB	
- Primary CCPCH Tx Power		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	
- CHOICE TDD option		1.28 Mcps TDD	
- Uplink synchronization parameters			
- Uplink synchronization step size		1	
- Uplink synchronization frequency		1	
- Synchronization parameters		Not present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6.11 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6 Parameter Set	
- TFCI coding		Reference to clause 6 Parameter Set	
- Puncturing Limit		Reference to clause 6 Parameter Set	
- Repetition Period			
- Repetition Length		null	
- Uplink DPCH timeslots and codes			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
CHOICE channel requirement	A2	Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- CHOICE *TDD option*		1.28 Mcps TDD	
- $PRX_{PDPCHdes}$		Reference to clause 6.11 Parameter set	
- CHOICE *UL OL PC info*		Individually signalled	
- CHOICE *TDD option*		1.28 Mcps TDD	
- TPC step size		1 dB	
- Primary CCPCH Tx Power		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control			
- CHOICE Timing Advance		Enabled	

Information Element	Condition	Value/remark	Version
- CHOICE TDD option		1.28 Mcps TDD	
- Uplink synchronization parameters			
- Uplink synchronization step size		1	
- Uplink synchronization frequency		1	
- Synchronization parameters		Not present	
- UL CCTrCH List			
- TFCS ID		1	
- UL Target SIR		Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6.11 Parameter set.	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6 Parameter Set	
- TFCI coding		Reference to clause 6 Parameter Set	
- Puncturing Limit		Reference to clause 6 Parameter Set	
- Repetition Period			
- Repetition Length		null	
- Uplink DPCH timeslots and codes			
- Dynamic SF usage		FALSE	
- First individual timeslot info			
- Timeslot number			
- CHOICE TDD option		1.28 Mcps TDD	
- Timeslot number		1 OR 2 OR 3	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps TDD	
- Midamble allocation mode		Default midamble	
- Midamble configuration		16	
- Midamble Shift		Not Present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Sysbols		Not present	
- First timeslot Code List		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes		(SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		No more timeslots	
- UL CCTrCH List to Remove		Not present	
E-DCH Info	A1	Not Present	Rel-6
E-DCH info	A2		Rel-6
- MAC-es/e reset indicator		TRUE	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
-E-RUCCH Info			
-T-RUCCH		200ms	
- N-RUCCH		3	
- T-WAIT		40ms	
-E-PUCH Info			
-E-TFCS info			
-Reference Beta Information QPSK list		1	
-Reference Code Rate		1	
-Reference Beta		0	
-Reference Beta Information 16QAM list		1	
-Reference Code Rate		6	
-Reference Beta		0	
-CHOICE TDD mode		1.28 Mcps TDD	
-SNPL Reporting Type		Type1	
-PRXdes_base		-100	

Information Element	Condition	Value/remark	Version
-Beacon PL Est.		FALSE	
-TPC step size		1	
-Uplink synchronisation parameters			
-Uplink synchronisation step size		1	
-Uplink synchronisation frequency		1	
-E-PUCH TS configuration list		1	
-TS number		1	
-Midamble shift and burst type			
-Midamble Allocation Mode		Default midamble	
-Midamble configuration		16	
-Midamble Shift		Not present	
-Minimum allowed code rate		0	
-Maximum allowed code rate		63	
Downlink HS-PDSCH Information	A1	Not Present	Rel-6
Downlink HS-PDSCH Information	A2		Rel-6
- HS-SCCH Info			
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps	
- HS-SCCH Set Configuration			
- Timeslot number		0	
- First Channelisation code		(16/5)	
- Second Channelisation code		(16/6)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/11)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$			
- TPC step size		1dB	
- Timeslot number		0	
- First Channelisation code		(16/7)	
- Second Channelisation code		(16/8)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/12)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$			
- TPC step size		1dB	
- Timeslot number		0	
- First Channelisation code		(16/9)	
- Second Channelisation code		(16/10)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- BLER target		-2.0	
- HS-SICH configuration			
- Timeslot number		1	
- Channelisation code		(16/14)	
- Midamble Allocation mode		Default midamble	
- Midamble configuration		8	
- Ack-Nack Power Offset		0	
- $PRX_{HS-SICH}$			
- TPC step size		1dB	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- HS-PDSCH Midamble Configuration			
- Midamble Allocation Mode		Default midamble	
- Midamble Configuration		8	
- Midamble Shift		Not present	

Information Element	Condition	Value/remark	Version
Downlink information common for all radio links	A1		
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE *mode*		TDD	
- TPC Step Size		1 dB	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD (no data)	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
Downlink information common for all radio links	A2		Rel-5
- Downlink DPCH info common for all RL			
- Timing indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		TRUE	
Downlink information for each radio link list	A1		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- CHOICE *mode*		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Not present	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCH List			
- TFCS ID		1	
- Time info			
- Activation time		(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter set	
- TFCI coding		Reference to clause 6.11 Parameter set	
- Puncturing limit		Reference to clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots and codes			
- First Individual timeslot info			
- Timeslot number			
- CHOICE *more timeslots*			
- CHOICE TDD option		1.28 McpsTDD	
- Timeslot number		4 OR 5 OR 6	
- Individual timeslot info			
- TFCI existence		TRUE	
- Midamble shift and			

Information Element	Condition	Value/remark	Version
burst type			
- CHOICE TDD option		1.28 Mcps TDD	
-Midamble		Default	
Allocation Mode			
- Midamble		16 Integer(2, 4, 6, 8, 10, 12, 14, 16)	
configuration			
- Midamble Shift		Not present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS		Not present	
Symbols			
- First timeslot			
channelisation codes			
- CHOICE codes		Consecutive codes	
representation			
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List			
- UL TPC TFCS Identity		1	
- DL CCTrCH List to Remove		Not present	
-SCCPCH information for		Not Present	R99 and
FACH			Rel-4 only
Downlink information for each radio link list	A2		
- Downlink information for each radio link			
- Choice mode		TDD	
- Primary CCPCH info			
- CHOICE *mode*		TDD	
- CHOICE TDD option		1.28 Mcps TDD	
- TSTD indicator		FALSE	
- Cell parameters ID		Not present	
- SCTD indicator		FALSE	
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCH List			
- TFCS ID		1	
- Time info			
- Activation time		(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter set	
- TFCI coding		Reference to clause 6.11 Parameter set	
- Puncturing limit		Reference to clause 6.11 Parameter set	
- Repetition period		1	
- Repetition length		NULL	
- Downlink DPCH timeslots			
and codes			
- First Individual timeslot			
info			
- Timeslot number			
- CHOICE *more timeslots*			
- CHOICE TDD option		1.28 McpsTDD	
- Timeslot number		4 OR 5 OR 6	
- Individual timeslot			
info			
- TFCI existence		TRUE	
- Midamble shift and			
burst type			
- CHOICE TDD option		1.28 Mcps TDD	

Information Element	Condition	Value/remark	Version
-Midamble Allocation Mode		Default	
- Midamble configuration		16 Integer(2, 4, 6, 8, 10, 12, 14, 16)	
- Midamble Shift		Not present	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not present	
- First timeslot channelisation codes			
- CHOICE codes representation		Consecutive codes	
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List			
- UL TPC TFCS Identity		1	
- DL CCTrCH List to Remove		Not present	
-SCCPCH information for FACH		Not Present	R99 and Rel-4 only
-E-AGCH info			
- CHOICE mode			
-CHOICE TDD option		1.28 Mcps TDD	Rel-7
-RDI Indicator		False	Rel-7
-TPC step size		1	Rel-7
-E-AGCH set configuration		1	Rel-7
-Timeslot number		6	Rel-7
-First Channelisation code		16/3	Rel-7
-Second Channelisation code		16/4	Rel-7
-Midamble Allocation mode		Default midamble	Rel-7
-Midamble configuration		4	Rel-7
- Midamble Shift		Not present	Rel-7
- E-AGCH BLER target		-2	Rel-7
-CHOICE *mode*		TDD	Rel-7
-E-HICH Information			Rel-7
-CHOICE TDD option		1.28 Mcps TDD	Rel-7
- $N_{E\text{-}HICH}$		4	Rel-7
-E-HICH set configuration		1	Rel-7
-EI		0	Rel-7
-Timeslot number		6	Rel-7
-Channelisation code		16/5	Rel-7
-Midamble Allocation mode		Default midamble	Rel-7
-Midamble configuration		4	Rel-7
-Midamble Shift		Not present	Rel-7

Condition	Explanation	Version
A1	This IE is needed for “Stand-alone SRBs mapped on DCH/DCH”	
A2	This IE is needed for "Stand-alone SRBs mapped on E-DCH and HS-DSCH "	Rel-6
NOTE: If not specified, then A1 will be the default condition		

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (7.68 Mcps TDD option)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	0	

Information Element	Value/remark	Version
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	0000 0000 0000 0001B	
New H-RNTI	Not Present	Rel-6
CHOICE mode	TDD	Rel-7
- New E-RNTI	Not Present	Rel-7
RRC State Indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	1	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	

Information Element	Value/remark	Version
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not Present	

Information Element	Value/remark	Version
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	

Information Element	Value/remark	Version
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		

Information Element	Value/remark	Version
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS ID	(This IE is repeated for TFC number.)	
- UL TFCS		
- TFC subset	Default value is the complete existing set of transport format combinations	
- Allowed Transport Format combination	0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS	(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- TFCS complete reconfigure information		
- CHOICE TFCS Size	Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC size	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- Number of TBs and TTI lists	(This IE is repeated for TFI number)	
- CHOICE mode	TDD	
- Transmission Time Interval	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator		
- CHOICE DL parameters	Same as UL	
Added or Reconfigured TrCH information list		
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
-DCH quality target		
- BLER Quality value	-63 (-6.3)	
Frequency info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX information	Not Present	Rel-7
HS-SCCH less information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	Not Present	
Uplink DPCH info		Rel-6

Information Element	Value/remark	Version
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE *TDD option*	7.68 Mcps	Rel-7
- UL target SIR	Reference to clause 6.11 Parameter set	
- CHOICE mode	TDD	
- CHOICE *UL OL PC info*	Individually signalled	
- CHOICE *TDD option*	7.68 Mcps	Rel-7
- Individual timeslot interference	Not Present	
info		
- Individual timeslot interference		
- DPCH Constant Value		
- Primary CCPCH Tx Power	Not Present	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to clause 6.11 Parameter Set	
- TFCI coding	Reference to clause 6.11 Parameter Set	
- Puncturing Limit	Reference to clause 6.11 Parameter Set	
- Repetition Period	Reference to clause 6.11 Parameter Set	
- Repetition Length	Reference to clause 6.11 Parameter Set	
- CHOICE TDD Option	7.68 Mcps	Rel-7
- Uplink DPCH timeslots and	Default is to use the old timeslots and codes	Rel-7
codes VHCR		
- CPCH SET Info	(no data)	R99 and Rel-4 only
E-DCH info	Not Present	Rel-6
Downlink HS-PDSCH Information	Not Present	Rel-6
Downlink information common for all radio links		
- Downlink DPCH info common for all RL		
- Timing indicator	Maintain	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- DPC mode	0 (single)	
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps (no data)	Rel-7
- Default DPCH Offset Value	Not Present	
Downlink information for each radio link list		
- Downlink information for each radio link		
- CHOICE mode	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps	Rel-7
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	PCCPCH timeslot	
- Cell parameters ID	0	
- SCTD indicator		
- CHOICE DPCH info	Downlink DPCH info for each RL	Rel-6
- Downlink DPCH info for each RL		
- CHOICE mode	7.68Mcps TDD	Rel-7
- DL CCTrCH List		
- TFCS ID	1	
- Time info		
- Activation time	(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration	infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to the present document	
- TFCI coding	TRUE	
- Puncturing limit	Reference to clause 6 Parameter set	
- Repetition period	1	
- Repetition length	Empty	
- Downlink DPCH timeslots		Rel-7
and codes VHCR		
- CHOICE *more timeslots*		

Information Element	Value/remark	Version
- CHOICE TDD option	7.68 Mcps	Rel-7
- Timeslot number	The number of a downlink timeslot that has unassigned codes in a frame.	
- Individual timeslot info		
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	7.68 Mcps	Rel-7
-CHOICE Burst Type		
-Type 1		
-Midamble Allocation Mode	Default	
- Midamble configuration burst type 1 and 3	8	
- First timeslot channelisation codes VHCR		
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	Not Present	
- DL CCTrCH List to Remove	Not Present	
- E-AGCH Info	Not Present	Rel-7
- CHOICE mode	TDD	Rel-7
- E-HICH information	Not Present	Rel-7

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	Not Present	
RRC State Indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	9 , Integer(3...9)	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access 3.84 Mcps TDD capability update requirement	FALSE	
- UE radio access 1.28 Mcps TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup list		
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	1	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	

Information Element	Value/remark	Version
- Logical channel identity	1	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity		
- Logical channel identity	1	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	2	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	

Information Element	Value/remark	Version
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	3	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	

Information Element	Value/remark	Version
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	4	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	

Information Element	Value/remark	Version
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.3.4 Parameter Set.	
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.10.3.4 Parameter Set	
- CTFC	Reference to clause 6.10.3.4 Parameter Set	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0, Integer(0.. 3)	
- CHOICE mode	TDD	
- TFC subset	Not present Default value is the complete existing set of transport format combinations	
- TFC subset list	Not present	
- Added or Reconfigured UL TrCH information list		
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC size	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- Number of TBs and TTI lists	(This IE is repeated for TFI number)	
- Transmission Time Interval	According to clause 6 for standalone 13.6 kbps signalling radio bearer	
- Number of Transport blocks	Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 Parameter Set	
- Type of channel coding	Reference to clause 6.10 Parameter Set	
- Coding Rate	Reference to clause 6.10 Parameter Set	
- Rate matching attribute	Reference to clause 6.10 Parameter Set	
- CRC size	Reference to clause 6.10 Parameter Set	
- DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- CHOICE DL parameters	Same as UL	
- UL DCH TFCS Identity	1	

Information Element	Value/remark	Version
- Shared Channel Indicator	FALSE	
- Added or Reconfigured TrCH information list		
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
-DCH quality target		
- BLER Quality value	-63 (-6.3)	
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present Default value is the existing maximum UL TX power	
CHOICE channel requirement	Not present	
Downlink information common for all radio links	Not present	
Downlink information for each radio link list	Not present	

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	0000 0000 0000 0001B	
RRC State Indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	9 , Integer(3...9)	
Capability update requirement		
- UE radio access FDD capability update requirement		
- UE radio access 3.84 Mcps TDD capability update requirement		
- UE radio access 1.28 Mcps TDD capability update requirement		
- System specific capability update requirement list		
CHOICE specification mode		
- Complete specification		
- Signalling RB information to setup list		
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	1	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	

Information Element	Value/remark	Version
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity		
- Logical channel identity	1	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	2	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200 Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	

Information Element	Value/remark	Version
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	3	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	

Information Element	Value/remark	Version
- RB identity	4	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		

Information Element	Value/remark	Version
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC	Reference to clause 6.11.5.4 Parameter Set	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0, Integer(0.. 3)	
- CHOICE mode	TDD	
- TFC subset	Not present. Default value is the complete existing set of transport format combinations	
- TFC subset list	Not present	
- DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- CHOICE DL parameters	Same as UL	
- UL DCH TFCS Identity	1	
- Shared Channel Indicator	FALSE	
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present. Default value is the existing maximum UL TX power	
CHOICE channel requirement	Not present	
Downlink information common for all radio links	Not present	
Downlink information for each radio link list	Not Present	

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	0000 0000 0000 0001B	
New H-RNTI	Not Present	Rel-6
CHOICE mode	TDD	Rel-7
- New E-RNTI	Not Present	Rel-7
RRC State Indicator	CELL_FACH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement		
- UE radio access 3.84 Mcps TDD capability update requirement		
- UE radio access 1.28 Mcps TDD capability update requirement		
- System specific capability update requirement list		
CHOICE specification mode		
- Complete specification		
- Signalling RB information to setup list		

Information Element	Value/remark	Version
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	1	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity		
- Logical channel identity	1	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	2	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
	Not present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	

Information Element	Value/remark	Version
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	3	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		

Information Element	Value/remark	Version
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	4	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No discard	
- MAX_DAT	15	
- Transmission window size	32	
- Timer_RST	500	
- Max_RST	1	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Window	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	32	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configure	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity		
- Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	

Information Element	Value/remark	Version
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	Not Present	
- Logical channel identity	4	
- UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set.	
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set	
- CTFC	Reference to clause 6.11.5.4 Parameter Set	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0, Integer(0.. 3)	
- CHOICE mode	TDD	
- TFC subset	Not present. Default value is the complete existing set of transport format combinations	
- TFC subset list	Not present	
- DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
-Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- CHOICE DL parameters	Same as UL	
- UL DCH TFCS Identity	1	
- Shared Channel Indicator	FALSE	
Frequency info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX information	Not Present	Rel-7
HS-SCCH less information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	Not Present. Default value is the existing maximum UL TX power	
Uplink DPCH info	Not Present	
E-DCH info	Not Present	Rel-6
Downlink HS-PDSCH information	Not Present	Rel-6
Downlink information common for all radio links	Not Present	
Downlink information for each radio link list	Not Present	

Contents of RRC CONNECTION SETUP COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION SETUP message.
START list	This IE is checked to see if it is present.
UE radio access capability	Not checked
UE radio access capability extension	Not checked
UE system specific capability	Not checked

Contents of SECURITY MODE COMMAND message: AM

Information Element	Condition	Value/remark
Message Type	A1, A2	
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3
Integrity check info		
- Message authentication code		Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message Sequence Number		Set to an arbitrarily selected integer between 0 and 15
Security capability		
- Ciphering algorithm capability		
- UEA0		If ciphering is not indicated to be active on IXIT statements in 3GPP TS 34.123-2 [3], set this IE to TRUE.
- UEA1		If ciphering is indicated to be active on IXIT statements in 3GPP TS 34.123-2 [3], set this IE to TRUE.
- Spare		FALSE
- Integrity protection algorithm capability		0000000000000010B (UIA1)
- UIA1		TRUE
- Spare		FALSE
Ciphering mode info		This presence of this IE is dependent on IXIT statements in 3GPP TS 34.123-2 [3]. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted.
- Ciphering mode command		Start/restart
- Ciphering algorithm		Use the same ciphering algorithm specified in "ciphering algorithm capability" IE in this message.
- Ciphering activation time for DPCH		Not Present
- Radio bearer downlink ciphering activation time info		
- Radio bearer activation time		
- RB identity		1
- RLC sequence number		Current RLC SN+2
- RB identity		2
- RLC sequence number		Current RLC SN+2
- RB identity		3
- RLC sequence number		Current RLC SN + 2
- RB identity		4
- RLC sequence number		Current RLC SN + 2
Integrity protection mode info		
- Integrity protection mode command		Start
- Downlink integrity protection activation info		Not Present
- Integrity protection algorithm		UIA1
- Integrity protection initialisation number		SS selects an arbitrary 32 bits number for FRESH
CN domain identity		Supported domain
UE system specific security capability	A1	Not Checked
UE system specific security capability	A2	
- Inter-RAT UE security capability		
- CHOICE *system*		GSM
- GSM security capability		The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the

Information Element	Condition	Value/remark
		RRC CONNECTION SETUP COMPLETE message.

Condition	Explanation
A1	UE not supporting GSM
A2	UE supporting GSM

Contents of SECURITY MODE COMPLETE message: AM

Information Element	Value/remark
Message Type	
RRC transaction identifier	The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message.
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
Uplink integrity protection activation info	Not checked.
Radio bearer uplink ciphering activation time info	If ciphering is not activated in SECURITY MODE COMMAND message, this IE must be absent. Else, SS checks this IE for the presence of activation times for all ciphered uplink RLC-UM and RLC-AM RBs.

Contents of UPLINK DIRECT TRANSFER message: AM

Information Element	Value/remark
Message Type	
Integrity check info	
- Message authentication code	This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message sequence number	This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value.
CN domain identity	Checked to see if set to supported CN domain as specified in the IXIT statements
NAS message	Set according to that indicated in specific message content clause
Measured results on RACH	Not checked

9.1.3 Default RRC Message Contents for MBSFN (TDD)

Contents of MBMS GENERAL INFORMATION message: UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message type			Rel-6
MBMS preferred frequency information		Not Present	Rel-6
MBMS timers and counters			Rel-6
- T318		Not Present (MD)	
MICH configuration information			Rel-6
- MICH Power offset		0dB	
- CHOICE *mode*		TDD	
- Timeslot Number		0	
- Midamble shift and burst type			
- CHOICE *TDD option*		3.84 Mcps	
- CHOICE Burst Type		MBSFN Burst Type	Rel-7

Information Element	Condition	Value/remark	Version
- CHOICE TDD option		3.84 Mcps TDD	
- Channelisation code		16/1	
- Repetition period/length		(16,2)	
- Offset		14	
- MBMS Notification indicator length		Not Present (MD - default value = 4)	
Cell group identity		'000000000001'	Rel-6
Default MSCH configuration information		Not Present	Rel-6
Indicate changes in MBMS Selected Services		Not Present (MD - default value = FALSE)	Rel-6
MBSFN inter frequency neighbour list			Rel-7
- MBSFN frequency			
- CHOICE *mode*		TDD	
- UARFCN (Nt)	A1	Refer to clause 5.1 "Test frequencies" for frequency "f2"	
- UARFCN (Nt)	A2	Refer to clause 5.1 "Test frequencies" for frequency "f1"	
- CHOICE *MBSFN services notification*		MBSFN services notified	
- no data			

Condition	Explanation
A1	This IE is needed for cells on frequency "f1" (by default Cell 31, Cell 32, Cell 37 and Cell 38)
A2	This IE is needed for cells on frequency "f2" (by default Cell 33, Cell 34, Cell 35 and Cell 36)

Contents of MBMS GENERAL INFORMATION message: UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message type			Rel-6
MBMS preferred frequency information		Not Present	Rel-6
MBMS timers and counters			Rel-6
- T318		Not Present (MD)	
MICH configuration information			Rel-6
- MICH Power offset		0dB	
- CHOICE *mode*		TDD	
- Timeslot Number		1	
- Midamble shift and burst type			
- CHOICE *TDD option*		1.28 Mcps	
- Codes list			Rel-7
- Channelisation code		16/1	
- MBSFN Special Time Slot		TS7	
- Repetition period/length		(16,2)	
- Offset		14	
- MBMS Notification indicator length		Not Present (MD - default value = 4)	
Cell group identity		'000000000001'	Rel-6
Default MSCH configuration information		Not Present	Rel-6
Indicate changes in MBMS Selected Services		Not Present (MD - default value = FALSE)	Rel-6
MBSFN inter frequency neighbour list			Rel-7
- MBSFN frequency			
- CHOICE *mode*		TDD	
- UARFCN (Nt)	A1	Refer to clause 5.1 "Test frequencies" for frequency "f2"	
- UARFCN (Nt)	A2	Refer to clause 5.1 "Test frequencies" for frequency "f1"	
- CHOICE *MBSFN services notification*		MBSFN services notified	
- no data			

Condition	Explanation
A1	This IE is needed for cells on frequency "f1" (by default Cell 31, Cell 32, Cell 37 and Cell 38)
A2	This IE is needed for cells on frequency "f2" (by default Cell 33, Cell 34, Cell 35 and Cell 36)

Contents of MBMS GENERAL INFORMATION message: UM (7.68 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message type			Rel-6
MBMS preferred frequency information		Not Present	Rel-6
MBMS timers and counters			Rel-6
- T318		Not Present (MD)	
MICH configuration information			Rel-6
- MICH Power offset		0dB	
- CHOICE *mode*		TDD	
- Timeslot Number		0	
- Midamble shift and burst type			
- CHOICE *TDD option*		7.68 Mcps	
- CHOICE Burst Type		MBSFN Burst Type	Rel-7
- CHOICE TDD option		7.68 Mcps TDD	
- Channelisation code		32/1	
- Repetition period/length		(16,2)	
- Offset		14	
- MBMS Notification indicator length		Not Present (MD - default value = 4)	
Cell group identity		'000000000001'	Rel-6
Default MSCH configuration information		Not Present	Rel-6
Indicate changes in MBMS Selected Services		Not Present (MD - default value = FALSE)	Rel-6
MBSFN inter frequency neighbour list			Rel-7
- MBSFN frequency			
- CHOICE *mode*		TDD	
- UARFCN (Nt)	A1	Refer to clause 5.1 "Test frequencies" for frequency "f2"	
- UARFCN (Nt)	A2	Refer to clause 5.1 "Test frequencies" for frequency "f1"	
- CHOICE *MBSFN services notification*		MBSFN services notified	
- no data			

Condition	Explanation
A1	This IE is needed for cells on frequency "f1" (by default Cell 31, Cell 32, Cell 37 and Cell 38)
A2	This IE is needed for cells on frequency "f2" (by default Cell 33, Cell 34, Cell 35 and Cell 36)

Contents of MBMS MODIFIED SERVICES INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
Modified service list	1 entry per modified service - maximum 8. If no services are modified in the current modification period this IE is Not Present	Rel-6
- MBMS Transmission identity		
- MBMS Service ID		
- MBMS Service ID	Set to the value of the service ID being modified (e.g. '000001')	
- CHOICE *PLMN identity*	SameAs-MIB	
- no data		
- MBMS Session ID	'01'	
- MBMS required UE action	Acquire PTM RB info	
- MBMS preferred frequency	Not Present	
- Continue MCCH reading	FALSE	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
MBMS re- acquire MCCH	Not Present	Rel-6
MBMS dynamic persistence level	Not Present	Rel-6
End of modified MCCH information	Not Present	Rel-6
MBMS number of neighbour cells	0	Rel-6
MBMS all unmodified p-t-m services	Not Present	Rel-6
MBMS p-t-m activation time	Set to the 11 LSB of the first SFN of the next modification period.	Rel-6
MIB Value tag	Not Present	Rel-7

Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM (Mixed Local/National carrier)

Information Element	Value/remark	Version
Message type		Rel-6
Unmodified service list	8 services by default. See NOTE 1.	Rel-6
- MBMS Transmission identity		
- MBMS Service ID	(National Service 5)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 6)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(Local Service 1)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(Local Service 2)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 1)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 2)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 3)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	

Information Element	Value/remark	Version
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 4)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7

Condition	Information Element	Value/remark	Explanation
A1	- MBMS Session ID	Not Present	Condition used when the session is currently not being transmitted
	- MBMS required UE action	'None'	
A2	- MBMS Session ID	'01'	Condition used when the session is currently ongoing
	- MBMS required UE action	'Acquire PTM RB info'	

NOTE 1: Any service (as identified by MBMS Service ID) which is included in MBMS MODIFIED SERVICES INFORMATION message in the current modification period shall not have an unmodified service entry in the list of services in this message.

NOTE 2: By default the Mixed Local/National Carrier (on frequency "f2") is broadcast by Cell 33, Cell 34, Cell 35 and Cell 36.

Contents of MBMS UNMODIFIED SERVICES INFORMATION message: UM (Dedicated National carrier)

Information Element	Value/remark	Version
Message type		Rel-6
Unmodified service list	8 services by default. See NOTE 1.	Rel-6
- MBMS Transmission identity		
- MBMS Service ID	(National Service 1)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 2)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 3)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 4)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	Not Present (MD)	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 5)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(National Service 6)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(Local Service 1)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	

Information Element	Value/remark	Version
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7
- MBMS Transmission identity		
- MBMS Service ID	(Local Service 2)	
- MBMS Service ID	Refer to clause 11.2.4 "MBSFN service availability"	
- CHOICE PLMN identity	SameAs-MIB	
- no data		
- MBMS Session ID	Value set according to table for condition A1 or A2	
- MBMS required UE action	Value set according to table for condition A1 or A2	
- MBMS preferred frequency	Not Present	
- MBSFN cluster frequency	1	Rel-7

Condition	Information Element	Value/remark	Explanation
A1	- MBMS Session ID	Not Present	Condition used when the session is currently not being transmitted
	- MBMS required UE action	'None'	
A2	- MBMS Session ID	'01'	Condition used when the session is currently ongoing
	- MBMS required UE action	'Acquire PTM RB info'	

NOTE 1: Any service (as identified by MBMS Service ID) which is included in MBMS MODIFIED SERVICES INFORMATION message in the current modification period shall not have an unmodified service entry in the list of services in this message.

NOTE 2: By default the Dedicated National Carrier (on frequency "f1") is broadcast by Cell 31, Cell 32, Cell 37 and Cell 38.

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (3.84 Mcps)

Information Element	Condition	Value/remark	Version
Message type			Rel-6
RB information list		2 entries in the list	Rel-6
- RB identity			
- MBMS Common RB identity		14	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
- RB identity			
- MBMS Common RB identity		15	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	

Information Element	Condition	Value/remark	Version
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
TrCh information for each TrCh		2 entries in the list	Rel-6
- Transport channel identity			
- MBMS Common TrCh identity		1	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.10 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.10 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.10 "Parameter Set"	
- CRC size		Reference to clause 6.10 "Parameter Set"	
- Transport channel identity			
- MBMS Common TrCh identity		2	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.10 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.10 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.10 "Parameter Set"	
- CRC size		Reference to clause 6.10 "Parameter Set"	
TrCh information for each CCTrCh		Not Present (Default TFCS applies for each CCTrCh)	Rel-6
PhyCh information		2 entries in list	Rel-6
- PhyCh identity			
- MBMS Common PhyCh identity		23	
- Secondary CCPCH info MBMS			
- CHOICE *mode*		1.28/3.84 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.10 "Parameter Set"	
- Puncturing limit		Reference to clause 6.10 "Parameter Set"	
- Downlink Timeslots and Codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		3.84 Mcps	
- Timeslot number		1	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		3.84 Mcps TDD	
- CHOICE Burst Type		MBSFN Burst Type	Rel-7
- no data			Rel-7
- CHOICE *TDD option*		3.84Mcps TDD	

Information Element	Condition	Value/remark	Version
- no data			
- First timeslot channelisation codes			
- CHOICE *codes representation*		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE *more timeslots*	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		3.84 Mcps	
- Timeslot number		8	
- Modulation		Reference to clause 6.10 "Parameter Set"	Rel-7
- PhyCh identity			Rel-6
- MBMS Common PhyCh identity		27	
- Secondary CCPCH info MBMS			Rel-6
- CHOICE *mode*		1.28/3.84 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.10 "Parameter Set"	
- Puncturing limit		Reference to clause 6.10 "Parameter Set"	
- Downlink Timeslots and Codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		3.84 Mcps	
- Timeslot number		4	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		3.84 Mcps TDD	
- CHOICE Burst Type		MBSFN Burst Type	Rel-7
- no data			Rel-7
- CHOICE *TDD option*		3.84Mcps TDD	
- no data			
- First timeslot channelisation codes			
- CHOICE *codes representation*		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE *more timeslots*	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		3.84 Mcps	
- Timeslot number		11	
- Modulation		Reference to clause 6.10 "Parameter Set"	Rel-7

Condition	Explanation
A1	This IE is needed for RBs configured to use one timeslot. Refer to clause 6.10 "Parameter Set"
A2	This IE is needed for RBs configured to use two timeslots. Refer to clause 6.10 "Parameter Set"

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (1.28 Mcps)

Information Element	Condition	Value/remark	Version
Message type			Rel-6
RB information list		2 entries in the list	Rel-6
- RB identity			
- MBMS Common RB identity		14	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
- RB identity			
- MBMS Common RB identity		15	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	

Information Element	Condition	Value/remark	Version
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
TrCh information for each TrCh		2 entries in the list	Rel-6
- Transport channel identity			
- MBMS Common TrCh identity		1	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.11 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.11 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.11 "Parameter Set"	
- CRC size		Reference to clause 6.11 "Parameter Set"	
- Transport channel identity			
- MBMS Common TrCh identity		2	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.11 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.11 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.11 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.11 "Parameter Set"	
- CRC size		Reference to clause 6.11 "Parameter Set"	
TrCh information for each CCTrCh		Not Present (Default TFCS applies for each CCTrCh)	Rel-6
PhyCh information		2 entries in list	Rel-6
- PhyCh identity			
- MBMS Common PhyCh identity		23	
- Secondary CCPCH info MBMS			
- CHOICE *mode*		1.28/3.84 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.11 "Parameter Set"	
- Puncturing limit		Reference to clause 6.11 "Parameter Set"	
- Downlink Timeslots and Codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		1.28 Mcps	
- Timeslot number		1	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		1.28 Mcps TDD	
- Midamble Allocation Mode		Common midamble	Rel-7
- Midamble configuration		2	Rel-7
- CHOICE *TDD option*		1.28Mcps TDD	

Information Element	Condition	Value/remark	Version
- Modulation		Reference to clause 6.11 "Parameter Set"	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not Present	
- First timeslot channelisation codes			
- CHOICE *codes representation*		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE *more timeslots*	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		1.28 Mcps	
- Timeslot number		2	
- MBSFN Special Time Slot		TS7	Rel-7
- Modulation		Reference to clause 6.11 "Parameter Set"	Rel-7
- PhyCh identity			Rel-6
- MBMS Common PhyCh identity		27	
- Secondary CCPCH info MBMS			Rel-6
- CHOICE *mode*		1.28/3.84 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.11 "Parameter Set"	
- Puncturing limit		Reference to clause 6.11 "Parameter Set"	
- Downlink Timeslots and Codes			
- First individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		1.28 Mcps	
- Timeslot number		4	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		1.28 Mcps TDD	
- Midamble Allocation Mode		Common midamble	
- Midamble configuration		2	
- CHOICE *TDD option*		1.28Mcps TDD	
- Modulation		Reference to clause 6.11 "Parameter Set"	
- SS-TPC Symbols		1	
- Additional TPC-SS Symbols		Not Present	
- First timeslot channelisation codes			
- CHOICE *codes representation*		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE *more timeslots*	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		1.28 Mcps	
- Timeslot number		4	
- MBSFN Special Time Slot		TS7	Rel-7
- Modulation		Reference to clause 6.11 "Parameter Set"	Rel-7
LCR TDD MBSFN information		Not Present	Rel-7

Condition	Explanation
A1	This IE is needed for RBs configured to use one timeslot. Refer to clause 6.11 "Parameter Set"
A2	This IE is needed for RBs configured to use two timeslots. Refer to clause 6.11 "Parameter Set"

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM (7.68 Mcps)

Information Element	Condition	Value/remark	Version
Message type			Rel-7
RB information list		2 entries in the list	Rel-7
- RB identity			
- MBMS Common RB identity		14	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
- RB identity			
- MBMS Common RB identity		15	
- PDCP info			
- Support for lossless SRNS relocation		Not Present	
- Max PDCP SN window size		Not Present	
- PDCP PDU header		absent	
- Header compression information		Not Present	

Information Element	Condition	Value/remark	Version
- RLC info			
- DL UM RLC LI size		15	
- DL Duplication Avoidance and Reordering info		Not Present	
- DL Out of sequence delivery info		Not Present	
TrCh information for each TrCh		2 entries in the list	Rel-7
- Transport channel identity			
- MBMS Common TrCh identity		1	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.10 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.10 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.10 "Parameter Set"	
- CRC size		Reference to clause 6.10 "Parameter Set"	
- Transport channel identity			
- MBMS Common TrCh identity		2	
- TFS			
- CHOICE *Transport channel type*		Common transport channels	
- Dynamic Transport Format Information			
- RLC Size		Reference to clause 6.10 "Parameter Set"	
- Number of TBs and TTI List		(This IE is repeated for each TFI)	
- Number of Transport blocks		Reference to clause 6.10 "Parameter Set"	
- CHOICE *mode*		TDD	
- Transmission Time Interval		Not Present	
- CHOICE *Logical Channel List*		ALL	
- no data			
- Semi-static Transport Format information			
- Transmission time interval		Reference to clause 6.10 "Parameter Set"	
- Type of channel coding		Turbo	
- Coding Rate		Not Present	
- Rate matching attribute		Reference to clause 6.10 "Parameter Set"	
- CRC size		Reference to clause 6.10 "Parameter Set"	
TrCh information for each CCTrCh		Not Present (Default TFCS applies for each CCTrCh)	Rel-7
PhyCh information		2 entries in list	Rel-7
- PhyCh identity			
- MBMS Common PhyCh identity		23	
- Secondary CCPCH info MBMS			
- CHOICE *mode*		7.68 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.10 "Parameter Set"	
- Puncturing limit		Reference to clause 6.10 "Parameter Set"	
- Downlink Timeslots and Codes VHCR			
- First Individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		7.68 Mcps option	
Timeslot number		1	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		7.68 Mcps TDD	
- CHOICE *Burst Type*		MBSFN Burst Type	
- no data		Default	
- CHOICE *TDD option*		7.68Mcps TDD	

Information Element	Condition	Value/remark	Version
- no data			
- First timeslot channelisation codes VHCR			
- CHOICE *codes representation*		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE more timeslots	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		7.68 Mcps	
- Timeslot number		8	
- Modulation		Reference to clause 6.10 "Parameter Set"	
- PhyCh identity			
- MBMS Common PhyCh identity		27	
- Secondary CCPCH info MBMS			
- CHOICE *mode*		7.68 Mcps TDD	
- Common timeslot info MBMS			
- 2nd interleaving mode		Frame	
- TFCI coding		Reference to clause 6.10 "Parameter Set"	
- Puncturing limit		Reference to clause 6.10 "Parameter Set"	
- Downlink Timeslots and Codes VHCR			
- First Individual timeslot info			
- Timeslot number			
- CHOICE *TDD option*		7.68 Mcps option	
- Timeslot number		4	
- TFCI existence		TRUE	
- Midamble Shift and burst type			
- CHOICE *TDD option*		7.68 Mcps TDD	
- CHOICE *Burst Type*		MBSFN Burst Type	
- no data		Default	
- CHOICE *TDD option*		7.68 Mcps TDD	
- no data			
- First timeslot channelisation codes VHCR			
- CHOICE codes representation		Consecutive codes	
- First channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- Last channelisation code		Reference clause 5.5.2 "Downlink physical channels code allocation for signalling"	
- CHOICE more timeslots	A1	No more timeslots	
- no data			
- CHOICE *more timeslots*	A2	Timeslot list	
- Additional timeslot list			
- CHOICE *parameters*		Same as last	
- Timeslot number			
- CHOICE *TDD option*		7.68 Mcps	
- Timeslot number		11	
- Modulation		Reference to clause 6.10 "Parameter Set"	

Condition	Explanation
A1	This IE is needed for RBs configured to use one timeslot. Refer to clause 6.10 "Parameter Set"
A2	This IE is needed for RBs configured to use two timeslots. Refer to clause 6.10 "Parameter Set"

Contents of MBMS CURRENT CELL P-T-M RB INFORMATION message: UM

Information Element	Condition	Value/remark	Version
Message type	A1, A2, A3		Rel-6
S-CCPCH list	A1	Not Present	Rel-6
S-CCPCH list	A2	Contains 1 S-CCPCH	Rel-6
S-CCPCH list	A3	Contains 2 S-CCPCH	Rel-6
- S-CCPCH identity		Not Present	
- Secondary CCPCH info	A2, A3	23	
- MBMS Soft Combining Timing Offset		Not Present	
- TrCh information common for all TrCh		Not Present (MD)	
- TrCH information list	A2, A3		
- TrCh information		1	
- RB information list			
- RB information			
- RB information		14	
- MBMS short transmission ID		Reference to the service which is being provided on this RB. See Note 1.	
- MBMS logical channel identity		1	
- L1 combining status		Not Present	
- MSCH configuration information		Not Present	
- S-CCPCH identity		Not Present	
- Secondary CCPCH info	A3	27	
- MBMS Soft Combining Timing Offset		Not Present	
- TrCh information common for all TrCh		Not Present (MD)	
- TrCH information list	A3		
- TrCh information		2	
- RB information list			
- RB information			
- RB information		15	
- MBMS short transmission ID		Reference to the service which is being provided on this RB. See Note 1.	
- MBMS logical channel identity		2	
- L1 combining status		Not Present	
- MSCH configuration information		Not Present	
S-CCPCH in SIB type 5		Not Present	Rel-6
MBSFN TDM Info List		Not Present	Rel-7

Condition	Explanation
A1	No services ongoing or starting
A2	1 service ongoing or starting
A3	2 services ongoing or starting

NOTE 1: MBMS short transmission ID is an index to a service in a list of services. The list is compiled by concatenating, in the following order, the lists of services from the MBMS MODIFIED SERVICES INFORMATION message and the MBMS UNMODIFIED SERVICES INFORMATION messages transmitted in the same modification period as this message.

9.2 Default Message Contents for RF

This clause contains the default values of common messages for RF test. The parameters of the UL/DL reference measurement channel 12.2 kbps, the DL reference measurement channel for BTFD, UE test loop mode 1 without Dummy DCCH transmission and UE test loop mode 2 with Dummy DCCH transmission are set to default message contents.

9.2.1 Default Message Contents for RF (FDD)

Contents of Activate RB Test Mode message

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	44h

Contents of Close UE Test Loop message (UE test loop mode 1 without Dummy DCCH transmission)

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	40h
UE test loop mode	00h
UE test loop mode 1 LB setup	03h 00h F4h 0Ah

Contents of Close UE Test Loop message (UE test loop mode 2 without Dummy DCCH transmission)

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	40h
UE test loop mode	01h

Contents of Open UE Test Loop message

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	42h

Contents of MBMS COMMON P-T-M RB INFORMATION message: UM

Information Element	Value/remark	Version
Message type		Rel-6
RB information list	One entry in the list	Rel-6
- RB identity	14	Rel-6
- PDCP info		
- Support for lossless SRNS relocation	Not Present	
- PDCP PDU header	Absent	
- Header compression information	Not Present	
- RLC info		
- DL UM RLC LI size	7	
- DL Duplication Avoidance and Reordering info	Not Present	
TrCh information for each TrCh	One entry in the list	Rel-6
- Transport channel identity	17	Rel-6
- TFS		
- CHOICE *Transport channel type*	Common transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
- Number of TBs List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
- CHOICE *Logical channel list*	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
- Type of channel coding	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
- Coding Rate	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
- Rate matching attribute	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2]	
- CRC size	Reference to TS34.121 [2] Annex C.12 DL reference parameters or as specified within test case in TS34.121 [2].	
TrCh information for each CCTrCh	One entry in the list	Rel-6
- CCTrCH identity	1	Rel-6
- TFCS		
- CHOICE *TFCI signalling*	Normal	
- TFCI Field 1 information		
- CHOICE *TFCS representation*	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from TS34.121 [2] Annex C.12 parameter set or as specified within test case in TS34.121 [2].	
- CTFC information	This IE is repeated for number of CTFCs from TS34.121 [2] Annex C.12	

Information Element	Value/remark	Version
	parameter set or as specified within test case in TS34.121 [2].	
- CTFC	Reference to TS34.121 [2] Annex C.12 parameter set or as specified within test case in TS34.121 [2].	
- Power offset information	Not Present	
PhyCh information	One entry in list	Rel-6
- PhyCh identity	13	Rel-6
- Secondary CCPCH info MBMS		
- CHOICE *mode*	FDD	
- Secondary scrambling code	Not Present	
- STTD indicator	FALSE	
- Spreading factor	Reference to TS34.121 [2] Annex C.12 DL reference parameters.	
- Code number	Reference to TS34.121 [2] Annex E.6.4 “Downlink physical channels code allocation for MBMS test cases”	
- Timing Offset	Not Present Absence of this IE is equivalent to default value 0.	

Contents of PAGING TYPE 1 message: TM (CS)

Information Element	Value/remark
Message Type	
Paging record list	
-Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Streaming Call
- CN domain identity	CS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of PAGING TYPE 1 message: TM (PS)

Information Element	Value/remark
Message Type	
Paging record list	
-Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Interactive Call
- CN domain identity	PS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of PAGING TYPE 2 message: TM (PS)

Information Element	Value/remark
Message Type	
RRC transaction identifier	Arbitrarily selects one integer between 0 to 3
Integrity check info	
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC message sequence number	SS provides the value of this IE, from its internal counter.
- Paging cause	Terminating Interactive Call
- CN domain identity	PS domain
- Paging record tyoe identifier	TMSI(GSM-MAP)/P-TMSI

Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)

Information Element	Condition	Value/remark	Version
Message Type	A1, A3, A4, A5, A6, A7, A8, A9		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI		Not Present	
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		Not Present	Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
CHOICE specification mode		Complete specification	Rel-6
- Signalling RB information to setup		Not Present	
- RAB information for setup list	A1, A3, A4, A5		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RAB information for setup list	A6, A7, A8, A9		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT315	
- RB information to setup list			
- RB information to setup			
- RB identity	A1	10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- One sided RLC re-establishment		FALSE	Rel-5
- RB mapping info			
- Information for each multiplexing			

Information Element	Condition	Value/remark	Version
option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity	A3, A4, A5	10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		Selected with Total RLC AM Buffer Size	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		400	
- Timer_poll		400	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size	A3	1280 bits	Rel-5
- DL RLC PDU size	A4	2880 bits	Rel-5
- DL RLC PDU size	A5	3840 bits	Rel-5
- In-sequence delivery	A3, A4, A5	TRUE	
- Receiving window size		Selected with Total RLC AM Buffer Size	
- Downlink RLC status info			
- Timer_status_prohibit		330	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	Rel-5
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel		6	

Information Element	Condition	Value/remark	Version
identity			
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RB identity	A6, A7, A8, A9	20	
- PDCP info		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		Selected with Total RLC AM Buffer Size	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		400	
- Timer_poll		400	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size	A6	1280 bits	Rel-5
- DL RLC PDU size	A7	2880 bits	Rel-5
- DL RLC PDU size	A8	3840 bits	Rel-5
- DL RLC PDU size	A9	336 bits	Rel-5
- In-sequence delivery	A6, A7, A8, A9	TRUE	
- Receiving window size		Selected with Total RLC AM Buffer Size	
- Downlink RLC status info			
- Timer_status_prohibit		330	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- One sided RLC re-establishment		FALSE	Rel-5
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit list	

Information Element	Condition	Value/remark	Version
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		7	
RB information to reconfigure list	A1, A3, A4, A5, A6, A7, A8, A9	Not Present	Rel-6
RB information to be affected list		Not Present	
Downlink counter synchronization info		Not Present	
PDCP ROHC target mode		Not Present	Rel-5
UL Transport channel information for all transport channels			
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		2 bit CTFC	
- CTFC information		4 TFCs	
- 2bit CTFC		0	
-Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		2	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		3	
- Power offset Information			
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		FDD	
- Gain factor ßc		8	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
Deleted UL TrCH information list		Not Present	
Added or Reconfigured TrCH information list	A1	1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	

Information Element	Condition	Value/remark	Version
- Dynamic Transport Format Information			
- RLC size		244 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		20	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		256	
- CRC size		16	
Added or Reconfigured TrCH information list	A3, A4, A5, A6, A7, A8, A9	1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information			
- RLC size		240 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		20	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		256	
- CRC size		16	
CHOICE mode	A1, A3, A4, A5, A6, A7, A8	Not Present	
DL Transport channel information common for all transport channel			
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Same as UL	
Deleted DL TrCH information list		Not Present	
CHOICE mode	A9	Not Present	
DL Transport channel information common for all transport channel			
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		DL DCH TFCS	
- DL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		4 bit CTFC	
- CTFC information		6 TFCs	
- 4bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	

Information Element	Condition	Value/remark	Version
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 4bit CTFC		3	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 4bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 4bit CTFC		4	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 4bit CTFC		2	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 4bit CTFC		5	
- Power offset Information			
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		FDD	
- Gain factor ßc		8	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
Deleted DL TrCH information list		Not Present	
Added or Reconfigured TrCH information list	A1	1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A3, A6	1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		1280 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel list		ALL	
- Semi-static Transport Format			

Information Element	Condition	Value/remark	Version
information			
- Transmission time interval		20	
- Type of channel coding		Turbo	
- Rate matching attribute		256	
- CRC size		16	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A4, A7	1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		2880 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel list		ALL	
- Semi-static Transport Format information			
- Transmission time interval		20	
- Type of channel coding		Turbo	
- Rate matching attribute		256	
- CRC size		16	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A5, A8	1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		3840 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel list		ALL	
- Semi-static Transport Format information			
- Transmission time interval		10	
- Type of channel coding		Turbo	
- Rate matching attribute		256	
- CRC size		16	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Added or Reconfigured TrCH information list	A9	1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Explicit	
- TFS			

Information Element	Condition	Value/remark	Version
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic transport format information			
- RLC Size		336 bits	
- Number of TBs and TTI List		3	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		4	
- CHOICE Logical channel list		ALL	
- Semi-static Transport Format information			
- Transmission time interval		20	
- Type of channel coding		Turbo	
- Rate matching attribute		143	
- CRC size		16	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Frequency info	A1, A3, A4, A5, A6, A7, A8, A9	Not Present	
Multi-frequency Info		Not Present	Rel-7
DTX-DRX timing information		Not Present	Rel-7
DRX Information		Not Present	Rel-7
HS-SCCH less Information		Not Present	Rel-7
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement		Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info			Rel-6
- Uplink DPCH power control info			
- CHOICE mode		FDD	
- DPCCH power offset		-40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- CHOICE mode		FDD	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		1	
- spreading factor	A1, A3, A4, A5, A6, A7, A8, A9	64	
- TFCI existence	A1, A3, A4, A5, A6, A7, A8, A9	TRUE	
- Number of FBI bit		Not Present(0)	
- Puncturing Limit		1	
CHOICE Mode		FDD	R99 and Rel-4 only
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
E-DCH Info		Not Present	Rel-6
Downlink HS-PDSCH Information		Not Present	Rel-5
Downlink information common for all radio links			

Information Element	Condition	Value/remark	Version
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		FDD	
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor	A1	128	
- Fixed or Flexible Position		Fixed	
- TFCI existence		TRUE	
- CHOICE SF		128	
- Number of bits for Pilot bits		8	
- Spreading factor	A3, A6, A9	32	
- Fixed or Flexible Position		Fixed	
- TFCI existence		TRUE	
- CHOICE SF		32	
- Spreading factor	A4, A7	16	
- Fixed or Flexible Position		Fixed	
- TFCI existence		TRUE	
- CHOICE SF		16	
- Spreading factor	A5, A8	8	
- Fixed or Flexible Position		Fixed	
- TFCI existence		TRUE	
- CHOICE SF		8	
- CHOICE mode	A1, A3, A4, A5, A6, A7, A8, A9	FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
Downlink information per radio link list			
- Downlink information for each radio link			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-5
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		Not Present	
- Spreading factor	A1	128	
- Code number		96	
- Spreading factor	A3, A6, A9	32	
- Code number		24	
- Spreading factor	A4, A7	16	

Information Element	Condition	Value/remark	Version
- Code number		12	
- Spreading factor - Code number	A5, A8	8 6	
- Scrambling code change	A1, A3, A4, A5, A6, A7, A8, A9	No change	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation	Version
A1	This IE is needed for "UE supports CS RAB for Test Loop Mode1 RMC 12.2/12.2 (TM)"	
A2	Not used	
A3	This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/64 (AM)"	
A4	This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/144 (AM)"	
A5	This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/384 (AM)"	
A6	This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/64 (AM)"	
A7	This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/144 (AM)"	
A8	This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/384 (AM)"	
A9	This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/64(Channel2) (AM)"	

Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New DSCH-RNTI	Not Present	R99 and Rel-4 only
New H-RNTI	Not Present	Rel-5
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
- Signalling RB information to setup	Not Present	
- RAB information for setup list		
- RAB information for setup		
- RAB info	(AM DTCH for PS domain)	
- RAB identity	0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	

Information Element	Value/remark	Version
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	useT315	
- RB information to setup		
- RB identity	20	
- PDCP info		
- Support for lossless SRNS relocation	FALSE	
- Max PDCP SN window size	Not present	
- PDCP PDU header	Absent	
- Header compression information	Not present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- CHOICE SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- DL RLC PDU size	Reference to clause 6 Parameter Set	Rel-5
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- Logical channel identity	Not Present	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	8	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	6	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	Not Present	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	7	
- CHOICE RLC size list	Explicit list	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	8	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	7	

Information Element	Value/remark	Version
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set	
- CTFC	Reference to clause 6.10.2.4 Parameter Set	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Gain factor βc	11 (below 64 kbps) 9 (higher than 64 kbps) (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Gain factor βd	15 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset $P_{p\text{-}m}$	Not Present	
Deleted UL TrCH information list	Not Present	
Added or Reconfigured UL TrCH information list	1	
Added or Reconfigured UL TrCH information	1 DCH added, 1 DCH reconfigured	
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 Parameter Set	
- Type of channel coding	Reference to clause 6.10 Parameter Set	
- Coding Rate	Reference to clause 6.10 Parameter Set	
- Rate matching attribute	Reference to clause 6.10 Parameter Set	
- CRC size	Reference to clause 6.10 Parameter Set	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport format information		
- RLC Size	Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List	(This IE is repeated for TFI number.)	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 Parameter Set	
- Type of channel coding	Reference to clause 6.10 Parameter Set	
- Coding Rate	Reference to clause 6.10 Parameter Set	

Information Element	Value/remark	Version
- Rate matching attribute	Reference to clause 6.10 Parameter Set	
- CRC size	Reference to clause 6.10 Parameter Set	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	FDD	
- CHOICE DL parameters	Explicit	
- DL DCH TFCS		
- CHOICE TFCI Signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure		
- CHOICE CTFC Size	Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set.	
- CTFC information	This IE is repeated for TFC numbers and reference to clause 6.10.2.4	
- CTFC	Reference to clause 6.10.2.4 Parameter Set	
- Power offset information	Not Present	
Added or Reconfigured DL TrCH information list	1	
Added or Reconfigured DL TrCH information	2 TrCHs(DCH for DCCH and DCH for DTCH)	
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL TrCH identity	5	
- DCH quality target		
- BLER Quality value	-20 (-2.0)	
- Downlink transport channel type	DCH	
- DL Transport channel identity	6	
- CHOICE DL parameters	Explicit	
- TFS		
- CHOICE Transport channel type	Dedicated transport channel	
- Dynamic transport format information		
- RLC Size	Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List	(This IE is repeated for TFI number.)	
- Dynamic transport format information		
- Transmission Time Interval	Not Present	
- Number of Transport blocks	Reference to clause 6.10 Parameter Set	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
- Transmission time interval	Reference to clause 6.10 Parameter Set	
- Type of channel coding	Reference to clause 6.10 Parameter Set	
- Coding Rate	Reference to clause 6.10 Parameter Set	
- Rate matching attribute	Reference to clause 6.10 Parameter Set	
- CRC size	Reference to clause 6.10 Parameter Set	
- DCH quality target		
- BLER Quality value	-20 (-2.0)	
Frequency info	Not Present	
Multi-frequency Info	Not present	Rel-7
DTX-DRX timing information	Not present	Rel-7
DRX Information	Not present	Rel-7
HS-SCCH less Information	Not present	Rel-7
MIMO parameters	Not present	Rel-7
Maximum allowed UL TX power	33dBm	
CHOICE channel requirement	Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	FDD	
- DPCCH power offset	-40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	
- PC Preamble	1 frame	

Information Element	Value/remark	Version
- SRB delay	7 frames	
- Power Control Algorithm	Algorithm1	
- TPC step size	0 (1dB)	
- Δ_{ACK}	Not Present	Rel-5
- Δ_{NACK}	Not Present	Rel-5
- Ack-Nack repetition factor	Not Present	Rel-5
- CHOICE mode	FDD	
- Scrambling code type	Long	
- Scrambling code number	0 (0 to 16777215)	
- Number of DPDCH	1	
- spreading factor	64	
- TFCI existence	TRUE	
- Number of FBI bit	Not Present(0)	
- Puncturing Limit	1	
CHOICE Mode	FDD	R99 and Rel-4 only
E-DCH Info	Not Present	Rel-6
- Downlink PDSCH information	Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information	Not Present	Rel-5
Downlink information common for all radio links		
- Downlink DPCH info common for all RL		
- Timing indicator	Maintain	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- CHOICE mode	FDD	
- DPC mode	0 (single)	
- CHOICE mode	FDD	
- Power offset $P_{Pilot-DPDCH}$	0	
- DL rate matching restriction information	Not Present	
- Spreading factor	Reference to clause 6.10 Parameter Set	
- Fixed or Flexible Position	Reference to clause 6.10 Parameter Set	
- TFCI existence	Reference to clause 6.10 Parameter Set	
- CHOICE SF	Reference to clause 6.10 Parameter Set	
- CHOICE mode	FDD	
- DPCH compressed mode info	Not Present	
- TX Diversity mode	None	
- SSDT information	Not Present	R99 and Rel-4 only
- Default DPCH Offset Value	Not Present	
- MAC-hs reset indicator	Not Present	Rel-5
- Post-verification period	Not Present	Rel-6
Downlink information per radio link list		
- Downlink information for each radio link		
- CHOICE mode	FDD	
- Primary CPICH info		
- Primary scrambling code	Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info	Not Present	R99 and Rel-4 only
- PDSCH code mapping	Not Present	R99 and Rel-4 only
- Downlink DPCH info for each RL		
- CHOICE mode	FDD	
- Primary CPICH usage for channel estimation	Primary CPICH may be used	
- DPCH frame offset	Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info	Not Present	
- DL channelisation code		
- Secondary scrambling code	Not present	
- Spreading factor	Reference to clause 6.10 Parameter Set	
- Code number	Depends upon radio bearer used.	
- Scrambling code change	No change	
- TPC combination index	0	
- SSDT Cell Identity	Not Present	R99 and

Information Element	Value/remark	Version
- Closed loop timing adjustment mode	Not Present	Rel-4 only
- SCCPCH information for FACH	Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RADIO BEARER SETUP message: AM or UM (UE supports CS RAB for Test Loop Mode 2)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New DSCH-RNTI	Not Present	R99 and Rel-4 only
New H-RNTI	Not Present	Rel-5
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info		
- RAB identity	0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	CS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT314	
- RB information to setup list		
- RB information to setup		
- RB identity	10	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	TM RLC	
- Transmission RLC discard	Not Present	
- Segmentation indication	FALSE	
- CHOICE Downlink RLC mode	TM RLC	
- Segmentation indication	FALSE	
- RB mapping info		
- Information for each multiplexing option		
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- Logical channel identity	Not Present	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	7	
- Downlink RLC logical channel info		
- Number of downlink RLC logical	1	

Information Element	Value/remark	Version
channels		
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	6	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- 2bit CTFC	0	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset P_{p-m}	Not Present	
- 2bit CTFC	2	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset P_{p-m}	Not Present	
- 2bit CTFC	1	
- Power offset Information		
- CHOICE Gain Factors	Computed Gain Factors	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset P_{p-m}	Not Present	
- 2bit CTFC	3	
- Power offset Information		
- CHOICE Gain Factors	Signalled Gain Factors	
- CHOICE mode	FDD	
- Gain factor ßc	8	
- Gain factor ßd	15	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset P_{p-m}	Not Present	
Deleted UL TrCH information list	Not Present	
Added or Reconfigured UL TrCH information list	1	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	260 bits	
- Number of TBs and TTI List	2	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	0	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	1	
- CHOICE Logical channel List	ALL	
- Semi-static Transport Format Information		
- Transmission time interval	20	
- Type of channel coding	Convolutional	

Information Element	Value/remark	Version
- Coding Rate	1/3	
- Rate matching attribute	256	
- CRC size	0	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	FDD	
- CHOICE DL parameters	Same as UL	
Deleted DL TrCH information list	Not Present	
Added or Reconfigured DL TrCH information list	1	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	6	
- CHOICE DL parameters		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	244 bits	
- Number of TBs and TTI List	2	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	0	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	1	
- CHOICE Logical channel List	ALL	
- Semi-static Transport Format Information		
- Transmission time interval	20	
- Type of channel coding	Convolutional	
- Coding Rate	1/3	
- Rate matching attribute	256	
- CRC size	16	
- DCH quality target		
- BLER Quality value	-20 (-2.0)	
Frequency info	Not Present	
Multi-frequency Info	Not present	Rel-7
DTX-DRX timing information	Not present	Rel-7
DRX Information	Not present	Rel-7
HS-SCCH less Information	Not present	Rel-7
MIMO parameters	Not present	Rel-7
Maximum allowed UL TX power	33dBm	
CHOICE channel requirement	Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	FDD	
- DPCCH power offset	-40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	
- PC Preamble	1 frame	
- SRB delay	7 frames	
- Power Control Algorithm	Algorithm1	
- TPC step size	0 (1dB)	
- Δ_{ACK}	Not Present	Rel-5
- Δ_{NACK}	Not Present	Rel-5
- Ack-Nack repetition factor	Not Present	Rel-5
- CHOICE mode	FDD	
- Scrambling code type	Long	
- Scrambling code number	0 (0 to 16777215)	
- Number of DPDCH	1	
- spreading factor	64	
- TFCI existence	TRUE	
- Number of FBI bit	Not Present(0)	
- Puncturing Limit	1	
CHOICE Mode	FDD	R99 and Rel-4 only
- Downlink PDSCH information	Not Present	R99 and Rel-4 only
E-DCH Info	Not Present	Rel-6

Information Element	Value/remark	Version
Downlink HS-PDSCH Information	Not Present	Rel-5
Downlink information common for all radio links		
- Downlink DPCH info common for all RL		
- Timing indicator	Maintain	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- CHOICE mode	FDD	
- DPC mode	0 (single)	
- CHOICE mode	FDD	
- Power offset $P_{Pilot-DPDCH}$	0	
- DL rate matching restriction information	Not Present	
- Spreading factor	128	
- Fixed or Flexible Position	Fixed	
- TFCI existence	TRUE	
- CHOICE SF	128	
- Number of bits for Pilot bits	8	
- CHOICE mode	FDD	
- DPCH compressed mode info	Not Present	
- TX Diversity mode	None	
- SSDT information	Not Present	R99 and Rel-4 only
- Default DPCH Offset Value	Not Present	
- MAC-hs reset indicator	Not Present	Rel-5
- Post-verification period	Not Present	Rel-6
Downlink information for per radio link list		
- Downlink information for each radio link		
- CHOICE mode	FDD	
- Primary CPICH info		
- Primary scrambling code	Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info	Not Present	R99 and Rel-4 only
- PDSCH code mapping	Not Present	R99 and Rel-4 only
- Downlink DPCH info for each RL		
- CHOICE mode	FDD	
- Primary CPICH usage for channel estimation	Primary CPICH may be used	
- DPCH frame offset	Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info	Not Present	
- DL channelisation code		
- Secondary scrambling code	Not Present	
- Spreading factor	128	
- Code number	96	
- Scrambling code change	No change	
- TPC combination index	0	
- SSDT Cell Identity	Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode	Not Present	
- SCCPCH information for FACH	Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information	Not Present	Rel-6

Information Element	Condition	Value/remark	Version
Message Type	A1,A2		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	

Information Element	Condition	Value/remark	Version
Ciphering mode info		Not Present	
Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI		Not Present	
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		Not Present	Rel-5
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
CHOICE specification mode		Complete specification	Rel-6
Signalling RB information to setup		Not Present	
RAB information for setup list			
- RAB information for setup			
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RB information to reconfigure list		Not Present	Rel-6
RB information to be affected list		Not Present	
Downlink counter synchronization info		Not Present	
UL Transport channel information for all transport channels			
- PRACH TFCS		Not Present	
- CHOICE mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size	A1	2 bit CTFC	

Information Element	Condition	Value/remark	Version
- CTFC information		4 TFCs	
- 2bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		2	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 2bit CTFC		3	
- Power offset Information			
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		FDD	
- Gain factor ßc		8	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- CHOICE CTFC Size	A2	4 bit CTFC	
- CTFC information		6 TFCs	
- 4bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		3	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		4	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		2	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		5	
- Power offset Information			
- CHOICE Gain Factors		Signalled Gain Factors	
- CHOICE mode		FDD	
- Gain factor ßc		8	

Information Element	Condition	Value/remark	Version
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
Deleted UL TrCH information list	A1,A2	Not Present	
Added or Reconfigured UL TrCH information list		1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information	A1		
- RLC size		260 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Dynamic Transport Format Information	A2		
- RLC size		260 bits	
- Number of TBs and TTI List		1	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- CHOICE Logical channel List		ALL	
- Dynamic Transport Format Information			
- RLC size		16 bits	
- Number of TBs and TTI List		1	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Dynamic Transport Format Information			
- RLC size		260 bits	
- Number of TBs and TTI List		1	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information	A1,A2		
- Transmission time interval		20	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		256	
- CRC size		0	
CHOICE mode		Not Present	
DL Transport channel information common for all transport channel			
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters	A1	Same as UL	
- CHOICE DL parameters	A2	DL DCH TFCS	
- DL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure information			
- CHOICE CTFC Size		4 bit CTFC	
- CTFC information		4 TFCs	
- 4bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		Computed Gain Factors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset P_{p-m}		Not Present	
- 4bit CTFC		2	
- Power offset Information			

Information Element	Condition	Value/remark	Version
- CHOICE Gain Factors - Reference TFC ID - CHOICE mode - Power offset P_{p-m} - 4bit CTFC - Power offset Information - CHOICE Gain Factors - Reference TFC ID - CHOICE mode - Power offset P_{p-m} - 4bit CTFC - Power offset Information - CHOICE Gain Factors - CHOICE mode - Gain factor ßc - Gain factor ßd - Reference TFC ID - CHOICE mode - Power offset P_{p-m}		Computed Gain Factors 0 FDD Not Present 1 Computed Gain Factors 0 FDD Not Present 3 Signalled Gain Factors FDD 8 15 0 FDD Not Present	
Deleted DL TrCH information list Added or Reconfigured DL TrCH information list - Added or Reconfigured DL TrCH information - Downlink transport channel type - DL Transport channel identity - CHOICE DL parameters - CHOICE Transport channel type	A1,A2	Not Present 1 DCH 6 Dedicated transport channels	
- Dynamic Transport Format Information - RLC size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel List	A1	 244 bits 2 Not Present 0 Not Present 1 ALL	
- Dynamic Transport Format Information - RLC size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel List - Dynamic Transport Format Information - RLC size - Number of TBs and TTI List - Transmission Time Interval - Number of Transport blocks - CHOICE Logical channel List	A2	 0 bits 1 Not Present 1 ALL 244 bits 1 Not Present 1 ALL	
- Semi-static Transport Format Information - Transmission time interval - Type of channel coding - Coding Rate - Rate matching attribute - CRC size - DCH quality target - BLER Quality value Frequency info Maximum allowed UL TX power CHOICE channel requirement Uplink DPCH info - Uplink DPCH power control info - CHOICE mode - DPCCH power offset	A1,A2	 20 Convolutional 1/3 256 16 -20 (-2.0) Not Present 33dBm Uplink DPCH info FDD -40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	Rel-5 and earlier Rel-6

Information Element	Condition	Value/remark	Version
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- CHOICE mode		FDD	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		1	
- spreading factor		64	
- TFCI existence		TRUE	
- Number of FBI bit		Not Present(0)	
- Puncturing Limit		1	
CHOICE Mode		FDD	R99 and Rel-4 only
- Downlink PDSCH information		Not Present	R99 and Rel-4 only
E-DCH Info		Not Present	Rel-6
Downlink HS-PDSCH Information		Not Present	Rel-5
Downlink information common for all radio links			
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		FDD	
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		128	
- Fixed or Flexible Position		Fixed	
- TFCI existence		TRUE	
- CHOICE SF		128	
- Number of bits for Pilot bits		8	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	Rel-5
- Post-verification period		Not Present	Rel-6
Downlink information for per radio link list			
- Downlink information for each radio link			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	

Information Element	Condition	Value/remark	Version
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		Not Present	
- Spreading factor		128	
- Code number		96	
- Scrambling code change		No change	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	This IE is needed for "UE supports CS RAB for Test Loop Mode2 RMC 12.2/12.2 (TM)"
A2	This IE is needed for "UE supports CS RAB for Test Loop Mode2 RMC 0 and 12.2 (TM)"

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		

Information Element	Value/remark	Version
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to clause TS 34.121 clause C.2.1 Parameter Set	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Gain factor βc	8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Gain factor βd	15 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset $P_{p\text{-}m}$	Not Present	
Deleted UL TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	FDD	
- CHOICE DL parameters	Explicit	
- DL DCH TFCS		
- CHOICE TFCI Signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to clause TS 34.121 clause C.3.1 Parameter Set	
- Power offset information	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information	(HS-DSCH for DTCH)	
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.121 [2] Annex C Fixed Reference Channels	

Information Element	Value/remark	Version
- CHOICE *Memory Partitioning*	Explicit	
- Memory size	Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes”.	
- Process Memory Size	Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.	
- Additional memory sizes for MIMO	Not Present	Rel-7
- CHOICE DL MAC header type	MAC-hs	Rel-7
- Added or reconfigured MAC-d flow		
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	
- MAC-d Flow Identity	0	
- T1	50	
- MAC-hs window size	16	
- MAC-d PDU size Info		
- MAC-d PDU size	Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- MAC-d PDU size index	0	
- MAC-hs queue to delete list	Not present	
- DCH quality target	Not present	
Frequency info	Not Present	
Multi-frequency Info	Not present	Rel-7
DTX-DRX timing information	Not present	Rel-7
DRX Information	Not present	Rel-7
HS-SCCH less Information	Not present	Rel-7
MIMO parameters	Not present	Rel-7
Maximum allowed UL TX power	33dBm	
CHOICE channel requirement	Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	FDD	
- DPCCH power offset	-40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	
- PC Preamble	1 frame	
- SRB delay	7 frames	
- Power Control Algorithm	Algorithm1	
- TPC step size	0 (1dB)	
- Δ_{ACK}	3	Rel-5
- Δ_{NACK}	3	Rel-5
- Ack-Nack repetition factor	1	Rel-5
- CHOICE mode	FDD	
- Scrambling code type	Long	
- Scrambling code number	0 (0 to 16777215)	
- Number of DPDCH	Not Present (1)	
- spreading factor	64	
- TFCI existence	TRUE	
- Number of FBI bit	Not Present(0)	
- Puncturing Limit	1	
CHOICE Mode	FDD	R99 and Rel-4 only
- Downlink PDSCH information	Not Present	R99 and Rel-4 only
E-DCH Info	Not Present	Rel-6
Downlink HS-PDSCH Information		
- HS-SCCH Info		
- CHOICE mode	FDD	
- DL Scrambling Code		
- HS-SCCH Channelisation Code Information		
- HS-SCCH Channelisation Code	2	
- HS-SCCH Channelisation Code	3	
- HS-SCCH Channelisation Code	6	
- HS-SCCH Channelisation Code	7	
- Measurement Feedback Info		
- CHOICE mode	FDD	
- POhsdsch	6 dB	Rel-5

Information Element	Value/remark	Version
- CQI Feedback cycle, k	2 ms	Rel-5
- CQI repetition factor	1	Rel-5
- Δ_{CQI}	5 (corresponds to 0dB in relative power offset)	Rel-5
- CHOICE mode	FDD	
- Downlink 64QAM configured	Not Present	Rel-7
Downlink information common for all radio links	Not Present	
Downlink information per radio link list		
- Downlink information for each radio link		
- CHOICE mode	FDD	
- Primary CPICH info		
- Primary scrambling code	Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info	Not Present	R99 and Rel-4 only
- PDSCH code mapping	Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator	TRUE	Rel-5
- Downlink DPCH info for each RL		
- CHOICE mode	FDD	
- Primary CPICH usage for channel estimation	Primary CPICH may be used	
- DPCH frame offset	Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info	Not Present	
- DL channelisation code		
- Secondary scrambling code	Not present	
- Spreading factor	128	
- Code number	96	
- Scrambling code change	No change	
- TPC combination index	0	
- SSDT Cell Identity	Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode	Not Present	
- SCCPCH information for FACH	Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RADIO BEARER SETUP message: BTFD RMC for Test Loop Mode 2

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present. For correct operation of test loop mode 2 this IE shall be omitted.	
Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New DSCH-RNTI	Not Present	R99 and Rel-4 only
New H-RNTI	Not Present	Rel-5
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE *specification mode*	Complete specification	Rel-5
- RAB information for setup		

Information Element	Value/remark	Version
- RAB info		
- RAB identity	0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	CS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT314	
- RB information to setup		
- RB identity	10	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	TM RLC	
- Transmission RLC discard	Not Present	
- Segmentation indication	FALSE	
- CHOICE Downlink RLC mode	TM RLC	
- Segmentation indication	FALSE	
- RB mapping info		
- Information for each multiplexing option		
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- Logical channel identity	Not Present	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	7	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	6	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected	Not Present	
Downlink counter synchronization info	Not Present	
	RMC for BTFD	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	ctfc6Bit	
- ctfc6Bit	22	
- ctfc6	0	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	11	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	1	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	12	
-powerOffsetInformation(OP)		
-gainFactorInformation	SignalledGainFactors	
-modeSpecificInfo	Fdd	
-fdd		
- Gain factor ßc	8	
- Gain factor ßd	15	
- Reference TFC ID	0	

Information Element	Value/remark	Version
- ctfc6	2	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	13	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	3	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	14	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	4	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	15	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	5	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	16	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	0	
- ctfc6	6	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	17	
-powerOffsetInformation(OP)		
-gainFactorInformation	SignalledGainFactors	
-modeSpecificInfo	Fdd	
-fdd		
- Gain factor ßc	11	
- Gain factor ßd	15	
- Reference TFC ID	1	
- ctfc6	7	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	18	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	8	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	19	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	9	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	20	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	

Information Element	Value/remark	Version
- Reference TFC ID	1	
- ctfc6	10	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
- ctfc6	21	
-powerOffsetInformation(OP)		
-gainFactorInformation	ComputedGainFactors	
- Reference TFC ID	1	
Added or Reconfigured UL TrCH information list	1	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	1	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
-DedicatedDynamicTF-Info		
RLC size	256	
-numberOfTbSizeList		
-NumberOfTransportBlocks	Zero	
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	216	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	171	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	160	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	146	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	130	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	115	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	107	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	51	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	12	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
-Semistatic Transport Format Information		
-Transmission Time interval	20 ms	
-channelCodingType	Convolutional	
-convolutional	1/3	
- Rate matching attribute	256	
- CRC size	0	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	FDD	

Information Element	Value/remark	Version
- CHOICE DL parameters	Explicit	
- DL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	Ctfc6Bit	
- ctfc6Bit	18	
- ctfc6	9	
- ctfc6	0	
- ctfc6	10	
- ctfc6	1	
- ctfc6	11	
- ctfc6	2	
- ctfc6	12	
- ctfc6	3	
- ctfc6	13	
- ctfc6	4	
- ctfc6	14	
- ctfc6	5	
- ctfc6	15	
- ctfc6	6	
- ctfc6	16	
- ctfc6	7	
- ctfc6	17	
- ctfc6	8	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	6	
- CHOICE DL parameters	Explicit	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
-DedicatedDynamicTF-Info		
RLC size	244	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	204	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	159	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	148	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	134	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	118	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	103	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
RLC size	95	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	

Information Element	Value/remark	Version
RLC size	39	
-numberOfTbSizeList		
-NumberOfTransportBlocks	One	
- Choice Logical channel List	ALL	
-Semistatic Transport Format Information		
-Transmission Time interval	20 ms	
-channelCodingType	Convolutional	
-convolutional	1/3	
- Rate matching attribute	256	
- CRC size	12	
- DCH quality target		
- BLER Quality value	-20 (-2.0)	
- Transparent mode signalling info	Not Present	
Frequency info	Not Present	
Multi-frequency Info	Not present	Rel-7
DTX-DRX timing information	Not present	Rel-7
DRX Information	Not present	Rel-7
HS-SCCH less Information	Not present	Rel-7
MIMO parameters	Not present	Rel-7
Maximum allowed UL TX power	33 dBm	
CHOICE channel requirement	Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- DPCCH power offset	-40 (-80dB) IE value will have no effect on the UE UL power when closed loop power control is active	
- PC Preamble	1 frame	
- SRB delay	7 frames	
- Power Control Algorithm	Algorithm1	
- TPC step size	0 (1dB)	
- Δ_{ACK}	Not Present	Rel-5
- Δ_{NACK}	Not Present	Rel-5
- Ack-Nack repetition factor	Not Present	Rel-5
- Scrambling code type	Long	
- Scrambling code number	0	
- Number of DPDCH	1	
- spreading factor	64	
- TFCI existence	TRUE	
- Number of FBI bit	Not Present(0)	
- Puncturing Limit	1	
CHOICE Mode	FDD	R99 and Rel-4 only
- Downlink PDSCH information	Not Present(0)	R99 and Rel-4 only
E-DCH Info	Not Present	Rel-6
Downlink HS-PDSCH Information	Not Present	Rel-5
Downlink information common for all radio links		
- Downlink DPCH info common for all RL	FDD	
- Timing indicator	Maintain	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- DPC mode	0 (single)	
- CHOICE mode	FDD	
- Power offset $P_{Pilot-DPDCH}$	0	
- DL rate matching restriction information	Not Present	
- Spreading factor	128	
- Number of bits for Pilot bits(SF=128,256)	4	
- Fixed or Flexible Position	Fixed	
- TFCI existence	FALSE	
- DPCH compressed mode info	Not Present	
- TX Diversity mode	None	
- SSDT information	Not Present	R99 and Rel-4 only
- Default DPCH Offset Value	Not Present	
Downlink information for each radio link list		
- Primary CPICH info		

Information Element	Value/remark	Version
- Primary scrambling code	Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info	Not Present	R99 and Rel-4 only
- PDSCH code mapping	Not Present	R99 and Rel-4 only
- Downlink DPCH info for each RL		
- Primary CPICH usage for channel estimation	Primary CPICH may be used	
- DPCH frame offset	Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info	Not Present	
- DL channelisation code		
- Secondary scrambling code	Not Present	
- Spreading factor	128	
- Code number	96	
- Scrambling code change	No change	
- TPC combination index	0	
- SSDT Cell Identity	Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode	Not Present	
- SCCPCH information for FACH	Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)

Information Element	Condition	Value/remark	Version
Message Type	A1, A2		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time	A1	Not Present	
Activation time	A2	(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI	A1, A2	Not Present	
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		'1010 1010 1010 1010'	Rel-5
New Primary E-RNTI		'1010 1010 1010 1010'	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
CHOICE specification mode		Complete specification	Rel-6
- Signalling RB information to setup		Not Present	
- RAB information for setup list			
- RAB information for setup			
- RAB info		(high-speed UM DTCH for PS domain)	
- RAB identity		0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		useT315	
- RB information to setup			
- RB identity		25	
- PDCP info		Not present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7	Rel-5
- DL Reception Window Size		Not present	Rel-6
- One sided RLC re-establishment		FALSE	
- Alternative E-bit interpretation		Not present	Rel-6
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		7	
- E-DCH MAC-d flow identity		2	
- DDI		5	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		336 bits	
- Include in scheduling info		TRUE	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		HS-DSCH	
- DL DCH Transport channel identity		Not Present	

Information Element	Condition	Value/remark	Version
- DL DSCH Transport channel identity		Not Present	
- CHOICE DL MAC header type		MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity		0	
- Logical channel identity		Not Present	
RB information to reconfigure list		Not Present	Rel-6
RB information to be affected	A1	Not Present	
RB information to be affected	A2		
- RB identity		1 (UM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		1	
- E-DCH MAC-d flow identity		1	
- DDI		1	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		96 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- RB identity		2 (AM DCCH for RRC)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		2	
- E-DCH MAC-d flow identity		1	
- DDI		2	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		96 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RB identity		3 (AM DCCH for NAS High Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		E-DCH	
- Logical channel identity		3	
- E-DCH MAC-d flow identity		1	
- DDI		3	
- RLC PDU size list		1 RLC PDU size	
- RLC PDU size		96 bits	
- Include in scheduling info		FALSE	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RB identity		4 (AM DCCH for NAS Low Priority)	
- RB mapping info			
- Information for each multiplexing option		1 RBMuxOption	

Information Element	Condition	Value/remark	Version
- RLC logical channel mapping indicator - Number of uplink RLC logical channels - Uplink transport channel type - Logical channel identity - E-DCH MAC-d flow identity - DDI - RLC PDU size list - RLC PDU size - Include in scheduling info - MAC logical channel priority - Downlink RLC logical channel info - Number of RLC logical channels - Downlink transport channel type - DL DCH Transport channel identity - DL DSCH Transport channel identity - Logical channel identity		Not Present 1 E-DCH 4 1 4 1 RLC PDU size 96 bits FALSE 4 1 DCH 10 Not Present 4	
Downlink counter synchronization info PDCP ROHC target mode	A1, A2	Not Present Not Present	 Rel-5
UL Transport channel information for all transport channels		Not Present	
Deleted UL TrCH information	A1	Not Present	
Deleted UL TrCH information - Uplink transport channel type - UL transport channel identity	A2	 DCH 5	
Added or Reconfigured TrCH information list - Added or Reconfigured UL TrCH information - Uplink transport channel type - CHOICE UL parameters - E-DCH Transmission Time Interval - HARQ info for E-DCH - HARQ RV Configuration - Added or reconfigured E-DCH MAC-d flow - E-DCH MAC-d flow identity - E-DCH MAC-d flow power offset - E-DCH MAC-d flow maximum number of retransmissions - E-DCH MAC-d flow multiplexing list - CHOICE transmission grant type	A1	1 TrCH added 1 E-DCH added E-DCH E-DCH 10 ms Rv0 2 0 7 Not Present Scheduled grant info	
Added or Reconfigured UL TrCH information list - Added or Reconfigured UL TrCH information - Uplink transport channel type - CHOICE UL parameters - E-DCH Transmission Time Interval - HARQ info for E-DCH - HARQ RV Configuration - Added or reconfigured E-DCH MAC-d flow - E-DCH MAC-d flow identity - E-DCH MAC-d flow power offset - E-DCH MAC-d flow maximum number of retransmissions - E-DCH MAC-d flow multiplexing list - CHOICE transmission grant type - Max MAC-e PDU contents size - 2 ms non-scheduled transmission grant HARQ process allocation - Added or reconfigured E-DCH MAC-d flow - E-DCH MAC-d flow identity - E-DCH MAC-d flow power offset - E-DCH MAC-d flow maximum number of retransmissions - E-DCH MAC-d flow multiplexing list - CHOICE transmission grant type	A2	1 TrCH added 1 E-DCH added with one DCCH MAC-d flow and one DTCH MAC-d flow E-DCH E-DCH 2ms Rv0 (for DCCH) 1 0 7 Not Present Non-scheduled grant info 114 bits Not Present (for DTCH) 2 0 7 Not Present Scheduled grant info	
CHOICE *mode*	A1, A2	Not Present	R99 and Rel-4

Information Element	Condition	Value/remark	Version
			only
DL Transport channel information common for all transport channels	A1	Not Present	
DL Transport channel information common for all transport channels	A2		
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Explicit	
- DL DCH TFCS			
- CHOICE TFCI Signalling		Normal	
- TFCI Field 1 Information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfigure			
- CHOICE CTFC Size		2 bit CTFC	
- CTFC information		2 TFCs	
- 2bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		computedGainFactors	
- Reference TFC ID		0	
- Power offset Pp-m		Not Present	
- 2bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		signalledGainFactors	
- CHOICE mode		FDD	
- Gain factor ßc		15	
- Gain factor ßd		15	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
Deleted TrCH information list	A1, A2	Not Present	
Added or Reconfigured TrCH information list	A1	1 TrCH added	
- Added or Reconfigured DL TrCH information		HS-DSCH for DTCH added	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- CHOICE *Memory Partitioning*		Explicit	
- Memory size		Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes".	
- Process Memory Size		Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.	
- Additional memory sizes for MIMO		Not Present	Rel-7
- CHOICE DL MAC header type		MAC-hs	Rel-7
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Added or Reconfigured DL TrCH information	A2	2 TrCHs (DCH for DCCH and HS-DSCH for DTCH)	
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		Explicit	
- TFS			

Information Element	Condition	Value/remark	Version
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport format information			
- RLC Size		96 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel list		ALL	
- Semi-static Transport Format information			
- Transmission time interval		40	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		256	
- CRC size		12	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
- Downlink transport channel type		HS-DSCH	
- DL Transport channel identity		Not Present	
- CHOICE DL parameters		HS-DSCH	
- HARQ Info			
- Number of Processes		Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- CHOICE *Memory Partitioning*		Explicit	
- Memory size		Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter "Number of HARQ Processes".	
- Process Memory Size		Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter "Number of SML's per HARQ Proc.".	
- Additional memory sizes for MIMO		Not Present	Rel-7
- CHOICE DL MAC header type		MAC-hs	Rel-7
- Added or reconfigured MAC-d flow			
- MAC-hs queue to add or reconfigure list		(one queue)	
- MAC-hs queue Id		0	
- MAC-d Flow Identity		0	
- T1		50	
- MAC-hs window size		16	
- MAC-d PDU size Info			
- MAC-d PDU size		Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- MAC-d PDU size index		0	
- MAC-hs queue to delete list		Not present	
- DCH quality target		Not present	
Frequency info	A1, A2	Not present	
Multi-frequency Info		Not present	Rel-7
DTX-DRX timing information		Not present	Rel-7
DRX Information		Not present	Rel-7
HS-SCCH less Information		Not present	Rel-7
MIMO parameters		Not present	Rel-7
Maximum allowed UL TX power		33dBm	
CHOICE channel requirement		Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info			Rel-6
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		3	
- Δ_{NACK}		3	
- Ack-Nack repetition factor		1	
- HARQ_preamble_mode		0	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	

Information Element	Condition	Value/remark	Version
- Number of DPDCH - spreading factor - TFCI existence - Number of FBI bit - Puncturing Limit	A1	Not Present(1) Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set Reference to clause 6.10 Parameter Set	
- Number of DPDCH - spreading factor - TFCI existence - Number of FBI bit - Puncturing Limit	A2	0 Not present FALSE Not present Not present	
E-DCH info - MAC-es/e reset indicator - E-DPCCH info - E-DPCCH/DPCCH power offset - Happy bit delay condition - E-TFCI boost info - E-TFCI BetaED SwitchE-DPDCH power interpolation	A1, A2	 TRUE 0 100 ms Not present Not present	Rel-6 Rel-7 Rel-7
- E-DPDCH info - E-TFCI table index - E-DCH minimum set E-TFCI - Reference E-TFCIs - Reference E-TFCI - Reference E-TFCI PO - Maximum channelisation codes - PLnon-max - Scheduling Information Configuration - Periodicity for Scheduling Info – no grant - Periodicity for Scheduling Info – grant - Power Offset for Scheduling Info - 3-Index-Step Threshold - 2-Index-Step Threshold	A1	 0 9 1 E-TFCI 11 4 2sf4 0.84 Not present Not present 0 Not present Not present	
- E-DPDCH info - E-TFCI table index - E-DCH minimum set E-TFCI - Reference E-TFCIs - Reference E-TFCI - Reference E-TFCI PO - Reference E-TFCI - Reference E-TFCI PO - Maximum channelisation codes - PLnon-max - Scheduling Information Configuration - Periodicity for Scheduling Info – no grant - Periodicity for Scheduling Info – grant - Power Offset for Scheduling Info - 3-Index-Step Threshold - 2-Index-Step Threshold	A2	 0 9 2 E-TFCI 11 4 83 16 2sf2and2sf4 0.84 Not present Not present 0 Not present Not present	
- Scheduled Transmission configuration - 2ms scheduled transmission grant HARQ process allocation - Serving Grant	A1, A2	 Not present Not present	
- UL 16QAM settings		Not present	Rel-7
CHOICE Mode - Downlink PDSCH information		FDD Not Present	R99 and Rel-4 only R99 and Rel-4 only
Downlink HS-PDSCH Information - HS-SCCH Info - CHOICE mode - DL Scrambling Code - HS-SCCH Channelisation Code Information - HS-SCCH Channelisation Code - HS-SCCH Channelisation Code		 FDD Not present 2 3	

Information Element	Condition	Value/remark	Version
- Measurement Feedback Info			
- CHOICE mode		FDD	
- POhsdsch		6 dB	
- CQI Feedback cycle, k		2 ms	
- CQI repetition factor		1	
- Δ_{CQI}		5 (corresponds to 0dB in relative power offset)	
- CHOICE mode		FDD	
- Downlink 64QAM configured		Not Present	Rel-7
Downlink information common for all radio links	A1	Not Present	
Downlink information common for all radio links	A2		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset PPilot-DPDCH		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		256	
- Fixed or Flexible Position		Fixed	
- TFCI existence		FALSE	
- CHOICE SF		256	
- Number of bits for Pilot bits		8	
- CHOICE mode		FDD	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- Default DPCH Offset Value		Not Present	
- MAC-hs reset indicator		Not Present	
- Post-verification period		Not Present	
Downlink information for each radio link list	A1, A2		
- Downlink information for each radio link			
- Choice mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Ref. to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		TRUE	
- Serving E-DCH radio link indicator		TRUE	
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		Not Present	
- Spreading factor	A1	Reference to clause 6.10 Parameter Set	
- Code number		96	
- Spreading factor	A2	256	
- Code number		192	
- Scrambling code change	A1, A2	No code change	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info			Rel-6
- E-AGCH Channelisation Code		14	
- CHOICE E-HICH Information			Rel-6
- E-HICH Information			

Information Element	Condition	Value/remark	Version
- DL Scrambling code		Not Present (default is primary)	
- Channelisation code		6	
- Signature sequence		1	
- CHOICE E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only
MBMS PL Service Restriction Information		Not Present	Rel-6

Condition	Explanation
A1	Not using E-DCH 4codes
A2	Using E-DCH 4codes

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA with F-DPCH)

Information Element	Value/remark	Version
Message Type		Rel-6
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	Rel-6
New U-RNTI	Not Present	Rel-6
New C-RNTI	Not Present	Rel-6
New H-RNTI	'1010 1010 1010 1010'	Rel-6
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	Rel-6
UTRAN DRX cycle length coefficient	Not Present	Rel-6
CN information info	Not Present	
URA identity	Not Present	
CHOICE Specification mode	Complete specification	Rel-6
- Signalling RB information to setup	Not Present	
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	useT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOption	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not present	
- DL DSCH Transport channel identity	Not present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	1	
- Logical channel identity	Not Present	

Information Element	Value/remark	Version
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected		Rel-6
- RB identity	1 (UM DCCH for RRC)	
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOption	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not present	
- DL DSCH Transport channel identity	Not present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	1	
- RB identity	2 (AM DCCH for RRC)	
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOption	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	2	
- RB identity	3 (AM DCCH for NAS High Priority)	
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOption	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	3	
- RB identity	4 (AM DCCH for NAS Low Priority)	
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOption	
- RLC logical channel mapping indicator	Not Present	
- Number of uplink RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		

Information Element	Value/remark	Version
- Number of RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- CHOICE DL MAC header type	MAC-hs	Rel-7
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	4	
Downlink counter synchronization info	Not Present	Rel-6
PDCP ROHC target mode	Not Present	Rel-6
UL Transport channel information for all transport channels		Rel-6
- PRACH TFCS	Not Present	
- CHOICE Mode	FDD	
- TFC subset	Not Present	
- UL DCH TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	2 TFCs	
- 2bit CTFC	0	
- Power offset Information		
- CHOICE Gain Factors	computedGainFactors	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset Pp-m	Not Present	
- 2bit CTFC	1	
- Power offset Information		
- CHOICE Gain Factors	signalledGainFactors	
- CHOICE mode	FDD	
- Gain factor ßc	15	
- Gain factor ßd	15	
- Reference TFC ID	0	
- CHOICE mode	FDD	
- Power offset Pp-m	Not Present	
Deleted UL TrCH information	Not Present	Rel-6
Added or Reconfigured UL TrCH information		Rel-6
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	96 bits	
- Number of TBs and TTI List	2	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	0	
- Transmission Time Interval	Not Present	
- Number of Transport blocks	1	
- CHOICE Logical channel List	ALL	
- Semi-static Transport Format Information		
- Transmission time interval	40	
- Type of channel coding	Convolutional	
- Coding Rate	1/3	
- Rate matching attribute	256	
- CRC size	12	
DL Transport channel information common for all transport channel	Not Present	Rel-6
Deleted DL TrCH information		Rel-6
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
Added or Reconfigured DL TrCH information	1 TrCH (HS-DSCH for DTCH and DCCH)	Rel-6
- Downlink transport channel type	HS-DSCH	
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	
- HARQ Info		

Information Element	Value/remark	Version
- Number of Processes	Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- CHOICE *Memory Partitioning*	Explicit	
- Memory size	Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes”.	
- Process Memory Size	Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.	
- Additional memory sizes for MIMO	Not Present	Rel-7
- CHOICE DL MAC header type	MAC-hs	Rel-7
- Added or reconfigured MAC-d flow		
- MAC-hs queue to add or reconfigure list	(two queues)	
- MAC-hs queue Id	0 (for DCCH)	
- MAC-d Flow Identity	0	
- T1	50	
- MAC-hs window size	16	
- MAC-d PDU size Info		
- MAC-d PDU size	96	
- MAC-d PDU size index	0	
- MAC-hs queue Id	1 (for DTCH)	
- MAC-d Flow Identity	1	
- T1	50	
- MAC-hs window size	16	
- MAC-d PDU size Info		
- MAC-d PDU size	Reference to TS34.121 [2] Annex C Fixed Reference Channels	
- MAC-d PDU size index	0	
- MAC-hs queue to delete list	Not present	
- DCH quality target	Not present	
Frequency info	Not present	
Multi-frequency Info	Not present	Rel-7
DTX-DRX timing information	Not present	Rel-7
DRX Information	Not present	Rel-7
HS-SCCH less Information	Not present	Rel-7
MIMO parameters	Not present	Rel-7
Maximum allowed UL TX power	33dBm	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- DPCCH power offset	-40 (-80dB)	
- PC Preamble	1 frame	
- SRB delay	7 frames	
- Power Control Algorithm	Algorithm1	
- TPC step size	0 (1dB)	
- Δ_{ACK}	3	
- Δ_{NACK}	3	
- Ack-Nack repetition factor	1	
- HARQ_preamble_mode	0	
- CHOICE mode	FDD	
- Scrambling code type	Long	
- Scrambling code number	0 (0 to 16777215)	
- Number of DPDCH	Not Present (1)	
- spreading factor	256	
- TFCI existence	TRUE	
- Number of FBI bit	Not Present(0)	
- Puncturing Limit	1	
E-DCH info	Not Present	Rel-6
Downlink HS-PDSCH Information		Rel-6
- HS-SCCH Info		
- CHOICE mode	FDD	
- DL Scrambling Code	Not present	
- HS-SCCH Channelisation Code Information		
- HS-SCCH Channelisation Code	2	
- Measurement Feedback Info		
- CHOICE mode	FDD	
- POhsdsch	6 dB	

Information Element	Value/remark	Version
- CQI Feedback cycle, k	2 ms	
- CQI repetition factor	1	
- Δ_{CQI}	5 (corresponds to 0dB in relative power offset)	
- CHOICE mode	FDD	
- Downlink 64QAM configured	Not Present	Rel-7
Downlink information common for all radio links		Rel-6
- Downlink F-DPCH info common for all RL		
- Timing Indication	Maintain	
- Timing maintained Synchronization indicator	FALSE	
- Downlink F-DPCH power control information		
- DPC mode	0 (single)	
- TPC command error rate target	0.04	
- CHOICE mode	FDD	
- DPCH compressed mode info	Not Present	
- TX Diversity mode	None	
- Default DPCH Offset Value	Not Present	
- MAC-hs reset indicator	Not Present	
Downlink information for each radio link list		Rel-6
- Downlink information for each radio link		
- Choice mode	FDD	
- Primary CPICH info		
- Primary scrambling code	Ref. to clause 6.1 "Default settings (FDD)"	
- Serving HS-DSCH radio link indicator	TRUE	
- Downlink DPCH info for each RL	Not Present	
- Downlink F-DPCH info for each RL		
- Primary CPICH usage for channel estimation	Primary CPICH may be used	
- F-DPCH frame offset	Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400	
- Secondary CPICH info	Not Present	
- Secondary scrambling code	Not Present	
- Code number	6	
- TPC combination index	0	
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RRC CONNECTION RELEASE message: UM

Information Element	**Value/remark**	**Version**
Message Type		
U-RNTI	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	R99, Rel-4
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
CHOICE identity type	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	Rel-5
- U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info	This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted.	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.	
N308	2 (for CELL_DCH state). Not Present (for UE in other connected mode states).	
Release cause	Normal event	
Rplmn information	Not Present	
Redirection info	Not Present	Rel-6

Contents of RRC CONNECTION SETUP message: UM

Information Element	Condition	Value/remark	Version
Message Type			
Initial UE identity		Select the same identity as in the IE "Initial UE Identity" in received "RRC CONNECTION REQUEST" message	
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Activation time		Not Present(Now)	
New U-RNTI			
- SRNC identity		0000 0000 0001B	
- S-RNTI		0000 0000 0000 0000 0001B	
New C-RNTI		Not Present	
New H-RNTI		Not Present	Rel-6
New Primary E-RNTI		Not Present	Rel-6
New Secondary E-RNTI		Not Present	Rel-6
RRC State Indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		9	
Capability update requirement			
- UE radio access FDD capability update requirement		TRUE	
- UE radio access TDD capability update requirement		FALSE	
- UE radio access 3.84 Mcps TDD capability update requirement		FALSE	Rel-4
- UE radio access 1.28 Mcps TDD capability update requirement		FALSE	Rel-4
- System specific capability update requirement list		GSM	
CHOICE *specification mode*		Complete specification	Rel-5
- Complete specification			Rel-5
- Signalling RB information to setup list		4 SRBs	
- Signalling RB information to setup		(UM DCCH for RRC)	
- RB identity		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		UM RLC	
- Transmission RLC discard		Not Present	
- CHOICE Downlink RLC mode		UM RLC	
- DL UM RLC LI size		7 bit	Rel-6
- One sided RLC re-establishment		FALSE	Rel-6
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		1	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		1	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		1	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		1	

Information Element	Condition	Value/remark	Version
- Signalling RB information to setup		(AM DCCH for RRC)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		96 bits	Rel-6
- DL RLC PDU size	A1	144 bits	Rel-6
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		2	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		2	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		2	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		2	
- Signalling RB information to setup		(AM DCCH for NAS_DT High priority)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No Discard	
- MAX_DAT		15	

Information Element	Condition	Value/remark	Version
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		96 bits	Rel-6
- DL RLC PDU size	A1	144 bits	Rel-6
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
-UL Transport channel identity		5	
- Logical channel identity		3	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		3	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		3	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		3	
- Signalling RB information to setup		(AM DCCH for NAS_DT Low priority)	
- RB identity		Not Present	
- CHOICE RLC info type			
- RLC info			
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- SDU discard mode		No Discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		1	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_PDU		Not Present	
- Poll_SDU		1	

Information Element	Condition	Value/remark	Version
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- DL RLC PDU size		96 bits	Rel-6
- DL RLC PDU size	A1	144 bits	Rel-6
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		Not Present	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2 RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- Logical channel identity		4	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		10	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
- RLC logical channel mapping indicator		Not Present	
- Number of RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		4	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		4	
- Downlink RLC logical channel info			
- Number of RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		4	
UL Transport channel information for all transport channels			
- PRACH TFCS		Not Present	
- CHOICE Mode		FDD	
- TFC subset		Not Present	
- UL DCH TFCS			
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- CHOICE TFCS representation		Complete reconfiguration	
- TFCS complete reconfiguration information			
- CHOICE CTFC Size		2 bit CTFC	
- CTFC information		2 TFCs	
- 2bit CTFC		0	
- Power offset Information			
- CHOICE Gain Factors		computedGainFactors	
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
- 2bit CTFC		1	
- Power offset Information			
- CHOICE Gain Factors		signalledGainFactors	
- CHOICE mode		FDD	
- Gain factor ßc		15	
- Gain factor ßd		15	

Information Element	Condition	Value/remark	Version
- Reference TFC ID		0	
- CHOICE mode		FDD	
- Power offset Pp-m		Not Present	
Added or Reconfigured UL TrCH information list		1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		5	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information			
- RLC size		96 bits	
- RLC size	A1	144 bits	
- Number of TBs and TTI List		2	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		0	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		40	
- Type of channel coding		Convolutional	
- Coding Rate		1/3	
- Rate matching attribute		256	
- CRC size		12	
- CRC size	A1	16	
DL Transport channel information common for all transport channel			
- SCCPCH TFCS		Not Present	
- CHOICE mode		FDD	
- CHOICE DL parameters		Same as UL	
Added or Reconfigured DL TrCH information list		1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		10	
- CHOICE DL parameters		SameAasUL	
- Uplink transport channel type		DCH	
- UL TrCH Identity		5	
- DCH quality target			
- BLER Quality value		-20 (-2.0)	
Frequency info		Not Present	
Maximum allowed UL TX power		Not Present	
CHOICE channel requirement		Uplink DPCH info	Rel-5 and earlier
Uplink DPCH info			Rel-6
- Uplink DPCH power control info			
- DPCCH power offset		-40 (-80dB)	
- PC Preamble		1 frame	
- SRB delay		7 frames	
- Power Control Algorithm		Algorithm1	
- TPC step size		0 (1dB)	
- Δ_{ACK}		Not Present	Rel-5
- Δ_{NACK}		Not Present	Rel-5
- Ack-Nack repetition factor		Not Present	Rel-5
- HARQ_preamble_mode		0	Rel-6
- CHOICE mode		FDD	
- Scrambling code type		Long	
- Scrambling code number		0 (0 to 16777215)	
- Number of DPDCH		Not Present (1)	
- Spreading factor		256	
- TFCI existence		TRUE	
- Number of FBI bit		Not Present(0)	
- Puncturing Limit		1	
E-DCH Info		Not Present	Rel-6
Downlink HS-PDSCH Information		Not Present	Rel-6
Downlink information common for all radio links			

Information Element	Condition	Value/remark	Version
- Downlink DPCH info common for all RL			
- Timing Indication		Initialize	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		FDD	
- DPC mode		0 (single)	
- CHOICE mode		FDD	
- Power offset $P_{Pilot-DPDCH}$		0	
- DL rate matching restriction information		Not Present	
- Spreading factor		256	
- Fixed or Flexible Position		Fixed	
- TFCI existence		FALSE	
- CHOICE SF			
- Number of bits for Pilot bits		8	
- DPCH compressed mode info		Not Present	
- TX Diversity mode		None	
- SSDT information		Not Present	R99 and Rel-4 only
- Default DPCH Offset Value		Arbitrary set to value 0..306688 by step of 512	
Downlink information for per radio links list			
-Downlink information for each radio links			
- CHOICE mode		FDD	
- Primary CPICH info			
- Primary scrambling code		Reference to clause 6.1 "Default settings (FDD)"	
- PDSCH with SHO DCH info		Not Present	R99 and Rel-4 only
- PDSCH code mapping		Not Present	R99 and Rel-4 only
- Serving HS-DSCH radio link indicator		FALSE	Rel-6
- Serving E-DCH radio link indicator		FALSE	Rel-6
- Downlink DPCH info for each RL			
- CHOICE mode		FDD	
- Primary CPICH usage for channel estimation		Primary CPICH may be used	
- DPCH frame offset		Set to value : Default DPCH Offset Value mod 38 400	
- Secondary CPICH info		Not Present	
- DL channelisation code			
- Secondary scrambling code		Not Present	
- Spreading factor		256	
- Code number		192	
- Scrambling code change		Not Present	
- TPC combination index		0	
- SSDT Cell Identity		Not Present	R99 and Rel-4 only
- Closed loop timing adjustment mode		Not Present	
- E-AGCH Info		Not Present	Rel-6
- E-HICH Information		Not Present	Rel-6
- E-RGCH Information		Not Present	Rel-6
- SCCPCH information for FACH		Not Present	R99 and Rel-4 only

Condition	Explanation
A1	UE supporting 64kbps(Chanel2)

Contents of SECURITY MODE COMMAND message: AM

Information Element	Condition	Value/remark
Message Type	A1, A2	

RRC transaction identifier		Arbitrarily selects an integer between 0 and 3
Integrity check info - Message authentication code - RRC Message Sequence Number		 Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. Set to an arbitrarily selected integer between 0 and 15
Security capability - Ciphering algorithm capability - UEA0 - UEA1 - Spare - Integrity protection algorithm capability - UIA1 - Spare		 If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. Spare 2-15 = FALSE 0000000000000010B (UIA1) TRUE Spare 0 and Spare 2-15 = FALSE
Ciphering mode info - Ciphering mode command - Ciphering algorithm - Ciphering activation time for DPCH - Radio bearer downlink ciphering activation time info - Radio bearer activation time - RB identity - RLC sequence number - RB identity - RLC sequence number - RB identity - RLC sequence number - RB identity - RLC sequence number		This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. Start/restart UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message. Not Present 1 Current RLC SN 2 Current RLC SN+3(or Calculated Value) 3 Current RLC SN 4 Current RLC SN
Integrity protection mode info - Integrity protection mode command - Downlink integrity protection activation info - Integrity protection algorithm - Integrity protection initialisation number		 Start Not Present UIA1 SS selects an arbitrary 32 bits number for FRESH. The first/ leftmost bit of the bit string contains the most significant bit of the FRESH.A1
CN domain identity		CS or PS
UE system specific security capability	A1	Not Present
UE system specific security capability - Inter-RAT UE security capability - CHOICE *system* - GSM security capability	A2	 GSM The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message.

Condition	Explanation
A1	UE not supporting GSM
A2	UE supporting GSM

9.2.2 Default Message Contents for RF (TDD)

Contents of Activate RB Test Mode message

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	44h

Contents of Close UE Test Loop message

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	40h
UE test loop mode	00h
UE test loop mode 1 LB setup	03h 00h F4h 0Ah

Contents of Open UE Test Loop message

Information Element	Value/remark
Protocol discriminator	F (Length 1/2)
Skip indicator	0 (Length 1/2)
Message Type	42h

Contents of PAGING TYPE 1 message: TM (CS)

Information Element	Value/remark
Message Type	
Paging record list	
-Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Streaming Call
- CN domain identity	CS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of PAGING TYPE 1 message: TM (PS)

Information Element	Value/remark
Message Type	
Paging record list	
-Paging record	
- CHOICE Used paging identity	CN identity
- Paging cause	Terminating Interactive Call
- CN domain identity	PS domain
- CHOICE UE identity	
- IMSI (GSM-MAP)	Set to the same octet string as in the IMSI stored in the USIM card
BCCH modification info	Not Present

Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A3		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the	

Information Element	Condition	Value/remark	Version
		MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI		Not Present	
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		Not Present	Rel-5
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
- Signalling RB information to setup		Not Present	
- RAB information for setup list	A1		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RAB information for setup list	A3		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			

Information Element	Condition	Value/remark	Version
- RB information to setup			
- RB identity		20	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		200	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel		Not Present	

Information Element	Condition	Value/remark	Version
identity			
- Logical channel identity		Not Present	
RB information to be affected list	A1,A3	Not Present	
Downlink counter synchronization info		Not Present	
UL Transport channel information for all transport channels	A1,A3		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
-Individual UL CCTrCH information			
- TFCS ID		(This IE is repeated for TFC number.)	
- Allowed Transport Format combination		0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS		(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- TFCS complete reconfigure information			
- CHOICE TFCS Size		Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information		Not Present	
Deleted UL TrCH information list		Not Present	
Added or Reconfigured UL TrCH information list	A1	1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information			
- RLC size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
CHOICE mode	A1, A3	TDD (no data)	
DL Transport channel information common for all transport channel	A1,A3		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- CHOICE DL parameters		Independent (Refer to clause 6)	
Deleted DL TrCH information list	A1,A3	Not Present	
Added or Reconfigured DL TrCH information list		1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			

Information Element	Condition	Value/remark	Version
- BLER Quality value		Reference to clause 6	
Frequency info	A1,A3	Not Present	
Maximum allowed UL TX power		30dBm	
CHOICE channel requirement		Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL Target SIR		Reference to clause 6 Parameter set.	
- CHOICE UL OL PC info		Individually signalled	
- CHOICE TDD option		3.84 Mcps	
- Individual timeslot interference info			
- Individual timeslot interference			
- DPCH Constant Value		Values are used for open loop power control, clause 8 in 3GPP TS 25.331 [34]	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS Id		1	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.10 Parameter Set	
- TFCI coding		Reference to clause 6.10 Parameter Set	
- Puncturing Limit		Reference to clause 6.10 Parameter Set	
- Repetition Period		Reference to clause 6.10 Parameter Set	
- Repetition Length		Reference to clause 6.10 Parameter Set	
- First individual timeslot info			
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps	
-CHOICE Burst Type			
-Type 1			
-Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- Channelisation code		(i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned.	
CHOICE Mode		TDD (no data)	
Downlink HS-PDSCH Information	A1,A3	Not Present	Rel-5
Downlink information common for all radio links	A1,A3		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- DPC mode		0 (single)	
- CHOICE TDD mode		3.84 Mcps (no data)	
- Default DPCH Offset Value		Not Present	
Downlink information for per radio link list	A1,A3		
- Downlink information for each radio link			
- CHOICE mode		TDD	

Information Element	Condition	Value/remark	Version
- Primary CCPCH info			
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		PCCPCH timeslot	
- Cell parameters ID		0	
- SCTD indicator			
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCH List			
- TFCS ID		1	
- Time info			
- Activation time		(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to the present document	
- TFCI coding		TRUE	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		Empty	
- Downlink DPCH timeslots and codes			
- Individual timeslot info			
- Timeslot number		The number of a downlink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		3.84 Mcps	
-CHOICE Burst Type			
-Type 1			
-Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes			
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- Bitmap		Bitmap of the codes that are being assigned in the slot.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List		Not Present	
-SCCPCH information for FACH		Not Present	R99 and Rel-4 only

Condition	Explanation
A1	This IE is needed for transparent mode. In the case of TX and RX test cases, this IE is selected.
A3	This IE is needed for acknowledged mode.
NOTE: In the case of Performance Requirement and RRM test cases, A1 or A3 is selected according to the combination of UL and DL channels or test requirements.	

Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A3		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant	

Information Element	Condition	Value/remark	Version
		bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI		Not Present	
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		Not Present	Rel-5
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
- Signalling RB information to setup		Not Present	
- RAB information for setup list	A1		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RAB information for setup list	A3		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT315	
- RB information to setup list			
- RB information to setup			

Information Element	Condition	Value/remark	Version
- RB identity		20	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	
- Transmission RLC discard			
- CHOICE SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		200	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RB information to be affected list	A1,A3	Not Present	

Information Element	Condition	Value/remark	Version
Downlink counter synchronization info		Not Present	
UL Transport channel information for all transport channels	A1,A3		
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
-Individual UL CCTrCH information			
- TFCS ID		(This IE is repeated for TFC number.)	
- Allowed Transport Format combination		0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS		(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- TFCS complete reconfigure information			
- CHOICE TFCS Size		Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information		Not Present	
Deleted UL TrCH information list		Not Present	
Added or Reconfigured UL TrCH information list	A1,A3	1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information			
- RLC size		Reference to clause 6 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6 Parameter Set	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		Reference to clause 6 Parameter Set	
- Type of channel coding		Reference to clause 6 Parameter Set	
- Coding Rate		Reference to clause 6 Parameter Set	
- Rate matching attribute		Reference to clause 6 Parameter Set	
- CRC size		Reference to clause 6 Parameter Set	
CHOICE mode	A1, A3	TDD (no data)	
DL Transport channel information common for all transport channel	A1,A3		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- CHOICE DL parameters		Independent (Refer to clause 6)	
Deleted DL TrCH information list	A1,A3	Not Present	
Added or Reconfigured DL TrCH information list		1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			
- BLER Quality value		Reference to clause 6	
Frequency info	A1,A3	Not Present	
Maximum allowed UL TX power		30dBm	
CHOICE channel requirement		Uplink DPCH info	

Information Element	Condition	Value/remark	Version
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL Target SIR		Reference to clause 6 Parameter set.	
- CHOICE UL OL PC info		Individually signalled	
- CHOICE TDD option		1.28 Mcps	
- TPC step size		0 (1 dB)	
- Primary CCPCH Tx Power		Not Present	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS Id		1	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6 Parameter Set	
- TFCI coding		Reference to clause 6 Parameter Set	
- Puncturing Limit		Reference to clause 6 Parameter Set	
- Repetition Period		Reference to clause 6 Parameter Set	
- Repetition Length		Reference to clause 6 Parameter Set	
- First individual timeslot info			
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps	
- Midamble allocation mode		Default	
- Midamble configuration		16	
- CHOICE TDD option		1.28 Mcps TDD	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- CHOICE Mode		TDD	
- First timeslot channelisation codes		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- Channelisation code		(i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned.	
CHOICE Mode		TDD (no data)	
Downlink HS-PDSCH Information	A1,A3	Not Present	Rel-5
Downlink information common for all radio links	A1,A3		
- Downlink DPCH info common for all RL			
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC step size		1 dB	
- CHOICE TDD mode		1.28 Mcps	
- TSTD indicator		TRUE	
- Default DPCH Offset Value		Not Present	
Downlink information for per radio link list	A1,A3		
- Downlink information for each radio link			
- CHOICE mode		TDD	
- Primary CCPCH info			
- CHOICE TDD option		1.28 Mcps	
- TSTD indicator		TRUE	
- Cell parameters ID		0	
- Block STTD indicator		FALSE	

Information Element	Condition	Value/remark	Version
- Downlink DPCH info for each RL			
- CHOICE mode		TDD	
- DL CCTrCH List			
- TFCS ID		1	
- Time info			
- Activation time		(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to the present document	
- TFCI coding		TRUE	
- Puncturing limit		Reference to clause 6 Parameter set	
- Repetition period		1	
- Repetition length		Empty	
- Downlink DPCH timeslots and codes			
- Individual timeslot info			
- Timeslot number		The number of a downlink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		1.28 Mcps	
-Midamble Allocation Mode		Default	
- Midamble configuration		16	
- Modulation		QPSK	
- SS-TPC Symbols		1	
- First timeslot channelisation codes			
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- Bitmap		Bitmap of the codes that are being assigned in the slot.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List		Not Present	
-SCCPCH information for FACH		Not Present	R99 and Rel-4 only

Condition	Explanation
A1	This IE is needed for CS RAB
A3	This IE is needed for PS RAB.

Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD)

Information Element	Condition	Value/remark	Version
Message Type	A1,A3		
RRC transaction identifier		Arbitrarily selects an integer between 0 and 3	
Integrity check info			
- message authentication code		SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number		SS provides the value of this IE, from its internal counter.	
Integrity protection mode info		Not Present	
Ciphering mode info		Not Present	
Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
New U-RNTI		Not Present	

Information Element	Condition	Value/remark	Version
New C-RNTI		Not Present	
New DSCH-RNTI		Not Present	R99 and Rel-4 only
New H-RNTI		Not Present	Rel-5
CHOICE mode		TDD	Rel-7
- New E-RNTI		Not Present	Rel-7
RRC State indicator		CELL_DCH	
UTRAN DRX cycle length coefficient		Not Present	
CN information info		Not Present	
URA identity		Not Present	
- Signalling RB information to setup		Not Present	
- RAB information for setup list	A1		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		CS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RB identity		10	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		TM RLC	
- Transmission RLC discard		Not Present	
- Segmentation indication		FALSE	
- CHOICE Downlink RLC mode		TM RLC	
- Segmentation indication		FALSE	
- RB mapping info			
- Information for each multiplexing option			
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		7	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RAB information for setup list	A3		
- RAB information for setup			
- RAB info			
- RAB identity		0000 0101B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity		PS domain	
- NAS Synchronization Indicator		Not Present	
- Re-establishment timer		UseT314	
- RB information to setup list			
- RB information to setup			
- RB identity		20	
- PDCP info		Not Present	
- CHOICE RLC info type		RLC info	
- CHOICE Uplink RLC mode		AM RLC	

Information Element	Condition	Value/remark	Version
- Transmission RLC discard			
- CHOICE SDU discard mode		No discard	
- MAX_DAT		15	
- Transmission window size		128	
- Timer_RST		500	
- Max_RST		4	
- Polling info			
- Timer_poll_prohibit		200	
- Timer_poll		200	
- Poll_SDU		1	
- Last transmission PDU poll		TRUE	
- Last retransmission PDU poll		TRUE	
- Poll_Windows		99	
- Timer_poll_periodic		Not Present	
- CHOICE Downlink RLC mode		AM RLC	
- In-sequence delivery		TRUE	
- Receiving window size		128	
- Downlink RLC status info			
- Timer_status_prohibit		200	
- Timer_EPC		200	
- Missing PDU indicator		TRUE	
- Timer_STATUS_periodic		Not Present	
- RB mapping info			
- Information for each multiplexing option		2RBMuxOptions	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- Logical channel identity		Not Present	
- CHOICE RLC size list		Configured	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		DCH	
- DL DCH Transport channel identity		6	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
- RLC logical channel mapping indicator		Not Present	
- Number of uplink RLC logical channels		1	
- Uplink transport channel type		RACH	
- UL Transport channel identity		Not Present	
- Logical channel identity		7	
- CHOICE RLC size list		Explicit List	
- RLC size index		Reference to clause 6 Parameter Set	
- MAC logical channel priority		8	
- Downlink RLC logical channel info			
- Number of downlink RLC logical channels		1	
- Downlink transport channel type		FACH	
- DL DCH Transport channel identity		Not Present	
- DL DSCH Transport channel identity		Not Present	
- Logical channel identity		Not Present	
RB information to be affected list	A1,A3	Not Present	
Downlink counter synchronization info		Not Present	
UL Transport channel information for all	A1,A3		

Information Element	Condition	Value/remark	Version
transport channels			
- PRACH TFCS		Not Present	
- CHOICE mode		TDD	
-Individual UL CCTrCH information			
- TFCS ID		(This IE is repeated for TFC number.)	
- Allowed Transport Format combination		0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS		(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling		Normal	
- TFCI Field 1 information			
- TFCS complete reconfigure information			
- CHOICE TFCS Size		Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information		Not Present	
- CHOICE mode		TDD	
- Individual UL CCTrCH information		Not Present	
Deleted UL TrCH information list		Not Present	
Added or Reconfigured UL TrCH information list	A1	1	
- Added or Reconfigured UL TrCH information			
- Uplink transport channel type		DCH	
- UL Transport channel identity		1	
- TFS			
- CHOICE Transport channel type		Dedicated transport channels	
- Dynamic Transport Format Information			
- RLC size		Reference to clause 6.10 Parameter Set	
- Number of TBs and TTI List		(This IE is repeated for TFI number.)	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		Reference to clause 6.10 Parameter Set	
- Transmission Time Interval		Not Present	
- Number of Transport blocks		1	
- CHOICE Logical channel List		ALL	
- Semi-static Transport Format Information			
- Transmission time interval		Reference to clause 6.10 Parameter Set	
- Type of channel coding		Reference to clause 6.10 Parameter Set	
- Coding Rate		Reference to clause 6.10 Parameter Set	
- Rate matching attribute		Reference to clause 6.10 Parameter Set	
- CRC size		Reference to clause 6.10 Parameter Set	
CHOICE mode	A1, A3	TDD (no data)	
DL Transport channel information common for all transport channel	A1,A3		
- SCCPCH TFCS		Not Present	
- CHOICE mode		TDD	
- CHOICE DL parameters		Independent (Refer to clause 6)	
Deleted DL TrCH information list	A1,A3	Not Present	
Added or Reconfigured DL TrCH information list		1	
- Added or Reconfigured DL TrCH information			
- Downlink transport channel type		DCH	
- DL Transport channel identity		6	
- CHOICE DL parameters		Same as UL	
- Uplink transport channel type		DCH	
- UL TrCH identity		1	
- DCH quality target			
- BLER Quality value		Reference to clause 6	
Frequency info	A1,A3	Not Present	
DTX-DRX timing information		Not Present	Rel-7
DTX-DRX information		Not Present	Rel-7
HS-SCCH less information		Not Present	Rel-7

Information Element	Condition	Value/remark	Version
MIMO parameters		Not Present	Rel-7
Maximum allowed UL TX power		30dBm	
CHOICE channel requirement		Uplink DPCH info	
- Uplink DPCH power control info			
- CHOICE mode		TDD	
- UL Target SIR		Reference to clause 6 Parameter set.	
- CHOICE UL OL PC info		Individually signalled	
- CHOICE TDD option		7.68 Mcps	
- Individual timeslot interference info			
- Individual timeslot interference			
- DPCH Constant Value		Values are used for open loop power control, clause 8 in 3GPP TS 25.331 [34]	
- CHOICE mode		TDD	
- Uplink Timing Advance Control		Not Present	
- UL CCTrCH List			
- TFCS Id		1	
- Time info			
- Activation time		(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration		Infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter Set	
- TFCI coding		Reference to clause 6.11 Parameter Set	
- Puncturing Limit		Reference to clause 6.11 Parameter Set	
- Repetition Period		Reference to clause 6.11 Parameter Set	
- Repetition Length		Reference to clause 6.11 Parameter Set	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Uplink DPCH timeslots and codes VHCR			Rel-7
- Dynamic SF usage		TRUE	
- First individual timeslot info			
- Timeslot number		The number of an uplink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
-CHOICE Burst Type			
-Type 1			
-Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		As defined in 3GPP TS 25.221 [28]	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- First timeslot code list		Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- Channelisation code		(i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots		The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove		Not Present	
CHOICE Mode		TDD (no data)	
Downlink HS-PDSCH Information	A1,A3	Not Present	Rel-5
Downlink information common for all radio links	A1,A3		
- CHOICE DPCH info		Downlink DPCH info common for all RL	Rel-6
- Timing indicator		Maintain	
- CFN-targetSFN frame offset		Not Present	
- Downlink DPCH power control information			
- CHOICE mode		TDD	
- TPC Step Size		1	
- MAC-d HFN initial value		Not Present	

Information Element	Condition	Value/remark	Version
- CHOICE mode		TDD	
- CHOICE mode		TDD	
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
- Default DPCH Offset Value		Not Present	
- Mac-hs reset indicator		Not Present	
Downlink information for per radio link list	A1,A3		
- Downlink information for each radio link			
- CHOICE mode		7.68 Mcps TDD	Rel-7
- Primary CCPCH info			
- CHOICE *SyncCase*		Sync Case 1	
- Timeslot		PCCPCH timeslot	
- Cell parameters ID		0	
- SCTD indicator			
- CHOICE DPCH info		Downlink DPCH info for each RL	Rel-6
- CHOICE mode		TDD	
- DL CCTrCH List			
- TFCS ID		1	
- Time info			
- Activation time		(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration		infinite	
- Common timeslot info			
- 2nd interleaving mode		Reference to clause 6.11 Parameter Set	
- TFCI coding		Reference to clause 6.11 Parameter Set	
- Puncturing limit		Reference to clause 6.11 Parameter Set	
- Repetition period		Reference to clause 6.11 Parameter Set	
- Repetition length		Reference to clause 6.11 Parameter Set	
- Downlink DPCH timeslots and codes VHCR			Rel-7
- Individual timeslot info			
- Timeslot number		The number of a downlink timeslot that has unassigned codes.	
- TFCI existence		TRUE	
- Midamble shift and burst type			
- CHOICE TDD option		7.68 Mcps TDD	Rel-7
-CHOICE Burst Type			
-Type 1			
-Midamble Allocation Mode		Default	
- Midamble configuration burst type 1 and 3		As defined in 3GPP TS 25.221 [28]	
- CHOICE TDD option		7.68 Mcps	Rel-7
- First timeslot channelisation codes VHCR			Rel-7
- First channelisation code		(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code		(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- Bitmap		Bitmap of the codes that are being assigned in the slot.	
- CHOICE more timeslots		The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List		Not Present	
- DL CCTrCH List to Remove		Not Present	
-SCCPCH information for FACH		Not Present	R99 and Rel-4 only
- E-AGCH Info		Not Present	Rel-6
- CHOICE E-HICH Information		Not Present	Rel-6
- CHOICE E-RGCH Information		Not Present	Rel-6
MBMS PL Service Restriction Information		Not Present	Rel-5

Condition	Explanation
A1	This IE is needed for transparent mode. In the case of TX and RX test cases, this IE is selected.

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
New Primary E-RNTI	Not Present	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		

Information Element	Value/remark	Version
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0 Integer(0.. 3)	
- CHOICE Gain Factors	Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode	TDD	
- Gain factor βd	8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID	0	
- CHOICE mode	TDD	
- TFC subset	Not Present	
- CHOICE Subset representation	Full transport format combination set	
- TFC subset list		
Deleted UL TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual DL CCTrCH information	1 CCTrCh	
- DL TFCS identity	1	
- CHOICE DL parameters	Independent	
- DL TFCS		
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to TS 34.122 [5] Annex C.3.1 Parameter Set	
- Power offset information	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	Rel-5
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels	Rel-5
- CHOICE *Memory Partitioning*	Explicit	Rel-5
- Memory size	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels parameter “Number of HARQ Processes”.	Rel-5
- Process Memory Size	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.	Rel-5
- Added or reconfigured MAC-d flow		Rel-5
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	Rel-5
- MAC-d Flow Identity	0	Rel-5
- T1	160	Rel-5
- MAC-hs window size	16	Rel-5
- MAC-d PDU size Info		Rel-5
- MAC-d PDU size	Reference to TS34.122 [2] Annex C.4 Fixed Reference Channels	Rel-5

Information Element	Value/remark	Version
- MAC-d PDU size index	0	Rel-5
- MAC-hs queue to delete list	Not present	Rel-5
- DCH quality target	Not present	
Frequency info	Not Present	
Maximum allowed UL TX power	30dBm	
CHOICE channel requirement	Uplink DPCH info	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- UL target SIR	Not present	
- CHOICE UL OL PC info	Broadcast UL OL PC info	
- CHOICE mode	TDD	
- Uplink Timing Advance Control		
- CHOICE Timing Advance	Enabled	
- CHOICE TDD option	3.84 Mcps TDD	
- UL Timing Advance	Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List	1 CCTrCh	
- TFCS Id	1	
- UL target SIR	+20dB	
- Activation time	Not present	
- Duration	Not present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS 34.122 clause C.2.1 Parameter Set	
- TFCI coding	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Puncturing Limit	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Repetition Period	1	
- Repetition Length	1	
- Uplink DPCH timeslots and codes		
- Dynamic SF usage	TRUE	
- Timeslot number	The number of an uplink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	3.84 Mcps	
- CHOICE Burst Type	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Midamble Allocation Mode	Default	
- Midamble configuration	Choose lowest possible Kcell value given burst type	
- CHOICE TDD option	3.84 Mcps TDD	
- First timeslot Code List	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS 34.122 clause C.2.1 Parameter Set.	
- Channelisation code	(i/SF) where i denotes an unassigned code matching the SF specified in TS 34.122 clause C.2.1 Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon the number of resources specified in TS 34.122 clause C.2.1 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove	Not present	
E-DCH Info	Not present	Rel-6
Downlink HS-PDSCH Information		Rel-5
- HS-SCCH Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	3.84 Mcps TDD	Rel-5
- Ack-Nack Power Offset	0dB	Rel-5
- HS-SICH Power Control Info		Rel-5
- UL SIR target	0dB	Rel-5
- HS-SICH Constant Value	-10dB	Rel-5
- $D_{hs\text{-}sync}$	Not present	Rel-6
- HS-SCCH Set Configuration	4	Rel-5
- Timeslot number	The timeslot in which HS-SCCH is to be configured	Rel-5

Information Element	Value/remark	Version
- Channelisation code	CC16/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- BLER target	-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	Rel-5
- HS-SICH configuration		
- Timeslot number	The timeslot in which HS-SICH has beeen configured	Rel-5
- Channelisation code	CC16/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- Measurement Feedback Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	3.84 Mcps TDD	Rel-5
- HS-PDSCH Timeslot Configuration		Rel-5
- HS-PDSCH Timeslot Configuration List	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Timeslot Number	The timeslot(s) in which HS-HS-DSCH is to configured	Rel-5
- CHOICE Burst Type	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Midamble Allocation Mode	Default	Rel-5
- Midamble configuration burst type 1 and 3	8	Rel-5
Downlink information common for all radio links	Not Present	
Downlink information per radio link list	1	
- Downlink information for each radio link		
- Choice mode	TDD	
- Primary CCPCH info		
- Choice mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	
- CHOICE SyncCase	Sync Case 1	
- Timeslot	Set to Timeslot containing PCCPCH	
- Cell parameters ID	10	
- SCTD indicator	FALSE	
- CHOICE DPCH info	Downlink DPCH info for each RL	
- CHOICE mode	TDD	
- DL CCTrCH List	1 CCTrCh	
- TFCS ID	1	
- Activation time	Not Present	
- Duration	Not Present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS 34.122 clause C.3.1 Parameter Set	
- TFCI coding	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Puncturing Limit	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Repetition Period	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Repetition Length	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Downlink DPCH timeslots and codes		
- Individual timeslot info		
- Timeslot number	The number of a downlink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	3.84 Mcps	
- CHOICE Burst Type	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Midamble Allocation Mode	Default	
- Midamble configuration	Set Kcell to lowest possible value given the number of codes defined in TS 34.122 clause C.3.1 Parameter Set	
- CHOICE TDD option	3.84 Mcps	

Information Element	Value/remark	Version
- First timeslot channelisation codes		
- CHOICE codes representation	Consecutive codes	
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in TS 34.122 clause C.3.1 Parameter Set.	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot as specified in TS 34.122 clause C.3.1 Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of TS 34.122 clause C.3.1 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	No Present	
- DL CCTrCH List to Remove	Not Present	
- E-AGCH Info	Not Present	Rel-6
- CHOICE E-HICH Information	Not Present	Rel-6
- CHOICE E-RGCH Information	Not Present	Rel-6
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	

Information Element	Value/remark	Version
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfiguration information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to clause TS 34.122 clause C.2.1 Parameter Set	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0 Integer(0.. 3)	
- CHOICE Gain Factors	Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode	TDD	
- Gain Factor β_d	15	
- Reference TFC ID	0 Integer(0.. 3)	
- CHOICE mode	TDD	
- TFC subset		
- CHOICE Subset representation	Full transport format combination set	
- TFC subset list	Not Present	
Deleted UL TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual DL CCTrCH information		
- DL TFCS Identity		
- TFCS ID	2	
- Shared Channel Indicator	FALSE	
- CHOICE DL parameters	Explicit	
- DL DCH TFCS		
- CHOICE TFCI Signalling	Normal	
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to clause TS 34.122 clause C.2.1 Parameter Set	
- Power offset information	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information	(HS-DSCH for DTCH)	

Information Element	Value/remark	Version
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.122 [5] Annex C Fixed Reference Channels	
- CHOICE *Memory Partitioning*	Implicit	
- Added or reconfigured MAC-d flow		
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	
- MAC-d Flow Identity	0	
- T1	50	
- MAC-hs window size	16	
- MAC-d PDU size Info		
- MAC-d PDU size	Reference to TS34.122 [5] Annex C Fixed Reference Channels	
- MAC-d PDU size index	0	
- MAC-hs queue to delete list	Not present	
- DCH quality target	Not present	
Frequency info	Not Present	
Maximum allowed UL TX power	33dBm	
CHOICE channel requirement	Uplink DPCH info	Rel-5 and earlier
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE TDD option	1.28 Mcps TDD	
- $PRX_{PDPCHdes}$	Integer (-120…-58 by step of 1)	
- CHOICE *UL OL PC info*		
- Broadcast UL OL PC info	Null	
- Uplink Timing Advance Control	Not Present	
- UL CCTrCH List		
- TFCS ID	1	
- UL Target SIR	Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6 Parameter set.	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Default value is "Frame"	
- TFCI coding	Reference to clause 6 Parameter set	
- Puncturing limit	Reference to clause 6 Parameter set	
- Repetition period	1	
- Repetition length		
- Uplink DPCH timeslots and code		
- Dynamic SF usage	FALSE	
- First individual timeslot info		
- Timeslot number		
- CHOICE TDD option	1.28 Mcps TDD	
- Timeslot number	1 OR 2 OR 3	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	1.28 Mcps TDD	
- Midamble allocation mode	Default midamble	
- Midamble configuration	16	
- Midamble Shift	Not Present	
- CHOICE TDD option	1.28 Mcps TDD	
- Modulation	QPSK	
- SS-TPC Symbols	1	
- Additional TPC-SS Symbols	Not present	
- First timeslot Code List	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- channelisation codes	(SF/ i) where i denotes an unassigned code	

Information Element	Value/remark	Version
	matching the SF specified in clause 6 Parameter Set.	
- CHOICE more timeslots	No more timeslots	
- UL CCTrCH List to Remove	Not present	
CHOICE Mode	TDD	R99 and Rel-4 only
- Downlink PDSCH information	Not Present	R99 and Rel-4 only
Downlink HS-PDSCH Information		
- HS-SCCH Info		
- CHOICE mode	TDD	
- CHOICE TDD option	1.28 Mcps	
- HS-SCCH Set Configuration		
- Timeslot number	0	
- First Channelisation code	(16/5)	
- Second Channelisation code	(16/6)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- BLER target	-2.0	
- HS-SICH configuration		
- Timeslot number	1	
- Channelisation code	(16/11)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- Ack-Nack Power Offset	0	
- $PRX_{HS-SICH}$		
- TPC step size	1dB	
- Timeslot number	0	
- First Channelisation code	(16/7)	
- Second Channelisation code	(16/8)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- BLER target	-2.0	
- HS-SICH configuration		
- Timeslot number	1	
- Channelisation code	(16/12)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- Ack-Nack Power Offset	0	
- $PRX_{HS-SICH}$		
- TPC step size	1dB	
- Timeslot number	0	
- First Channelisation code	(16/9)	
- Second Channelisation code	(16/10)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- BLER target	-2.0	
- HS-SICH configuration		
- Timeslot number	1	
- Channelisation code	(16/13)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- Ack-Nack Power Offset	0	
- $PRX_{HS-SICH}$		
- TPC step size	1dB	
- Timeslot number	0	
- First Channelisation code	(16/11)	
- Second Channelisation code	(16/12)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- BLER target	-2.0	
- HS-SICH configuration		
- Timeslot number	1	

Information Element	Value/remark	Version
- Channelisation code	(16/14)	
- Midamble Allocation mode	Default midamble	
- Midamble configuration	8	
- Ack-Nack Power Offset	0	
- $PRX_{HS-SICH}$		
- TPC step size	1dB	
Downlink information common for all radio links	Not Present	
Downlink information per radio link list		
- Downlink information for each radio link		
- CHOICE mode	TDD	
- Downlink information for each radio link		
- Choice mode	2 Integer(1.8)	
- Primary CCPCH info		
- Choice mode	Now	
- Choice TDD Option	Infinite	
- TSTD indicator		
- Cell parameters ID	Default value is "Frame"	
- SCTD indicator	Reference to clause 6 Parameter set	
- Downlink DPCH info for each RL	Reference to clause 6 Parameter set	
- CHOICE mode	1	
- DL CCTrCh List	NULL	
- TFCS ID		
- Time info		
- Activation time		
- Duration	1.28 Mcps TDD	
- Common timeslot info	4 OR 5 OR 6	
- 2nd interleaving mode	TRUE	
- TFCI coding		
- Puncturing limit	1.28 Mcps TDD	
- Repetition period	Default midamble	
- Repetition length	16	
- Downlink DPCH timeslots and codes	Not Present	
- First individual timeslot info	1.28 Mcps TDD	
- Timeslot number	QPSK	
- CHOICE TDD option	1	
- Timeslot number	Not present	
- TFCI existence	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set.	
- Midamble shift and burst type		
- CHOICE TDD option	Reference to clause 6.11 Parameter Set	
- Midamble allocation mode	No more timeslots	
- Midamble configuration	This list is not required for 1.28 Mcps TDD and is to be ignored by the UE.	
- Midamble Shift		
- CHOICE TDD option	1	
- Modulation	FALSE	
- SS-TPC Symbols	Not present	
- Additional TPC-SS Sysbols	Not Present	
- First timeslot channelisation codes	TDD	
- CHOICE codes representation		
- Channelisation codes bitmap	2 Integer(1.8)	
- CHOICE more timeslots		
- UL CCTrCH TPC List	Now	
- UL TPC TFCS Identity	Infinite	
- TFCS ID		
- Shared Channel Indicator	Default value is "Frame"	
- DL CCTrCH List to Remove	Reference to clause 6 Parameter set	
- SCCPCH Information for FACH	Reference to clause 6 Parameter set	R99 and Rel-4 only

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
CHOICE mode	TDD	Rel-7
New E-RNTI	Not Present	Rel-7
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information		
- UL TFCS Identity		
- TFCS ID	1	
- Shared Channel Indicator	FALSE	
- UL TFCS		
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		

Information Element	Value/remark	Version
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure information		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Power offset information		
- CHOICE Gain Factors	Computed Gain Factors(The last TFC is set to Signalled Gain Factors)	
- Reference TFC ID	0 Integer(0.. 3)	
- CHOICE Gain Factors	Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors)	
- CHOICE mode	TDD	
- Gain factor βd	8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors)	
- Reference TFC ID	0	
- CHOICE mode	TDD	
- TFC subset	Not Present	
- CHOICE Subset representation	Full transport format combination set	
- TFC subset list		
Deleted UL TrCH information list	Not Present	
Added or Reconfigured TrCH information list	Not Present	
CHOICE mode	Not Present	
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- Individual DL CCTrCH information	1 CCTrCh	
- DL TFCS identity	1	
- CHOICE DL parameters	Independent	
- DL TFCS		
- TFCI Field 1 Information		
- CHOICE TFCS representation	Complete reconfiguration	
- TFCS complete reconfigure		
- CHOICE CTFC Size	2 bit CTFC	
- CTFC information	4 TFCs	
- CTFC	Reference to TS 34.122 [5] Annex C.3.1 Parameter Set	
- Power offset information	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	Rel-5
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels	Rel-5
- CHOICE *Memory Partitioning*	Explicit	Rel-5
- Memory size	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels parameter “Number of HARQ Processes”.	Rel-5
- Process Memory Size	Reference to TS34.122 [5] Annex C.4 Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.	Rel-5
- Added or reconfigured MAC-d flow		Rel-5
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	Rel-5
- MAC-d Flow Identity	0	Rel-5
- T1	160	Rel-5
- MAC-hs window size	16	Rel-5
- MAC-d PDU size Info		Rel-5
- MAC-d PDU size	Reference to TS34.122 [2] Annex C.4 Fixed Reference Channels	Rel-5
- MAC-d PDU size index	0	Rel-5
- MAC-hs queue to delete list	Not present	Rel-5

Information Element	Value/remark	Version
- DCH quality target	Not present	
Frequency info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX information	Not Present	Rel-7
HS-SCCH less information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	30dBm	
CHOICE channel requirement	Uplink DPCH info	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- UL target SIR	Not present	
- CHOICE UL OL PC info	Broadcast UL OL PC info	
- CHOICE mode	TDD	
- Uplink Timing Advance Control		
- CHOICE Timing Advance	Enabled	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- UL Timing Advance	Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List	1 CCTrCh	
- TFCS Id	1	
- UL target SIR	+20dB	
- Activation time	Not present	
- Duration	Not present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS 34.122 clause C.2.1 Parameter Set	
- TFCI coding	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Puncturing Limit	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Repetition Period	1	
- Repetition Length	1	
- CHOICE mode	7.68 Mcps TDD	Rel-7
- Uplink DPCH timeslots and codes VHCR		Rel-7
- Dynamic SF usage	TRUE	
- Timeslot number	The number of an uplink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- CHOICE Burst Type	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Midamble Allocation Mode	Default	
- Midamble configuration	Choose lowest possible Kcell value given burst type	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- First timeslot Code List	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS 34.122 clause C.2.1 Parameter Set.	
- Channelisation code	(i/SF) where i denotes an unassigned code matching the SF specified in TS 34.122 clause C.2.1 Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon the number of resources specified in TS 34.122 clause C.2.1 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove	Not present	
E-DCH Info	Not present	Rel-6
Downlink HS-PDSCH Information		Rel-5
- HS-SCCH Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- Ack-Nack Power Offset	0dB	Rel-5
- HS-SICH Power Control Info		Rel-5
- UL SIR target	0dB	Rel-5
- HS-SICH Constant Value	-10dB	Rel-5

Information Element	Value/remark	Version
- $D_{hs\text{-}sync}$	Not present	Rel-6
- HS-SCCH Set Configuration	4	Rel-5
- Timeslot number	The timeslot in which HS-SCCH is to be configured	Rel-5
- Channelisation code	CC32/x where x is a previously unassigned channelisation code in this TS	Rel-7
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- BLER target	-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	Rel-5
- HS-SICH configuration		
- Timeslot number	The timeslot in which HS-SICH has beeen configured	Rel-5
- Channelisation code	CC32/x where x is a previously unassigned channelisation code in this TS	Rel-7
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- Measurement Feedback Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- HS-PDSCH Timeslot Configuration VHCR		Rel-5
- HS-PDSCH Timeslot Configuration List	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Timeslot Number	The timeslot(s) in which HS-HS-DSCH is to configured	Rel-5
- CHOICE Burst Type	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Midamble Allocation Mode	Default	Rel-5
- Midamble configuration burst type 1 and 3	8	Rel-5
Downlink information common for all radio links	Not Present	
Downlink information for each radio link list	1	
- Downlink information for each radio link		
- Choice mode	7.68 Mcps TDD	Rel-7
- Primary CCPCH info		
- Choice mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	Rel-7
- CHOICE SyncCase	Sync Case 1	
- Timeslot	Set to Timeslot containing PCCPCH	
- Cell parameters ID	10	
- SCTD indicator	FALSE	
- CHOICE DPCH info	Downlink DPCH info for each RL	
- CHOICE mode	TDD	
- DL CCTrCH List	1 CCTrCh	
- TFCS ID	1	
- Activation time	Not Present	
- Duration	Not Present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS 34.122 clause C.3.1 Parameter Set	
- TFCI coding	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Puncturing Limit	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Repetition Period	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Repetition Length	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Downlink DPCH timeslots and codes VHCR		Rel-7
- Individual timeslot info		
- Timeslot number	The number of a downlink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	7.68 Mcps	
- CHOICE Burst Type	Reference to TS 34.122 clause C.3.1 Parameter Set	
- Midamble Allocation Mode	Default	
- Midamble configuration	Set Kcell to lowest possible value given the number	

Information Element	Value/remark	Version
	of codes defined in TS 34.122 clause C.3.1 Parameter Set	
- CHOICE TDD option	7.68 Mcps	Rel-7
- First timeslot channelisation codes VHCR		Rel-7
- CHOICE codes representation	Consecutive codes	
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in TS 34.122 clause C.3.1 Parameter Set.	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot as specified in TS 34.122 clause C.3.1 Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of TS 34.122 clause C.3.1 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	No Present	
- DL CCTrCH List to Remove	Not Present	
- E-AGCH Info	Not Present	Rel-6
- CHOICE E-HICH Information	Not Present	Rel-6
- CHOICE E-RGCH Information	Not Present	Rel-6
MBMS PL Service Restriction Information	Not Present	Rel-6

Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA) (3.84Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
New Primary E-RNTI	'1010 1010 1010 1010'	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		

Information Element	Value/remark	Version
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels	Not Present	
Deleted UL TrCH information list	Not Present	
Added or Reconfigured TrCH information list		
- Uplink transport channel type	E-DCH	
- CHOICE UL parameters	E-DCH	
- CHOICE mode	TDD	
- HARQ info for E-DCH		
- CHOICE UL parameters	E-DCH	
- HARQ RV Configuration	rvtable	
- Added or reconfigured E-DCH MAC-d flow		
- E-DCH MAC-d flow identity	2	
- E-DCH MAC-d flow power offset	0	
- E-DCH MAC-d flow maximum number of retransmissions	7	
- E-DCH MAC-d flow multiplexing list	Not Present	
- CHOICE transmission grant type	Scheduled grant info	
CHOICE mode	Not Present	R99 and Rel-4 only
DL Transport channel information common for all transport channel	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	Rel-5
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- CHOICE *Memory Partitioning*	Explicit	Rel-5
- Memory size	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- Process Memory Size	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- Added or reconfigured MAC-d flow		Rel-5
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	Rel-5
- MAC-d Flow Identity	0	Rel-5
- T1	50	Rel-5
- MAC-hs window size	16	Rel-5
- MAC-d PDU size Info		Rel-5
- MAC-d PDU size	Reference to TS34.122 [2] Annex C Fixed Reference Channels	Rel-5
- MAC-d PDU size index	0	Rel-5
- MAC-hs queue to delete list	Not present	Rel-5
- DCH quality target	Not present	
Frequency info	Not Present	
Maximum allowed UL TX power	30dBm	
CHOICE channel requirement	Uplink DPCH info	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	TDD	

Information Element	Value/remark	Version
- UL target SIR	Not present	
- CHOICE UL OL PC info	Broadcast UL OL PC info	
- CHOICE mode	TDD	
- Uplink Timing Advance Control		
- CHOICE Timing Advance	Enabled	
- CHOICE TDD option	3.84 Mcps TDD	
- UL Timing Advance	Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List	1 CCTrCh	
- TFCS Id	1	
- UL target SIR	+20dB	
- Activation time	Not present	
- Duration	Not present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS 34.122 clause C.2.1 Parameter Set	
- TFCI coding	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Puncturing Limit	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Repetition Period	1	
- Repetition Length	1	
- Uplink DPCH timeslots and codes		
- Dynamic SF usage	TRUE	
- Timeslot number	The number of an uplink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	3.84 Mcps	
- CHOICE Burst Type	Reference to TS 34.122 clause C.2.1 Parameter Set	
- Midamble Allocation Mode	Default	
- Midamble configuration	Choose lowest possible Kcell value given burst type	
- CHOICE TDD option	3.84 Mcps TDD	
- First timeslot Code List	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS 34.122 clause C.2.1 Parameter Set.	
- Channelisation code	(i/SF) where i denotes an unassigned code matching the SF specified in TS 34.122 clause C.2.1 Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon the number of resources specified in TS 34.122 clause C.2.1 Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove	Not present	
E-DCH Info		Rel-6
- MAC-es/e reset indicator	TRUE	
- CHOICE mode	TDD	
- CHOICE TDD mode	3.84 TDD	
- E-RUCCH info		
- E-RUCCH constant value	0dB	
- E-RUCCH persistance scaling	0.9	
- T-RUCCH	100ms	
- E-RUCCH timeslot number	Not Present	
- E-RUCCH midamble	Not Present	
- T-adv	Not Present	
- T-SCHED	Not Present	
- CHOICE TDD option	3.84Mcps TDD	
- CHOICE SF	Not present	
- E-PUCH info		
- E-TFCS information		
- Reference Beta Information QPSK list	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference Code Rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference beta	Reference to TS34.122 Annex C Fixed Reference	

Information Element	Value/remark	Version
	Channels	
- Reference Beta Information 16QAM list	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference Code Rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference beta	Reference to TS34.122 Annex C Fixed Reference Channels	
- CHOICE TDD mode	3.84Mcps TDD	
- $N_{E\text{-}UCCH}$	Not Present	
- E-PUCH constant value	0dB	
- E-PUCH TS configuration list	Reference to TS34.122 Annex C Fixed Reference Channels	
- TS number	Reference to TS34.122 Annex C Fixed Reference Channels	
- CHOICE *Burst Type*	Reference to TS34.122 Annex C Fixed Reference Channels	
- Midmable configuration	Reference to TS34.122 Annex C Fixed Reference Channels	
- E-PUCH code hopping	TRUE	
- E-PUCH TPC step size	1dB	
- Minimum allowed code rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Maximum allowed code rate	Reference to TS34.122 Annex C Fixed Reference Channels	
Downlink HS-PDSCH Information		Rel-5
- HS-SCCH Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	3.84 Mcps TDD	Rel-5
- Ack-Nack Power Offset	0dB	Rel-5
- HS-SICH Power Control Info		Rel-5
- UL SIR target	0dB	Rel-5
- HS-SICH Constant Value	-10dB	Rel-5
- $D_{hs\text{-}sync}$	Not present	Rel-6
- HS-SCCH Set Configuration	4	Rel-5
- Timeslot number	The timeslot in which HS-SCCH is to be configured	Rel-5
- Channelisation code	CC16/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- BLER target	-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	Rel-5
- HS-SICH configuration		
- Timeslot number	The timeslot in which HS-SICH has beeen configured	Rel-5
- Channelisation code	CC16/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- Measurement Feedback Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	3.84 Mcps TDD	Rel-5
- HS-PDSCH Timeslot Configuration		Rel-5
- HS-PDSCH Timeslot Configuration List	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Timeslot Number	The timeslot(s) in which HS-HS-DSCH is to configured	Rel-5
- CHOICE Burst Type	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Midamble Allocation Mode	Default	Rel-5
- Midamble configuration burst type 1 and 3	8	Rel-5
Downlink information common for all radio links	Not Present	
Downlink information per radio link list	1	
- Downlink information for each radio link		
- Choice mode	TDD	
- Primary CCPCH info		
- Choice mode	TDD	
- CHOICE TDD option	3.84 Mcps TDD	

Information Element	Value/remark	Version
- CHOICE SyncCase	Sync Case 1	
- Timeslot	Set to Timeslot containing PCCPCH	
- Cell parameters ID	10	
- SCTD indicator	FALSE	
- CHOICE DPCH info	Downlink DPCH info for each RL	

Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA) (7.68Mcps TDD)

Information Element	Value/remark	Version
Message Type		
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info		
- message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC message sequence number	SS provides the value of this IE, from its internal counter.	
Integrity protection mode info	Not Present	
Ciphering mode info	Not Present	
Activation time	Not Present	
New U-RNTI	Not Present	
New C-RNTI	Not Present	
New H-RNTI	'1010 1010 1010 1010'	Rel-5
New Primary E-RNTI	'1010 1010 1010 1010'	Rel-6
New Secondary E-RNTI	Not Present	Rel-6
RRC State indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	Not Present	
CN information info	Not Present	
URA identity	Not Present	
CHOICE specification mode	Complete specification	Rel-6
Signalling RB information to setup	Not Present	
RAB information for setup list		
- RAB information for setup		
- RAB info	(high-speed UM DTCH for PS domain)	
- RAB identity	0000 0110B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity.	
- CN domain identity	PS domain	
- NAS Synchronization Indicator	Not Present	
- Re-establishment timer	UseT315	
- RB information to setup		
- RB identity	25	
- PDCP info	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- DL UM RLC LI size	7	Rel-5
- One sided RLC re-establishment	FALSE	Rel-5
- RB mapping info		
- Information for each multiplexing option	1 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Downlink RLC logical channel info		
- Number of downlink RLC logical channels	1	
- Downlink transport channel type	HS-DSCH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- DL HS-DSCH MAC-d flow identity	0	
- Logical channel identity	Not Present	
RB information to reconfigure list	Not Present	Rel-6
RB information to be affected list	Not Present	
Downlink counter synchronization info	Not Present	
PDCP ROHC target mode	Not Present	Rel-5
UL Transport channel information for all transport channels	Not Present	
Deleted UL TrCH information list	Not Present	

Information Element	Value/remark	Version
Added or Reconfigured TrCH information list		
- Uplink transport channel type	E-DCH	
- CHOICE UL parameters	E-DCH	
- CHOICE mode	TDD	
- HARQ info for E-DCH		
- CHOICE UL parameters	E-DCH	
- HARQ RV Configuration	rvtable	
- Added or reconfigured E-DCH MAC-d flow		
- E-DCH MAC-d flow identity	2	
- E-DCH MAC-d flow power offset	0	
- E-DCH MAC-d flow maximum number of retransmissions	7	
- E-DCH MAC-d flow multiplexing list	Not Present	
- CHOICE transmission grant type	Scheduled grant info	
CHOICE mode	Not Present	R99 and Rel-4 only
DL Transport channel information common for all transport channel	Not Present	
Deleted DL TrCH information	Not Present	
Added or Reconfigured DL TrCH information list	1 TrCHs added	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	HS-DSCH	Rel-5
- DL Transport channel identity	Not Present	
- CHOICE DL parameters	HS-DSCH	Rel-5
- HARQ Info		Rel-5
- Number of Processes	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- CHOICE *Memory Partitioning*	Explicit	Rel-5
- Memory size	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- Process Memory Size	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- Added or reconfigured MAC-d flow		Rel-5
- MAC-hs queue to add or reconfigure list	(one queue)	Rel-5
- MAC-hs queue Id	0	Rel-5
- MAC-d Flow Identity	0	Rel-5
- T1	50	Rel-5
- MAC-hs window size	16	Rel-5
- MAC-d PDU size Info		Rel-5
- MAC-d PDU size	Reference to TS34.122 Annex C Fixed Reference Channels	Rel-5
- MAC-d PDU size index	0	Rel-5
- MAC-hs queue to delete list	Not present	Rel-5
- DCH quality target	Not present	
Frequency info	Not Present	
Maximum allowed UL TX power	30dBm	
CHOICE channel requirement	Uplink DPCH info	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- UL target SIR	Not present	
- CHOICE UL OL PC info	Broadcast UL OL PC info	
- CHOICE mode	TDD	
- Uplink Timing Advance Control		
- CHOICE Timing Advance	Enabled	
- CHOICE TDD option	7.68 Mcps TDD	
- UL Timing Advance	Determined by observed timing deviation of the RACH at the node B	
- UL CCTrCH List	1 CCTrCh	
- TFCS Id	1	
- UL target SIR	+20dB	
- Activation time	Not present	
- Duration	Not present	
- Common timeslot info		
- 2nd interleaving mode	Reference to TS34.122 Annex C Fixed Reference Channels	

Information Element	Value/remark	Version
- TFCI coding	Reference to TS34.122 Annex C Fixed Reference Channels	
- Puncturing Limit	Reference to TS34.122 Annex C Fixed Reference Channels	
- Repetition Period	1	
- Repetition Length	1	
- Uplink DPCH timeslots and codes		
- Dynamic SF usage	TRUE	
- Timeslot number	The number of an uplink timeslot that has unassigned codes.	
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	7.68 Mcps	
- CHOICE Burst Type	Reference to TS34.122 Annex C Fixed Reference Channels	
- Midamble Allocation Mode	Default	
- Midamble configuration	Choose lowest possible Kcell value given burst type	
- CHOICE TDD option	7.68 Mcps TDD	
- First timeslot Code List	Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS 34.122 clause C Parameter Set.	
- Channelisation code	(i/SF) where i denotes an unassigned code matching the SF specified in TS 34.122 clause C Parameter Set.	
- CHOICE more timeslots	The presence of this IE depends upon the number of resources specified in TS 34.122 clause C Parameter Set and the number of slots in which they are being assigned.	
- UL CCTrCH List to Remove	Not present	
E-DCH Info		Rel-6
- MAC-es/e reset indicator	TRUE	
- CHOICE mode	TDD	
- CHOICE TDD mode	7.68 TDD	
- E-RUCCH info		
- E-RUCCH constant value	0dB	
- E-RUCCH persistance scaling	0.9	
- T-RUCCH	100ms	
- E-RUCCH timeslot number	Not Present	
- E-RUCCH midamble	Not Present	
- T-adv	Not Present	
- T-SCHED	Not Present	
- CHOICE TDD option	7.68Mcps TDD	
- CHOICE SF	Not present	
- E-PUCH info		
- E-TFCS information		
- Reference Beta Information QPSK list	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference Code Rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference beta	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference Beta Information 16QAM list	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference Code Rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Reference beta	Reference to TS34.122 Annex C Fixed Reference Channels	
- CHOICE TDD mode	7.68Mcps TDD	
- $N_{E\text{-}UCCH}$	Not Present	
- E-PUCH constant value	0dB	
- E-PUCH TS configuration list	Reference to TS34.122 Annex C Fixed Reference Channels	
- TS number	Reference to TS34.122 Annex C Fixed Reference Channels	
- CHOICE *Burst Type*	Reference to TS34.122 Annex C Fixed Reference	

Information Element	Value/remark	Version
	Channels	
- Midmable configuration	Reference to TS34.122 Annex C Fixed Reference Channels	
- E-PUCH code hopping	TRUE	
- E-PUCH TPC step size	1dB	
- Minimum allowed code rate	Reference to TS34.122 Annex C Fixed Reference Channels	
- Maximum allowed code rate	Reference to TS34.122 Annex C Fixed Reference Channels	
Downlink HS-PDSCH Information		Rel-5
- HS-SCCH Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	7.68 Mcps TDD	Rel-5
- Ack-Nack Power Offset	0dB	Rel-5
- HS-SICH Power Control Info		Rel-5
- UL SIR target	0dB	Rel-5
- HS-SICH Constant Value	-10dB	Rel-5
- $D_{hs\text{-}sync}$	Not present	Rel-6
- HS-SCCH Set Configuration	4	Rel-5
- Timeslot number	The timeslot in which HS-SCCH is to be configured	Rel-5
- Channelisation code	CC32/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- BLER target	-2.4 (note that this equates to a BLER target of 0.4%, log10(0.004) = -2.4)	Rel-5
- HS-SICH configuration		
- Timeslot number	The timeslot in which HS-SICH has beeen configured	Rel-5
- Channelisation code	CC32/x where x is a previously unassigned channelisation code in this TS	Rel-5
- Midamble Allocation mode	Default	Rel-5
- Midamble configuration	8	Rel-5
- Measurement Feedback Info		Rel-5
- CHOICE mode	TDD	Rel-5
- CHOICE TDD option	7.68 Mcps TDD	Rel-5
- HS-PDSCH Timeslot Configuration		Rel-5
- HS-PDSCH Timeslot Configuration List	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Timeslot Number	The timeslot(s) in which HS-HS-DSCH is to configured	Rel-5
- CHOICE Burst Type	Reference to TS 34.122 clause C.4.1 Parameter Set	Rel-5
- Midamble Allocation Mode	Default	Rel-5
- Midamble configuration burst type 1 and 3	8	Rel-5
Downlink information common for all radio links	Not Present	
Downlink information per radio link list	1	
- Downlink information for each radio link		
- Choice mode	TDD	
- Primary CCPCH info		
- Choice mode	TDD	
- CHOICE TDD option	7.68 Mcps TDD	
- CHOICE SyncCase	Sync Case 1	
- Timeslot	Set to Timeslot containing PCCPCH	
- Cell parameters ID	10	
- SCTD indicator	FALSE	
- CHOICE DPCH info	Downlink DPCH info for each RL	

Contents of RRC CONNECTION RELEASE message: UM

Information Element	Value/remark	Version
Message Type		
U-RNTI	This IE is set to the following value when the message is transmitted on the DCCCH. When transmitted on CDCCH, this is absent.	R99, Rel-4
- SRNC identity	0000 0000 0001B	

- S-RNTI	0000 0000 0000 0000 0001B	
CHOICE identity type	This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent.	Rel-5
- U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
- Group identity	[FFS]	
- Group release information	[FFS]	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Integrity check info	This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted.	
- Message authentication code	SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.	
- RRC Message sequence number	SS provides the value of this IE, from its internal counter.	
N308	2 (for CELL_DCH state). Not Present (for UE in other connected mode states).	
Release cause	Normal event	
Rplmn information	Not Present	

Contents of RRC CONNECTION SETUP message: UM (3.84 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	Not Present	
RRC State Indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup list	4 SRBs	
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	

Information Element	Value/remark	Version
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	

Information Element	Value/remark	Version
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
-UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		

Information Element	Value/remark	Version
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE Mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS ID	(This IE is repeated for TFC number.)	
- UL TFCS		
- TFC subset	Default value is the complete existing set of transport format combinations	
- Allowed Transport Format combination	0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS	(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		

Information Element	Value/remark	Version
- TFCS complete reconfigure information		
- CHOICE TFCS Size	Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured UL TrCH information list	1	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	According to clause 6	
- Number of TBs and TTI List	(This IE is repeated for TFI number)	
- CHOICE mode	TDD	
- Transmission Time Interval	According to clause 6	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- CHOICE DL parameters	Same as UL	
Added or Reconfigured DL TrCH information list	1	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL TrCH Identity	5	
- DCH quality target		
- BLER Quality value	Reference to the present document	
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present	
CHOICE channel requirement	Uplink DPCH info	
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE *TDD option*	3.84 Mcps	
- UL target SIR	Reference to clause 6 Parameter set	
- CHOICE mode	TDD	
- CHOICE *UL OL PC info*	Individually signalled	
- CHOICE *TDD option*	3.84 Mcps	
- Individual timeslot interference info	Not Present	
- Individual timeslot interference		
- DPCH Constant Value		
- Primary CCPCH Tx Power	Not Present	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to clause 6.10 Parameter Set	
- TFCI coding	Reference to clause 6.10 Parameter Set	
- Puncturing Limit	Reference to clause 6.10 Parameter Set	
- Repetition Period	Reference to clause 6.10 Parameter Set	
- Repetition Length	Reference to clause 6.10 Parameter Set	
- Uplink DPCH timeslots and codes	Default is to use the old timeslots and codes	
- CPCH SET Info	(no data)	R99 and Rel-4 only
Downlink information common for all radio links		
- Downlink DPCH info common for all RL		
- Timing Indication	Initialize	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		

Information Element	Value/remark	Version
- DPC mode	0 (single)	
- CHOICE mode	TDD	
- CHOICE TDD option	3.84 Mcps (no data)	
- Default DPCH Offset Value	Arbitrary set to value 0..306688 by step of 512	
Downlink information for per radio links list		
-Downlink information for each radio links		
- CHOICE mode	TDD	
- Primary CCPCH info		
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	PCCPCH timeslot	
- Cell parameters ID	0	
- SCTD indicator		
- Downlink DPCH info for each RL		
- CHOICE mode	TDD	
- DL CCTrCH List		
- TFCS ID	1	
- Time info		
- Activation time	(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration	infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to the present document	
- TFCI coding	TRUE	
- Puncturing limit	Reference to clause 6 Parameter set	
- Repetition period	1	
- Repetition length	Empty	
- Downlink DPCH timeslots and codes		
- CHOICE *more timeslots*		
- CHOICE TDD option	3.84 Mcps	
- Timeslot number	The number of a downlink timeslot that has unassigned codes in a frame.	
- Individual timeslot info		
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	3.84 Mcps	
-CHOICE Burst Type		
-Type 1		
-Midamble Allocation Mode	Default	
- Midamble configuration burst type 1 and 3	As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes		
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	Not Present	
-SCCPCH information for FACH	Not Present	R99 and Rel-4 only

Contents of RRC CONNECTION SETUP message: UM (1.28 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	Not Present	

Information Element	Value/remark	Version
RRC State Indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup list	4 SRBs	
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	

Information Element	Value/remark	Version
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	

Information Element	Value/remark	Version
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
-UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	

Information Element	Value/remark	Version
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE Mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS ID	(This IE is repeated for TFC number.)	
- UL TFCS		
- TFC subset	Default value is the complete existing set of transport format combinations	
- Allowed Transport Format combination	0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS	(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- TFCS complete reconfigure information		
- CHOICE TFCS Size	Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured UL TrCH information list	1	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	According to clause 6	
- Number of TBs and TTI List	(This IE is repeated for TFI number)	
- CHOICE mode	TDD	
- Transmission Time Interval	According to clause 6	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- CHOICE DL parameters	Same as UL	
Added or Reconfigured DL TrCH information list	1	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL TrCH Identity	5	
- DCH quality target		
- BLER Quality value	Reference to the present document	

Information Element	Value/remark	Version
Frequency info	Not Present	
Maximum allowed UL TX power	Not Present	
CHOICE channel requirement	Uplink DPCH info	
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE *TDD option*	1.28 Mcps	
- $PRX_{PDPCHdes}$	Reference to clause 6 Parameter set	
- CHOICE mode	TDD	
- CHOICE *UL OL PC info*	Individually signalled	
- CHOICE *TDD option*	1.28 Mcps	
- TPC step size	Not Present	
- Primary CCPCH Tx Power	Not Present	
- Primary CCPCH Tx Power	Not Present	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to clause 6 Parameter Set	
- TFCI coding	Reference to clause 6 Parameter Set	
- Puncturing Limit	Reference to clause 6 Parameter Set	
- Repetition Period	Reference to clause 6 Parameter Set	
- Repetition Length	Reference to clause 6 Parameter Set	
- Uplink DPCH timeslots and codes	Default is to use the old timeslots and codes	
- CPCH SET Info	(no data)	R99 and Rel-4 only
Downlink information common for all radio links		
- Downlink DPCH info common for all RL		
- Timing Indication	Initialize	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- DPC mode	0 (single)	
- CHOICE mode	TDD	
- CHOICE TDD option	1.28 Mcps	
- TSTD indicator	TRUE	
- Default DPCH Offset Value	Arbitrary set to value 0..306688 by step of 512	
Downlink information for per radio links list		
-Downlink information for each radio links		
- CHOICE mode	TDD	
- Primary CCPCH info		
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	PCCPCH timeslot	
- Cell parameters ID	0	
- SCTD indicator		
- Downlink DPCH info for each RL		
- CHOICE mode	TDD	
- DL CCTrCH List		
- TFCS ID	1	
- Time info		
- Activation time	(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration	infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to the present document	
- TFCI coding	TRUE	
- Puncturing limit	Reference to clause 6 Parameter set	
- Repetition period	1	
- Repetition length	Empty	
- Downlink DPCH timeslots and codes		
- CHOICE *more timeslots*		
- CHOICE TDD option	1.28 Mcps	
- Timeslot number	The number of a downlink timeslot that has unassigned codes in a subframe.	
- Individual timeslot info		
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	1.28 Mcps	

Information Element	Value/remark	Version
-CHOICE Burst Type		
-Midamble Allocation Mode	Default	
- Midamble configuration	As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes		
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	Not Present	
-SCCPCH information for FACH	Not Present	R99 and Rel-4 only

Contents of RRC CONNECTION SETUP message: UM (7.68 Mcps TDD)

Information Element	Value/remark	Version
Message Type		
Initial UE identity	Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message	
RRC transaction identifier	Arbitrarily selects an integer between 0 and 3	
Activation time	Not Present(Now)	
New U-RNTI		
- SRNC identity	0000 0000 0001B	
- S-RNTI	0000 0000 0000 0000 0001B	
New C-RNTI	Not Present	
New H-RNTI	Not Present	Rel-6
CHOICE mode	TDD	Rel-7
- New E-RNTI	Not Present	Rel-7
RRC State Indicator	CELL_DCH	
UTRAN DRX cycle length coefficient	9	
Capability update requirement		
- UE radio access FDD capability update requirement	FALSE	
- UE radio access TDD capability update requirement	TRUE	
- System specific capability update requirement list	GSM	
CHOICE *specification mode*	Complete specification	Rel-5
- Complete specification		Rel-5
- Signalling RB information to setup list	4 SRBs	
- Signalling RB information to setup	(UM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type	RLC info	
- CHOICE Uplink RLC mode	UM RLC	
- Transmission RLC discard	Not Present	
- CHOICE Downlink RLC mode	UM RLC	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	

Information Element	Value/remark	Version
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	1	
- CHOICE RLC size list	Configured	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	1	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	1	
- Signalling RB information to setup	(AM DCCH for RRC)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	2	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	2	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	2	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	

Information Element	Value/remark	Version
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	2	
- Signalling RB information to setup	(AM DCCH for NAS_DT High priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		
- SDU discard mode	No Discard	
- MAX_DAT	415	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
-UL Transport channel identity	5	
- Logical channel identity	3	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	3	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	3	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	3	
- Signalling RB information to setup	(AM DCCH for NAS_DT Low priority)	
- RB identity	Not Present	
- CHOICE RLC info type		
- RLC info		
- CHOICE Uplink RLC mode	AM RLC	
- Transmission RLC discard		

Information Element	Value/remark	Version
- SDU discard mode	No Discard	
- MAX_DAT	15	
- Transmission window size	128	
- Timer_RST	500	
- Max_RST	4	
- Polling info		
- Timer_poll_prohibit	200	
- Timer_poll	200	
- Poll_PDU	Not Present	
- Poll_SDU	1	
- Last transmission PDU poll	TRUE	
- Last retransmission PDU poll	TRUE	
- Poll_Windows	99	
- Timer_poll_periodic	Not Present	
- CHOICE Downlink RLC mode	AM RLC	
- In-sequence delivery	TRUE	
- Receiving window size	128	
- Downlink RLC status info		
- Timer_status_prohibit	200	
- Timer_EPC	Not Present	
- Missing PDU indicator	TRUE	
- Timer_STATUS_periodic	Not Present	
- RB mapping info		
- Information for each multiplexing option	2 RBMuxOptions	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- Logical channel identity	4	
- CHOICE RLC size list	Configured	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	DCH	
- DL DCH Transport channel identity	10	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
- RLC logical channel mapping indicator	Not Present	
- Number of RLC logical channels	1	
- Uplink transport channel type	RACH	
- UL Transport channel identity	Not Present	
- Logical channel identity	4	
- CHOICE RLC size list	Explicit List	
- RLC size index	Reference to clause 6 Parameter Set	
- MAC logical channel priority	4	
- Downlink RLC logical channel info		
- Number of RLC logical channels	1	
- Downlink transport channel type	FACH	
- DL DCH Transport channel identity	Not Present	
- DL DSCH Transport channel identity	Not Present	
- Logical channel identity	4	
UL Transport channel information for all transport channels		
- PRACH TFCS	Not Present	
- CHOICE Mode	TDD	
-Individual UL CCTrCH information		
- UL TFCS ID	(This IE is repeated for TFC number.)	
- UL TFCS		
- TFC subset	Default value is the complete existing set of transport format combinations	
- Allowed Transport Format combination	0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.)	
- PRACH TFCS	(This IE is repeated for TFC number.)	
- CHOICE TFCI signalling	Normal	
- TFCI Field 1 information		
- TFCS complete reconfigure information		

Information Element	Value/remark	Version
- CHOICE TFCS Size	Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set	
- CTFC information	Not Present	
- CHOICE mode	TDD	
- Individual UL CCTrCH information	Not Present	
Deleted TrCH information list	Not Present	
Added or Reconfigured UL TrCH information list	1	
- Added or Reconfigured UL TrCH information		
- Uplink transport channel type	DCH	
- UL Transport channel identity	5	
- TFS		
- CHOICE Transport channel type	Dedicated transport channels	
- Dynamic Transport Format Information		
- RLC size	According to clause 6	
- Number of TBs and TTI List	(This IE is repeated for TFI number)	
- CHOICE mode	TDD	
- Transmission Time Interval	According to clause 6	
- CHOICE Logical channel list	All	
- Semi-static Transport Format information		
DL Transport channel information common for all transport channel		
- SCCPCH TFCS	Not Present	
- CHOICE mode	TDD	
- CHOICE DL parameters	Same as UL	
Added or Reconfigured DL TrCH information list	1	
- Added or Reconfigured DL TrCH information		
- Downlink transport channel type	DCH	
- DL Transport channel identity	10	
- CHOICE DL parameters	Same as UL	
- Uplink transport channel type	DCH	
- UL TrCH Identity	5	
- DCH quality target		
- BLER Quality value	Reference to the present document	
Frequency info	Not Present	
DTX-DRX timing information	Not Present	Rel-7
DTX-DRX information	Not Present	Rel-7
HS-SCCH less information	Not Present	Rel-7
MIMO parameters	Not Present	Rel-7
Maximum allowed UL TX power	Not Present	
Uplink DPCH info		Rel-6
- Uplink DPCH power control info		
- CHOICE mode	TDD	
- CHOICE *TDD option*	7.68 Mcps	Rel-7
- UL target SIR	Reference to clause 6 Parameter set	
- CHOICE mode	TDD	
- CHOICE *UL OL PC info*	Individually signalled	
- CHOICE *TDD option*	7.68 Mcps	Rel-7
- Individual timeslot interference info	Not Present	
- Individual timeslot interference		
- DPCH Constant Value		
- Primary CCPCH Tx Power	Not Present	
- Time info		
- Activation time	(256+CFN-(CFN MOD 8 + 8))MOD 256	
- Duration	Infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to clause 6.11 Parameter Set	
- TFCI coding	Reference to clause 6.11 Parameter Set	
- Puncturing Limit	Reference to clause 6.11 Parameter Set	
- Repetition Period	Reference to clause 6.11 Parameter Set	
- Repetition Length	Reference to clause 6.11 Parameter Set	
- CHOICE TDD Option	7.68 Mcps	Rel-7
- Uplink DPCH timeslots and codes VHCR	Default is to use the old timeslots and codes	Rel-7
- CPCH SET Info	(no data)	R99 and Rel-4 only
Downlink information common for all radio links		

Information Element	Value/remark	Version
- Downlink DPCH info common for all RL		
- Timing Indication	Initialize	
- CFN-targetSFN frame offset	Not Present	
- Downlink DPCH power control information		
- DPC mode	0 (single)	
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps (no data)	Rel-7
- Default DPCH Offset Value	Not Present	
Downlink information for per radio links list		
-Downlink information for each radio links		
- CHOICE mode	TDD	
- Primary CCPCH info		
- CHOICE mode	TDD	
- CHOICE TDD option	7.68 Mcps	Rel-7
- CHOICE *SyncCase*	Sync Case 1	
- Timeslot	PCCPCH timeslot	
- Cell parameters ID	0	
- SCTD indicator		
- CHOICE DPCH info	Downlink DPCH info for each RL	Rel-6
- Downlink DPCH info for each RL		
- CHOICE mode	TDD	
- DL CCTrCH List		
- TFCS ID	1	
- Time info		
- Activation time	(256+CFN-(CFN mod 8 + 8))mod 256	
- Duration	infinite	
- Common timeslot info		
- 2nd interleaving mode	Reference to the present document	
- TFCI coding	TRUE	
- Puncturing limit	Reference to clause 6 Parameter set	
- Repetition period	1	
- Repetition length	Empty	
- Downlink DPCH timeslots and codes VHCR		Rel-7
- CHOICE *more timeslots*		
- CHOICE TDD option	7.68 Mcps	Rel-7
- Timeslot number	The number of a downlink timeslot that has unassigned codes in a frame.	
- Individual timeslot info		
- TFCI existence	TRUE	
- Midamble shift and burst type		
- CHOICE TDD option	7.68 Mcps	
-CHOICE Burst Type		
-Type 1		
-Midamble Allocation Mode	Default	
- Midamble configuration burst type 1 and 3	As defined in 3GPP TS 25.221 [28]	
- First timeslot channelisation codes		
- First channelisation code	(i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set..	
- Last channelisation code	(j/SF) where j is the highest numbered code that is being assigned in the slot.	
- CHOICE more timeslots	The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot.	
- UL CCTrCH TPC List	Not Present	
-SCCPCH information for FACH	Not Present	R99 and Rel-4 only

Contents of SECURITY MODE COMMAND message: AM

Information Element	Condition	Value/remark
Message Type	A1, A2	
RRC transaction identifier		Arbitrarily selects an integer between 0

Information Element	Condition	Value/remark
		and 3
Integrity check info		
- Message authentication code		Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I.
- RRC Message Sequence Number		Set to an arbitrarily selected integer between 0 and 15
Security capability		
- Ciphering algorithm capability		
- UEA0		If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE.
- UEA1		If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE.
- Spare		Spare 2-15 = FALSE
- Integrity protection algorithm capability		0000000000000010B (UIA1)
- UIA1		TRUE
- Spare		Spare 0 and Spare 2-15 = FALSE
Ciphering mode info		This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted.
- Ciphering mode command		Start/restart
- Ciphering algorithm		UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message.Use the same ciphering algorithm specified in "ciphering
- Ciphering activation time for DPCH		Not Present
- Radio bearer downlink ciphering activation time info		
- Radio bearer activation time		
- RB identity		1
- RLC sequence number		Current RLC SN
- RB identity		2
- RLC sequence number		Current RLC SN+3(or Calculated Value)
- RB identity		3
- RLC sequence number		Current RLC SN
- RB identity		4
- RLC sequence number		Current RLC SN
Integrity protection mode info		
- Integrity protection mode command		Start
- Downlink integrity protection activation info		Not Present
- Integrity protection algorithm		UIA1
- Integrity protection initialisation number		SS selects an arbitrary 32 bits number for FRESH
CN domain identity		CS or PS
UE system specific security capability	A1	Not Checked
UE system specific security capability	A2	
- Inter-RAT UE security capability		
- CHOICE *system*		GSM
- GSM security capability		The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message.

Condition	Explanation
A1	UE not supporting GSM
A2	UE supporting GSM

10 A-GPS GPS Scenarios and Assistance Data

10.1 General

This clause defines the GPS scenarios and assistance data IEs which shall be available for use as specified in all A-GPS test cases in 3GPP TS 34.171 [41] and 3GPP TS 34.123-1 [1].

The information elements are given with reference to 3GPP TS 25.331 [34], where the details are defined.

Clauses 10.2 and 10.3 list the assistance data IEs required for performance testing of UE-based mode detailed in 3GPP TS 34.171 [41], and clauses 10.4 and 10.5 list the assistance data available for performance testing of UE-assisted mode detailed in 3GPP TS 34.171 [41]. Clause 10.6 lists the values of the assistance data IE fields for performance testing detailed in 3GPP TS 34.171 [41].

Clause 10.7 details the GPS scenario and the values of the assistance data IE fields for signalling testing detailed in 3GPP TS 34.123-1 [1], clause 17.2.

The A-GPS minimum performance requirements are defined by assuming that all relevant and valid assistance data is received by the UE in order to perform GPS measurements and/or position calculation. This clause does not include nor consider delays occurring in the various signalling interfaces of the network.

10.1.1 Satellite constellations and assistance data for performance testing

The satellite constellations for performance testing shall consist of 24 satellites. Almanac assistance data shall be available for all these 24 satellites. At least 9 of the satellites shall be visible to the UE (that is above 5 degrees elevation with respect to the UE). Other assistance data shall be available for 9 of these visible satellites. In each test, signals are generated for only a sub-set of these satellites for which other assistance data is available. The number of satellites in this sub-set is specified in the test. The satellites in this sub-set shall all be above 15 degrees elevation with respect to the UE. The HDOP for the test shall be calculated using this sub-set of satellites. The selection of satellites for this sub-set shall be random and consistent with achieving the required HDOP for the test.

10.1.2 GPS Scenarios for performance testing

This section defines the GPS scenarios that shall be used for all Assisted GPS performance tests defined in TS 34.171 [41].

They have been selected to be consistent with achieving the required HDOP for the Test Cases and so that for each test instance the reference location shall change sufficiently such that the UE shall have to use the new assistance data.

The satellites to be simulated in each test case are specified in clause 10.1.2.5.

The viable running time during which the scenario maintains the required HDOP or HDOPs is given. Once this time has been reached the scenario shall be restarted from its nominal start time.

10.1.2.1 GPS Scenario #1

The following GPS scenario #1 shall be used during the TTFF tests defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #1 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 1 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is calculated as a random offset from the reference location using the method described in section 10.1.2.4. The reference location is: latitude: 33 degrees 45 minutes 0.019 seconds north, longitude: 84 degrees 23 minutes 0.011 seconds west, (Atlanta USA), height: = 300m.

Nominal start time: 22nd January 2005 (Saturday) 00:08:00.

Viable running time to maintain specified HDOP values: 19 minutes.

Visible satellites available for simulation and for which Assistance Data (other than Almanac) shall be generated: PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

10.1.2.2 GPS Scenario #2

The following GPS scenario #2 shall be used during the TTFF tests defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #2 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 2 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is calculated as a random offset from the reference location using the method described in section 10.1.2.4. The reference location is: latitude: 37 degrees 48 minutes 59.988 seconds south, longitude: 144 degrees 58 minutes 0.013 seconds east, (Melbourne Australia), height: = 100m.

Nominal start time: 22nd January 2004 (Thursday) 00:08:00.

Viable running time to maintain specified HDOP values: 19 minutes.

Visible satellites available for simulation and for which Assistance Data (other than Almanac) shall be generated: PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

10.1.2.3 GPS Scenario #3

The following GPS scenario #3 shall be used during the Moving Scenario and Periodic Update test case defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #3 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 3 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is given as a trajectory as shown in Figure 5.6.1 of TS 34.171 [41]. The reference location is at the centre of the trajectory and is at: latitude: 37 degrees 48 minutes 59.988 seconds south, longitude: 144 degrees 58 minutes 0.013 seconds east, (Melbourne Australia), height: = 100m.

Start time: 22nd January 2004 (Thursday) 00:08:00.

Start location: at the point between l_{11} and l_{12} in Figure 5.6.1 of TS 34.171 [41], going in a clock-wise direction.

Visible satellites available for simulation and for which Assistance Data (other than Almanac) shall be generated: PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31.

Viable running time to maintain specified HDOP values: 19 minutes.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

10.1.2.4 UE Location for TTFF test cases

This section defines the method for generating the random UE locations that are required to be used for the TTFF tests defined in TS 34.171 [41].

For every Test Instance in each TTFF test case, the UE location shall be randomly selected to be within 3 km of the Reference Location. The Altitude of the UE shall be randomly selected between 0 m to 500 m above WGS-84 reference ellipsoid. These values shall have uniform random distributions.

The UE location is calculated as an offset from the Reference Location.

10.1.2.4.1 UE Location Offset

The UE location offset shall be calculated by selecting the next pair of random numbers, representing a pair of latitude and longitude offsets in degrees, from a standard uniform random number generator, with the following properties:

The ranges of the latitude and longitude offsets values shall be such that when translated onto the surface of the earth they shall lie within a 3km radius circle, centred on the Reference location specified for the GPS scenario under consideration. For the purposes of this calculation make the following assumptions:

a) Over the 3km radius circle at the Reference location the earth is flat and the meridians and parallels form a rectangular grid

b) The earth is spherical with a radius of 6371141m (equal to the WGS 84 value at 35 degrees latitude)

The resolution used for the latitude and longitude offsets values shall be 90/2E23 for the latitude offset values and 360/2E24 for the longitude offset values, representing the coding resolution in degrees specified in TS23.032 [42].

10.1.2.4.2 UE Altitude

The UE altitude value shall be calculated by selecting the next random number from a standard uniform random number generator, in the range 0 to 500, representing meters. The resolution used for the random number shall be 1, representing 1 meter.

10.1.2.5 Satellites to be simulated in each test case

The satellites to be simulated in each test case have been selected in order to achieve the required HDOP for that test case.

Satellites to be simulated

Test case	PRNs GPS #1	PRNs GPS #2	PRNs GPS #3
Sensitivity Coarse Time Assistance	2, 6, 10, 17, 18, 21, 26, 29	3, 11, 14, 15, 22, 23, 25, 31	-
Sensitivity Fine Time Assistance	2, 6, 10, 17, 18, 21, 26, 29	3, 11, 14, 15, 22, 23, 25, 31	-
Nominal Accuracy	2, 6, 10, 17, 18, 21, 26, 29	3, 11, 14, 15, 22, 23, 25, 31	-
Dynamic Range	2, 6, 10, 17, 26, 29	3, 14, 15, 22, 25, 31	-
Multi-path Performance	2, 6, 17, 21, 26	3, 14, 15, 22, 25	-
Moving Scenario and Periodic Update Performance	-	-	3, 14, 15, 22, 25

10.2 Information elements required for normal UE based testing

The following A-GPS assistance data IEs and fields shall be present for each test. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

a) UE positioning GPS reference time IE. This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
UE positioning GPS reference time	
	GPS Week
	GPS TOW msec
	GPS TOW Assist
	SatID
	TLM Message
	TLM Reserved
	Alert
	Anti-Spoof

b) UE positioning GPS reference UE position IE. This information element is defined in clause 10.3.8.4c of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Reference Location	Ellipsoid point with Altitude and uncertainty ellipsoid

c) UE positioning GPS navigation model IE. This information element is defined in clause 10.3.7.94 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Navigation Model	All satellite information

d) UE positioning GPS ionospheric model IE. This information element is defined in clause 10.3.7.92 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Ionospheric Model	All

10.3 Information elements required for UE based Sensitivity Fine Time Assistance test case

The A-GPS assistance data IEs and fields that shall be present for the Sensitivity Fine Time Assistance test case shall be those specified in clause 10.2 with the following exception. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

UE positioning GPS reference time IE. This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
UE positioning GPS reference time	
	GPS Week
	GPS TOW msec
	UTRAN GPS reference time
	>UTRAN GPS timing of cell frames
	>CHOICE mode
	>>FDD
	>>>Primary CPICH Info
	>SFN
	SFN-TOW Uncertainty
	TUTRAN-GPS drift rate
	GPS TOW Assist
	SatID
	TLM Message
	TLM Reserved
	Alert
	Anti-Spoof

10.4 Information elements available for normal UE assisted testing

The following A-GPS assistance data IEs and fields shall be available for use in each test. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

a) UE positioning GPS reference time IE. This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
UE positioning GPS reference time	
	GPS Week
	GPS TOW msec
	GPS TOW Assist
	SatID
	TLM Message
	TLM Reserved
	Alert
	Anti-Spoof

b) UE positioning GPS reference UE position IE. This information element is defined in clause 10.3.8.4c of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Reference Location	Ellipsoid point with Altitude and uncertainty ellipsoid

c) UE positioning GPS almanac This information element is defined in clause 10.3.7.89 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
UE positioning GPS almanac	
	Almanac Reference Week
	All Satellite information

d) UE positioning GPS navigation model IE. This information element is defined in clause 10.3.7.94 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Navigation Model	All satellite information

e) UE positioning GPS acquisition assistance IE. This information element is defined in clause 10.3.7.88 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Acquisition Assistance	
	GPS TOW msec
	Satellite information
	>SatID
	>Doppler (0th order term)
	>Extra Doppler
	>>Doppler (1st order term)
	>>Doppler Uncertainty
	>Code Phase
	>Integer Code Phase
	>GPS Bit number
	>Code Phase Search Window
	>Azimuth and Elevation
	>> Azimuth
	>> Elevation

10.5 Information elements available for UE assisted Sensitivity Fine Time Assistance test case

The A-GPS assistance data IEs and fields that shall be available for use for the Sensitivity Fine Time Assistance test case shall be those specified in clause 10.4 with the following exceptions. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

a) UE positioning GPS reference time IE. This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
UE positioning GPS reference time	
	GPS Week
	GPS TOW msec
	UTRAN GPS reference time
	>UTRAN GPS timing of cell frames
	>CHOICE mode
	>>FDD
	>>>Primary CPICH Info
	>SFN

	SFN-TOW Uncertainty
	TUTRAN-GPS drift rate
	GPS TOW Assist
	SatID
	TLM Message
	TLM Reserved
	Alert
	Anti-Spoof

b) UE positioning GPS acquisition assistance IE. This information element is defined in clause 10.3.7.88 of 3GPP TS 25.331 [34].

Name of the IE	Fields of the IE
Acquisition Assistance	
	GPS TOW msec
	UTRAN GPS reference time
	>UTRAN GPS timing of cell frames
	>CHOICE mode
	>>FDD
	>>>Primary CPICH Info
	>SFN
	Satellite information
	>SatID
	>Doppler (0th order term)
	>Extra Doppler
	>>Doppler (1st order term)
	>>Doppler Uncertainty
	>Code Phase
	>Integer Code Phase
	>GPS Bit number
	>Code Phase Search Window
	>Azimuth and Elevation
	>> Azimuth
	>> Elevation

10.6 Contents of Information elements for performance testing

10.6.1 General

This section defines the assistance data values that shall be used for all Assisted GPS performance tests defined in TS 34.171 [41]. It is given for GPS scenarios #1, #2 and #3 where it is different for each scenario; otherwise it is marked "All" where the same value is used for all scenarios.

Where assistance data is required on a per-satellite basis, or where the values of the data also varies with time it is specified in comma-separated-variable files with suffixes XX in the GPS data perf zip file specified in annex C.2, where XX is 01, 02 and 03 for GPS scenarios #1, #2 and #3 respectively. These files specify the values to be used for each satellite, indexed by satellite PRN, and, where applicable, the values to be used indexed by both time and satellite PRN.

Assistance data that is marked as "time varying" is specified and used in 80 ms increments. Interpolation between these values shall not be used.

Assistance data Information Elements and fields that are not specified shall not be used.

10.6.2 IE Random Offset Values

This section defines the methods for generating the random offsets that are required to be applied to one or two assistance data IEs for certain tests defined in TS 34.171 [41].

10.6.2.1 GPS TOW msec

For every Test Instance in each TTFF test case, the IE GPS TOW msec shall have a random offset, relative to GPS system time, within the allowed error range of Coarse Time Assistance defined in the test case. This offset value shall have a uniform random distribution.

NOTE: For the Moving Scenario and Periodic Update Test Case the value of the IE GPS TOW msec shall be set to the nominal value, i.e. no offset shall be used.

The offset value shall be calculated by selecting the next random number from a standard uniform random number generator, in the range specified for the GPS Coarse Time assistance error range in the Test Requirements, Test parameters table for the test under consideration. The resolution used for the random number shall be 0.01, representing 10ms.

10.6.2.1 UTRAN GPS timing of cell frames

In addition, for every Fine Time Assistance Test Instance the IE UTRAN GPS timing of cell frames shall have a random offset, relative to the true value of the relationship between the two time references, within the allowed error range of Fine Time Assistance defined in the test case. This offset value shall have a uniform random distribution.

The offset value shall be calculated by selecting the next random number from a standard uniform random number generator with the following properties:

The range shall be the number of UMTS chips whose duration is less than the range specified for the GPS Fine Time assistance error range in the Test Requirements, Test parameters table for the test under consideration.

The resolution used for the random number shall be 1, representing 1 UMTS chip.

10.6.3 Assistance Data Reference Time

Contents of UE positioning GPS reference time IE

Reference Time (Fields occurring once per message)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
GPS Week	weeks	282	230	230
GPS TOW msec	msec	518880000. Start time. Add number of ms as required. (Note 1)	346080000. Start time. Add number of ms as required. (Note 1)	346080000. Start time. Add number of ms as required. (Note 1)
UTRAN GPS reference time		Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Absent
>UTRAN GPS timing of cell frames		Note 2	Note 2	-
>CHOICE mode		Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Present for Sensitivity Fine Time Assistance test case. Absent otherwise	-
>>FDD		-	-	-
>>>Primary CPICH Info		100	100	-
>SFN		Note 2	Note 2	-
SFN-TOW Uncertainty		lessThan10	lessThan10	-
TUTRAN-GPS drift rate		0	0	-

Note 1: GPS TOW msec

This is the value in ms of GPS TOW msec when the GPS scenario is initially started in the GPS simulator. For all TTFF test cases, each time a GPS scenario is used, the GPS start time shall be advanced by 120 seconds from the value last used so that, at the time the fix is made, it is at least 2 minutes later than the previous fix made with that scenario.

The actual value of GPS TOW msec to be used in the Reference Time IE (before the addition of the random offset, if applicable) shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value. The accuracy shall be such that the Maximum Test System Uncertainty for Coarse Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

For all TTFF test cases a random offset is then added to the value of GPS TOW msec as described in clause 10.6.2

Note 2: UTRAN GPS timing of cell frames and SFN

The values of UTRAN GPS timing of cell frames (before the addition of the random offset) and SFN shall be calculated at the time the IE is required. The accuracy of the relationship between the two fields shall be such that the Maximum Test System Uncertainty for Fine Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

A random offset is then added to the value of UTRAN GPS timing of cell frames as described in clause 10.6.2

Satellite Information

Parameter	Units	Value/remark GPS All
Number of satellites	---	9

Reference Time - GPS TOW Assist (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
SatID		PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31

Reference Time - GPS TOW Assist (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS All
TLM Message	Bit string	10922
TLM Reserved	Bit string	2
Alert	Boolean	0
Anti-Spoof	Boolean	1

10.6.4 Assistance Data Reference Position

Contents of UE positioning GPS reference UE position IE

The uncertainty of the semi-major axis is 3 km. The uncertainty of the semi-minor axis is 3 km. The orientation of the major axis is 0 degrees. The uncertainty of the altitude information is 500 m. The confidence factor is 68%.

Reference Position

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
Type of Shape	Bit field	Ellipsoid point with altitude and uncertainty Ellipsoid	Ellipsoid point with altitude and uncertainty Ellipsoid	Ellipsoid point with altitude and uncertainty Ellipsoid
Degrees of latitude	degrees	33.750005	-37.816663	-37.816663
Degrees of longitude	degrees	-84.383517	144.966670	144.966670
Altitude	m	+300	+100	+100
Uncertainty semi-major	m	3000	3000	3000
Uncertainty semi-minor	m	3000	3000	3000
Orientation of major axis	degrees	0	0	0
Uncertainty altitude	m	500	500	500
Confidence	%	68	68	68

10.6.5 Assistance Data Navigation Model

Contents of UE positioning GPS navigation model IE

Satellite Information

Parameter	Units	Value/remark GPS All
Number of satellites	---	9

Navigation Model (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
SatID	---	PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31
Satellite Status	Boolean	0 (Note)	0 (Note)	0 (Note)

NOTE: For consistency Satellite Status is also given in file: Navigation model XX.csv

Ephemeris and Clock Correction parameters (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS All
C/A or P on L2	Boolean	See file: Navigation model XX.csv
URA Index	Boolean	See file: Navigation model XX.csv
SV Health	Boolean	See file: Navigation model XX.csv
IODC	---	See file: Navigation model XX.csv
L2 P Data Flag	Boolean	See file: Navigation model XX.csv
SF 1 Reserved	---	See file: Navigation model XX.csv
T_{GD}	sec	See file: Navigation model XX.csv
t_{oc}	sec	See file: Navigation model XX.csv
af_2	sec/sec^2	See file: Navigation model XX.csv
af_1	sec/sec	See file: Navigation model XX.csv
af_0	sec	See file: Navigation model XX.csv
C_{rs}	meters	See file: Navigation model XX.csv
Δn	semi-circles/sec	See file: Navigation model XX.csv
M_0	semi-circles	See file: Navigation model XX.csv
C_{uc}	radians	See file: Navigation model XX.csv
e	---	See file: Navigation model XX.csv
C_{us}	radians	See file: Navigation model XX.csv
$(A)^{1/2}$	$meters^{1/2}$	See file: Navigation model XX.csv
t_{oe}	sec	See file: Navigation model XX.csv
Fit Interval Flag	Boolean	See file: Navigation model XX.csv
AODO	sec	See file: Navigation model XX.csv
C_{ic}	radians	See file: Navigation model XX.csv
$OMEGA_0$	semi-circles	See file: Navigation model XX.csv
C_{is}	radians	See file: Navigation model XX.csv
i_0	semi-circles	See file: Navigation model XX.csv
C_{rc}	meters	See file: Navigation model XX.csv
ω	semi-circles	See file: Navigation model XX.csv
OMEGAdot	semi-circles/sec	See file: Navigation model XX.csv
Idot	semi-circles/sec	See file: Navigation model XX.csv

10.6.6 Assistance Data Ionospheric Model

Contents of UE positioning GPS ionospheric model IE

Ionospheric Model

Parameter	Units	Value/remark GPS All
α_0	seconds	4.6566129 10E-9
α_1	sec/semi-circle	1.4901161 10E-8
α_2	$sec/(semi\text{-}circle)^2$	-5.96046 10E-8
α_3	$sec/(semi\text{-}circle)^3$	-5.96046 10E-8
β_0	seconds	79872
β_1	sec/semi-circle	65536
β_2	$sec/(semi\text{-}circle)^2$	-65536
β_3	$sec/(semi\text{-}circle)^3$	-393216

10.6.7 Assistance Data Almanac

Contents of UE positioning GPS almanac

Almanac (Field occurring once per message)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
WN_a	weeks	27	230	230

Satellite Information

Parameter	Units	Value/remark GPS All
Number of satellites	---	24

Almanac (Fields occurring once per satellite)

Parameter	Units	Value/remark
DataID	---	See file: Almanac XX.csv

Almanac (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
SatID	---	PRNs: 1, 2, 4, 5, 6, 7, 9, 10, 11, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 29, 30	PRNs: 1, 2, 3, 4, 5, 6, 7, 8, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 25, 27, 28, 30, 31	PRNs: 1, 2, 3, 4, 5, 6, 7, 8, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 25, 27, 28, 30, 31

Almanac (Fields occurring once per satellite)

Parameter	Units	Value/remark
e	dimensionless	See file: Almanac XX.csv
t_{oa}	sec	See file: Almanac XX.csv
δi	semi-circles	See file: Almanac XX.csv
OMEGADOT	semi-circles/sec	See file: Almanac XX.csv
SV Health	Boolean	See file: Almanac XX.csv
$A^{1/2}$	$meters^{1/2}$	See file: Almanac XX.csv
$OMEGA_0$	semi-circles	See file: Almanac XX.csv
M_0	semi-circles	See file: Almanac XX.csv
ω	semi-circles	See file: Almanac XX.csv
af_0	seconds	See file: Almanac XX.csv
af_1	sec/sec	See file: Almanac XX.csv

10.6.8 Assistance Data Acquisition Assistance

Contents of UE positioning GPS acquisition assistance IE

GPS Acquisition Assistance (Fields occurring once per message)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
GPS TOW msec	msec	51888000 ms. Start time. Add number of ms as required. (Note 1)	346080000 ms. Start time. Add number of ms as required. (Note 1)	346080000 ms. Start time. Add number of ms as required. (Note 1)
UTRAN GPS reference time		Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Absent
>UTRAN GPS timing of cell frames		Note 2	Note 2	-
>CHOICE mode		Present for Sensitivity Fine Time Assistance test case. Absent otherwise	Present for Sensitivity Fine Time Assistance test case. Absent otherwise	-
>>FDD		-	-	-
>>>Primary CPICH Info		100	100	-
>SFN		Note 2	Note 2	-

Note 1: GPS TOW msec

This is the value in ms of GPS TOW msec when the GPS scenario is initially started in the GPS simulator. For all TTFF test cases, each time a GPS scenario is used, the GPS start time shall be advanced by 120 seconds from the value last used so that, at the time the fix is made, it is at least 2 minutes later than the previous fix made with that scenario.

The actual value of GPS TOW msec to be used in the Acquisition Assistance IE (before the addition of the random offset, if applicable) shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value. The accuracy shall be such that the Maximum Test System Uncertainty for Coarse Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

For all TTFF test cases a random offset is then added to the value of GPS TOW msec as described in clause 10.6.2

This “final GPS TOW msec” value is then also used to determine the value of the Acquisition Assistance parameters marked as “Time varying” in clause 10.6.8

Note 2: UTRAN GPS timing of cell frames and SFN

The values of UTRAN GPS timing of cell frames (before the addition of the random offset) and SFN shall be calculated at the time the IE is required. The accuracy of the relationship between the two fields shall be such that the Maximum Test System Uncertainty for Fine Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

A random offset is then added to the value of UTRAN GPS timing of cell frames as described in clause 10.6.2

Satellite Information

Parameter	Units	Value/remark GPS All
Number of satellites	---	9

GPS Acquisition Assistance (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS #1	Value/remark GPS #2	Value/remark GPS #3
SatID	---	PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31	PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31

GPS Acquisition Assistance (Fields occurring once per satellite)

Parameter	Units	Value/remark GPS All
Doppler (0th order term)	Hz	Time varying. See file: Acquisition assist XX.csv (Note)
Doppler (1st order term)	Hz/sec	Time varying. See file: Acquisition assist XX.csv (Note)
Doppler Uncertainty	Hz	Time varying. See file: Acquisition assist XX.csv (Note)
Code Phase	chips	Time varying. See file: Acquisition assist XX.csv (Note)
Integer Code Phase	---	Time varying. See file: Acquisition assist XX.csv (Note)
GPS Bit number	---	Time varying. See file: Acquisition assist XX.csv (Note)
Code Phase Search Window	chips	Time varying. See file: Acquisition assist XX.csv (Note)
Azimuth	deg	Time varying. See file: Acquisition assist XX.csv (Note)
Elevation	deg	Time varying. See file: Acquisition assist XX.csv (Note)
Note: Acquisition Assistance parameters This field is “Time varying” and its value depends on the “final GPS TOW msec” as described in clause 10.6.8. The value of this field to be used shall be determined by taking the “final GPS TOW msec” value and selecting the nearest field value in the Acquisition assist.csv file corresponding to the value of “final current GPS TOW msec”.		

10.7 GPS Scenario and values of Information Elements for signalling testing

10.7.1 General

This clause defines the GPS scenario and the associated assistance data values that shall be used for all Assisted GPS signalling tests defined in 3GPP TS 34.123-1 [1] clause 17.2.

Where assistance data is required on a per-satellite basis, or where the values of the data also varies with time it is specified in comma-separated-variable files in the GPS data sig zip file specified in annex C.1. These files specify the values to be used for each satellite, indexed by satellite PRN, and, where applicable, the values to be used indexed by both time and satellite PRN.

Assistance data that is marked as "time varying", and the GPS TOW msec field are only specified and used in 1 second increments. Interpolation between these values shall not be used.

The accuracy of the GPS TOW msec and assistance data that is marked as "time varying" in the provided assistance data shall be within ±2 s relative to the GPS time in the system simulator.

Assistance data Information Elements and fields that are not specified shall not be used.

10.7.2 GPS Scenario

The following GPS scenario shall be used. The assistance data specified in the following clauses is consistent with this GPS scenario:

- Yuma Almanac data: see file Tokyo Yuma.txt in the GPS data sig zip file specified in annex C.1.
- UE location and Reference location: static at latitude: 35 degrees 40 minutes north, longitude: 139 degrees 45 minutes east, (Tokyo) height: = 50m.
- Start time: 12th September 2003 21:30:00.
- Visible satellites simulated: PRNs: 4, 6, 9, 10, 13, 22.
- Ionospheric model: see values in clause 10.7.6.

10.7.3 Assistance Data Reference Time

Contents of UE positioning GPS reference time IE

Reference Time (Fields occurring once per message)

Parameter	Units	Value/remark
GPS Week	weeks	211
GPS TOW msec	msec	509 400 s. Start time. Add integer number of 1 seconds as required (see note)
NOTE: This is the value in seconds of GPS TOW msec when the GPS scenario is started in the GPS simulator. The value of GPS TOW msec to be used in the Reference Time IE shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value, rounded up to the next 1 second interval.		

10.7.4 Assistance Data Reference Position

Contents of UE positioning GPS reference UE position IE

Reference Position

Parameter	Units	Value/remark
Type of Shape	Bit field	Ellipsoid point with altitude and uncertainty Ellipsoid
Degrees of latitude	degrees	+35.666667
Degrees of longitude	degrees	+139.75
Altitude	m	+50
Uncertainty semi-major	m	3 000
Uncertainty semi-minor	m	3 000
Orientation of major axis	degrees	0
Uncertainty altitude	m	500
Confidence	%	68

10.7.5 Assistance Data Navigation Model

Contents of UE positioning GPS navigation model IE

Satellite Information

Parameter	Units	Value/remark
Number of Satellites	---	6

Navigation Model (Fields occurring once per satellite)

Parameter	Units	Value/remark
SatID	---	PRNs: 4, 6, 9, 10, 13, 22.
Satellite Status	Boolean	0 (see note)
NOTE: For consistency Satellite Status is also given in file: Navigation model.csv.		

Ephemeris and Clock Correction parameters (Fields occurring once per satellite)

Parameter	Units	Value/remark
C/A or P on L2	Boolean	See file: Navigation model.csv
URA Index	Boolean	See file: Navigation model.csv
SV Health	Boolean	See file: Navigation model.csv
IODC	---	See file: Navigation model.csv
L2 P Data Flag	Boolean	See file: Navigation model.csv
SF 1 Reserved	---	See file: Navigation model.csv
T_{GD}	sec	See file: Navigation model.csv
t_{oc}	sec	See file: Navigation model.csv
af_2	sec/sec^2	See file: Navigation model.csv
af_1	sec/sec	See file: Navigation model.csv

af_0	sec	See file: Navigation model.csv
C_{rs}	meters	See file: Navigation model.csv
Δn	semi-circles/sec	See file: Navigation model.csv
M_0	semi-circles	See file: Navigation model.csv
C_{uc}	radians	See file: Navigation model.csv
e	---	See file: Navigation model.csv
C_{us}	radians	See file: Navigation model.csv
$(A)^{1/2}$	$meters^{1/2}$	See file: Navigation model.csv
t_{oe}	sec	See file: Navigation model.csv
Fit Interval Flag	Boolean	See file: Navigation model.csv
AODO	sec	See file: Navigation model.csv
C_{ic}	radians	See file: Navigation model.csv
$OMEGA_0$	semi-circles	See file: Navigation model.csv
C_{is}	radians	See file: Navigation model.csv
i_0	semi-circles	See file: Navigation model.csv
C_{rc}	meters	See file: Navigation model.csv
ω	semi-circles	See file: Navigation model.csv
OMEGAdot	semi-circles/sec	See file: Navigation model.csv
Idot	semi-circles/sec	See file: Navigation model.csv

10.7.6 Assistance Data Ionospheric Model

Contents of UE positioning GPS ionospheric model IE

Ionospheric Model

Parameter	Units	Value/remark
α_0	seconds	4.6566129 10E-9
α_1	sec/semi-circle	1.4901161 10E-8
α_2	$sec/(semi\text{-}circle)^2$	-5.96046 10E-8
α_3	$sec/(semi\text{-}circle)^3$	-5.96046 10E-8
β_0	seconds	79872
β_1	sec/semi-circle	65536
β_2	$sec/(semi\text{-}circle)^2$	-65536
β_3	$sec/(semi\text{-}circle)^3$	-393216

10.7.7 Assistance Data Almanac

Contents of UE positioning GPS almanac

Almanac (Field occurring once per message)

Parameter	Units	Value/remark
WN_a	weeks	212

Satellite Information

Parameter	Units	Value/remark
Number of Satellites	---	24

Almanac (Fields occurring once per satellite)

Parameter	Units	Value/remark
DataID	---	See file: Almanac.csv
SatID	---	PRNs: 1 to 24
e	dimensionless	See file: Almanac.csv

t_{oa}	sec	See file: Almanac.csv
δi	semi-circles	See file: Almanac.csv
OMEGADOT	semi-circles/sec	See file: Almanac.csv
SV Health	Boolean	See file: Almanac.csv
$A^{1/2}$	$meters^{1/2}$	See file: Almanac.csv
$OMEGA_0$	semi-circles	See file: Almanac.csv
M_0	semi-circles	See file: Almanac.csv
ω	semi-circles	See file: Almanac.csv
af_0	seconds	See file: Almanac.csv
af_1	sec/sec	See file: Almanac.csv

10.7.8 Assistance Data Acquisition Assistance

Contents of UE positioning GPS acquisition assistance IE

GPS Acquisition Assist (Field occurring once per message)

Parameter	Units	Value/remark
GPS TOW msec	msec	509 400 s. Start time. Add integer number of 1 seconds as required (see note)
NOTE: This is the value in seconds of GPS TOW msec when the GPS scenario is started in the GPS simulator. The value of GPS TOW msec to be used in the Acquisition Assistance IE shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value, rounded up to the next 1 second interval. This "current GPS TOW msec" is then also used to determine the value of the Acquisition Assistance parameters marked as "Time varying" in clause 10.7.8.		

Satellite Information

Parameter	Units	Value/remark
Number of Satellites	---	6

GPS Acquisition Assist (Fields occurring once per satellite)

Parameter	Units	Value/remark
SatID	---	PRNs: 4, 6, 9, 10, 13, 22.
Doppler (0th order term)	Hz	Time varying. See file: Acquisition assist .csv (see note)
Doppler (1st order term)	Hz/sec	Time varying. See file: Acquisition assist .csv (see note)
Doppler Uncertainty	Hz	Time varying. See file: Acquisition assist .csv (see note)
Code Phase	chips	Time varying. See file: Acquisition assist .csv (see note)
Integer Code Phase	---	Time varying. See file: Acquisition assist .csv (see note)
GPS Bit number	---	Time varying. See file: Acquisition assist .csv (see note)
Code Phase Search Window	chips	Time varying. See file: Acquisition assist .csv (see note)
Azimuth	deg	Time varying. See file: Acquisition assist .csv (see note)
Elevation	deg	Time varying. See file: Acquisition assist .csv (see note)
NOTE: This field is "Time varying" and its value depends on the "current GPS TOW msec" as described in clause 10.7.3. The value of this field to be used shall be determined by taking the "current GPS TOW msec" value and selecting the field value in the Acquisition assist.csv file corresponding to the value of "current GPS TOW msec".		

11 MBMS configurations for signalling test

Clause 11.1 specifies MCCH configurations for MBMS and MBSFN FDD mode. Clause 11.2 specifies MCCH configurations for MBSFN for 3.84 Mcps and 7.68 Mcps TDD

11.1 MCCH configurations

11.1.1 MCCH configuration parameters

MCCH is configured stand-alone on a separate SCCPCH for test. Four typical MCCH scheduling configurations are included in the clause. The MCCH RAB is found in 6.10.2.4.3.8.

11.1.1.1 Default1 MCCH information scheduling (mp 5.12s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	512 frames	5.12 s
Repetition period (rp)	128 frames	1.28 s
Access information period (aip)	64 frames	0.64 s
MCCH configuration (number of mp-rp-aip)	1-4-8	
MCCH data rate	7.6 kbs	
Max. total lengths of MCCH PER-encoded messages	1 k octets	1216 octets
Max. MBMS neighbouring cells	15	
Max. MBMS services	12	

11.1.1.2 Default2 MCCH information scheduling (mp 2.56s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	256 frames	2.56 s
Repetition period (rp)	128 frames	1.28 s
Access information period (aip)	64 frames	0.64 s
MCCH configuration (number of mp-rp-aip)	1-2-4	
MCCH data rate	7.6 kbs	
Max. total lengths of MCCH PER-encoded messages	1 k octets	1216 octets
Max. MBMS neighbouring cells	8	
Max. MBMS services	12	

11.1.1.3 Longest MCCH information scheduling (mp 10.24s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	1024 frames	10.24 s
Repetition period (rp)	256 frames	2.56 s
Access information period (aip)	128 frames	1.28 s
MCCH configuration (number of mp-rp-aip)	1-4-8	
MCCH data rate	7.6 kbs	
Max. total lengths of MCCH PER-encoded messages	2 k octets	2432 octets
Max. MBMS neighbouring cells	15	
Max. MBMS services	16	

11.1.1.4 Shortest MCCH information scheduling (mp 1.28s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	128 frames	1.28 s
Repetition period (rp)	64 frames	0.64 s
Access information period (aip)	16 frames	0.16 s
MCCH configuration (number of mp-rp-aip)	1-2-8	
MCCH data rate	7.6 kbs	
Max. total lengths of MCCH PER-encoded messages	0.5 k octets	608 octets
Max. MBMS neighbouring cells	8	
Max. MBMS services	12	

11.1.2 MCCH messages transmission in test

The clause provides rules for MCCH messages transmission for MBMS test.

A whole set of MCCH messages is repeatedly transmitted.

The sending of the whole set of critical MCCH information messages is started at the first TTI of a repetition period if no ACCESS INFORMATION message is sent.

No ACCESS INFORMATION messages shall be transmitted in the modification period unless explicitly mentioned in the Test Procedure or Expected Sequence.

In case an ACCESS INFORMATION message transmission is explicitly mentioned in the test procedure or expected sequence these shall be transmitted starting at the first frame of the second access information period of the modification period, after MBMS MODIFIED SERVICES INFORMATION, unless stated otherwise. If the test case procedure describes more than one ACCESS INFORMATION message in the same modification period, this means that an ACCESS INFORMATION message is transmitted in every acess info period from the first message until the end of the modification period, unless stated otherwise.

If an ACCESS INFORMATION message is to be sent in the access information period in a repetition period the critical MCCH messages are transmitted in the next frame after the ACCESS INFORMATION message.

If an ACCESS INFORMATION message is to be sent in the access information period that is not the first access information period in a repetition period,

> And if a critical MCCH message is segmented into several RLC PDUs with consecutive sequence numbers and is occasionally, only partially transmitted at the end of the preceding access information period, the remaining RLC PDUs shall be transmitted after ACCESS INFORMATION (as out of sequence delivery) in the next access information period within the repetition period.

If an MBMS service is changed, this will be notified on MICH during one entire modification period before the change occurs. The service should then appear in MBMS MODIFIED SERVICES INFORMATION for one modification period, and then in the next modification period move to MBMS UNMODIFIED SERVICES INFORMATION. The MBMS MODIFIED SERVICES INFORMATION message should be transmitted once per repetition period throughout the modification period. All MCCH messages will contain the same content during and after this service change, except for MBMS MODIFIED SERVICES INFORMATION and MBMS UNMODIFIED SERVICES INFORMATION, unless stated otherwise in the test procedure.

11.1.3 Combinations and transmission order of critical MCCH messages

Combination Id	Ordered message combinations	comment
C1	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION	No session ongoing or PTP session ongoing.
C2	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON RB INFORMATION + MBMS CURRENT CELL p-t-m RB INFORMATION	PTM sessions are ongoing, no service modification and no neighbouring cells are defined
C3	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON RB INFORMATION + MBMS CURRENT CELL p-t-m RB INFORMATION + MBMS NEIGHBOURING CELL p-t-m RB INFORMATION (per neighbouring cell)	PTM sessions are ongoing, no service modification
C4	MBMS MODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON RB INFORMATION + MBMS CURRENT CELL p-t-m RB INFORMATION + MBMS UNMODIFIED SERVICES INFORMATION	PTM sessions are ongoing or starting, service modification indicated, no neighbouring cells (for one modification period)
C5	MBMS MODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON RB INFORMATION + MBMS CURRENT CELL p-t-m RB INFORMATION + MBMS NEIGHBOURING CELL p-t-m RB INFORMATION (per neighbouring cell related to modified services) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS NEIGHBOURING CELL p-t-m RB INFORMATION (per neighbouring cell not related to modified services)	PTM sessions are ongoing or starting, service modification indicated (for one modification period)
C6	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION (empty services_list) + MBMS GENERAL INFORMATION	No MBMS services
C7	MBMS MODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS UNMODIFIED SERVICES INFORMATION	PTP sessions are starting or another required UE action. MBMS UNMODIFIED SERVICES INFORMATION may or may not contain services.

NOTE 1: PTM test cases shall use message combinations C2 and C4 by default, according to the rules in clause 11.1.2, unless stated otherwise. If MBMS neighbouring cells are configured in the PTM test case then combinations C3 and C5 shall be used unless stated otherwise. PTP and counting test cases shall use message combinations C6 and C7, unless stated otherwise.

NOTE 2: If combination C6 is used in the initial condition, then the list of services in MBMS UNMODIFIED SERVICES INFORMATION will be empty. Then if any service is modified this will be added to the list, so the list of services will grow during the test (e.g. C6->C4->C2 or C6->C5->C3 or C6->C7->C6). If combination C1 is used in the initial condition then a total of 12 services will always be included in MBMS UNMODIFIED SERVICES INFORMATION and MBMS MODIFIED SERVICES INFORMATION (e.g. C1->C4->C2 or C1->C5->C3 or C1->C7->C6).

11.2 MCCH configurations for MBSFN (3.84 Mcps and 7.68 Mcps TDD)

11.2.1 MCCH configuration parameters

MCCH is configured stand-alone on a separate SCCPCH for testing MBSFN. Four typical MCCH scheduling configurations are included in this clause. The MCCH RB is found in clause 6.10.3.4.4.12 (3.84 Mcps TDD) or 6.11.6.4.4.12 (7.68 Mcps TDD).

11.2.1.1 Default1 MCCH information scheduling (mp 5.12s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	512 frames	5.12s. Modification period coefficient = 9
Repetition period (rp)	128 frames	1.28s. Repetition period coefficient = 2
MCCH configuration (number of mp-rp)	1-4	
MCCH data rate	7.2 kbs	
Max. total lengths of MCCH PER-encoded messages	1 k octets	1152 octets

11.1.1.2 Default2 MCCH information scheduling (mp 2.56s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	256 frames	2.56s. Modification period coefficient = 8
Repetition period (rp)	128 frames	1.28s. Repetition period coefficient = 1
MCCH configuration (number of mp-rp)	1-2	
MCCH data rate	7.2 kbs	
Max. total lengths of MCCH PER-encoded messages	1 k octets	1152 octets

11.1.1.3 Longest MCCH information scheduling (mp 10.24s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	1024 frames	10.24s. Modification period coefficient = 10
Repetition period (rp)	256 frames	2.56s. Repetition period coefficient = 2
MCCH configuration (number of mp-rp)	1-4	
MCCH data rate	7.2 kbs	
Max. total lengths of MCCH PER-encoded messages	2 k octets	2304 octets

11.1.1.4 Shortest MCCH information scheduling (mp 1.28s)

MCCH configuration parameters	Values	Comments
Modification period (mp)	128 frames	1.28 s. Modification period coefficient = 7
Repetition period (rp)	64 frames	0.64 s. Repetition period coefficient = 1
MCCH configuration (number of mp-rp)	1-2	
MCCH data rate	7.2 kbs	
Max. total lengths of MCCH PER-encoded messages	0.5 k octets	576 octets

11.2.2 MCCH messages transmission in test

This clause provides rules for MCCH message transmission for MBMS testing on MBSFN clusters.

A whole set of MCCH messages is repeatedly transmitted.

The sending of the whole set of critical MCCH information messages is started at the first TTI of a repetition period.

If an MBMS service is changed, this will be notified on MICH during one entire modification period before the change occurs. The service should then appear in MBMS MODIFIED SERVICES INFORMATION for one modification period, and then in the next modification period move to MBMS UNMODIFIED SERVICES INFORMATION. The

MBMS MODIFIED SERVICES INFORMATION message should be transmitted once per repetition period throughout the modification period. All MCCH messages will contain the same content during and after this service change, except for MBMS MODIFIED SERVICES INFORMATION and MBMS UNMODIFIED SERVICES INFORMATION, unless stated otherwise in the test procedure.

11.2.3 Combinations and transmission order of critical MCCH messages

Combination Id	Ordered message combinations	comment
C1	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION	No session ongoing.
C2	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON P-T-M RB INFORMATION + MBMS CURRENT CELL P-T-M RB INFORMATION	PTM sessions are ongoing, no service modification
C3	Reserved	
C4	MBMS MODIFIED SERVICES INFORMATION + MBMS GENERAL INFORMATION + MBMS COMMON RB INFORMATION + MBMS CURRENT CELL P-T-M RB INFORMATION + MBMS UNMODIFIED SERVICES INFORMATION	PTM sessions are ongoing or starting, service modification indicated (for one modification period)
C5	Reserved	
C6	MBMS MODIFIED SERVICES INFORMATION (empty services_list) + MBMS UNMODIFIED SERVICES INFORMATION (empty services_list) + MBMS GENERAL INFORMATION	No MBMS services
C7	Reserved	

NOTE 1: MBSFN test cases shall use message combinations C2 and C4 by default, according to the rules in clause 11.2.2, unless stated otherwise.

NOTE 2: If combination C6 is used in the initial condition, then the list of services in MBMS UNMODIFIED SERVICES INFORMATION will be empty. Then if any service is modified this will be added to the list, so the list of services will grow during the test (e.g. C6->C4->C2). If combination C1 is used in the initial condition then a total of 8 services will always be included in MBMS UNMODIFIED SERVICES INFORMATION and MBMS MODIFIED SERVICES INFORMATION (e.g. C1->C4->C2).

11.2.4 MBSFN service availability

By default a total of 14 MBMS services are defined. However, in each cell only a selection of these services are available.

The default cell environment comprising Cell 31 to Cell 38 is configured to simulate four geographical service areas. In each service area there is one cell (cluster) on carrier frequency f1 (referred to as the Dedicated National Carrier) providing only nationally available services and another cell (cluster) on carrier frequency f2 (referred to as the Mixed Local/National Carrier) providing a mix of national and local services. By default all cells will provide notification of all services available in that service area. The default allocation of cells to carrier frequencies is defined by a combination of clause 6.1 and clause 5.1.2.

Cell 31, Cell 32, Cell 37 and Cell 38 will by default provide Dedicated National Carrier services. Cell 33, Cell 34, Cell 35 and Cell 36 will by default provide Mixed Local/National Carrier services. The default service availability in each cell is given in Table 11.2.4-1.

Table 11.2.4-1 - Default Service Availability in Cell 31 - Cell 38

MBMS Service ID	Cell 31	Cell 32	Cell 33	Cell 34	Cell 35	Cell 36	Cell 37	Cell 38	Comments
000001	X	X					X	X	National service 1
000002	X	X					X	X	National service 2
000003	X	X					X	X	National service 3
000004	X	X					X	X	National service 4
000005			X	X	X	X			National service 5
000006			X	X	X	X			National service 6
010001			X						Service Area 1 local service 1
010002			X						Service Area 1 local service 2
020001				X					Service Area 2 local service 1
020002				X					Service Area 2 local service 2
030001					X				Service Area 3 local service 1
030002					X				Service Area 3 local service 2
040001						X			Service Area 4 local service 1
040002						X			Service Area 4 local service 2

Annex A (informative): Void

Annex B (informative): Void

Annex C (normative): GPS Data Files

C.1 GPS data files for signalling tests

The GPS data files for use in signalling tests defined in 3GPP TS 34.123-1 [1] are contained in archive GPS Data Sig V2.zip which accompanies the present document.

C.2 GPS data files for performance tests

The GPS data files for use in performance tests defined in 3GPP TS 34.171 [41]are contained in archive GPS Data Perf V4.zip which accompanies the present document.

Annex D (informative): Change history

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
TP-08				Approval of the specification		2.0.0	3.0.0	
TP-09	TP-000131	001		RRC Message Contents: RLCSize	C	3.0.1	3.1.0	T1-000190
TP-09	TP-000131	002		RRC Message Contents: RLCParam	C	3.0.1	3.1.0	T1-000191
TP-09	TP-000131	003		RRC Message Contents: PCPreamble	C	3.0.1	3.1.0	T1-000192
TP-09	TP-000131	004		RRC Message Contents: RBIdentity	C	3.0.1	3.1.0	T1-000193
TP-09	TP-000131	005		RRC Message Contents: TrCHParam	C	3.0.1	3.1.0	T1-000194
TP-09	TP-000131	006		RRC Message Contents: UECapability	C	3.0.1	3.1.0	T1-000195
TP-09	TP-000131	007		RRC Message Contents: RBMapping	C	3.0.1	3.1.0	T1-000196
TP-09	TP-000131	008		RRC Message Contents: PagingCause	C	3.0.1	3.1.0	T1-000197
TP-09	TP-000131	009		RRC Message Contents: CipheringAndIntegrity	C	3.0.1	3.1.0	T1-000198
TP-09	TP-000131	010		RRC Message Contents: RLCInfo	C	3.0.1	3.1.0	T1-000199
TP-09	TP-000131	011		RRC Message Contents: CompressedMode	C	3.0.1	3.1.0	T1-000200
TP-09	TP-000131	012		RRC Message Contents: SIB	C	3.0.1	3.1.0	T1-000201
TP-09	TP-000131	013		RRC Message Contents: PhyCH	D	3.0.1	3.1.0	T1-000202
TP-09	TP-000131	014		RRC Message Contents: Measurement	C	3.0.1	3.1.0	T1-000203
TP-09	TP-000131	015		RRC Message Contents: TFCS	C	3.0.1	3.1.0	T1-000204
TP-09	TP-000131	016		RRC Message Contents: DPCHFrameOffset	C	3.0.1	3.1.0	T1-000205
TP-09	TP-000131	017		Test USIM Parameters	F	3.0.1	3.1.0	T1-000215
TP-09	TP-000131	018		Correction to definition of the test algorithm for authentication (clause 8.1.2)	F	3.0.1	3.1.0	T1-000164
TP-09	TP-000131	019		Reference Radio Bearer Configurations	F	3.0.1	3.1.0	T1-000212
TP-09	TP-000131	020		TDD Single mode	F	3.0.1	3.1.0	T1-000220
TP-10	TP-000215	021		Common generic procedure for AS testing	B	3.1.0	3.2.0	T1-000294
TP-10	TP-000215	022		Requirements for the system simulator for support of Tcell parameter	F	3.1.0	3.2.0	T1-000303
TP-10	TP-000215	023		Minimum Performance Levels	F	3.1.0	3.2.0	T1-000306
TP-10	TP-000215	024		Downlink signal conditions and propagation conditions	D	3.1.0	3.2.0	T1-000307
TP-10	TP-000215	025		Updating 34.108 v3.1.0 to TDD single mode	F	3.1.0	3.2.0	T1-000281
TP-10	TP-000215	026		Application of integrity mode protection to signalling message by default	F	3.1.0	3.2.0	T1-000296
TP-10	TP-000215	027		Updates to the default message contents in clause 9	C	3.1.0	3.2.0	T1-000282
TP-10	TP-000215	028		Updates to System Information Block (SIB) and Master Information Block (MIB) messages	C	3.1.0	3.2.0	T1-000283
TP-10	TP-000215	029		Application of ciphering during conformance testing	C	3.1.0	3.2.0	T1-000285
TP-10	TP-000215	030		Addition for System Information parameters (34.108 clause 6.1)	F	3.1.0	3.2.0	T1-000304
TP-10	TP-000215	031		Correction for Generic Setup Procedures (34.108 clause 7.2)	F	3.1.0	3.2.0	T1-000305
TP-11	TP-010018	032		Default radio conditions for multi-cell environment	F	3.2.0	3.3.0	T1-010078
TP-11	TP-010018	033		Correction for Generic Setup Procedures (34.108 clause 7.2)	F	3.2.0	3.3.0	T1-010079
TP-11	TP-010018	034		Corrections for Test USIM Parameters(34.108 clause 8)	F	3.2.0	3.3.0	T1-010080
TP-11	TP-010018	035		Correction of clause number in TS 34.108.	D	3.2.0	3.3.0	T1-010081
TP-11	TP-010018	036		Update of authentication test algorithm	C	3.2.0	3.3.0	T1-010082
TP-11	TP-010018	037		Updates to clause 9 of TS 34.108 v3.2.0	F	3.2.0	3.3.0	T1-010084
TP-11	TP-010018	038		Updating to TDD single mode	F	3.2.0	3.3.0	T1-010088
TP-11	TP-010018	039		Simulated network environments for TDD mode (SIB)	F	3.2.0	3.3.0	T1-010089
TP-12	TP-010118	040		Corrections to clause 6.10 FDD parameters	F	3.3.0	3.4.0	T1-010205
TP-12	TP-010118	041		Corrections to clause 6.10 TDD parameters	F	3.3.0	3.4.0	T1-010206
TP-12	TP-010118	042		Adding section for radio bearer configurations intended for functional testing	D	3.3.0	3.4.0	T1-010210
TP-12	TP-010118	043		Update of list of abbreviations	D	3.3.0	3.4.0	T1-010211
TP-12	TP-010118	044		Updates to clause 6.1 and 9	F	3.3.0	3.4.0	T1-010212
TP-12	TP-010118	045		Updates to clause 7.4	F	3.3.0	3.4.0	T1-010213
TP-12	TP-010118	046		clause 6.1: System Information Blocks for TDD Mode	F	3.3.0	3.4.0	T1-010214
TP-12	TP-010118	047		Editorial corrections and removal of a reference document	F	3.3.0	3.4.0	T1-010215
TP-13	TP-010215	048		Correction to reference	F	3.4.0	3.5.0	T1-010275
TP-13	TP-010215	049		Editorial modification for References	F	3.4.0	3.5.0	T1-010276
TP-13	TP-010215	050		Some corrections in clause 5	F	3.4.0	3.5.0	T1-010277
TP-13	TP-010215	051		Update to Scope Statement	F	3.4.0	3.5.0	T1-010278
TP-13	TP-010215	052		Clause 6.10 Definition of RB configurations, TDD parameters	F	3.4.0	3.5.0	T1-010279
TP-13	TP-010215	053		Updates to clause 6.1, clause 7.4 and clause 9	F	3.4.0	3.5.0	T1-010280
TP-13	TP-010215	054		Clause 6.1: Default radio conditions for Signalling tests	F	3.4.0	3.5.0	T1-010281

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
TP-13	TP-010215	055		Correction of Radio Bearer Configurations for FDD Mode	F	3.4.0	3.5.0	T1-010282
TP-13	TP-010215	056		Correction of Radio Bearer Configurations for TDD Mode	F	3.4.0	3.5.0	T1-010283
TP-13	TP-010215	057		Changes to Signalling Radio Bearer (SRB) numbering	F	3.4.0	3.5.0	T1-010284
TP-13	TP-010215	058		Missing bearers in tables 6.10.2.1.1 and 6.10.3.1.1	F	3.4.0	3.5.0	T1-010285
TP-13	TP-010215	059		Correction of system information block 5	F	3.4.0	3.5.0	T1-010286
TP-13	TP-010215	060		Introducing of 1.28 Mcps TDD Mode in clauses 4, 5 and 6	F	3.4.0	4.0.0	T1-010287
TP-13	TP-010215	061		Introduction of System Information Blocks for 1.28 Mcps TDD Mode	F	3.4.0	4.0.0	T1-010288
TP-13	TP-010215	062		Introduction of typical radio parameters for 1.28 McpsTDD	F	3.4.0	4.0.0	T1-010289
TP-13	TP-010215	063		Clause 6.11 RBs for RLC and PDCP testing	F	3.4.0	3.5.0	T1-010290
TP-14	TP-010285	065	1	Correction to 6.1 Contents of System Information Blocks	A	4.0.0	4.1.0	T1-010475
TP-14	TP-010285	067	1	Corrections to clause 6.1, 7.4 and 9	A	4.0.0	4.1.0	T1-010473
TP-14	TP-010258	069		Reference Radio Conditions	A	4.0.0	4.1.0	T1-010461
TP-14	TP-010258	071		Modification of Test procedures for RF tests	A	4.0.0	4.1.0	T1-010463
TP-14	TP-010258	073		Default message contents for RF tests	A	4.0.0	4.1.0	T1-010465
TP-14	TP-010258	075		Correction to 6.10 Reference Radio Bearer configurations	A	4.0.0	4.1.0	T1-010467
TP-14	TP-010258	077		Definition of default value of rate matching attribute	A	4.0.0	4.1.0	T1-010469
TP-14	TP-010258	079		Update of clause 7.4 and 6.10	A	4.0.0	4.1.0	T1-010471
TP-14	TP-010292	081		Correction on introduction of clause 6.10	A	4.0.0	4.1.0	--
TP-15	TP-020038	083		Replacement of Block STTD by Space Code Transmit Diversity (SCTD) (Rel-4)	A	4.1.0	4.2.0	T1-020092
TP-15	TP-020038	085		Update of reference radio conditions (Rel-4)	A	4.1.0	4.2.0	T1-020098
TP-15	TP-020038	087		Update of system reference configurations and default messages (Rel-4)	A	4.1.0	4.2.0	T1-020100
TP-15	TP-020038	089		Corrections to 34108-410	A	4.1.0	4.2.0	T1-020102
TP-15	TP-020038	091		Introduction of new Reference RABs (Rel-4)	A	4.1.0	4.2.0	T1-020195
TP-15	TP-020038	094		Update of SIBs for TDD (both modes) in TS 34.108 (Rel4)	F	4.1.0	4.2.0	T1-020107
TP-15	TP-020038	095		Clarification of bit rate of Interactive/Background PS RAB function (Rel-4)	A	4.1.0	4.2.0	T1-020184
				Correction of CR implementation errors in clauses: 6.10.2.2 and 6.10.2.4.1.58.2.1.1		4.2.0	4.2.1	
TP-16	TP-020141	108		Clause 7(reference) Update of generic setup procedures to use 13.6 kbps SRB in RRC connection establishment TDD (3.84 Mcps and 1.28 Mcps)	F	4.2.1	4.3.0	T1-020289
TP-16	TP-020141	109		Correction to clause 7.3.3.4 RADIO BEARER SETUP message	A	4.2.1	4.3.0	T1-020291
TP-16	TP-020141	110		Change of RM attribute of DL:3.4 kbps SRBs for DCCH in for REL4	A	4.2.1	4.3.0	T1-020292
TP-16	TP-020141	111		New additional RAB configuration (R1-020669) for REL4	A	4.2.1	4.3.0	T1-020293
TP-16	TP-020141	112		Correction of Puncturing Limit for RABs for REL4	A	4.2.1	4.3.0	T1-020294
TP-16	TP-020141	113		Test USIM	A	4.2.1	4.3.0	T1-020295
TP-16	TP-020141	114		Clause 6.1 (SIBs)Rel 4 (3.84 Mcps and 1.28 Mcps TDD)	F	4.2.1	4.3.0	T1-020296
TP-16	TP-020141	115		Clause 6.10 References for TDD about Clarification of bit rate of Interactive/Background PS RAB	A	4.2.1	4.3.0	T1-020297
TP-16	TP-020141	116		Correction to default message in clause 9 for Rel4	A	4.2.1	4.3.0	T1-020298
TP-16	TP-020141	117		Correction to clause 6.1 for Rel4	A	4.2.1	4.3.0	T1-020299
TP-16	TP-020141	118		WCDMA1800 additions for Rel4	A	4.2.1	4.3.0	T1-020300
TP-16	TP-020141	119		Clause 9.1 Default message contents for TDD (3.84 Mcps and 1.28 Mcps) R4	F	4.2.1	4.3.0	T1-020301
TP-16	TP-020141	121		Update of generic setup procedures to use 13.6 kbps SRB in RRC connection establishment	A	4.2.1	4.3.0	T1-020434
TP-17	TP-020184	123	-	Alignment of reference configurations on S-CCPCH with default system information messages	A	4.3.0	4.4.0	T1-020503
TP-17	TP-020184	125	-	Addition of reference compressed mode pattern	A	4.3.0	4.4.0	T1-020505
TP-17	TP-020184	127	-	Corrections to default message contents as T1S-020347rev1	A	4.3.0	4.4.0	T1-020507
TP-17	TP-020184	129	-	Additional default message contents for RF Testing	A	4.3.0	4.4.0	T1-020509
TP-17	TP-020184	131	-	Corrections related to SIB11, SIB12 and to the MEASUREMENT CONTROL message	A	4.3.0	4.4.0	T1-020527
TP-17	TP-020184	133	-	Corrections to clause 6.1 (T1S-020349rev1)	A	4.3.0	4.4.0	T1-020530
TP-17	TP-020184	135	-	Introduction of reference configurations on S-CCPCH and PRACH with two interactive PS domain RABs	A	4.3.0	4.4.0	T1-020539
TP-17	TP-020184	137	-	Removal of reference radio bearer configurations for unidirectional streaming CS RABa above 64 kbps	A	4.3.0	4.4.0	T1-020541

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
TP-17	TP-020184	140	-	Some corrections and updates in clause 6.1 for TDD mode	F	4.3.0	4.4.0	T1-020576
TP-17	TP-020184	142	-	Inclusion of default message contents for RF in clause 9.2 for TDD mode	F	4.3.0	4.4.0	T1-020578
TP-18	TP-020293	144	-	Correction to default messages in 9.1 and 9.2	A	4.4.0	4.5.0	T1-020658
TP-18	TP-020293	146	-	Corrections in the TDD test frequencies according to core specs	A	4.4.0	4.5.0	T1-020674
TP-18	TP-020293	148	-	Addition of alternative configuration using Turbo Coding for Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH	A	4.4.0	4.5.0	T1-020694
TP-18	TP-020293	150	-	Correction to content of clause 6.10.2.	A	4.4.0	4.5.0	T1-020709
TP-18	TP-020293	152	-	Correction to SIB 11/12 definition	A	4.4.0	4.5.0	T1-020712
TP-18	TP-020293	154	-	Reference Measurement Channels	A	4.4.0	4.5.0	T1-020768
TP-18	TP-020293	156	-	Transferring system information definition using ASN.1 description to PRD	A	4.4.0	4.5.0	T1-020778
TP-18	TP-020293	158	-	Correction to RLC RAB TFCS	A	4.4.0	4.5.0	T1-020780
TP-18	TP-020293	160	-	Default Message contents : Correction from CRs approved in RP17meeting	A	4.4.0	4.5.0	T1-020783
TP-18	TP-020293	162	-	Corrections to SIB1 to SIB6	A	4.4.0	4.5.0	T1-020799
TP-18	TP-020293	164	-	Correction to RAB configurations as revision of T1S020756	A	4.4.0	4.5.0	T1-020801
TP-18	TP-020293	166	-	Parameter addition for Reference RABs based on LS from RAN2	A	4.4.0	4.5.0	T1-020803
TP-18	TP-020293	168	-	Addition to clause 7.4 for multi call as T1S-020577rev2 (revision to T1S020820)	A	4.4.0	4.5.0	T1-020818
TP-18	TP-020293	169	-	RAB Combinations for IMS Services	F	4.4.0	4.5.0	T1-020819
TP-18	TP-020293	171	-	Correction to Contents of the Scheduling Block Syste Information in clause 6.1.3.	F	4.4.0	4.5.0	T1-020844
TP-19	TP-030044	173	-	RAB Removal from Rel 4 TS 34.108 as T1S030002rev1	A	4.5.0	4.6.0	T1-030037
TP-19	TP-030044	175	-	Combine all Radio Bearer Setup messages into one table	A	4.5.0	4.6.0	T1-030040
TP-19	TP-030044	177	-	Corrections to SB and SIB configurations in clause 6.1 as T1S030046rev1	A	4.5.0	4.6.0	T1-030042
TP-19	TP-030044	179	-	Correction to TS 34.108 Rel-4 ; PAGING TYPE1 message (Packet in PS)	A	4.5.0	4.6.0	T1-030044
TP-19	TP-030044	181	-	Clarification of autentication test algorithm and GSM cipher key	A	4.5.0	4.6.0	T1-030046
TP-19	TP-030044	183	-	Addition of simulated network environment for inter-RAT test cases	A	4.5.0	4.6.0	T1-030048
TP-19	TP-030044	185	-	Corrections to SIB1 to align with default values for LAC and RAC in 51.010-1.	A	4.5.0	4.6.0	T1-030050
TP-19	TP-030044	187	-	Addition of default inter-RAT handover messages	A	4.5.0	4.6.0	T1-030052
TP-19	TP-030044	189	-	Correction of activation time IEs in default messages	A	4.5.0	4.6.0	T1-030054
TP-19	TP-030044	191	-	Correction to default SECURITY MODE COMMAND message	A	4.5.0	4.6.0	T1-030056
TP-19	TP-030044	193	-	Addition of option for UL CM only in default reference CM patterns	A	4.5.0	4.6.0	T1-030058
TP-19	TP-030044	195	-	Introduction of a reference RB configuration for RMC for BTFD tests (Rel4)	A	4.5.0	4.6.0	T1-030060
TP-19	TP-030044	197	-	Update of the RRC connection request messages in 34.108 Rel4	A	4.5.0	4.6.0	T1-030063
TP-19	TP-030043	198	-	Introduction of Conversational PS RABs in Rel 4 TS 34.108 as T1S030003rev1	F	4.5.0	4.6.0	T1-030107
TP-19	TP-030043	200	-	Update of default parameters for 1 to 8 cell environments (TDD), clause 6.1.4, Rel 4	A	4.5.0	4.6.0	T1-030208
TP-19	TP-030043	202	-	Update of Multi-cell environment for default radio conditions (TDD), clause 6.1.6 (Inclusion of cell 4), Rel 4	A	4.5.0	4.6.0	T1-030210
TP-19	TP-030043	204	-	Modification to Generic Registration Procedures	A	4.5.0	4.6.0	T1-030222
TP-19	TP-030043	206	-	Update of default configurations to enable testing of low end UE	A	4.5.0	4.6.0	T1-030228
TP-20	TP-030098	208	-	Reinstate parameters for Interactive or background /UL:64 kbps / PS RAB	A	4.6.0	4.7.0	T1-030437
TP-20	TP-030098	210	-	Correction to Figure 7.4.1.1 (Rel-4)	A	4.6.0	4.7.0	T1-030483
TP-20	TP-030098	212	-	Update of SIB 11 and 12 in clause 6.1.0b in TS 34.108 (TDD)	A	4.6.0	4.7.0	T1-030507
TP-20	TP-030098	214	-	Update of Default parameters for 1 to 8 cell environments in TS 34.108 (TDD)	A	4.6.0	4.7.0	T1-030509
TP-20	TP-030098	216	-	Correction of default messages according to 25331 CR1823	A	4.6.0	4.7.0	T1-030632
TP-20	TP-030098	218	-	Clause 8.2: Definition of default values for authentication key K on test USIM	A	4.6.0	4.7.0	T1-030644
TP-20	TP-030098	219	-	Update of Reconfiguration messages	A	4.6.0	4.7.0	T1-030692

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
TP-20	TP-030098	221	-	Correction to RADIO BEARER RELEASE and RRC CONNECTION SETUP messages (Revision of T1-030569)	A	4.6.0	4.7.0	T1-030699
TP-20	TP-030140	226	-	Correction to default SIB5 (FDD)	A	4.6.0.	4.7.0	T1-030745
TP-21	TP-030191	228	-	CR to 34.108, Rel-4, Clarification of seg_count in 6.1.0a.3	A	4.7.0	4.8.0	T1-030827
TP-21	TP-030191	230	-	General correction in clause 7.4 for Common generic procedures for AS testing	A	4.7.0	4.8.0	T1-030976
TP-21	TP-030191	233	-	Incorrect activation time in CELL_FACH state .	A	4.7.0	4.8.0	T1-031064
TP-21	TP-030191	235	-	Incorrect Transport channel Parameters	A	4.7.0	4.8.0	T1-031066
TP-21	TP-030191	237	-	Corrections to TS 34.108 common procedures in clause 7.4 of Rel-4 of TS 34.108	A	4.7.0	4.8.0	T1-031095
TP-21	TP-030191	239	-	Removal of RLC AM in the Default Message Content	A	4.7.0	4.8.0	T1-031151
TP-21	TP-030191	242	-	CR 34.108 Rel-4: Manual attach in State 7 Registrated Idle Mode on CS/PS	A	4.7.0	4.8.0	T1-031175
TP-21	TP-030191	244	-	URA Identity in Cell Update Confirm and URA Update Confirm	A	4.7.0	4.8.0	T1-031179
TP-21	TP-030191	246	-	CR to 34.108 R4; Correction to specification to reflect a change already approved in TTCN CR T1-030396	A	4.7.0	4.8.0	T1-031241
TP-21	TP-030191	248	-	CR to 34.108 REL-4; Correction to clause 7.3 Test procedures for RF test	A	4.7.0	4.8.0	T1-031251
TP-21	TP-030191	240	-	RB configuration for the support of wideband AMR speech telephony services	F	4.7.0	4.8.0	T1-031154
TP-22	TP-030279	251	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031659
TP-22	TP-030279	252	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031660
TP-22	TP-030279	253	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031661
TP-22	TP-030279	254	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031662
TP-22	TP-030279	255	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031663
TP-22	TP-030279	256	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031664
TP-22	TP-030279	257	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031665
TP-22	TP-030279	258	1	Addition of Default message contents for TDD	F	4.8.0	4.9.0	T1-031666
TP-22	TP-030279	260	2	CR on PAGING TYPE 1, RRC CONNECTION REQUEST and RRC CONNECTION SETUP messages for MT RR Connection	A	4.8.0	4.9.0	T1-031596
TP-22	TP-030279	262		CR 34.108 Rel-4: EFRPLMNACT (RPLMN Last used Access Technology) removed	A	4.8.0	4.9.0	T1-031381
TP-22	TP-030279	264	1	Update of default messages for RRC CONNECTION SETUP and SECURITY MODE COMMAND	A	4.8.0	4.9.0	T1-031547
TP-22	TP-030279	266	1	Description and corrections of channels for minimum performance levels, TDD mode.	F	4.8.0	4.9.0	T1-031645
TP-22	TP-030279	268	1	Test frequencies of UMTS800MHz band VI	A	4.8.0	4.9.0	T1-031555
TP-22	TP-030279	269		CR 34.108 Rel-4: Addition of Bearer combination for Interactive/background UL 64 kbps DL 768 kbps for Rel-5	F	4.8.0	4.9.0	T1-031441
TP-22	TP-030279	271	1	Update of generic test procedure for TX, RX and Performance Requirement	A	4.8.0	4.9.0	T1-031610
TP-22	TP-030279	273	1	Introduction of generic test procedure for RRM handover test cases	A	4.8.0	4.9.0	T1-031608
TP-22	TP-030279	275	1	Correction of CM TGD parameter	A	4.8.0	4.9.0	T1-031591
TP-22	TP-030279	277	1	Corrections to default message contents of Radio Bearer Release	F	4.8.0	4.9.0	T1-031594
TP-22	TP-030279	279	1	Modification to default DPCCH_Power_offset value	A	4.8.0	4.9.0	T1-031598
TP-22	TP-030279	283		Correction of TFCS for radio bearer combination 6.10.2.4.1.51b	A	4.8.0	4.9.0	T1-031527
TP-23	TP-040037	284	-	New Radio Bearer Setup (FDD) message for RF (Revision of T1-040258)	F	4.9.0	4.10.0	T1-040417
TP-23	TP-040037	287	-	Corrections to default message contents of RRC Connection Setup message -> 2nd change not implemented (not implementable)	A	4.9.0	4.10.0	T1-040080
TP-23	TP-040037	289	-	Correction to Default parameters for Cells 1 to 8 in MultiPLMN cell environments - Rel-4	A	4.9.0	4.10.0	T1-040095
TP-23	TP-040037	291	-	Corrections to TDD HCR RABs	A	4.9.0	4.10.0	T1-040103
TP-23	TP-040037	296	-	LCR Corrections to TDD RABs merge of T1-040104 , T1-040201 and T1-040203	F	4.9.0	4.10.0	T1-040299
TP-23	TP-040037	298	-	Correction to handling of Entered Parameter IE in default contents for Initial Direct Transfer	A	4.9.0	4.10.0	T1-040411
TP-23	TP-040037	300	-	The diverse operation in TDD mode updating according to the core specification	A	4.9.0	4.10.0	T1-040368
TP-23	TP-040037	302	-	correction of measurement control default message contents for TDD -> Not implemented (not implementable)	F	4.9.0	4.10.0	T1-040370
TP-23	TP-040037	303	-	correction of RADIO BEARER SETUP default message	F	4.9.0	4.10.0	T1-040371

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
				contents for 1.28 Mcps TDD				
TP-23	TP-040037	304	-	Correction of RADIO BEARER RELEASE default message contents for TDD: AM or UM (1.28 Mcps TDD)	F	4.9.0	4.10.0	T1-040372
TP-23	TP-040037	305	-	Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD) -> Not implemented (not implementable)	F	4.9.0	4.10.0	T1-040373
TP-23	TP-040037	292	-	New I/B UL:64 DL:768 kbps PS RAB misplaced	F	4.10.0	5.0.0	T1-040109
TP-23	TP-040037	294	-	Generic setup procedure and default message contents for HSDPA (as of T1-040069rev1)	F	4.10.0	5.0.0	T1-040271
TP-23	TP-040037	295	-	Baseline radio bearer combination for HSDPA support	B	4.10.0	5.0.0	T1-040273
TP-24	TP-040112	308	-	Correction to IEs "START" and "ul_CounterSynchronisationInfo".	F	5.0.0	5.1.0	T1-040512
TP-24	TP-040112	309	-	Correction to HSDPA reference radio bearer configurations	F	5.0.0	5.1.0	T1-040522
TP-24	TP-040112	310	-	Addition of test procedure for HSDPA RF testing	F	5.0.0	5.1.0	T1-040546
TP-24	TP-040112	315	-	Corrections to default RRC messages	F	5.0.0	5.1.0	T1-040593
TP-24	TP-040112	318	-	Change of default LAC/RAC for inter-RAT test cases	A	5.0.0	5.1.0	T1-040656
TP-24	TP-040112	319	-	Contents of Physical channel Reconfiguration message modified to incorporate transition to URA_PCH or CELL_PCH	F	5.0.0	5.1.0	T1-040673
TP-24	TP-040112	320	-	Correction of reference test frequencies for UMTS800(band VI)	F	5.0.0	5.1.0	T1-040701
TP-24	TP-040112	325	-	Update of generic setup procedures in clauses 7.3.4 and 7.3.5.	A	5.0.0	5.1.0	T1-040754
TP-24	TP-040112	326	-	Physical channel parameters for AM RLC 7 bit Length Indicator TestCases (Rel-5)	F	5.0.0	5.1.0	T1-040902
TP-24	TP-040112	327	-	Corrections to the default contents of Security Mode Command (Rel-5)	F	5.0.0	5.1.0	T1-040903
TP-24	TP-040112	330	-	Corrections to Contents of Scheduling Block 1 (FDD)	F	5.0.0	5.1.0	T1-040909
TP-24	TP-040112	331	-	Corrections to Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM	F	5.0.0	5.1.0	T1-040911
TP-24	TP-040112	332	-	Corrections to Contents of RRC CONNECTION SETUP message: UM	F	5.0.0	5.1.0	T1-040913
TP-24	TP-040112	333	-	RADIO BEARER SETUP message (FDD) for Test Loop Mode2.	F	5.0.0	5.1.0	T1-040917
TP-24	TP-040112	335	-	Changes to establish one version of 34.108 covering all releases	A	5.0.0	5.1.0	T1-040931
TP-24	TP-040112	338	-	Addition of generic test procedure for AS test cases using the test loop	A	5.0.0	5.1.0	T1-040934
TP-24	TP-040112	339	-	Corrections to LCR TDD RABs	F	5.0.0	5.1.0	T1-040935
TP-25	TP-040157	343	-	Correction to generic test procedure in clause 7.4.2.6a.	F	5.1.0	5.2.0	T1-041040
TP-25	TP-040157	344	-	Addition of default messages for Signalling (FDD)	F	5.1.0	5.2.0	T1-041044
TP-25	TP-040157	345	-	Minor change to terminology in SRB tables of clause 6.10	F	5.1.0	5.2.0	T1-041140
TP-25	TP-040157	346	-	Default Message Content for System Information Block type 5 (FDD) and type 6 (FDD)	F	5.1.0	5.2.0	T1-041154
TP-25	TP-040157	347	-	Corrections to DCCH Transport channel Parameters for HSDPA RAB	D	5.1.0	5.2.0	T1-041171
TP-25	TP-040157	348	-	Corrections to clause 9	F	5.1.0	5.2.0	T1-041223
TP-25	TP-040157	349	-	Corrections to HCR TDD RAB combinations	F	5.1.0	5.2.0	T1-041235
TP-25	TP-040157	350	-	Adding missing clause 6.10.2.4.1.62.1	F	5.1.0	5.2.0	T1-041252
TP-25	TP-040157	351	-	Modification of AICH power offset in SysInfo 5 and 6.	F	5.1.0	5.2.0	T1-041253
TP-25	TP-040157	352	-	Correction to Default Message Content for Radio Bearer Setup Message.	F	5.1.0	5.2.0	T1-041259
TP-25	TP-040157	353	-	Correction to Default Message Content for Radio Bearer Reconfiguration Message for Condition A6	F	5.1.0	5.2.0	T1-041266
TP-25	TP-040157	354	-	CR to 34.108: introduction of default RB SETUP message from cell_FACH state for HSDPA	F	5.1.0	5.2.0	T1-041298
TP-25	TP-040157	355	-	Corrections to Contents of RADIO BEARER SETUP message: BTFD RMC	F	5.1.0	5.2.0	T1-041317
TP-25	TP-040157	340	-	Resolution of downlink code conflict between OCNS DPCH and S-CCPCH	F	5.1.0	5.2.0	T1-041327
TP-25	TP-040157	361	-	Correction to test procedure for test cases using Cell_PCH or URA_PCH state	F	5.1.0	5.2.0	T1-041346
TP-25	TP-040157	362	-	Removal of DCCH dummy transmission for RF testing	F	5.1.0	5.2.0	T1-041350
TP-25	TP-040157	341	-	Correct title to test procedure for test cases using Cell_PCH or URA_PCH state	F	5.1.0	5.2.0	T1-041354
TP-25	TP-040157	363	-	Addition of intra frequency cell to cell environments	F	5.1.0	5.2.0	T1-041356
TP-25	TP-040157	342	-	Correct primary scrambling code usage in default message contents in clause 9.2.1	F	5.1.0	5.2.0	T1-041365
TP-25	TP-040157	356	-	HSDPA downlink code allocation	F	5.1.0	5.2.0	T1-041374
TP-25	TP-040157	357	-	Correction to test procedure for test cases using	F	5.1.0	5.2.0	T1-041376

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
				CELL_FACH state				
TP-25	TP-040157	358	-	Varying DPCH Power Offset according to data transmission rate	F	5.1.0	5.2.0	T1-041416
TP-25	TP-040157	359	-	Corrections to default message for RADIO BEARER SETUP message in clause 9.2.1 (HSDPA RF)	F	5.1.0	5.2.0	T1-041418
TP-25	TP-040157	360	-	Test SIB schedule for two S-CCPCH or two PRACH in 34.108	F	5.1.0	5.2.0	T1-041422
TP-25	TP-040157	364	-	Correction to Default Message Content for Radio Bearer Setup Message re: RM Attribute values	F	5.1.0	5.2.0	T1-041433
TP-26	TP-040233	365	-	CR to 34.108 Rel-5:Correction to default value of Qrxlevmin	F	5.2.0	5.3.0	T1-041532
TP-26	TP-040233	366	-	CR to 34.108 Rel-5: Corrections of the values in 6.11.5.4 for LCR TDD	F	5.2.0	5.3.0	T1-041573
TP-26	TP-040233	367	-	Alignment of Prose to TTCN for SCH power level	F	5.2.0	5.3.0	T1-041584
TP-26	TP-040233	368	-	Addition of new HSDPA RAB configurations with UL 64 kbps	F	5.2.0	5.3.0	T1-041651
TP-26	TP-040233	369	-	Correction to initial conditions and references in clause 7.3	F	5.2.0	5.3.0	T1-041654
TP-26	TP-040233	370	-	Introduction of reference radio bearer combination for PS streaming and downlink rate up to 128 kbps	F	5.2.0	5.3.0	T1-041685
TP-26	TP-040233	371	-	Correction of clause 6.1 (Simulated network environment)	F	5.2.0	5.3.0	T1-041686
TP-26	TP-040233	372	-	Correction to generic Call Setup procedure for mobile terminating circuit switched calls	F	5.2.0	5.3.0	T1-041699
TP-26	TP-040233	373	-	CR to 34.108 Rel-5; Corrections to the default RADIO BEARER SETUP message for HSDPA	F	5.2.0	5.3.0	T1-041754
TP-26	TP-040233	374	-	Physical layer multiplexing configuration in case of AMR and two PS RABs	F	5.2.0	5.3.0	T1-041801
TP-26	TP-040233	375	-	Addition of new HSDPA RAB configurations	F	5.2.0	5.3.0	T1-041802
TP-26	TP-040233	376	-	Introduction of information for tests for Performance requirements for A-GPS.	B	5.2.0	5.3.0	T1-041850
TP-26	TP-040233	377	-	Introduction of UMTS-850 MHz band V	F	5.2.0	5.3.0	T1-041874
TP-26	TP-040233	378	-	CR to TS 34.108 Rel-5; Adding a new test condition for RADIO BEARER RELEASE Procedure (Revision of T1-041716).	F	5.2.0	5.3.0	T1-041933
TP-26	TP-040233	379	-	Update of Reference Radio Bearer for Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB for DL SF=256	F	5.2.0	5.3.0	T1-041942
TP-26	TP-040233	380	-	CR to 34.108: Correction to the maximum bit rate for HS-PDSCH	F	5.2.0	5.3.0	T1-041943
TP-26	TP-040233	381	-	Alignment of Prose to TTCN for RRC Connection Release (Cell DCH state) and RRC Connection Setup Message (Cell FACH State).	F	5.2.0	5.3.0	T1-041965
TP-27	TP-050032	382	-	Updates from core specification changes	F	5.3.0	5.4.0	T1-050095
TP-27	TP-050032	383	-	Correction to Hand over test procedure in CELL_DCH	F	5.3.0	5.4.0	T1-050350
TP-27	TP-050032	384	-	CR to 34.108: Changes to test frequencies for UMTS 850 Band	B	5.3.0	5.4.0	T1-050380
TP-27	TP-050032	385	-	Correction to default SIB configurations	F	5.3.0	5.4.0	T1-050019
TP-27	TP-050032	386	-	Editorial corrections in HSDPA RAB configurations 6.10.2.4.5.2 and 6.10.2.4.5.4.	D	5.3.0	5.4.0	T1-050052
TP-27	TP-050032	387	-	CR to 34.108 Rel-5: Update to the contents of PHYSICAL CHANNEL RECONFIGURATION message for 1.28 Mcps TDD	F	5.3.0	5.4.0	T1-050064
TP-27	TP-050032	388	-	CR to 34.108 Rel-5: Update to the contents of TRANSPORT CHANNEL RECONFIGURATION message for 1.28 Mcps TDD	F	5.3.0	5.4.0	T1-050065
TP-27	TP-050032	389	-	CR to 34.108 Rel-5: Update to the contents of RRC CONNECTION REQUEST message for TDD	F	5.3.0	5.4.0	T1-050066
TP-27	TP-050032	390	-	Correction to the HSDPA RB Identity in Radio Bearer Setup & Radio Bearer Release message contents	F	5.3.0	5.4.0	T1-050072
TP-27	TP-050032	391	-	CR to TS 34.108 v5.3.0 - Correction to Default RADIO BEARER RELEASE message (FDD)	F	5.3.0	5.4.0	T1-050202
TP-27	TP-050032	392	-	Addition of reference radio bearer configuration for MAC-hs testing	F	5.3.0	5.4.0	T1-050239
TP-27	TP-050032	393	-	CR to 34.108 Rel-5: Update to the contents of RRC CONNECTION REQUEST message for TDD	F	5.3.0	5.4.0	T1-050295
TP-27	TP-050032	394	-	CR to 34.108 Rel-5: Update to the contents of Default System Information Block Messages for TDD	F	5.3.0	5.4.0	T1-050296
TP-27	TP-050032	395	-	CR to 34.108 Rel-5: Add the contents of SIB 5 & 6 for HCR TDD	F	5.3.0	5.4.0	T1-050297
TP-27	TP-050032	396	-	Correction to TFCS ordering	F	5.3.0	5.4.0	T1-050451r1
TP-27	TP-050032	397	-	Addition of GPS scenario and A-GPS assistance data values for signalling tests to 34.108	F	5.3.0	5.4.0	T1-050458

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
TP-27	TP-050032	398	-	CR to TS 34.108 Rel-5; Correction to the physical channel parameters (Revison of T1-050176)	F	5.3.0	5.4.0	T1-050469
RP-28	RP-050267	399	-	Additional call setup procedures for inter RAT RRM testing	F	5.4.0	5.5.0	R5-050618
RP-28	RP-050267	400	-	CR to 34.108: Correction to RADIO BEARER SETUP message for BTFD RMC	F	5.4.0	5.5.0	R5-050704
RP-28	RP-050267	401	-	CR to 34.108: Correction to reference radio conditions for GSM	F	5.4.0	5.5.0	R5-050811
RP-28	RP-050267	402	-	Addition of RADIO BEARER SETUP Messages for Auxiliary Measurement	F	5.4.0	5.5.0	R5-050856
RP-28	RP-050267	404	-	CR 34.108 Addition of specific message content to A-GPS performance test procedures in clause 7.5	F	5.4.0	5.5.0	R5-050709
RP-28	RP-050267	405	-	CR to 34.108 Rel-5: Clarification of generic setup procedures in section 7.3.4	F	5.4.0	5.5.0	R5-050663
RP-28	RP-050267	406	-	Removal of TGPL2	F	5.4.0	5.5.0	R5-050513
RP-28	RP-050267	407	-	Addition of compressed mode pattern for Inter Frequency FDD measurement & Inter RAT measurement GSM	F	5.4.0	5.5.0	R5-050525
RP-28	RP-050267	408	-	Correction to MIB, PLMN and Cell Value Tag Value Definition to 34.108	F	5.4.0	5.5.0	R5-050608
RP-28	RP-050267	409	-	CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 11 (3.84 Mcps and 1.28 Mcps TDD) in section 6.1.0b	F	5.4.0	5.5.0	R5-050613
RP-28	RP-050267	410	-	CR to 34.108 Rel-5: Corrections to the usage of 'Cell info' IE in System Information Block type 11 in section 6.1.4 for TDD cell	F	5.4.0	5.5.0	R5-050619
RP-28	RP-050267	411	-	CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 5 (1.28 Mcps TDD)	F	5.4.0	5.5.0	R5-050620
RP-28	RP-050267	412	-	Update to clause 8 Test USIM Parameters	F	5.4.0	5.5.0	R5-050638
RP-28	RP-050267	413	-	CR to 34.108 Rel-5: Update of SIB3, SIB4, SIB11 and SIB12 for TDD in section 6.1.0b	F	5.4.0	5.5.0	R5-050662
RP-28	RP-050267	414	-	CR to 34.108: Correction to TFCS	F	5.4.0	5.5.0	R5-050677
RP-28	RP-050267	415	-	CR to TS34.108 Rel-5; Correction to the physical channel parameter	F	5.4.0	5.5.0	R5-050724
RP-28	RP-050267	416	-	Correction to default SIB configurations	F	5.4.0	5.5.0	R5-050947
RP-28	RP-050267	417	-	CR to 34.108: Missing Rel-5 IE's in the default Radio Bearer Setup message at section 9.1.1.	F	5.4.0	5.5.0	R5-050600
RP-28	RP-050267	418	-	CR to TS34.108 Rel-5; Clarification of the reference TFCS for three RB multiplexing option (condition A9)	F	5.4.0	5.5.0	R5-050913
RP-28	RP-050268	419	-	Addition of new HSDPA Streaming RAB configurations	F	5.4.0	5.5.0	R5-050880
RP-28	RP-050268	420	-	CR to 34.108 Rel-5: Content Correction of RRC CONNECTION SETUP message for LCR TDD in 9.1.2	F	5.4.0	5.5.0	R5-050585
RP-28	RP-050268	421	-	Add Default RADIO BEARER RELEASE message (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050680
RP-28	RP-050268	422	-	Add Default Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050681
RP-28	RP-050268	423	-	Add Default Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050682
RP-28	RP-050268	424	-	Add Default Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050683
RP-28	RP-050268	425	-	Add Default Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050684
RP-28	RP-050268	426	-	Add Default Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050685
RP-28	RP-050268	427	-	Add Default Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050686
RP-28	RP-050268	428	-	Add Default Contents of MEASUREMENT REPORT message: AM (intra/inter-frequency measurement (3.84 Mcps TDD)	F	5.4.0	5.5.0	R5-050956
RP-28	RP-050268	430	-	Correction to RADIO BEARER SETUP message for HSDPA RF testing	F	5.4.0	5.5.0	R5-050879
RP-28	RP-050349	403	-	Addition of GPS scenario and assistance data for A-GPS performance tests in 34.108	B	5.4.0	5.5.0	R5-050836
RP-28	RP-050350	429	-	Corrections to section 10.7 and GPS data file for 34.108	F	5.4.0	5.5.0	R5-050969
RP-29	RP-050600	431	-	Feature Clean Up: Removal of DRAC from section 9 of 34.108	F	5.5.0	6.0.0	R5-051312

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-29	RP-050600	432	-	Feature Clean Up: Removal of SSDT from 34.108	F	5.5.0	6.0.0	R5-051356
RP-29	RP-050600	433	-	Feature Clean Up: Removal of 80 ms TTI for DCH for all cases except when the UE supports SF512 from 34.108	F	5.5.0	6.0.0	R5-051379
RP-29	RP-050600	434	-	Feature Clean Up: Removal of CPCH from section 4 of 34.108	C	5.5.0	6.0.0	R5-051543
RP-29	RP-050600	435	-	Feature Clean Up: Removal of CPCH from section 6 of 34.108	C	5.5.0	6.0.0	R5-051544
RP-29	RP-050600	436	-	Feature Clean Up: Removal of CPCH from section 7 & 8 of 34.108	C	5.5.0	6.0.0	R5-051545
RP-29	RP-050600	437	-	Feature Clean Up: Removal of CPCH from section 9 of 34.108	C	5.5.0	6.0.0	R5-051546
RP-29	RP-050600	438	-	Feature Clean Up: Removal of DSCH (FDD mode) from 34.108	F	5.5.0	6.0.0	R5-051548
RP-29	RP-050600	439	-	Modification to PS setup procedure for inter RAT RRM testing	F	5.5.0	6.0.0	R5-051161
RP-29	RP-050600	440	-	CR to 34.108: RRC CONNECTION SETUP exception for HSDPA testing	F	5.5.0	6.0.0	R5-051430
RP-29	RP-050600	441	-	CR to 34.108: Correction to the RADIO BEARER SETUP message for HSDPA testing	F	5.5.0	6.0.0	R5-051112
RP-29	RP-050512	442	-	Changes to GPS Scenarios and Assistance data in TS 34.108	F	5.5.0	6.0.0	R5-051076
RP-29	RP-050514	443	-	CR to 34.108 Rel-5: Correction of contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD) in 9.2.2	F	5.5.0	6.0.0	R5-051212
RP-29	RP-050514	444	-	CR to 34.108 Rel-5: SIB default schedule in 6.1.0a - Default Master Information Block and Scheduling Block messages	F	5.5.0	6.0.0	R5-051213
RP-29	RP-050514	445	-	CR to 34.108 Rel-5: Corrections to the IE"Midamble shift and burst type" of System Information Block type 5/6 (3.84Mcps TDD) in section 6.1.0b	F	5.5.0	6.0.0	R5-051222
RP-29	RP-050514	446	-	CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 5 (3.84 Mcps TDD) in section 6.1.1	F	5.5.0	6.0.0	R5-051344
RP-29	RP-050514	447	-	CR to 34.108 Rel-5: Corrections to the value of Sintrasearch and Sintersearch in "Cell selection and reselection quality measure" of System Information Block type 3/4 (1.28Mcps TDD and 3.84Mcps TDD) in section 6.1.0b	F	5.5.0	6.0.0	R5-051536
RP-29	RP-050600	448	-	Use 'Same as UL' for the Added or Reconfigured DL TrCH information of the added or reconfigured PS RAB	F	5.5.0	6.0.0	R5-051041
RP-29	RP-050600	449	-	Correction to the default contents for Radio Bearer Setup message	F	5.5.0	6.0.0	R5-051044
RP-29	RP-050600	450	-	Corrections to default parameters of UL:384kbps PS Bearer	F	5.5.0	6.0.0	R5-051058
RP-29	RP-050600	451	-	Correction to NB AMR Radio Bearer Configurations	F	5.5.0	6.0.0	R5-051318
RP-29	RP-050600	452	-	Correction to default contents of Cell Update and Initial Direct transfer message for Rel-5	F	5.5.0	6.0.0	R5-051325
RP-29	RP-050600	453	-	Correction to DPCCH Power Offset IE in default contents for RRC Connection Setup and Radio Bearer Setup message	F	5.5.0	6.0.0	R5-051365
RP-29	RP-050515	454	-	Using Test USIM for VSTK generation of VGCS/VBS ciphering	B	5.5.0	6.0.0	R5-051553
RP-29	RP-050600	455	-	Correction to default contents of RADIO BEARER SETUP MESSAGE for the IE "Number of Processes"	F	5.5.0	6.0.0	R5-051324
RP-29	RP-050600	456	-	Correction of DL channelisation code in RADIO BEARER SETUP for HSDPA configurations	F	5.5.0	6.0.0	R5-051345
RP-29	RP-050513	457	-	Clarifiation of reference radio bearer configuration for MAC-hs test case 7.1.5.2.	F	5.5.0	6.0.0	R5-051164
RP-29	RP-050600	458	-	Replacement of the technical content of 34.108 Rel-5 by a pointer to Rel-6 document	F	5.5.0	6.0.0	R5-051584
RP-29	RP-050600	459	-	Introduction of HSDPA + Wideband AMR radio bearer combination	F	5.5.0	6.0.0	R5-051588
RP-29	RP-050600	460	-	Feature Clean Up: Removal of DRAC from section 6 of 34.108	F	5.5.0	6.0.0	R5-051311
RP-30	RP-050767	461	-	Correction to 34.108: RF Radio Bearer setup message for HSDPA testing	F	6.0.0	6.1.0	R5-052320
RP-30	RP-050720	462	-	Changes, additions and corrections to GPS scenarios and assistance data in TS 34.108	F	6.0.0	6.1.0	R5-052017
RP-30	RP-050716	463	-	Generic test procedure for EDCH RF testing	F	6.0.0	6.1.0	R5-052345
RP-30	RP-050769	464	-	Correction of UE test states for RF testing.	F	6.0.0	6.1.0	R5-052302
RP-30	RP-050769	465	-	Removal of temporary BLER measurement configuration	F	6.0.0	6.1.0	R5-051933

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-30	RP-050780	466	-	Introduction of UMTS1700 for TS34.108	B	6.0.0	6.1.0	R5-052329
RP-30	RP-050767	467	-	Re-definition of reference radio bearer configuration for MAC-hs test case 7.1.5.2	F	6.0.0	6.1.0	R5-051857
RP-30	RP-050767	468	-	Introduction of additional HSDPA radio bearer combination	F	6.0.0	6.1.0	R5-052159
RP-30	RP-050717	469	-	CR to 34.108; Correction to default message content in INITIAL DIRECT TRANSFER	F	6.0.0	6.1.0	R5-052179
RP-30	RP-050716	470	-	CR to 34.108 Rel-6; Generic Setup Procedure and Default RRC Messages for Enhanced Uplink Tests	F	6.0.0	6.1.0	R5-052192
RP-30	RP-050716	471	-	Generic reference bearer radio configurations for E-DCH testing	F	6.0.0	6.1.0	R5-052167
RP-30	RP-050716	472	-	Addition of basic radio bearer combinations for E-DCH testing	F	6.0.0	6.1.0	R5-052144
RP-30	RP-050718	473	-	CR to TS34.108; Correction to the default system information block type3 for DSAC	F	6.0.0	6.1.0	R5-052145
RP-30	RP-050769	474	-	Editorial corrections to TS 34.108	D	6.0.0	6.1.0	R5-051840
RP-30	RP-050769	475	-	Removal of deprecated values from the default contents for RRC Connection Setup (Cell FACH) and RRC Connection Release (Cell DCH) messages.	F	6.0.0	6.1.0	R5-051848
RP-30	RP-050769	476	-	Correction to the default RRC message contents for the IE “UARFCN uplink (Nu)”	F	6.0.0	6.1.0	R5-052155
RP-30	RP-050769	477	-	Correction of references to IB UL:8 DL:8 kbps transport channel parameters in reference radio bearer configuration 6.10.2.4.1.58a	F	6.0.0	6.1.0	R5-051858
RP-30	RP-050769	478	-	Correction of UE states tables for Generic setup procedures.	F	6.0.0	6.1.0	R5-051942
RP-30	RP-050769	479	-	Corrections to default message contents of ‘HANDOVER FROM UTRAN COMMAND-GSM’ message.	F	6.0.0	6.1.0	R5-051955
RP-30	RP-050776	480	-	Addition of multi-rate AMR-NB configuration with SRB#5	F	6.0.0	6.1.0	R5-052176
RP-30	RP-050769	481	-	Proposed CR to 34.108 [R6 version, R99 affected] to change slot format for AMR 5.9 mono rate RAB	F	6.0.0	6.1.0	R5-052055
RP-30	RP-050769	482	-	Correction to puncturing limit in radio bearer configuration 6.10.2.4.1.38a	F	6.0.0	6.1.0	R5-052096
RP-30	RP-050777	483	-	Introduction of third RAB sublfow to WB-AMR test configurations	F	6.0.0	6.1.0	R5-052140
RP-30	RP-050833	484	-	Correction to RADIO BEARER SETUP message for HSDPA RF testing	F	6.0.0	6.1.0	-
RP-30	RP-050832	485	-	Correction to test procedure for HSDPA RF testing	F	6.0.0	6.1.0	-
				2006-01: Editorial conversion to make file compatible with Word 2000.		6.1.0	6.1.1	
RP-31	RP-060154	486	-	Corrections to GPS data files for signalling tests.	F	6.1.1	6.2.0	R5-060522
RP-31	RP-060163	487	-	Corrections for reference RABs	F	6.1.1	6.2.0	R5-060273
RP-31	RP-060144	488	-	Corrections to the RADIO BEARER SETUP message for Enhanced uplink	F	6.1.1	6.2.0	R5-060335
RP-31	RP-060150	489	-	Correction to default message content in INITIAL DIRECT TRANSFER	F	6.1.0	6.2.0	R5-060396
RP-31	RP-060154	490	-	Corrections to default message contents for signaling	F	6.1.1	6.2.0	R5-060274
RP-31	RP-060163	491	-	Update of RB configuration 6.10.2.4.1.4b to increase test coverage for multi-mode AMR configurations	F	6.1.1	6.2.0	R5-060122
RP-31	RP-060144	492	-	Correction of UE RRC states table for common procedures (section 7.4.1)	F	6.1.1	6.2.0	R5-060268
RP-31	RP-060154	493	-	Correction to DPCCH power offset value in RADIO BEARER SETUP messages	F	6.1.1	6.2.0	R5-060459
RP-31	RP-060153	494	-	Introduction of Band VII and Band VIII to TS34.108 Chapter 5 and introduction of new UARFCN scheme	F	6.1.1	6.2.0	R5-060440
RP-31	RP-060154	495	-	Clarification of RB Test Mode State for RF testing	F	6.1.1	6.2.0	R5-060446
RP-31	RP-060144	496	-	Common message content for E-DCH RF testing	F	6.1.1	6.2.0	R5-060439
RP-31	RP-060144	497	-	Generic test procedure for E-DCH RF testing	F	6.1.1	6.2.0	R5-060438
RP-31	RP-060166	498	-	Correction to GPS Assistance Data for Performance tests	F	6.1.1	6.2.0	R5-060007
RP-31	RP-060154	499	-	Corrections to default message contents for RF	F	6.1.1	6.2.0	R5-060257
RP-31	RP-060163	500	-	Adding of RB Configuration 6.11.4b to clause 6.11 to increase test coverage for Interactive or background PS RAB UL 0kbps/DL 0kbps.	F	6.1.1	6.2.0	R5-060571
RP-32	RP-060331	501	-	Addition of release information to A-GPS MEASUREMENT CONTROL message IEs	F	6.2.0	6.3.0	R5-061454
RP-32	RP-060331	502	-	Change to altitude of simulated UE position	F	6.2.0	6.3.0	R5-061410
RP-32	RP-060331	503	-	Clarification of A-GPS simulated satellites	F	6.2.0	6.3.0	R5-061223
RP-32	RP-060337	504	-	Clarification to loopback on HSDPA call set up procedure	F	6.2.0	6.3.0	R5-061046
RP-32	RP-060337	505	-	Removal of alternative procedure for HSDPA RF testing	F	6.2.0	6.3.0	R5-061212

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-32	RP-060337	506	-	Removal of alternative RB Setup message for HSDPA RF testing	F	6.2.0	6.3.0	R5-061213
RP-32	RP-060338	507	-	CR for 34.108:Addition of RADIO BEARER SETUP default message contents for LCR TDD HSDPA RF testing	F	6.2.0	6.3.0	R5-061438
RP-32	RP-060332	508	-	Correction to RB setup message for E-DCH	F	6.2.0	6.3.0	R5-061443
RP-32	RP-060334	509	-	CR to 34.108: Addition of band IV to test frequencies for UMTS 1.7/2.1 GHz	F	6.2.0	6.3.0	R5-061188
RP-32	RP-060 336	510	-	Addition of radio bearer setup and release for HCR HSDPA testing to 34.108	F	6.2.0	6.3.0	R5-061148
RP-32	RP-060337	511	-	HS-SCCH and HS-PDSCH power levels in signaling tests	F	6.2.0	6.3.0	R5-061207
RP-32	RP-060338	512	-	CR for 34.108:Addition of the combinations on DPCH and HS-PDSCH for LCR TDD	F	6.2.0	6.3.0	R5-061310
RP-32	RP-060338	513	-	CR for 34.108:Correction of RADIO BEARER SETUP default message contents for LCR TDD HSDPA	F	6.2.0	6.3.0	R5-061053
RP-32	RP-060338	514	-	CR for 34.108:Correction of RADIO BEARER RELEASE default message contents for LCR TDD HSDPA	F	6.2.0	6.3.0	R5-061054
RP-32	RP-060332	515	-	Corrections to the default RADIO BEARER SETUP message for Enhanced uplink	F	6.2.0	6.3.0	R5-061314
RP-32	RP-060332	516	-	Generalize E-DCH radio bearer names and correction to section numbering for 6.10.2.4.6.3.2.1.1.2.	F	6.2.0	6.3.0	R5-061315
RP-32	RP-060332	517	-	Addition of conversational radio bearer combinations for E-DCH/HS-DSCH testing	F	6.2.0	6.3.0	R5-061266
RP-32	RP-060332	518	-	E-HICH/E-RGCH and E-AGCH codes used in Radio Bearer Setup for signaling	F	6.2.0	6.3.0	R5-061386
RP-32	RP-060332	519	-	Introduction of additional WB-AMR RAB combination for E-DCH/HS-DSCH testing	F	6.2.0	6.3.0	R5-061339
RP-32	RP-060336	520	-	Addition of combinations on DPCH and HS-DSCH for HCR to 34.108	F	6.2.0	6.3.0	R5-061149
RP-32	RP-060322	521	-	Corrections to the values for IE based on calculated ASN.1 value to 34.108 clause 9	F	6.2.0	6.3.0	R5-061369
RP-32	RP-060328	522	-	CR to 34.108 Rel-6: Supplement to the UTRAN mobility information procedure in TDD	F	6.2.0	6.3.0	R5-061392
RP-32	RP-060322	523	-	Correction to specific message contents for UE Capability Information confirm message	F	6.2.0	6.3.0	R5-061139
RP-32	RP-060322	524	-	Corrections to the values for IE based on calculated ASN.1 value to 34.108 clause 6	F	6.2.0	6.3.0	R5-061281
RP-32	RP-060322	525	-	Addition of a new section for downlink physical channels code allocation for signalling in FDD	F	6.2.0	6.3.0	R5-061385
RP-33	RP-060560	526	-	Editorial changes in 34.108	F	6.3.0	6.4.0	R5-062092
RP-33	RP-060549	527	-	CR to 34.108: Correction of reference test frequencies for UMTS800 (Band VI)	F	6.3.0	6.4.0	R5-062440
RP-33	RP-060549	528	-	Correction to SIB11 in 6.1	F	6.3.0	6.4.0	R5-062427
RP-33	RP-060549	529	-	Correction to SECURITY MODE COMMAND message in 9.2.1	F	6.3.0	6.4.0	R5-062403
RP-33	RP-060562	530	-	Correction to RB Setup default message for E-DCH RF testing	F	6.3.0	6.4.0	R5-062208
RP-33	RP-060568	531	-	CR to 34.108: Correction the contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)	F	6.3.0	6.4.0	R5-062511
RP-33	RP-060566	532	-	Addition of HSDPA cases to radio bearer setup and radio bearer release in section 9.1.2	F	6.3.0	6.4.0	R5-062253
RP-33	RP-060567	533	-	Corrections to the default PHYSICAL CHANNEL RECONFIGURATION message	F	6.3.0	6.4.0	R5-062291
RP-33	RP-060567	534	-	Corrections to specification of HARQ RV sequence and retransmissions for 34.123-1 test cases configuring HSDPA	F	6.3.0	6.4.0	R5-062544
RP-33	RP-060562	535	-	Corrections to the default RADIO BEARER SETUP message	F	6.3.0	6.4.0	R5-062320
RP-33	RP-060562	536	-	New Test RABS for MAC-E/Es test cases	F	6.3.0	6.4.0	R5-062328
RP-33	RP-060562	537	-	Correction to radio bearer configuration 6.10.2.4.6 and 6.10.2.4.8	F	6.3.0	6.4.0	R5-062199
RP-33	RP-060562	538	-	Correction to 34.108 Section 9.1 : Corrections to Radio Bearer Setup for A12	F	6.3.0	6.4.0	R5-062348
RP-33	RP-060549	539	-	Clarification to section 6.10 and 6.11	F	6.3.0	6.4.0	R5-062194
RP-33	RP-060549	540	-	Correction to 34.108 Section 6.1 : Inclusion of System Information Block Type 5bis	F	6.3.0	6.4.0	R5-062398
RP-33	RP-060549	541		Corrections to maximum data rate for combinations on PRACH (FDD)	F	6.3.0	6.4.0	R5-062350
RP-34	RP-060739	542	-	Assistance Data change for A-GPS Minimum Performance Test	F	6.4.0	6.5.0	R5-063400

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-34	RP-060731	543	-	Correction to SECURITY MODE COMMAND message in 9.2.2	F	6.4.0	6.5.0	R5-063296
RP-34	RP-060735	544	-	Addition of PAGING TYPE 2 message in 9.2	F	6.4.0	6.5.0	R5-063401
RP-34	RP-060731	545	-	Correction to RADIO BEARER SETUP message in 9.2	F	6.4.0	6.5.0	R5-063402
RP-34	RP-060743	546	-	Correction to the set of Reference E-TFCIs in RB Setup default message (RF)	F	6.4.0	6.5.0	R5-063233
RP-34	RP-060731	547	-	Correction the Default System Information Block type6 Messages in 6.1.0b	F	6.4.0	6.5.0	R5-063294
RP-34	RP-060743	548	-	CR to 34.108, correction of RADIO BEARER SETUP for EDCH	F	6.4.0	6.5.0	R5-063545
RP-34	RP-060743	549	-	Correction to the set of Reference E-TFCIs in RB Setup default message (SIG)	F	6.4.0	6.5.0	R5-063240
RP-34	RP-060750	550	-	Introduction of radio bearers for MTCH	F	6.4.0	6.5.0	R5-063543
RP-34	RP-060749	551	-	Introduction of FDD interband testing in TS 34.108	F	6.4.0	6.5.0	R5-063255
RP-34	RP-060735	552	-	Correction to Radio Bearer Setup message –Mac-hs reset indicator	F	6.4.0	6.5.0	R5-063339
RP-34	RP-060731	553	-	CR to 34.108, Add MCC value for Band VI test	F	6.4.0	6.5.0	R5-063340
RP-34	RP-060731	554	-	CR to 34.108, Modify MCC value in IMSI of test USIM for Band VI test	F	6.4.0	6.5.0	R5-063341
RP-34	RP-060731	555	-	Correction to default content for System Information Block Type 7	F	6.4.0	6.5.0	R5-063079
RP-34	RP-060731	556	-	Clarification of FDD test channels used for signaling test cases	F	6.4.0	6.5.0	R5-063260
RP-35	RP-070111	557		34.108 v6.6.0 pointer to Release 7 version	F	6.5.0	7.0.0	R5-070338
RP-35	RP-070105	558		Correction of IE “DL UM RLC LI size” in RF default messages for HSDPA	F	6.5.0	7.0.0	R5-070548
RP-35	RP-070096	559		Signalled Reference E-TFCIs for E-DCH RF tests	F	6.5.0	7.0.0	R5-070113
RP-35	RP-070096	560		Correction to RB setup message used for E-DCH tests	F	6.5.0	7.0.0	R5-070221
RP-35	RP-070096	561		Correction of IE “DL UM RLC LI size” in RF default messages for E-DCH	F	6.5.0	7.0.0	R5-070549
RP-35	RP-070090	562		Generic test procedure for MBMS RF test case	F	6.5.0	7.0.0	R5-070553
RP-35	RP-070094	563		Introduction of FDD Mode Test frequencies for Operating Band X (Extended 1.7/2.1 GHz)	F	6.5.0	7.0.0	R5-070160
RP-35	RP-070094	564		Introduction of FDD Band X (Extended 1.7/2.1 GHz) to Contents of System Information Block type 5bis	F	6.5.0	7.0.0	R5-070161
RP-35	RP-070104	565		Correction to contents of System Information Block type 5 (1.28 Mcps TDD)	F	6.5.0	7.0.0	R5-070109
RP-35	RP-070104	566		Remove DCH information from RRC Connection Setup message to Cell_FACH state	F	6.5.0	7.0.0	R5-070353
RP-35	RP-070096	567		Corrections to 34.108 Radio Bearer Setup Message: AM or UM	F	6.5.0	7.0.0	R5-070038
RP-35	RP-070096	568		Introduction of radio bearers for Stand-alone SRBs for DCCH on E-DCH and HS-DSCH	F	6.5.0	7.0.0	R5-070356
RP-35	RP-070096	569		Introduction of RRC Connection setup message for Stand-alone SRBs for DCCH on E-DCH and HS-DSCH	F	6.5.0	7.0.0	R5-070357
RP-35	RP-070086	570		MBMS test - MCCH configurations	F	6.5.0	7.0.0	R5-070429
RP-35	RP-070086	571		Generic setup procedures and default values for MBMS signalling testing	F	6.5.0	7.0.0	R5-070422
RP-35	RP-070086	572		Introduction of signalling radio bearer for MCCH	F	6.5.0	7.0.0	R5-070147
RP-36	RP-070360	573		Definition of MCCH default message content	F	7.0.0	7.1.0	R5-071469
RP-36	RP-070354	574		Addition of Additional Dynamic Transport Format Information for CCCH and Additional RACH TFCS for CCCH in SIB5	F	7.0.0	7.1.0	R5-071536
RP-36	RP-070354	575		Remove MCC codes 440/441 for Band VI test	F	7.0.0	7.1.0	R5-071036
RP-36	RP-070350	576		Modify the Physical channel parameters for HS-DSCH for 1.28 Mcps TDD.	F	7.0.0	7.1.0	R5-071119
RP-36	RP-070357	577		Modification to RB setup message used for E-DCH test cases	F	7.0.0	7.1.0	R5-071126
RP-36	RP-070357	578		Correction to RB setup message used for E-DCH RF test cases	F	7.0.0	7.1.0	R5-071127
RP-36	RP-070364	579		Addition of 7.68 Mcps TDD TX diversity modes to 34.108 (section 6.7)	F	7.0.0	7.1.0	R5-071270
RP-36	RP-070364	580		Addition of default message contents for RF tests (7.68Mcps TDD)	F	7.0.0	7.1.0	R5-071277
RP-36	RP-070364	581		Addition of default message contents for signalling tests (7.68Mcps TDD)	F	7.0.0	7.1.0	R5-071278
RP-36	RP-070364	582		Addition of default system information for 7,68 Mcps TDD to 34.108	F	7.0.0	7.1.0	R5-071279
RP-36	RP-070364	583		Addition of default system information for 7,68Mcps TDD to 34.108 for SCCPCH configuration with Stand-alone SRB	F	7.0.0	7.1.0	R5-071280
RP-36	RP-070364	584		Addition of Reference Radio Bearer configurations used	F	7.0.0	7.1.0	R5-071284

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
				in Radio Bearer testing for 7.68 Mcps TDD				
RP-36	RP-070364	585		Addition Standard TDD reference test frequencies (7.68 Mcps option) to 34.108	F	7.0.0	7.1.0	R5-071287
RP-36	RP-070363	586		Introduction of the generic test procedure for MBMS RF/RRM test cases	F	7.0.0	7.1.0	R5-071378
RP-36	RP-070350	587		Adding Radio Bearer Configurations for LCR TDD HSDPA	F	7.0.0	7.1.0	R5-071423
RP-36	RP-070344	588		Clarification of two default values for IE Scrambling code change	F	7.0.0	7.1.0	R5-071435
RP-36	RP-070356	589		Editorial corrections in the reference list	F	7.0.0	7.1.0	R5-071443
RP-36	RP-070344	590		Correction to the list of RAB combinations on DPCH and HS-PDSCH	F	7.0.0	7.1.0	R5-071448
RP-36	RP-070360	591		Introduce SIB scheduling and contents for MBMS test	F	7.0.0	7.1.0	R5-071458
RP-36	RP-070360	592		Update to generic setup procedures and default values for MBMS signalling testing	F	7.0.0	7.1.0	R5-071459
RP-36	RP-070360	593		Clarification of MBMS test case default behaviour	F	7.0.0	7.1.0	R5-071460
RP-36	RP-070360	594		Downlink physical channels code allocation for MBMS test	F	7.0.0	7.1.0	R5-071487
RP-36	RP-070354	595		Changes to inclusion of Start Value in Cell Update Message from Rel-6	F	7.0.0	7.1.0	R5-071490
RP-36	RP-070354	596		Addition of Additional Transport Format and TFCS for CCCH on RACH	F	7.0.0	7.1.0	R5-071529
RP-37	RP-070600	597	-	UE performance requirements for high speed train	F	7.1.0	7.2.0	R5-072280
RP-37	RP-070596	598	-	CR to 34.108: New Requirements for Fast L1 Synchronization	F	7.1.0	7.2.0	R5-072368
RP-37	RP-070609	600	-	Addition of MTCH and MCCH to combinations on SCCPCH for 3.84 Mcps TDD	F	7.1.0	7.2.0	R5-072296
RP-37	RP-070609	601	-	Addition of MTCH and MCCH to combinations on SCCPCH for 7.68 Mcps TDD	F	7.1.0	7.2.0	R5-072297
RP-37	RP-070604	602	-	Corrections of Test Procedures for MBMS Testing (IGMP and MLD)	F	7.1.0	7.2.0	R5-072066
RP-37	RP-070604	603	-	Addition of RB combinations for PTP MBMS Testing	F	7.1.0	7.2.0	R5-072067
RP-37	RP-070604	604	-	Correction to default MBMS MCCH messages in 34.108	F	7.1.0	7.2.0	R5-072166
RP-37	RP-070596	605	-	Correction to SIB5 contents for additional RACH TFCS for CCCH	F	7.1.0	7.2.0	R5-072186
RP-37	RP-070596	606	-	Correction to default RB combinations on PRACH for additional RACH TFCS for CCCH	F	7.1.0	7.2.0	R5-072187
RP-37	RP-070608	607	-	SIB5 configuration for MBMS RF/RRM generic procedure	F	7.1.0	7.2.0	R5-072379
RP-37	RP-070604	608	-	Default message content for MBMS NEIGHBOURING CELL P-T-M RB INFORMATION	F	7.1.0	7.2.0	R5-072468
RP-37	RP-070604	609	-	Addition of PTP RB for MBMS	F	7.1.0	7.2.0	R5-072521
RP-37	RP-070604	610	-	Addition of new SIB5 configuration to have MCCH mapped on to an S-CCPCH also used for non-MBMS purposes	F	7.1.0	7.2.0	R5-072469
RP-37	RP-070593	611	-	Addition of RB combination for RoHCTesting	F	7.1.0	7.2.0	R5-072520
RP-37	RP-070611	612	-	Introduce SIB scheduling and contents for LCR TDD MBMS test	F	7.1.0	7.2.0	R5-072534
RP-37	RP-070611	613	-	Introduction of radio bearers of MTCH for LCR TDD MBMS test	F	7.1.0	7.2.0	R5-072535
RP-37	RP-070604	614	-	Addition of MBMS PTP RB Setup message contents	F	7.1.0	7.2.0	R5-072496
RP-37	RP-070601	615	-	Introduction of F-DPCH Support Indicator	F	7.1.0	7.2.0	R5-072568
RP-37	RP-070612	616	-	Correction of HS-PDSCH for 64QAM Enhancement	F	7.1.0	7.2.0	R5-072559
RP-37	RP-070613	617	-	Correction of HS-PDSCH for MIMO Enhancement	F	7.1.0	7.2.0	R5-072560
RP-37	RP-070614	618	-	Correction of Maximum Data Rate for E-DPDCH	F	7.1.0	7.2.0	R5-072561
RP-37	RP-070609	620	-	Addition of message contents required for 3.84Mcps and 7.68Mcps TDD to the default messages contents for signalling	F	7.1.0	7.2.0	R5-072295
RP-37	RP-070600	619	-	Production of 34.108 Rel-7 pointer version to point to Rel-8 of the spec	F	7.1.0	7.2.0	R5-072590
RP-37	RP-070599	599	-	Introduction of FDD Mode Test frequencies for Operating Band XI (UMTS1500)	F	7.1.0	8.0.0	R5-072273
RP-38	RP-070858	621		Addition of exceptional messages for RF test procedure	F	8.0.0	8.1.0	R5-073202
RP-38	RP-070869	622		Correction to DL RLC PDU size in 9.2 for RB setup and RRC connection setup	F	8.0.0	8.1.0	R5-073203
RP-38	RP-070858	623		Correction to DL RLC PDU size in 9.1 for RB setup and RRC connection setup	F	8.0.0	8.1.0	R5-073204
RP-38	RP-070863	624		Addition of new IE to RRC connection setup message	F	8.0.0	8.1.0	R5-073390
RP-38	RP-070858	625		Addition of test frequencies for low and high ranges for serving cell	F	8.0.0	8.1.0	R5-073311
RP-38	RP-070877	626		Update of generic setup procedure for MBMS RF/RRM	F	8.0.0	8.1.0	R5-073300

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
				testing				
RP-38	RP-070877	627		Addition of specific message content for MBMS RF/RRM Testing	F	8.0.0	8.1.0	R5-073342
RP-38	RP-070886	628		Addition of combinations on HS-PDSCH and E-PUCH for 3.84Mcps TDD	F	8.0.0	8.1.0	R5-073227
RP-38	RP-070879	629		Addition of message contents required for 1.28Mcps TDD to the default messages contents for signalling	F	8.0.0	8.1.0	R5-073089
RP-38	RP-070887	630		Update of the default RADIO BEARER SETUP message to support enhanced data rates	F	8.0.0	8.1.0	R5-073436
RP-38	RP-070887	631		Update of radio bearer configuration for Enhanced Layer 2	F	8.0.0	8.1.0	R5-073402
RP-38	RP-070887	632		Introduction of new RB configuration to be used by MAC-ehs test cases	F	8.0.0	8.1.0	R5-073403
RP-38	RP-070871	633		Correction conditions table for RADIO BEARER SETUP for MBMS PtP	F	8.0.0	8.1.0	R5-073449
RP-38	RP-070871	634		Update of generic setup procedure for MBMS protocol testing	F	8.0.0	8.1.0	R5-073446
RP-38	RP-070871	635		Correction to the SIB5 when MCCH mapped on to an S-CCPCH also used for non-MBMS purposes	F	8.0.0	8.1.0	R5-073465
RP-38	RP-070871	636		Corrections to MBMS specific message content and addition of MBMS ACCESS INFORMATION	F	8.0.0	8.1.0	R5-073454
RP-38	RP-070858	637		Addition of general SS requirements for UL and DL RF signal levels	F	8.0.0	8.1.0	R5-073269
RP-38	RP-070858	638		Correction to SIB1	F	8.0.0	8.1.0	R5-073418
RP-38	RP-070858	639		Editorial moving the word "Default1" or "Default2" in the correct column	F	8.0.0	8.1.0	R5-073425
RP-38	RP-070858	640		CR wrongly indicated to be to 34.108 and hence not implemented	-	-	-	R5-073425
RP-39	RP-080095	0641		Correction to RRC Connection setup and RB setup messages for 64kbps(Channel2)	F	8.1.0	8.2.0	R5-080247
RP-39	RP-080103	0642		Addition of default message contents for MBMS RF/RRM testing	F	8.1.0	8.2.0	R5-080149
RP-39	RP-080108	0643		CR to 34.108: Introduction of the UE E-DCH 16-QAM feature	F	8.1.0	8.2.0	R5-080118
RP-39	RP-080106	0644		Addition of combinations on E-DCH to 6.11 of 34.108 for 3.84Mcps TDD	F	8.1.0	8.2.0	R5-080346
RP-39	RP-080106	0645		Update of radio bearer setup in default message contents to include E-DCH conditions for 3.84Mcps and 7.68Mcps TDD	F	8.1.0	8.2.0	R5-080350
RP-39	RP-080115	0646		CR for 34.108:Correction of CELL UPDATE CONFIRM default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080099
RP-39	RP-080115	0647		CR for 34.108:Correction of PHYSICAL CHANNEL RECONFIGURATION default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080100
RP-39	RP-080115	0648		CR for 34.108:Correction of RADIO BEARER RECONFIGURATION default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080101
RP-39	RP-080115	0649		CR for 34.108:Correction of RADIO BEARER RELEASE default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080102
RP-39	RP-080115	0650		CR for 34.108:Correction of RADIO BEARER SETUP default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080103
RP-39	RP-080115	0651		CR for 34.108:Correction of TRANSPORT CHANNEL RECONFIGURATION default message contents for LCR TDD EDCH	F	8.1.0	8.2.0	R5-080104
RP-39	RP-080115	0652		Correction to the cotents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD option)	F	8.1.0	8.2.0	R5-080190
RP-39	RP-080108	0653		Introduction of Default Radio Bearer Setup message contents for UL 16QAM	F	8.1.0	8.2.0	R5-080480
RP-39	RP-080107	0654		Addition of HARQ transmission parameters for 64QAM	F	8.1.0	8.2.0	R5-080482
RP-39	RP-080109	0655		Update of the default RADIO BEARER SETUP message to support enhanced data rates	F	8.1.0	8.2.0	R5-080531
RP-39	RP-080109	0656		Addition of enhanced L2 configurations for DL SRBs on HS	F	8.1.0	8.2.0	R5-080287
RP-39	RP-080109	0657		Correction of physical layer parameters for radio bearer configuration 6.10.2.4.5 and HSPA	F	8.1.0	8.2.0	R5-080289
RP-39	RP-080109	0658		Addition of enhanced L2 configurations to conversational UM PS radio bearer 6.10.2.4.6.6	F	8.1.0	8.2.0	R5-080484
RP-39	RP-080111	0659		Addition of RB setup message contents for CPC	F	8.1.0	8.2.0	R5-080512
RP-39	RP-080105	0660		Addition of HARQ transmission parameters for MIMO	F	8.1.0	8.2.0	R5-080519
RP-39	RP-080095	0661		Correction to RB identity in RRC Connection Setup	F	8.1.0	8.2.0	R5-080036
RP-39	RP-080095	0662		Addition of new RB Combination for MBMS Testing	F	8.1.0	8.2.0	R5-080600

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-39	RP-080095	0663		Corrections to default content of MBMS GENERAL INFORMATION message	F	8.1.0	8.2.0	R5-080160
RP-39	RP-080095	0664		Correction to MBMS selected service test procedure for MBMS Modification Request message	F	8.1.0	8.2.0	R5-080521
RP-39	RP-080109	0665		Update of the default RRC CONNECTION SETUP message to support enhanced data rates	F	8.1.0	8.2.0	R5-080523r1
RP-40	RP-080372	0666		CR to 34.108: Correction to Contents of RADIO BEARER SETUUP message: AM or UM (E-DCH and HSDPA) for 16QAM	F	8.2.0	8.3.0	R5-081140
RP-40	RP-080370	0667		CR to 34.108: Introduction of Operating Bands XII XIII and XIV (UMTS700 MHz)	F	8.2.0	8.3.0	R5-081150
RP-40	RP-080364	0668		Correction to RRC Connection setup messages for 64kbps(Channel2)	F	8.2.0	8.3.0	R5-081435
RP-40	RP-080364	0669		Additional IEs for 7.8.4 of 34.121-1	F	8.2.0	8.3.0	R5-081436
RP-40	RP-080364	0670		Additional test procedure for HSDPA with F-DPCH RF Performance Requirement for 7.8.5 of 34.121-1	F	8.2.0	8.3.0	R5-081279
RP-40	RP-080364	0671		Additional RB setup message for 7.8.5 of 34.121-1	F	8.2.0	8.3.0	R5-081280
RP-40	RP-080371	0672		Update of radio bearer setup in default message contents to include E-DCH conditions for 3.84Mcps TDD	F	8.2.0	8.3.0	R5-081336
RP-40	RP-080377	0673		CR for 34.108: Baseline radio bearer combination for LCR TDD EDCH	F	8.2.0	8.3.0	R5-081116
RP-40	RP-080379	0674		Addition of MBSFN RABs and Signalling RB for 3.84 Mcps TDD	F	8.2.0	8.3.0	R5-081034
RP-40	RP-080379	0675		Addition of MBSFN RABs and Signalling RB for 7.68 Mcps TDD	F	8.2.0	8.3.0	R5-081035
RP-40	RP-080379	0676		Initial additions to common test environment definition for HCR and VHCR TDD MBSFN	F	8.2.0	8.3.0	R5-081039
RP-40	RP-080363	0677		Clarification of TDD test channels used for signaling test cases	F	8.2.0	8.3.0	R5-081354
RP-40	RP-080373	0678		Addition of new Test Bearer for MAC CPC testing	F	8.2.0	8.3.0	R5-081219
RP-40	RP-080373	0679		CPC: Corrections to Radio Bearer Setup message contents for conditions A20 A21	F	8.2.0	8.3.0	R5-081388
RP-40	RP-080365	0680		Update of radio bearer configurations for HSPA	F	8.2.0	8.3.0	R5-081194
RP-40	RP-080364	0681		Correction to the Contents of RADIO BEARER SETUP message: AM or UM for MBMS PtP Radio Bearer Setup	F	8.2.0	8.3.0	R5-081205
RP-40	RP-080365	0682		Addition of new Test Bearer for MAC-ehs testing	F	8.2.0	8.3.0	R5-081218
RP-40	RP-080365	0683		Addition of new Test Bearer for Improved L2 testing	F	8.2.0	8.3.0	R5-081220
RP-40	RP-080365	0684		Corrections to the default RADIO BEARER SETUP message	F	8.2.0	8.3.0	R5-081393
RP-40	RP-080364	0685		Correction to RB Combination for MBMS Testing	F	8.2.0	8.3.0	R5-081298
RP-40	RP-080364	0686		Removal of MBMS dispersion indicator in default message content	F	8.2.0	8.3.0	R5-081508
RP-41	RP-080561	0687		Addition of combinations on HS-DSCH and E-PUCH to typical radio parameter sets for 7.68Mcps TDD	F	8.3.0	8.4.0	R5-083075
RP-41	RP-080568	0688		Simulated network environment description for MBSFN tests	F	8.3.0	8.4.0	R5-083105
RP-41	RP-080568	0689		Add default SIB configuration for 3.84 amd 7.68 Mcps TDD MBSFN	F	8.3.0	8.4.0	R5-083106
RP-41	RP-080568	0690		Add default MCCH configurations for 3.84 Mcps and 7.68 Mcps TDD MBSFN	F	8.3.0	8.4.0	R5-083107
RP-41	RP-080568	0691		Default MBMS RRC message contents for 3.84 and 7.68 Mcps TDD MBSFN	F	8.3.0	8.4.0	R5-083108
RP-41	RP-080568	0692		Radio Bearer configurations for 7bit and 15bit UM RLC TCs for HCR and VHCR TDD MBSFN	F	8.3.0	8.4.0	R5-083109
RP-41	RP-080557	0693		Corrections to the default RADIO BEARER SETUP message for condition A22	F	8.3.0	8.4.0	R5-083179
RP-41	RP-080570	0694		Correction of HS-PDSCH for 64QAM Enhancement	F	8.3.0	8.4.0	R5-083314
RP-41	RP-080554	0695		Correction to Contents of default RADIO BEARER SET-UP messages	F	8.3.0	8.4.0	R5-083336
RP-41	RP-080557	0696		Corrections to 34.108: Radio access bearer establishment procedure for packet switched sessions (procedure P26)	F	8.3.0	8.4.0	R5-083338
RP-41	RP-080557	0697		Corrections to Radio Bearer Setup message for Rel-7	F	8.3.0	8.4.0	R5-083340
RP-41	RP-080557	0698		Corrections to RRC Connection Setup message for Rel-7	F	8.3.0	8.4.0	R5-083341
RP-41	RP-080557	0699		New Radio Bearer Combination for testing flexible size SRB in 8.2.2.57	F	8.3.0	8.4.0	R5-083342
RP-41	RP-080567	0700		New Radio Bearer Combinations for CS over HSPA	F	8.3.0	8.4.0	R5-083360
RP-41	RP-080568	0701		Add setup test procedures and Access Control Class settings for MBSFN MBMS testing	F	8.3.0	8.4.0	R5-083441
RP-41	RP-080555	0702		Correction to uplink beta factors in default messages	F	8.3.0	8.4.0	R5-083566

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-41	RP-080567	0703		New Radio Bearer Setup message contents for CS-HSPA RABS	F	8.3.0	8.4.0	R5-083591
RP-41	RP-080570	0704		Correction of HS-PDSCH physical layer categories for LCR TDD 64QAM Enhancement	F	8.3.0	8.4.0	R5-083645
RP-41	RP-080561	0705		Addition of radio bearer set-up default messages for RF testing for E-DCH and HSDPA operation for 3.84Mcps TDD and 7.68Mcps TDD	F	8.3.0	8.4.0	R5-083825
RP-42	RP-080954	0706		Correction to E-RGCH Info in Radio Bearer Setup	F	8.4.0	8.5.0	R5-085037
RP-42	RP-080952	0707		Correction to System Information Block type 11	F	8.4.0	8.5.0	R5-085039
RP-42	RP-080964	0708		Addition of Elementary Files (EFs) needed in Network Selection Enhancements tests	F	8.4.0	8.5.0	R5-085064
RP-42	RP-080968	0709		Addition of MBSFN RABs and Signalling RB for 1.28 Mcps TDD	F	8.4.0	8.5.0	R5-085130
RP-42	RP-080968	0710		Simulated network environment description for MBSFN tests	F	8.4.0	8.5.0	R5-085131
RP-42	RP-080968	0711		Default MBMS RRC message contents for LCR TDD MBSFN	F	8.4.0	8.5.0	R5-085132
RP-42	RP-080954	0712		Correction to A9 condition in RB set-up message	F	8.4.0	8.5.0	R5-085230
RP-42	RP-080967	0713		Introduction of radio bearer parameters for Improved L2 UL	F	8.4.0	8.5.0	R5-085422
RP-42	RP-080956	0714		Addition of Rel-7 IEs to default messages specified in 34.108	F	8.4.0	8.5.0	R5-085424
RP-42	RP-080956	0715		Correction of UL DPPCH slot format for default Radio Bearer Setup message for DTX/DRX	F	8.4.0	8.5.0	R5-085425
RP-42	RP-080954	0716		Addition of RF E-DCH test procedure for E-DPDCH with 2SF2+2SF4	F	8.4.0	8.5.0	R5-085726
RP-42	RP-080954	0717		Correction to Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)	F	8.4.0	8.5.0	R5-085727
RP-43	RP-090203	0718	-	Correction to E-DCH Transmission Time Interval	F	8.5.0	8.6.0	R5-090167
RP-43	RP-090203	0719	-	Correction to Radio Bearer Setup message (A17 & A18)	F	8.5.0	8.6.0	R5-090168
RP-43	RP-090216	0720	-	Addition 1.28 Mcps TDD contents in 6.11.5	F	8.5.0	8.6.0	R5-090312
RP-43	RP-090203	0721	-	Correction to the default RADIO BEARER SETUP message for condition A22	F	8.5.0	8.6.0	R5-090313
RP-43	RP-090203	0722	-	Correction 1.28 Mcps TDD default RADIO BEARER SETUP message in 9.1.2	F	8.5.0	8.6.0	R5-090314
RP-43	RP-090216	0723	-	Addition 1.28 Mcps TDD contents in 9.1.3	F	8.5.0	8.6.0	R5-090316
RP-43	RP-090217	0724	-	Supported Channels for MBSFN FDD	F	8.5.0	8.6.0	R5-090368
RP-43	RP-090217	0725	-	Simulated network environment for MBSFN FDD	F	8.5.0	8.6.0	R5-090369
RP-43	RP-090217	0726	-	Generic test procedure for MBSFN FDD	F	8.5.0	8.6.0	R5-090371
RP-43	RP-090217	0727	-	MBSFN FDD configurations for signalling test	F	8.5.0	8.6.0	R5-090373
RP-43	RP-090215	0728	-	Introduction of radio bearer parameters for testing Improved L2 UL	F	8.5.0	8.6.0	R5-090445
RP-43	RP-090212	0729	-	RRC Connection Setup Default Message for HS-DSCH Reception in CELL_FACH state	F	8.5.0	8.6.0	R5-090452
RP-43	RP-090212	0730	-	New Default SIB5/SIB5bis for Enhanced CELL FACH	F	8.5.0	8.6.0	R5-090456
RP-43	RP-090212	0731	-	Addition of Radio Bearer Setup Condition for Enhanced CELL_FACH	F	8.5.0	8.6.0	R5-090459
RP-43	RP-090203	0732	-	Addition of Rel-7 IE's to RRConnectionRequest and correction to Rel-7 IEs for Cell update and URA update under default messages specified in 34.108	F	8.5.0	8.6.0	R5-090465
RP-43	RP-090203	0733	-	Addition of Code Tree Allocation Table for 64QAM HSDPA Test Cases	F	8.5.0	8.6.0	R5-090466
RP-43	RP-090203	0734	-	Correction to Radio Bearer Setup for 64QAM test cases	F	8.5.0	8.6.0	R5-090467
RP-43	RP-090215	0735	-	Additions to the default RRC messages for improved L2 in uplink	F	8.5.0	8.6.0	R5-090559
RP-43	RP-090218	0736	-	Addition of radio bearer parameters for UE HS-DSCH Physical Layer category 19 and 20	F	8.5.0	8.6.0	R5-090562
RP-43	RP-090203	0737	-	Correction of radio bearer parameters for UE HS-DSCH Physical Layer category 15 and 18	F	8.5.0	8.6.0	R5-090563

Meeting-1st-Level	Doc-1st-Level	CR	Rev	Subject	Cat	Version-Current	Version-New	Doc-2nd-Level
RP-43	RP-090203	0738	-	Correction to the RAB Setup for CS over HSPA (A23 set of parameters) in 34.108	F	8.5.0	8.6.0	R5-090564
RP-43	RP-090215	0739	2	Update of Radio Bearer Configurations for flexible RLC	F	8.5.0	8.6.0	R5-090727
RP-43	RP-090212	0740	-	Introduction of radio bearer parameters for testing HS-DSCH Reception in CELL_FACH	F	8.5.0	8.6.0	R5-090728

History

Document history		
V8.0.0	October 2007	Publication
V8.1.0	January 2008	Publication
V8.2.0	April 2008	Publication
V8.3.0	July 2008	Publication
V8.4.0	October 2008	Publication
V8.5.0	January 2009	Publication
V8.6.0	April 2009	Publication